Elementary Real Analysis

Martin Klazar

(KAM MFF UK Praha)

Preliminary version of July 17, 2026

dedicated to the memory of Jiří Matoušek (1963–2015)

Was sich überhaupt sagen lässt, lässt sich klar sagen; ...[1]

L. Wittgenstein [80, Vorwort (Foreword)]

[1] We omit the well known conclusion of Wittgenstein's sentence "Was sich überhaupt sagen lässt, lässt sich klar sagen; und wovon man nicht reden kann, darüber muss man schweigen". "*What can be said at all, can be said clearly;* and what one cannot talk about, must be left in silence".

Contents

(^{c} ... means a relatively complete section.)

Introduction

This text extends to a book my lecture notes for the course *Mathematical Analysis 1*, which I was teaching in the School of Computer Science of MFF UK[2] in Praha for several years. When I am done with it, I hope to produce the Czech version. The book has fourteen chapters and two appendices. Each chapter begins with a summary and is divided into sections, which are further divided into passages. Each passage contains at least one exercise. Solutions and hints to these exercises are in Appendix B. Appendix A reviews notation and auxiliary and related notions.

The book is dedicated to the memory of Jiří Matoušek. He was my colleague in the Department of Applied Mathematics of MFF UK, and one of the greatest Czech mathematicians and computer scientists. He was preparing lectures for the analysis course in the school year 2014/15, but illness did not allow him to deliver them.

Praha and Louny, August 2024 to ??? 2026 Martin Klazar

[2]See the index of notation starting on p. 378.

Highlights

There exist many lecture notes, textbooks, and monographs on elementary real analysis; such as [2, 4, 12, 13, 18, 32, 40, 43, 47, 52, 55, 62, 65, 66, 75, 82]. Why write another text? Did not already *Nicolas Bourbaki (?–?)* write it all in [12, 14]? Simply put, I want to write it in a better way. (Many authors of the cited works were clearly motivated similarly.) Below I list some points of interest in my book.

Chapter 1. Five numeric domains. The chapter started as an overview of real numbers. In the final form, it provides a detailed development of the five numeric domains $\mathbb{N}_0$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{R}^*$. There are three novelties. Section 1.2 contains a rigorous construction of the structure $\mathbb{N}_0$ of natural numbers; in our approach, the principle of induction is a theorem following from more fundamental set-theoretic axioms. In Section 1.7 we treat the structure $\mathbb{R}^*$ of extended real numbers. Of the five algebraic characterization Theorems 1.2.5, 1.3.6, 1.4.7, 1.5.1, and 1.7.11 for the five domains, only the one for $\mathbb{R}$ is well known, and the last one for $\mathbb{R}^*$ is completely new. Thus, up to isomorphism, $\mathbb{N}_0$ is the unique simple ordered semiring, $\mathbb{Z}$ is the unique simple ordered ring, $\mathbb{Q}$ is the unique simple ordered field, $\mathbb{R}$ is the unique complete ordered field, and $\mathbb{R}^*$ is the limit closure of the linear order $\mathbb{R}$. In Section 1.1 we review terminology, notation, and notions related to functions between sets and relations on sets.

Chapter 2. Limits of real sequences. In Definition 2.1.14 we define simultaneously finite and infinite limits of real sequences. Equivalent definitions of limits were given in Definitions 1.4.22 and 1.7.3.

Chapter 1

Five numeric domains

The arena in which elementary real analysis takes place is the complete ordered field of real numbers $\mathbb{R}$. In this chapter, we build the hierarchy of five numeric domains $\mathbb{N}_0$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{R}^*$. It starts with the natural numbers $\mathbb{N}_0$ and ends with the extended reals $\mathbb{R}^*$. For each domain, we obtain an algebraic characterization theorem.

- Section 1.2 is devoted to natural numbers $\mathbb{N}_0$. We define their set ω in the framework of ZFC set theory via the axiom of infinity, see Appendix A.2. Theorem 1.2.5 states that $\mathbb{N}_0$ is the up to isomorphism unique ordered semiring such that every element in it can be obtained as a sum of ones. The principle of induction for natural numbers is, in our approach, a theorem following from more fundamental set-theoretic axioms.

- Section 1.3 concerns the integers $\mathbb{Z}$. Theorem 1.3.6 states that $\mathbb{Z}$ is the up to isomorphism unique ordered ring such that every element in it can be obtained as a sum of ones and minus ones.

- In Section 1.4 we build the fractions $\mathbb{Q}$. Theorem 1.4.7 states that $\mathbb{Q}$ is the up to isomorphism unique ordered field such that every element in it can be obtained as a sum of ones and minus ones, or as a ratio of previously obtained elements.

- Section 1.5 introduces, by means of Dedekind's cuts, the real numbers $\mathbb{R}$. Theorem 1.5.1 states that $\mathbb{R}$ is the up to isomorphism unique ordered field in which every nonempty and upper-bounded set has a supremum. In the short Section 1.6 we show that $\mathbb{R}$ is an uncountable set.

- In the novel Section 1.7 we introduce the structure of extended reals $\mathbb{R}^*$. By Theorem 1.7.11, $\mathbb{R}^*$ is the up to isomorphism unique limit closure of the linear order $\mathbb{R}$. Theorem 1.7.14 describes the partial operations in the structure $\mathbb{R}^*$.

In Appendix A.5 we review another extension of $\mathbb{R}$, the complex numbers $\mathbb{C}$. Theorem A.5.2 states that $\mathbb{C}$ is the up to isomorphism unique field that extends the field $\mathbb{R}$ and has degree 2 over $\mathbb{R}$. The reader may wish to compare how these basic numeric domains are developed in the book *Numbers* [29].

Main actors in the arena are real functions of one real variable. Section 1.1 is therefore devoted to an overview of functions and relations. In Definitions 1.1.1–1.1.3 we introduce functions and their congruence. Then we overview various types of functions, homomorphisms, images and preimages of sets by functions, inverses of functions and compositions of functions, set systems, the axiom of choice, equivalence relations and partitions, linear and partial orders, infima and suprema, and well orderings.

1.1 Functions and relations

We define functions and their congruence.

- *Functions and congruence of functions.* See Definition A.1.3 for k-tuples

$$\langle x_1,\, x_2,\, \dots,\, x_k\rangle,\ k \in \mathbb{N} = \{1,\, 2,\, \dots\}\,.$$

The Cartesian product of sets A and B is the set

$$A \times B := \{\langle a,\, b\rangle\colon\ a \in A,\, b \in B\}\,.$$

Any set $C \subset A \times B$ is a binary relation between A and B. Instead of $\langle a,\, b\rangle \in C$ we write $a\,C\,b$, for example $2 < 5$. If $A = B$, we speak of a relation on A.

Definition 1.1.1 *A relation C between A and B is functional if for every element $a \in A$ there is exactly one element $b \in B$ such that $a\,C\,b$.*

Definition 1.1.2 *Let A and B be sets. A function or a map f from A to B, written $f\colon A \to B$, is any triple*

$$f = \langle A,\, B,\, G_f\rangle$$

such that G_f, the graph of f, is a functional relation between A and B. We call A the domain of f and denote it by $M(f)$, and B is the range of f. If $a \in A$, we write $f(a)$ for the unique $b \in B$ such that $\langle a, b\rangle \in G_f$. We call $f(a)$ the value of the function f at the argument a.

Thus $G_f = \{\langle x, f(x)\rangle\colon\ x \in A\}$. Instead of $f\colon A \to B$ we also write

$$A \ni a \mapsto \varphi(a) \in B\,,$$

where $\varphi(a)$ is a formula producing the value $f(a)$.

Any description of a family of mathematical objects should include conditions under which two objects are regarded as practically identical, even if they differ as sets. We call the resulting relation the congruence of the family. For example, isomorphisms of algebraic and combinatorial structures are congruences. Congruences are usually equivalences (Definition 1.1.33). We introduce the congruence of functions.

Definition 1.1.3 *Two functions are congruent, which means practically identical, if they differ at most in their ranges.*

Exercise 1.1.4 *Functions $f = \langle A, B, G_f \rangle$ and $g = \langle C, D, G_g \rangle$ are congruent if and only if $G_f = G_g$.*

• *Partial functions.* Let A and B be sets. A relation C between A and B is a partial function from A to B if for every $a \in A$ there is at most one $b \in B$ such that $a\,C\,b$. Partial functions are common in recursion theory: if an algorithm does not terminate on an input x, then the corresponding function is not defined at x. Also, the subtraction of natural numbers $m - n$ and the division a/b in a field F, which we introduce later, are partial functions. The former is not defined if $m < n$, and the latter if $b = 0_F$.

Exercise 1.1.5 *What is the relation of functions and partial functions?*

• *Empty functions.* What is an empty function? For example, the function

$$f_\emptyset := \langle \emptyset, \emptyset, \emptyset \rangle .$$

A function is empty if it is congruent to $f_\emptyset$.

Exercise 1.1.6 *Empty functions are functions of the form $\langle \emptyset, X, \emptyset \rangle$.*

• *Images, preimages, and restrictions.* Let $f\colon A \to B$ be a function and C be *any* set. We define sets

$$\begin{aligned} f[C] &:= \{f(a)\colon\ a \in C \cap A\}\ (\subset B),\ \text{the image of } C \text{ by } f, \text{ and} \\ f^{-1}[C] &:= \{a \in A\colon\ f(a) \in C\}\ (\subset A),\ \text{the preimage of } C \text{ by } f\,. \end{aligned}$$

C is arbitrary, it is practical to drop the usual restrictions $C \subset A$ and $C \subset B$. The set $f[A]$ $(\subset B)$ is the image of the function f.

Exercise 1.1.7 *Is it true that $f^{-1}[f[C]] = C$ and that $f[f^{-1}[C]] = C$?*

Let $f\colon A \to B$ be a function and C be any set. The restriction of f to C is the function $f \,|\, C\colon A \cap C \to B$ such that for every $x \in A \cap C$,

$$(f \,|\, C)(x) := f(x)\,.$$

In this situation, we say that f extends $f \,|\, C$. It is clear that if functions f_1 and f_2 are congruent, then for any set C the restrictions $f_1 \,|\, C$ and $f_2 \,|\, C$ are congruent as well. We say that a function f is a subfunction of a function g if $G_f \subset G_g$. If f and g are functions, C is any set and the restrictions $f \,|\, C$ and $g \,|\, C$ are congruent, we say that $f = g$ on C.

Exercise 1.1.8 *Every empty function is a subfunction of every function.*

• *Sequences, words, and operations.* Let $\omega = \{0, 1, \dots\}$ be the set of natural numbers introduced in Section 1.2, and let $\mathbb{N} := \omega \setminus \{0\} = \{1, 2, \dots\}$. For $n \in \mathbb{N}$, let $[n] := \{1, 2, \dots, n\}$. We set $[0] := \emptyset$. Let X be any set.

Definition 1.1.9 *Three important families of functions are sequences, words, and operations.*

1. *Functions $a\colon \mathbb{N} \to X$ are sequences in X.*
2. *Functions $u\colon [n] \to X$, for $n \in \omega$, are words over the alphabet X.*
3. *Functions $o\colon X \times X \to X$ are binary operations on X.*

For a sequence a in X and an index $n \in \mathbb{N}$, we set $a_n := a(n)$. We invoke a sequence in X by writing

$$(a_n) \subset X\,.$$

A sequence $(b_n) \subset X$ is a subsequence of (a_n) if there exists a sequence of natural numbers $1 \le m_1 < m_2 < \dots$ such that $b_n = a_{m_n}$ for every $n \in \mathbb{N}$.

A word u over X is written as

$$u = u_1 u_2 \dots u_n\,,$$

with $u_i := u(i)$, $i \in [n]$. A word $v = v_1 v_2 \dots v_m$ is a subword of u if $v_1 = u_i$, $v_2 = u_{i+1}$, $\dots$, $v_m = u_{i+m-1}$ for some $i \in [n]$ (hence $m \le n$). For $n = 0$ we have the empty word over X

$$u_\emptyset := \langle \emptyset,\, X,\, \emptyset \rangle\,,$$

which is an empty function; $u_\emptyset$ is a subword of any word. The elements in the alphabet X, and sometimes also the terms u_i in u, are called letters. The set of words over X is denoted by X^* and the length n ($\in \omega$) of the word u is denoted by $|u|$. If $u = u_1 \dots u_m$ and $v = v_1 \dots v_n$ are words in X^*, their concatenation is the word $uv = w = w_1 \dots w_{m+n} \in X^*$ given by

$$w_i = u_i \text{ for } i \in [m] \text{ and } w_i = v_{i-m} \text{ for } i \in [m+n] \setminus [m]\,.$$

Iterating the concatenation operation, we can speak of the concatenation of several (but always finitely many) words. In view of Exercises 1.1.12 and 1.1.13, only the order of these words is relevant.

The value

$$o(\langle a,\, b \rangle) = c$$

of an operation $o\colon X \times X \to X$ is written as $a\,o\,b = c$, for example $1 + 1 = 2$. An element $b \in X$ such that $b\,o\,a = a$ for every $a \in X$ is called neutral to o. The operation o is commutative if always $a\,o\,b = b\,o\,a$. It is associative if always $(a\,o\,b)\,o\,c = a\,o\,(b\,o\,c)$. Another operation p on X is distributive to o if always $a\,p\,(b\,o\,c) = (a\,p\,b)\,o\,(a\,p\,c)$. We follow the usual convention that, in the absence of brackets, the operation denoted by $\cdot$ has precedence over the operation denoted by $+$. The distributivity of $\cdot$ to $+$ is then written as $a \cdot (b + c) = a \cdot b + a \cdot c$. We often abbreviate $a \cdot b$ by ab. A function $o\colon X \to X$ is called a unary operation on X.

Exercise 1.1.10 *Neutral elements to commutative operations are uniquely determined.*

Exercise 1.1.11 *Count words of length five over a three-element alphabet.*

Exercise 1.1.12 *Let X be any alphabet. The concatenation operation on X^* is associative.*

Exercise 1.1.13 *However, it is not commutative.*

- *Injective and other functions.* Let $f\colon X \to Y$ be a function.

1. We say that f is injective or an injection if $x \neq x' \Rightarrow f(x) \neq f(x')$.
2. We say that f is surjective or onto or a surjection if $f[X] = Y$.
3. The function f is bijective or a bijection if f is surjective and injective.
4. The function f is constant if it has exactly one value.
5. The function f is an identity function if $f(x) = x$ for every $x \in X$.

More narrowly, the identity on a set X is the map $\mathrm{id}_X\colon X \to X$ given by $\mathrm{id}_X(x) = x$.

Exercise 1.1.14 *When is the identity function from X to Y bijective?*

Exercise 1.1.15 *Suppose that two functions are congruent by Definition 1.1.3. Is it true that they both are, or are not, injective? Same question for surjective, bijective, constant, and identity functions.*

We reveal one answer: bijective functions are not preserved by congruence. Thus, bijectiveness is not an essential property of functions.

- *Homomorphisms.* We make use of them in the following sections.

Definition 1.1.16 *Suppose that p_i for $i \in I$ are operations on a set X and that q_i for $i \in I$ are operations on a set Y. A homomorphism to these operations is a map $f\colon X \to Y$ such that for every $a, b \in X$ and $i \in I$,*

$$f(a\, p_i\, b) = f(a)\, q_i\, f(b)\,.$$

If f is bijective, we call it an isomorphism to the mentioned operations.

Exercise 1.1.17 *If $f\colon X \to Y$ is an isomorphism to operations p on X and q on Y and $a \in X$ is neutral to p, then $f(a)$ is neutral to q. What if f is just a homomorphism?*

Exercise 1.1.18 *Let f, p, and q be as in the previous exercise. Then also $f^{-1}\colon Y \to X$ is an isomorphism to the operations q and p.*

Exercise 1.1.19 *Let f, p, and q be as before and let $g\colon Y \to Z$ be an isomorphism to operations q and r, where r is an operation on Z. Then $g(f)$ is an isomorphism to p and r.*

• *Inverses and compositions.* Let $f\colon X \to Y$ be an injective map. The inverse function f^{-1} of f is the function

$$f^{-1}\colon f[X] \to X, \text{ given by } f^{-1}(y) = x \iff f(x) = y\ .$$

Non-injective functions do not have inverses.

Exercise 1.1.20 *Is it a problem that $f^{-1}[A]$ means both the preimage of A by f, and the image of A by f^{-1}?*

Exercise 1.1.21 *Let f be injective. Are f and $(f^{-1})^{-1}$ congruent? Are they equal as sets? Same questions for f^{-1} and $((f^{-1})^{-1})^{-1}$.*

We introduce the kind of composition of functions that is used in mathematical analysis.

Definition 1.1.22 *Let $g\colon X \to Y$ and $f\colon A \to B$ be any pair of functions. Their composition $f(g)\colon X' \to B$ has the domain*

$$X' = \{x \in X\colon\ g(x) \in A\} \quad (= g^{-1}[A])$$

and values $f(g)(x) := f(g(x))$. We call g the inner function, and f the outer function.

For example, if

$$g(x) = 1 - x\colon \mathbb{R} \to \mathbb{R} \text{ and } f(x) = \sqrt{x}\colon [0, +\infty) \to \mathbb{R}\,,$$

then $f(g)(x) = \sqrt{1-x}\colon (-\infty, 1] \to \mathbb{R}$ and $g(f)(x) = 1 - \sqrt{x}\colon [0, +\infty) \to \mathbb{R}$.

Exercise 1.1.23 *Let f_1, f_2 and g_1, g_2 be pairs of congruent functions. Are the compositions $f_1(g_1)$ and $f_2(g_2)$ congruent?*

Exercise 1.1.24 *Composition of two injections is an injection. The composition $f(g)$ of surjections $g\colon X \to Y$ and $f\colon Y \to B$ is a surjection. In general, composition of two surjections need not be a surjection.*

Exercise 1.1.25 *For every three functions f, g, and h, the two compositions $f(g(h))$ and $f(g)(h)$ are equal as sets.*

Exercise 1.1.26 *For every map $h\colon X \to Z$ there exists a set Y and maps $g\colon X \to Y$ and $f\colon Y \to Z$ such that $h = f(g)$, g is surjective, and f is injective.*

Exercise 1.1.27 *A map $f\colon X \to Y$ is a bijection $\iff$ there exists a map $g\colon Y \to X$ such that $f(g)$ is id_Y and $g(f)$ is id_X.*

Exercise 1.1.28 *Which constant functions can be inverted?*

• *Set systems.* Set systems are just maps . Let $I \neq \emptyset$ be a set. A set system

$$S = \{A_i\colon\ i \in I\}$$

indexed by I is just a map $f\colon I \to B$ to some set B, and the set corresponding to $i \in I$ is $A_i := f(i)$. The union of S is

$$\textstyle\bigcup_{i\in I} A_i = \bigcup\{A_i:\ i \in I\} := \bigcup f[I]\,.$$

The intersection of S is defined by

$$\textstyle\bigcap_{i\in I} A_i = \bigcap\{A_i:\ i \in I\} := \bigcap f[I]\,.$$

Exercise 1.1.29 *Explain the notation $\bigcup_{i\geq 1} A_i$, $\bigcup_{i=1}^{\infty} A_i$, and $\bigcap_{i=0}^{\infty} A_i$.*

• *The axiom of choice.* In mathematics, this set-theoretic axiom plays an important role.

Axiom 1.1.30 (AC) *The axiom of choice, abbreviated* AC, *is any of the three following statements which are equivalent.*

1. *Every set system $\{A_i\colon i \in I\}$ of nonempty sets has a selector, a map $S\colon I \to \bigcup_{i\in I} A_i$ such that $S(i) \in A_i$ for every $i \in I$.*
2. *For every set X of nonempty and mutually disjoint sets there is a set Y such that $Z \cap Y$ is a one-element set for every set $Z \in X$.*
3. *For every surjection $f\colon A \to B$ there is a function $g\colon B \to A$ such that $f(g)$ is id_B.*

Exercise 1.1.31 *These three formulations of* AC *are mutually equivalent, in the sense that each can be easily derived from any other.*

Exercise 1.1.32 *In part 2 of Axiom 1.1.30, the assumption of disjointness cannot be omitted.*

What is AC good for? It formalizes the intuitively clear, but practically unrealizable, act of selecting an element from each of *infinitely many* nonempty sets $\{A_i\colon\ i \in I\}$. If this set system is finite, we do not need AC for making the selection. The existence of the selector then follows from other set-theoretic axioms.

• *Equivalence relations.* A relation R on a set A is reflexive if $a\,R\,a$ for every a in A. It is irreflexive if $a\,R\,a$ for no $a \in A$. It is symmetric if $a\,R\,b$ implies $b\,R\,a$ for every $a, b \in A$. It is transitive if $a\,R\,b$ and $b\,R\,c$ imply $a\,R\,c$ for every $a, b, c \in A$.

Definition 1.1.33 *An equivalence relation on a set is a reflexive, symmetric, and transitive relation on the set.*

Definition 1.1.34 *A partition of a set B is any set A such that the elements of A are nonempty and mutually disjoint, and $\bigcup A = B$. The elements of A are called blocks of the partition.*

Let R be an equivalence relation on a set A. For any $a \in A$ we define

$$[a]_R := \{b \in A\colon\ b\,R\,a\}\ \ (\subset A)\,.$$

We call the set $[a]_R$ the block of the element a in the relation R. It is clear that if $a\,R\,b$ then $[a]_R = [b]_R$. We set

$$A/R := \{[a]_R\colon\ a \in A\}\,.$$

Exercise 1.1.35 *If R is an equivalence relation on a set A then A/R is a partition of A. Elements $b, c \in A$ are in one block of A/R iff $b\,R\,c$.*

Let X be a partition of Y. We define the relation Y/X on Y by

$$x\,(Y/X)\,y \iff (\exists\, Z \in X\colon x \in Z \wedge y \in Z)\,.$$

Exercise 1.1.36 *If X is a partition of Y then Y/X is an equivalence relation on Y. Elements $x, y \in Y$ are in one block of X iff $x\,(Y/X)\,y$.*

Exercise 1.1.37 *If R is an equivalence relation on A and B is a partition of A then*

$$A/(A/R) = R \ \text{ and } \ A/(A/B) = B\,.$$

• *Linear orders.* A relation R on a set A is trichotomic if for every $a, b \in A$ it is true that $a\,R\,b$ or $b\,R\,a$ or $a = b$. R is dichotomic if for every $a, b \in A$ it is true that $a\,R\,b$ or $b\,R\,a$. R is asymmetric if never $a\,R\,b$ and $b\,R\,a$. Finally, R is weakly asymmetric if $a\,R\,b$ and $b\,R\,a$ imply $a = b$.

Definition 1.1.38 *A relation on a nonempty set is a linear order if it is irreflexive, transitive, and trichotomic.*

Exercise 1.1.39 *Every linear order is asymmetric.*

We denote a linear order $<$ on a set A as a pair $\langle A, <\rangle$. Let $a, b \in A$. Notation $a \le b$ means that $a < b$ or $a = b$. Notation $a > b$ is synonymous with $b < a$, and $a \ge b$ is synonymous with $b \le a$. Relations $<$ and $>$ are strict linear orders. Relations $\le$ and $\ge$ are non-strict linear orders.

Exercise 1.1.40 *Any non-strict linear order is reflexive, transitive, dichotomic and weakly asymmetric.*

Exercise 1.1.41 *Let $\langle A, <\rangle$ be a linear order. Then for every pair of elements $a, b \in A$ exactly one of $a < b$, $b < a$, and $a = b$ holds.*

Linear orders $\langle A, <\rangle$ and $\langle B, \prec\rangle$ are isomorphic if there exists a bijection $f\colon A \to B$ such that for any $a, a' \in A$ we have $a < a' \iff f(a) \prec f(a')$.

Exercise 1.1.42 *In the previous definition, $\iff$ can be weakened to $\Rightarrow$.*

• *Suprema and infima.* Let $\langle A, <\rangle$ be a linear order and let $B \subset A$. We say that B is bounded from above if for some $h \in A$ we have $h \ge b$ for every $b \in B$. Then h is an upper bound of B. We denote the set of upper bounds of B by $H(B)$ ($\subset A$). We similarly define boundedness from below, lower bounds, and the set $D(B)$ ($\subset A$) of lower bounds of B. An element $m \in B$ is the maximum of B if $m \in H(B)$. We similarly define the minimum of B. We denote these elements by $\max(B)$ and $\min(B)$. If $B = \emptyset$, they are not defined.

Exercise 1.1.43 *Show that maxima and minima are unique.*

Exercise 1.1.44 *Any nonempty finite subset in any linear order has both maximum and minimum.*

We define suprema and infima in linear orders.

Definition 1.1.45 *Let $\langle A, <\rangle$ be a linear order and $B \subset A$. The elements*

$$\sup(B) := \min(H(B)) \ (\in A) \ \text{ and } \ \inf(B) := \max(D(B)) \ (\in A)$$

are called the supremum of B and the infimum of B, respectively.

Suprema and infima need not exist. In contrast to maxima and minima, suprema and infima may lie outside the considered set.

Exercise 1.1.46 *Suprema and infima are unique.*

Exercise 1.1.47 *Prove the following proposition. State and prove the analogous result for infima.*

Proposition 1.1.48 *Let $\langle A, <\rangle$ be a linear order, $B \subset A$, and $c \in A$. Then $c = \sup(B) \iff b \leq c$ for every $b \in B$ and for every $a \in A$ with $a < c$ there exists $b \in B$ such that $a < b$.*

• *Well orderings.* This is an important and useful type of linear orders.

Definition 1.1.49 *A linear order $\langle A, <\rangle$ is a well ordering if every nonempty subset of A has a minimum element.*

Exercise 1.1.50 *The standard linear order $\langle \omega, <\rangle$ of natural numbers is a well ordering.*

Exercise 1.1.51 *The standard linear order $\langle \mathbb{Z}, <\rangle$ of integers is not a well ordering.*

Exercise 1.1.52 *A linear order $\langle A, <\rangle$ is a well ordering $\iff$ there do not exist elements $a_n \in A$, $n \in \mathbb{N}$, such that $a_1 > a_2 > \cdots$.*

• *Partial orders.* Partial orders, often called just orders, generalize linear orders.

Definition 1.1.53 *A relation on a set is an order if it is irreflexive and transitive.*

In an order $\langle A, <\rangle$, we may have (distinct) incomparable elements $a, b \in A$ such that neither $a < b$ nor $b < a$ holds. Like for linear orders, we associate with $\langle A, <\rangle$ the non-strict order $\langle A, \leq\rangle$ by setting $a \leq b$ iff $a < b$ or $a = b$. If $B \subset A$ and $b \in B$, we say that b is a maximal element in B, if $b < c \in B$ for no c. Minimal elements are defined similarly.

Exercise 1.1.54 *Every order is asymmetric. Every non-strict order is reflexive, transitive and weakly asymmetric.*

Exercise 1.1.55 *For every set x and subset $y \subset \mathcal{P}(x)$ of the power set of x we have the non-strict order $\langle y, \subset\rangle$.*

1.2 Natural numbers

We introduce the basic mathematical structure of natural numbers. We define their set in the framework of ZFC set theory described in Appendix A.2. The main result is the algebraic characterization of natural numbers in Theorem 1.2.5 as the up to isomorphism unique simple ordered semiring.

• *Ordered semirings and simple semirings.* We introduce the algebraic structure that fits the natural numbers.

Definition 1.2.1 *An* <u>*ordered semiring*</u> *is an algebraic structure*

$$S = \langle S,\, 0_S,\, 1_S,\, +,\, \cdot,\, < \rangle$$

such that S *is a set,* $0_S, 1_S \in S$ *and* $0_S \neq 1_S$*,* $+$ *and* $\cdot$ *are operations on* S*,* $<$ *is a relation on* S*, and the following holds.* 0_S *is neutral to* $+$*, and* 1_S *to* $\cdot$*, operations* $+$ *and* $\cdot$ *are commutative and associative,* $\cdot$ *is distributive to* $+$*, the relation* $<$ *is a linear order,* $0_S < 1_S$*, and for every* $a, b, c \in S$ *we have*

$$a < b \Rightarrow a + c < b + c\,.$$

We call this implication the <u>*first order axiom*</u>*. The inequality* $0_S < 1_S$ *is the* <u>*second order axiom*</u>*. If we omit* $<$*, we get the structure of a* <u>*semiring*</u>*.*

A subset of S that contains 0_S and 1_S, and is closed under the operations $+$ and $\cdot$, induces a <u>sub-semiring</u> of the semiring S. If S and S' are semirings, a map $f\colon S \to S'$ is a <u>semiring homomorphism</u> if it is a homomorphism to their operations and, moreover, $f(0_S) = 0_{S'}$ and $f(1_S) = 1_{S'}$. If S and S' are ordered and f in addition is bijective and preserves the orders, we say that f is an <u>isomorphism of ordered semirings</u>.

Exercise 1.2.2 *Neutral elements* 0_S *and* 1_S *are in every semiring uniquely determined.*

Exercise 1.2.3 *Every ordered semiring is infinite. Semirings may be finite.*

Exercise 1.2.4 *Deduce from the first order axiom its non-strict version. If* $a \le b$*, then* $a + c \le b + c$ *for every* c*.*

A semiring S is <u>simple</u> if for any set $X \subset S$ the following holds. If $0_S \in X$ and if for any $x \in X$ also $x + 1_S \in X$, then $X = S$.

Theorem 1.2.5 *There exists a simple ordered semiring. Every two simple ordered semirings are isomorphic.*

Our main goal is to prove Theorem 1.2.5. The proof goes as follows. We construct the natural numbers as sets $0 = \emptyset$, $1 = \{\emptyset\}$, $2 = \{\emptyset, \{\emptyset\}\}$, $\ldots$ and show in Proposition 1.2.36 that they form a simple ordered semiring $\mathbb{N}_0$. In Proposition 1.2.57, we use a semiring homomorphism f_S from the natural numbers to a semiring S, to show that any two simple ordered semirings are isomorphic. In the theorem, the adjective "ordered" cannot be omitted: there exist both finite and infinite simple semirings. We arrive at the following class perspective on natural numbers.

Corollary 1.2.6 *The class*

$$\text{NATURAL NUMBERS} := \{x\colon\ x \textit{ is a simple ordered semiring}\}$$

contains the "standard" natural numbers $\mathbb{N}_0$ *and every two sets in it are isomorphic as ordered semirings.*

For classes, see Appendices A.2 and A.3.

• *Neutral elements and isomorphisms.* It is useful to note the following fact.

Exercise 1.2.7 *Prove the next proposition.*

Proposition 1.2.8 *Suppose that S and S' are semirings, and that $f\colon S \to S'$ is an isomorphism to their operations.* Then

$$f(0_S) = 0_{S'} \ \text{ and } \ f(1_S) = 1_{S'}\,.$$

• *Multiplying by zero.* In natural numbers, we are used to equalities $0 \cdot n = n \cdot 0 = 0$ for every number n. We show that in general semirings S it is possible that $a \cdot 0_S \ne 0_S$.

Exercise 1.2.9 *Let S be any semiring. We define $S_\infty := S \cup \{\infty\}$, for a new element $\infty \notin S$. For every $a \in S_\infty$ we set $a + \infty = \infty + a := \infty$ and $a \cdot \infty = \infty \cdot a := \infty$. In particular, $0_S \cdot \infty = \infty \cdot 0_S = \infty \ne 0_S$. Show that S_∞ is a semiring.*

We show that in simple or ordered semirings S, it is always $0_S \cdot a = 0_S$.

Proposition 1.2.10 *Let S be a semiring.* Then

$$0_S \cdot 0_S = 0_S\,.$$

Proof. Indeed, writing $0 = 0_S$ and $1 = 1_S$, by Exercise 1.2.11 we have

$$0 = 0 \cdot 1 = 0 \cdot (0 + 1) = 0 \cdot 0 + 0 \cdot 1 = 0 \cdot 0 + 0 = 0 \cdot 0\,.$$

□

Exercise 1.2.11 *Justify the above five equalities.*

Now we can treat simple semirings.

Proposition 1.2.12 *Let S be a simple semiring.* Then *for every $a \in S$ we have*

$$0_S \cdot a = a \cdot 0_S = 0_S\,.$$

Proof. We again write $0 = 0_S$ and $1 = 1_S$. Let $X \subset S$ be the set of $a \in S$ such that $0 \cdot a = 0$. We have $0 \in X$ by Proposition 1.2.10. Since S is simple, it suffices to show that if $a \in X$, then $a + 1 \in X$. Let $a \in X$. Then, indeed,

$$0 \cdot (a + 1) = 0 \cdot a + 0 \cdot 1 = 0 + 0 = 0$$

and $a + 1 \in X$. □

We turn to ordered semirings.

Proposition 1.2.13 *Let S be an ordered semiring. Then for every $a \in S$ we have*

$$0_S \cdot a = a \cdot 0_S = 0_S\,.$$

Proof. Let $a \in S$. We denote $b = 0_S \cdot a$. The element b is idempotent to $+$:

$$b + b = 0_S \cdot a + 0_S \cdot a = (0_S + 0_S) \cdot a = 0_S \cdot a = b\,.$$

If $b \neq 0_S$, then the trichotomy of $<$ implies that either $b < 0_S$ or $b > 0_S$. Adding b to both sides of the inequalities and using the first order axiom, we get the contradiction that

$$b = b + b < 0_S + b = b \text{ or } b = b + b > 0_S + b = b\,.$$

Thus $b = 0_S$. □

• *The set of natural numbers.* To define an algebraic structure built on the set of natural numbers, we first define this set. Recall from Definition A.2.4 that a set X is inductive if (i) $\emptyset \in X$ and if (ii) for every set b it is true that if $b \in X$ then $b \cup \{b\} \in X$.

Definition 1.2.14 *The set ω of natural numbers is defined with the help of Axiom A.2.5 of infinity, by which there exists an inductive set x_0, and Axiom A.2.3 of separation as the set*

$$\omega := \{x \in x_0\colon\ \forall\, y\,(y \textit{ is inductive} \to x \in y)\}\,.$$

Exercise 1.2.15 *The sets $0 := \emptyset$, $1 := \{\emptyset\}$, and $2 := \{\emptyset, \{\emptyset\}\}$ are natural numbers.*

• *Properties of natural numbers.* We begin with the principle of induction for natural numbers.

Lemma 1.2.16 *The set ω is an inductive set and is a subset of every inductive set.*

Proof. This follows from Definition 1.2.14. □

Here is the principle of induction for ω.

Theorem 1.2.17 *Let $X \subset \omega$ be an inductive set. Then $X = \omega$.*

Proof. Since X is an inductive set, Lemma 1.2.16 implies that $\omega \subset X$. Thus $X = \omega$. □

We show, among other, that every natural number is a subset of ω.

Proposition 1.2.18 *Let $m, n \in \omega$. The following holds.*

1. $0 \in m$ *or* $m = 0$.

2. If $m \neq 0$*, then* $m = l \cup \{l\}$ *for a unique* $l \in \omega$.

3. $m \cup \{m\} \in \omega$.

4. $m \subset \omega$.

5. If $m \in n$*, then* $m \subset n$.

6. If $m \in n$*, then* $m \cup \{m\} \in n$ *or* $m \cup \{m\} = n$.

Proof. 1. Let X be the set of $m \in \omega$ with the stated property. Obviously, $\emptyset \in X$. Let $m \in X$. If $\emptyset \in m$, then also $\emptyset \in m \cup \{m\}$. If $m = \emptyset$, then also $\emptyset \in m \cup \{m\} = \{\emptyset\}$. Thus $m \cup \{m\} \in X$ and $X = \omega$ by Theorem 1.2.17.

2. We first prove the uniqueness. If $x \cup \{x\} = y \cup \{y\}$ for sets x, y such that $x \neq y$, then the axiom of extensionality implies that $x \in y \in x$, which contradicts the Axiom A.2.2 of foundation. Let X be the set of $m \in \omega$ such that $m = \emptyset$ or m has the stated representation. Then $\emptyset \in X$ by definition. Let $m \in X$. Then $m \in \omega$ and $m \cup \{m\}$ has trivially the stated representation. Thus $m \cup \{m\} \in X$ and $X = \omega$ by Theorem 1.2.17.

3. This follows from Lemma 1.2.16.

4. Let X be the set of $m \in \omega$ such that $m \subset \omega$. Then $\emptyset \in X$ because $\emptyset$ is a subset of every set. Let $m \in X$. Then $m \in \omega$, $m \subset \omega$, and we deduce that also $m \cup \{m\} \subset \omega$. Thus $m \cup \{m\} \in X$ and $X = \omega$ by Theorem 1.2.17.

5. Let X be the set of $n \in \omega$ such that the implication $m \in n \Rightarrow m \subset n$ holds. Then $\emptyset \in X$ because $\emptyset$ has no elements. Let $n \in X$ and $m \in n \cup \{n\}$. If $m \in n$ then $m \subset n \cup \{n\}$ because $m \subset n$ by the assumption that $n \in X$. If $m = n$ then again $m = n \subset n \cup \{n\}$. Thus $n \cup \{n\} \in X$ and $X = \omega$ by Theorem 1.2.17.

6. Let X be the set of $n \in \omega$ such that the implication $m \in n \Rightarrow m \cup \{m\} \in n$ or $m \cup \{m\} = n$ holds. Then $\emptyset \in X$ because $\emptyset$ has no elements. Let $n \in X$ and $m \in n \cup \{n\}$. If $m \in n$ then $m \cup \{m\}$ is an element of n or equals n by the assumption that $n \in X$. In either case $m \cup \{m\} \in n \cup \{n\}$. If $m = n$ then $m \cup \{m\} = n \cup \{n\}$. Thus $n \cup \{n\} \in X$ and $X = \omega$ by Theorem 1.2.17. □

We use parts 2 and 3 in the following definition.

Definition 1.2.19 *Let* $m, n \in \omega$ *with* $m \neq 0$*. We define the natural numbers* $m - 1 := k$*, where* $m = k \cup \{k\}$*, and* $n \oplus 1 := n \cup \{n\}$.

Exercise 1.2.20 *Let* $m, n \in \omega$ *with* $m \neq 0$*. Then* $(m - 1) \oplus 1 = m$ *and* $(n \oplus 1) - 1 = n$.

• *The linear order on natural numbers.* We define the relation $<$ on ω by $m < n \iff m \in n$.

Proposition 1.2.21 *The relation* $<$ *is a linear order on* ω.

Proof. The irreflexivity of $<$ follows from Exercise 1.2.22. We show that $<$ is transitive. Let $k \in l \in m \in \omega$. By part 3 of Proposition 1.2.18, we have $l \subset m$, so that $k \in m$. We show that $<$ is trichotomic. Let $n \in \omega$ and let X be the set of $m \in \omega$ such that $n \in m$ or $n = m$ or $m \in n$. Then $\emptyset \in X$ by part 1 of Proposition 1.2.18. Let $m \in X$ and consider the set $p = m \cup \{m\}$. If $n \in m$ or $n = m$ then $n \in p$. If $m \in n$ then $p \in n$ or $p = n$ by part 4 of Proposition 1.2.18. Thus $p \in X$ and $X = \omega$ by Theorem 1.2.17. $\square$

Exercise 1.2.22 *Show that the relation $<$ on ω is irreflexive.*

We show that $\langle \omega, < \rangle$ is a well ordering. We need an auxiliary proposition.

Proposition 1.2.23 *For every $m \in \omega$, the linear order $\langle m, < \rangle$ is a well ordering.*

Proof. Let X be the set of $m \in \omega$ with the stated property. Then $\emptyset \in X$ because $\emptyset$ has no nonempty subset. Let $m \in X$ and $p = m \cup \{m\}$. Let $\emptyset \neq A \subset p$. If $A \cap m \neq \emptyset$, then $\min(A) = \min(A \cap m)$ by the assumption that $m \in X$ and because m is the maximum of p. If $A \cap m = \emptyset$, then $A = \{m\}$ and m is the minimum of A. Thus $p \in X$ and $X = \omega$ by Theorem 1.2.17. $\square$

The following theorem is basic.

Theorem 1.2.24 *The linear order $\langle \omega, < \rangle$ is a well ordering.*

Proof. Let $\emptyset \neq A \subset \omega$. We take any $m \in A$. If $m \cap A = \emptyset$, then $\min(A) = m$. If $m \cap A \neq \emptyset$, then $\min(A) = \min(m \cap A)$ by Proposition 1.2.23. $\square$

• *Inductive definitions of functions.* We state and prove a rigorous form of inductive definitions of functions from ω to ω. We write ω^2 for $\omega \times \omega$.

Theorem 1.2.25 *Let $m_0 \in \omega$ and $F\colon \omega^2 \to \omega$. Then there exists a unique function $f\colon \omega \to \omega$ such that*

$$f(0) = m_0 \text{ and, for every } n \in \omega \text{ with } n \neq 0, \text{ we have } f(n) = F(n,\, f(n-1))\,.$$

Proof. We prove the uniqueness and then the existence. Suppose that $f \neq g$ are two functions from ω to ω with the displayed property. Let $n \in \omega$ be the $<$-minimum number such that $f(n) \neq g(n)$ (Theorem 1.2.24). Then $n \neq 0$ because $f(0) = g(0) = m_0$, and $f(n-1) = g(n-1)$. We have the contradiction

$$f(n) = F(n,\, f(n-1)) = F(n,\, g(n-1)) = g(n)\,.$$

In order to prove the existence of f, we consider the set X of $m \in \omega$ such that there exists a function $f\colon m \to \omega$ with the property that $f(0) = m_0$ and $f(n) = F(n, f(n-1))$ for every $n \in m$ with $n \neq 0$. Then $0 \in X$ because the empty function to ω has, trivially, the stated property. Let $m \in X$. If $m = 0$, then $m \oplus 1 = 1 \in X$ because the function $f\colon 1 \to \omega$ with $f(0) = m_0$ has the stated property. Let $m \neq 0$ and $f\colon m \to \omega$ be a function with the stated property. We define $g\colon m \oplus 1 \to \omega$ as an extension of f by the value

$$g(m) = F(m,\ f(m-1))\,.$$

It follows that g has the stated property. Thus, $m \oplus 1 \in X$. By Theorem 1.2.17, $X = \omega$. For $m \in \omega$, we denote by X_m the set of maps from m to ω with the stated property. We have just proven that $X_m \neq \emptyset$ for every $m \in \omega$.

Two things are easy to see: (i) if $m \in n$ and $f \in X_n$, then $f \mid m \in X_m$ and (ii) every X_m is a one-element set (by the uniqueness argument at the beginning). Using the axiom of choice, we choose for every $m \in \omega$ a function $f_m \in X_m$. (It is of no help that each choice is unique; we have to make infinitely many choices and cannot avoid AC.) It follows from (i) and (ii) that

$$G_f = \bigcup_{m \in \omega} G_{f_m}$$

is (the graph of) the desired function f. $\square$

We include one exercise on inductive definitions. In the next passage, we use them to define addition and multiplication of natural numbers.

Exercise 1.2.26 *Let $m_0 = 0$ and $F(k,l) = l$ if k is odd, and $F(k,l) = l+1$ if k is even. Find an explicit formula for the function f provided by Theorem 1.2.25.*

• *Addition and multiplication on ω. The algebraic structure* $\mathbb{N}_0$. We define these two operations by means of Theorem 1.2.25. Addition arises as an iteration of the operation $\oplus 1$.

Proposition 1.2.27 *There exists a unique operation $+\colon \omega^2 \to \omega$ such that for every $m, n \in \omega$ with $n \neq 0$ we have*

$$m + 0 = m \ \text{ and } \ m + n = (m + (n-1)) \oplus 1\,.$$

Proof. We fix $m \in \omega$ and set $m_0 = m$ and $F(k,l) = l \oplus 1$. Theorem 1.2.25 provides the stated function $m + n$ for the fixed m. $\square$

Exercise 1.2.28 *Show that $m + 1 = m \oplus 1$ for every $m \in \omega$.*

Multiplication is iterated addition.

Proposition 1.2.29 *There exists a unique operation $\cdot\colon \omega^2 \to \omega$ such that for every $m, n \in \omega$ with $n \neq 0$ we have*

$$m \cdot 0 = 0 \ \text{ and } \ m \cdot n = (m \cdot (n-1)) + m\,.$$

Proof. We fix $m \in \omega$ and set $m_0 = 0$ and $F(k,l) = l + m$. Theorem 1.2.25 provides the function $m \cdot n$ for the fixed m. $\square$

Exercise 1.2.30 *Prove, using the above definition of multiplication of natural numbers, that* $3 \cdot 4 = 12$.

We define the algebraic structure of natural numbers.

Definition 1.2.31 *Recall the set*

$$\omega = \{0,\, 1,\, 2,\, \dots\} = \{\emptyset,\, \{\emptyset\},\, \{\emptyset,\, \{\emptyset\}\},\, \dots\}$$

of Definition 1.2.14. The algebraic structure of <u>*natural numbers*</u>

$$\mathbb{N}_0 := \langle \omega,\, 0_{\mathbb{N}_0},\, 1_{\mathbb{N}_0},\, +,\, \cdot,\, < \rangle$$

consists of the set ω*, the elements* <u>*zero*</u> $0_{\mathbb{N}_0} := \emptyset$ *and* <u>*one*</u> $1_{\mathbb{N}_0} := \{\emptyset\}$ *in* ω*, the operations of addition* $+$ *and multiplication* $\cdot$ *on* ω *introduced in Propositions 1.2.27 and 1.2.29, and the linear order* $<$ *on* ω *defined above as* $m < n \iff m \in n$.

We usually write just 0 and 1 instead of $0_{\mathbb{N}_0}$ and $1_{\mathbb{N}_0}$. For natural numbers, we distinguish in notation between the base set ω and the algebraic structure $\mathbb{N}_0$ on ω. For the numeric domains of integers, fractions, real numbers, extended reals, and complex numbers, we use the same symbol $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{R}^*$, and $\mathbb{C}$, respectively, to denote the base set and the algebraic structure built on it.

• *Natural numbers form a simple ordered semiring.* To prove it, we employ four lemmas. In view of Exercise 1.2.28, instead of $\oplus 1$ we write just $+1$.

Lemma 1.2.32 *Let* $m, n \in \omega$*.* <u>*Then*</u> *the following holds.*

1. $0 + m = m$.
2. *If* $m, n \neq 0$*, then* $(m+n) - 1 = (m-1) + n = m + (n-1)$.

Proof. 1. For $m = 0$ this follows from the definition of $+$. Let $m \neq 0$. Then, by the definition of $+$, induction on m, and Exercise 1.2.20,

$$0 + m = (0 + (m-1)) + 1 = (m-1) + 1 = m\,.$$

2. We proceed by induction on n. Let $n = 1$. Then, by the definition of $+$ and Exercise 1.2.20,

$$(m-1) + n = (m-1) + 1 = m = m + 0 = m + (1-1) = m + (n-1)\,.$$

Also, $(m+n) - 1 = (m+1) - 1 = m$ by Exercise 1.2.20. Let $n > 1$. Then, by the definition of $+$ and induction,

$$\begin{aligned} (m-1) + n &= ((m-1) + (n-1)) + 1 \\ &= (m + ((n-1) - 1)) + 1 = m + (n-1)\,. \end{aligned}$$

Also, $(m+n) - 1 = ((m + (n-1)) + 1) - 1 = m + (n-1)$ by the definition of $+$ and Exercise 1.2.20. $\square$

Lemma 1.2.33 *Let $l, m, n \in \omega$. Then the following holds.*

1. $(l + m) + n = (l + n) + m$.
2. *If* $m, n \neq 0$, *then* $(l + (m - 1)) + n = (l + (n - 1)) + m$.

Proof. 1. For a fixed $l \in \omega$, we proceed by induction on m and n. If one of them is 0, the stated identity holds by the definition of $+$. Let $m, n \neq 0$. Then

$$\begin{aligned}(l+m)+n &= ((l+(m-1))+1)+n \\ &= ((l+1)+(m-1))+n = (((l+1)+(m-1))+(n-1))+1 \\ &= (((l+1)+(n-1))+(m-1))+1 \\ &= ((l+1)+(n-1))+m = ((l+(n-1))+1)+m \\ &= (l+n)+m\,.\end{aligned}$$

In the first equality, we use the definition of $+$. In the second equality, we use induction on m. The third equality follows from the definition of $+$. The fourth equality follows from induction on m (or n). In the fifth equality, we use the definition of $+$. In the sixth equality, we use induction on n. The last seventh equality follows from the definition of $+$.

2. Using the definition of $+$ and part 1, we have

$$\begin{aligned}(l+(m-1))+n &= ((l+(m-1))+(n-1))+1 \\ &= ((l+(n-1))+(m-1))+1 = ((l+(n-1))+m\,.\end{aligned}$$

□

Lemma 1.2.34 *Let $m, n \in \omega$. Then the following holds.*

1. $0 \cdot m = 0$.
2. *If* $m, n \neq 0$, *then* $(n-1) \cdot m + m = n \cdot (m-1) + n$.

Proof. 1. For $m = 0$ this follows from the definition of $\cdot$. Let $m \neq 0$. Then, by the definitions of $\cdot$ and $+$, and induction on m,

$$0 \cdot m = (0 \cdot (m-1)) + 0 = 0 + 0 = 0\,.$$

2. We proceed by induction on m. Let $m = 1$. Then

$$\begin{aligned}(n-1)\cdot m + m &= ((n-1)\cdot(1-1)+(n-1))+1 \\ &= ((n-1)\cdot 0+(n-1))+1 = (0+(n-1))+1 \\ &= (n-1)+1 = n = n\cdot 0+n = n\cdot(m-1)+n\,.\end{aligned}$$

In the first equality, we use the definition of $\cdot$. The second equality uses that $1 - 1 = 0$. The third equality uses the definition of $\cdot$. The fourth equality uses part 1 of Lemma 1.2.32. The fifth equality uses Exercise 1.2.20. The sixth equality uses the definition of $\cdot$ and part 1 of Lemma 1.2.32. The last, seventh equality uses that $1-1=0$.

Let $m>1$. Then

$$\begin{aligned}(n-1)\cdot m+m &= ((n-1)\cdot(m-1)+(n-1))+m\\ &= ((n-1)\cdot(m-1)+(m-1))+n\\ &= (n\cdot((m-1)-1)+n)+n\\ &= n\cdot(m-1)+n\,.\end{aligned}$$

The first equality follows from the definition of $\cdot$. The second equality follows from part 2 of Lemma 1.2.33. In the third equality, we use induction on m. In the last, fourth equality, we use the definition of $\cdot$. □

Lemma 1.2.35 *Let $m,n\in\omega$ and $m<n$. Then $m+1<n+1$.*

Proof. Let $m\in n$. Then $m\cup\{m\}\in n\cup\{n\}$ by part 6 of Proposition 1.2.18. □

We are ready to prove the first claim in Theorem 1.2.5.

Proposition 1.2.36 *The algebraic structure*

$$\mathbb{N}_0=\langle\omega,\,0,\,1,\,+,\,\cdot,\,<\rangle$$

introduced in Definition 1.2.31 is a simple ordered semiring.

Proof. The commutativity of $+$ follows at once if we set $l=0$ in part 1 of Lemma 1.2.33. We prove that $\cdot$ is commutative. Let $m,n\in\omega$. If one of them is 0, then $m\cdot n=n\cdot m=0$ by the definition of $\cdot$ and part 1 of Lemma 1.2.34. Let $m,n\neq 0$. Then, by the definition of $\cdot$, induction on n, and by part 2 of Lemma 1.2.34,

$$\begin{aligned}m\cdot n &= (m\cdot(n-1))+m=((n-1)\cdot m)+m\\ &= (n\cdot(m-1))+n=n\cdot m\,.\end{aligned}$$

We prove the neutrality of 0 and 1. Let $m\in\omega$. Then $m+0=m$ and

$$m\cdot 1=m\cdot(1-1)+m=m\cdot 0+m=0+m=m+0=m\,,$$

with the six equalities justified in Exercise 1.2.37. Hence, 0 and 1 are neutral to $+$ and $\cdot$, respectively.

We prove that $+$ is associative. Let $k,l,m\in\omega$. If one of them is zero, then $(k+l)+m=k+(l+m)$ holds due to the neutrality of 0. Let $k,l,m\neq 0$. Then

$$\begin{aligned}(k+l)+m &= ((k+l)+(m-1))+1\\ &= (k+(l+(m-1)))+1\\ &= (k+((l+m)-1))+1=k+(l+m)\,.\end{aligned}$$

The first equality follows from the definition of +. In the second equality, we use induction on m. The third equality follows from part 2 of Lemma 1.2.32. The last, fourth equality follows from the definition of +.

We prove that $\cdot$ is distributive to +. Let $k, l, m \in \omega$. If one of l and m is 0, then $k \cdot (l + m) = k \cdot l + k \cdot m$ holds by the neutrality of 0, and by $0n = n0 = 0$. Let $l, m \neq 0$. Then

$$\begin{aligned} k \cdot (l + m) &= k \cdot ((l + m) - 1) + k \\ &= k \cdot (l + (m - 1)) + k \\ &= (k \cdot l + k \cdot (m - 1)) + k \\ &= k \cdot l + (k \cdot (m - 1) + k) = k \cdot l + k \cdot m\,. \end{aligned}$$

The first equality follows from the definition of $\cdot$. The second equality follows from part 2 of Lemma 1.2.32. In the third equality, we use induction on m. The fourth equality follows from the associativity of +. The last, fifth equality follows from the definition of $\cdot$.

We prove that $\cdot$ is associative. Let $k, l, m \in \omega$. If one of them is zero, then $(k \cdot l) \cdot m = k \cdot (l \cdot m) = 0$ because $0n = n0 = 0$. Let $k, l, m \neq 0$. Then

$$\begin{aligned} (k \cdot l) \cdot m &= (k \cdot l) \cdot (m - 1) + k \cdot l \\ &= k \cdot (l \cdot (m - 1)) + k \cdot l \\ &= k \cdot (l \cdot (m - 1) + l) = k \cdot (l \cdot m)\,. \end{aligned}$$

The first equality follows from the definition of $\cdot$. In the second equality, we use induction on m. The third equality follows from the distributivity of $\cdot$. The last, fourth equality follows from the definition of $\cdot$.

We prove that the two order axioms hold in $\mathbb{N}_0$. Let $k, l, m \in \omega$ with $k < l$. If $m = 0$, then $k + m < l + m$ by the neutrality of 0. Let $m \neq 0$. Then, by the definition of +, induction on m, and by Lemma 1.2.35,

$$k + m = (k + (m - 1)) + 1 < (l + (m - 1)) + 1 = l + m\,,$$

which proves the first order axiom. The second order axiom $0 < 1$ is obvious.

Finally, let $0 \in X \subset \omega$ and let X be closed to adding 1. Since $m + 1 = m \cup \{m\}$, the set X is inductive and $X = \omega$ by Theorem 1.2.17. The semiring $\mathbb{N}_0$ is simple. □

Exercise 1.2.37 *Justify the six equalities in the computations of* $m + 0$ *and* $m \cdot 1$*.*

- *Subtraction of natural numbers.* We define a useful partial operation on ω.

Proposition 1.2.38 *There exists a unique partial operation of* <u>*subtraction*</u> $-$ *on* ω *such that for every* $m, n \in \omega$ *with* $m \geq n$ *we have* $n + (m - n) = m$*. If* $m < n$*, the value* $m - n$ *is not defined.*

Proof. Let $n, m \in \omega$ with $n \in m$ or $n = m$. We prove that there is a unique number $k \in \omega$ such that $n + k = m$; then we define $m - n := k$. We prove the uniqueness and then the existence. If $k, l \in \omega$ are distinct, then, for example, $l < k$. Using the first order axiom, we deduce that

$$n + l < n + k\,,$$

which shows that the map $\omega \ni k \mapsto n + k \in \omega$ is injective and subtraction is unique. Suppose that n and m are as stated, but k does not exist, and n is $<$-minimum. Then $n \neq 0$ because $0 + m = m$, but $(n - 1) + l = m$ for some $l \in \omega$. Clearly, $l \neq 0$ because $n - 1 < m$. As we know already from part 2 of Lemma 1.2.32,

$$n + (l - 1) = (n - 1) + l = m\,.$$

□

For the later distributive law for integers, we prove the distributivity of multiplication of natural numbers to subtraction.

Corollary 1.2.39 *Let $l, m, n \in \omega$ with $m \leq n$. Then*

$$l \cdot (n - m) = l \cdot n - l \cdot m\,.$$

Proof. Let $k = n - m \in \omega$ be the unique number such that $m + k = n$. By the distributivity of $\cdot$,

$$l \cdot m + l \cdot k = l \cdot (m + k) = l \cdot n\,.$$

Thus $l \cdot k = l \cdot n - l \cdot m$. □

In the next exercises, we collect properties of subtraction.

Exercise 1.2.40 *Prove that $5 - 3 = 2$.*

Exercise 1.2.41 *Let $l, m, n \in \omega$. If $l \geq n$, then $(l + m) - n = (l - n) + m$. If $l \geq m + n$, then $l - (m + n) = (l - m) - n$ and $(l - m) - n = (l - n) - m$. If $l \geq m \geq n$, then $l - (m - n) = (l - m) + n$.*

• *Countable and uncountable sets. The number of elements of a finite set.* First, we define (in)finiteness and (un)countability.

Definition 1.2.42 *A set X is countable if there exists a bijection between the sets X and ω. The set X is finite if there exists a bijection between X and m for some $m \in \omega$. If X is not finite, then it is infinite. We call X at most countable if it is finite or countable. X is uncountable if it is not at most countable.*

Exercise 1.2.43 *Show that every $k \in \mathbb{N}$ has a unique expression as*

$$k = (2l + 1) \cdot 2^m$$

for some $l, m \in \omega$. Use it to define a bijection $f\colon \omega \to \omega^2$, which shows that the set ω^2 is countable.

We proceed to the definition of the number of elements of a finite set.

Lemma 1.2.44 *Let $m < n$ be in ω. Then there exists a bijection $f\colon n \to n$ such that $f(n-1) = m$.*

Proof. We prove it by induction on n. Let $n = 1$. Then $m = 0 = n - 1$ and we take the only bijection $f\colon 1 \to 1$, with $f(0) = 0$. Let $n > 1$ and $m < n$. We distinguish two cases.

The first case is that $m < n-1$. By induction, we take a bijection $g\colon n-1 \to n-1$ such that $g(n-2) = m$. The required bijection $f\colon n \to n$ is defined by $f(i) := g(i)$ for $i \in n-1 \setminus \{n-2\}$, $f(n-2) := n-1$, and $f(n-1) := m$.

The last, second case is that $m = n-1$. By induction, we take any bijection $g\colon n-1 \to n-1$ (with any prescribed value at $n-2$), and set $f(i) := g(i)$ for $i \in n-1$ and $f(n-1) = n-1$. $\square$

It follows from this lemma that different natural numbers have different cardinalities.

Proposition 1.2.45 *Let $m < n$ be in ω. Then there does not exist any bijection from m to n.*

Proof. For the contrary, let $m < n$ in ω be such that $f\colon m \to n$ is a bijection and m is minimum. Then $m > 0$ because there is no surjection $s\colon \emptyset \to A$ with $A \neq \emptyset$. Let $l \in m$ be such that $f(l) = n-1$. We use Lemma 1.2.44 and take a bijection $g\colon m \to m$ such that $g(m-1) = l$. Then the map $h\colon m-1 \to n-1$, defined by $h(i) = f(g(i))$ for $i \in m-1$, is a bijection from $m-1$ to $n-1$, in contradiction with the minimality of m. $\square$

Corollary 1.2.46 *For every finite set X there exists a unique number $m \in \omega$ that is in bijection with X.*

Proof. By the definition of finite sets and by Proposition 1.2.45. $\square$

Now we can introduce the cardinalities of finite sets.

Definition 1.2.47 *Let X be a finite set. We call the unique number $m \in \omega$ that is in bijection with X the cardinality of X or the number of elements of X.*

• *Hereditarily finite sets.* Although the set $\{\omega\}$ has just one element and $2 = \{\emptyset, \{\emptyset\}\}$ has two, the former set is in a clear sense larger than the latter set. We introduce besides the dichotomy of finite vs. infinite sets another distinction.

Definition 1.2.48 *A set x is hereditarily finite, abbreviated* HF*, if for every $n \in \omega$ and every chain of sets*

$$x_n \in x_{n-1} \in \cdots \in x_0 = x\,,$$

the set x_n is finite.

Thus $\{\omega\}$ is not HF but 2 is.

Exercise 1.2.49 *Show that there is no function f such that $M(f) = \omega$ and $f(m+1) \in f(m)$ for every $m \in \omega$.*

Exercise 1.2.50 *Every natural number is* HF*.*

• *Semiring homomorphisms f_S.* We introduce semiring homomorphisms f_S from $\mathbb{N}_0$ to any semiring S.

Definition 1.2.51 *Let*

$$S = \langle S,\, 0_S,\, 1_S,\, \oplus,\, \odot \rangle$$

be a semiring. We define by induction on ω a map $f_S\colon \omega \to S$. First, $f_S(0) := 0_S$. For $m \in \omega$ with $m > 0$ we define

$$f_S(m) := f_S(m-1) \oplus 1_S\,.$$

A rigorous proof of existence (and uniqueness) of f_S would follow lines similar to those of the proof of Theorem 1.2.25. Clearly, $f_S(0) = 0_S$ and $f_S(1) = 1_S$.

We prove four lemmas on maps f_S. In the lemmas, $S = \langle S, 0_S, 1_S, \oplus, \odot \rangle$ is a semiring and $f_S\colon \omega \to S$ is the map in Definition 1.2.51. The lemmas show that f_S is a semiring homomorphism, and that if S is simple and ordered, then f_S is an isomorphism of ordered semirings.

Lemma 1.2.52 *For every $m, n \in \omega$, we have $f_S(m+n) = f_S(m) \oplus f_S(n)$.*

Proof. By induction on n. For $n = 0$ the equality holds for every $m \in \omega$ due to the neutrality of 0_S. Let $n > 0$. Then, writing f for f_S,

$$\begin{aligned} f(m+n) &= f((m+1)+(n-1)) = f(m+1) \oplus f(n-1) \\ &= (f(m) \oplus 1_S) \oplus f(n-1) \\ &= f(m) \oplus (f(n-1) \oplus 1_S) = f(m) \oplus f(n)\,. \end{aligned}$$

The first equality follows from the properties of $\mathbb{N}_0$. The second equality follows from induction. The third equality follows from the definition of f. The fourth equality follows from the properties of $\oplus$. The last, fifth equality follows from the definition of f. □

Lemma 1.2.53 *For every $m, n \in \omega$, we have $f_S(m \cdot n) = f_S(m) \odot f_S(n)$.*

Proof. We again write f for f_S and first prove the equality for $n = 0$ by induction on m. For $m = 0$ we have

$$f(0 \cdot 0) = f(0) = 0_S = 0_S \odot 0_S = f(0) \odot f(0)\,,$$

due to Proposition 1.2.10 in the third equality. If $m > 0$, then

$$\begin{aligned} f(m \cdot 0) &= f(0) = 0_S = 0_S \oplus 0_S = f(m-1) \odot 0_S \oplus 1_S \odot 0_S \\ &= (f(m-1) \oplus 1_S) \odot 0_S = f(m) \odot f(0)\,. \end{aligned}$$

The first equality follows from the properties of $\mathbb{N}_0$. In the second equality, we use the definition of f. In the third equality, we use the neutrality of 0_S. The fourth equality follows from induction and the neutrality of 1_S. The fifth equality follows from the distributive law in S. In the last, sixth equality, we use the definition of f. The equality therefore holds for every $m \in \omega$ and $n = 0$.

For $n > 0$ we proceed by induction on n. Let $m, n \in \omega$ and $n > 0$. Then

$$\begin{aligned} f(m \cdot n) &= f(m \cdot (n-1) + m) = f(m \cdot (n-1)) \oplus f(m) \\ &= f(m) \odot f(n-1) \oplus f(m) = f(m) \odot (f(n-1) \oplus 1_S) \\ &= f(m) \odot f(n)\,. \end{aligned}$$

In the first equality, we use the properties of $\mathbb{N}_0$. In the second equality, we use Lemma 1.2.52. In the third equality, we use induction. The fourth equality follows from the properties of S. The last, fifth equality follows from the definition of f. $\square$

Lemma 1.2.54 *Let S be an ordered semiring, with the linear order $\prec$. Then for every $m, n \in \omega$,*

$$f_S(m) \prec f_S(n) \iff m < n\,.$$

Proof. We again write f for f_S. By the trichotomy of $<$, it suffices to prove the implication

$$m < n \Rightarrow f(m) \prec f(n)\,.$$

Let $m = 0$. We proceed by induction on n. If $n = 1$ then the implication holds because

$$f(0) = 0_S \prec 1_S = f(1)$$

by Exercise 1.2.55. Let $n > 1$. Then the inductive assumption

$$f(0) = 0_S \prec f(n-1) \rightsquigarrow f(0) = 0_S \prec 1_S = 0_S \oplus 1_S \prec f(n-1) \oplus 1_S = f(n)\,,$$

and $f(0) \prec f(n)$ by the two order axioms for $\prec$, the transitivity of $\prec$, and the definition of f.

In the general case, we show the above displayed implication by induction on m. We proved the case $m = 0$ and assume that $0 < m < n$. Then, since $m - 1 < n - 1$,

$$f(m-1) \prec f(n-1) \rightsquigarrow f(m-1) \oplus 1_S \prec f(n-1) \oplus 1_S \iff f(m) \prec f(n)\,,$$

by the inductive assumption, the first order axiom for $\prec$, and the definition of the map $f = f_S$. $\square$

Exercise 1.2.55 *Why is* $0_S \prec 1_S$*?*

• *Simple ordered semirings are mutually isomorphic.* We employ one more lemma.

Lemma 1.2.56 *Let* S *be a simple ordered semiring, with the linear order denoted by* $\prec$*. Then the map*

$$f_S \colon \mathbb{N}_0 = \langle \omega,\, 0,\, 1,\, +,\, \cdot,\, < \rangle \to S = \langle S,\, 0_S,\, 1_S,\, \oplus,\, \odot,\, \prec \rangle$$

is an isomorphism of ordered semirings.

Proof. We only need to prove that $f_S \colon \omega \to S$ is a bijection; the three required properties of f_S were proven in the three previous lemmas. Lemma 1.2.54 shows that f_S is injective. To show that $f_S[\omega] = S$, we use that S is simple and observe that $f_S[\omega]$ ($\subset S$) contains 0_S and is closed to adding 1_S. Indeed, $0_S = f_S(0)$ and if $x = f_S(m)$ for some $m \in \omega$, then

$$x \oplus 1_S = f_S(m) \oplus 1_S = f_S(m+1)$$

by the definition of f_S. □

We are ready to prove the second claim in Theorem 1.2.5 and thereby complete its proof. By this, we conclude the section on natural numbers. Recall Proposition 1.2.8.

Proposition 1.2.57 *Suppose that*

$$S = \langle S,\, 0_S,\, 1_S,\, +,\, \cdot,\, < \rangle \ \textit{ and } \ S' = \langle S',\, 0_{S'},\, 1_{S'},\, \oplus,\, \odot,\, \prec \rangle$$

are two simple ordered semirings. Then S *and* S' *are isomorphic, which means that there exists a bijection* $f \colon S \to S'$ *with three properties.*

1. *For every* $a, b \in S$*, we have* $f(a+b) = f(a) \oplus f(b)$*.*
2. *For every* $a, b \in S$*, we have* $f(a \cdot b) = f(a) \odot f(b)$*.*
3. *For every* $a, b \in S$*, we have* $a < b \iff f(a) \prec f(b)$*.*

Proof. By Lemma 1.2.56 and Exercises 1.1.18 and 1.1.19, the map

$$f \colon S \to S',\ f := f_{S'}(f_S^{-1})\,,$$

is the desired isomorphism. □

Still, one more corollary.

Exercise 1.2.58 *Prove the next corollary*

Corollary 1.2.59 *Let*

$$S = \langle S,\, 0_S,\, 1_S,\, +,\, \cdot,\, < \rangle$$

be a simple ordered semiring. Then the linear order $\langle S, < \rangle$ *is a well ordering.*

1.3 Integers

We build on the natural numbers $\mathbb{N}_0$ constructed previously and obtain the integers $\mathbb{Z}$. The main result is their algebraic characterization in Theorem 1.3.6 as the up to isomorphism unique simple ordered ring.

• *Ordered rings and simple rings.* We review the algebraic structure that fits the integers. Recall Definition 1.2.1 of ordered semirings.

Definition 1.3.1 *An ordered ring*

$$R = \langle R,\, 0_R,\, 1_R,\, +,\, \cdot,\, < \rangle$$

is an ordered semiring with two additions/changes. (i) *For every $a \in R$ there is $b \in R$ satisfying $a + b = 0_R$. We call b the additive inverse of a and denote it by $-a$.* (ii) *The second order axiom now takes the form that for every $a, b, c \in R$ with $c > 0_R$,*

$$a < b \Rightarrow a \cdot c < b \cdot c\,.$$

If we omit $<$, we have the structure of a ring.

A subset of R that contains 0_R and 1_R, is closed under the operations $+$ and $\cdot$, and under additive inverses, induces a subring of the ring R. If R and R' are rings, a ring homomorphism from R to R' is any map $f\colon R \to R'$ that is a homomorphism to their operations and that satisfies $f(1_R) = 1_{R'}$. If R and R' are ordered and f in addition is bijective and preserves the orders, we say that f is an isomorphism of ordered rings.

Exercise 1.3.2 *In any ring, additive inverses are uniquely determined.*

Exercise 1.3.3 *Let R be a ring and $a, b \in R$. Then $-a = (-1_R)\cdot a$, $-(-a) = a$, $a \cdot (-b) = -(a \cdot b)$, and $-(a + b) = (-a) + (-b)$.*

Exercise 1.3.4 *Show that in ordered rings the second order axiom implies its semiring version, that is, $0_R < 1_R$ in every ordered ring R.*

Exercise 1.3.5 *Show that ring homomorphisms preserve additive neutral elements and additive inverses.*

A ring R is simple if for any set $X \subset R$ the following holds. If $0_R \in X$ and if for any $x \in X$ also $x + 1_R \in X$ and $x + (-1_R) \in X$, then $X = R$.

Theorem 1.3.6 *There exists a simple ordered ring. Every two simple ordered rings are isomorphic.*

Our main goal is to prove Theorem 1.3.6. We prove the former claim in Proposition 1.3.27, and the latter claim in Proposition 1.3.36. As for semirings, the adjective "ordered" cannot be omitted in the theorem. We again have the class of structures that are integers.

Corollary 1.3.7 *The class*

$$\text{INTEGERS} := \{x\colon\ x \textit{ is a simple ordered ring}\}$$

contains the "standard" integers $\mathbb{Z}$ and every two sets in it are isomorphic as ordered rings.

• *Again, multiplying by zero.* We show that in every ring R, $a \cdot 0_R = 0_R$, and we prove some more results on rings.

Proposition 1.3.8 *The following holds.*

1. *Let R be a ring and $a \in R$. Then*

$$a \cdot 0_R = 0_R \cdot a = 0_R\,.$$

2. *Let R be an ordered ring and $a, b \in R \setminus \{0_R\}$. Then $a \cdot b \neq 0_R$.*

3. *Let R be an ordered ring, let $a, b, c \in R$ with $a \neq 0_R$, and let $a \cdot b = a \cdot c$. Then $b = c$.*

4. *Every ordered ring is infinite.*

Proof. 1. Let $a \in R$ be any element. We write $0 = 0_R$ and compute

$$\begin{aligned} 0 &= a \cdot 0 + (-(a \cdot 0)) = a \cdot (0 + 0) + (-(a \cdot 0)) \\ &= (a \cdot 0 + a \cdot 0) + (-(a \cdot 0)) = a \cdot 0 + (a \cdot 0 + (-(a \cdot 0))) \\ &= a \cdot 0 + 0 = a \cdot 0\,. \end{aligned}$$

We leave justifications of these equalities as Exercise 1.3.10.

2. We first prove that for any $a, b, c \in R$,

$$a < b \wedge c < 0_R \Rightarrow a \cdot c > b \cdot c\,.$$

Indeed, by multiplying $a < b$ with $-c$ $(> 0_R)$, we obtain, with the help of Exercise 1.3.3, that

$$-(a \cdot c) < -(b \cdot c)\,.$$

We add $a \cdot c + b \cdot c$ and obtain $b \cdot c < a \cdot c$.

Now suppose that, for example, $a < 0_R$ and $b < 0_R$. Using the previous result and part 1, we have

$$a \cdot b > 0_R \cdot b = 0_R\,,$$

which means that $a \cdot b \neq 0_R$ (Exercise 1.1.41). In the other three cases, we get that $a \cdot b \neq 0_R$ similarly.

3. From $a \cdot b = a \cdot c$ we get

$$a \cdot (b + (-c)) = 0_R\,.$$

By part 2, $b + (-c) = 0_R$ and $b = c$.

4. This follows from Exercise 1.2.3. $\square$

Rings enjoying property 2, that the product of any two nonzero elements is nonzero, are called (integral) domains. Thus, every ordered ring is an ordered domain. From now on, we will speak of ordered domains instead of ordered rings.

Exercise 1.3.9 *Present examples of simple rings that are not domains.*

Exercise 1.3.10 *Justify the six equalities in the proof that $a \cdot 0_R = 0_R$.*

• *Subtraction and units.* We know how to partially subtract natural numbers. In rings, subtraction is defined for every pair of elements.

Definition 1.3.11 *Let R be a ring and $a, b \in R$. We define the difference of a and b as $a - b := a + (-b)$.*

Exercise 1.3.12 *Let R be a ring and $b \in R$. Then $b - b = 0_R$ and $b - 0_R = b$.*

Exercise 1.3.13 *Let R be a ring and $a, b, c \in R$. Then*

$$a - (b + c) = (a - b) - c \text{ and } a - (b - c) = (a - b) + c\,.$$

Recall that an element $a \in R$ of a ring R is a unit if $a \cdot b = 1_R$ for some $b \in R$, that is, if a has a multiplicative inverse. The set of units in R is denoted by $R^\times$.

Exercise 1.3.14 *What are the units in the ring of integers (which we introduce below)?*

• *The set of integers and the linear order on it.* We define the set of integers. Recall that $\mathbb{N} = \omega \setminus \{0\}$. For $m \in \mathbb{N}$ we define $-m := \langle 0, m \rangle$. Let

$$-\mathbb{N} := \{-m :\ m \in \mathbb{N}\}\,.$$

Definition 1.3.15 *The set of integers $\mathbb{Z}$ is the disjoint union*

$$\mathbb{Z} := -\mathbb{N} \cup \omega\,.$$

The elements of $-\mathbb{N}$ are the negative integers. The elements of ω are called, in the context of $\mathbb{Z}$, the nonnegative integers. If $n \in \omega$, we define the absolute value of n by $|n| := n$. If $-n \in -\mathbb{N}$, we set $|-n| := n$.

Exercise 1.3.16 *The union in the definition of $\mathbb{Z}$ is really disjoint.*

Exercise 1.3.17 *Every integer is* HF.

We define a linear order on $\mathbb{Z}$.

Definition 1.3.18 *Let $m, n \in \mathbb{Z}$. We set $m < n$ if and only if $m, n \in \omega$ and $m < n$ in $\mathbb{N}_0$, or if $m \in -\mathbb{N}$ and $n \in \omega$, or if $m, n \in -\mathbb{N}$ and $|m| > |n|$ in $\mathbb{N}_0$.*

Note that on ω this relation coincides with the linear order in $\mathbb{N}_0$ and that $m < 0 < n$ for every $m \in -\mathbb{N}$ and $n \in \mathbb{N}$.

Exercise 1.3.19 *Show that $<$ is a linear order on $\mathbb{Z}$. Is it a well ordering?*

• *Addition and multiplication of integers. The algebraic structure $\mathbb{Z}$.* We begin with addition, which is more complicated than multiplication. We use the partial operation $-$ of subtraction on ω introduced in Proposition 1.2.38. In the next two definitions, operations on the right sides of $:=$ are always in $\mathbb{N}_0$.

Definition 1.3.20 *Let $m, n \in \mathbb{Z}$. We define the sum $m + n$ as follows.*

1. *If $m, n \geq 0$, then $m + n$ is as in $\mathbb{N}_0$.*
2. *If $m < 0 \leq n$, then $m + n := n - |m|$ if $|m| \leq n$, and $m + n := -(|m| - n)$ if $|m| > n$.*
3. *If $n < 0 \leq m$, then exchange m and n in part 2.*
4. *If $m, n < 0$, then $m + n := -(|m| + |n|)$.*

The multiplication on $\mathbb{Z}$ is, up to signs, the same as in $\mathbb{N}_0$.

Definition 1.3.21 *Let $m, n \in \mathbb{Z}$. We define the product $m \cdot n$ as follows.*

1. *If $m = 0$ or $n = 0$, then $m \cdot n := 0$.*
2. *If $m, n < 0$ or $m, n > 0$, then $m \cdot n := |m| \cdot |n|$.*
3. *If $m < 0 < n$ or $n < 0 < m$, then $m \cdot n := -(|m| \cdot |n|)$.*

It is easy to see that the operations $+$ and $\cdot$ are commutative, and that 0 is neutral to $+$ and 1 to $\cdot$. Also, $0 + 0 = 0$, if $m \in \mathbb{N}$ then $m + (-m) = 0$, and if $-m \in -\mathbb{N}$ then $-m + m = 0$. Hence every integer has an additive inverse.

We define the algebraic structure $\mathbb{Z}$.

Definition 1.3.22 *The algebraic structure of <u>integers</u>*

$$\mathbb{Z} := \langle \mathbb{Z},\, 0_{\mathbb{Z}},\, 1_{\mathbb{Z}},\, +,\, \cdot,\, < \rangle$$

consists of the set $\mathbb{Z}$ introduced in Definition 1.3.15, the elements $0_{\mathbb{Z}} := 0$ and $1_{\mathbb{Z}} := 1$ in ω, the operations of addition $+$ and multiplication $\cdot$ on $\mathbb{Z}$ introduced in Definitions 1.3.20 and 1.3.21, and the linear order $<$ on $\mathbb{Z}$ introduced in Definition 1.3.18.

We usually write just 0 and 1 instead of $0_{\mathbb{Z}}$ and $1_{\mathbb{Z}}$. These integers we learn in elementary school. In our heads, we add and multiply integers according to the rules in Definitions 1.3.20 and 1.3.21. At the end of this section, we outline in exercises a rival "scientific" construction of integers based on abstract differences of natural numbers.

Exercise 1.3.23 *Compute, according to the definitions,* $7+(-9)$, $(-2)+(-3)$, *and* $(-2)\cdot 3$. *Compare by* $<$ *the integers* -2 *and* -3, *and* -10 *and* 1.

• $\mathbb{Z}$ *is a simple ordered domain.* $\mathbb{N}_0$ *is an ordered sub-semiring of* $\mathbb{Z}$. We employ three lemmas.

Lemma 1.3.24 *Let* $m \in \mathbb{Z}$ *and* $m > 1$. *<u>Then</u>*

$$-m+1 = -(m-1)\,.$$

Proof. This follows from part 2 of Definition 1.3.20. □

Lemma 1.3.25 *Let* $m, n \in \mathbb{Z}$. *<u>Then</u>*

$$m+(n+1) = (m+1)+n = (m+n)+1\,.$$

Proof. Due to the commutativity of $+$, it suffices to show that for every $m, n \in \mathbb{Z}$,

$$(m+1)+n = (m+n)+1\,.$$

We check this identity by dividing it in eleven cases.

1. Let $m \ge 0$ and $n \ge 0$. Then $(m+1)+n = (m+n)+1$ by the properties of $\mathbb{N}_0$.

2. Let $m \ge 0$ and $-m \le n < 0$. Then $(m+1)+n = m+1-|n| = m-|n|+1 = (m+n)+1$ by Definition 1.3.20 and Exercise 1.2.41.

3. Let $m \ge 0$ and $n = -(m+1)$. Then $(m+1)+n = (m+1)-|n| = 0 = -1+1 = (m+n)+1$ by Definition 1.3.20.

4. Let $m \ge 0$ and $n < -(m+1)$. Then $(m+1)+n = -(|n|-(m+1)) = -(|n|-m-1) = -(|m+n|-1) = (m+n)+1$ by Definition 1.3.20 and Exercise 1.2.41.

5. Let $m = -1$ and $n \ge 1$. Then $(m+1)+n = 0+n = n = (n-1)+1 = (m+n)+1$ by Definition 1.3.20.

6. Let $m = -1$ and $n = 0$. Then $(m+1)+n = 0 = -1+1 = (m+n)+1$ by Definition 1.3.20.

7. Let $m = -1$ and $n < 0$. Then $(m+1)+n = n = -(|m+n|-1) = (m+n)+1$ by Definition 1.3.20.

8. Let $m < -1$ and $n \geq |m|$. Then $(m+1)+n = -(|m|-1)+n = n-(|m|-1) = (n-|m|)+1 = (m+n)+1$ by Definition 1.3.20 and Exercise 1.2.41.

9. Let $m < -1$ and $n = |m|-1$. Then $(m+1)+n = -(|m|-1)+n = 0 = -1+1 = (m+n)+1$ by Definition 1.3.20.

10. Let $m < -1$ and $0 \leq n < |m|-1$. Then $(m+1)+n = -(|m|-1)+n = -(|m|-1-n) = -(|m|-n-1) = -(|m|-n)+1 = (m+n)+1$ by Definition 1.3.20 and Exercise 1.2.41.

11. Finally, let $m < -1$ and $n < 0$. Then $(m+1)+n = -(|m|-1)+n = -(|m|-1+|n|) = -(|m+n|-1) = (m+n)+1$ by Definition 1.3.20.

□

Lemma 1.3.26 *Let $m, n \in \mathbb{Z}$. Then $m < n$ if and only if $m+l = n$ for some $l \in \mathbb{N}$.*

Proof. Let $m, n \in \mathbb{Z}$ and $m < n$. If $m, n \in \omega$, then $m+(n-m) = n$ for some $n-m \in \omega$ by Proposition 1.2.38, and $n-m > 0$. If $m < 0 \leq n$, then $m+(|m|+n) = n$ with $|m| \in \mathbb{N}$, so that $|m|+n > 0$. If $m, n < 0$, then $|n| < |m|$, so that $|n|+(|m|-|n|) = |m|$ with $|m|-|n| > 0$. We have, by Exercise 1.2.41,

$$m+(|m|-|n|) = -(|m|-(|m|-|n|)) = -|n| = n\,.$$

Let $m, n \in \mathbb{Z}$ and let $m+l = n$ for some $l \in \mathbb{N}$. We prove by induction on l that $m < n$. Let $l = 1$. If $m \in \omega$, then $m \in m \cup \{m\} = m+1 = n$ and $m < n$. If $m = -1$, then $n = m+1 = 0$ and $m < n$. If $m < -1$, then $n = m+1 = -(|m|-1)$ and $|n| < |m|$, so that $m < n$. Let $l > 1$ and $n' = m+(l-1)$. Then $m < n'$ by induction, $n' < n'+1 = n$ by Lemma 1.3.25 and by the case $l = 1$, and $m < n$ by the transitivity of $<$. □

We are ready to prove the first claim in Theorem 1.3.6.

Proposition 1.3.27 *The algebraic structure of integers*

$$\mathbb{Z} = \langle \mathbb{Z},\, 0,\, 1,\, +,\, \cdot,\, < \rangle$$

introduced in Definition 1.3.22 is a simple ordered domain.

Proof. We already noted that $+$ and $\cdot$ are commutative, that 0 and 1 is neutral to $+$ and $\cdot$, respectively, and that every integer has an additive inverse. The associativity of multiplication in $\mathbb{Z}$ follows at once from the associativity of multiplication in $\mathbb{N}_0$ because the sign of the result depends only on the number of minus signs of the three factors. The distributivity of $\cdot$ to $+$ is also clear, more or less. Let $l, m, n \in \mathbb{Z}$. If $m, n < 0$ or $m, n \geq 0$, then the distributive identity

$$l \cdot (m+n) = l \cdot m + l \cdot n$$

holds because it holds in $\mathbb{N}_0$. If $m < 0 \leq n$ or $n < 0 \leq m$, then the distributive identity follows from Corollary 1.2.39.

Least clear is the associativity of $+$. Suppose for the contrary that $l, m, n \in \mathbb{Z}$ are such that

$$(l + m) + n \neq l + (m + n)$$

and that $|l| + |m| + |n|$ is minimum. Then one of the three numbers is positive. Let, for example, $m > 0$. Then

$$(l + (m - 1)) + n = l + ((m - 1) + n)\,.$$

We add 1 to both sides of the equation, use the commutativity of $+$, apply four times Lemma 1.3.25, and obtain the contradiction that

$$(l + m) + n = l + (m + n)$$

after all. The other two cases, $l > 0$ or $n > 0$, are similar.

We show that the two order axioms hold in $\mathbb{Z}$. Let $k, m, n \in \mathbb{Z}$ with $m < n$. By Lemma 1.3.26, $m + l = n$ for some $l \in \mathbb{N}$. Thus, since $+$ is commutative and associative, also $(m + k) + l = n + k$. By the same lemma, $m + k < n + k$, which proves the first order axiom. Let also $k > 0$. Then, by the distributivity of $\cdot$,

$$m \cdot k + l \cdot k = n \cdot k\,.$$

Since $l \cdot k > 0$ by Definition 1.3.21, we have $m \cdot k < n \cdot k$ by Lemma 1.3.26, which proves the second order axiom.

Finally, we show that the ring $\mathbb{Z}$ is simple. Let $0 \in X \subset \mathbb{Z}$ and let X be closed to adding 1 and -1. Since $\mathbb{N}_0$ is simple, $\omega \subset X$. Since any $-n$ in $-\mathbb{N}$ arises by adding -1 exactly n times to 0, also $-\mathbb{N} \subset X$. Thus $X = \mathbb{Z}$. □

Exercise 1.3.28 *Prove the next proposition.*

Proposition 1.3.29 *The subset $\omega \subset \mathbb{Z}$ induces a sub-semiring of $\mathbb{Z}$.*

• *Ring homomorphisms f_R.* To define isomorphisms of simple ordered domains, we use the same technique as for semirings.

Definition 1.3.30 *Let*

$$R = \langle R,\, 0_R,\, 1_R,\, \oplus,\, \odot \rangle$$

be a ring. We define a map $f_R\colon \mathbb{Z} \to R$ simply as an extension of the map $f_R\colon \omega \to R$ in Definition 1.2.51 by the formula

$$f_R(-m) := -f_R(m),\ \ m \in \mathbb{N}\,.$$

Exercise 1.3.31 *Express the element $f_R(-3)$ in terms of the operations in R.*

We again prove four lemmas on f_R. In the lemmas, $R = \langle R, 0_R, 1_R, \oplus, \odot\rangle$ is a ring and $f_R\colon \mathbb{Z} \to R$ is the map in Definition 1.3.30. The lemmas show that f_R is a ring homomorphism, and that if R is simple and ordered, then f_R is an isomorphism of ordered domains.

Lemma 1.3.32 *For every $m, n \in \mathbb{Z}$, we have $f_R(m+n) = f_R(m) \oplus f_R(n)$.*

Proof. We write f for f_R. Let $m, n \in \mathbb{Z}$. If $m, n \geq 0$, then the equality holds by Lemma 1.2.52. If $m < 0 \leq n$ and $|m| > n$, then

$$\begin{aligned} f(m+n) &= f(-(|m|-n)) = -f(|m|-n) = -(f(|m|) \ominus f(n)) \\ &= -f(|m|) \oplus f(n) = f(m) \oplus f(n)\,. \end{aligned}$$

The first equality follows from Definition 1.3.20. The second equality follows from the definition of f. The third equality follows from Lemma 1.2.52 and the fact that $(m-n)+n = m$. In the fourth equality, we use Exercise 1.3.3. In the last, fifth equality, we use the definition of f.

Let $m < 0 \leq n$ and $|m| \leq n$. Then

$$f(m+n) = f(n-|m|) = f(n) \ominus f(|m|) = -f(|m|) \oplus f(n) = f(m) \oplus f(n)\,,$$

with similar justifications. The case $n < 0 \leq m$ is symmetric. Finally, if $m, n < 0$, then

$$\begin{aligned} f(m+n) &= f(-(|m|+|n|)) = -f(|m|+|n|) = -(f(|m|) \oplus f(|n|)) \\ &= -f(|m|) \oplus (-f(|n|)) = f(m) \oplus f(n)\,, \end{aligned}$$

again with clear justifications using Lemma 1.2.52. □

Lemma 1.3.33 *For every $m, n \in \mathbb{Z}$, we have $f_R(m \cdot n) = f_R(m) \odot f_R(n)$.*

Proof. We write f for f_R. Let $m, n \in \mathbb{Z}$. If $m = 0$ or $n = 0$, then

$$f(m \cdot n) = f(0) = 0_R = f_R(m) \odot f_R(n)$$

by part 1 of Proposition 1.3.8 because $f_R(m) = 0_R$ or $f_R(n) = 0_R$. If $m, n > 0$ or $m, n < 0$, then

$$\begin{aligned} f(m \cdot n) &= f(|m| \cdot |n|) = f(|m|) \odot f(|n|) \\ &= (\pm f(m)) \odot (\pm f(n)) = f(m) \odot f(n)\,, \end{aligned}$$

with equal signs. If $m < 0 < n$ or $n < 0 < m$, then

$$\begin{aligned} f(m \cdot n) &= f(-(|m| \cdot |n|)) = -f(|m| \cdot |n|) = -(f(|m|) \odot f(|n|)) \\ &= -((\pm f(m)) \odot (\mp f(n))) = f(m) \odot f(n)\,, \end{aligned}$$

again with equal signs. In the second, respectively third, equality we use Lemma 1.2.53. □

Lemma 1.3.34 *Let R be an ordered ring, with the linear order $\prec$. Then for every $m, n \in \mathbb{Z}$,*

$$m < n \iff f_R(m) \prec f_R(n)\,.$$

Proof. We again write f for f_R. By the trichotomy of $<$, it suffices to prove the implication

$$m < n \Rightarrow f(m) \prec f(n)\,.$$

Let $m, n \in \mathbb{Z}$ and $m < n$. If $m, n \ge 0$, the implication holds by Lemma 1.2.54. Let $m < 0 \le n < |m|$. Then

$$f(m) = -f(|m|) \prec -f(n) \preceq f(n)\,.$$

The second inequality follows from $f(n) \prec f(|m|)$ (due to Lemma 1.2.54) via the first order axiom by adding to it $-f(n) + (-f(|m|))$. To obtain the last non-strict inequality, we add to $0_R = f(0) \preceq f(n)$ (due to Lemma 1.2.54) the element $-f(n)$ and get $-f(n) \preceq 0_R$ (cf. Exercise 1.2.4). Now $0_R \preceq f(n)$ and we use the transitivity of $\preceq$.

Let $m < 0 < |m| \le n$. Then by similar arguments,

$$f(m) = -f(|m|) \prec f(0) \prec f(n)\,.$$

Finally, let $m < n < 0$. Then

$$f(m) = -f(|m|) \prec -f(|n|) = f(n)$$

by Lemma 1.2.54 and the first order axiom, because $|n| < |m|$. □

• *Simple ordered domains are mutually isomorphic.* We again employ one more lemma.

Lemma 1.3.35 *Let R be a simple ordered domain, with the linear order denoted by $\prec$. Then the map*

$$f_R \colon \mathbb{Z} = \langle \mathbb{Z},\, 0,\, 1,\, +,\, \cdot,\, < \rangle \to R = \langle R,\, 0_R,\, 1_R,\, \oplus,\, \odot,\, \prec \rangle$$

is an isomorphism of ordered domains.

Proof. We only need to prove that $f_R \colon \mathbb{Z} \to R$ is a bijection; the three required properties of f_R are proven in the three previous lemmas. Lemma 1.3.34 shows that f_R is injective. To prove that f_R is surjective, it suffices to show, since R is simple, that $f_R[\mathbb{Z}]$ contains 0_R and is closed to adding and subtracting 1_R. The former is clear, $0_R = f_R(0)$. Let $x = f_R(m)$ for some $m \in \mathbb{Z}$. Then, by Lemma 1.3.32,

$$x \oplus \ominus 1_R = f_R(m) \oplus \ominus 1_R = f_R(m \pm 1) \in f_R[\mathbb{Z}]$$

and we are done. □

We are ready to prove the second claim in Theorem 1.3.6 and thereby complete its proof. Recall Proposition 1.2.8.

Proposition 1.3.36 *Suppose that*

$$R = \langle R,\, 0_R,\, 1_R,\, +,\, \cdot,\, < \rangle \ \textit{ and } \ R' = \langle R',\, 0_{R'},\, 1_{R'},\, \oplus,\, \odot,\, \prec \rangle$$

are two simple ordered domains. Then R and R' are isomorphic, which means that there exists a bijection $f\colon R \to R'$ with three properties.

1. *For every $a, b \in R$, we have $f(a+b) = f(a) \oplus f(b)$.*
2. *For every $a, b \in R$, we have $f(a \cdot b) = f(a) \odot f(b)$.*
3. *For every $a, b \in R$, we have $a < b \iff f(a) \prec f(b)$.*

Proof. By Lemma 1.3.35 and Exercises 1.1.18 and 1.1.19, the map

$$f\colon R \to R',\ f := f_{R'}(f_R^{-1})\,,$$

is the desired isomorphism. □

Exercise 1.3.37 *Is this isomorphism unique?*

• *Difference integers.* We outline an alternative definition of integers. For $m, n \in \omega$, we set $m \ominus n := \langle m, n \rangle$ and define a relation $\sim$ on ω^2:

$$m \ominus n \sim m' \ominus n' \iff m + n' = m' + n\,.$$

Exercise 1.3.38 *Show that $\sim$ is an equivalence relation.*

We set $\mathbb{Z}' := \omega^2/\sim$ and call the elements of $\mathbb{Z}'$ difference integers. It is clear that they are infinite sets, cf. Exercise 1.3.17. We define two elements in $\mathbb{Z}'$, two arithmetic operations on $\mathbb{Z}'$, and a relation on $\mathbb{Z}'$.

- $0_{\mathbb{Z}'} := [0 \ominus 0]_\sim$ and $1_{\mathbb{Z}'} := [1 \ominus 0]_\sim$.
- $m \ominus n + m' \ominus n' := (m + m') \ominus (n + n')$.
- $m \ominus n \cdot m' \ominus n' := (mm' + nn') \ominus (mn' + nm')$.
- $m \ominus n < m' \ominus n' \iff m + n' < m' + n$.

Exercise 1.3.39 *Show that the operations $+$ and $\cdot$ and the relation $<$ on ω^2 do not depend on the choice of representatives of blocks, and therefore are operations and relation on $\mathbb{Z}'$.*

Definition 1.3.40 *Thus we define the algebraic structure of difference integers*

$$\mathbb{Z}' := \langle \mathbb{Z}',\, 0_{\mathbb{Z}'},\, 1_{\mathbb{Z}'},\, +,\, \cdot,\, < \rangle\,.$$

It is easy to see that $0_{\mathbb{Z}'}$ and $1_{\mathbb{Z}'}$ is neutral to $+$ and $\cdot$, respectively, that $+$ is commutative and associative, and that $\cdot$ is commutative. The identity

$$m \ominus n + n \ominus m = (m+n) \ominus (m+n) \sim 0 \ominus 0$$

shows that every difference integer has an additive inverse.

Exercise 1.3.41 *Show that $\cdot$ is associative, and distributive to $+$.*

Exercise 1.3.42 *Show that $<$ is a linear order.*

Exercise 1.3.43 *Show that $<$ satisfies the two order axioms.*

Thus $\mathbb{Z}'$ is an ordered domain (ring). Since $[m \ominus n]_\sim$ is the sum of m copies of $1_{\mathbb{Z}'} = [1 \ominus 0]_\sim$ and n copies of $-1_{\mathbb{Z}'} = [0 \ominus 1]_\sim$, we see that $\mathbb{Z}'$ is a simple ordered domain. Thus

$$\mathbb{Z}' \in \text{INTEGERS}\,.$$

By Corollary 1.3.7, $\mathbb{Z}$ and $\mathbb{Z}'$ are isomorphic simple ordered rings. We close this section with the remark that the integers in Definition 1.3.15 are hereditarily finite sets, but each difference integer is a countable set.

1.4 Fractions

We continue to build the hierarchy of numeric domains and introduce the fractions $\mathbb{Q}$. The main result is their algebraic characterization in Theorem 1.4.7 as the up to isomorphism unique simple ordered field.

• *Ordered fields and simple fields.* Ordered fields are better known than ordered semirings or ordered rings. Recall Definition 1.3.1 of the latter.

Definition 1.4.1 *An ordered field*

$$F = \langle F,\, 0_F,\, 1_F,\, +,\, \cdot,\, < \rangle$$

is an ordered ring such that for every $a \in F \setminus \{0_F\}$ there is $b \in F$ satisfying $a \cdot b = 1_F$. We call b the multiplicative inverse of a and denote it by a^{-1}. If we omit $<$, we have the structure of a field

A subset of F that contains 0_F and 1_F, is closed under the operations $+$ and $\cdot$, and under additive and multiplicative inverses, induces a subfield of the field F. If F and G are fields, a field homomorphism is any map $f\colon F \to G$ that is a ring homomorphism from F to G. If F and G are ordered and f is also bijective and preserves the orders, we say that f is an isomorphism of ordered fields.

Exercise 1.4.2 *Let F be a field and $a \in F$. Then $a \cdot 0_F = 0_F$.*

Exercise 1.4.3 *Every field is a domain.*

Exercise 1.4.4 *In every field, multiplicative inverses are unique.*

Exercise 1.4.5 *Let F be a field and $a, b \in F\backslash\{0_F\}$. Then $a^{-1} \neq 0_F$, $(a^{-1})^{-1} = a$, and $(a \cdot b)^{-1} = a^{-1} \cdot b^{-1}$.*

Proposition 1.4.6 *Every field homomorphism $f\colon F \to G$ is injective.*

Proof. Suppose, for the contrary, that $f(a) = f(b)$ for some elements $a \neq b$ in F. By Exercise 1.3.5, $f(a-b) = 0_G$. Let $c := a - b$ ($\neq 0_F$). Then, by Exercise 1.4.2,

$$1_G = f(1_F) = f(c \cdot c^{-1}) = f(c) \cdot f(c^{-1}) = 0_G \cdot f(c^{-1}) = 0_G\,.$$

This contradicts the requirement $1_G \neq 0_G$ in Definition 1.2.1. □

So instead of field homomorphism, we will speak of field embeddings because for any field homomorphism $f\colon F \to G$, its image $f[F]$ induces a subfield of G that is isomorphic via f^{-1} to the field F.

A field F is simple if for any set $X \subset F$ the following holds. If $0_F \in X$ and if for any $x, y \in X$ with $y \neq 0_F$ also $x + 1_F \in X$, $x - 1_F \in X$, and $x \cdot y^{-1} \in X$, then $X = F$.

Theorem 1.4.7 *There exists a simple ordered field. Every two simple ordered fields are isomorphic.*

As before, our main goal is to prove Theorem 1.4.7. We prove the former claim in Proposition 1.4.40, and the latter claim in Proposition 1.4.57. As for natural numbers and integers, the word "ordered" cannot be omitted in the theorem. We have the following class.

Corollary 1.4.8 *The class*

$$\mathrm{FRACTIONS} := \{x\colon\ x \textit{ is a simple ordered field}\}$$

contains the "standard" fractions $\mathbb{Q}$ and every two sets in it are isomorphic as ordered fields.

• *Ordered fields are almost normed.* We write "almost" because the values of the "norm" (absolute value) are elements of the field and not, as is standard, of $\mathbb{R}$, which is not yet defined. We introduced the absolute value for integers in Definition 1.3.15. If F is an ordered field and $x \in F$, we define the absolute value of x by $|x| := x$ if $x \geq 0_F$, and by $|x| := -x$ if $x < 0_F$.

Proposition 1.4.9 *The absolute value in an ordered field*

$$F = \langle F,\, 0_F,\, 1_F,\, +,\, \cdot,\, < \rangle$$

has three properties of norms.

1. *Always* $|x| \ge 0_F$ *and* $|x| = 0_F$ *iff* $x = 0_F$.

2. (multiplicativity) $|x \cdot y| = |x| \cdot |y|$.

3. (triangle inequality, abbreviated TI) $|x + y| \le |x| + |y|$.

Proof. Property 1. Clearly, $|0_F| = 0_F$. If $x \ne 0_F$, then also $-x \ne 0_F$ and $|x| \ne 0_F$. If $x < 0_F$, then the first order axiom gives, by adding $-x$, that $0_F < -x$. Property 2 follows from Exercise 1.3.3.

3. We prove the triangle inequality. Let $x, y \in F$. If $x, y \ge 0_F$, then $|x + y| = x + y = |x| + |y|$ because $x + y \ge 0_F$ as well (Exercise 1.2.4). If $x, y < 0_F$, then

$$|x + y| = -(x + y) = (-x) + (-y) = |x| + |y|$$

by Exercise 1.3.3 because $x+y < 0_F$ as well (by the first order axiom). Suppose that $x < 0_F < -x \le y$. Then

$$|x + y| = x + y < y = |y| < |y| + |x|$$

and we are done by the transitivity of $\le$. The first equality is due to $0_F \le x+y$, which follows by Exercise 1.2.4 from $-x \le y$. The second inequality follows by the first order axiom from $x < 0_F$. The third equality follows from $y \ge 0_F$. The last, fourth inequality follows from $0_F < |x|$ by the first order axiom.

Let $x < 0_F \le y < -x$. Then

$$|x + y| = -(x + y) = -x + (-y) = |x| + (-y) \le |x| \le |x| + |y|$$

and we are done by the transitivity of $\le$. The first equality is due to $x+y < 0_F$, which follows by the first order axiom from $y < -x$. The second equality follows from Exercise 1.3.3. The third equality follows from $x < 0_F$. The fourth inequality follows from $-y \le 0_F$ by Exercise 1.2.4. The last, fifth inequality follows from $0_F \le |y|$ by Exercise 1.2.4.

The case $y < 0_F \le x$ is by this handled as well because TI is symmetric in x and y. $\square$

In the next exercises, we collect several properties of absolute value.

Exercise 1.4.10 *Let F be an ordered field and $x, y \in F$. Then $|x| = |-x|$, $||x|| = |x|$, $x \ne 0_F \Rightarrow |x^{-1}| = |x|^{-1}$, and $|x + y| \ge |x| - |y|$.*

Exercise 1.4.11 *Let F be an ordered field. We define a map $d\colon F^2 \to F$ by $d(x, y) = |x - y|$. Show that d has the properties of metric stated in Definition A.6.1.*

• *Division in fields and Archimedean ordered fields.* We introduce in any field the partial operation of division.

Definition 1.4.12 *Let F be a field and $a, b \in F$ with $b \neq 0_F$. We define the* ratio *of a and b as*

$$a/b := a \cdot b^{-1}\,.$$

The expression $a/0_F$ is not defined. We say that a/b arises by dividing *a by b.*

In a simple field, all elements can be obtained from zero by adding and subtracting one and by division.

Exercise 1.4.13 *Let F be a field and $a, b \in F \setminus \{0_F\}$. Then $a/1_F = a$, $1_F/a = a^{-1}$ and $(a/b)^{-1} = b/a$.*

Exercise 1.4.14 *Let F be a field and $a, b, c, d \in F \setminus \{0_F\}$. Then $(a/b)/(c/d) = (a \cdot d)/(b \cdot c)$.*

Recall the semi-ring homomorphism $f_F \colon \omega \to F$ in Definition 1.2.51.

Definition 1.4.15 *An ordered field*

$$F = \langle F,\, 0_F,\, 1_F,\, +,\, \cdot,\, < \rangle$$

is Archimedean *if for any element $x \in F$ there exists a number $m \in \omega$ such that $|x| \leq f_F(m)$.*

The term refers to *Archimedes of Syracuse (*$\approx$ 287 *to* $\approx$ 212 BCE*)*. A field is Archimedean if it does not contain infinitely large elements. Equivalently, by the next proposition, if it does not contain infinitesimals. These are elements $x \in F \setminus \{0_F\}$ such that $0_F < |x| < f_F(m)^{-1} = 1_F/f_F(m)$ for every $m \in \mathbb{N}$.

Exercise 1.4.16 *Prove the next proposition.*

Proposition 1.4.17 *Let F be an ordered field. The following three claims are mutually equivalent, in the sense that each can be simply derived from the other.*

1. *F is Archimedean.*
2. *For every $x \in F$ there exists $m \in \omega$ such that $x \leq f_F(m)$.*
3. *For every $x \in F \setminus \{0_F\}$ there exists $n \in \omega$ such that*

$$0_F < 1_F/f_F(n) < |x|\,.$$

Proposition 1.4.18 *Every simple ordered field is Archimedean.*

Proof. We only outline the proof. We can write every positive element $x \in F$ as the ratio $f_F(m)/f_F(n)$ with $m, n \in \omega$ and $n > 0$. Then

$$x = f_F(m)/f_F(n) \leq f_F(m)\,.$$

$\square$

• *Complete ordered fields.* Completeness discriminates between the fields $\mathbb{Q}$ and $\mathbb{R}$. The former is not complete, the latter is. Recall the suprema and infima of sets in linear orders, defined in Section 1.1.

Definition 1.4.19 *Let*

$$F = \langle F,\, 0_F,\, 1_F,\, +,\, \cdot,\, < \rangle$$

be an ordered field. We say that F is complete *if for every set $\emptyset \ne X \subset F$ that is bounded from above, which means that $x \le y$ for every $x \in X$ and some $y \in F$, the supremum $\sup(X) \in F$ exists in the linear order $\langle F, < \rangle$.*

Exercise 1.4.20 *In any complete ordered field, every nonempty and lower-bounded set has an infimum.*

• *The main theorem on complete ordered fields.* We first define Cauchy sequences and limits.

Definition 1.4.21 *In an ordered field F, we say that a sequence $(a_n) \subset F$ is* Cauchy *if for every $e \in F$ with $e > 0_F$ there is an $n_0 \in \mathbb{N}$ such that if $m, n \ge n_0$, then*

$$|a_m - a_n| \le e\,.$$

Definition 1.4.22 *We say that $a \in F$ is a* limit *of a sequence $(a_n) \subset F$, and write $\lim a_n = a$, if for every $e \in F$ with $e > 0_F$ there is an $n_0 \in \mathbb{N}$ such that if $n \ge n_0$, then*

$$|a - a_n| \le e\,.$$

In Definition 1.7.3, we provide an alternative definition of limits of sequences, which works in linear orders. In the case of ordered fields, both definitions are equivalent.

Exercise 1.4.23 *Every Cauchy sequence (a_n) in an ordered field F is bounded. In more detail, there exists $c \in F$ such that $|a_n| \le c$ for every $n \in \mathbb{N}$.*

Exercise 1.4.24 *Limits are unique. That is, if $\lim a_n = a$ and $\lim a_n = b$, then $a = b$.*

Exercise 1.4.25 *If a sequence $(a_n) \subset F$, where F is an ordered field, has a limit, then (a_n) is Cauchy.*

We employ a lemma, called the *infinite Erdős–Szekeres lemma*. It is named after *Paul (Pál) Erdős (1913–1996)* and *George (György) Szekeres (1911–2005)*. A sequence $(a_n) \subset A$ in a linear order $\langle A, < \rangle$ is monotone if

$$a_1 \le a_2 \le a_3 \le \dots$$

or if the same holds with $\le$ replaced by $\ge$.

Lemma 1.4.26 *Any sequence in any linear order $\langle A, < \rangle$ has a monotone subsequence.*

Proof. Let $(a_n) \subset A$. We define

$$X = \{n \in \mathbb{N}\colon\ \forall\, m\colon\ m > n \Rightarrow a_n > a_m\}\ \ (\subset \mathbb{N})\,.$$

If the set X is infinite, $X = \{n_1 < n_2 < \dots\}$, then

$$a_{n_1} > a_{n_2} > \dots$$

is a monotone subsequence of (a_n). Suppose that X is finite. If $X = \emptyset$, we take any $n_1 \in \mathbb{N}$. If $X \neq \emptyset$, we take any $n_1 \in \mathbb{N}$ with $n_1 > \max(X)$. Since $n_1 \notin X$, there exists an index $n_2 > n_1$ such that $a_{n_1} \le a_{n_2}$. Since $n_2 \notin X$, there exists an index $n_3 > n_2$ such that $a_{n_2} \le a_{n_3}$. Continuing in this way, we obtain the monotone subsequence of (a_n)

$$a_{n_1} \le a_{n_2} \le a_{n_3} \le \dots\,.$$

□

Proposition 1.4.27 *Let F be a complete ordered field, and $a_1, a_2, \dots$ and b be elements in F such that $a_1 \le a_2 \le \dots \le a_n \le \dots \le b$. Then*

$$\lim a_n = \sup(\{a_n\colon\ n \in \mathbb{N}\})\ \ (\le b)\,.$$

If we reverse the inequalities to $\ge$, then the same equality holds with sup *replaced by* inf*.*

Proof. We denote the stated supremum by c $(\in F)$. Let an $e \in F$ with $e > 0_F$ be given. By the definition of supremum, there exists $m \in \mathbb{N}$ such that $c - e < a_m \le c$. By the assumption on (a_n) and the definition of c, these inequalities still hold if m is replaced with any $n \ge m$. Thus for every $n \ge m$,

$$|a_n - c| < e\,,$$

and $\lim a_n = c$. In the second part with $\ge$s and inf, we argue similarly. □

The main theorem on complete ordered fields follows.

Theorem 1.4.28 *Let F be a complete ordered field. Then the following holds.*

1. *F is Archimedean.*
2. *Every Cauchy sequence in F has a (unique) limit.*

Proof. 1. It suffices to show that the set $f_F[\omega]$ $(\subset F)$ is not bounded from above. We assume, for the contrary, that it is and set $a := \sup(f_F[\omega])$. Then, by Proposition 1.1.48, there exists $b \in f_F[\omega]$ such that

$$a - 1_F < b \le a\,.$$

Then $b + 1_F \in f_F[\omega] \le a$ and $a < b + 1_F$, which is a contradiction.

2. Let $(a_n) \subset F$ be a Cauchy sequence. Let (b_n) be a monotone subsequence of (a_n) guaranteed by Lemma 1.4.26. We write $b_n = a_{m_n}$. By Exercise 1.4.23, the sequence (b_n) is bounded, and

$$\lim b_n = \begin{matrix}\sup\\ \inf\end{matrix}(\{b_n\colon\ n \in \mathbb{N}\}) =: b$$

by Proposition 1.4.27. We show that $\lim a_n = b$. Let $e \in F$ with $e > 0_F$ be given. We take $n_0 \in \mathbb{N}$ such that if $m, n \ge n_0$, then

$$|a_m - a_n| \le e/2_F \ \text{ and } \ |b_n - b| = |a_{m_n} - b| \le e/2_F\,.$$

Then for every $n \ge n_0$ we have

$$|a_n - b| \le |a_n - a_{m_n}| + |a_{m_n} - b| \le e/2_F + e/2_F = e\,.$$

We see that $\lim a_n = b$. We used that $m_n \ge n$. □

The only weak point of this nice theorem is that, as we will see in the next section, it applies only to a uniquely determined (up to isomorphism) object, the real numbers $\mathbb{R}$.

• *The set of fractions and the linear order on it.* Finally, we begin to define the algebraic structure of fractions $\mathbb{Q}$. We obtain $\mathbb{Q}$ by the standard algebraic construction as the field of fraction of the domain $\mathbb{Z}$.

For $m, n \in \mathbb{Z}$ we define $\frac{m}{n} = m/n := \langle m, n\rangle$ and set

$$Z = \left\{\tfrac{m}{n}\colon\ m \in \mathbb{Z},\, n \in \mathbb{Z} \setminus \{0\}\right\}.$$

The elements of Z are called <u>protofractions</u>. In a protofraction $\frac{m}{n}$, the integer m is the <u>numerator</u>, and n is the <u>denominator</u>.

We define on Z a relation $\sim$ by

$$k/l \sim m/n \iff k \cdot n = l \cdot m$$

(the multiplication is in $\mathbb{Z}$).

Exercise 1.4.29 *Show that $\sim$ is an equivalence relation on Z.*

For the next definition, recall relations of equivalence and the notation $\cdots/\!\sim$ and $[a]_\sim$ introduced in Section 1.1.

Definition 1.4.30 *The set of equivalence blocks*

$$\mathbb{Q} := Z/\!\sim$$

is the set of fractions *or* rational numbers. *We often refer to the blocks* $[m/n]_\sim$ *via the protofraction representatives* m/n *or* $\frac{m}{n}$.

Exercise 1.4.31 *Every fraction has a protofraction representative with positive denominator.*

For the next exercise recall that two numbers $m, n \in \mathbb{Z}$ are coprime, if their only common divisors are -1 and 1 (cf. Exercise 1.3.14). A protofraction $\frac{m}{n}$ is in lowest terms if $n > 0$ and m and n are coprime. We denote the set of protofractions in lowest terms by Z_0 ($\subset Z$).

Exercise 1.4.32 *There exists a unique bijection*

$$f\colon \mathbb{Q} \to Z_0 \;\text{ such that }\; f([m/n]_\sim) \in [m/n]_\sim\,.$$

So we have unique and distinct representatives of fractions by protofractions in their lowest terms.

We define a linear order on $\mathbb{Q}$. All comparisons by $<$ in the next definition are in $\mathbb{Z}$.

Definition 1.4.33 *We define a relation* $<_Z$ *on* Z *as follows. Let* a/b *and* c/d *be protofractions. For* $bd > 0$, *we set* $a/b <_Z c/d \iff ad < bc$. *For* $bd < 0$, *we set* $a/b <_Z c/d \iff ad > bc$.

Proposition 1.4.34 *The following holds.*

1. *The relation* $<_Z$ *on* Z *is irreflexive and transitive, and has the property that for every two protofractions* $\frac{a}{b}$ *and* $\frac{c}{d}$, *exactly one of* $\frac{a}{b} <_Z \frac{c}{d}$, $\frac{a}{b} >_Z \frac{c}{d}$, *and* $\frac{a}{b} \sim \frac{c}{d}$ *holds.*

2. *If* $\frac{a}{b} \sim \frac{a'}{b'}$ *and* $\frac{c}{d} \sim \frac{c'}{d'}$, *then* $\frac{a}{b} <_Z \frac{c}{d} \iff \frac{a'}{b'} <_Z \frac{c'}{d'}$.

Using this proposition, we introduce a linear order $<_\mathbb{Q}$ on $\mathbb{Q}$.

Definition 1.4.35 *Let* $\alpha, \beta \in \mathbb{Q}$ *be two distinct fractions. We define*

$$\alpha <_\mathbb{Q} \beta \iff a/b <_Z c/d \text{ for any } a/b \in \alpha \text{ and } c/d \in \beta\,.$$

By Proposition 1.4.34, this definition is correct, and $\langle \mathbb{Q}, <_\mathbb{Q} \rangle$ *is a linear order.*

Proof. (Proposition 1.4.34) 1. Irreflexivity of $<_Z$ is clear. Let $\frac{a}{b} <_Z \frac{c}{d}$ and $\frac{c}{d} <_Z \frac{e}{f}$. Suppose that $bd, df > 0$. Then $ad < bc$ and $cf < de$. It follows that

$$ad^2 f < dfbc = bdcf < bd^2 e \;\text{ and }\; af < be\,.$$

Since $bf > 0$, this means that $\frac{a}{b} <_Z \frac{e}{f}$. The other three cases are similar. Suppose, for example, that $bd > 0$ but $df < 0$. Then $ad < bc$ and $cf > de$. It follows that

$$ad^2f > dfbc = bdcf > bd^2e \text{ and } af > be\,.$$

Since $bf < 0$, this means that again $\frac{a}{b} <_Z \frac{e}{f}$. In the remaining two cases we proceed similarly. This proves that $<_Z$ is transitive.

Let $\frac{a}{b}, \frac{c}{d} \in Z$. If $ad = bc$, then $\frac{a}{b} \sim \frac{c}{d}$. If $ad \neq bc$, then according to if $ad < bc$ or $ad > bc$, and according to the sign of bd, we have exactly one of $\frac{a}{b} <_Z \frac{c}{d}$ and $\frac{a}{b} >_Z \frac{c}{d}$. This proves the last property of $<_Z$.

2. Let $\frac{a}{b} \sim \frac{a'}{b'}$, $\frac{c}{d} \sim \frac{c'}{d'}$, and let $\frac{a}{b} <_Z \frac{c}{d}$. It suffices to show that $\frac{a'}{b'} <_Z \frac{c'}{d'}$ as well. Thus $ab' = ba'$ and $cd' = dc'$. Let $bd > 0$. Then $ad < bc$. If $b'd' > 0$, we have by the second order axiom that

$$bda'd' = adb'd' < bcb'd' = bdb'c'\,.$$

hence, again by the second order axiom, $a'd' < b'c'$ and $\frac{a'}{b'} <_Z \frac{c'}{d'}$. In the remaining three cases when $b'd' < 0$ or/and $bd < 0$ we argue similarly. □

• *Addition and multiplication of fractions. The algebraic structure* $\mathbb{Q}$. We define, in the standard way, the arithmetic on $\mathbb{Q}$. All arithmetic operation on the right-hand side of := in the next definition are in $\mathbb{Z}$.

Definition 1.4.36 *Let $\alpha = [a/b]_\sim$ and $\beta = [c/d]_\sim$ be two fractions.*

1. *We define $\alpha + \beta := [(ad + cb)/bd]_\sim$.*
2. *We define $\alpha \cdot \beta := [ac/bd]_\sim$.*

The next exercise shows that the definition is correct: the value of + and · does not depend on the protofraction representatives.

Exercise 1.4.37 *Let $a/b \sim a'/b'$ and $c/d \sim c'/d'$ be two pairs of equivalent protofractions. Then $a/b + c/d \sim a'/b' + c'/d'$ and $a/b \cdot c/d \sim a'/b' \cdot c'/d'$.*

We define the algebraic structure $\mathbb{Q}$.

Definition 1.4.38 *The algebraic structure of fractions*

$$\mathbb{Q} = \langle \mathbb{Q},\, 0_{\mathbb{Q}},\, 1_{\mathbb{Q}},\, +,\, \cdot,\, <_{\mathbb{Q}} \rangle$$

consists of the set $\mathbb{Q}$ introduced in Definition 1.4.30, the elements $0_{\mathbb{Q}} := [0/1]_\sim$ and $1_{\mathbb{Q}} := [1/1]_\sim$ in $\mathbb{Q}$, the operations of addition + and multiplications · on $\mathbb{Q}$ introduced in Definition 1.4.36, and the linear order $<_{\mathbb{Q}}$ on $\mathbb{Q}$ introduced in Definition 1.4.33.

We write just 0 and 1 instead of $0_{\mathbb{Q}}$ and $1_{\mathbb{Q}}$. Instead of $<_{\mathbb{Q}}$ we write just $<$.

Exercise 1.4.39 *Show that $0_{\mathbb{Q}} \neq 1_{\mathbb{Q}}$.*

• $\mathbb{Q}$ *is a simple ordered field.* $\mathbb{Z}$ *embeds as a subring in* $\mathbb{Q}$. We prove the first claim in Theorem 1.4.7.

Proposition 1.4.40 *The algebraic structure* $\mathbb{Q}$ *introduced in Definition 1.4.38 is a simple ordered field.*

Proof. The element 0 is neutral to $+$: $\frac{0}{1}+\frac{a}{b}=\frac{0b+1a}{1b}=\frac{a}{b}$. Similarly, 1 is neutral to $\cdot$: $\frac{1}{1}\cdot\frac{a}{b}=\frac{1a}{1b}=\frac{a}{b}$. The commutativity of $+$ and $\cdot$ follows from the commutativity of these operations in $\mathbb{Z}$. The same holds for the associativity of $\cdot$. We prove the associativity of $+$. By the distributive law in $\mathbb{Z}$ we have

$$\left(\tfrac{a}{b}+\tfrac{c}{d}\right)+\tfrac{e}{f}=\tfrac{(ad+bc)f+bde}{bdf}=\tfrac{adf+b(cf+de)}{bdf}=\tfrac{a}{b}+\left(\tfrac{c}{d}+\tfrac{e}{f}\right).$$

The distributive law holds in $\mathbb{Q}$ too:

$$\tfrac{a}{b}\cdot\left(\tfrac{c}{d}+\tfrac{e}{f}\right)=\tfrac{a(cf+de)}{bdf}\sim\tfrac{acbf+bdae}{b^2df}=\tfrac{a}{b}\cdot\tfrac{c}{d}+\tfrac{a}{b}\cdot\tfrac{e}{f}\,.$$

The additive inverse of $\frac{a}{b}$ is $\frac{-a}{b}$:

$$\tfrac{a}{b}+\tfrac{-a}{b}=\tfrac{ab+b(-a)}{b^2}=\tfrac{0}{b^2}\sim\tfrac{0}{1}\,.$$

The multiplicative inverse of $\frac{a}{b}\not\sim\frac{0}{1}$, which means that $a\neq 0$, is $\frac{b}{a}$:

$$\tfrac{a}{b}\cdot\tfrac{b}{a}=\tfrac{ab}{ba}\sim\tfrac{1}{1}\,.$$

We show that the two order axioms hold in $\mathbb{Q}$. Let $\frac{a}{b}<\frac{c}{d}$ and $\frac{e}{f}$ be three protofraction with $b,d,f>0$ (we use Exercise 1.4.31). Thus $ad<bc$. Then also

$$\tfrac{a}{b}+\tfrac{e}{f}<\tfrac{c}{d}+\tfrac{e}{f}$$

because $(af+be)df<(cf+de)bf \iff adf^2<bcf^2 \iff ad<bc$ in $\mathbb{Z}$. Suppose that in addition $\frac{e}{f}>\frac{0}{1}$, which means that $e>0$. Then

$$\tfrac{a}{b}\cdot\tfrac{e}{f}<\tfrac{c}{d}\cdot\tfrac{e}{f}$$

because $aedf<bfce \iff ad<bc$ in $\mathbb{Z}$.

Finally, we show that the field $\mathbb{Q}$ is simple. Let $\alpha=[m/n]_\sim$ be any fraction. We can easily obtain the fractions $\beta=[m/1]_\sim$ and $\gamma=[n/1]_\sim$ by adding and subtracting $1_\mathbb{Q}$ to and from $0_\mathbb{Q}$. Then we express α as $\alpha=\beta/\gamma$. $\square$

Exercise 1.4.41 *Prove the next proposition.*

Proposition 1.4.42 *The map* $F\colon\mathbb{Z}\to\mathbb{Q}$*, given by*

$$F(m):=[m/1]_\sim\,,$$

is an injective homomorphism of rings.

Thus the set
$$F[\mathbb{Z}] = \{[m/1]_\sim :\ m \in \mathbb{Z}\} \quad (\subset \mathbb{Q})$$
induces a subring of $\mathbb{Q}$ isomorphic to the ring $\mathbb{Z}$. We usually abuse notation and pretend that $\mathbb{Z} \subset \mathbb{Q}$.

Exercise 1.4.43 *Deduce from this embedding of $\mathbb{Z}$ in $\mathbb{Q}$, without using the order, that $\mathbb{Z}$ is a domain.*

• *Incompleteness of fractions.* We show that the ordered field $\mathbb{Q}$ is not complete. We deduce it from the fact that the equation
$$x^2 = 2$$
has no solution in $\mathbb{Q}$.

Proposition 1.4.44 *For every $\alpha \in \mathbb{Q}$ we have $\alpha^2 \neq 2$.*

Proof. For the contrary, let $\frac{m}{n} \in Z$ be such that $([m/n]_\sim)^2 = [2/1]_\sim$. We may assume that $m, n \in \mathbb{N}$ (Exercise 1.4.45). Thus $m^2 = 2n^2$ and by Theorem 1.2.24, we may assume that m is minimum. It follows (Exercise 1.4.46) that m is even and $m = 2l$ with $l \in \mathbb{N}$. Then
$$(2l)^2 = 2n^2 \ \text{ and } \ n^2 = 2l^2 \,.$$
If $n \geq m$, then by Exercise 1.2.4 we have
$$m^2 = 2n^2 \geq 2nm \geq 2m^2 \ \text{ and } \ 0 \geq m^2 \,,$$
which is a contradiction with $m^2 \in \mathbb{N}$. Thus $n < m$, but this is also a contradiction, with the minimality of m. The protofraction $\frac{m}{n}$ does not exist. □

Exercise 1.4.45 *Why can we assume in the proof that $m, n > 0$?*

Exercise 1.4.46 *Why is m even?*

Corollary 1.4.47 *The ordered field $\mathbb{Q}$ is not complete. For example, the set*
$$X = \{\alpha \in \mathbb{Q} :\ \alpha^2 < 2\} \quad (\subset \mathbb{Q})$$
is nonempty and bounded from above, but $\sup(X)$ does not exist in the linear order $\langle \mathbb{Q}, < \rangle$.

Proof. We have $1 \in X$ and $x < 2$ for every $x \in X$. For the contrary, let
$$s := \sup(X) \quad (\in \mathbb{Q}) \,.$$

Clearly, $s > 0$. Let $s^2 > 2$. Then there is an $r \in \mathbb{Q}$ such that $0 < r < s$ and still $(s-r)^2 > 2$. Hence for every $x \in X$ we have

$$(s-r)^2 > 2 > x^2 \text{ and } s - r > x\,.$$

This contradicts the fact that s is the smallest upper bound of X. Let $s^2 < 2$. Then there is a fraction $r > 0$ such that still $(s+r)^2 < 2$. Thus

$$s < s + r \in X\,,$$

which contradicts the fact that s is an upper bound of X. The trichotomy of $<$ implies that $s^2 = 2$, but this is excluded by the previous theorem. We deduce that the supremum s does not exist. $\Box$

Exercise 1.4.48 *Provide concrete values for the fractions $r = r(s)$.*

• *Field embeddings $f_F\colon \mathbb{Q} \to F$.* We proceed as for the structures $\mathbb{N}_0$ and $\mathbb{Z}$ and define isomorphisms of $\mathbb{Q}$ with simple ordered fields. To avoid dividing by zero, we assume from the beginning that the field F is ordered.

Definition 1.4.49 *Let*

$$F = \langle F,\, 0_F,\, 1_F,\, \oplus,\, \odot,\, \prec\rangle$$

be an ordered field. We define a function $f_F\colon \mathbb{Q} \to F$ by means of the ring homomorphism $f_F\colon \mathbb{Z} \to F$ introduced in Definition 1.3.30. We set

$$f_F(\alpha) = f_F(a) \oslash f_F(b) := f_F(a) \odot f_F(b)^{-1},\ \alpha = [a/b]_\sim \in \mathbb{Q}\,.$$

In contrast to the structures $\mathbb{N}_0$ and $\mathbb{Z}$, now we have to show that the definition of f_F is correct.

Lemma 1.4.50 *Let F be as in the previous definition. Then the ring homomorphism $f_F\colon \mathbb{Z} \to F$ has the property that $f_F(m) \neq 0_F$ for every $m \in \mathbb{Z} \setminus \{0\}$.*

Proof. By Lemma 1.3.34, the inequality $m < 0$, respectively $0 < m$, is preserved by f_F. We have

$$f_F(m) \prec f_F(0) = 0_F, \text{ respectively } f_F(0) = 0_F \prec f_F(m)\,,$$

which means that for nonzero $m \in \mathbb{Z}$ also $f_F(m) \neq 0_F$. $\Box$

Lemma 1.4.51 *Let F be as in the previous definition, $f_F\colon \mathbb{Z} \to F$ be the ring homomorphism in Definition 1.3.30, and let $\frac{a}{b} \sim \frac{c}{d}$ be two equivalent protofractions. Then always*

$$f_F(a) \oslash f_F(b) = f_F(c) \oslash f_F(d) \quad (\in F)\,.$$

Proof. $\frac{a}{b} \sim \frac{c}{d}$ means that $a \cdot d = b \cdot c$, multiplied in $\mathbb{Z}$. By Lemma 1.3.33,

$$f_F(a) \odot f_F(d) = f_F(a \cdot d) = f_F(b \cdot c) = f_F(b) \odot f_F(c).$$

The stated equality follows via multiplication by $f_F(d)^{-1} \odot f_F(b)^{-1}$. This element is defined due to the previous lemma. $\square$

We prove four lemmas. In them,

$$F = \langle F,\, 0_F,\, 1_F,\, \oplus,\, \odot,\, \prec \rangle$$

is an ordered field and $f_F \colon \mathbb{Q} \to F$ is as in Definition 1.4.49. The lemmas show that f_F is a field embedding, and that if F is simple, then f_F is an isomorphism of ordered fields.

Lemma 1.4.52 *For every $\alpha, \beta \in \mathbb{Q}$, we have $f_F(\alpha + \beta) = f_F(\alpha) \oplus f_F(\beta)$.*

Proof. We write f for f_F. Let $\alpha = [a/b]_\sim$ and $\beta = [c/d]_\sim$ be two fractions. We compute

$$\begin{aligned} f(\alpha + \beta) &= f([(ad + bc)/bd]_\sim) = f(ad + bc) \oslash f(bd) \\ &= ((f(a) \odot f(d) \oplus f(b) \odot f(c)) \oslash (f(b) \odot f(d)) \\ &= (f(a) \oslash f(b)) \oplus (f(c) \oslash f(d)) = f(\alpha) \oplus f(\beta)\,. \end{aligned}$$

The first equality follows from the definition of $+$ in $\mathbb{Q}$. The second equality is due to the definition of f. In the third equality, we use Lemmas 1.3.32 and 1.3.33. The fourth equality is a computation in the field F. In the last, fifth equality, we use the definition of f. $\square$

Lemma 1.4.53 *For every $\alpha, \beta \in \mathbb{Q}$, we have $f_F(\alpha \cdot \beta) = f_F(\alpha) \odot f_F(\beta)$.*

Proof. We write f for f_F. Let $\alpha = [a/b]_\sim$ and $\beta = [c/d]_\sim$ be two fractions. We compute

$$\begin{aligned} f(\alpha \cdot \beta) &= f([ac/bd]_\sim) = f(ac) \oslash f(bd) = ((f(a) \odot f(c)) \oslash (f(b) \odot f(d)) \\ &= (f(a) \oslash f(b)) \odot (f(c) \oslash f(d)) = f(\alpha) \odot f(\beta)\,. \end{aligned}$$

The first equality follows from the definition of $\cdot$ in $\mathbb{Q}$. The second equality is due to the definition of f. In the third equality, we use Lemma 1.3.33. The fourth equality is a computation in the field F. In the last, fifth equality, we use the definition of f. $\square$

Lemma 1.4.54 *For every $\alpha, \beta \in \mathbb{Q}$ we have*

$$\alpha < \beta \iff f_F(\alpha) \prec f_F(\beta)\,.$$

Proof. We write f for f_F. By the trichotomy of $<$, it suffices to prove the implication

$$\alpha < \beta \Rightarrow f(\alpha) \prec f(\beta)\,.$$

Let $\alpha = [a/b]_\sim$ and $\beta = [c/d]_\sim$ be two fractions such that $\alpha < \beta$ and $b, d > 0$ (Exercise 1.4.31). Thus $ad < bc$ and, by Lemma 1.3.34, $f(b), f(d) \succ 0_F = f(0)$. We have

$$f(a) \odot f(d) = f(ad) \prec f(bc) = f(b) \odot f(c)\,.$$

The first inequality follows from Lemma 1.4.53. The second inequality follows from Lemma 1.3.34; we view f as the ring homomorphism $f_F\colon \mathbb{Z} \to F$. The last, third inequality follows from Lemma 1.4.53. The second order axiom yields, after multiplying the last displayed inequality by the element $f(b)^{-1} \odot f(d)^{-1} \succ 0_F$ (Exercise 1.4.55), that

$$f(\alpha) = f(a) \oslash f(b) \prec f(c) \oslash f(d) = f(\beta)\,.$$

$\Box$

Exercise 1.4.55 *In any ordered field, the multiplicative inverse of a positive element is positive, and so is the product of two positive elements.*

- *Simple ordered fields are mutually isomorphic.* We employ one more lemma.

Lemma 1.4.56 *Let F be in addition simple. Then the map*

$$f_F\colon \mathbb{Q} = \langle \mathbb{Q},\, 0,\, 1,\, +,\, \cdot,\, < \rangle \to F = \langle F,\, 0_R,\, 1_R,\, \oplus,\, \odot,\, \prec \rangle$$

is an isomorphism of ordered fields.

Proof. We only need to prove that $f_F\colon \mathbb{Q} \to F$ is a bijection; the three required properties of f_F are proven in the three previous lemmas. The map f_F is injective by Lemma 1.4.54 or by Proposition 1.4.6. We prove that f_F is surjective. It suffices to show, since F is simple, that $f_F[\mathbb{Q}]$ contains 0_F and is closed under adding and subtracting 1_F, and under division. Clearly, $0_F = f_F(0)$. Let $x = f_F(\alpha)$ and $y = f_F(\beta) \ne 0_F$ for some $\alpha, \beta \in \mathbb{Q}$ with $\beta \ne 0$. Then, by Lemma 1.4.52,

$$x \oplus \ominus 1_F = f_F(\alpha) \oplus \ominus 1_F = f_F(\alpha \pm 1) \in f_F[\mathbb{Q}]\,.$$

By Lemma 1.4.53,

$$x \oslash y = f_F(\alpha) \oslash f_F(\beta) = f_F(\alpha/\beta) \in f_F[\mathbb{Q}]\,.$$

Thus $f_F[\mathbb{Q}] = F$ and we are done. $\Box$

We are ready to prove the second claim in Theorem 1.4.7 and thereby complete its proof. Recall Proposition 1.2.8.

Proposition 1.4.57 *Suppose that*

$$F = \langle F,\, 0_F,\, 1_F,\, +,\, \cdot,\, < \rangle \ \textit{and}\ G = \langle G,\, 0_G,\, 1_G,\, \oplus,\, \odot,\, \prec \rangle$$

are two simple ordered fields. <u>*Then*</u> *F and G are isomorphic, which means that there exists a bijection $f\colon F \to G$ with three properties.*

1. *For every $a, b \in F$, we have $f(a + b) = f(a) \oplus f(b)$.*
2. *For every $a, b \in F$, we have $f(a \cdot b) = f(a) \odot f(b)$.*
3. *For every $a, b \in F$, we have $a < b \iff f(a) \prec f(b)$.*

Proof. By Lemma 1.4.56 and Exercises 1.1.18 and 1.1.19, the map

$$f\colon F \to G,\ f := f_G(f_F^{-1}),$$

is the desired isomorphism. □

Exercise 1.4.58 *Is this isomorphism unique?*

1.5 Real numbers

We describe in modern terms the epochal discovery due to *Richard Dedekind (1831–1916).* In 1858, as a lecturer in mathematical analysis in Zurich, he defined real numbers as certain sets of fractions. Now we call them (Dedekind) cuts. In our modernization, we define arithmetic of cuts via a map from rational Cauchy sequences to cuts. The main result is Theorem 1.5.1 characterizing real numbers as the up to isomorphism unique complete ordered field.

• *Complete ordered fields.* Recall Definition 1.4.19 of complete ordered fields. As for $\mathbb{N}_0$, $\mathbb{Z}$, and $\mathbb{Q}$, we prove the next theorem in two steps. In Proposition 1.5.23 we prove the existential part. The uniqueness of real numbers is proven in Proposition 1.5.41.

Theorem 1.5.1 *There exists a complete ordered field. Every two complete ordered fields are isomorphic.*

We have the class of algebraic structures that are real numbers.

Corollary 1.5.2 *The class*

$$\text{REAL NUMBERS} := \{x\colon\ x \textit{ is a complete ordered field}\}$$

contains the "standard" real numbers $\mathbb{R}$ and every two sets in it are isomorphic as ordered fields.

Another definition of real numbers emerged around 1872 due to *Georg Cantor (1845–1918)* and *Eduard Heine (1821–1881)*; Heine actually published Cantor's findings. Somewhat earlier similar results had been obtained by *Charles Méray (1835–1911).* In this approach, real numbers form the set

$$\mathbb{R} = \mathcal{S}/\sim\,,$$

where $\mathcal{S}$ is the set of Cauchy sequences $(a_n) \subset \mathbb{Q}$ (recall Definition 1.4.21) and $\sim$ is the following equivalence relation on $\mathcal{S}$:

$$(a_n) \sim (b_n) \iff \lim(a_n - b_n) = 0$$

(recall Definition 1.4.22).

Exercise 1.5.3 *Show that $\sim$ is an equivalence relation.*

• *Hereditarily at most countable sets.* In the next section, we show that the set of real numbers is, in any definition, uncountable. In the Cantor–Heine–Méray definition, even every real number is an uncountable set. In contrast, every Dedekind cut is a countable, even HMC, set.

Definition 1.5.4 *A set x is hereditarily at most countable, abbreviated* HMC*, if for every $n \in \omega$ and every chain of sets*

$$x_n \in x_{n-1} \in \cdots \in x_0 = x\,,$$

the set x_n is at most countable.

Thus $\{\omega\}$ is HMC.

Exercise 1.5.5 *The set of fractions $\mathbb{Q}$ is* HMC*.*

• *Cuts. A linear order on cuts.* We define real numbers as cuts.

Definition 1.5.6 *The set of real numbers is*

$$\mathbb{R} := \{X \in \mathcal{P}(\mathbb{Q}) \colon\ X \textit{ is a cut}\}\,,$$

where a cut is any set $X \subset \mathbb{Q}$ such that (i) $\emptyset \neq X \neq \mathbb{Q}$, (ii) *for every $a, b \in \mathbb{Q}$ with $a < b \in X$ also $a \in X$, and* (iii) *X has no largest element. We denote the set of cuts also by $\mathcal{C}$.*

Exercise 1.5.7 *Every cut is* HMC*.*

The linear order on $\mathbb{R}$ is the strict inclusion.

Definition 1.5.8 *For two distinct cuts $\alpha, \beta \in \mathbb{R}$, we set $\alpha < \beta \iff \alpha \subset \beta$.*

Exercise 1.5.9 *Show that $\langle \mathbb{R}, < \rangle$ is a linear order.*

In Dedekind's definition, the completeness of real numbers is easy to show.

Theorem 1.5.10 *Let a set X with $\emptyset \neq X \subset \mathbb{R}$ be bounded from above in the linear order $\langle \mathbb{R}, < \rangle$. Then the supremum $\sup(X)$ exists.*

Proof. We set $\alpha = \bigcup X$ and prove that (i) α is a cut, (ii) α is an upper bound of X, and (iii) α is the least upper bound of X.

(i) Clearly, $\alpha \neq \emptyset$. Let $\beta \in \mathbb{R}$ be such that $\gamma \leq \beta$ for every $\gamma \in X$. We take any $a \in \mathbb{Q} \setminus \beta$ and conclude that $a \notin \alpha$. Thus $\alpha \neq \mathbb{Q}$. Let $a, b \in \mathbb{Q}$ be such that $a < b \in \alpha$. Then $b \in \beta \in X$ for some cut β, and $a \in \beta$. Thus $a \in \alpha$. Finally, let $a \in \alpha$. Then $a \in \beta \in X$ for some cut β, and $a < b \in \beta$ for some fraction b. Thus $a < b \in \alpha$, and a is not the maximum of α. We see that α is a cut.

(ii) This is obvious, $\beta \subset \alpha$ for every $\beta \in X$.

(iii) Let β be any cut with $\beta < \alpha$. Thus $\beta \subset \alpha$ but $\beta \neq \alpha$. We take a fraction $a \in \alpha \setminus \beta$. Then $a \in \gamma \in X$ for some cut γ and

$$\beta \subset \{b \in \mathbb{Q} \colon\ b < a\} \subset \gamma\,.$$

Hence $\beta < \gamma$ and β is not an upper bound of X. We see that α is the least upper bound of X. $\square$

Exercise 1.5.11 *Prove that in the linear order $\langle \mathbb{R}, < \rangle$ every nonempty and lower-bounded set has an infimum.*

• *The map Φ.* We define arithmetic operations on $\mathbb{R}$ by means of a map Φ from $\mathcal{S}$ to $\mathcal{C}$. Recall that $\mathcal{S}$ denotes the set of Cauchy sequences in $\mathbb{Q}$ and that $\mathcal{C} = \mathbb{R}$ is the set of cuts.

Definition 1.5.12 *Let $(a_n) \in \mathcal{S}$,*

$$A := \{b \in \mathbb{Q} \colon\ \exists\, n_0 \colon\ n \geq n_0 \Rightarrow b \leq a_n\} \quad (\subset \mathbb{Q})\,,$$

and let $B := \{\max(A)\}$ if this maximum exists, and $B := \emptyset$ otherwise. We define the map $\Phi \colon \mathcal{S} \to \mathcal{C}$ by

$$\Phi(a_n) = \Phi((a_n)) := A \setminus B\,.$$

By the next exercise, every value of Φ is a cut.

Exercise 1.5.13 *Show that $\Phi(a_n)$ is a cut for every $(a_n) \in \mathcal{S}$.*

The next lemma is interesting by itself, it is a germ of a theorem on limits of monotone sequences (of real numbers).

Lemma 1.5.14 *Let $a_1, a_2, \ldots$ and b be fractions such that*

$$a_1 \geq a_2 \geq \cdots \geq a_n \geq \cdots \geq b\,.$$

Then (a_n) is a Cauchy sequence.

Proof. Let an $e \in \mathbb{Q}$ with $e > 0$ be given. If the Cauchy condition for (a_n) and e does not hold, then there is a sequence $m_1 < m_2 < \dots$ of indices in $\mathbb{N}$ such that $a_{m_n} - a_{m_{n+1}} \geq e$ for every $n \in \mathbb{N}$. But then

$$a_1 - b \geq a_{m_1} - a_{m_{n+1}} = \textstyle\sum_{j=1}^{n}(a_{m_j} - a_{m_{j+1}}) \geq ne$$

for every $n \in \mathbb{N}$, which is a contradiction. $\square$

Φ has the following properties. Recall the equivalence $\sim$ on $\mathcal{S}$ defined earlier.

Proposition 1.5.15 *The following holds.*

1. *Let $(a_n), (b_n) \in \mathcal{S}$. We have $(a_n) \sim (b_n) \iff \Phi(a_n) = \Phi(b_n)$.*
2. *Φ is a surjection.*

Proof. 1. Let $(a_n), (b_n) \in \mathcal{S}$. Suppose that $(a_n) \sim (b_n)$. Let $c \in \Phi(a_n)$. We take any $c' \in \Phi(a_n)$ with $c' > c$, which is possible because $\Phi(a_n)$ does not have a maximum element. Then $c < c' \leq a_n$ for every large n. It follows that $c \leq b_n$ for every large n. Thus $c \in \Phi(b_n)$ and $\Phi(a_n) \subset \Phi(b_n)$. The same argument shows the opposite inclusion and $\Phi(a_n) = \Phi(b_n)$. Let $(a_n) \not\sim (b_n)$. It follows that there exist fractions $e > e'$ such that $a_n \geq e > e' \geq b_n$ for every large n, or $b_n \geq e > e' \geq a_n$ for every large n. Then any fraction $c \in (e', e)$ has the property that $c \in \Phi(a_n) \setminus \Phi(b_n)$ in the former case, and $c \in \Phi(b_n) \setminus \Phi(a_n)$ in the latter case. Thus $\Phi(a_n) \neq \Phi(b_n)$.

2. Let $C \in \mathcal{C}$. We define by induction on $\mathbb{N}$ a sequence $(a_n) \in \mathcal{S}$ such that $\Phi(a_n) = C$. We take any $a_1 \in \mathbb{Q}$ satisfying $a_1 > b$ for every $b \in C$ and set $k := 1$. If fractions $a_1, a_2, \dots, a_n$ are already defined, we set $c := a_n - \frac{1}{k}$ and define a_{n+1} as follows. If $c > b$ for every $b \in C$, we set $a_{n+1} := c$. Else, when $c \leq b$ for some $b \in C$, we set $a_{n+1} := a_n$ and $k := k + 1$. The sequence (a_n) $(\subset \mathbb{Q})$ is Cauchy by Lemma 1.5.14. We show that $\Phi(a_n) = C$. By the definition of (a_n), for every $b \in C$ and $n \in \mathbb{N}$ we have $b < a_n$. Thus $C \subset \Phi(a_n)$. Let $b' \in \mathbb{Q} \setminus C$; then $b' > b$ for every $b \in C$. We show that $b' \notin \Phi(a_n)$. If b' is not $\min(\mathbb{Q} \setminus C)$, then by the definition of (a_n) we have $a_n < b'$ for every large n and $b' \notin \Phi(a_n)$. If $b' = \min(\mathbb{Q} \setminus C)$, then $C = \{b \in \mathbb{Q}\colon\ b < b'\}$ and b' is the largest fraction b such that $b \leq a_n$ for every large n (in fact, $b' \leq a_n$ for every n). But then b' is deleted from the set A by the definition of (a_n) and $b' \notin \Phi(a_n)$. $\square$

• *Addition and multiplication of cuts. The algebraic structure* $\mathbb{R}$. We define addition and multiplication on $\mathbb{R}$. First we employ a lemma.

Lemma 1.5.16 *Let $(a_n), (b_n), (a'_n), (b'_n) \in \mathcal{S}$. Then the following holds.*

1. *$(a_n + b_n), (a_n b_n) \in \mathcal{S}$.*
2. *If $(a_n) \sim (a'_n)$ and $(b_n) \sim (b'_n)$, then $(a_n + b_n) \sim (a'_n + b'_n)$ and $(a_n b_n) \sim (a'_n b'_n)$.*

Proof. 1. Let $e \in \mathbb{Q}$ with $e > 0$ be given. Then for every large m and n,

$$|a_m + b_m - (a_n + b_n)| \leq |a_m - a_n| + |b_m - b_n| \leq \varepsilon/2 + \varepsilon/2 = \varepsilon$$

and we see that $(a_n + b_n)$ is Cauchy. As for the product, we use that $|a_n|, |b_n| \leq c$ for every n and some constant $c > 0$ (Exercise 1.4.23). Then for every large m and n,

$$|a_m b_m - a_n b_n| \leq |a_m| \cdot |b_m - b_n| + |a_m - a_n| \cdot |b_n| \leq c\tfrac{\varepsilon}{2c} + \tfrac{\varepsilon}{2c}c = \varepsilon$$

and we see that $(a_n \cdot b_n)$ is Cauchy.

2. Exercise 1.5.17. $\square$

Exercise 1.5.17 *Prove part 2 of Lemma 1.5.16.*

Definition 1.5.18 *Let $\alpha, \beta \in \mathbb{R} = \mathcal{C}$ be two cuts. We define their sum and product as follows.*

1. $\alpha + \beta := \Phi(a_n + b_n)$ *for any* $(a_n) \in \Phi^{-1}(\alpha)$ *and* $(b_n) \in \Phi^{-1}(\beta)$.
2. $\alpha \cdot \beta := \Phi(a_n b_n)$ *for any* $(a_n) \in \Phi^{-1}(\alpha)$ *and* $(b_n) \in \Phi^{-1}(\beta)$.

The sequences $(a_n + b_n)$ and $(a_n b_n)$ are Cauchy by part 1 of Lemma 1.5.16. The preimages are nonempty by part 2 of Proposition 1.5.15. The result of the sum and product does not depend on the choice of (a_n) and (b_n) by part 2 of Lemma 1.5.16 and by part 1 of Proposition 1.5.15.

Definition 1.5.19 *The algebraic structure of* <u>*real numbers*</u>

$$\mathbb{R} := \langle \mathcal{C},\, 0_{\mathbb{R}},\, 1_{\mathbb{R}},\, +,\, \cdot,\, < \rangle$$

consists of the set of cuts $\mathcal{C}$ introduced in Definition 1.5.6, the cuts

$$0_{\mathbb{R}} := \{a \in \mathbb{Q}\colon\ a < 0\} \ \text{ and } \ 1_{\mathbb{R}} := \{a \in \mathbb{Q}\colon\ a < 1\}\,,$$

the operations of addition $+$ and multiplication $\cdot$ on $\mathcal{C}$ introduced in Definition 1.5.18, and the linear order $<$ on $\mathcal{C}$ introduced in Definition 1.5.8.

We usually write just 0 and 1 instead of $0_{\mathbb{R}}$ and $1_{\mathbb{R}}$.

• *Real numbers form a complete ordered field. $\mathbb{Q}$ naturally embeds in $\mathbb{R}$.* We employ two lemmas. By $\mathbb{Q}^{\mathbb{N}}$ we denote the set of sequences $(a_n) \subset \mathbb{Q}$.

Lemma 1.5.20 *The algebraic structure*

$$\mathbb{Q}^{\mathbb{N}} := \langle \mathbb{Q}^{\mathbb{N}},\, \overline{0},\, \overline{1},\, \oplus,\, \odot \rangle\,,$$

where $\overline{0} := (0, 0, \dots)$, $\overline{1} := (1, 1, \dots)$, and the operations $\oplus$ and $\odot$ are defined component-wise from $+$ and $\cdot$ in $\mathbb{Q}$, respectively, is a ring.

Proof. It is easy to see that $\overline{0}$ is neutral to $\oplus$, and $\overline{1}$ to $\odot$. Likewise, the commutativity and associativity of $\oplus$ and $\odot$, as well as the distributivity of $\odot$ to $\oplus$, are inherited from the structure $\mathbb{Q}$. Any sequence $(a_n) \in \mathbb{Q}^{\mathbb{N}}$ has an additive inverse, namely the sequence $(-a_n)$. $\square$

Exercise 1.5.21 *The ring $\mathbb{Q}^{\mathbb{N}}$ is not a domain.*

Lemma 1.5.22 *Let $\alpha = \Phi(a_n)$ and $\beta = \Phi(b_n)$ be two cuts. <u>Then</u>*

$$\alpha < \beta \iff \exists\, e \in \mathbb{Q},\, e > 0\, \exists\, n_0\colon\ n \geq n_0 \Rightarrow a_n \leq b_n - e\,.$$

Proof. Let $\alpha < \beta$. We take two fractions $c < c'$ in $\beta\backslash\alpha$. Then $a_n < c < c' \leq b_m$ for infinitely many n and every large m. Since (a_n) is Cauchy, we see that $a_n \leq b_n - e$ for every large n and some fraction $e > 0$. Let $\alpha \not< \beta$. Then either $\beta < \alpha$ or $\alpha = \beta$. In the former case, the right-hand side of the equivalence does not hold because it holds with a_n and b_n exchanged. In the latter case, $(a_n) \sim (b_n)$ by part 1 of Proposition 1.5.15. Then $\lim(a_n - b_n) = 0$ and the right-hand side of the equivalence again does not hold. $\square$

We prove the first claim in Theorem 1.5.1.

Proposition 1.5.23 *The structure of real numbers*

$$\mathbb{R} = \langle \mathbb{R},\, 0,\, 1,\, +,\, \cdot,\, < \rangle = \langle \mathcal{C},\, 0,\, 1,\, +,\, \cdot,\, < \rangle$$

introduced in Definition 1.5.19 is a complete ordered field.

Proof. Completeness of the linear order $\langle \mathcal{C}, < \rangle$ was proven in Theorem 1.5.10. It follows from part 1 in Proposition 1.5.15, Lemma 1.5.16, Definition 1.5.18, and Lemma 1.5.20 that the structure

$$\langle \mathcal{C},\, 0,\, 1,\, +,\, \cdot \rangle$$

is a ring. It remains to prove the existence of multiplicative inverses and the two order axioms.

Let $\alpha \in \mathbb{R} = \mathcal{C}$ with $\alpha \neq 0_{\mathbb{R}}$. Using part 2 of Proposition 1.5.15, we take any sequence $(a_n) \in \mathcal{S}$ such that $\Phi(a_n) = \alpha$. Since $\Phi(0, 0, \dots) = 0_{\mathbb{R}}$, part 1 of Proposition 1.5.15 shows that $(a_n) \not\sim (0, 0, \dots)$. Since (a_n) is Cauchy, it follows that there is a fraction $e > 0$ such that $|a_n| \geq e$ for every large n. For $n \in \mathbb{N}$, we define $b_n := 0$ if $a_n = 0$, and $b_n := a_n^{-1}$ if $a_n \neq 0$. By Exercise 1.5.24, $(b_n) \in \mathcal{S}$. We set $\beta := \Phi(b_n)$. By Definition 1.5.18 and part 1 of Proposition 1.5.15 we have

$$\alpha \cdot \beta = \Phi(a_n b_n) = \Phi(1,\, 1,\, \dots) = 1_{\mathbb{R}}\,,$$

because $(a_n b_n) \sim (1, 1, \dots)$. Thus $\beta = \alpha^{-1}$.

Let $\alpha = \Phi(a_n)$, $\beta = \Phi(b_n)$, and $\gamma = \Phi(c_n)$ be three cuts such that $\alpha < \beta$. By Lemma 1.5.22, $a_n \leq b_n - e$ for every large n and some fraction $e > 0$. Then

$a_n + c_n \le b_n + c_n - e$ for the same n. We see that $\alpha + \gamma < \beta + \gamma$ by the same lemma. Let additionally $\gamma > 0_{\mathbb{R}}$. Then $c_n \ge e'$ for every large n and some fraction $e' > 0$. Thus for every large n,

$$a_n c_n \le (b_n - e)c_n = b_n c_n - ec_n \le b_n c_n - ee'$$

and $\alpha \cdot \gamma < \beta \cdot \gamma$ by Lemma 1.5.22. $\Box$

Exercise 1.5.24 *Show that the sequence of reciprocals* $(b_n) = (a_n^{-1})$ *in the proof is Cauchy.*

The field embedding $f_{\mathbb{R}} \colon \mathbb{Q} \to \mathbb{R}$ introduced in Definition 1.4.49 embeds the field $\mathbb{Q}$ in the field $\mathbb{R}$. We describe a more natural embedding.

Exercise 1.5.25 *Prove the next proposition.*

Proposition 1.5.26 *The map* $F \colon \mathbb{Q} \to \mathbb{R}$ *given by*

$$F(\alpha) := \{\beta \in \mathbb{Q} \colon\ \beta < \alpha\},\ \alpha \in \mathbb{Q}\,,$$

is a field embedding of $\mathbb{Q}$ *in* $\mathbb{R}$.

• *Square roots.* We show that in a complete ordered field, every nonnegative element has a square root.

Proposition 1.5.27 *Let*

$$F = \langle F,\, 0_F,\, 1_F,\, \oplus,\, \odot,\, \prec\rangle$$

be a complete ordered field and let $\alpha \in F$ *with* $\alpha \succeq 0_F$. *Then there exists a unique element* $\beta \in F$ *such that* $\beta \succeq 0_F$ *and* $\beta^2 = \alpha$.

Proof. We consider the nonempty and upper-bounded set

$$X = \{\gamma \in F \colon\ \gamma^2 \preceq \alpha\}$$

and define $\beta := \sup(X)$. Since $0_F \in X$, it follows that $\beta \succeq 0_F$. The proof of Corollary 1.4.47 shows that $\beta^2 = \alpha$. We leave the uniqueness of β to Exercise 1.5.28 $\Box$

We call the number β the root of α and denote it by $\sqrt{\alpha}$. By Exercise 1.5.29, negative real numbers do not have a root.

Exercise 1.5.28 *Let* F *be a field. If* $a, b \in F$ *are such that* $a^2 = b^2$, *then* $a = b$ *or* $a = -b$.

Exercise 1.5.29 *Let* F *be an ordered field and* $a \in F$. *Then* $a^2 \ge 0_F$.

Exercise 1.5.30 *Let $f\colon F \to G$ be a field embedding between complete ordered fields and let $a, b \in F$. Then $a <_F b \iff f(a) <_G f(b)$.*

• *Complete ordered fields are mutually isomorphic.* In defining the isomorphisms, we proceed differently compared to the structures $\mathbb{N}_0$, $\mathbb{Z}$, and $\mathbb{Q}$. We define them right now.

Definition 1.5.31 *Let*

$$F = \langle F,\, 0_F,\, 1_F,\, +,\, \cdot,\, < \rangle \ \text{and}\ G = \langle G,\, 0_G,\, 1_G,\, \oplus,\, \odot,\, \prec \rangle$$

be two complete ordered fields. We define a function $\varphi\colon F \to G$ by means of the field embeddings $f_F\colon \mathbb{Q} \to F$ and $f_G\colon \mathbb{Q} \to G$ introduced in Definition 1.4.49. For $\alpha \in F$ we set

$$\varphi(\alpha) := \lim f_G(a_n)\,,$$

where $(a_n) \subset \mathbb{Q}$ is any sequence such that $\lim f_F(a_n) = \alpha$.

First, we show in two lemmas that the definition of the map φ is correct. Let F be an ordered field and $X \subset F$. We say that X is dense in F, if for every $\alpha, e \in F$ with $e > 0_F$ there is an element $x \in X$ such that $|\alpha - x| \le e$.

Exercise 1.5.32 *Let F be an Archimedean ordered field and $X \subset F$. Then X is dense in $F \iff$ every element $\alpha \in F$ is the limit*

$$\alpha = \lim x_n$$

for some sequence $(x_n) \subset X$.

Lemma 1.5.33 *Let F be an Archimedean ordered field and $f_F\colon \mathbb{Q} \to F$ be the field embedding introduced in Definition 1.4.49. Then the image $f_F[\mathbb{Q}]$ is dense in F.*

Proof. Let $e, \alpha \in F$ with $e > 0_F$ be given. Since F is Archimedean, there exist fractions $u \le v$ in $\mathbb{Q}$ such that

$$f_F(u) \le \alpha \le f_F(v)\,,$$

and there exists $n \in \omega$ such that $f_F((v-u)/2^n) \le e$. We halve n times the interval $[u, v]$, always select a half $u' \le v'$ in $\mathbb{Q}$ such that

$$f_F(u') \le \alpha \le f_F(v')\,,$$

and obtain fractions $u'' \le v''$ such that

$$f_F(u'') \le \alpha \le f_F(v'') \wedge f_F(v'') - f_F(u'') \le e\,.$$

Then $|\alpha - f_F(v'')| \le e$. □

Lemma 1.5.34 *Let F, G, f_F, and f_G be as in the previous definition and let $\alpha \in F$. Let $(a_n) \subset \mathbb{Q}$ be a sequence such that $\lim f_F(a_n) = \alpha$. Then the limit*

$$\lim f_G(a_n) \quad (\in G)$$

exists and does not depend on the sequence (a_n).

Proof. Suppose that the sequence $(f_G(a_n))$ $(\subset G)$ is not Cauchy. This means that, since G is Archimedean (by part 1 of Theorem 1.4.28), there exists a fraction $\beta > 0$ and two sequences of indices $m_1 < m_2 < \ldots$ and $n_1 < n_2 < \ldots$ in $\mathbb{N}$ such that

$$\forall\, k\colon\ |f_G(a_{m_k}) - f_G(a_{n_k})| \geq f_G(\beta)\,.$$

By the fact that

$$f_F(f_G^{-1})\colon f_G[\mathbb{Q}] \to F$$

is a field embedding, and by Lemma 1.4.54, the above displayed inequalities hold for every k even when the lower index G is replaced with F. By Exercise 1.4.25, this is a contradiction. The sequence $(f_G(a_n))$ is therefore Cauchy and by part 2 of Theorem 1.4.28, the limit

$$\lim\ f_G(a_n)$$

exists. A similar argument shows that for any sequence $(b_n) \subset \mathbb{Q}$ we have

$$\lim f_G(b_n) \neq \lim f_G(a_n) \Rightarrow \lim\ f_F(b_n) \neq \lim\ f_F(a_n)\,.$$

It follows that $\lim f_G(a_n)$ indeed does not depend on the sequence (a_n). $\square$

The two lemmas show that $M(\varphi) = F$ and that the values of φ are correctly defined.

Now we prove three lemmas. In them, F, G, f_F, f_G, and φ are as in Definition 1.5.31.

Lemma 1.5.35 *For every $\alpha, \beta \in F$, we have $\varphi(\alpha + \beta) = \varphi(\alpha) \oplus \varphi(\beta)$.*

Proof. Let $\alpha, \beta \in F$ and let $(a_n), (b_n) \subset \mathbb{Q}$ be sequences such that $\lim f_F(a_n) = \alpha$ and $\lim f_F(b_n) = \beta$. By Exercise 1.5.36,

$$\lim\ f_F(a_n + b_n) = \lim\,(f_F(a_n) + f_F(b_n)) = \lim\ f_F(a_n) + \lim\ f_F(b_n) = \alpha + \beta\,.$$

Hence, by the definition of φ,

$$\begin{aligned}\varphi(\alpha + \beta) &= \lim\ f_G(a_n + b_n) = \lim\,(f_G(a_n) \oplus f_G(b_n)) \\ &= \lim\ f_G(a_n) \oplus \lim\ f_G(b_n) = \varphi(\alpha) \oplus \varphi(\beta)\,.\end{aligned}$$

The first equality follows from the definition of φ. The second equality follows from the fact that $f_G\colon \mathbb{Q} \to G$ is a field embedding. In the third equality, we use Exercise 1.5.36. In the last, fourth equality, we use the definition of φ. $\square$

Exercise 1.5.36 *Let F be an ordered field. Show that for every two sequences $(u_n), (v_n) \subset F$, the equality*

$$\lim(u_n + v_n) = \lim u_n + \lim v_n \quad (\in F)$$

holds if the last two limits exist.

Lemma 1.5.37 *For every $\alpha, \beta \in F$, we have $\varphi(\alpha \cdot \beta) = \varphi(\alpha) \odot \varphi(\beta)$.*

Proof. The proof is similar to the previous one, only Exercise 1.5.36 is replaced with Exercise 1.5.38. $\square$

Exercise 1.5.38 *Let F be an ordered field. Show that for every two sequences $(u_n), (v_n) \subset F$, the equality*

$$\lim(u_n \cdot v_n) = \lim u_n \cdot \lim v_n \quad (\in F)$$

holds if the last two limits exist.

Lemma 1.5.39 *For every $\alpha, \beta \in F$, we have $\alpha < \beta \iff \varphi(\alpha) \prec \varphi(\beta)$.*

Proof. Due to the trichotomy of $<$, it suffices to prove just the implication $\Rightarrow$. Let $\alpha < \beta$ be in F and $(a_n), (b_n) \subset \mathbb{Q}$ be sequences such that $\lim f_F(a_n) = \alpha$ and $\lim f_F(b_n) = \beta$. By Lemma 1.5.33, there exist $u < v$ in $\mathbb{Q}$ such that in F,

$$\alpha < f_F(u) < f_F(v) < \beta\,.$$

By the definition of limits,

$$f_F(a_n) < f_F(u) < f_F(v) < f_F(b_n)$$

for every large n. By Lemma 1.4.54, in $\mathbb{Q}$ we have

$$a_n < u < v < b_n$$

for the same n. By Exercise 1.5.40 and Lemma 1.4.54,

$$\varphi(\alpha) = \lim f_G(a_n) \preceq f_G(u) \prec f_G(v) \preceq \lim f_G(b_n) = \varphi(\beta)\,.$$

Thus $\varphi(\alpha) \prec \varphi(\beta)$. $\square$

Exercise 1.5.40 *Let F be an ordered field, $\alpha \in F$, and let $(a_n) \subset \mathbb{Q}$ be such that $f_F(a_n) \le \alpha$ for every large n. Then in F,*

$$\lim f_F(a_n) \le \alpha\,,$$

if this limit exists.

We prove the second claim in Theorem 1.5.1. Recall Proposition 1.2.8.

Proposition 1.5.41 *Let F, G, f_F, f_G, and φ be as in Definition 1.5.31. Then $\varphi\colon F \to G$ is an isomorphism of ordered fields, which means that φ is a bijection and the following holds.*

1. *For every $a, b \in F$, we have $\varphi(a + b) = \varphi(a) \oplus \varphi(b)$.*
2. *For every $a, b \in F$, we have $\varphi(a \cdot b) = \varphi(a) \odot \varphi(b)$.*
3. *For every $a, b \in F$, we have $a < b \iff \varphi(a) \prec \varphi(b)$.*

Proof. The three properties of φ are proven in the three previous lemmas. It remains to show that φ is bijective. Injectivity of φ follows from Lemma 1.5.39 or from Proposition 1.4.6. Let $\beta \in G$. By Lemma 1.5.33 and Exercise 1.5.32, there exists a sequence $(a_n) \subset \mathbb{Q}$ such that $\lim f_G(a_n) = \beta$. By Lemma 1.5.34, the limit

$$\alpha := \lim\, f_F(a_n) \ \ (\in F)$$

exists. It follows that $\varphi(\alpha) = \beta$. □

1.6 Real numbers are uncountable

We show that real numbers form an uncountable set. By Definition 1.2.42, this means that the set $\mathbb{R}$ is infinite and that there is no bijection between the sets $\mathbb{R}$ and ω.

• *Nested intervals and the uncountability of* $\mathbb{R}$. We deduce the uncountability of $\mathbb{R}$ from the next theorem.

Theorem 1.6.1 *Let $(a_n), (b_n) \subset \mathbb{R}$ be two sequences such that $a_m \le b_n$ for every $m, n \in \mathbb{N}$. Then*

$$\textstyle\bigcap_{n\ge1}[a_n, b_n] \ne \emptyset\,.$$

Proof. Using Theorem 1.5.10, we define

$$\alpha = \sup(\{a_n\colon\ n \in \mathbb{N}\}) \ \text{ and } \ \beta = \inf(\{b_n\colon\ n \in \mathbb{N}\}) \ \ (\in \mathbb{R})\,.$$

It follows that $\bigcap_{n\ge1}[a_n, b_n] = [\alpha, \beta]$ (Exercise 1.6.2). If we show that $\alpha \le \beta$, then $[\alpha, \beta] \ne \emptyset$, and we are done. We deduce it from Proposition 1.1.48 and from the assumption that $a_m \le b_n$ holds for every $m, n \in \mathbb{N}$. Suppose that $\beta < \alpha$. By Proposition 1.1.48 and the definition of α, we have $a_m > \beta$ for some m. This contradicts the definition of β because $a_m \le b_n$ for every n. □

Exercise 1.6.2 *Prove that $\bigcap_{n\ge1}[a_n, b_n] = [\alpha, \beta]$.*

Exercise 1.6.3 *If in addition $b_n - a_n \to 0$ as $n \to \infty$, then $\bigcap_{n\geq 1}[a_n, b_n] = \{c\}$ for some $c \in \mathbb{R}$.*

It is clear that the set $\mathbb{R}$ is infinite. We show that $\mathbb{R}$ is uncountable by proving that there is no map from $\mathbb{N}$ onto $\mathbb{R}$ (Exercise 1.6.5).

Corollary 1.6.4 *For every sequence $(a_n) \subset \mathbb{R}$ there exists $b \in \mathbb{R}$ such that $b \neq a_n$ for every $n \in \mathbb{N}$. Thus the set $\mathbb{R}$ is uncountable.*

Proof. Let $(a_n) \subset \mathbb{R}$. We take any real interval $I_1 = [c_1, d_1]$ with $c_1 < d_1$ such that $a_1 \notin I_1$. Suppose that intervals $I_j = [c_j, d_j]$, $j = 1, 2, \ldots, n$, with $c_j < d_j$ are already defined such that

$$I_1 \supset I_2 \supset \cdots \supset I_n \ \text{ and } \ a_j \notin I_j\,.$$

There obviously exists an interval $I_{n+1} = [c_{n+1}, d_{n+1}] \subset I_n$ with $c_{n+1} < d_{n+1}$ such that $a_{n+1} \notin I_{n+1}$ (Exercise 1.6.6). So we can continue the construction of intervals I_j without end. By Theorem 1.6.1, there is a number $b \in \mathbb{R}$ such that $b \in I_n$ for every n. Then $b \neq a_n$ for every n. $\square$

Exercise 1.6.5 *How does the uncountability of $\mathbb{R}$ follow from the fact that there is no surjection from $\mathbb{N}$ to $\mathbb{R}$?*

Exercise 1.6.6 *How do we define the interval I_{n+1}?*

1.7 Extended reals

The complete ordered field $\mathbb{R}$ is not completely complete because not every sequence $(a_n) \subset \mathbb{R}$ has a subsequence with a limit. To fix it, we extend $\mathbb{R}$ to the structure of extended reals $\mathbb{R}^*$. With $\mathbb{R}^*$ we are in uncharted waters: as far as we know, this is the first time $\mathbb{R}^*$ is treated as an algebraic structure on par with the other structures $\mathbb{N}_0$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$. The main results are Theorem 1.7.11, by which $\mathbb{R}^*$ is the up to isomorphism unique limit closure of the linear order $\langle \mathbb{R}, < \rangle$, and Theorem 1.7.14 describing the structure of the partial complete ordered field $\mathbb{R}^*$ obtained by limit transitions.

• *Limits in linear orders.* We introduce limits in linear orders. Let $X := \langle X, < \rangle$ be a linear order. A set $I \subset X$ is an interval in X if for every $a, b, c \in X$ such that $a < b < c$ and $a, c \in I$ also $b \in I$.

Definition 1.7.1 *Let $X = \langle X, < \rangle$ be a linear order and let $a, b \in X$. The open intervals in X are the intervals*

$$\begin{aligned}
\{x\} \quad & \ldots \quad \textit{if } X = \{x\}\,, \\
(a,\, b)_X = (a,\, b) \quad & := \quad \{x \in X\colon\ a < x < b\}\,, \\
(a,\, \max(X)]_X = (a,\, \max(X)] \quad & := \quad \{x \in X\colon\ a < x \leq \max(X)\}\,, \ \textit{and} \\
[\min(X),\, a)_X = [\min(X),\, a) \quad & := \quad \{x \in X\colon\ \min(X) \leq x < a\}\,.
\end{aligned}$$

The intervals of the latter two types are, of course, defined only if X has a maximum, respectively, a minimum.

Exercise 1.7.2 *The open intervals in X form a base of a topology on X.*

Definition 1.7.3 *Let $\langle X, < \rangle$ be a linear order, $(a_n) \subset X$, and $L \in X$. If for every open interval I in X with $L \in I$ there exists an index $n_0 \in \mathbb{N}$ such that $a_n \in I$ for every $n \geq n_0$, we write $\lim a_n = L$ or $\lim_{n\to\infty} a_n = L$ and say that the sequence (a_n) has the limit L.*

We show that limits in linear orders are unique.

Proposition 1.7.4 *Let $\langle X, < \rangle$ be a linear order, $(a_n) \subset X$, and let $a, b \in X$ be such that $\lim a_n = a$ and $\lim a_n = b$. Then $a = b$.*

Proof. This trivially holds if $|X| = 1$. Suppose that X has at least two distinct elements. It suffices to show that any two distinct elements a and b in X can be separated by disjoint open intervals I and J: $a \in I$, $b \in J$, and $I \cap J = \emptyset$. In other words, the topology in Exercise 1.7.2 is Hausdorff. Let $a < b$ be in X. Suppose that $a = \min(X)$ and $b = \max(X)$. (i) If there is no $c \in X$ with $a < c < b$, then the disjoint separating intervals are $[a, b) = \{a\}$ and $(a, b] = \{b\}$. (ii) If there exists such c, the intervals are $[a, c)$ and $(c, b]$. Suppose that $a' < a$ for some $a' \in X$ and $b = \max(X)$. In case (i), the intervals are (a', b) and $(a, b] = \{b\}$. In case (ii), the intervals are (a', c) and $(c, b]$. The case when $a = \min(X)$ and $b < b'$ for some $b' \in X$ is symmetric. Finally, suppose that $a' < a$ and $b < b'$ for some $a', b' \in X$. In case (i), the intervals are (a', b) and (a, b'). In case (ii), the intervals are (a', c) and (c, b'). □

Proposition 1.7.5 *Let*

$$F := \langle F,\, 0_F,\, 1_F,\, +,\, \cdot,\, < \rangle$$

be an ordered field. Then the two definitions of limits that can be applied in F, Definition 1.4.22 and Definition 1.7.3, are equivalent.

Proof. Let $L \in F$ and $(a_n) \subset F$. Suppose that $\lim a_n = L$ by the former definition, and that $I \ni L$ is an open interval in the linear order $\langle F, < \rangle$. Thus $I = (a, b)$ for some $a < b$ in F because $|F| > 1$, and $\min(F)$ and $\max(F)$ do not exist. We set

$$e := \min(b - L,\ L - a)/2_F\,.$$

There is $n_0 \in \mathbb{N}$ such that $|a_n - L| \leq e$ for every $n \geq n_0$. It follows that for the same n we have $a_n \in I$. So $\lim a_n = L$ also by the latter definition.

Suppose that $\lim a_n = L$ by the latter definition, and that $e \in F$ with $e > 0_F$ is given. We consider the open interval $I := (L - e, L + e)$. Since for any $c \in F$,

$$c \in I \iff |c - L| < e\,,$$

we see that $\lim a_n = L$ also by the former definition. $\square$

• *Limit closures of linear orders. The linear order $\langle \mathbb{R}^*, < \rangle$.* Let $X := \langle X, <_X \rangle$ and $Y := \langle Y, <_Y \rangle$ be two linear orders.

Definition 1.7.6 *We say that the linear order Y is a limit closure of the linear order X, if three conditions hold.*

1. *Every sequence $(a_n) \subset Y$ has a subsequence with a limit in Y.*
2. *There exists an increasing map $\mu\colon X \to Y$ such that for every $y \in Y$ there is a sequence $(a_n) \subset X$ with $\lim \mu(a_n) = y$.*
3. *The set $Y \setminus \mu[X]$ is inclusion-wise minimal with respect to condition 1.*

The map μ is increasing if for every $a, b \in X$,

$$a <_X b \iff \mu(a) <_Y \mu(b)\,.$$

Exercise 1.7.7 *The equivalence in the definition of increasing maps can be weakened to $\Rightarrow$.*

We define the set of extended reals and a linear order on it. Recall Definition 1.5.8 of the linear order $\langle \mathbb{R}, < \rangle$.

Definition 1.7.8 *Let $-\infty := 0 = \emptyset$, $+\infty := 1 = \{\emptyset\}$, and let*

$$\mathbb{R}^* := \mathbb{R} \cup \{-\infty,\, +\infty\}\,.$$

We call the set $\mathbb{R}^*$ extended reals, $-\infty$ is the minus infinity, and $+\infty$ is the plus infinity. We define the linear order

$$\mathbb{R}^* := \langle \mathbb{R}^*,\, <_{\mathbb{R}^*} \rangle$$

as an extension of the linear order $\langle \mathbb{R}, < \rangle$ by the comparisons $-\infty < a$, $a < +\infty$, and $-\infty < +\infty$ for every $a \in \mathbb{R}$. We usually write just $<$ instead of $<_{\mathbb{R}^*}$.

Note that $\min(\mathbb{R}^*) = -\infty$ and $\max(\mathbb{R}^*) = +\infty$.

Exercise 1.7.9 *Show that $\langle \mathbb{R}^*, < \rangle$ is a linear order.*

The linear order $\langle \mathbb{R}^*, < \rangle$ is better behaved than $\langle \mathbb{R}, < \rangle$. Recall that $H(X)$ denotes the set of upper bounds of a set X in a linear order.

Proposition 1.7.10 *In the linear order $\langle \mathbb{R}^*, < \rangle$, every set has an infimum and a supremum.*

Proof. Let $X \subset \mathbb{R}^*$. We show that $\sup(X)$ exists. One reduces infima to suprema by means of the map $\mathbb{R}^* \ni A \mapsto -A \in \mathbb{R}^*$. It is clear that

$$\begin{aligned}\sup(\emptyset) &= \min(H(\emptyset)) = \min(\mathbb{R}^*) = -\infty \text{ and} \\ \sup(\{-\infty\}) &= \min(H(\{-\infty\})) = \min(\mathbb{R}^*) = -\infty\,.\end{aligned}$$

If $+\infty \in X$, then $\sup(X) = \min(H(X)) = \min(\{+\infty\}) = +\infty$. Let $X \neq \emptyset, \{-\infty\}$, and let $+\infty \notin X$. We set $X' = X \setminus \{-\infty\}$. Then $\emptyset \neq X' \subset \mathbb{R}$. If X' is not bounded from above in the linear order $\langle \mathbb{R}, < \rangle$, then $\sup(X) = \min(H(X)) = \min(\{+\infty\}) = +\infty$. If X' is bounded from above, then

$$\sup_{\langle \mathbb{R}^*, < \rangle}(X) = \sup_{\langle \mathbb{R}, < \rangle}(X') \quad (\in \mathbb{R})$$

exists by Theorem 1.5.10. $\square$

• *The limit closure of the real numbers.* We show that $\langle \mathbb{R}^*, < \rangle$ is the up to isomorphism unique limit closure of the linear order $\mathbb{R}$.

Theorem 1.7.11 *The linear order $\langle \mathbb{R}^*, < \rangle$ introduced in Definition 1.7.8 is a limit closure of the linear order $\langle \mathbb{R}, < \rangle$. Every limit closure of $\langle \mathbb{R}, < \rangle$ is isomorphic as a linear order to $\langle \mathbb{R}^*, < \rangle$.*

Proof. First, we prove that $\langle \mathbb{R}^*, < \rangle$ is a limit closure of $\langle \mathbb{R}, < \rangle$. The map μ in Definition 1.7.6 is now the identity $\mu(a) = a$. The satisfaction of condition 2 in the definition is clear: if $A \in \mathbb{R}^*$ is in $\mathbb{R}$, then A is the limit of the constant sequence $(A, A, \dots)$ $(\subset \mathbb{R})$, and if $A = \pm\infty$, then (with the same sign)

$$\lim \pm n = A\,.$$

We show that every sequence $(A_n) \subset \mathbb{R}^*$ has in the linear order $\langle \mathbb{R}^*, < \rangle$ a subsequence with limit in $\mathbb{R}^*$, so that condition 1 holds. Let $S = (A_n) \subset \mathbb{R}^*$ be any sequence of extended reals. If $A_n = \pm\infty$ for infinitely many n, then S has a constant subsequence

$$(A,\, A,\, \dots) = (\pm\infty,\, \pm\infty,\, \dots)$$

(with equal signs) that has the limit A. So we may assume that $S = (A_n) \subset \mathbb{R}$. If S is unbounded from above, then for every $k \in \mathbb{N}$ there exists an index $n \in \mathbb{N}$ such that $A_n \geq k$. It is clear that there is a sequence of indices $m_1 < m_2 < \dots$ in $\mathbb{N}$ such that $A_{m_n} \geq n$ for every $n \in \mathbb{N}$. Then the subsequence (A_{m_n}) has the limit

$$\lim_{n \to \infty} A_{m_n} = +\infty\,.$$

Similarly, if S in unbounded from below, then S has a subsequence with the limit $-\infty$. Suppose that $S = (A_n) \subset \mathbb{R}$ is bounded. Then S has a subsequence with a limit in $\mathbb{R}$ by Proposition 1.4.27. Finally, we show that condition 3 holds. This is clear, if we denote by $\mathbb{R}'$ the $\mathbb{R}^*$ with one or both infinities omitted, then any sequence $(a_n) \subset \mathbb{R}$ with $\lim a_n = I$ being the omitted infinity I has no subsequence with a limit in $\mathbb{R}'$ because $\lim b_n = I$ for every subsequence (b_n) of (a_n).

Second, we prove the isomorphism claim. Let $\langle \mathbb{R}', < \rangle$ be a limit closure of the linear order of real numbers. For simplicity of notation, we assume that $\mathbb{R} \subset \mathbb{R}'$, and that $<$ extends the linear order on $\mathbb{R}$. The sequences $(-n), (n) \subset \mathbb{R}$, which have no subsequence with a limit in $\mathbb{R}$, show that there exist distinct elements $A, B \in \mathbb{R}' \setminus \mathbb{R}$ such that

- $\{A\} < \mathbb{R} < \{B\}$ and
- there is no $C \in \mathbb{R}' \setminus \mathbb{R}$ such that $\{A\} < \{C\} < \mathbb{R}$ or $\mathbb{R} < \{C\} < \{B\}$ (else A or B would not be a limit of any real sequence, and condition 2 would be violated).

Clearly, the linear order $\langle \mathbb{R} \cup \{A, B\}, < \rangle$ is isomorphic to $\langle \mathbb{R}^*, < \rangle$. Condition 3 shows that $\mathbb{R}' = \mathbb{R} \cup \{A, B\}$. $\square$

The next example shows that without condition 3 in Definition 1.7.6, we lose the uniqueness of limit closures.

Exercise 1.7.12 *Consider the linear order $\langle \mathbb{R}', < \rangle$, where $\mathbb{R}' := \mathbb{R}^* \cup \{\infty\}$, that extends $\langle \mathbb{R}^*, < \rangle$ by the comparisons*

$$\{-\infty\} \cup (-\infty,\, 0) < \{\infty\} < [0,\, +\infty) \cup \{+\infty\}\,.$$

Then $\langle \mathbb{R}', < \rangle$ satisfies, with respect to $\langle \mathbb{R}, < \rangle$, conditions 1 and 2 in Definition 1.7.6.

• $\mathbb{R}^*$ *as a partial complete ordered field.* Of course, we want more structure on $\mathbb{R}^*$ than just the linear order. We extend $+$ and $\cdot$ from $\mathbb{R}$ to $\mathbb{R}^*$ by limit transitions as follows. Limits are in the sense of Definition 1.7.3.

Definition 1.7.13 *Let $A, B \in \mathbb{R}^*$.*

1. *If for any two sequences $(a_n), (b_n) \subset \mathbb{R}$ with $\lim a_n = A$ and $\lim b_n = B$, the limit*
$$C := \lim\,(a_n + b_n) \quad (\in \mathbb{R}^*)$$
always exists and is unique, we define $A + B := C$. If it is not the case, the expression $A + B$ is undefined.
2. *If for any two sequences $(a_n), (b_n) \subset \mathbb{R}$ with $\lim a_n = A$ and $\lim b_n = B$, the limit*
$$C := \lim\,(a_n \cdot b_n) \quad (\in \mathbb{R}^*)$$
always exists and is unique, we define $A \cdot B := C$. If it is not the case, the expression $A \cdot B$ is undefined.

In the next theorem we describe the algebraic structure spawned by the previous definition. In Section 1.1 we defined commutative, associative, and distributive operations on a set. Now we extend these definitions to partial operations in the natural way: the corresponding identities hold if all involved expressions are defined. In other words, commutativity, associativity, or distributivity is not violated

Theorem 1.7.14 *The algebraic structure of* <u>*extended reals*</u>

$$\mathbb{R}^* := \langle \mathbb{R}^*, 0, 1, +, \cdot, < \rangle$$

consists of the set $\mathbb{R}^*$ *introduced in Definition 1.7.8, the real numbers* 0 *and* 1*, the linear order* $<$ *on* $\mathbb{R}^*$ *introduced in the same definition, and the partial operations* $+$ *and* $\cdot$ *on* $\mathbb{R}^*$ *introduced in Definition 1.7.13. The following holds* $(A, B \in \mathbb{R}^*)$*.*

1. $+$ *and* $\cdot$ *extend the same operations on* $\mathbb{R}$*.*
2. *The two expressions* $+\infty + (-\infty)$ *and* $-\infty + (+\infty)$ *are undefined.*
3. *The four expressions* $0 \cdot (-\infty)$, $(-\infty) \cdot 0$, $0 \cdot (+\infty)$, *and* $(+\infty) \cdot 0$ *are undefined.*
4. *Except for these six expressions, every other sum* $A + B$ *and product* $A \cdot B$ *is defined. The values are, when infinity is involved, as follows.*
5. $a + A = A + a = A$ *for every* $a \in \mathbb{R}$ *and* $A = \pm\infty$, $+\infty + (+\infty) = +\infty$, *and* $-\infty + (-\infty) = -\infty$*.*
6. $a \cdot A = A \cdot a = A$ *for every real* $a > 0$ *and* $A = \pm\infty$, $a \cdot A = A \cdot a = -A$ *for every real* $a < 0$ *and* $A = \pm\infty$, $(\pm\infty) \cdot (\pm\infty) = +\infty$, *and* $(\mp\infty) \cdot (\pm\infty) = -\infty$ *(both top or both bottom signs).*
7. *Thus* $+$ *and* $\cdot$ *are commutative and associative, and* $\cdot$ *is distributive to* $+$*.*
8. $-\infty$ *and* $+\infty$ *have neither additive nor multiplicative inverse. All other additive and multiplicative inverses are as in* $\mathbb{R}$*.*
9. *In the linear order* $\langle \mathbb{R}^*, < \rangle$*, every subset of* $\mathbb{R}^*$ *has an infimum and a supremum.*
10. *Let* $C = \pm\infty$*. The inequalities* $A < B$ *with* $A, B \neq -C$ *are not preserved under addition of* C*. In all other cases, the first order axiom holds in* $\mathbb{R}^*$*, provided that involved expressions are defined.*
11. *Let* $C = +\infty$*. The inequalities* $A < B$ *such that* $A, B \neq 0$ *and* A *and* B *have equal signs are not preserved under multiplication by* C*. In all other cases, the second order axiom holds in* $\mathbb{R}^*$*, provided that involved expressions are defined.*

Proof. 1. This follows from the fact that if $(a_n), (b_n) \subset \mathbb{R}$ and $a, b \in \mathbb{R}$ are such that $\lim a_n = a$ and $\lim b_n = b$, then

$$\lim (a_n + b_n) = a + b \text{ and } \lim (a_n \cdot b_n) = a \cdot b\,.$$

We prove it in Theorem 2.6.2.

2. Indeed, $\lim n = \lim n^2 = +\infty$ and $\lim(-n) = -\infty$, but $\lim(n + (-n)) = 0$ and $\lim(n^2 + (-n)) = +\infty$.

3. Indeed, $\lim(\pm n) = \lim(\pm n^2) = \pm\infty$ and $\lim n^{-1} = 0$, but $\lim(\pm n) \cdot n^{-1} = \pm 1$ and $\lim(\pm n^2) \cdot n^{-1} = \pm\infty$ (all top or all bottom signs).

4, 5, and 6. We prove these results in Theorem 2.6.2.

7. It follows from parts 5 and 6 that the commutativity of $+$ and $\cdot$ is not violated in $\mathbb{R}^*$. We show that the associativity of $+$ and $\cdot$ is not violated either. Let $A, B, C \in \mathbb{R}^*$ and at least one of them be $\pm\infty$. We check the equalities

$$(A + B) + C = A + (B + C) \text{ and } (A \cdot B) \cdot C = A \cdot (B \cdot C)\,.$$

We prove the associativity of addition. If two of A, B, and C are infinities with different signs, then neither side is defined. Else, the same infinity is correctly equated. We prove the associativity of multiplication. If one of A, B, and C is zero, then neither side is defined. Else, the same infinity with the sign equal to the product of the signs of A, B, and C is correctly equated.

We show that the distributive law is not violated and check the equality

$$A \cdot (B + C) = A \cdot B + A \cdot C\,.$$

Let $A = \pm\infty$. We assume that $B, C \neq 0$ and have the same sign s; else the right side is not defined. Then the same infinity, with the sign equal to the product of the sign of A and the sign s, is correctly equated. Let $A \in \mathbb{R}$. We may assume that $A \neq 0$ and that $B + C \neq \pm\infty + (\mp\infty)$ (both top or both bottom signs). It follows that the same infinity is correctly equated.

8. The nonexistence of additive and multiplicative inverses of infinities is obvious. Part 1 implies that for other elements in $\mathbb{R}^*$ these inverses remain the same as in $\mathbb{R}$.

9. We proved it in Proposition 1.7.10.

10 and 11. This is moved to Exercise 1.7.15. $\square$

We also call this structure the partial complete ordered field of extended reals.

Exercise 1.7.15 *Prove parts 10 and 11 of the theorem.*

Exercise 1.7.16 *Compute values of the expressions* $2 + (-\infty)$, $+\infty + (-\infty)$, $2 \cdot (-\infty)$, *and* $(+\infty) \cdot (-\infty)$.

• *Subtraction and division in* $\mathbb{R}^*$. We define these two partial operations on $\mathbb{R}^*$ in the same way as $+$ and $\cdot$ by limit transitions.

Definition 1.7.17 *Let* $A, B \in \mathbb{R}^*$.

1. *If for any two sequences* $(a_n), (b_n) \subset \mathbb{R}$ *with* $\lim a_n = A$ *and* $\lim b_n = B$, *the limit*
$$C := \lim (a_n - b_n) \quad (\in \mathbb{R}^*)$$
always exists and is unique, we define $A - B := C$. *If it is not the case, the expression* $A - B$ *is undefined.*

2. *If for any two sequences* $(a_n), (b_n) \subset \mathbb{R}$ *with* $\lim a_n = A$ *and* $\lim b_n = B$ *with* $B \neq 0$, *the limit*
$$C := \lim (a_n / b_n) \quad (\in \mathbb{R}^*)$$
always exists and is unique, we define $A/B := C$. *If it is not the case or if* $B = 0$, *the expression* A/B *is undefined.*

In the next proposition, we describe the values of subtraction $-$ and division $/$ determined by this definition.

Proposition 1.7.18 *The following holds* $(A, B \in \mathbb{R}^*)$.

1. $-$ *and* $/$ *extend the same partial operations in* $\mathbb{R}^*$.
2. *The two expressions* $-\infty - (-\infty)$ *and* $+\infty - (+\infty)$ *are undefined.*
3. *The four expressions* $(\pm\infty)/(\pm\infty)$ *and the expressions* $A/0$ *are undefined.*
4. *Except for the expressions in parts 2 and 3, every other difference* $A - B$ *and ratio* A/B *is defined. The values are, when infinity is involved, as follows.*
5. $a - A = -A$ *and* $A - a = A$ *for every* $a \in \mathbb{R}$ *and* $A = \pm\infty$, $+\infty - (-\infty) = +\infty$, *and* $-\infty - (+\infty) = -\infty$.
6. $A/a = A$ *for every real* $a > 0$ *and* $A = \pm\infty$, $A/a = -A$ *for every real* $a < 0$ *and* $A = \pm\infty$, *and* $a/(\pm\infty) = 0$ *for every* $a \in \mathbb{R}$.

Proof. 1. This follows from the fact that if $(a_n), (b_n) \subset \mathbb{R}$ and $a, b \in \mathbb{R}$ are such that $\lim a_n = a$ and $\lim b_n = b$, then
$$\lim (a_n - b_n) = a - b \text{ and } \lim (a_n / b_n) = a/b \quad (b \neq 0)\,.$$
We prove it in Theorem 2.6.2.

2. Indeed, $\lim(\pm n) = \lim(\pm n^2) = \pm\infty$, but $\lim((\pm n) - (\pm n)) = 0$ and $\lim((\pm n^2) - (\pm n)) = \pm\infty$ (all top or all bottom signs).

3. Indeed, $\lim(\pm n) = \lim(\pm n^2) = \pm\infty$ (all top or all bottom signs), but $\lim((\pm n)/(\pm n)) = \pm 1$ and $\lim((\pm n^2)/(\pm n)) = \pm\infty$ (all combinations of signs on the left-hand sides). Division by 0 is undefined for another reason.

4, 5, and 6. We prove these results in Theorem 2.6.2. □

In the next chapter, we show in Theorems **??** and 2.1.9 that the respective identities for differences and ratios

$$A - (B - C) = (A - B) - C \text{ and } (A - B) - C = (A - C) - B\,,$$

and

$$A/C + B/C = (A + B)/C \text{ and } (A/C) \cdot (B/D) = (A \cdot B)/(C \cdot D)\,,$$

are not violated in $\mathbb{R}^*$.

Exercise 1.7.19 *Compute values of the expressions* $2 - (-\infty)$, $+\infty - (-\infty)$, $2/(-\infty)$, *and* $(+\infty)/(-\infty)$.

We conclude the first chapter with the summary of indefinite expressions. These are the expressions in $\mathbb{R}^*$ involving the partial operations $+$, $\cdot$, $-$, and $/$ that are undefined.

Definition 1.7.20 *The indefinite expressions in* $\mathbb{R}^*$ *are exactly the two sums of infinities with different signs, the four products of* 0 *and an infinity, the two differences of infinities with equal signs, the four ratios of infinities, and the ratios* $A/0$ *with* $A \in \mathbb{R}^*$.

Chapter 2

Limits of real sequences

The second chapter is devoted to the limits of sequences of real numbers. We treat finite and infinite limits on equal footing, and in Section 2.1 we therefore introduce the arithmetic of infinities. In Theorem 2.1.8 we determine how much the algebraic structure

$$\mathbb{R}^* = \langle \mathbb{R}^*,\, 0,\, 1,\, +,\, \cdot,\, < \rangle, \text{ where } \mathbb{R}^* = \mathbb{R} \cup \{-\infty,\, +\infty\}\,,$$

differs from a complete ordered field. We define neighborhoods of points and infinities. Definition 2.1.14 of limits uses neighborhoods. In Definition 2.1.20 we introduce the robustness of properties of real sequences. In Proposition 2.1.29 we show that $\lim n^{1/n} = 1$.

Section **??** deals with subsequences. In Theorem **??** we show that any sequence has a monotone subsequence. The famous Erdős–Szekeres Theorem **??** is a finite version. In Theorem 2.2.11 we present two dualities for limits. 1. (a_n) has no limit $\iff$ (a_n) has two subsequences with different limits. 2. It is not true that $\lim a_n = A \iff (a_n)$ has a subsequence (b_n) such that $\lim b_n = B \neq A$. In Theorems 2.2.14 and 2.2.15 we obtain results on partitions of sequences into subsequences. The former theorem states that if such a partition has finitely many subsequences that all have the same common limit, then it is the limit of the whole sequence. The latter theorem shows that every sequence can be partitioned into infinitely many subsequences that all have the same common limit.

Section 2.3 introduces $\liminf$ and $\limsup$ of a real sequence. Theorem 2.3.4 shows that these quantities are always defined. Theorem 2.3.6 describes their basic properties.

Section 2.4 contains four existence theorems for limits. By Theorem 2.1.32 every monotone sequence has a limit. Theorem 2.4.9 is a generalization to quasi-monotone sequences. Theorem 2.4.4 and Corollary 2.4.10 are robust versions of these theorems. The Bolzano–Weierstrass Theorem 2.4.12 states that every bounded sequence has a convergent subsequence; the proof uses Theorem **??** on monotone subsequences. A supplement is that every unbounded sequence has a subsequence with the limit $\pm\infty$. In Theorem 2.4.18 we show that convergent sequences coincide with Cauchy sequences.

Section 2.5 is concerned with the fifth existential result on limits, the additive and multiplicative Fekete's lemmas (Theorem 2.5.3 and Corollary 2.5.8). We present five applications of the lemmas in extremal and enumerative combinatorics in Propositions 2.5.11, 2.5.15, 2.5.20, 2.5.24, and 2.5.27.

Section 2.6 deals with the interactions of limits and arithmetic operations. The main result is Theorem 2.6.2 on limits of sums, products, and ratios. In two supplementary Propositions 2.6.4 and 2.6.5, we consider situations not covered by the theorem. The proof of the former proposition is left to exercises.

In Section 2.7 in Definition 2.7.3, we introduce f-recurrent sequences

$$a_{n+k} = f(a_n,\, a_{n+1},\, \dots,\, a_{n+k-1})\,,$$

where $f = f(x_1, x_2, \dots, x_k)$ is a real function with k variables. In Proposition 2.7.6, Theorem 2.7.10, and Corollary 2.7.11 we describe the usual method for finding their limits. We illustrate it with several examples in Propositions 2.7.15, 2.7.17, and 2.7.18.

Section 2.8 is devoted to interactions of limits and order. In Theorem 2.8.1 we strengthen the standard limit-versus-order theorem. Our squeeze Theorem 2.8.9 is more general than the standard theorem, which is given in Corollary 2.8.10.

2.1 Infinities, neighborhoods, limits

We introduce the algebraic structure of extended reals. Then we define neighborhoods of points and infinities, as well as finite and infinite limits of real sequences.

• *Notation.* For logical and set-theoretic notation, see Appendix A.1. Letters i, j, k, l, m, m_0, m_1, …, and n, n_0, n_1, …, possibly with primes, are variables ranging in $\mathbb{N} = \{1, 2, \dots\}$. By a, b, c, d, e, δ, ε and θ, possibly with indices and primes, we denote real numbers. <u>Always $\delta, \varepsilon, \theta > 0$</u>. Recall that a sequence $(a_n) \subset \mathbb{R}$ is a function $a\colon \mathbb{N} \to \mathbb{R}$ with $a_n = a(n)$. Sets of real numbers are denoted by M and N. Recall that the absolute value of $a \in \mathbb{R}$ is $|a| = a$ if $a \ge 0$, and $|a| = -a$ if $a < 0$.

Exercise 2.1.1 *Recall the proof of the triangle inequality, abbreviated* TI*,*

$$|a + b| \le |a| + |b|\,.$$

Theorem 3.5.39 is an infinite TI. In Proposition A.5.8, we give two proofs of the TI in $\mathbb{C}$.

• *The linear order and tree partial operations on extended reals.* We add to $\mathbb{R}$ two new elements, the <u>infinities</u> $+\infty$ and $-\infty$, and obtain the set of <u>extended reals</u>

$$\mathbb{R}^* := \mathbb{R} \cup \{+\infty,\, -\infty\}\,.$$

Elements in $\mathbb{R}^*$ are denoted by A, B, K, and L. If an expression contains k symbols $\pm\infty$, $\mp\infty$, or other symbols involving $\pm$ and $\mp$, then selecting same signs means the two choices of either all upper or all lower signs. Selecting any signs means the 2^k choices of all signs.

We extend the linear order $\langle \mathbb{R}, < \rangle$ to a relation $<$ on $\mathbb{R}^*$ by setting $-\infty < a$ and $a < +\infty$ for every $a \in \mathbb{R}$, and by setting $-\infty < +\infty$.

Exercise 2.1.2 *Show that $\langle \mathbb{R}^*, < \rangle$ is a linear order.*

We extend the operations of addition and multiplication on $\mathbb{R}$, and the partial operation of division on $\mathbb{R}$ to three partial operations $+$, $\cdot$, and $/$ on $\mathbb{R}^*$.

Definition 2.1.3 *These extensions are defined as follows.*

1. *Addition. We set $\pm\infty + a = a + (\pm\infty) := \pm\infty$ for every $a \in \mathbb{R}$, with same signs. Similarly, $\pm\infty + (\pm\infty) := \pm\infty$, with same signs. The two expressions $\pm\infty + (\mp\infty)$ with same signs are not defined.*
2. *Multiplication. We set $a \cdot (\pm\infty) = (\pm\infty) \cdot a := \pm\infty$ for every real $a > 0$, with same signs. Similarly, $a \cdot (\pm\infty) = (\pm\infty) \cdot a := \mp\infty$ for every real $a < 0$, with same signs. Also, $(\pm\infty) \cdot (\pm\infty) := +\infty$ and $(\pm\infty) \cdot (\mp\infty) := -\infty$, with same signs. The four expressions $0 \cdot (\pm\infty)$ and $(\pm\infty) \cdot 0$ are not defined.*
3. *Division. We set $(\pm\infty)/a := \pm\infty$ for every real $a > 0$, with same signs. Similarly, $(\pm\infty)/a := \mp\infty$ for every real $a < 0$, with same signs. Also, $a/(\pm\infty) := 0$ for every real a. The expressions $A/0$ and $\pm\infty/(\pm\infty)$, for every $A \in \mathbb{R}^*$ and with any signs, are not defined.*

Another operation is the change of sign: $-(\pm\infty) := \mp\infty$, with same signs. Let us recapitulate the undefined expressions, called indefinite expressions:

$$\pm\infty + (\mp\infty),\ 0 \cdot (\pm\infty),\ (\pm\infty) \cdot 0,\ \tfrac{\pm\infty}{\pm\infty},\ \text{and}\ \tfrac{A}{0},\ \text{where}\ A \in \mathbb{R}^*,$$

with same signs in the first expression, and any signs in the other expressions.

Exercise 2.1.4 *Compute $\frac{-\infty}{-2}$, $(-\infty) - (+\infty)$, $-\infty + 10$, and $\frac{\pm\infty}{0}$.*

• *The algebraic structure of extended reals $\mathbb{R}^*$.* We introduce an algebraic structure that extends the complete ordered field of real numbers $\mathbb{R}$ and fits the infinities.

Definition 2.1.5 *The algebraic structure of extended reals*

$$\mathbb{R}^* := \langle \mathbb{R}^*, 0_{\mathbb{R}^*}, 1_{\mathbb{R}^*}, +, \cdot, < \rangle$$

consists of the set $\mathbb{R}^* = \mathbb{R} \cup \{-\infty, +\infty\}$, the elements $0_{\mathbb{R}^*} = 0_{\mathbb{R}}$ and $1_{\mathbb{R}^*} = 1_{\mathbb{R}}$ in $\mathbb{R}$, the partial operations of addition $+$ and multiplication $\cdot$ on $\mathbb{R}^*$ in Definition 2.1.3, and the linear order $<$ on $\mathbb{R}^*$ defined above.

We usually write just 0 and 1 instead of $0_{\mathbb{R}^*}$ and $1_{\mathbb{R}^*}$.

By adding the infinities, the linear order $\langle \mathbb{R}, < \rangle$ is considerably improved.

Proposition 2.1.6 *In the linear order $\langle \mathbb{R}^*, < \rangle$, every set has an infimum and supremum.*

Proof. Let $X \subset \mathbb{R}^*$. We show that $\sup(X)$ exists. One reduces infima to suprema by means of the map $\mathbb{R}^* \ni A \mapsto -A \in \mathbb{R}^*$. It is clear that

$$\begin{aligned}\sup(\emptyset) &= \min(H(\emptyset)) = \min(\mathbb{R}^*) = -\infty \text{ and} \\ \sup(\{-\infty\}) &= \min(H(\{-\infty\})) = \min(\mathbb{R}^*) = -\infty\,.\end{aligned}$$

If $+\infty \in X$, then $\sup(X) = \min(H(X)) = \min(\{+\infty\}) = +\infty$. Let $X \neq \emptyset, \{-\infty\}$, and let $+\infty \notin X$. We set $X' = X \setminus \{-\infty\}$. Then $\emptyset \neq X' \subset \mathbb{R}$. If X' is not bounded from above in the linear order $\langle \mathbb{R}, < \rangle$, then $\sup(X) = \min(H(X)) = \min(\{+\infty\}) = +\infty$. If X' is bounded from above, then

$$\sup_{\langle \mathbb{R}^*, < \rangle}(X) = \sup_{\langle \mathbb{R}, < \rangle}(X') \;\; (\in \mathbb{R})$$

exists by Theorem 1.5.10. $\Box$

Exercise 2.1.7 *Find all sets $X \subset \mathbb{R}^*$ such that* $\sup(X) = -\infty$.

Recall that addition and multiplication on $\mathbb{R}^*$ are partial operations: the expressions $(+\infty) + (-\infty)$, $(-\infty) + (+\infty)$, $0 \cdot (\pm\infty)$, and $(\pm\infty) \cdot 0$ are not defined. We determine how much of the structure of a complete ordered field is lost in $\mathbb{R}^*$.

Theorem 2.1.8 *The properties of complete ordered fields are stated in Definitions 1.2.1, 1.3.1, 1.4.1, and 1.4.19. Those violated in $\mathbb{R}^*$ are listed below.*

1. *$-\infty$ and $+\infty$ do not have additive and multiplicative inverses.*
2. *Let $K = \pm\infty$. The inequalities $A < B$ with $A, B \neq -K$ are not preserved under addition of K.*
3. *Let $K = +\infty$. The inequalities $A < B$ such that $A, B \neq 0$ and A and B have equal signs are not preserved under multiplication by K.*

No other property of complete ordered fields is violated in $\mathbb{R}^*$ $-$ it holds if all arithmetic expressions involved are defined.

Proof. 1. The claim is clear. Every $a \in \mathbb{R}$ has an additive inverse $-a$ and every $a \in \mathbb{R} \setminus \{0\}$ has a multiplicative inverse a^{-1}. 2. If A, B, and K are as stated, then $A + K = B + K = K$. In all other cases, it is easy to see that $A + K < B + K$, if these sums are defined. 3. If A, B, and K are as stated, then $A \cdot K = B \cdot K$. In all other cases, it is easy to see that $A \cdot K < B \cdot K$, if these products are defined.

It is easy to check that 0 and 1 is neutral to $+$ and $\cdot$, respectively, in $\mathbb{R}^*$. It is clear that the commutativity of $+$ and $\cdot$ is not violated in $\mathbb{R}^*$. We show that the associativity of $+$ and $\cdot$ is not violated. Let $A, B, K \in \mathbb{R}^*$ and at least one of them be $\pm\infty$. We check the equalities

$$(A + B) + K = A + (B + K) \text{ and } (A \cdot B) \cdot K = A \cdot (B \cdot K)\,.$$

Associativity of addition. If two of A, B and K are infinities with different signs then neither side is defined. Else the same infinity is correctly equated. Associativity of multiplication. If one of A, B and K is zero then neither side is defined. Else the same infinity with the sign equal to the product of the signs of A, B and K is correctly equated.

We show that the distributive law is not violated. We check the equality

$$A \cdot (B + K) = A \cdot B + A \cdot K\ .$$

Let $A = \pm\infty$. We assume that $B, K \neq 0$ and have the same sign s; else the right side is not defined. Then the same infinity with the sign equal to the product of the sign of A and the sign s is correctly equated. Let $A \in \mathbb{R}$. We may assume that $A \neq 0$ and that $B + K \neq \pm\infty + (\mp\infty)$ (same signs). It follows that the same infinity is correctly equated.

$\langle\mathbb{R}^*, <\rangle$ is a linear order by Exercise 2.1.2. This linear order is complete in a strong form by Proposition 2.1.6. $\Box$

We show that two familiar identities involving division generalize to $\mathbb{R}^*$.

Theorem 2.1.9 *For every $A, B, K, L \in \mathbb{R}^*$ we have*

$$A/K + B/K = (A + B)/K \text{ and } (A/K) \cdot (B/L) = (A \cdot B)/(K \cdot L)\,,$$

provided that the involved arithmetic expressions are defined.

Proof. Let $A, B, K, L \in \mathbb{R}^*$ and one of them be infinity. We check the first equality. If $K = \pm\infty$ then the left side is not defined or we have equality $0 + 0 = 0$. Let $K \in \mathbb{R} \setminus \{0\}$. We may assume that $A + B \neq \pm\infty + (\mp\infty)$ (same signs) and see that the same infinity is correctly equated.

We check the second equality. We may assume that $K, L \neq 0$. If A or B is infinity, then we may assume that $A, B \neq 0$ and $K, L \in \mathbb{R}$. Then the same infinity is equated. Let $A, B \in \mathbb{R}$. Then we have equality $0 = 0$. $\Box$

• *Neighborhoods of points and infinities.* Recall the notation for real intervals. For example,

$$(a,\, b] = \{x \in \mathbb{R}\colon\ a < x \leq b\} \text{ and } (-\infty,\, a) = \{x \in \mathbb{R}\colon\ x < a\}\,.$$

For $b \in \mathbb{R}$ and $\varepsilon > 0$ we define the set

$$U(b,\, \varepsilon) := (b - \varepsilon,\, b + \varepsilon)$$

and call it the ε-neighborhood of b. Similarly,

$$U(-\infty, \varepsilon) := (-\infty, -\tfrac{1}{\varepsilon}) \text{ and } U(+\infty, \varepsilon) := (\tfrac{1}{\varepsilon}, +\infty)$$

are ε-neighborhoods of infinities. In the next four exercises, we review basic properties of neighborhoods.

Exercise 2.1.10 *Let $A \in \mathbb{R}^*$, $c \in U(A, \varepsilon)$ and $c < b < A + \varepsilon$ or $A - \varepsilon < b < c$. Then $b \in U(A, \varepsilon)$.*

For $M, N \subset \mathbb{R}$, the notation $M < N$ means that $x < y$ for every $x \in M$ and $y \in N$.

Exercise 2.1.11 *Let $A < B$ be in $\mathbb{R}^*$. Then there is an ε such that $U(A, \varepsilon) < U(B, \varepsilon)$. In particular, $U(A, \varepsilon) \cap U(B, \varepsilon) = \emptyset$ for some ε.*

Exercise 2.1.12 *Let $A \in \mathbb{R}^*$. If $\varepsilon \leq \delta$ then $U(A, \varepsilon) \subset U(A, \delta)$.*

Exercise 2.1.13 $\bigcap_{k=1}^{\infty} U(b, \tfrac{1}{k}) = \{b\}$ *and* $\bigcap_{k=1}^{\infty} U(\pm\infty, \tfrac{1}{k}) = \emptyset$.

• *Finite and infinite limits of real sequences.* We denote sequences of real numbers by (a_n), (b_n), and (c_n). We call them real sequences or just sequences. We introduced limits in ordered fields already in Definition 1.4.22. Now we slightly extend this definition in the case of the ordered field $\mathbb{R}$ by allowing the limit to lie in $\mathbb{R}^*$.

Definition 2.1.14 *Let $(a_n) \subset \mathbb{R}$ and $L \in \mathbb{R}^*$. If for every $\varepsilon > 0$ there is an n_0 such that for every $n \geq n_0$ we have*

$$a_n \in U(L, \varepsilon)\,,$$

we write that $\lim a_n = L$ or $\lim_{n\to\infty} a_n = L$ or $a_n \to L$, and say that the sequence (a_n) has the limit L.

If $L \in \mathbb{R}$, we say that (a_n) has a finite limit or that (a_n) converges. If $L = \pm\infty$, we say that (a_n) has an infinite limit. A sequence diverges if it has no limit or an infinite limit. An eventually constant sequence (a_n) with $a_n = a$ for $n \geq n_0$ converges and $\lim a_n = a$.

Exercise 2.1.15 *Show that in the case of ordered field $\mathbb{R}$, finite limits in Definition 2.1.14 are equivalent with Definition 1.4.22.*

Exercise 2.1.16 *Suppose that $(a_n) \subset \mathbb{R}$, $(m_n) \subset \mathbb{N}$ and that the sequence (b_n) arises from (a_n) by replacing each term a_n by m_n copies $a_n, a_n, \ldots, a_n$. Then $\lim a_n = \lim b_n$ whenever one limit exists.*

Proposition 2.1.17 $\lim a_n = -\infty \iff$ *for every $c < 0$ there exists n_0 such that if $n \geq n_0$, then $a_n \leq c$.*

Proof. For the implication $\Rightarrow$ it suffices to take $\varepsilon > 0$ such that $-1/\varepsilon \leq c$. For the implication $\Leftarrow$ it suffices to take $c < 0$ such that $c \leq -1/\varepsilon$. $\Box$

Exercise 2.1.18 *State and prove the analogous equivalence for the limit $+\infty$.*

We show that finite and infinite limits are unique (cf. Exercise 1.4.24).

Proposition 2.1.19 *Let $K, L \in \mathbb{R}^*$, $\lim a_n = K$ and $\lim a_n = L$. Then $K = L$.*

Proof. Let K and L be limits of (a_n) and ε be arbitrary. By Definition 2.1.14, if $n \geq n_0$ then $a_n \in U(K, \varepsilon)$ and $a_n \in U(L, \varepsilon)$. Thus $U(K, \varepsilon) \cap U(L, \varepsilon) \neq \emptyset$ for every ε. By Exercise 2.1.11, $K = L$. $\Box$

• *Robust properties of sequences.* Let $\mathbb{R}^{\mathbb{N}}$ be the set of real sequences. Any set $V \subset \mathbb{R}^{\mathbb{N}}$ is called a property of real sequences.

Definition 2.1.20 *A property $V \subset \mathbb{R}^{\mathbb{N}}$ is robust if for every two sequences (a_n) and (b_n) such that $a_n \neq b_n$ for only finitely many indices n, we have $(a_n) \in V \iff (b_n) \in V$.*

Exercise 2.1.21 *For every $L \in \mathbb{R}^*$, the property $\{(a_n)\colon\ \lim a_n = L\}$ is robust.*

The next exercise shows that the notion of robustness is itself robust.

Exercise 2.1.22 *Prove the following.*

1. *If $V \subset \mathbb{R}^{\mathbb{N}}$ is robust, then so is $\mathbb{R}^{\mathbb{N}} \setminus V$.*
2. *If $X \subset \mathcal{P}(\mathbb{R}^{\mathbb{N}})$ is such that every $Y \in X$ is robust, then $\bigcup X$ is robust.*
3. *If $\emptyset \neq X \subset \mathcal{P}(\mathbb{R}^{\mathbb{N}})$ is such that every $Y \in X$ is robust, then $\bigcap X$ is robust.*

Exercise 2.1.23 *Which of the following properties of sequences (a_n) is robust?*

1. *(a_n) converges.*
2. *$a_1 \leq a_2 \leq \cdots$.*
3. *There exists an index m such that $a_m \geq a_{m+1} \geq \cdots$.*
4. *There exist indices $m_1 < m_2 < \ldots$ such that $a_{m_1} > a_{m_2} > \cdots$.*
5. *$\inf(\{a_n\colon\ n \in \mathbb{N}\}) = -1$.*
6. *$\inf(\{a_n\colon\ n \in \mathbb{N}\}) = -\infty$.*

• *Three interesting limits.* For $a \in \mathbb{R}$, we define the upper integer part $\lceil a \rceil$ $(\in \mathbb{Z})$ of a as the smallest integer v such that $v \geq a$. Similarly, the lower integer part $\lfloor a \rfloor$ $(\in \mathbb{Z})$ of a is the largest $v \in \mathbb{Z}$ such that $v \leq a$. Many limits can be reduced to the next limit.

Proposition 2.1.24 $\lim \frac{1}{n} = 0$.

Proof. Let an $\varepsilon > 0$ be given and $n_0 = \lceil \frac{1}{\varepsilon} \rceil + 1$. Then for every $n \geq n_0$,

$$0 < \tfrac{1}{n} \leq \tfrac{1}{n_0} = \tfrac{1}{\lceil 1/\varepsilon \rceil + 1} < \tfrac{1}{1/\varepsilon} = \varepsilon\,.$$

Thus if $n \geq n_0$ then $\frac{1}{n} \in U(0, \varepsilon)$ and $\lim \frac{1}{n} = 0$. $\square$

Exercise 2.1.25 *Show that* $\lceil \frac{1}{\varepsilon} \rceil + 1$ *is, for every* $\varepsilon > 0$*, the minimum value of* $n_0 \in \mathbb{N}$ *such that* $\frac{1}{n} \in U(0, \varepsilon)$ *for every* $n \geq n_0$*.*

We compute the next limit via an algebraic transformation.

Proposition 2.1.26 $\lim(\sqrt[3]{n} - \sqrt{n}) = -\infty$.

Proof. We use Proposition 2.1.17. Let a $c < 0$ be given. We take any $n_0 \geq \max(4c^2, 2^6)$. Then for every $n \geq n_0$ we have

$$\overbrace{\sqrt[3]{n} - \sqrt{n}}^{\text{nontrivial}} = \overbrace{n^{1/2} \cdot \underbrace{(n^{-1/6} - 1)}_{\cdots\, \leq\, -1/2}}^{\text{trivial}} \leq \underbrace{-n^{1/2}}_{\cdots\, \leq\, -2|c|} /\, 2 \leq -2|c|/2 = c\,,$$

and the limit is $-\infty$. The first upper bracket says that in this form the limit is non-trivial, in the indefinite form $+\infty - (+\infty)$. We transform it algebraically in the trivial form $(+\infty) \cdot (0 - 1) = (+\infty)(-1) = -\infty$. Lower brackets show upper bounds for enclosed expressions for $n \geq n_0$. $\square$

Exercise 2.1.27 *Find the limit* $\lim_{n \to \infty} \frac{\sqrt[3]{n} - \sqrt{n}}{\sqrt[4]{n}}$.

Limit of a sequence is <u>nontrivial</u> if in the original form it leads to an indefinite expression. Else it is <u>trivial</u>. For example, the limits $\lim(2^n + 3^n)$ and $\lim \frac{4}{5n-3}$ are trivial, but $\lim(2^n - 3^n) = +\infty - (+\infty)$ and $\lim \frac{4n+7}{5n-3} = \frac{+\infty}{+\infty}$ are nontrivial. Non-trivial limits can often, but not always, be computed by algebraically transforming them into trivial limits, as in Proposition 2.1.26. The next limit of $n^{1/n}$ is nontrivial because $n \to +\infty$, $\frac{1}{n} \to 0$ and $(+\infty)^0$ is an indefinite power expression (see Exercise 5.1.17). No algebraic transformation works in this case, and we have to compute the limit from the definition. We show that the exponent prevails and $n^{1/n} \to 1$. We use the well known <u>binomial theorem</u> in the next exercise.

Exercise 2.1.28 *Let* $a, b \in \mathbb{R}$ *and* $n \in \mathbb{N}_0$*. Then*

$$(a + b)^n = \textstyle\sum_{j=0}^{n} \binom{n}{j} a^j b^{n-j}\,.$$

Here $\binom{n}{j} := \frac{1}{j!} n(n-1) \ldots (n-j+1)$ *for* $j \in \mathbb{N}$ *and* $\binom{n}{0} := 1$*.*

We assume that the reader is familiar with the real powers a^b with real base $a > 0$ and exponent $b \in \mathbb{Q}$. We introduce them in Section 5.1.

Proposition 2.1.29 $\lim_{n\to\infty} n^{1/n} = 1$.

Proof. Always $n^{1/n} \ge 1$. If $n^{1/n} \not\to 1$, there would be a number $c > 0$ and a sequence of integers $2 \le n_1 < n_2 < \dots$ such that $n_i^{1/n_i} \ge 1 + c$ for every i (Exercise 2.1.30). Raising this inequality to the power n_i (Exercise 2.1.31) and using Exercise 2.1.28 we get

$$\begin{aligned} n_i &\ge (1+c)^{n_i} = \textstyle\sum_{j=0}^{n_i} \binom{n_i}{j} c^j = 1 + \binom{n_i}{1} c + \binom{n_i}{2} c^2 + \dots + \binom{n_i}{n_i} c^{n_i} \\ &\ge \textstyle\binom{n_i}{2} c^2 = \frac{1}{2} n_i(n_i - 1) \cdot c^2 \,. \end{aligned}$$

So for every i,

$$n_i \ge \tfrac{1}{2} n_i(n_i - 1) \cdot c^2 \ \text{ and } \ 1 + 2/c^2 \ge n_i \,.$$

This is impossible, the sequence $n_1 < n_2 < \dots$ is not bounded from above. □

Exercise 2.1.30 *Explain why there is the sequence* (n_i).

Exercise 2.1.31 *Explain why* $n_i^{1/n_i} \ge 1 + c \Rightarrow n_i \ge (1 + c)^{n_i}$.

• *Limits of monotone sequences.* In the next passage, we determine the limits of sequences (q^n) with $q \in \mathbb{R}$. We need the existence of limits of monotone sequences for this.

Theorem 2.1.32 *Let $a_1 \le a_2 \le \dots$ be real numbers. Then*

$$\lim_{n\to\infty} a_n = \sup(\{a_n \colon\ n \in \mathbb{N}\}) \ \ (\in \mathbb{R}^*) \,.$$

The same holds when $\le$ is replaced with $\ge$, and sup *with* inf. *Infima and suprema are taken in the linear order $\langle \mathbb{R}^*, < \rangle$.*

Proof. We prove the case with $\le$, the case with $\ge$ is similar. Let A be the supremum and let an $\varepsilon > 0$ be given. We take any $c \in U(A, \varepsilon)$ with $c < A$. Then $c < a_m$ for some m and $c < a_m \le a_n \le A$ for every $n \ge m$. By Exercise 2.1.10, $a_n \in U(A, \varepsilon)$ for the same n. We see that $\lim a_n = A$. □

We repeated the argument from the proof of part 2 of Theorem 1.4.28.

Exercise 2.1.33 *Is the assumption on (a_n) a robust property of sequences?*

Exercise 2.1.34 *Reduce the $\ge$ case to the $\le$ case.*

• *Limits of geometric sequences.* A geometric sequence is any sequence of the form

$$(q^n) = (q, q^2, q^3, \dots), \ q \in \mathbb{R} \,.$$

Exercise 2.1.35 *Let* $(a_n) \subset \mathbb{R}$*. Then* $\lim a_n = 0 \iff \lim |a_n| = 0$.

Proposition 2.1.36 *Let* $q \in \mathbb{R}$*. The limit* $L = \lim q^n$ *has the following values.*

1. *If* $|q| < 1$ *then* $L = 0$.
2. *If* $q = 1$ *then* $L = 1$.
3. *If* $q \leq -1$ *then* L *does not exist.*
4. *If* $q > 1$ *then* $L = +\infty$.

Proof. 1. By Exercise 2.1.35, we can assume that $q \in (0, 1)$. We show that $\lim q^n = 0$. Let $L = \inf(\{q^n : n \in \mathbb{N}\})$ $(\in [0, 1])$. Since $q > q^2 > q^3 > \cdots > 0$, $\lim q^n = L$ by Theorem 2.1.32. It remains to show that $L = 0$. Let $L > 0$. Then $L/q > q^n \geq L$ for some $n \in \mathbb{N}$. We get the contradiction $L > q^{n+1}$.

2. The constant sequence $(1, 1, \dots)$ has limit 1.

3. For $q \leq -1$, the sequence (q^n) does not have a limit: if $m, n \in \mathbb{N}$ have different parity, then $|q^m - q^n| \geq 2$.

4. Let $q > 1$. We proceed as in part 1. Let $L = \sup(\{q^n : n \in \mathbb{N}\})$ $(\in (1, +\infty) \cup \{+\infty\})$, taken in the linear order $\langle \mathbb{R}^*, < \rangle$. Since $q < q^2 < q^3 < \dots$, $\lim q^n = L$ by Theorem 2.1.32. It remains to show that $L = +\infty$. Let $L < +\infty$. So $L/q < q^n \leq L$ for some $n \in \mathbb{N}$. We get the contradiction that $L < q^{n+1}$. $\square$

2.2 Subsequences

We obtain some results on the relation of subsequence.

• *Subsequences.* Recall the definition of subsequences after Definition 1.1.9. If (a_n) is a subsequence of (b_n), we write $(a_n) \preceq (b_n)$. In Definition 2.2.6 we introduce weak subsequences. The subsequence $(a_k, a_{k+1}, \dots)$ of (a_n) is a <u>tail</u> of (a_n).

Exercise 2.2.1 *The relation* $\preceq$ *on* $\mathbb{R}^{\mathbb{N}}$ *is reflexive and transitive.*

Exercise 2.2.2 *Find distinct sequences* (a_n) *and* (b_n) *such that* $(a_n) \preceq (b_n)$ *and* $(b_n) \preceq (a_n)$.

Proposition 2.2.3 *Let* $(b_n) \preceq (a_n)$ *and* $\lim a_n = L$*.* <u>*Then*</u> $\lim b_n = L$.

Proof. This follows from Definition 2.1.14 and the definition of a subsequence. The numbers m_n in the latter definition satisfy $m_n \geq n$. $\square$

• *Orderings of sets of natural numbers.* A subsequence is obtained from the given sequence by deleting some terms. We can delete only so many terms that infinitely many of them remain. We formalize it in the next proposition. A map $f \colon A \to \mathbb{N}$, where $A \subset \mathbb{N}$, <u>increases</u> if $f(m) < f(n)$ for every $m, n \in A$ with $m < n$.

Proposition 2.2.4 *Let $B \subset \mathbb{N}$. If B is finite, then there exist a unique number $m \in \omega$ and a unique increasing bijection $f\colon [m] \to B$. If B is infinite, then there exists a unique increasing bijection $f\colon \mathbb{N} \to B$. In both cases, we call f the ordering of B.*

Proof. For finite B, the number $m \in \omega$ is the number of elements of B. The map f is defined by the inductive formula

$$f(n) = \min(B \setminus f[\,[n-1]\,])\,,$$

where $n \in [m]$, respectively $n \in \mathbb{N}$. □

We can restate subsequences in terms of orderings of sets,

Proposition 2.2.5 $(b_n) \preceq (a_n) \iff$ *there is a unique infinite set $B \subset \mathbb{N}$ such that $b_n = a_{f(n)}$ for every $n \in \mathbb{N}$, where f is the ordering of B.*

Proof. Suppose that (b_n) is a subsequence of (a_n), so that $b_n = a_{m_n}$ for a sequence of natural numbers $1 \le m_1 < m_2 < \cdots$. It follows that $B = \{m_n\colon\ n \in \mathbb{N}\}$. Clearly, $f(n) = m_n$.

Suppose that the right-hand side of the equivalence holds. For $n \in \mathbb{N}$ we set $m_n = f(n)$. Then $m_1 < m_2 < \ldots$ because f is increasing and $b_n = a_{m_n}$. So (b_n) is a subsequence of (a_n). □

We call the set B $(\subset \mathbb{N})$ the support of the subsequence (b_n) of (a_n).

Definition 2.2.6 *We say that (b_n) is a weak subsequence of (a_n) if there is a sequence $(m_n) \subset \mathbb{N}$ such that $\lim m_n = +\infty$ and $b_n = a_{m_n}$ for every $n \in \mathbb{N}$. We write $(b_n) \preceq^* (a_n)$.*

Exercise 2.2.7 *Generalize Proposition 2.2.3: if $(b_n) \preceq^* (a_n)$ and $\lim a_n = L$ then $\lim b_n = L$.*

Exercise 2.2.8 *If $(b_n) \preceq^* (a_n)$ then there is a sequence (c_n) such that $(c_n) \preceq (b_n)$ and $(c_n) \preceq (a_n)$.*

• *Limit dualities.* We show that the existence of the limit of a sequence is prevented by the existence of limits of certain subsequences. The next theorem and exercise are three dualities of this form. We begin with an existential theorem on limits of subsequences.

Theorem 2.2.9 *Every sequence $(a_n) \subset \mathbb{R}$ has a subsequence (b_n) such that $\lim b_n$ $(\in \mathbb{R}^*)$ exists.*

Corollary 2.2.10 *Any sequence (a_n) has a subsequence (b_n) such that $\lim b_n$ exists.*

Proof. This follows from Theorem 2.4.12. □

Theorem 2.2.11 (two limit dualities) *Let $(a_n) \subset \mathbb{R}$ and $A \in \mathbb{R}^*$. The following holds.*

1. $\lim a_n$ *does not exist* $\iff$ *two subsequences of* (a_n) *have different limits.*
2. *It is not true that* $\lim a_n = A \iff$ *a subsequence* (b_n) *of* (a_n) *exists such that* $\lim b_n \neq A$.

Proof. 1. The implication $\neg \Rightarrow \neg$ follows from Proposition 2.2.3. We prove the implication $\Rightarrow$. Suppose that (a_n) does not have a limit. By Corollary 2.2.10 there is a $(b_n) \preceq (a_n)$ with $\lim b_n = B$. Since B is not a limit of (a_n), there exists an ε and a sequence $(c_n) \preceq (a_n)$ such that $c_n \notin U(B, \varepsilon)$ for every n. By Corollary 2.2.10 there is a $(d_n) \preceq (c_n)$ such that $\lim d_n = K$. Then $(d_n) \preceq (a_n)$ and $K \neq B$. Hence the required subsequences are (b_n) and (d_n).

2. The implication $\neg \Rightarrow \neg$ again follows from Proposition 2.2.3. We prove the implication $\Rightarrow$. Let $\neg(\lim a_n = A)$. Hence there is an ε and a $(b_n) \preceq (a_n)$ such that $b_n \notin U(A, \varepsilon)$ for every n. By Corollary 2.2.10 there is a $(c_n) \preceq (b_n)$ such that $\lim c_n = B$. Then $(c_n) \preceq (a_n)$ and $B \neq A$. Hence (c_n) is the required subsequence. □

So if a sequence does not have a limit, it is always possible to prove it by presenting two subsequences with different limits. For example,

$$(a_n) = ((-1)^n) = (-1,\, 1,\, -1,\, 1,\, -1,\, \dots)$$

does not have a limit because $(1, 1, \dots) \preceq (a_n)$, $(-1, -1, \dots) \preceq (a_n)$ and these constant subsequences have different limits 1 and -1.

Exercise 2.2.12 *In part 1 of the theorem, the two subsequences may have disjoint supports.*

Exercise 2.2.13 *A sequence diverges* $\iff$ *it has two subsequences with two different limits, or a subsequence with an infinite limit.*

• *Partitions into subsequences.* Let $(a_n) \subset \mathbb{R}$. A partition of (a_n) into k ($\in \mathbb{N}$) subsequences is a partition $\{B_j\colon\ j \in [k]\}$ of $\mathbb{N}$ with k infinite blocks B_j and the corresponding subsequences $(b_{n,j}) = (a_{f_j(n)})$, $n = 1, 2, \dots$, where f_j is the ordering of B_j. Partitions of (a_n) into infinitely many subsequences are defined similarly.

Theorem 2.2.14 (finite partitions) *Let $L \in \mathbb{R}^*$. If a sequence (a_n) has a partition into k subsequences such that each has limit L, then* $\lim a_n = L$.

Proof. Let $m_{1,j} < m_{2,j} < \dots$, $j \in [k]$, be indices of these k subsequences and let an ε be given. Then there exists k numbers $n_j \in \mathbb{N}$ such that $a_{m_{n,j}} \in U(L, \varepsilon)$ whenever $n \geq n_j$ and $j \in [k]$. Let

$$n_0 = \max(\{m_{n_1,1},\, m_{n_2,2},\, \dots,\, m_{n_k,k}\})\,.$$

Then for every $n \geq n_0$ we have $a_n \in U(L, \varepsilon)$ because $n = m_{i,j}$ for unique $j \in [k]$ and $i \geq n_j$. Hence $a_n \to L$. □

We use this theorem in the proof of Theorem 3.5.31 on sums of Leibnizian series, and in the proof of Theorem 7.4.1 on derivatives of composite functions. It does not hold for infinite partitions. In fact, for them we have an opposite result.

Theorem 2.2.15 (infinite partitions) *Every real sequence can be partitioned into infinitely many subsequences such that each has the same limit.*

Proof. Let $(a_n) \subset \mathbb{R}$ be arbitrary. By Corollary 2.2.10 there is an $L \in \mathbb{R}^*$ and an infinite set $B_0 \subset \mathbb{N}$ such that $\lim b_n = L$, where $(b_n) = (a_{f(n)})$ for the ordering f of B_0. By Exercise 2.2.16 we may assume that the complement $\mathbb{N} \setminus B_0 = \{c_1 < c_2 < \dots\}$ is infinite. We take any infinite partition $\{C_j\colon\ j \in \mathbb{N}\}$ of B_0 with infinite blocks C_j and set $B_j = C_j \cup \{c_j\}$. Then

$$\{B_j\colon\ j \in \mathbb{N}\}$$

is an infinite partition of $\mathbb{N}$ with infinite blocks. The corresponding subsequences $(b_{n,j}) = (a_{f_j(n)})$ of (a_n), where f_j is the ordering of B_j, form the desired partition of (a_n): it is clear that $\lim_{n\to\infty} b_{n,j} = L$ for every $j \in \mathbb{N}$. □

Exercise 2.2.16 *Why can we assume that the complement $\mathbb{N} \setminus B_0$ is infinite?*

Exercise 2.2.17 *Show that every bounded sequence can be partitioned in infinitely many convergent subsequences with the same limit.*

Exercise 2.2.18 *Show that every sequence that does not have a limit can be partitioned in infinitely many subsequences such that one of them has a limit A, and other have a common limit $B \neq A$.*

2.3 Limes inferior and limes superior

In Latin, this means “the lowest limit” and “the highest limit”, respectively. Limes inferior, briefly liminf, and limes superior, briefly limsup, of a real sequence always exist, which is an advantage over limits.

- *Limit points of sequences* are limits of subsequences.

Definition 2.3.1 (limit points) $A \in \mathbb{R}^*$ *is a limit point of a sequence (a_n) if $A = \lim b_n$ for a subsequence $(b_n) \preceq (a_n)$. We denote the set of limit points of (a_n) by $L(a_n)$ ($\subset \mathbb{R}^*$).*

For example, the sequence $(a_n) = (n - 1 + (-1)^n n + \frac{1}{n})$ has limit points $L(a_n) = \{-1, +\infty\}$.

Exercise 2.3.2 *Every real sequence has at least one limit point.*

• *Limes inferior and limes superior of a sequence.* We have already revealed that these are the smallest and largest limit points of the sequence, respectively.

Definition 2.3.3 (liminf and limsup) *Let (a_n) be a real sequence. We define* $\liminf a_n := \min(L(a_n))$ *and* $\limsup a_n := \max(L(a_n))$. *The minimum and maximum are taken in the linear order $\langle \mathbb{R}^*, < \rangle$.*

We show that these minima and maxima always exist.

Theorem 2.3.4 (liminf and limsup exist) *The following holds.*

1. *The set of limit points $L(a_n) \neq \emptyset$ for every sequence $(a_n) \subset \mathbb{R}$.*

2. *In the linear order $\langle \mathbb{R}^*, < \rangle$, the set $L(a_n)$ has minimum and maximum.*

Proof. 1. Let $(a_n) \subset \mathbb{R}$. Then $L(a_n) \neq \emptyset$ by Exercise 2.3.2.

2. We show that $\max(L(a_n))$ exists, the minimum is treated similarly. Let $A = \sup(L(a_n))$ be taken in the linear order $\langle \mathbb{R}^*, < \rangle$ (Proposition 2.1.6). We show that $A \in L(a_n)$. If $A = -\infty$, then $L(a_n) = \{-\infty\}$ and we are done, $A \in L(a_n)$.

Let $A > -\infty$. We claim that there is a sequence $(b_n) \subset L(a_n) \cap \mathbb{R}$ such that $\lim b_n = A$. For $A < +\infty$ and for $A = +\infty \notin L(a_n)$ it follows from the definition of supremum. If $A = +\infty \in L(a_n)$, we are done. Since every number b_n is the limit of a subsequence of (a_n), it is easy to find a subsequence (a_{m_n}) such that $a_{m_n} \in U(b_n, 1/n)$ for every n. Then $\lim a_{m_n} = \lim b_n = A$ and $A \in L(a_n)$. $\Box$

Clearly, if $\lim a_n$ exists, then $L(a_n) = \{\lim a_n\}$. We obtain some more properties of liminfs and limsups.

Proposition 2.3.5 ($\liminf \overset{?}{=} \limsup$) *The following holds.*

1. *Always* $\liminf a_n \leq \limsup a_n$.

2. *The inequality in part 1 holds as an equality $\iff$ the limit* $\lim a_n$ *exists. Then* $\liminf a_n = \limsup\, a_n = \lim a_n$.

Proof. Let $(a_n) \subset \mathbb{R}$. 1. This is obvious as $\liminf a_n = \min(L(a_n))$ and $\limsup a_n = \max(L(a_n))$.

2. If equality holds, then $L(a_n)$ has just one element and (a_n) does not have two subsequences with different limits. By part 2 of Theorem 2.2.11, $\lim a_n$ exists and equals to $\liminf a_n = \limsup a_n$. If $\liminf a_n \neq \limsup a_n$, then the sequence (a_n) has two subsequences with different limits and $\lim a_n$ does not exist. $\Box$

Theorem 2.3.6 (on liminfs and limsups) *Let $(a_n) \subset \mathbb{R}$, $A = \liminf a_n$ and $B = \limsup a_n$. The following holds.*

1. *If $A = -\infty$ <u>then</u> for every c we have $a_n \leq c$ for infinitely many n. If $A = +\infty$ <u>then</u> $\lim a_n = +\infty$.*

2. *If $A \in \mathbb{R}$ <u>then</u> for every ε we have $a_n \le A + \varepsilon$ for infinitely many n, and $a_n \ge A - \varepsilon$ for every $n \ge n_0$.*

3. *If $B = +\infty$ <u>then</u> for every c we have $a_n \ge c$ for infinitely many n. If $B = -\infty$ <u>then</u> $\lim a_n = -\infty$.*

4. *If $B \in \mathbb{R}$ <u>then</u> for every ε we have $a_n \ge B - \varepsilon$ for infinitely many n, and $a_n \le B + \varepsilon$ for every $n \ge n_0$.*

Proof. We prove parts 1 and 2. Proofs of parts 3 and 4 are left for the exercise below. 1. Let $A = -\infty$. Then for some $(b_n) \preceq (a_n)$ we have $\lim b_n = -\infty$ and the claim follows. Let $A = +\infty$. Then $L(a_n) = \{+\infty\}$ and by Proposition 2.3.5 we have $\lim a_n = +\infty$.

2. Let $A \in \mathbb{R}$ and let an ε be given. Since there is a $(b_n) \preceq (a_n)$ with $\lim b_n = A$, we have for infinitely many n that $a_n \le A+\varepsilon$. If we had $a_n < A-\varepsilon$ for infinitely many n, a sequence $(c_n) \preceq (a_n)$ would exist with $\lim c_n \le A - \varepsilon$. This is impossible because $A = \min(L(a_n))$. Hence for every $n \ge n_0$ we have $a_n \ge A - \varepsilon$. $\square$

We relate liminfs and limsups to infima and suprema. These are taken in the linear order $\langle \mathbb{R}^*, < \rangle$.

Proposition 2.3.7 (limits of infima and suprema) *Let $(a_n) \subset \mathbb{R}$ be any sequence. <u>Then</u>*

$$\begin{aligned} \liminf_{n\to\infty} a_n &= \lim_{n\to\infty} \inf(\{a_m\colon\ m \ge n\}) \ \text{ and} \\ \limsup_{n\to\infty} a_n &= \lim_{n\to\infty} \sup(\{a_m\colon\ m \ge n\})\,, \end{aligned}$$

where we extend the notion of a limit by the definition

$$\lim_{n\to\infty} (\pm\infty,\, \pm\infty,\, \dots) = \pm\infty,$$

with same signs.

Proof. We prove the former formula, the proof of the latter is similar. Let $A_n = \inf(\{a_m\colon\ m \ge n\})$ $(\in \mathbb{R} \cup \{-\infty\})$. Clearly, $A_1 \le A_2 \le \cdots$. It is easy to see that if $A_1 = -\infty$, then $A_n = -\infty$ for every n, and that then (a_n) has a subsequence (a_{m_n}) with $\lim a_{m_n} = -\infty$. So $\liminf a_n = -\infty$, and the equality holds.

Let $A_1 \in \mathbb{R}$. Then $A_n \in \mathbb{R}$ for every n. We set $A = \sup(\{A_n\colon\ n \in \mathbb{N}\})$ $(\in \mathbb{R} \cup \{+\infty\})$. Clearly, if $A = +\infty$, then $\lim a_n = \lim A_n = +\infty$ (Theorem 2.1.32) and the equality again holds.

Finally, suppose that $A \in \mathbb{R}$. Since $A_1 \le A_2 \le \dots$, it is easy to see that $\lim A_n = A$ (Theorem 2.1.32). We show that $\liminf a_n = A$. Let an $\varepsilon > 0$ be given. Then $a_n \le A+\varepsilon$ for infinitely many n, for else we would have $A_n \ge A+\varepsilon$ for $n \ge n_0$. Thus, $\liminf a_n \le A + \varepsilon$. On the other hand, $A_{n_0} \ge A - \varepsilon$ for some n_0, so that $\liminf a_n \ge A - \varepsilon$. We see that $\liminf a_n = A = \lim A_n$. $\square$

Finally, we illustrate the use of liminfs and limsups by a number-theoretic result. Recall that the function $\tau(n)$ counts divisors of n. For example, $\tau(6) = |\{1, 2, 3, 6\}| = 4$. One can prove that

$$\limsup \frac{\log(\tau(n))}{(\log 2)(\log n)/(\log\log n)} = 1 \text{ and } \liminf \tau(n) = 2\,.$$

Exercise 2.3.8 *Prove the latter equality.*

Exercise 2.3.9 *Prove parts 3 and 4 of the last theorem.*

Exercise 2.3.10 *Find a sequence* (a_n) *such that* $L(a_n) = \mathbb{R}^*$.

Exercise 2.3.11 *Why for no sequence* $L(a_n) = [-1, 1] \setminus \{0\}$*?*

Exercise 2.3.12 *Find* $\liminf a_n$ *and* $\limsup a_n$ *if* $a_n = n(1 + (-1)^n)$.

2.4 Four existence theorems on limits

We prove four theorems on the existence of limits of real sequences: Theorems 2.4.4, 2.4.9, 2.4.12, and 2.4.18.

• *Monotonicity and boundedness.* We say that a sequence (a_n) weakly increases (respectively weakly decreases) if for every n we have $a_n \le a_{n+1}$ (respectively $a_n \ge a_{n+1}$). It increases (respectively decreases) if for every n we have $a_n < a_{n+1}$ (respectively, $a_n > a_{n+1}$). It is monotone if it weakly decreases or weakly increases. It is strictly monotone if it decreases or increases.

We say that (a_n) is bounded from above if there is a c such that for every n we have $a_n \le c$. Reversing the inequality, we get boundedness from below. A sequence (a_n) is bounded if it is bounded both from above and from below.

Exercise 2.4.1 (a_n) *weakly increases iff* $m \le n \Rightarrow a_m \le a_n$. *State and prove analogous equivalences for weakly decreasing, increasing and decreasing sequences.*

Exercise 2.4.2 (a_n) *is bounded iff there is a* c *such that* $|a_n| \le c$ *for every* n.

Exercise 2.4.3 *Which of the above nine underlined properties of sequences are robust?*

• *Limits of monotone sequences.* We proved the existence of limits of monotone sequences already in Theorem 2.1.32. Now we give a robust version.

Theorem 2.4.4 (monotone sequences 2) *Let* (a_n) *be a real sequence that has a monotone tail* $T = (a_m, a_{m+1}, \dots)$. *Then, with* $A = \{a_m, a_{m+1}, \dots\}$,

$$\lim_{n\to\infty} a_n = \sup(A), \ \textit{resp.} \ \lim_{n\to\infty} a_n = \inf(A)\,,$$

depending on whether T weakly increases or decreases. Infima and suprema are taken in $\langle \mathbb{R}^*, < \rangle$.

Proof. Clearly, the limit of the tail $(a_m, a_{m+1}, \dots)$ is the limit of (a_n). $\Box$

Exercise 2.4.5 *The assumption on (a_n) is a robust property of sequences.*

• *Limits of quasi-monotone sequences.* Monotone sequences can be generalized. A sequence $(a_n) \subset \mathbb{R}$ goes up (respectively goes down) if for every index n the set of indices m such that $a_m < a_n$ (respectively $a_m > a_n$) is finite. We say that (a_n) is quasi-monotone if it goes up or down.

Exercise 2.4.6 *Every monotone sequence is quasi-monotone.*

Exercise 2.4.7 *Find a quasi-monotone sequence (a_n) such that for no m the tail $(a_m, a_{m+1}, \dots)$ is monotone.*

Exercise 2.4.8 *Write the property of quasi-monotonicity in terns of quantifiers, logical connectives, brackets, variables, and inequalities, which are applied to natural and real numbers.*

Theorem 2.4.9 (quasi-monotone sequences). *Every quasi-monotone sequence (a_n) has a limit.*

Proof. We assume that (a_n) goes up, the case with (a_n) going down is similar. Let $A = \limsup a_n$ and let an $\varepsilon > 0$ be given. Then (a_n) has a subsequence (a_{m_n}) with $\lim a_{m_n} = A$ and we have $a_n < A + \varepsilon$ for every $n \ge n_0$. We take an n' such that $a_{m_{n'}} \in U(A, \varepsilon)$. Since (a_n) goes up, we can take an $n_1 \ge n_0$ such that $a_{m_{n'}} \le a_n < A + \varepsilon$ for every $n \ge n_1$. By Exercise 2.1.10, $a_n \in U(A, \varepsilon)$ for the same n. Hence $\lim a_n = A$. $\Box$

Here is the robust strengthening. The proof is clear and we omit it.

Corollary 2.4.10 (robust version) *Every sequence (a_n) that has a quasi-monotone tail has a limit.*

Exercise 2.4.11 *The assumption in the corollary defines a robust property of sequences.*

Quasi-monotone sequences were introduced by the British mathematician *Godfrey H. Hardy (1877–1947).*

• *The Bolzano–Weierstrass theorem.* Part 1 of the next theorem is known as the Bolzano–Weierstrass theorem.

Theorem 2.4.12 (BW theorem in three parts) *Let (a_n) be any real sequence. One of the following three cases occurs.*

1. *(the Bolzano–Weierstrass theorem) The sequence* (a_n) *is bounded and has a convergent subsequence.*

2. *The sequence* (a_n) *is not bounded from above and has a subsequence* (b_n) *such that* $\lim b_n = +\infty$.

3. *The sequence is not bounded from below and has a subsequence* (b_n) *such that* $\lim b_n = -\infty$.

Proof. 1. Let (a_n) be bounded and $(b_n) \preceq (a_n)$ be a monotone subsequence guaranteed by Theorem **??**. Then (b_n) is bounded and, by Theorem 2.1.32, has a finite limit.

2. Suppose that (a_n) is not bounded from above. We inductively define a subsequence (a_{m_n}) of (a_n) such that $a_{m_n} \geq n$. Then $\lim a_{m_n} = +\infty$.

We take any m_1 such that $a_{m_1} \geq 1$. Suppose that indices $m_1 < m_2 < \cdots < m_n$ are defined such that $a_{m_i} \geq i$ for $i = 1, 2, \ldots, n$. Since (a_n) is not bounded from above, there exists $j \in \mathbb{N}$ such that

$$a_j \geq 1 + \max(\{a_1,\, a_2,\, \ldots,\, a_{m_n}\} \cup \{n+1\})\,.$$

Then $j > m_n$ and $a_j \geq n + 1$. We set $m_{n+1} = j$.

3. This left to Exercise 2.4.13. □

Exercise 2.4.13 *Prove part 3 of the theorem.*

Exercise 2.4.14 *Let* $a \leq b$ *be real numbers. Then every sequence* $(a_n) \subset [a, b]$ *has a subsequence* (a_{m_n}) *such that* $\lim a_{m_n} \in [a, b]$.

Karl Weierstrass (1815–1897) was a German mathematician.

• *Cauchy sequences.* We met rational Cauchy sequences in the definition of $\mathbb{R}$ in Section 1.5. Real Cauchy sequences are defined in the same way.

Definition 2.4.15 (real Cauchy sequences) $(a_n) \subset \mathbb{R}$ *is* Cauchy *if for every* ε *there is an* n_0 *such that for every* $m, n \geq n_0$ *we have*

$$|a_m - a_n| \leq \varepsilon\,.$$

Exercise 2.4.16 *Cauchy sequences form a robust property of sequences.*

Exercise 2.4.17 *Every Cauchy sequence is bounded.*

Theorem 2.4.18 (metric completeness of $\mathbb{R}$**)** *A real sequence converges if and only if it is Cauchy.*

Proof. Implication $\Rightarrow$. Let $\lim a_n = a$ and ε be given. For every large n we have $|a_n - a| \le \frac{\varepsilon}{2}$. Using TI (Exercise 2.1.1) we have for every large m and n that

$$|a_m - a_n| \le |a_m - a| + |a - a_n| \le \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} = \varepsilon\,.$$

Hence (a_n) is Cauchy.

Implication $\Leftarrow$. Let (a_n) be Cauchy. By Exercise 2.4.17, (a_n) is bounded. By the Bolzano–Weierstrass theorem it has a convergent subsequence (a_{m_n}) with a limit a. Thus for a given ε we have for every large m and n that $|a_{m_n} - a| \le \frac{\varepsilon}{2}$ and $|a_m - a_n| \le \frac{\varepsilon}{2}$. By TI we have for the same large n, since $m_n \ge n$, that

$$|a_n - a| \le |a_n - a_{m_n}| + |a_{m_n} - a| \le \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} = \varepsilon\,.$$

Hence $\lim a_n = a$. $\Box$

Interestingly, A.-L. Cauchy lived in Prague in the Austrian Empire for several years in political exile.

Exercise 2.4.19 *There is a Cauchy sequence* $(a_n) \subset \mathbb{Q}$ *such that* $\lim a_n \notin \mathbb{Q}$.

The previous theorem therefore does not hold in the ordered field $\mathbb{Q}$. This is not surprising — we know that $\mathbb{Q}$ is not complete.

Exercise 2.4.20 *Where did we use the completeness of* $\mathbb{R}$ *in the previous proof?*

2.5 Fekete's lemma in combinatorics

This is in fact the fifth result ensuring the existence of limits of sequences. We state its additive, resp. multiplicative, form as a theorem, resp. a corollary.

• *Additive Fekete's lemma.* The lemma is credited to the Hungarian–Israeli mathematician *Michael Fekete (1886–1957).*

Exercise 2.5.1 *"fekete" means ... .*

Exercise 2.5.2 *Solve the previous exercise by methods available in 1984.*

A sequence $(a_n) \subset \mathbb{R}$ is superadditive, respectively subadditive, if for every two indices m and n we have $a_{m+n} \ge a_m + a_n$, respectively $a_{m+n} \le a_m + a_n$.

Theorem 2.5.3 (Fekete's lemma) *Let* $(a_n) \subset \mathbb{R}$ *and* $M = \{a_n/n\colon\ n \in \mathbb{N}\}$. *If* (a_n) *is superadditive, resp. subadditive, then*

$$\lim \tfrac{a_n}{n} = \sup(M),\ \ \textit{resp.}\ \ \lim \tfrac{a_n}{n} = \inf(M)\,.$$

The supremum and infimum are taken in $\langle \mathbb{R}^*, < \rangle$.

Proof. Let (a_n) be superadditive; the other case is similar. Let an ε be given. We take a number $c \in U(\sup(M), \varepsilon)$ with $c < \sup(M)$. Then $\frac{a_m}{m} > c$ for some m. We write any $n \in \mathbb{N}$ with $n \geq m$ as $n = km + l$, where $k \in \mathbb{N}$, $l \in \mathbb{N}_0$ and $0 \leq l < m$. We have $a_n \geq ka_m + a_l$ (Exercise 2.5.4), so that

$$\tfrac{a_n}{n} \geq \tfrac{ka_m}{km+l} + \tfrac{a_l}{n} = \tfrac{a_m/m}{1+l/km} + \tfrac{a_l}{n}\,.$$

For $n \to \infty$ also $k \to \infty$. So $1 + \frac{l}{km} \to 1$ and $\frac{a_l}{n} \to 0$. Hence for every δ we have for large n that $\frac{a_n}{n} \geq \frac{a_m}{m} - \delta$. Thus there is an $n_0 \geq m$ such that for every $n \geq n_0$,

$$c < a_n/n \leq \sup(M)\,.$$

By Exercise 2.1.10, $\frac{a_n}{n} \in U(\sup(M), \varepsilon)$ for the same n. Hence $\frac{a_n}{n} \to \sup(M)$. $\square$

Exercise 2.5.4 *If (a_n) is superadditive and $n = km + l$, then $a_n \geq ka_m + a_l$.*

• *Multiplicative Fekete's lemma.* We say that a sequence $(a_n) \subset (0, +\infty)$ is <u>supermultiplicative</u>, resp. <u>submultiplicative</u>, if for every two indices m and n we have $a_{m+n} \geq a_m a_n$, resp. $a_{m+n} \leq a_m a_n$. The real power a^b and functions $\log x$ and $\exp x$ used in the the next corollary are introduced in Section 5.1.

Exercise 2.5.5 *Prove the next two lemmas.*

Lemma 2.5.6 *Let $(c_n) \subset \mathbb{R}$ and $\lim c_n = K$, where $K \in \mathbb{R} \cup \{+\infty\}$. <u>Then</u> $\lim \exp(c_n) = \exp(K)$, where we define $\exp(+\infty) = +\infty$.*

Lemma 2.5.7 *Let $\emptyset \neq X \subset \mathbb{R}$ with $A = \sup(X) \in \mathbb{R} \cup \{+\infty\}$ and let $Y = \{\exp x\colon\ x \in X\}$. <u>Then</u> $\sup(Y) = \exp(A)$.*

Corollary 2.5.8 (multiplicative Fekete's lemma) *Let $(a_n) \subset (0, +\infty)$ and let $M = \{a_n^{1/n}\colon\ n \in \mathbb{N}\}$. If (a_n) is supermultiplicative, resp. submultiplicative, <u>then</u>*

$$\lim a_n^{1/n} = \sup(M), \ \ \textit{resp.}\ \ \lim a_n^{1/n} = \inf(M)\,.$$

Suprema and infima are taken in $\langle \mathbb{R}^*, < \rangle$.

Proof. We assume that (a_n) is supermultiplicative (the submultiplicative case is similar) and set $b_n = \log a_n$. The sequence (b_n) is superadditive: $b_{m+n} = \log(a_{m+n}) \geq \log(a_m a_n) = \log(a_m) + \log(a_n) = b_m + b_n$. By Theorem 2.5.3,

$$\lim_{n \to \infty} b_n/n = \sup(\{b_n/n\colon\ n \in \mathbb{N}\}) = B \ \ (\in \mathbb{R} \cup \{+\infty\})\,.$$

Since $b_n/n = \log\left(a_n^{1/n}\right)$ and $a_n^{1/n} = \exp(b_n/n)$, using Lemma 2.5.6 and 2.5.7 in the second and third equality, respectively, we obtain

$$\lim_{n \to \infty} a_n^{1/n} = \lim_{n \to \infty} \exp(b_n/n) = \exp(B) = \sup(M)\,.$$

$\square$

• *Fekete's lemma in combinatorics.* We present five applications of Fekete's lemma: two in extremal combinatorics and three in enumerative combinatorics.

• *Extremal functions of words.* Let $u = a_1a_2 \dots a_j$ and $v = b_1b_2 \dots b_k$ with $j, k \in \mathbb{N}$ be two words. We write $u \preceq v$ and say that u is <u>contained</u> in v if there exist j indices $1 \le i_1 < i_2 < \dots < i_j \le k$ such that for every $l, l' \in [j]$ we have

$$b_{i_l} = b_{i_{l'}} \iff a_l = a_{l'} \,.$$

In other words, a subsequence in v has the same equality pattern of letters as the word u. Let $r \in \mathbb{N}$. The word v is r-<u>sparse</u> if for any two indices $1 \le l < l' \le k$ with $b_l = b_{l'}$ we have $l' - l \ge r$.

Definition 2.5.9 ($\mathrm{ex}(u, n)$) *Suppose that* $u = a_1a_2 \dots a_j$ *is a word which uses* $r = |\{a_1, a_2, \dots, a_j\}|$ *distinct letters. Its* <u>*extremal function*</u> $\mathrm{ex}(u, n)\colon \mathbb{N} \to \mathbb{N}_0$ *is defined by*

$$\mathrm{ex}(u,\, n) := \max(\{|v|\colon\ v \in [n]^* \text{ is an } r\text{-sparse word such that } u \not\preceq v\})\,.$$

Exercise 2.5.10 $\mathrm{ex}(u, n)$ *is correctly defined for every* $u \ne \emptyset$ *and every* $n \in \mathbb{N}$.

A word u is <u>irreducible</u> if it cannot be written as a concatenation $u = vw$ of nonempty words v and w over two disjoint alphabets. For example, *abab* is an irreducible word, but *aabb* is not irreducible. Here is the first application of Fekete's lemma.

Proposition 2.5.11 (1st application) *Let* $u \ne \emptyset$ *be an irreducible word.* <u>*Then*</u> *the limit*

$$L(u) := \lim_{n\to\infty} \frac{\mathrm{ex}(u,\, n)}{n} \quad (\in [0,\, +\infty) \cup \{+\infty\})$$

exists.

Proof. In view of Theorem 2.5.3, it suffices to show that $\mathrm{ex}(u, n)$ is superadditive. Let r ($\in \mathbb{N}$) be the number of distinct letters in u, let $m, n \in \mathbb{N}$, and let v, respectively w, be a word witnessing the value $\mathrm{ex}(u, m)$, respectively $\mathrm{ex}(u, n)$. Let the word w' be obtained from w by renaming the elements of $[n]$ using the map $i \mapsto i+m$. So w' is over the alphabet $\{m+1, m+2, \dots, m+n\}$, $u \not\preceq w'$ and $|w'| = |w|$. The concatenated word vw' over $[m+n]$ is r-sparse and $u \not\preceq vw'$, due to the irreducibility of u. Hence

$$\mathrm{ex}(u,\, m) + \mathrm{ex}(u,\, n) = |v| + |w'| = |vw'| \le \mathrm{ex}(u,\, m+n)\,.$$

$\square$

For some short words u, the extremal function can be determined exactly.

Exercise 2.5.12 *Prove that* $\mathrm{ex}(abab, n) = 2n - 1$.

Thus $L(abab) = 2$. In contrast, $L(ababa) = +\infty$, but this is quite hard to prove. See the survey article [48] for more information on extremal functions of words.

• *Szemerédi's theorem.* An arithmetic progression with length $k \in \mathbb{N}$, abbreviated k-AP, is any set of integers of the form

$$X = \{a + jd\colon\ j \in [k]\} \ \ (a \in \mathbb{Z}\,, d \in \mathbb{N})\,.$$

Definition 2.5.13 ($r_k(n)$) *Let $k \in \mathbb{N}$. For $n \in \mathbb{N}$ we define*

$$r_k(n) = \max(\{|A|\colon\ A \subset [n] \text{ and contains no } k\text{-AP}\})\,.$$

Exercise 2.5.14 *Prove that $r_1(n) = 0$ and $r_2(n) = 1$ for every n.*

For $k \geq 3$, the problem of determining or estimating $r_k(n)$ becomes quite hard. The second application of Fekete's lemma is, however, easy to obtain.

Proposition 2.5.15 (2nd application) *Let $k \in \mathbb{N}$. Then the limit*

$$r_k := \lim_{n\to\infty} \frac{r_k(n)}{n} \quad (\in [0,\ 1])$$

exists.

Proof. In view of Theorem 2.5.3, it suffices to show that $r_k(n)$ is subadditive. Let $m, n \in \mathbb{N}$ and let $A \subset [m+n]$ be a set witnessing the value $r_k(m+n)$. We consider the sets $A' = A \cap [m]$ and $A'' = \{a \in [n]\colon\ a+m \in A\}$. Clearly, $|A| = |A'| + |A''|$, $A' \subset [m]$, $A'' \subset [n]$ and both A' and A'' avoid k-AP. Thus,

$$r_k(m+n) = |A| = |A'| + |A''| \leq r_k(m) + r_k(n)\,.$$

□

The following simply formulated theorem is one of the most important results in combinatorics in the 20th century.

Theorem 2.5.16 (E. Szemerédi, 1975) *For every $k \geq 3$ we have $r_k = 0$.*

The proof is quite complicated; see [74, 76]. *Endre Szemerédi (1940)* is a Hungarian mathematician.

• *Paths aka self-avoiding walks.* Let $G = \langle V, E\rangle$ with $E \subset \binom{V}{2}$ be a graph with the set of vertices V and the set of edges E. Here $\binom{V}{2} := \{e\colon\ e \subset V \wedge |e| = 2\}$. A path in G of length n $(\in \mathbb{N}_0)$ is any $n+1$-tuple

$$w = \langle v_0,\ v_1,\ \ldots,\ v_n\rangle$$

of mutually distinct vertices $v_i \in V$ such that $\{v_{i-1}, v_i\} \in E$ for every $i \in [n]$. The graph $G = \langle V, E\rangle$ is locally finite if for every vertex $v \in V$ there exist only finitely many edges $e \in E$ such that $v \in e$. If there is $r \in \mathbb{N}_0$ such that for every vertex $v \in V$ the number of such edges equals r, we call G r-regular.

Let $G = \langle V, E\rangle$ be a (typically infinite) graph. An automorphism of G is any bijection

$$f\colon V \to V$$

such that for every edge $e \in E$ we have $f[e], f^{-1}[e] \in E$. We say that G is transitive if for every two vertices $u, v \in V$ the graph G has an automorphism f such that $f(u) = v$. A LFT graph is a locally finite and transitive graph.

Exercise 2.5.17 *Every* LFT *graph is r-regular for some $r \in \mathbb{N}_0$.*

Let $G = \langle V, E\rangle$ be an LFT graph, $v \in V$ and $n \in \mathbb{N}$. We denote by $p_G(v, n)$ ($\in \mathbb{N}_0$) the number of paths in G of length n starting at v.

Exercise 2.5.18 *Show that the number $p_G(v, n)$ is finite and independent of v.*

For a LFT graph G, we define $p_G(n)$ ($\in \mathbb{N}_0$) to be the common value of $p_G(v, n)$ for any starting vertex v.

Exercise 2.5.19 *Let $r \in \mathbb{N}$ and G be an r-regular* LFT *graph. Show that $p_G(n) \le r(r-1)^{n-1}$ for every $n \in \mathbb{N}$.*

In the third application, Fekete's lemma proves the existence of growth constants $\kappa(G)$ of LFT graphs G.

Proposition 2.5.20 (3rd application) *Let $r \in \mathbb{N}$ and let $G = \langle V, E\rangle$ be an r-regular* LFT *graph. Then the limit*

$$\kappa(G) := \lim_{n\to\infty} p_G(n)^{1/n} \quad (\in [0,\, r-1])$$

exists.

Proof. In view of Corollary 2.5.8, it suffices to prove that $p_G(n)$ is submultiplicative. The bound $\kappa(G) \in [0, r-1]$ follows from Exercise 2.5.19. We fix an initial vertex $v \in V$ of paths and take, for every $u \in V$, an automorphism $f_u\colon V \to V$ of G such that $f_u(u) = v$. We denote by P_n, for $n \in \mathbb{N}$, the finite set of paths in G of length n starting at v. Let $m, n \in \mathbb{N}$. We consider the map $F\colon P_{m+n} \to P_m \times P_n$ given by $F(w) = \langle w', f_u[w'']\rangle$, where

$$w = \langle u_0 = v,\, u_1,\, \ldots,\, u_{m+1},\, \ldots,\, u_{m+n+1}\rangle\,,$$

$w' = \langle u_0, \ldots, u_{m+1}\rangle$, $w'' = \langle u_{m+1}, \ldots, u_{m+n+1}\rangle$ and $u = u_{m+1}$. It is easy to see that F is injective. Hence

$$p_G(m+n) = |P_{m+n}| \le |P_m \times P_n| = |P_m| \cdot |P_n| = p_G(m) \cdot p_G(n).$$

$\square$

A very interesting result in enumerative combinatorics is the determination of $\kappa(H)$ for the graph H of the hexagonal lattice. H is obtained by tiling the plane $\mathbb{R}^2$ by congruent regular hexagons. In 2012, the Russian mathematician *Stanislav Smirnov (1970)* together with the French mathematician *Hugo Duminil-Copin (1985)* proved [28] that

$$\kappa(H) = \sqrt{2+\sqrt{2}} = 2\cos(\pi/8)\,.$$

Exercise 2.5.21 *Define the graph H in set-theoretic terms and show that H is a 3-regular* LFT *graph.*

[39] is a survey article on growth constants of graphs.

• *Meanders.* A matching is any graph $M = \langle V, E\rangle$ such that $V \subset \mathbb{Z}$ is finite with even cardinality and E is a partition of V. A matching $M = \langle V, E\rangle$ is non-crossing if for no two edges $e, e' \in E$ we have

$$\min(e) < \min(e') < \max(e) < \max(e')\,.$$

Definition 2.5.22 (meanders) *A meander is any triple*

$$M = \langle V,\, E,\, E'\rangle$$

such that $\langle V, E\rangle$ and $\langle V, E'\rangle$ are two non-crossing matchings on a common vertex set V such that for every partition $\{A, B\}$ of V, some edge in $E \cup E'$ joins A and B.

We can draw any meander $\langle V, E, E'\rangle$ in the plane: the vertices in V are located on the x-axis and the edges in E, resp. E', are represented by half-circles drawn above, resp. below, the x-axis. The last condition in Definition 2.5.22 is (equivalent to) connectivity: the resulting union of semicircles is a connected and closed non-self-intersecting plane curve (circuit). It is more common to introduce meanders in these geometric terms — see, for example, the beginning of [3, 25]. For $n \in \mathbb{N}$, we define $m(n)$ to be the number of meanders $\langle [2n], E, E'\rangle$.

Exercise 2.5.23 *Let $C_n := \frac{1}{n+1}\binom{2n}{n} \le 4^n$ be the n-th Catalan number. Show that $m(n) \le C_n^2 \le 16^n$.*

The fourth application of Fekete's lemma establishes the existence of the growth constant μ of meanders.

Proposition 2.5.24 (4th application) *The limit*

$$\mu := \lim_{n\to\infty} m(n)^{1/n} \quad (\in [0,\, 16])$$

exists.

Proof. In view of Corollary 2.5.8, it suffices to show that $m(n)$ is supermultiplicative. The bound $\mu \in [0,16]$ follows from Exercise 2.5.23.

We denote by M_n the set of meanders with the vertex set $[2n]$ and define an injection $f\colon M_m \times M_n \to M_{m+n}$ — by this we will be done. Let $M \in M_m$ and $M' \in M_n$, with $M = \langle [2m], E, E' \rangle$. We shift M' by the vector $\langle 2m, 0 \rangle$ to the meander

$$M'' = \langle \{2m+1,\, 2m+2,\, \dots,\, 2m+2n\},\, E_0,\, E_0' \rangle\,.$$

We take the edges e_1 and e_2 such that $2m \in e_1 \in E$ and $2m+1 \in e_2 \in E_0$, and define new edges $e_3 = (e_1 \cup e_2) \setminus \{2m, 2m+1\}$ and $e_4 = \{2m, 2m+1\}$. We set $E_1 = ((E \cup E_0) \setminus \{e_1, e_2\}) \cup \{e_3, e_4\}$. It follows that

$$N = \langle [2m+2n],\, E_1,\, E' \cup E_0' \rangle \in M_{m+n}\,.$$

We define $f(M, M') = N$. Since M and M' can be recovered from N, the map f is an injection. $\square$

Exercise 2.5.25 *How does one recover M and M' from N?*

In the article [3] the best currently known bounds

$$11.380 \le \mu \le 12.901$$

are obtained. Another interesting article on meanders is [25]. It is not known if there is an algorithm that computes the function $n \mapsto m(n)$ in polynomially many (in n) bit operations.

• *Pattern-free permutations.* Let $m \in \mathbb{N}$. An $\underline{m\text{-permutation}}$ $p = a_1 a_2 \dots a_m$ is any word over the alphabet $[m]$ of length m that uses every letter $i \in [m]$ exactly once. A $\underline{\text{permutation}}$ is an m-permutation for some m. We denote the set of m-permutations by S_m.

Exercise 2.5.26 *Show that $|S_m| = m! = \prod_{i=1}^{m} i$.*

We introduce a containment for permutations. It is similar to the containment of words. If $q = b_1 b_2 \dots b_n$ is an n-permutation, we write $p \preceq q$ and say that p is $\underline{\text{contained}}$ in q if there exist m indices $1 \le i_1 < i_2 < \dots < i_m \le n$ such that for every $l, l' \in [m]$ we have

$$b_{i_l} < b_{i_{l'}} \iff a_l < a_{l'}\,.$$

In other words, the word q contains a subsequence with the same comparison pattern as p. For $n \in \mathbb{N}$ and an m-permutation p, we define $\pi(p, n)$ to be the number of n-permutations not containing p, that is,

$$\pi(p,\, n) = |\{q \in S_n\colon\ p \not\preceq q\}|\,.$$

In the fifth application of Fekete's lemma, we prove the existence of growth constants for pattern-free permutations. Finiteness of these constants follows from the Marcus–Tardos Theorem 2.5.28 below.

Proposition 2.5.27 (5th application) *For every permutation p, the finite limit*

$$\pi(p) := \lim_{n\to\infty} \pi(p,\, n)^{1/n} \quad (\in [0,\, +\infty))$$

exists.

Proof. In view of Corollary 2.5.8, it suffices to show that $\pi(p,n)$ is supermultiplicative. The bound $\pi(p) < +\infty$ follows from Theorem 2.5.28.

We denote by $\mathrm{per}(p,n)$ the set of n-permutations avoiding the permutation p and define an injection

$$f\colon \mathrm{per}(p,m) \times \mathrm{per}(p,n) \to \mathrm{per}(p,m+n)$$

— by this we will be done. A permutation $q = b_1b_2\dots b_l$ is $\oplus$-irreducible if there is no i with $1 \le i < l$ such that $b_j < b_{j'}$ whenever $1 \le j \le i < j' \le l$. For example, 312 is $\oplus$-irreducible, but 2143 is not. $\ominus$-irreducible permutations are defined similarly, by replacing the inequality $b_j < b_{j'}$ with $b_j > b_{j'}$. It is easy to see that every permutation is $\oplus$-irreducible or $\ominus$-irreducible. We assume that the forbidden permutation p is $\ominus$-irreducible; the other case is very similar.

Let $m,n \in \mathbb{N}$, $q = a_1a_2\dots a_m \in \mathrm{per}(p,m)$ and $r = b_1b_2\dots b_n \in \mathrm{per}(p,n)$. Let

$$s = c_1c_2\dots c_{m+n}\,,$$

where $c_i = a_i + n$ for $1 \le i \le m$ and $c_{i+m} = b_i$ for $1 \le i \le n$. It follows, due to $\ominus$-irreducibility of p, that $s \in \mathrm{per}(p,m+n)$. We set $f(q,r) = s$. Since we easily recover q and r from s, the map f is an injection. $\Box$

It is instructive to compare this proof and that of Proposition 2.5.11.

For some time, it was an open problem to show that the value $+\infty$ cannot occur as a permutation growth constant. In 2004 this was confirmed in [56] by the American mathematician *Adam Marcus (1979)* and the Hungarian mathematician and computer scientist *Gábor Tardos (1964)*.

Theorem 2.5.28 (Marcus–Tardos) *For every permutation p there is a constant $c \in \mathbb{N}$ such that $\pi(p,n) \le c^n$ for every $n \in \mathbb{N}$.*

We close this section with the remark that the existence of the (necessarily finite) permutation growth constants for the generalization of the counting function $\pi(p,n)$ with more than one forbidden permutation p is still an open problem.

Exercise 2.5.29 *Where does the argument in the proof of Proposition 2.5.27 fail in the case of several forbidden permutations?*

2.6 Arithmetic of limits

We investigate the interplay of limits with arithmetic operations in $\mathbb{R}^*$.

• *Arithmetic of limits.* Recall that (a_n), (b_n) and (c_n) denote real sequences, that always $\varepsilon, \delta, \theta > 0$ and that $\mathbb{R}^* = \mathbb{R} \cup \{-\infty, +\infty\}$, with elements denoted by A, B, K and L. Recall the arithmetic on $\mathbb{R}^*$ introduced in Section 2.1.

Exercise 2.6.1 (variants of the triangle inequality) *For every $a, b \in \mathbb{R}$,*

$$|a+b| \geq |a| - |b| \ \textit{and}\ |a-b| \geq |a| - |b|\,.$$

The next theorem is the main tool for computing limits.

Theorem 2.6.2 (arithmetic of limits) *Let $(a_n), (b_n) \subset \mathbb{R}$ be sequences with limits $\lim a_n = K$ and $\lim b_n = L$. Then*

$$\lim(a_n + b_n) = K + L,\ \lim a_n b_n = KL \ \textit{and}\ \lim \tfrac{a_n}{b_n} = \tfrac{K}{L}\,,$$

provided that the expression on the right-hand side is not indeterminate.

Proof. Sum. Let $K, L \in \mathbb{R}$ and an ε be given. For every large n we have that $|a_n - K| \leq \frac{\varepsilon}{2}$ and $|b_n - L| \leq \frac{\varepsilon}{2}$. By the TI, we have

$$|(a_n + b_n) - (K + L)| \leq |a_n - K| + |b_n - L| \leq \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} = \varepsilon$$

for the same n. Hence $a_n + b_n \to K + L$.

Let $K = L = \pm\infty$ and an ε be given. For every large n the numbers a_n and b_n have the same sign as K and $|a_n|, |b_n| \geq \frac{1}{2\varepsilon}$. Thus for these n the sum $a_n + b_n$ has the same sign as K and $|a_n + b_n| = |a_n| + |b_n| \geq \frac{1}{2\varepsilon} + \frac{1}{2\varepsilon} = \frac{1}{\varepsilon}$. Hence $a_n + b_n \to K + L = K = L$.

Let $K = \pm\infty$, $L \in \mathbb{R}$ and an ε be given. For every large n the number a_n has the same sign as K, $|a_n| \geq \frac{1}{\varepsilon} + |L| + 1$ and $|b_n - L| \leq 1$, thus $|b_n| \leq |L| + 1$. For these n the sum $a_n + b_n$ has the same sign as K and, by Exercise 2.6.1, $|a_n + b_n| \geq |a_n| - |b_n| \geq \frac{1}{\varepsilon} + |L| + 1 - |L| - 1 = \frac{1}{\varepsilon}$. Hence $a_n + b_n \to K + L = K$. The cases $K \in \mathbb{R}$ and $L = \pm\infty$ follow from the commutativity of addition.

Product. Let $K, L \in \mathbb{R}$ and an $\varepsilon \leq 1$ be given. For every large n one has that $|a_n - K| \leq \frac{\varepsilon}{2|L|+1}$, thus $|a_n| \leq |K| + 1$, and $|b_n - L| \leq \frac{\varepsilon}{2|K|+2}$. By TI, we have

$$\begin{aligned} |a_n b_n - KL| &\leq |a_n| \cdot |b_n - L| + |L| \cdot |a_n - K| \\ &\leq (|K| + 1) \cdot \tfrac{\varepsilon}{2|K|+2} + |L| \cdot \tfrac{\varepsilon}{2|L|+1} \leq \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} = \varepsilon \end{aligned}$$

for the same n. Hence $a_n b_n \to KL$.

Let $K = \pm\infty$, $L = \pm\infty$ and an ε be given. For every large n the number a_n has the same sign as K, b_n as L and $|a_n|, |b_n| \geq \frac{1}{\sqrt{\varepsilon}}$. Thus for these n the product $a_n b_n$ has the same sign as KL and $|a_n b_n| = |a_n| \cdot |b_n| \geq \frac{1}{\sqrt{\varepsilon}} \cdot \frac{1}{\sqrt{\varepsilon}} = \frac{1}{\varepsilon}$. Hence $a_n b_n \to KL$.

Let $K = \pm\infty$, $L \in \mathbb{R} \setminus \{0\}$ ($L = 0$ yields an indefinite expression) and let an ε be given. For every large n the number a_n has the same sign as K, $|a_n| \ge \frac{2}{\varepsilon|L|}$ and $|b_n - L| \le \frac{|L|}{2}$, thus $|b_n| \ge \frac{|L|}{2}$. So for these n the product $a_n b_n$ has the same sign as KL and $|a_n b_n| = |a_n| \cdot |b_n| \ge \frac{2}{\varepsilon|L|} \cdot \frac{|L|}{2} = \frac{1}{\varepsilon}$. Hence $a_n b_n \to KL$. The cases $K \in \mathbb{R} \setminus \{0\}$ and $L = \pm\infty$ follow from the commutativity of multiplication.

Ratio. Let $K \in \mathbb{R}$, $L \in \mathbb{R} \setminus \{0\}$ ($L = 0$ yields an indefinite expression) and an ε be given. For every large n it holds that $|a_n - K| \le \min(\{1, \frac{\varepsilon L^2}{4(|L|+1)}\})$ and $|b_n - L| \le \min(\{1, \frac{\varepsilon L^2}{4(|K|+1)}, \frac{|L|}{2}\})$, thus $|a_n| \le |K| + 1$, $|b_n| \le |L| + 1$ and $|b_n| \ge \frac{|L|}{2}$. By TI, we have

$$\left|\frac{a_n}{b_n} - \frac{K}{L}\right| = \left|\frac{a_n L - b_n K}{b_n L}\right| \le \frac{|a_n| \cdot |L - b_n| + |b_n| \cdot |a_n - K|}{|b_n| \cdot |L|} \le \frac{\varepsilon L^2/4 + \varepsilon L^2/4}{L^2/2} = \varepsilon$$

for the same n. Hence $\frac{a_n}{b_n} \to \frac{K}{L}$.

Let $K = \pm\infty$, $L \in \mathbb{R} \setminus \{0\}$ and an ε be given. For every large n the number a_n has the same sign as K, $|a_n| \ge \frac{|L|+1}{\varepsilon}$ and $|b_n - L| \le 1$, thus $|b_n| \le |L| + 1$. So for these n the ratio $\frac{a_n}{b_n}$ has the same sign as $\frac{K}{L}$ and $\left|\frac{a_n}{b_n}\right| = \frac{|a_n|}{|b_n|} \ge \frac{|L|+1}{\varepsilon(|L|+1)} = \frac{1}{\varepsilon}$. Hence $\frac{a_n}{b_n} \to \frac{K}{L}$.

Let $K \in \mathbb{R}$, $L = \pm\infty$ and an ε be given. For every large n one has that $|a_n - K| \le 1$, thus $|a_n| \le |K| + 1$, and $|b_n| \ge \frac{|K|+1}{\varepsilon}$. For these n it holds that $\left|\frac{a_n}{b_n} - 0\right| = \left|\frac{a_n}{b_n}\right| = \frac{|a_n|}{|b_n|} \le \frac{|K|+1}{(|K|+1)/\varepsilon} = \varepsilon$. Hence $\frac{a_n}{b_n} \to \frac{K}{L} = 0$. $\square$

If $\lim a_n = K$, $\lim b_n = L$ and $\frac{K}{L}$ is not an indefinite expression, then $L \ne 0$. There are only finitely many n with $b_n = 0$. The corresponding undefined ratios $\frac{a_n}{b_n}$ may be ignored or defined arbitrarily.

• *Two more results on the arithmetic of limits.* The previous theorem does not describe the arithmetic of limits completely. Even if one of the two sequences does not have a limit, the sequence of sums, products, or ratios may still have a unique limit. We present six cases when this happens, and leave proofs for them as an exercise.

Exercise 2.6.3 *Prove the following proposition.*

Proposition 2.6.4 (more on AL) *Let $(a_n), (b_n) \subset \mathbb{R}$. The following implications hold.*

1. *If (a_n) is bounded and $L = \lim b_n = \pm\infty$ then $\lim(a_n + b_n) = L$.*
2. *If (a_n) is bounded and $\lim b_n = 0$ then $\lim a_n b_n = 0$.*
3. *If $a_n \ge c > 0$ for every $n \ge n_0$ and $L = \lim b_n = \pm\infty$ then $\lim a_n b_n = L$.*
4. *If (a_n) is bounded and $\lim b_n = \pm\infty$ then $\lim \frac{a_n}{b_n} = 0$.*
5. *If $a_n \ge c > 0$ and $b_n > 0$ for every $n \ge n_0$, and if $\lim b_n = 0$ then $\lim \frac{a_n}{b_n} = +\infty$.*

6. *If $0 < a_n \le c$ for every $n \ge n_0$ and $L = \lim b_n = \pm\infty$ <u>then</u> $\lim \frac{b_n}{a_n} = L$.*

In parts 3 and 5 we may also have $a_n \le c < 0$, in part 6 we may have $c \le a_n < 0$ and in part 5 we may have $b_n < 0$. It is not hard to state precisely and prove these modifications.

If $a_n \to K$, $b_n \to L$ and the expression $K + L$ is indefinite, then $\lim(a_n + b_n)$ is far from being uniquely determined. Similarly for KL and $\frac{K}{L}$. We make it more precise in the next proposition.

Proposition 2.6.5 (on indefiniteness) *The following holds.*

1. *Indefiniteness of $(+\infty) + (-\infty)$. If $a_n \to +\infty$ and $A \in \mathbb{R}^*$, <u>then</u> $a_n + b_n \to A$ for some (b_n) with $b_n \to -\infty$.*
2. *Indefiniteness of $(+\infty) \cdot 0$. If $a_n \to +\infty$ and $A \in \mathbb{R}^*$, <u>then</u> $a_n b_n \to A$ for some (b_n) with $b_n \to 0$.*
3. *Indefiniteness of $\frac{0}{0}$. If $a_n \to 0$, $a_n \ne 0$ and $A \in \mathbb{R}^*$, <u>then</u> $\frac{a_n}{b_n} \to A$ for some (b_n) with $b_n \to 0$.*
4. *Indefiniteness of $\frac{A}{0}$ with $A \ne 0$. If $a_n \to A$ and $B \in \{-\infty, +\infty\}$, <u>then</u> $\frac{a_n}{b_n} \to B$ for some (b_n) with $b_n \to 0$.*
5. *Indefiniteness of $\frac{\pm\infty}{\pm\infty}$. If $a_n \to +\infty$ and $A \in \mathbb{R}^*$, <u>then</u> $\frac{a_n}{b_n} \to A$ for some (b_n) with $b_n \to \pm\infty$.*

Proof. 1. If $A = +\infty$, we set $b_n = -\frac{a_n}{2}$. If $A \in \mathbb{R}$, we set $b_n = -a_n + A$. If $A = -\infty$, we set $b_n = -2a_n$.

2. If $A = +\infty$, we set $b_n = (1 + |a_n|)^{-1/2}$. If $A \in \mathbb{R}$, we set $b_n = \frac{A}{1+|a_n|}$. If $A = -\infty$, we set $b_n = -(1 + |a_n|)^{-1/2}$.

3. If $A = +\infty$ we set $b_n = \mathrm{sgn}(a_n)a_n^2$. If $A \in \mathbb{R} \setminus \{0\}$, we set $b_n = \frac{a_n}{A}$. If $A = 0$, we set $b_n = \sqrt{|a_n|}$. If $A = -\infty$ we set $b_n = -\mathrm{sgn}(a_n)a_n^2$.

4. If $B = +\infty$, we set $b_n = 1$ if $a_n = 0$ and $b_n = \mathrm{sgn}(a_n)n^{-1}$ otherwise. If $B = -\infty$, we set $b_n = 1$ if $a_n = 0$ and $b_n = -\mathrm{sgn}(a_n)n^{-1}$ otherwise.

5. If $A = +\infty$, we set $b_n = \sqrt{1 + |a_n|}$. If $A \in \mathbb{R} \setminus \{0\}$, we set $b_n = \frac{\sqrt{1+a_n^2}}{A}$. If $A = 0$, we set $b_n = 1 + a_n^2$. If $A = -\infty$, we set $b_n = -\sqrt{1 + |a_n|}$. $\square$

Exercise 2.6.6 *What can be said in part 3 when we assume instead of $a_n \ne 0$ for every n that, contrary-wise, $a_n = 0$ for infinitely many n?*

2.7 Limits of recurrent sequences

This section begins with an example of the determination of the limit of a recurrent sequence. We define f-recurrent sequences and make explicit the *method* *of equations* that finds their limits. We illustrate this method with several examples, including the sequence (F_{n+1}/F_n) of ratios of consecutive Fibonacci numbers.

• *A recurrent sequence.* We begin with the well known inequality between the arithmetic and geometric mean.

Exercise 2.7.1 (AG inequality) *For every $a, b \ge 0$ we have $\frac{a+b}{2} \ge \sqrt{ab}$.*

Proposition 2.7.2 (a recurrent limit) *Let $(a_n) \subset \mathbb{Q}$ be given by $a_1 = 1$ and $a_n = \frac{a_{n-1}}{2} + \frac{1}{a_{n-1}}$ for $n \ge 2$. Then $\lim a_n = \sqrt{2}$.*

Proof. We show that $a_2 \ge a_3 \ge \dots \ge 0$. From this, it follows by Theorem 2.4.4 that (a_n) has a finite nonnegative limit. Clearly, a_n is defined for every n. For $n \ge 2$, we get by the AG inequality that

$$a_n = \tfrac{a_{n-1}}{2} + \tfrac{1}{a_{n-1}} \ge 2\sqrt{\tfrac{a_{n-1}}{2} \cdot \tfrac{1}{a_{n-1}}} = \sqrt{2}\,.$$

Then for $n \ge 3$ we have $a_{n-1} \ge a_n \iff \frac{a_{n-1}}{2} \ge \frac{1}{a_{n-1}} \iff a_{n-1} \ge \sqrt{2}$ which is true. Thus (a_n) has a weakly decreasing tail. Let $a = \lim a_n$ $(\ge \sqrt{2})$. By the arithmetic of limits and limits of subsequences,

$$a = \lim a_n = \tfrac{\lim a_{n-1}}{2} + \tfrac{1}{\lim a_{n-1}} = \tfrac{a}{2} + \tfrac{1}{a}\,.$$

Hence $\frac{a}{2} = \frac{1}{a} \iff a^2 = 2$, and $a = \sqrt{2}$. □

Such computations are rigorous only when the limit is proven to exist. For example, the sequence given by $a_1 = 1$ and $a_n = -a_{n-1}$ for $n \ge 2$, does not converge to 0, although the equation $L = -L$ has in $\mathbb{R}^*$ the only solution $L = 0$. The sequence alternates, $(a_n) = (1, -1, 1, -1, \dots)$, and has no limit.

• *f-recurrent sequences.* What is a recurrent sequence? To explain it, we have to employ functions of several variables.

Definition 2.7.3 (f-recurrent sequences) *Let $k \in \mathbb{N}$, let $M \subset \mathbb{R}^k$ and let $f \colon M \to \mathbb{R}$. A sequence (a_n) is f-recurrent if for every $n \in \mathbb{N}$,*

$$\langle a_n,\, a_{n+1},\, \dots,\, a_{n+k-1}\rangle \in M \wedge a_{n+k} = f(a_n,\, a_{n+1},\, \dots,\, a_{n+k-1})\,.$$

We write $f(a_n, a_{n+1}, \dots, a_{n+k-1})$ instead of $f(\langle a_n, a_{n+1}, \dots, a_{n+k-1}\rangle)$.

Exercise 2.7.4 *The* P*-recurrent sequences in Section 8.2 are not f-recurrent for any function f. Extend Definition 2.7.3 so that it includes* P*-recurrent sequences.*

The well known Fibonacci sequence

$$(F_n) := (1,\, 1,\, 2,\, 3,\, 5,\, 8,\, 13,\, 21,\, 34 \dots)$$

is an f-recurrent sequence; $f = f(x_1, x_2) = x_1 + x_2$.

Exercise 2.7.5 *Prove that* $\lim F_n = +\infty$

• *The method of equations.* We are interested in the limits of f-recurrent sequences. They can be determined by the method we explain.

Definition 2.7.6 (limit fix points) *Let k, M, and f be as in Definition 2.7.3 and let $L \in \mathbb{R}^*$. We call L a limit fix point of f if for some sequence (a_n) three conditions hold.*

1. $\lim a_n = L$.
2. *For every n we have $\langle a_n, a_{n+1}, \dots, a_{n+k-1}\rangle \in M$.*
3. $\lim_{n\to\infty} f(a_n, a_{n+1}, \dots, a_{n+k-1}) = L$.

We denote the set of limit fix points of f by $\mathrm{LFP}(f)$.

Limit fix points of f are, in a sense, exactly the solutions $L \in \mathbb{R}^*$ of the equation $f(L, L, \dots, L) = L$.

Exercise 2.7.7 *Prove the next proposition.*

Proposition 2.7.8 (on LFP) *Let $L \in \mathbb{R}^*$ and let (a_n) be an f-recurrent sequence with $\lim a_n = L$. Then L is a limit fix point of f.*

A function $f\colon M \to \mathbb{R}$, where $M \subset \mathbb{R}^k$, is continuous if for every point $\overline{b} = \langle b_1, b_2, \dots, b_k\rangle \in M$ and every k-tuple of real sequences

$$\langle (a_{n,1}),\ (a_{n,2}),\ \dots,\ (a_{n,k})\rangle$$

with $\lim a_{n,i} = b_i$, $i = 1, 2, \dots, k$, we have $\lim f(a_{n,1}, a_{n,2}, \dots, a_{n,k}) = f(\overline{b})$.

Exercise 2.7.9 *State the equivalent ε-δ definition of continuity of f and prove the equivalence.*

Theorem 2.7.10 (LFP of continuous functions) *Let k be in $\mathbb{N}$, let $M \subset \mathbb{R}^k$ and let $f\colon M \to \mathbb{R}$ be continuous. Then*

$$\{L \in \mathrm{LFP}(f)\colon\ \langle L,\, L,\, \dots,\, L\rangle \in M\} = \{L \in \mathbb{R}\colon\ f(L,\, L,\, \dots,\, L) = L\}\,.$$

Proof. Suppose that L is in the set on the right-hand side. Then the constant sequence $(a_n) = (L, L, \dots)$ shows that L is an element of the set on the left-hand side.

Suppose that L is in the set on the left-hand side. Then there is a sequence (a_n) with $\lim a_n = L$ such that

$$\langle a_n,\, a_{n+1},\, \dots,\, a_{n+k-1}\rangle \in M$$

for every n and $\lim f(a_n, a_{n+1}, \dots, a_{n+k-1}) = L$. Since $\langle L, L, \dots, L\rangle \in M$ and f is continuous, the last limit also equals $f(L, L, \dots, L)$. Hence L is an element of the set on the right-hand side. □

The previous proposition and theorem yield a method for determining finite limits of f-recurrent sequences. The proof of the corollary is immediate.

Corollary 2.7.11 (method of equations) *Let $(a_n) \subset \mathbb{R}$ be an f-recurrent sequence with $\lim a_n = L \in \mathbb{R}$, where $f\colon M \to \mathbb{R}$ with $M \subset \mathbb{R}^k$ is continuous, and let $\langle L, L, \dots, L\rangle \in M$. Then L is a solution of the equation*

$$f(L,\, L,\, \dots,\, L) = L\,.$$

Infinite limits $L = \pm\infty$, as well as finite limits $L \in \mathbb{R}$ lying outside M (that is, $\langle L, L, \dots, L\rangle \notin M$), have to be handled differently.

• *The initial example.* Equipped with the last proposition and corollary, we revisit the initial example. In it we have $k = 1$ and the continuous function

$$f = f(x) = x/2 + 1/x\colon \mathbb{R} \setminus \{0\} \to \mathbb{R}\,.$$

It is easy to see that $0 \notin \mathrm{LFP}(f)$ and $\pm\infty \in \mathrm{LFP}(f)$. So by Theorem 2.7.10 we have

$$\mathrm{LFP}(f) = \{-\infty, +\infty, -\sqrt{2}, \sqrt{2}\}\,,$$

because the solutions of the equation $f(L) = L \iff L/2 + 1/L = L$ are exactly $L = \pm\sqrt{2}$. If (a_n) is an f-recurrent sequence with $\lim a_n = L$, then by Proposition 2.7.8, $L = \pm\infty$ or $L = \pm\sqrt{2}$.

Exercise 2.7.12 *Make it more precise in the next generalization of Proposition 2.7.2.*

Proposition 2.7.13 (a generalization) *Let $f(x) = \frac{x}{2} + \frac{1}{x}$. For every number $a \in \mathbb{R} \setminus \{0\}$ the unique f-recurrent sequence (a_n) starting from $a_1 = a$ has*

$$\lim a_n = \mathrm{sgn}(a) \cdot \sqrt{2}\,.$$

• *On the Fibonacci sequence.* The Fibonacci sequence (F_n) was recalled above. We determine the limit $\lim \frac{F_{n+1}}{F_n}$ by the method of equations.

Exercise 2.7.14 *Prove by induction on n that $F_{n+1}F_{n-1} - F_n^2 = (-1)^n$.*

Let $\phi := \frac{1}{2}(1 + \sqrt{5})$ be the golden ratio and $\psi := \frac{1}{2}(1 - \sqrt{5})$ ($\in [-0.6, -0.7]$) be its conjugate.

Proposition 2.7.15 (golden ratio and Fibonacci) *It is true that*

$$\lim_{n\to\infty} \frac{F_{n+1}}{F_n} = \phi\,.$$

Proof. For $n \in \mathbb{N}$, let $g_n = \frac{F_{n+1}}{F_n}$. Since $g_{n+1} = \frac{F_{n+2}}{F_{n+1}} = 1 + \frac{F_n}{F_{n+1}} = 1 + \frac{1}{g_n}$, the sequence (g_n) is f-recurrent for the continuous function $f = f(x) = 1 + \frac{1}{x}$. By Exercise 2.7.14 ($n \geq 2$),

$$g_n - g_{n-1} = \tfrac{F_{n+1}}{F_n} - \tfrac{F_n}{F_{n-1}} = \tfrac{(-1)^n}{F_n F_{n-1}} .$$

It follows that ($n \geq 3$)

$$g_n - g_{n-2} = g_n - g_{n-1} + g_{n-1} - g_{n-2} = \tfrac{(-1)^n F_{n-2} + (-1)^{n-1} F_n}{F_n F_{n-1} F_{n-2}} = \tfrac{(-1)^{n-1}}{F_n F_{n-2}} .$$

Thus

$$1 = g_1 < g_3 < g_5 < \cdots < \cdots < g_6 < g_4 < g_2 = 2$$

and $g_n - g_{n-1} \to 0$. We see that (g_n) converges because it is Cauchy. Since $f(L) = L \iff L^2 - L - 1 = 0$, Corollary 2.7.11 implies that $\lim g_n = \phi$. $\square$

We can obtain the limit $F_{n+1}/F_n \to \phi$ more easily from the next theorem. The point of the proposition is to illustrate the use of the method of equations for sequences that are not quasi-monotone; one must then resort to the Cauchy condition.

Theorem 2.7.16 (Binet's formula) *The n-th Fibonacci number*

$$F_n = \frac{\phi^n - \psi^n}{\sqrt{5}} .$$

Proof. It is easy to check this formula for $n = 1$ and 2. Let G_n be the ratio on the right-hand side. For every n we have

$$G_{n+2} - G_{n+1} - G_n = \tfrac{(\phi^2 - \phi - 1)\phi^n - (\psi^2 - \psi - 1)\psi^n}{\sqrt{5}} = \tfrac{0 - 0}{\sqrt{5}} = 0$$

because $(x - \phi)(x - \psi) = x^2 - x - 1$. Hence $F_n = G_n$ for every n. $\square$

Jacques P. M. Binet (1786–1856) was a French mathematician, physicist, and astronomer.

• *Two more examples of applications of the method of equations.* In the first, we determine for $a \geq 0$ the value $v(a)$ of the nested radicals

$$v(a) = \sqrt{a + \sqrt{a + \sqrt{a + \ldots}}} .$$

Let $f_a = f_a(x) = \sqrt{a + x}$. We interpret $v(a)$ as $v(a) = \lim a_n$ where (a_n) is the f_a-recurrent sequence starting from $a_1 = 0$.

Proposition 2.7.17 (nested radicals) *Then* $v(a) = \frac{1}{2}(1 + \sqrt{1 + 4a})$ *for* $a > 0$ *and* $v(0) = 0$.

Proof. The value $v(0) = 0$ is clear. Let $a > 0$. We have $k = 1$, $M = [-a, +\infty)$ (the domain of f_a), and f_a is continuous. It follows that

$$\mathrm{LFP}(f_a) = \{+\infty,\, r_a\} \ \text{ where } \ r_a = \tfrac{1}{2}(1 + \sqrt{1 + 4a})$$

because $L = r_a$ is the only solution of the equation

$$f_a(L) = L \iff \sqrt{a + L} = L \iff L^2 - L - a = 0 \wedge L \geq 0\,.$$

Let (a_n) be the f_a-recurrent sequence with $a_1 = 0$. Then $a_1 < a_2 = \sqrt{a}$. If $a_n < a_{n+1}$, then also

$$a_{n+1} = \sqrt{a + a_n} < \sqrt{a + a_{n+1}} = a_{n+2}$$

and we see that (a_n) increases. Clearly, $a_1 < r_a$. If $a_n < r_a$, then also $a_{n+1} < r_a$ because

$$a_{n+1} = \sqrt{a + a_n} < r_a \iff a + a_n < r_a^2 = r_a + a \iff a_n < r_a\,.$$

So (a_n) is bounded from above by r_a. By Theorem 2.1.32 and Corollary 2.7.11, $v(a) = \lim a_n = r_a$. $\square$

Note that $v(1) = \phi$.

In the second example, we find the limits of generalized Fibonacci sequences.

Proposition 2.7.18 (gen. Fibonacci sequences) *Let $f = f(x_1, x_2) = x_1 + x_2$, $a, b \in \mathbb{R}$ and let (a_n) be the f-recurrent sequence starting from $a_1 = a$ and $a_2 = b$; for $a = b = 1$ we get $(a_n) = (F_n)$. The following holds.*

1. *If $b < -a/\phi$ then $\lim a_n = -\infty$.*
2. *If $b = -a/\phi$ then $\lim a_n = 0$.*
3. *If $b > -a/\phi$ then $\lim a_n = +\infty$.*

Proof. We have $k = 2$, $M = \mathbb{R}^2$ (the domain of f), and f is continuous. It follows that

$$\mathrm{LFP}(f) = \{-\infty,\, 0,\, +\infty\}$$

because $f(L, L) = L \iff 2L = L$ has the only solution $L = 0$. We determine for which a and b each of these three possible limits occurs.

We compute

$$(a_n) = (a,\, b,\, a + b,\, a + 2b,\, 2a + 3b,\, 3a + 5b,\, 5a + 8b,\, \dots)\,.$$

So for $n \geq 3$ induction gives that $a_n = aF_{n-2} + bF_{n-1}$, where (F_n) is the Fibonacci sequence. We use the expression

$$a_n = F_{n-2} \cdot \left(a + \tfrac{F_{n-1}}{F_{n-2}} \cdot b\right).$$

Since $\frac{F_{n-1}}{F_{n-2}} \to \phi$ by Proposition 2.7.15 and $\lim F_{n-2} = +\infty$ by Exercise 2.7.5, we get the limits in cases 1 and 3. Suppose that case 2 occurs and $a + \phi b = 0$. Using Theorem 2.7.16 and item 1 of Proposition 2.1.36, we see that ($n \ge 3$)

$$a_n = \tfrac{1}{\sqrt{5}}\big(a\phi^{n-2} - a\psi^{n-2} + b\phi^{n-1} - b\psi^{n-1}\big) = -\tfrac{1}{\sqrt{5}}\big(a\psi^{n-2} + b\psi^{n-1}\big) \to 0$$

because $|\psi| < 1$. $\square$

Exercise 2.7.19

$$\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{\ddots}}}} = ?$$

2.8 Limits and order

We investigate interactions between limits and the linear order $\langle \mathbb{R}^*, < \rangle$.

• *Strengthening a standard theorem.* If we can compare terms in two sequences, we can compare the limits, and vice versa. However, which terms in the sequences are being compared?

Theorem 2.8.1 (limits versus order) *Let* $\lim a_n = K$ *and* $\lim b_n = L$. *The following holds.*

1. *If* $K < L$ *<u>then</u>* $a_m < b_n$ *for every two indices* $m, n \ge n_0$.
2. *If for every index* n_0 *there exist indices* $m, n \ge n_0$ *such that* $a_m \ge b_n$, *<u>then</u>* $K \ge L$.

Proof. 1. Let $K < L$. By Exercise 2.1.11 there is an ε such that $U(K,\varepsilon) < U(L,\varepsilon)$. By the definition of limits, $a_m \in U(K,\varepsilon)$ and $b_n \in U(L,\varepsilon)$ for every $m, n \ge n_0$. So $m, n \ge n_0 \Rightarrow a_m < b_n$.

2. The implication $\varphi \Rightarrow \psi$ is equivalent to the contrapositive $\neg\psi \Rightarrow \neg\varphi$. The contrapositive of the implication in part 1 is the implication in part 2. $\square$

Strangely, Theorem 2.8.1 is always presented in unnecessarily weak forms. Part 1 as: if $K < L$ then there is an n_0 such that $a_n < b_n$ for every $n \ge n_0$. Part 2 as: if $a_n \le b_n$ for every $n \ge n_0$, then $K \le L$. I was teaching these weakish variants of Theorem 2.8.1 for many years.

Exercise 2.8.2 *Why is the first part of Theorem 2.8.1 with possibly* $m \ne n$ *stronger than the standard version with* $m = n$?

Strict inequalities may not be preserved in limits; they may turn into equalities. This is another reason why non-strict equalities are safer than strict ones.

Exercise 2.8.3 *Find convergent sequences* (a_n) *and* (b_n) *such that* $a_m < b_n$ *for all* m *and* n*, but* $\lim a_n = \lim b_n$.

Exercise 2.8.4 *Prove the next strengthening of Theorem 2.8.1. State the corresponding part 2.*

Proposition 2.8.5 (a strengthening) *Let* $\lim a_n = K$*,* $\lim b_n = L$*, and let* $K < L$*. Then* $a_m \le a < b \le b_n$ *for every* $m, n \ge n_0$ *for some index* n_0 *and some real numbers* $a < b$.

• *Intervals.* We remind the notion of real intervals.

Definition 2.8.6 (intervals) *An interval, or a real convex set, is any set* $I \subset \mathbb{R}$ *such that if* $a < b < c$ *with* $a, c \in I$*, then always* $b \in I$*. An interval is nontrivial if it has at least two elements.*

In the next proposition we work in the linear order $\langle \mathbb{R}, < \rangle$.

Proposition 2.8.7 (on intervals) *Intervals are exactly the sets* $\emptyset$*,* $\{a\}$*,* $\mathbb{R}$*,* (a, b)*,* $(-\infty, a)$*,* $(a, +\infty)$*,* $(a, b]$*,* $[a, b)$*,* $[a, b]$*,* $(-\infty, a]$ *and* $[a, +\infty)$*, where* $a < b$ *are real numbers.*

Proof. The transitivity of $<$ shows that all stated sets are convex. We show that no other real convex sets exist. Let $X \subset \mathbb{R}$ be a convex set different from $\emptyset$, $\mathbb{R}$ and $\{a\}$, and let $a \in \mathbb{R} \setminus X$. The convexity of X implies that $a \in H(X)$ (upper bounds of X) or $a \in D(X)$ (lower bounds of X). We discuss only the former case because the latter case reduces to the former by the reversal of inequalities.

So let $H(X) \ne \emptyset$. We set $b = \sup(X)$. Let $D(X) = \emptyset$. If $b \in X$ then $X = (-\infty, b]$. If $b \notin X$ then $X = (-\infty, b)$. Let $D(X) \ne \emptyset$. Then we set $c = \inf(X)$, clearly $c < b$. If $b \notin X$ and $c \notin X$ then $X = (c, b)$. If $b \notin X$ and $c \in X$ then $X = [c, b)$. If $b \in X$ and $c \notin X$ then $X = (c, b]$. Finally, if $b \in X$ and $c \in X$ then $X = [c, b]$. □

Exercise 2.8.8 *Are there nonempty finite intervals?*

• *The squeeze theorem.* In Czech textbooks, it is the *two-cops-theorem.* We strengthen it by employing just one cop.

Theorem 2.8.9 (one cop) *Let* $\lim a_n = b \in \mathbb{R}$ *and let* $(b_n) \subset \mathbb{R}^n$ *be such that* $\lim |a_n - b_n| = 0$*. Then also* $\lim b_n = b$.

Proof. Let ε be given. Then $a_n \in U(b, \frac{\varepsilon}{2})$ and $|a_n - b_n| \le \frac{\varepsilon}{2}$ for every large n. By TI we have $b_n \in U(b, \varepsilon)$ for the same n. Hence $\lim b_n = b$. □

The classical two-cops-theorem is the next corollary. We denote by $I(a, b)$ the interval $[a, b]$ if $a \le b$, and the interval $[b, a]$ if $b \le a$.

Corollary 2.8.10 (two cops) *Let* $\lim a_n = \lim b_n = b$ *and let* (c_n) *be such that* $c_n \in I(a_n, b_n)$ *for every large* n. *Then also* $\lim c_n = b$.

Proof. Clearly, $\lim |a_n - b_n| = 0$. Since for large n we have

$$|a_n - c_n|,\ |b_n - c_n| \le |a_n - b_n|\,,$$

either of the sequences (a_n) and (b_n) can serve as a cop for (c_n) in the previous theorem. $\square$

Exercise 2.8.11 *State and prove the variants of Theorem 2.8.9 with the cop going to* $\pm\infty$.

Chapter 3

Infinite series

Infinite series are an important application of limits of real sequences. We treat their theory in three sections. In Section 3.1 we develop a theory of *absolutely convergent set series*. These are maps $h\colon X \to \mathbb{R}$ defined on at most countable sets X such that for countable X and every bijection $f\colon \mathbb{N} \to X$, the finite (and necessarily unique) sum

$$\lim_{n\to\infty} \sum_{j=1}^{n} h(f(j)) \ \ (\in \mathbb{R})$$

exists. In Section 3.2 we apply this theory to generalize Pólya's theorem. This theorem determines the limits of probabilities that walks in the grid graph $\mathbb{Z}^d$, starting at the origin, visit the given vertex. In Section 3.3, we specialize set series to the classical case $X = \mathbb{N}$ and relax the finiteness requirement by allowing unique sums $\pm\infty$. We call such series *abscon series*. The last Section 3.4 is devoted to *classical conditionally convergent series* $\sum_{n\in\mathbb{N}} a_n$ that may have non-unique finite sums.

3.1 Absolutely convergent set series

In this section, we consider series of a general form such that the index set may be any at most countable set.

• *Absolutely convergent set series.* We begin by defining set series.

Definition 3.1.1 *A* <u>*set series*</u> *is any map $h\colon X \to \mathbb{R}$ such that the domain X is at most countable. We use notation $\sum_{x\in X} h(x)$.*

Next we define absolutely convergent set series.

Definition 3.1.2 *A set series $\sum_{x\in X} h(x)$* <u>*absolutely converges*</u> *if two equivalent conditions hold.*

1. *There exists a constant $c > 0$ such that for every finite set $Y \subset X$ we have $\sum_{x\in Y} |h(x)| \le c$.*

2. *In the case that X is countable, for every bijection $f\colon \mathbb{N} \to X$ there exists the finite limit*
$$\lim_{n\to\infty} \textstyle\sum_{j=1}^{n} h(f(j)) \quad (\in \mathbb{R})\,.$$

Exercise 3.1.3 *Every finite set series absolutely converges.*

Exercise 3.1.4 *If $\sum_{x\in X} h(x)$ absolutely converges, then so does the set series $\sum_{x\in X} |h(x)|$.*

Let $S = \sum_{x\in X} h(x)$ be a set series and $Y \subset X$. We call $\sum_{x\in Y} h(x)$ the subseries of S.

Exercise 3.1.5 *Every subseries of an absolutely convergent series absolutely converges.*

Theorem 3.1.6 *The two conditions in Definition 3.1.2 are equivalent.*

Proof. Let $\sum_{x\in X} h(x)$ be a set series. The implication $1 \Rightarrow 2$. Let condition 1 hold, X be countable, and $f\colon \mathbb{N} \to X$ be a bijection. For $n \in \mathbb{N}$, let
$$t_n = \textstyle\sum_{j=1}^{n} |h(f(j))| \text{ and } s_n = \sum_{j=1}^{n} h(f(j))\,.$$
The sequence (t_n) is convergent because it weakly increases and, by the assumption, is bounded from above by c (Theorem 2.1.32). Thus (t_n) is Cauchy (Theorem 2.4.18), and for any given ε, there is n_0 such that for every $m \ge n \ge n_0$ we have
$$\textstyle\sum_{j=n+1}^{m} |h(f(j))| \le \varepsilon\,.$$
Using TI, we obtain for the same m and n the bound
$$\big|\textstyle\sum_{j=n+1}^{m} h(f(j))\big| \le \sum_{j=n+1}^{m} |h(f(j))| \le \varepsilon\,.$$
The sequence (s_n) is Cauchy and therefore convergent (Theorem 2.4.18).

The implication $\neg 1 \Rightarrow \neg 2$. We assume that X is countable and that condition 1 does not hold. We find a bijection $f\colon \mathbb{N} \to X$ such that
$$\lim_{n\to\infty} \textstyle\sum_{j=1}^{n} h(f(j)) = \pm\infty\,.$$
Let $f_0\colon \mathbb{N} \to X$ be any bijection. It follows from the assumption that
$$\lim \textstyle\sum_{j=1}^{n} |h(f_0(n))| = +\infty\,.$$
Thus there is an injection $f_1\colon \mathbb{N} \to X$ such that $\lim \sum_{j=1}^{n} h(f_1(j)) = \pm\infty$; the image $f_1[\mathbb{N}]$ is the subset of X on which h has the same sign $+$ or $-$. Let $Y = X \setminus f_1[\mathbb{N}]$. It is clear that by inserting the elements of Y, one by one, sufficiently sparsely into the sequence $(f_1(n))$ we can transform f_1 into a bijection $f\colon \mathbb{N} \to X$ such that
$$\lim_{n\to\infty} \textstyle\sum_{j=1}^{n} h(f(j)) = \lim_{n\to\infty} \sum_{j=1}^{n} h(f_1(j)) = \pm\infty\,.$$
□

- *Sums.* We define sums of absolutely convergent set series.

Proposition 3.1.7 *Let $\sum_{x\in X} h(x)$ be absolutely convergent and X be countable. Then there is a number $s\in\mathbb{R}$ such that for every bijection $f\colon \mathbb{N}\to X$,*

$$\lim_{n\to\infty} \textstyle\sum_{j=1}^{n} h(f(j)) = s\,.$$

Proof. For the contrary, let $f, g\colon \mathbb{N}\to X$ be bijections and $s, t\in\mathbb{R}$ be numbers such that

$$s = \lim_{n\to\infty} \textstyle\sum_{j=1}^{n} h(f(j)) \neq \lim_{n\to\infty} \sum_{j=1}^{n} h(g(j)) = t\,.$$

We obtain a bijection $e\colon \mathbb{N}\to X$ such that the limit

$$\lim_{n\to\infty} \textstyle\sum_{j=1}^{n} h(e(j))$$

does not exist, which contradicts the assumption of absolute convergence. Let $\varepsilon = \frac{1}{3}|s-t|$. We define a bijection $e\colon \mathbb{N}\to X$ and a sequence of integers $0 < n_1 < n_2 < \dots$ such that for every $i\in\mathbb{N}$,

$$\textstyle\sum_{j=1}^{n_{2i-1}} h(e(j)) \in U(s,\,\varepsilon) \ \text{ and } \ \sum_{j=1}^{n_{2i}} h(e(j)) \in U(t,\,\varepsilon)\,.$$

Then e is as desired.

We take $n_1\in\mathbb{N}$ such that

$$\textstyle\sum_{j=1}^{n_1} h(f(j)) \in U(s,\,\varepsilon)$$

and set $e(m) = f(m)$ for $m\le n_1$. We take $n_2 > n_1$ such that

$$f[\,[n_1]\,] \subset g[\,[n_2]\,] \wedge \textstyle\sum_{j=1}^{n_2} h(g(j)) \in U(t,\,\varepsilon)\,.$$

For $m\in[n_2]\backslash[n_1]$, the values $e(m)$ run, in some order, through $g[\,[n_2]\,]\backslash f[\,[n_1]\,]$. We take $n_3 > n_2$ such that

$$g[\,[n_2]\,] \subset f[\,[n_3]\,] \wedge \textstyle\sum_{j=1}^{n_3} h(f(j)) \in U(s,\varepsilon)\,.$$

For $m\in[n_3]\backslash[n_2]$, the values $u(m)$ run, in some order, through $f[\,[n_3]\,]\backslash g[\,[n_2]\,]$. Continuing in this way indefinitely, we obtain the bijection e. □

Exercise 3.1.8 *Let the maps f and g be as in the previous proof. We define a surjection $e\colon \mathbb{N}\to X$ by interleaving them as*

$$(e(n)) = (f(1),\, g(1),\, f(2),\, g(2),\, \dots)$$

(e is not injective). Does $\lim \sum_{j=1}^{n} h(e(j))$ exist?

Definition 3.1.9 *Let $S = \sum_{x\in X} h(x)$ be absolutely convergent and X be countable. The sum of S is the unique limit*

$$\lim_{n\to\infty} \textstyle\sum_{j=1}^{n} h(f(j))\,,$$

for any bijection $f\colon \mathbb{N}\to X$. We denote the sum of S again by $\sum_{x\in X} h(x)$, and by $h(X)$.

The definition is correct due to Pro;position 3.1.7. If $X = \{x_1, x_2, \dots, x_n\}$ ($x_i \ne x_j$ for $i \ne j$) is finite and nonempty, we define the sum as the usual finite sum

$$h(X) = \sum_{x \in X} h(x) := h(x_1) + h(x_2) + \cdots + h(x_n)\,.$$

We set $h(\emptyset) = \sum_{x \in \emptyset} h(x) := 0$. The adjective "absolute" in Definition 3.1.2 refers not so much to absolute values but to the independence of sums on the order of summation.

Exercise 3.1.10 *Find an absolutely convergent set series $S = \sum_{x \in X} h(x)$ and a subseries $\sum_{x \in Y} h(x)$ of S such that $h(X) = 0$ and $h(Y) = 1000$.*

• *Approximating sums.* We show that sums of absolutely convergent set series can be arbitrarily tightly approximated by finite sums.

Proposition 3.1.11 *Let $S = \sum_{x \in X} h(x)$ be absolutely convergent. Then for every $\varepsilon > 0$ there exists a finite set $Y \subset X$, denoted by $Y(\varepsilon, S)$, such that for every finite set Z with $Y \subset Z \subset X$, we have*

$$|\sum_{x \in Z} h(x) - h(X)| \le \varepsilon\,.$$

Proof. Let an $\varepsilon > 0$ be given. For finite X we take the set Y according to Exercise 3.1.12. Let X be countable and $f\colon \mathbb{N} \to X$ be any bijection. Let $n_0 \in \mathbb{N}$ be so large that

$$\sum_{n > n_0} |h(f(n))| \le \tfrac{\varepsilon}{2} \ \text{ and } \ \left|\sum_{n=1}^{n_0} h(f(n)) - h(X)\right| \le \tfrac{\varepsilon}{2}\,.$$

Let $Y = f[\,[n_0]\,]$. Then, for any finite set Z with $Y \subset Z \subset X$, we have

$$\begin{aligned} \left|\sum_{x \in Z} h(x) - h(X)\right| &\le \left|\sum_{x \in Y} h(x) - h(X)\right| + \left|\sum_{x \in Z \setminus Y} h(x)\right| \\ &\le \left|\sum_{x \in Y} h(x) - h(X)\right| + \sum_{x \in Z \setminus Y} |h(x)| \\ &\le \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} = \varepsilon\,. \end{aligned}$$

The first two inequalities follow from TI. The third inequality follows from the definition of Y. The last fourth equality is trivial. □

Exercise 3.1.12 *How do we define $Y = Y(\varepsilon, S)$ for finite X?*

• *Linear combinations of set series.*
• *The grouping construction.* Recall the definition of partition in Section 1.1. Let $S = \sum_{x \in X} h(x)$ be a set series and Y be a partition of X. If every subseries $\sum_{x \in Z} h(x)$ for $Z \in Y$ absolutely converges, we call the series $S_Y = \sum_{Z \in Y} h(Z)$ the grouping of S.

Theorem 3.1.13 (sums of groupings) *Let $S = \sum_{x \in X} h(x)$ be an absolutely convergent set series and Y be a partition of X. Then the grouping series S_Y absolutely converges and has sum $h(X)$.*

Proof. Note that S_Y is correctly defined because $\sum_{x\in Z} h(x)$ for $Z \in Y$ is a subseries of an absolutely convergent series S. We first show that S_Y absolutely converges. Let

$$c \equiv \sup(\{\textstyle\sum_{x\in Z} |r(x)|\colon\ Z \subset X \text{ and is finite}\}) \quad (\in [0, +\infty))$$

and let $Y' = \{Z_1, \dots, Z_n\} \subset Y$ be a finite set. For every $i \in [n]$ we use Proposition 3.1.11 and take a finite set $Z_i' \equiv Y(2^{-i}, R_{Z_i})$ $(\subset Z_i)$. We set $Z_0 \equiv Z_1' \cup \dots \cup Z_n'$ $(\subset X)$; this is a disjoint union. Then $\sum_{Z\in Y'} |S(R_Z)|$ is

$$\textstyle\sum_{i=1}^n |S(R_{Z_i}) - S(R_{Z_i'}) + S(R_{Z_i'})| \le \sum_{i=1}^n 2^{-i} + \sum_{x\in Z_0} |r(x)| \le 1 + c\,.$$

Hence $\sum_{Z\in Y} S(R_Z) \in \mathfrak{S}$.

Let an ε be given. We show that $|S(R) - S(R')| \le \varepsilon$. We use Proposition 3.1.11 and take finite sets $X' \equiv Y(\frac{\varepsilon}{3}, R)$ $(\subset X)$ and $Y' \equiv Y(\frac{\varepsilon}{3}, R')$ $(\subset Y)$. Then we take a finite superset of blocks $\{Z_1, \dots, Z_n\}$ of Y', that is $Y' \subset \{Z_1, \dots, Z_n\} \subset Y$ and $n \in \mathbb{N}$, such that $X' \subset \bigcup_{i=1}^n Z_i$. For every $i \in [n]$ we use Proposition 3.1.11 and take a finite set $Z_i' \equiv Y(2^{-i}\frac{\varepsilon}{3}, R_{Z_i})$ $(\subset Z_i)$. Finally, for every $i \in [n]$ we set $Z_i'' \equiv Z_i' \cup (X' \cap Z_i)$ $(\subset Z_i)$ and take the disjoint union $X_0 \equiv \bigcup_{i=1}^n Z_i''$ $(\subset X)$. Then X_0 is finite, $X' \subset X_0$, $Z_i' \subset Z_i''$ and

$$\begin{aligned} |S(R) - S(R')| \;&\le\; |S(R) - \textstyle\sum_{x\in X_0} r(x)| + \sum_{i=1}^n |\sum_{x\in Z_i''} r(x) - S(R_{Z_i})| + \\ &+\; |\textstyle\sum_{i=1}^n S(R_{Z_i}) - S(R')| \le \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon\,. \end{aligned}$$

Hence $S(R) = S(R')$. □

Exercise 3.1.14 *Explain the last three bounds* $\dots \le \frac{\varepsilon}{3}$.

• *Congruence of* AK *series.* $R = \sum_{x\in X} r(x)$ and $R' = \sum_{x\in Y} s(x)$ in $\mathfrak{S}$ are congruent, in symbols $R \sim R'$, if there is a bijection $f\colon X \to Y$ such that for every $x \in X$ we have $r(x) = s(f(x))$.

Exercise 3.1.15 *Show that* $\sim$ *is an equivalence relation on* $\mathfrak{S}$. *(Equivalence relation on a class is defined in the same way as on a set.)*

Exercise 3.1.16 *If* $R, R' \in \mathfrak{S}$ *are congruent then* $S(R) = S(R')$ $(\in \mathbb{R})$.

• *Binary sums and products of* AK *series.* We introduce two binary operations on $\mathfrak{S}$.

Theorem 3.1.17 (binary sums on $\mathfrak{S}$) *Suppose that* $R = \sum_{x\in X} r(x)$ *and* $R' = \sum_{y\in Y} s(y)$ *are in* $\mathfrak{S}$. *We set* $Z \equiv X \times \{0\} \cup Y \times \{1\}$ *and define*

$$R + R' = \textstyle\sum_{z\in Z} t(z)$$

by setting $t(z) \equiv r(x)$ *if* $z = (x, 0)$, *and* $t(z) \equiv s(y)$ *if* $z = (y, 1)$. *Then* $R + R' \in \mathfrak{S}$ *and* $S(R + R') = S(R) + S(R')$. *We call* $R + R'$ *the binary sum of* R *and* R'.

Proof. First we show that $R + R' \in \mathfrak{S}$. Let c be a constant witnessing that both $R \in \mathfrak{S}$ and $R' \in \mathfrak{S}$, and let $W \subset Z$ be a finite set. Then $W = (X' \times \{0\}) \cup (Y' \times \{1\})$, where $X' \subset X$ and $Y' \subset Y$ are finite sets, and

$$\textstyle\sum_{z \in W} |t(z)| = \sum_{x \in X'} |r(x)| + \sum_{y \in Y'} |s(y)| \le c + c = 2c\,.$$

Hence $R + R' \in \mathfrak{S}$.

We prove that $S(R + R') = S(R) + S(R')$. Let $r \equiv S(R)$, $s \equiv S(R')$ and $t \equiv S(R + R')$, and let an ε be given. We show that $|t - (r + s)| \le \varepsilon$. We use Proposition 3.1.11 and take finite sets $X' \equiv Y(\frac{\varepsilon}{3}, R)$ $(\subset X)$, $Y' \equiv Y(\frac{\varepsilon}{3}, S)$ $(\subset Y)$ and $Z' \equiv Y(\frac{\varepsilon}{3}, R + S)$ $(\subset X \times \{0\} \cup Y \times \{1\})$. We take finite sets X'' and Y'' such that $X' \subset X'' \subset X$, $Y' \subset Y'' \subset Y$ and $Z' \subset X'' \times \{0\} \cup Y'' \times \{1\} \equiv W$. Then $|t - (r + s)|$ is at most

$$\textstyle|t - \sum_{z \in W} t(z)| + |\sum_{x \in X''} r(x) - r| + |\sum_{y \in Y''} s(y) - s| \le \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon\,.$$

Hence $t = r + s$. □

Exercise 3.1.18 *Explain the last tree bounds* $\cdots \le \frac{\varepsilon}{3}$.

Exercise 3.1.19 *If* $Q \sim Q'$ *and* $R \sim R'$ *then* $Q + R \sim Q' + R'$.

Theorem 3.1.20 (products on $\mathfrak{S}$) *Suppose that* $R = \sum_{x \in X} r(x)$ *and* $R' = \sum_{y \in Y} s(y)$ *are in* $\mathfrak{S}$. *We define*

$$\textstyle R \cdot R' \equiv \sum_{(x,\,y) \in X \times Y} r(x)s(y)\,.$$

Then $R \cdot R' \in \mathfrak{S}$ *and* $S(R \cdot R') = S(R)S(R')$. *We call* $R \cdot R'$ *the product of* R *and* R'.

Proof. First we show that $R \cdot R' \in \mathfrak{S}$. We take a constant c witnessing that R and R' are AK series. Let $Z \subset X \times Y$ be a finite set. We take finite sets $X' \subset X$ and $Y' \subset Y$ such that $Z \subset X' \times Y'$. Then

$$\textstyle\sum_{(x,\,y) \in Z} |r(x)s(y)| \le \sum_{x \in X'} |r(x)| \cdot \sum_{y \in Y'} |s(y)| \le c \cdot c = c^2\,.$$

Hence $R \cdot R' \in \mathfrak{S}$.

We prove that $S(R \cdot R') = S(R)S(R')$. Let $r \equiv S(R)$, $s \equiv S(R')$ and $t \equiv S(R \cdot R')$, and let an $\varepsilon \le 1$ be given. We show that $|t - rs| \le \varepsilon$. We use Proposition 3.1.11 and take finite sets $X' \equiv Y(\frac{\varepsilon}{3(|s|+1)}, R)$ $(\subset X)$, $Y' \equiv Y(\frac{\varepsilon}{3(|r|+1)}, S)$ $(\subset Y)$ and $Z \equiv Y(\frac{\varepsilon}{3}, R \cdot R')$ $(\subset X \times Y)$. We take finite sets X'' and Y'' such that $X' \subset X'' \subset X$, $Y' \subset Y'' \subset Y$ and $Z \subset X'' \times Y''$. Then $|t - rs|$ is at most

$$\begin{aligned}&\textstyle|t - \sum_{(x,\,y) \in X'' \times Y''} r(x)s(y)| + |\sum_{x \in X''} r(x) \cdot \sum_{y \in Y''} s(y) - rs| \le \\ &\le \tfrac{\varepsilon}{3} + |(r + \delta)(s + \theta) - rs| \text{ where } |\delta| \le \tfrac{\varepsilon}{3(|s|+1)} \text{ and } |\theta| \le \tfrac{\varepsilon}{3(|r|+1)}\,.\end{aligned}$$

Hence $|t - rs| \le \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon$. This holds for every ε and $t = rs$. □

Exercise 3.1.21 *Explain the last tree bounds* $\cdots \le \frac{\varepsilon}{3}$.

Exercise 3.1.22 *If* $Q \sim Q'$ *and* $R \sim R'$ *then* $Q \cdot R \sim Q' \cdot R'$.

• *A semiring of factorized* AK *series.* We show that AK series, when factorized by the congruence $\sim$, form a semiring with respect to binary sums and to products. We set $\mathfrak{T} \equiv \mathfrak{S}/\!\sim$ and call this class factorized AK series. A semiring is the ring structure on a set or a class, with the existence of additive inverses dropped. Let $0_{\mathfrak{T}} \equiv \emptyset$ be the empty AK series and $1_{\mathfrak{T}} \equiv \left[\sum_{x \in \{1\}} 1\right]_{\sim}$ be the AK series with just a single summand 1.

Theorem 3.1.23 (semiring $\mathfrak{T}_{\mathrm{SR}}$) *The structure*

$$\mathfrak{T}_{\mathrm{SR}} \equiv \langle \mathfrak{T},\, 0_{\mathfrak{T}},\, 1_{\mathfrak{T}},\, +,\, \cdot \rangle$$

is a semiring. In more detail, $+$ *and* $\cdot$ *are commutative and associative operations on* $\mathfrak{T}$*, the elements* $0_{\mathfrak{T}}$ *and* $1_{\mathfrak{T}}$ *of* $\mathfrak{T}$ *are neutral to* $+$ *and* $\cdot$*, respectively, and* $\cdot$ *is distributive to* $+$.

Proof. Exercises 3.1.19 and 3.1.22 show that $+$ and $\cdot$ operate on the class $\mathfrak{T}$. First we show that $+$ is commutative. Let $R = \sum_{x \in X} r(x)$ and $R' = \sum_{x \in Y} s(x)$ be in $\mathfrak{S}$, and let $Z \equiv X \times \{0\} \cup Y \times \{1\}$ and $W \equiv X \times \{1\} \cup Y \times \{0\}$. We take the bijection $f\colon Z \to W$ that sends $(x, 0) \in Z$ to $(x, 1) \in W$, and $(y, 1) \in Z$ to $(y, 0) \in W$. Then we see that for $R + R' = \sum_{z \in Z} t(z)$ and $R' + R = \sum_{z \in W} t(z')$ it holds for every $z \in Z$ that $t(z) = t'(f(z))$ because $t(z) = r(x) = t'(f(z))$ if $z = (x, 0)$, and $t(z) = s(y) = t'(f(z))$ if $z = (y, 1)$. Hence $R + R' \sim R' + R$. The similar proof of the associativity of $+$ is relegated to Exercise 3.1.24. We leave the proofs of commutativity and associativity of $\cdot$ to respective Exercises 3.1.25 and 3.1.26.

We prove the neutrality of $0_{\mathfrak{T}}$ to $+$, and $1_{\mathfrak{T}}$ to $\cdot$. Let $R = \sum_{x \in X} r(x)$ be in $\mathfrak{S}$. Since $X \times \{0\} \cup \emptyset \times \{1\} = X \times \{0\}$, the bijection sending $x \in X$ to $(x, 0) \in X \times \{0\}$ proves that $R \sim R + 0_{\mathfrak{T}}$. Similarly, the bijection sending $x \in X$ to $(x, 1) \in X \times \{1\}$ proves that $R \sim R \cdot 1_{\mathfrak{T}}$.

Finally, we show that $\cdot$ is distributive to $+$. Let $R = \sum_{x \in X} r(x)$, $R' = \sum_{x \in Y} s(x)$ and $R'' = \sum_{x \in Z} t(x)$ be in $\mathfrak{S}$, and let

$$W \equiv X \times (Y \times \{0\} \cup Z \times \{1\}) \ \text{ and } \ W' \equiv (X \times Y) \times \{0\} \cup (X \times Z) \times \{1\}\,.$$

We take the bijection $f\colon W \to W'$ sending $(x, (y, 0)) \in W$ to $((x, y), 0) \in W'$, and $(x, (z, 1)) \in W$ to $((x, z), 1) \in W'$. Let

$$\textstyle\sum_{w \in W} u(w) \equiv R \cdot (R' + R'') \ \text{ and } \ \sum_{w \in W'} u'(w) \equiv (R \cdot R') + (R \cdot R'')\,.$$

Then for every $w \in W$ we have $u(w) = u'(f(w))$ because $u(w) = r(x)s(y) = u'(f(w))$ if $w = (x, (y, 0))$, and $u(w) = r(x)t(z) = u'(f(w))$ if $w = (x, (z, 1))$. Hence $R \cdot (R' + R'') \sim (R \cdot R') + (R \cdot R'')$. $\square$

Exercise 3.1.24 *Give the bijection proving that* $R + (R' + R'') \sim (R + R') + R''$.

Exercise 3.1.25 *Give the bijection proving that $R \cdot R' \sim R' \cdot R$.*

Exercise 3.1.26 *Give the bijection proving that $R \cdot (R' \cdot R'') \sim (R \cdot R') \cdot R''$.*

3.2 Generalizing Pólya's theorem

3.3 Classical abscon series

• *Some definitions.* A classical series is a sequence $(a_n) \subset \mathbb{R}$, denoted by

$$\textstyle\sum_{n=1}^{\infty} a_n,\ \sum_{n \ge 1} a_n,\ \sum a_n \ \text{ or } \ a_1 + a_2 + \dots\,.$$

Terms a_n of the sequence are called summands of the series. For $n \in \mathbb{N}$, the n-th partial sum of the series is $s_n := \sum_{j=1}^{n} a_j$. If $\lim s_n \in \mathbb{R}$, we say that the series $\sum a_n$ converges. If $\sum a_n$ is a series and $f\colon \mathbb{N} \to \mathbb{N}$ is a bijection, we call the series $\sum a_{f(n)}$ the reordering of $\sum a_n$. A subseries of $\sum a_n$ is any series $\sum b_n$ such that (b_n) is a subsequence of (a_n).

Definition 3.3.1 (abscon series) *A series $\sum a_n$ is an abscon series if for every bijection $f\colon \mathbb{N} \to \mathbb{N}$ the limit*

$$L = \lim_{n \to \infty} \textstyle\sum_{j=1}^{n} a_{f(j)} \quad (\in \mathbb{R}^*)$$

exists and does not depend on f.

In other words, a series is abscon iff in every reordering of it the sequence of partial sums has the same limit. The limit L is then the sum of the abscon series $\sum a_n$. We denote sums by the same symbols as series. This is standard ambiguous classical notation; we will make an effort to always indicate by the words "series" and "sum" which meaning is intended. Abscon series are more general than the classical absolutely convergent series because infinite sums are allowed. For the same reason, abscon series are a strictly larger family of series than the absolutely convergent set series $\sum_{x \in \mathbb{N}} h(x)$.

Exercise 3.3.2 *If a set series $\sum_{n \in \mathbb{N}} a_n$ is absolutely convergent, then $\sum a_n$ is an abscon series.*

Exercise 3.3.3 *Let $\sum a_n$ be a series with $a_n \in \{0, 1\}$. Is it an abscon series?*

• *An equivalent definition of abscon series.* We first obtain an analog of condition 1 of Definition 3.1.2. For a series $\sum a_n$ we set $A = \{n \in \mathbb{N}\colon\ a_n > 0\}$, $B = \{n \in \mathbb{N}\colon\ a_n < 0\}$, and define the set series

$$\textstyle\sum^{+} a_n := \sum_{n \in A} a_n \ \text{ and } \ \sum^{-} a_n := \sum_{n \in B} a_n\,.$$

Proposition 3.3.4 (equivalent definition) *A series $\sum a_n$ is abscon $\iff$ at least one of the set series $\sum^{+} a_n$ and $\sum^{-} a_n$ absolutely converges.*

Proof. Let $\sum a_n$ be a series. Implication $\Leftarrow$. Suppose that both $\sum^+ a_n$ and $\sum^- a_n$ absolutely converge, and that $c \ge 0$ is the constant in condition 1 of Definition 3.1.2 that works for both $\sum^+ a_n$ and $\sum^- a_n$. Then for any finite set $X \subset \mathbb{N}$,

$$\textstyle\sum_{n\in X} |a_n| = \sum_{n\in X\cap A} |a_n| + \sum_{n\in X\cap B} |a_n| \le c + c = 2c\,.$$

We see that $\sum_{n\in\mathbb{N}} a_n$ absolutely converges and therefore $\sum a_n$ is abscon series by Exercise 3.3.2.

Suppose that $\sum^+ a_n$ absolutely converges, with the corresponding constant $c \ge 0$, but that $\sum^- a_n$ does not; the other case is similar. Let $f\colon \mathbb{N} \to \mathbb{N}$ be any bijection and let a $d < 0$ be given. We take sufficiently large n_0 such that

$$\textstyle\sum_{n\in f[\,[n_0]\,]\cap B} a_n \le d - c\,.$$

Then for every $n \ge n_0$ we have

$$\textstyle\sum_{j=1}^{n} a_{f(j)} = \sum_{n\in f[\,[n]\,]\cap A} a_n + \sum_{n\in f[\,[n]\,]\cap B} a_n \le c + (d-c) = d\,.$$

We see that $\lim \sum_{j=1}^{n} a_{f(j)} = -\infty$.

Implication $\neg \Leftarrow \neg$. We suppose that neither $\sum^+ a_n$ nor $\sum^- a_n$ absolutely converges. Then we are done by Exercise 3.3.5, $\square$

Exercise 3.3.5 *If neither $\sum^+ a_n$ nor $\sum^- a_n$ absolutely converges, then there exist bijections $f, g\colon \mathbb{N} \to \mathbb{N}$ such that*

$$\textstyle\lim \sum_{j=1}^{n} a_{f(j)} = +\infty \ \text{ and } \ \lim \sum_{j=1}^{n} a_{g(j)} = -\infty\,.$$

- *Operations with abscon series.*
- *Abscon series with nonnegative summands.*
- *Abscon series with positive and negative summands.*

3.4 Classical conditionally convergent series

- *The Riemann theorem.* We show that if a series has a non-unique finite sum, then it can be reordered to have any sum in $\mathbb{R}^*$.

Definition 3.4.1 (CC series) *Let $\sum a_n$ be a classical series. We say that it is conditionally convergent, abbreviated* CC, *if it is convergent but is not an abscon series.*

One of the simplest examples of a CC series is in the next exercise.

Exercise 3.4.2 *The series*

$$1 - 1 + \tfrac{1}{2} - \tfrac{1}{2} + \tfrac{1}{3} - \tfrac{1}{3} + \dots$$

is CC.

Theorem 3.4.3 (Riemann) *Let $\sum a_n$ be a series. The next three claims on $\sum a_n$ are equivalent.*

1. *$\sum a_n$ is* CC *series.*
2. *The limit $\lim a_n = 0$, the sum $\sum^+ a_n = +\infty$ and the sum $\sum^- a_n = -\infty$.*
3. *For every element $L \in \mathbb{R}^*$ there is a bijection $f\colon \mathbb{N} \to \mathbb{N}$ such that*

$$\lim_{n\to\infty} \textstyle\sum_{j=1}^n a_{f(j)} = L\,.$$

Proof.

$\square$

Theorem 3.5.30, which we attributed to Riemann, completely characterizes the sequences $(a_n) \subset \mathbb{R}$ with the property that for every $A \in \mathbb{R}^*$ there is a bijection $\pi\colon \mathbb{N} \to \mathbb{N}$ such that the sum

$$\textstyle\sum_{n=1}^\infty a_{\pi(n)} = A\,.$$

In this extending section we obtain an analogous characterization for infinite products, that is, we replace addition with multiplication.

• *Riemannian infinite products.* Recall Definition 3.5.17 of infinite products. For any sequence $(a_n) \subset \mathbb{R}$ we denote by (a_{z_n}) the subsequence of terms a_n with $a_n \in (0, 1)$, and by (a_{k_n}) the subsequence of terms a_n with $a_n \geq 1$.

Definition 3.4.4 (Riemannian infinite products) *An infinite product*

$$\prod_{n=1}^\infty a_n$$

is <u>*Riemannian*</u> *if the following conditions hold.*

1. *We have $\lim a_n = 1$, $a_n \neq 0$ for every $n \in \mathbb{N}$ and $a_n < 0$ for only finitely many $n \in \mathbb{N}$.*
2. *$\sum \log(a_{z_n}) = -\infty$.*
3. *$\sum \log(a_{k_n}) = +\infty$.*

Exercise 3.4.5 *Logarithms of positive terms in a Riemannian infinite product form a Riemannian series.*

• *Riemann's theorem for infinite products.* We have devised the following analog of Theorem 3.5.30.

Theorem 3.4.6 (getting any product) *Let $\prod_{n=1}^\infty a_n$ be an infinite product. The following properties are equivalent.*

1. *$\prod_{n=1}^\infty a_n$ is Riemannian.*

2. *For every nonnegative element $A \in \mathbb{R}^*$ there exists a bijection $\pi\colon \mathbb{N} \to \mathbb{N}$ such that*
$$\prod_{n=1}^\infty a_{\pi(n)} = A\,,$$
or the same holds for nonpositive elements of $\mathbb{R}^*$.

There do not exist two bijections $\pi, \rho\colon \mathbb{N} \to \mathbb{N}$ such that
$$\prod_{n=1}^\infty a_{\pi(n)} = \lim_{n\to\infty} \prod_{j=1}^n a_{\pi(j)} < 0 < \lim_{n\to\infty} \prod_{j=1}^n a_{\rho(j)} = \prod_{n=1}^\infty a_{\rho(n)}\,.$$

Proof. Implication $1 \Rightarrow 2$. Let $\prod_{n=1}^\infty a_n$ be a Riemannian infinite product. We may assume that the negative terms in it are $a_1, a_2, \ldots, a_k$, with $k \in \mathbb{N}_0$. For $n \in \mathbb{N}$ we set
$$b_n \equiv \log(a_{n+k}) \quad (a_{n+k} > 0)\,.$$
Let
$$c \equiv \textstyle\prod_{j=1}^k a_j \quad (\in \mathbb{R} \setminus \{0\})\,,$$
where $c \equiv 1$ if $k = 0$. Let an $A \in \mathbb{R}^*$ be given, with $A \le 0$ if $c < 0$ and $A \ge 0$ if $c > 0$. We set
$$B \equiv \log(|A|) - \log|c|\,,$$
where $\log(|\pm\infty|) \equiv +\infty$ and $\log 0 \equiv -\infty$. By Exercise 3.4.5 and Theorem 3.5.30, there is a bijection $\rho\colon \mathbb{N} \to \mathbb{N}$ such that we have the sum
$$\textstyle\sum_{n=1}^\infty b_{\rho(n)} = B\,.$$
We define the bijection $\pi\colon \mathbb{N} \to \mathbb{N}$ by
$$\pi(1) = 1,\ \pi(2) = 2,\ \ldots,\ \pi(k) = k,\ \pi(k+1) = k+\rho(1),\ \pi(k+2) = k+\rho(2),\ \ldots\,.$$
Using the continuity of the exponential function and its limits
$$\lim_{x\to-\infty} \exp x = 0 \ \text{ and } \ \lim_{x\to+\infty} \exp x = +\infty$$
we get that $\prod_{n=1}^\infty a_{\pi(n)}$ equals
$$c \lim\nolimits_{n\to\infty} \exp\big(\textstyle\sum_{j=1}^n b_{\rho(j)}\big) = c\exp(\log|A| - \log|c|) = \frac{c}{|c|}\cdot|A| = A\,,$$
as required. But see Exercise 3.4.7.

Implication $\neg 1 \Rightarrow \neg 2$. Let $\prod_{n=1}^\infty a_n$ be an infinite product that is not Riemannian. If $a_n = 0$ for some $n \in \mathbb{N}$ then $\prod_{n=1}^\infty a_{\pi(n)} = 0$ for every permutation π of $\mathbb{N}$. If $a_n \ne 0$ for every n but $a_n < 0$ for infinitely many n, then for every permutation π of $\mathbb{N}$ the permuted partial products
$$\textstyle\prod_{j=1}^n a_{\pi(j)}$$

change sign infinitely often and cannot have any nonzero limit. If $a_n \neq 0$ for every n and $\lim a_n$ does not exist or is not 1, then this persists for any reordering of (a_n) and the proof of Proposition 3.5.19 shows that the permuted partial products cannot have any nonzero limit. Suppose that $a_n < 0$ for only finitely many n. If $\sum \log(a_{z_n}) > -\infty$ then no permutation π of $\mathbb{N}$ gives as infinite product 0, and if $\sum \log(a_{k_n}) < +\infty$ then no permutation π of $\mathbb{N}$ gives as infinite product $\pm\infty$.

We show that permutations π and ρ as stated do not exist. We may assume that $a_n \neq 0$ for every n. If $a_n < 0$ for infinitely many n then, as we know, by reordering we can get as an infinite product only 0. If $a_n < 0$ for finitely many n, let c be the product of the negative terms. Then for every reordering the permuted partial products have, eventually, the same sign as c. Thus we cannot get two infinite products with different signs. $\square$

Exercise 3.4.7 *Check that in the cases $A = 0$ and $A = \pm\infty$ the computation in the proof is really correct.*

Exercise 3.4.8 *Obtain an analog of Proposition 3.5.29 for infinite products.*

3.5 Classical infinite series

In Section 3.1 we introduced AK series. Now we give a standard introduction to infinite series.

• *Series in general.* We begin with basic notions of the theory of infinite series.

Definition 3.5.1 (infinite series) *An (infinite) series is a real sequence (a_n). We denote it by*

$$\textstyle\sum a_n \text{ or } \sum_{n=1}^{\infty} a_n \text{ or } a_1 + a_2 + \cdots .$$

The numbers a_n are the summands of the series. The sum of $\sum a_n$ is the limit

$$\lim_{n\to\infty}(a_1 + a_2 + \cdots + a_n) \quad (\in \mathbb{R}^*)\,,$$

if it exists. We denote the sum again by $\sum a_n$ or $\sum_{n=1}^{\infty} a_n$ or $a_1 + a_2 + \cdots$. Terms of the sequence

$$(s_n) \equiv (a_1 + \cdots + a_n)$$

are called partial sums of the series, so that the sum is $\lim s_n$.

A series with finite sum converges. If it has an infinite or no sum, it diverges. The notation for series generalizes the usual notation

$$\textstyle\sum_{j=1}^{n} a_j = a_1 + a_2 + \cdots + a_n$$

for finite sums.

Let $S \equiv \sum a_n$ be a series. For $m \in \mathbb{N}$, the series

$$\textstyle\sum_{n=m}^{\infty} a_n \text{ or } a_m + a_{m+1} + \ldots$$

is the standard series $\sum b_n$ with $b_n \equiv a_{m+n-1}$, $n \in \mathbb{N}$, and is called a <u>tail</u> of S. A <u>subseries</u> of S is any series $\sum b_n$ of the form $b_n \equiv a_{f(n)}$, $n \in \mathbb{N}$, where $f\colon \mathbb{N} \to B$ is the ordering of an infinite set $B \subset \mathbb{N}$. In other words, $\sum b_n$ is a subsequence of the sequence (series) S.

Every subsequence of a sequence with a limit has the same limit. For series the situation is different.

Proposition 3.5.2 (on subseries) *For any pair of elements $A, B \in \mathbb{R}^*$ there exist a series $\sum a_n$ and a subseries $\sum b_n$ of it such that the sums are*

$$\textstyle\sum a_n = A \text{ and } \sum b_n = B\,.$$

Proof. For $A, B \in \mathbb{R}$ we set $a_1 \equiv 2^{-1}(A - B)$, $a_2 \equiv 2^{-1}B$, $a_3 \equiv 2^{-2}(A - B)$, $a_4 \equiv 2^{-2}B$, $\ldots$ and $\sum b_n \equiv \sum a_{2n}$. For $A = \pm\infty$ and $B \in \mathbb{R}$ we set $a_1 \equiv \pm 1$, $a_2 \equiv 2^{-1}B$, $a_3 \equiv \pm 1$, $a_4 \equiv 2^{-2}B$, $\ldots$ (equal signs) and $\sum b_n \equiv \sum a_{2n}$. For positive $A \in \mathbb{R}$ and $B = -\infty$ we set $a_1 \equiv \frac{A}{1}$, $a_2 \equiv -\frac{A}{2}$, $a_3 \equiv \frac{A}{2}$, $a_4 \equiv -\frac{A}{3}$, $\ldots$ and $\sum b_n \equiv \sum a_{2n}$. The remaining cases when $A \in \mathbb{R}$ and $B = \pm\infty$ are left in the next exercise. For $A = -\infty$ and $B = +\infty$ we set $a_1 \equiv -2$, $a_2 \equiv 1$, $a_3 \equiv -2$, $a_4 \equiv 1$, $\ldots$ and $\sum b_n \equiv \sum a_{2n}$. The remaining cases with $A, B \in \{-\infty, +\infty\}$ are similar. $\square$

Exercise 3.5.3 *Resolve the remaining three cases with $A \in \mathbb{R}$ and $B = \pm\infty$, namely $A > 0$, $B = +\infty$ and $A \le 0$, $B = \pm\infty$.*

Exercise 3.5.4 *Convergence of a series is a robust property of sequences.*

However, the sum is sensitive to changes of summands.

Exercise 3.5.5 *In every convergent series any change of any single summand changes the sum.*

Exercise 3.5.6 *Find a convergent series that has a divergent subseries.*

Exercise 3.5.7 *A series converges iff its every tail converges. A series has a sum $\pm\infty$ iff its every tail has the same infinite sum.*

Exercise 3.5.8 *Suppose that the series $S \equiv a_1 + a_2 + \ldots$ is such that $a_n \ge 0$ for every $n \ge n_0$. Then S has a sum that is not $-\infty$. Similarly, every series with almost all summands non-positive has a sum that is not $+\infty$.*

Exercise 3.5.9 $\sum_{n=1}^{\infty} 1 = +\infty$.

Let $\sum a_n$ be a series. We assume that A is a set such that

$$A \subset \{n \in \mathbb{N}\colon\ a_n = 0\}$$

and that f is the ordering of $\mathbb{N} \setminus A$. If $\mathbb{N} \setminus A$ is finite and $m = |\mathbb{N} \setminus A|$, then the A-deletion of zeros from $\sum a_n$ is the finite sum $\sum_{i=1}^{m} a_{f(i)}$. If $\mathbb{N} \setminus A$ is infinite then it is the series $\sum b_n$ with $b_n = a_{f(n)}$. We show that this operation affect neither existence nor value of the sum. We begin with the finite case.

Proposition 3.5.10 (deletion of zeros 1) *Let $\sum a_n$ be a series and let the finite sum $\sum_{i=1}^{m} a_{f(i)}$ arise from $\sum a_n$ by A-deletion of zeros. Then we have the equality of sums*

$$\sum a_n = \sum_{i=1}^{m} a_{f(i)}\,.$$

Proof. The sequence (s_n) of partial sums of the series $\sum a_n$ is evenually constantly equal to $\sum_{i=1}^{m} a_{f(i)}$. □

Proposition 3.5.11 (deletion of zeros 2) *Let $\sum a_n$ be a series and let the series $\sum b_n$ arise from $\sum a_n$ by A-deletion of zeros. Then we have the equality of sums*

$$\sum a_n = \sum b_n\,,$$

whenever one sum exists.

Proof. Let $\sum a_n$ and $\sum b_n$ be as stated, f be the ordering of $\mathbb{N} \setminus A$ and (s_n) be partial sums of $\sum a_n$. Then for every n we have

$$\sum_{j=1}^{n} b_j = \sum_{j=1}^{n} a_{f(j)} = s_{f(n)}\,,$$

and $s_m = s_{f(n)}$ for every m such that $f(n) \le m < f(n+1)$. Thus, by Exercise 2.1.16,

$$\sum b_n = \lim_{n\to\infty} \sum_{j=1}^{n} b_j = \lim_{m\to\infty} s_m = \sum a_n\,,$$

whenever either limit exists. □

Let $\sum a_n$ be a series. Its reordering is any series $\sum b_n$ such that $b_n = a_{f(n)}$ for a bijection $f\colon \mathbb{N} \to \mathbb{N}$.

Proposition 3.5.12 (commutativity) *Let $\sum a_n$ be a series. If $\{n\colon\ a_n < 0\}$ or $\{n\colon\ a_n > 0\}$ is finite, then all reorderings of $\sum a_n$ have the same sum.*

Proof. Suppose that $I = \{n\colon\ a_n > 0\}$ is finite. Let $f, g\colon \mathbb{N} \to \mathbb{N}$ be any bijections and let

$$s_n \equiv \sum_{i=1}^{n} a_{f(i)} \ \text{ and }\ t_n \equiv \sum_{i=1}^{n} a_{g(i)}\,.$$

We take $m \in \mathbb{N}$ such that $f[\,[m]\,] \supset I$ and $g[\,[m]\,] \supset I$. Then both (s_n) and (t_n) weakly decrease starting from the index m, and by Theorem 2.4.4 both have a limit. It is not hard to see that for every $n_1 \geq m$ there exist two indices $n_2, n_3 \geq m$ such that $t_{n_2} \leq s_{n_1}$ and $s_{n_3} \leq t_{n_1}$. It follows that $\lim s_n = \lim t_n$. The other case when $\sum a_n$ has only finitely many negative summands is treated similarly. $\square$

The equality $\lim a_n = 0$ is the necessary convergence condition (NCC) of the series $\sum a_n$.

Proposition 3.5.13 (NCC) *If a series $\sum a_n$ converges then $\lim a_n = 0$.*

Proof. Suppose that $s \equiv \lim s_n = \lim(a_1 + \cdots + a_n)$ is in $\mathbb{R}$. Then $\lim a_n = \lim(s_n - s_{n-1}) = \lim s_n - \lim s_{n-1} = s - s = 0$. $\square$

Thus if $\lim a_n$ does not exist or is not 0 then $\sum a_n$ diverges. Exercise 3.5.9 shows that NCC does not generalize to infinite sums.

Exercise 3.5.14 *Explain the last four equalities in the previous proof.*

Let $\sum a_n$ and $\sum b_n$ be series and $c, d \in \mathbb{R}$. Their linear combination

$$c \sum a_n + d \sum b_n$$

is the series $\sum(ca_n + db_n)$.

Proposition 3.5.15 (lin. combinations of series) *Suppose that the series $\sum a_n$ and $\sum b_n$ have sums A and B, respectively. Then the linear combination*

$$c \sum a_n + d \sum b_n$$

has sum $cA + dB$, if this expression is not indefinite.

Proof. This follows at once from Definition 3.5.1 and Theorem 2.6.2. $\square$

Exercise 3.5.16 *L. Euler proved that $\sum n^{-2} = \frac{\pi^2}{6}$. Using a linear combination of series find the sum $\sum(-1)^{n+1}n^{-2}$.*

• *Infinite products.* We briefly mention the analog of sums of series for the operation of multiplication of real numbers.

Definition 3.5.17 (infinite products) *For any sequence $(a_n) \subset \mathbb{R}$ we define*

$$\prod_{n=1}^{\infty} a_n \equiv \lim_{n \to \infty} a_1 a_2 \dots a_n \quad (\in \mathbb{R}^*)\,,$$

if the limit of partial products exists, and call it the (infinite) product of the numbers a_n.

As for series, $\prod_{n=1}^{\infty} a_n$ also means the sequence (a_n). If $\prod_{n=1}^{\infty} a_n \in \mathbb{R}$, we say that the infinite product converges. We can and will consider more general infinite products of sequences $(z_n) \subset \mathbb{C}$, but in the complex domain we do not allow infinite products equal to $\pm\infty$.

Exercise 3.5.18 *What is* $\prod_{n=1}^{\infty}\left(1+\frac{1}{n}\right)$*?*

We have the following NCC for infinite products.

Proposition 3.5.19 (NCC for infinite products) *Let* $\prod_{n=1}^{\infty} a_n$ *be a convergent infinite product. Then*

$$a_n = 0 \textit{ for some } n \textit{ or } \liminf_{n\to\infty} |a_n| \le 1\,.$$

Proof. If $a_n = 0$ for some n then $\prod_{n=1}^{\infty} a_n = 0$. Suppose that

$$\lim_{n\to\infty} a_1 a_2 \ldots a_n = a \in \mathbb{R}$$

and that $a_n \ne 0$ for every n. The first case is that $a \ne 0$. Then

$$\lim a_n = \lim \tfrac{a_1 a_2 \ldots a_n}{a_1 a_2 \ldots a_{n-1}} = \tfrac{\lim a_1 a_2 \ldots a_n}{\lim a_1 a_2 \ldots a_{n-1}} = \tfrac{a}{a} = 1\,,$$

so that

$$\liminf |a_n| = \lim |a_n| = \lim a_n = 1 \le 1\,.$$

Let $a = 0$. Then it is clear that we cannot have $|a_n| \le 1$ for only finitely many n. Thus $\liminf |a_n| \le 1$. $\Box$

We will not hide from the reader that the standard terminology of infinite products regards the zero products

$$\prod_{n=1}^{\infty} a_n = 0$$

as divergent.

Exercise 3.5.20 *State for infinite products the standard* NCC.

It could be an interesting project to work out analogues of results about series for infinite products. Here we realize it only for Riemann's Theorem 3.5.30 in Section.

- *The harmonic series is the series* $\sum \frac{1}{n}$. We show that it has the sum $+\infty$.

Exercise 3.5.21 *If a sequence* (a_n) *weakly increases and has a subsequence with the limit* $+\infty$*, then* $\lim a_n = +\infty$.

Exercise 3.5.22 *If series* $\sum a_n$ *and* $\sum b_n$ *satisfy* $a_n \ge b_n$ *for every* $n \ge n_0$ *and* $\sum b_n = +\infty$*, then* $\sum a_n = +\infty$.

Proposition 3.5.23 ($\sum\frac{1}{n}=+\infty$) *The harmonic series sums to* $+\infty$*.*

Proof. We consider the series

$$\sum b_n \equiv \tfrac{1}{2}+\tfrac{1}{4}+\tfrac{1}{4}+\tfrac{1}{8}+\tfrac{1}{8}+\tfrac{1}{8}+\tfrac{1}{8}+\tfrac{1}{16}+\cdots\,,$$

where in general $b_{2^k}=b_{2^k+1}=\dots=b_{2^{k+1}-1}=\frac{1}{2^{k+1}}$. Then $\frac{1}{n}\ge b_n$ for every n. The partial sums (s_n) of $\sum b_n$ increase. For every $k\in\mathbb{N}_0$,

$$s_{2^{k+1}-1}=\tfrac{1}{2}+2\cdot\tfrac{1}{4}+4\cdot\tfrac{1}{8}+\cdots+2^k\cdot\tfrac{1}{2^{k+1}}=\tfrac{k+1}{2}\,.$$

By Exercise 3.5.21, $\sum b_n=\lim s_n=+\infty$. By Exercise 3.5.22, $\sum\frac{1}{n}=+\infty$. $\square$

By Proposition 3.5.12 every reordering of the series $\sum\frac{1}{n}$ has sum $+\infty$. The partial sums

$$(h_n)\equiv\left(\textstyle\sum_{i=1}^n\frac{1}{i}\right)=\left(1,\,\tfrac{3}{2},\,\tfrac{11}{6},\,\tfrac{25}{12},\,\tfrac{137}{60},\,\dots\right)\ \ (\subset\mathbb{Q})$$

are called harmonic numbers. Already in 1350 the French medieval philosopher *Nicolas Oresme (1320 to 1325 – 1382)* proved Proposition 3.5.23, that is, that $h_n\to+\infty$.

Theorem 3.5.24 (growth of h_n) *There is a* $c\ge 0$ *such that for every* n*,*

$$h_n=\log n+\gamma+\Delta(n)$$

where $|\Delta(n)|\le\frac{c}{n}$ *and* $\gamma=0.57721\dots$ *is so called Euler's constant.*

We prove this asymptotics in lecture 14 with the help of integrals. Theorem 4.8.5, which will be proven in *MA 1*$^+$, provides much more precise asymptotic estimates of h_n.

Exercise 3.5.25 *Prove that* $h_n\in\mathbb{N}$ *only for* $n=1$*. Hint:* $m=(2l-1)2^k$*.*

Exercise 3.5.26 (an open problem) *Prove that Euler's constant* γ *is an irrational number.*

• *Riemannian series.* In the first lecture we encountered the series

$$1-1+\tfrac{1}{2}-\tfrac{1}{2}+\tfrac{1}{3}-\tfrac{1}{3}+\cdots+\tfrac{1}{n}-\tfrac{1}{n}+\dots$$

with the sum 0. We described a reordering of it with a positive sum. We show in Theorem 3.5.30 that this and similar series can be reordered to have any sum. We begin with less extensive changes of sums. For a series $\sum a_n$ we denote by $k_1<k_2<\dots$ the indices n such that $a_n\ge 0$, and by $z_1<z_2<\dots$ the indices n such that $a_n<0$. If there are infinitely many k_n, by Proposition 3.5.12 all reorderings of the series $\sum a_{k_n}$ have the same sum. The same holds for the infinite series $\sum a_{z_n}$.

Before we state and prove two propositions and a theorem on reorderings of series, we explain two mutually inverse operations related to real sequences, decomposition and composition. A segment is a k-tuple $U = \langle b_1, \dots, b_k\rangle$ of real numbers b_i; we denote its length k by $|U|$ ($\in \mathbb{N}$). Let $S = (a_n)$ be a real sequence. An initial segment of S is any nonempty segment $U = \langle a_1, \dots, a_k\rangle$. We then write $S \setminus U$ for the tail $(a_{k+1}, a_{k+2}, \dots)$. A decomposition of S in segments

$$S = U_1 U_2 \dots$$

is a sequence $(U_1, U_2, \dots)$ of segments such that $U_1 = \langle a_1, \dots, a_{k_1}\rangle$, $U_2 = \langle a_{k_1+1}, \dots, a_{k_1+k_2}\rangle$, ... ($k_i \in \mathbb{N}$). Reversely, if $U_i = \langle u_{i,1}, \dots, u_{i,k_i}\rangle$, $i, k_i \in \mathbb{N}$, are segments then the real sequence

$$(b_n) = U_1 U_2 \dots$$

obtained by the composition of the segments U_i is defined by setting $b_n \equiv a_{i,\,j}$, where $k_0 \equiv 0$, the index i is given by

$$k_0 + k_1 + \dots + k_{i-1} < n \le k_0 + k_1 + \dots + k_i \ \text{ and } \ j \equiv n - k_0 - k_1 - \dots - k_{i-1}\,.$$

Proposition 3.5.27 (getting sums $\pm\infty$) *$\sum a_n$ can be reordered to the sum $-\infty$ $\iff$ the sum $\sum a_{z_n} = -\infty$. Similarly, $\sum a_n$ has a reordering with the sum $+\infty$ $\iff$ the sum $\sum a_{k_n} = +\infty$.*

Proof. We prove the former equivalence and leave the latter for Exercise 3.5.28. Let $\sum a_{z_n} = -\infty$. In particular the number of indices z_n is infinite. We may assume that also the number of indices k_n is infinite; if their number is finite, by Proposition 3.5.12 every reordering of $\sum a_n$ has sum $-\infty$. We define a bijection $f\colon \mathbb{N} \to \mathbb{N}$ for which the sum $\sum a_{f(n)} = -\infty$. It is the "limit" $\lim_{k\to\infty} P_k$ of certain injective sequences $P_k = (m_{n,k}) \subset \mathbb{N}$, $k \in \mathbb{N}_0$. Their terms are indices k_n and z_n. We set $P_0 \equiv (z_n)$. We take an initial segment U_1 of P_0 such that $\sum_{n\in U_1} a_n \le -1 - a_{k_1}$. We insert in P_0 after U_1 the index k_1 and get the sequence P_1. We take an initial segment U_2 of P_1 such that $|U_2| > |U_1|$ and $\sum_{n\in U_2} a_n \le -2 - a_{k_2}$. We insert in P_1 after U_2 the index k_2 and get the sequence P_2. And so on. Since $|U_1| < |U_2| < \dots$, the sequences P_0, P_1, P_2, ... converge, in the obvious sense, to the sought for bijection f.

Suppose that it is not true that $\sum a_{z_n} = -\infty$. Then either the number of indices z_n is finite or the sum $\sum a_{z_n} \in \mathbb{R}$. In the former case by Proposition 3.5.12 all reorderings of $\sum a_n$ have the same sum different from $-\infty$. In the latter case there exists a c such that for every n it holds that $\sum_{i=1}^{n} a_{z_i} \ge c$. It follows that no reordering of $\sum a_n$ has the sum $-\infty$. □

Exercise 3.5.28 *Reduce the latter equivalence to the former.*

Proposition 3.5.29 (getting no sum) *A series $\sum a_n$ has a reordering with no sum $\iff$ the sum $\sum a_{z_n} = -\infty$ and the sum $\sum a_{k_n} = +\infty$.*

Proof. We suppose that $\sum a_{z_n} = -\infty$ and $\sum a_{k_n} = +\infty$. We take an initial segment U_1 of (z_n) such that $\sum_{n\in U_1} a_n \le -1$, and an initial segment V_1 of (k_n) such that $\sum_{n\in U_1\cup V_1} a_n \ge 1$. We take an initial segment U_2 of $(z_n)\setminus U_1$ such that

$$\textstyle\sum_{n\in U_1\cup V_1\cup U_2} a_n \le -1\,,$$

and an initial segment V_2 of $(k_n)\setminus V_1$ such that

$$\textstyle\sum_{n\in U_1\cup V_1\cup U_2\cup V_2} a_n \ge 1\,.$$

We continue in this way indefinitely. The composed sequence

$$(p_n) \equiv U_1V_1U_2V_2\dots \quad (\subset \mathbb{N})$$

is a bijection from $\mathbb{N}$ to $\mathbb{N}$ and the series $\sum a_{p_n}$ does not have a sum because both $\sum_{i=1}^n a_{p_i} \le -1$ and $\sum_{i=1}^n a_{p_i} \ge 1$ hold for infinitely many n.

Suppose, for example, that the sum $\sum a_{z_n} \in \mathbb{R}$. For $\sum a_{k_n} \in \mathbb{R}$ we argue similarly. If $\sum a_{k_n} = +\infty$, then it follows that every reordering of $\sum a_n$ has the sum $+\infty$. If also $\sum a_{k_n} \in \mathbb{R}$, then the series $\sum a_n$ is, in the terminology introduced below, an abscon series, and by Proposition 3.5.42 all reorderings of it have the same finite sum. $\square$

A series $\sum a_n$ is <u>Riemannian</u> if $\lim a_n = 0$, the sum $\sum a_{k_n} = +\infty$ and the sum $\sum a_{z_n} = -\infty$.

Theorem 3.5.30 (getting any sum) *Let $\sum a_n$ be a series. The following claims are equivalent.*

1. *$\sum a_n$ is Riemannian.*
2. *For every $A \in \mathbb{R}^*$ there exists a reordering of $\sum a_n$ with the sum A.*

Proof. Implication $1 \Rightarrow 2$. Let $\sum a_n$ be Riemannian. The two cases $A = \pm\infty$ are dealt with in Proposition 3.5.27. The case of no sum is dealt with in Proposition 3.5.29. Let $A \in \mathbb{R}$. Both sequences (k_n) and (z_n) are infinite and every index n lies in exactly one of them. We define two decompositions

$$(k_n) = U_1U_2\dots \ \text{ and } \ (z_n) = V_1V_2\dots$$

in segments U_i and V_i. U_1 is the shortest initial segment of (k_n) such that $\sum_{n\in U_1} a_n \ge A$. V_1 is the shortest initial segment of (z_n) such that

$$\textstyle\sum_{n\in U_1\cup V_1} a_n \le A\,.$$

U_2 is the shortest initial segment of $(k_n)\setminus U_1$ such that

$$\textstyle\sum_{n\in U_1\cup V_1\cup U_2} a_n \ge A\,.$$

V_2 is the shortest initial segment of $(z_n)\setminus V_1$ such that

$$\textstyle\sum_{n\in U_1\cup V_1\cup U_2\cup V_2} a_n \le A\,.$$

We continue in this way indefinitely. We show that the composed sequence

$$(p_n) \equiv U_1 V_1 U_2 V_2 U_3 \dots \quad (\subset \mathbb{N})$$

is a bijection from $\mathbb{N}$ to $\mathbb{N}$ and that the sum $\sum a_{p_n} = A$.

First we remark that the definition of the segments U_i and V_i is correct because every tail of $\sum a_{k_n}$ (resp. $\sum a_{z_n}$) has sum $+\infty$ (resp. $-\infty$), see Exercise 3.5.7. It is also clear that $p\colon \mathbb{N} \to \mathbb{N}$ is a bijection because

$$\{(k_1, k_2, \dots), (z_1, z_2, \dots)\}$$

is a partition of $\mathbb{N}$ in two infinite sets. We show that $\sum a_{p_n} = A$. Let $k_i \equiv |U_i|$ and $l_i \equiv |V_i|$, $i \in \mathbb{N}$, and let u_i (resp. v_i) be the last term of U_i (resp. V_i). For $i \geq 2$ we set $S_i \equiv \sum_{j=1}^{i-1}(k_j + l_j)$. Let $s_n \equiv \sum_{j=1}^{n} a_{p_j}$. It follows from the definition of U_i and V_i that for every $i \geq 2$ we have

$$S_i < n < S_i + k_i \Rightarrow A + a_{v_{i-1}} \leq s_n < A,\ n = S_i + k_i \Rightarrow A \leq s_n < A + a_{u_i}$$

and similarly

$$S_i + k_i < n < S_{i+1} \Rightarrow A \leq s_n < A + a_{u_i},\ n = S_{i+1} \Rightarrow A + a_{v_i} < s_n \leq A\,.$$

But $\lim_{i\to\infty} a_{u_i} = \lim_{i\to\infty} a_{v_i} = 0$, hence $\lim s_n = A$ and $\sum a_{p_n} = A$.

Implication $\neg 1 \Rightarrow \neg 2$. Let $\sum a_n$ be not Riemannian. If the sum

$$\textstyle\sum_{a_{k_n}} < +\infty$$

then no reordering of the series yields the sum $+\infty$. If the sum

$$\textstyle\sum_{a_{z_n}} > -\infty$$

then no reordering of the series yields the sum $-\infty$. If $\lim a_n$ does not exist or is not 0, then this persists in any reordering and the series cannot be reordered to have a finite sum $\square$

The previous theorem is due to the German mathematician *Bernhard Riemann (1826–1866).*

- *Leibnizian series.* The series

$$1 - 1 + \tfrac{1}{2} - \tfrac{1}{2} + \tfrac{1}{3} - \tfrac{1}{3} + \dots + \tfrac{1}{n} - \tfrac{1}{n} + \dots$$

in the first lecture is not only Riemannian, but it is also a Leibnizian series. These are series of the form $\sum(-1)^{n-1}a_n$, where $a_1 \geq a_2 \geq \dots \geq 0$ and $\lim a_n = 0$.

Theorem 3.5.31 (on Leibnizian series) *Every Leibnizian series*

$$\textstyle\sum(-1)^{n-1}a_n$$

has a finite sum $s \in \mathbb{R}$. Moreover, for every n we have bounds

$$a_1 - a_2 + a_3 - \dots - a_{2n} \leq s \leq a_1 - a_2 + a_3 - \dots + a_{2n-1}\,.$$

Proof. Let $\sum(-1)^{n-1}a_n$ be Leibnizian and $s_n \equiv \sum_{j=1}^{n}(-1)^{j-1}a_j$. We show by induction on k and l that always

$$s_1 \ge s_3 \ge \cdots \ge s_{2k-1} \ge s_{2l} \ge s_{2l-2} \ge \cdots \ge s_2\,.$$

By Theorem 2.1.32 and since $s_{2n} = s_{2n-1} + a_{2n}$, we have the common limit $s \equiv \lim s_{2n-1} = \lim s_{2n}$. By Theorem 2.2.14, $\lim s_n = s$. Since $s_{2n-1} \ge s_m \ge s_{2n}$ for every $m \ge 2n$, using part 2 of Theorem 2.8.1 we get with $m \to \infty$ the stated inequalities for s. $\square$

This theorem is credited to the German mathematician, philosopher and diplomat *Gottfried W. Leibniz (1646–1716).* It is a remarkable result, partial sums are both lower and upper bounds on the sum.

Exercise 3.5.32 *Extend the previous theorem to series with Leibnizian tails.*

We prove the next theorem with two examples of sums of Leibnizian series in Chapter 9, see Corollaries 10.1.25 and.

Theorem 3.5.33 (two sums) *We have sums*

$$\sum \tfrac{(-1)^{n-1}}{n} = 1 - \tfrac{1}{2} + \tfrac{1}{3} - \cdots = \log 2 \ \ \textit{and} \ \ \sum \tfrac{(-1)^{n-1}}{2n-1} = 1 - \tfrac{1}{3} + \tfrac{1}{5} - \cdots = \tfrac{\pi}{4}\,.$$

See preprints [59, 68] for summations like

$$\sum \tfrac{1}{n(4n-3)} = \tfrac{\pi}{6} + \log 2,\ \sum \tfrac{1}{n(4n-1)} = 3\log 2 - \tfrac{\pi}{2} \ \text{ and } \ \sum \tfrac{1}{n(2n-1)(4n-3)} = \tfrac{\pi}{3}\,.$$

• *Grouping of summands.* This is a simple and useful operation on series. If $\sum a_n$ is a series and $S \equiv (m_n) \subset \mathbb{N}$ is a sequence, we set $m_0 \equiv 0$, define the series $\sum b_n$ by

$$b_n \equiv \textstyle\sum_{j=m_0+m_1+\cdots+m_{n-1}+1}^{m_0+m_1+\cdots+m_{n-1}+m_n} a_j$$

and call it the S-grouping of $\sum a_n$.

Theorem 3.5.34 (grouping of series) *Suppose that $\sum a_n$ is a series,*

$$S \equiv (m_n) \subset \mathbb{N}$$

is a sequence and that $\sum b_n$ is the S-grouping of $\sum a_n$. The following holds.

1. *If the sum $\sum a_n$ exists, then the equality of sums $\sum b_n = \sum a_n$ holds.*
2. *If $\lim a_n = 0$, the sequence S is bounded and the sum $\sum b_n$ exists, then the equality of sums $\sum a_n = \sum b_n$ holds.*
3. *If $\sum a_n$ and S are bounded sequences and the infinite sum $\sum b_n$ exists, then the equality of sums $\sum a_n = \sum b_n$ holds.*

Proof. 1. The sequence of partial sums of $\sum b_n$ is always a subsequence of the sequence of partial sums of $\sum a_n$.

2. We suppose that $\lim a_n = 0$, $m_n \le m$ ($\in \mathbb{N}$) for every n and that the sum $\sum b_n = s \in \mathbb{R}$. The case that $\sum b_n = \pm\infty$ is treated in part 3; the assumptions in part 2 imply the assumptions for part 3. Let an ε be given. We take an n_0 such that for every $n \ge n_0$ we have

$$m|a_n| \le \tfrac{\varepsilon}{2} \text{ and } |\textstyle\sum_{j=1}^{n} b_j - s| \le \tfrac{\varepsilon}{2}\,.$$

We set $n_1 \equiv \sum_{n=1}^{n_0} m_n$. Then certainly $n_1 \ge n_0$ and for any given $n \ge n_1$ we take the unique $k \in \mathbb{N}$ with $k \ge n_0$ such that $\sum_{n=1}^{k} m_n \le n < \sum_{n=1}^{k+1} m_n$. Then, with $l \equiv m_1 + m_2 + \cdots + m_k$, we have for every $n \ge n_1$ that

$$\big|s - \textstyle\sum_{j=1}^{n} a_j\big| \le \big|s - \sum_{j=1}^{k} b_j\big| + \sum_{j=l+1}^{n} |a_j| \le \frac{\varepsilon}{2} + m_{k+1} \max_{l+1 \le j \le n} |a_j|$$

which is at most $\frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$. Hence the sum $\sum a_n = s$.

3. We suppose that $|a_n|, m_n \le m$ ($\in \mathbb{N}$) for every n and that the sum $\sum b_n = -\infty$; the case $\sum b_n = +\infty$ is treated similarly. Let a c be given. We take an n_0 such that for every $n \ge n_0$ we have $\sum_{j=1}^{n} b_j \le c - m^2$. We define n_1 as before, and for any given $n \ge n_1$ we define $k = k(n)$ and $l = l(n)$ as before. Then for every $n \ge n_1$,

$$\textstyle\sum_{j=1}^{n} a_j = \sum_{j=1}^{k} b_j + \sum_{j=l+1}^{n} a_j \le c - m^2 + m_{k+1} \max_{l+1 \le j \le n} |a_j|$$

which is at most $c - m^2 + m^2 = c$. Hence the sum $\sum a_n = -\infty$. $\square$

Corollary 3.5.35 *We have the sums*

$$\sum \frac{1}{(2n-1)2n} = \log 2 \;\; \textit{and} \;\; \sum \frac{1}{(4n-3)(4n-1))} = \frac{\pi}{8}\,.$$

Proof. We obtain these sums by applying part 2 of Theorem 3.5.34 with $S = (2, 2, \dots)$ to the two series in Theorem 3.5.33. We divide the latter obtained series by 2 and use Proposition 3.5.15. $\square$

Cancellation of summands is at work: $\frac{1}{(2n-1)2n}$ is much smaller than any of the two summands in $\frac{1}{2n-1} - \frac{1}{2n}$. Thus we transform by grouping the Riemannian series $\sum \frac{(-1)^{n-1}}{n}$ in the better behaved abscon series (defined next) $\sum \frac{1}{(2n-1)2n}$. Same for the second series $\sum \frac{(-1)^{n-1}}{2n-1}$.

Exercise 3.5.36 *Show by an example that part 2 of Theorem 3.5.34 in general does not hold if we omit the assumption* $\lim a_n = 0$.

Exercise 3.5.37 *Show by an example that part 2 of Theorem 3.5.34 in general does not hold if we omit the assumption of boundedness of* S.

• *Abscon series.* In Section 3.1 we considered a more general version of this kind of series. Here we treat them standardly.

Definition 3.5.38 (abscon series) *$\sum a_n$ is an absolutely convergent series, or an abscon series, if the sum*

$$\sum |a_n| < +\infty .$$

Every abscon series is an AK series. In the next theorem we present an infinite triangle inequality for abscon series.

Theorem 3.5.39 (infinite triangle inequality) *If $\sum a_n$ is an abscon series, then $\sum a_n$ converges and*

$$|\sum a_n| \le \sum |a_n| .$$

Proof. Let $\sum a_n$ be an abscon series with partial sums (s_n) and let $\sum |a_n|$ have partial sums (t_n). By Theorem 2.4.18 the sequence (t_n) is Cauchy. Hence for any given ε for every two large indices $m \le n$ it holds that

$$|t_n - t_m| = \big||a_{m+1}| + |a_{m+2}| + \cdots + |a_n|\big| = |a_{m+1}| + |a_{m+2}| + \cdots + |a_n| \le \varepsilon$$

(for $m = n$ these sums are zero). By the Δ-inequality we have for the same indices $m \le n$ that

$$|s_n - s_m| = |a_{m+1} + a_{m+2} + \cdots + a_n| \le |a_{m+1}| + |a_{m+2}| + \cdots + |a_n| \le \varepsilon .$$

Hence (s_n) is Cauchy. By Theorem 2.4.18 the sequence (s_n) converges. Hence the series $\sum a_n$ converges.

By the Δ-inequality it holds for every n that $|s_n| \le t_n$, equivalently

$$-t_n \le s_n \le t_n .$$

Sending $n \to \infty$ gives by Theorem 2.8.1 for sums the inequalities

$$-\sum |a_n| \le \sum a_n \le \sum |a_n| .$$

Hence $\big|\sum a_n\big| \le \sum |a_n|$. □

Exercise 3.5.40 *Every reordering of an abscon series is an abscon series.*

Exercise 3.5.41 *Every subseries of an abscon series is an abscon series.*

Proposition 3.5.42 (on abscon series) *$\sum a_n$ is an abscon series $\iff$ all reorderings of it have the same finite sum.*

Proof. Suppose that $\sum a_n$ is an abscon series. Let $f, g\colon \mathbb{N} \to \mathbb{N}$ be bijections and let an ε be given. We take an n_0 such that $\sum_{n \ge n_0} |a_n| \le \varepsilon$. Let n_1 be such that $f[\,[n_1]\,] \supset [n_0]$ and $g[\,[n_1]\,] \supset [n_0]$. Then for every $m, n \ge n_1$,

$$\left| \sum_{j=1}^m a_{f(j)} - \sum_{j=1}^n a_{g(j)} \right| \le \sum_{k \in A} |a_k| + \sum_{k \in B} |a_k| \le \varepsilon + \varepsilon = 2\varepsilon\,,$$

where $A, B \subset \mathbb{N} \setminus [n_0]$ are finite sets. Thus every reordering of $\sum a_n$ has a finite sum, because the sequence of partial sums is Cauchy ($f = g$), and all these sums are equal ($f \ne g$).

If $\sum a_n$ is not an abscon series then the sum $\sum a_{z_n} = -\infty$ or the sum $\sum a_{k_n} = +\infty$, and by Proposition 3.5.27 some reordering does not have finite sum. $\Box$

• *The comparison criterion.* Absolute convergence of a series is usually shown by comparing it with another abscon series.

Proposition 3.5.43 (comparison criterion) *Suppose that*

$$\sum a_n \text{ and } \sum b_n$$

are series such that $a_n \ge 0$ for every n, $|b_n| \le a_n$ for every $n \ge n_0$ and the sum $\sum a_n < +\infty$. Then $\sum b_n$ is an abscon series.

Proof. Let $s_n \equiv \sum_{i=1}^n a_i$ and $t_n \equiv \sum_{i=1}^n |b_i|$. Then for every $n \ge n_0$,

$$0 \le t_n \le \sum_{i=1}^{n_0} |b_i| + s_n\,.$$

Hence there is a constant $c \ge 0$ such that $0 \le t_n \le c$ for every n and $\sum b_n$ is an abscon series. $\Box$

Exercise 3.5.44 *Suppose that (c_n) is a bounded sequence. Prove that the series $\sum \frac{c_n}{n(n+1)}$ is abscon.*

• *The Cauchy product of series.* We need this operation to prove the exponential identity in Theorem 5.1.4. In this context it is convenient to work instead of $\mathbb{N}$ with the index set $\mathbb{N}_0$ and with series of the form

$$\sum_{n=0}^\infty a_n = a_0 + a_1 + \cdots\,.$$

We understand $a_0 + a_1 + \ldots$ as the standard series $b_1 + b_2 + \ldots$ in which $b_n \equiv a_{n-1}$. The Cauchy product of the series $\sum_{n=0}^\infty a_n$ and $\sum_{n=0}^\infty b_n$ is the series

$$\sum_{n=0}^\infty a_n \odot \sum_{n=0}^\infty b_n \equiv \sum_{n=0}^\infty \sum_{j=0}^n a_j b_{n-j}\,.$$

Theorem 3.5.45 (Cauchy product of series) *Any two abscon series with sums s and t have abscon Cauchy product with the sum st.*

Proof. Let $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ be abscon series with the respective sums s and t. For $n \in \mathbb{N}_0$ let $s_n \equiv \sum_{j=0}^{n} a_j$, $t_n \equiv \sum_{j=0}^{n} b_j$, $z_n \equiv \sum_{j=0}^{n} \sum_{i=0}^{j} a_i b_{j-i}$, $s' \equiv \sum_{n=0}^{\infty} |a_n|$ (sum) and $t' \equiv \sum_{n=0}^{\infty} |b_n|$ (sum). Then

$$|st - z_n| \le |st - s_n t_n| + |s_n t_n - z_n| \equiv A_n + B_n\,.$$

By Definition 3.5.1 and Theorem 2.6.2, $\lim A_n = 0$. We show that also $\lim B_n = 0$. We have

$$B_n = \Big| \sum_{\substack{0 \le i,j \le n \\ i+j>n}} a_i b_j \Big| \le s' \sum_{\frac{n}{2} < j \le n} |b_j| + t' \sum_{\frac{n}{2} < i \le n} |a_i| \equiv s' C_n + t' D_n\,.$$

Since the sequences $(\sum_{i=0}^{n} |a_i|)$ and $(\sum_{j=0}^{n} |b_j|)$ are Cauchy, we have $\lim C_n = \lim D_n = 0$ and $\lim B_n = 0$. Thus the sum of $\sum_{n=0}^{\infty} a_n \odot \sum_{n=0}^{\infty} b_n$ is st. It remains to show that $z' \equiv \sum_{n=0}^{\infty} |z_n| < +\infty$. Since

$$|z_n| \le \sum_{j=0}^{n} \sum_{i=0}^{j} |a_i| \cdot |b_{j-i}|\,,$$

our previous result implies that $z' \le s't'$ and we are done. $\square$

Exercise 3.5.46 *Let* $F(x) = \sum_{n=0}^{\infty} a_n x^n$ *and* $G(x) = \sum_{n=0}^{\infty} b_n x^n$ *be formal power series with real coefficients and let* $\sum_{n=0}^{\infty} c_n \equiv \sum_{n=0}^{\infty} a_n \odot \sum_{n=0}^{\infty} b_n$. *Then* $F(x) \cdot G(x) = \sum_{n=0}^{\infty} c_n x^n$.

- *A geometric series* is any series of the form

$$\sum_{n=0}^{\infty} q^n = 1 + q + q^2 + \cdots + q^n + \cdots\,,\ q \in \mathbb{R}\,.$$

The number q is the quotient of the geometric series.

Theorem 3.5.47 (sums of geometric series) *The geometric series has sum* $\frac{1}{1-q}$ *if* $-1 < q < 1$ *and* $+\infty$ *if* $q \ge 1$. *For* $q \le -1$ *the sum does not exist.*

Proof. For every $q \in \mathbb{R} \setminus \{1\}$ and $n \in \mathbb{N}$ we have the identity

$$s_n \equiv 1 + q + q^2 + \cdots + q^{n-1} = \tfrac{1-q^n}{1-q} = \tfrac{1}{1-q} + \tfrac{q^n}{q-1}\,.$$

For $q < -1$ Theorem 2.6.2 shows that $\lim s_{2n-1} = +\infty$ and $\lim s_{2n} = -\infty$. Hence $\lim s_n$ does not exist. For $q = -1$ similarly $s_{2n-1} = 1$ and $s_{2n} = 0$, the sum again does not exist. For $-1 < q < 1$ we have $\lim q^n = 0$. Thus by Theorem 2.6.2, $\lim s_n = \frac{1}{1-q}$. For $q = 1$ we have $s_n = n$ and the sum is $+\infty$. For $q > 1$ we have $\lim q^n = +\infty$ and the sum is again $+\infty$ by Theorem 2.6.2. $\square$

As an application we get that, for example,

$$27.272727\cdots = 27(1 + 10^{-2} + 10^{-4} + \dots) = 27 \cdot \tfrac{1}{1-10^{-2}} = \tfrac{27\cdot 100}{99} = \tfrac{300}{11}\,.$$

Exercise 3.5.48 *Let* $q \in (-1, 1)$ *and* $m \in \mathbb{Z}$, *with* $q \ne 0$ *if* $m < 0$. *Then the sum* $\sum_{n \ge m} q^n = \frac{q^m}{1-q}$.

Exercise 3.5.49 *Which geometric series are abskon?*

• *Root and ratio tests.* These two classical convergence criteria for series with nonnegative summands use bounds by geometric series $1 + x + x^2 + \cdots$ for $x \in [0, 1)$.

Proposition 3.5.50 (root test) *Let $a_n \geq 0$ for every n. Then the sum*

$$\sum a_n \begin{cases} = +\infty & \ldots \quad \limsup a_n^{1/n} > 1 \quad \textit{and} \\ < +\infty & \ldots \quad \limsup a_n^{1/n} < 1\,. \end{cases}$$

Proof. Let $\limsup a_n^{1/n} > 1$. Then for some $c > 1$ we have $a_n \geq c^n > 1$ for infinitely many n. By Exercise 3.5.22, the sum $\sum a_n = +\infty$.

Let $\limsup a_n^{1/n} < 1$. Then for some $c \in [0, 1)$ and n_0 we have $0 \leq a_n \leq c^n$ for every $n \geq n_0$. With $s_n \equiv \sum_{j=1}^{n} a_j$ we have

$$s_1 \leq s_2 \leq \cdots \leq s_n \leq \cdots \leq a_1 + \cdots + a_{n_0 - 1} + \tfrac{1}{1-c}$$

and, by Theorem 2.1.32, the sum $\sum a_n < +\infty$. □

Exercise 3.5.51 *Show that with nonstrict inequalities $\cdots \geq 1$ or $\cdots \leq 1$ the root test is not valid.*

Proposition 3.5.52 (ratio test) *Let always $a_n > 0$. Then the sum*

$$\sum a_n \begin{cases} = +\infty & \ldots \quad \liminf \frac{a_{n+1}}{a_n} > 1 \quad \textit{and} \\ < +\infty & \ldots \quad \limsup \frac{a_{n+1}}{a_n} < 1\,. \end{cases}$$

Proof. Let $\liminf \frac{a_{n+1}}{a_n} > 1$. Then for some n_0 and $c > 1$ we have $\frac{a_{n+1}}{a_n} \geq c$ for every $n \geq n_0$. It follows that $a_n \geq a_{n_0} > 0$ for every $n \geq n_0$. Hence $\sum a_n = +\infty$ by Exercise 3.5.22.

Let $\limsup \frac{a_{n+1}}{a_n} < 1$. Then for some n_0 and $c \in (0, 1)$ we have $0 < \frac{a_{n+1}}{a_n} \leq c$ for every $n \geq n_0$. Thus $0 < a_n \leq c^{n-n_0} a_{n_0}$ for every $n \geq n_0$. With $s_n \equiv \sum_{j=1}^{n} a_j$ we have

$$s_1 \leq s_2 \leq \cdots \leq s_n \leq \cdots \leq a_1 + \cdots + a_{n_0 - 1} + \tfrac{a_{n_0}}{1-c}\,.$$

By Theorem 2.1.32, the sum $\sum a_n < +\infty$. □

Proposition 3.5.11 shows that the ratio test extends to series with nonnegative summands.

Exercise 3.5.53 *Show that with the nonstrict inequalities $\cdots \geq 1$ or $\cdots \leq 1$ the ratio test is not valid.*

Exercise 3.5.54 *Find a convergent series $\sum a_n$ with positive summands and $\limsup \frac{a_{n+1}}{a_n} = +\infty$.*

• *The zeta function (series)* $\zeta(s)$. We use real exponentiation a^b which we introduce soon in Section 5.1.

Definition 3.5.55 (series $\zeta(s)$) *For $s \in \mathbb{R}$ the zeta series is $\zeta(s) \equiv \sum \frac{1}{n^s}$.*

We determine the convergence of $\zeta(s)$ by the Cauchy condensation criterion.

Theorem 3.5.56 (CCC) *Let $a_1 \ge a_2 \ge \cdots \ge 0$ be real numbers. Then*

$$\sum a_n \text{ converges iff } R \equiv \sum 2^n \cdot a_{2^n} \text{ converges}.$$

Proof. Suppose that R has sum $+\infty$. Hence also the series $\frac{1}{2}R = \sum 2^{n-1} \cdot a_{2^n}$ has sum $+\infty$. We have the inequalities

$$a_2 \ge a_2,\ a_3 + a_4 \ge 2a_4,\ \dots,\ \sum_{j=2^{k-1}+1}^{2^k} a_j \ge 2^{k-1} a_{2^k},\ \cdots$$

and summing them we get $\sum a_n = +\infty$.

Suppose that R converges. We have the inequalities

$$a_2 + a_3 \le 2a_2,\ a_4 + a_5 + a_6 + a_7 \le 4a_4,\ \dots,\ \sum_{j=2^k}^{2^{k+1}-1} a_j \le 2^k a_{2^k},\ \cdots$$

and summing them we get that $\sum a_n$ converges. □

The proof of convergence of $\zeta(s)$ for $s > 1$ is a nice application of CCC.

Theorem 3.5.57 (convergence of $\zeta(s)$) *For $s \le 1$ the series $\zeta(s)$ has sum $+\infty$. For $s > 1$ the zeta series converges.*

Proof. The former claim is Exercise 3.5.58. Let $s > 1$. The series R in CCC for $\zeta(s)$ is

$$\sum \frac{2^n}{(2^n)^s} = \sum \frac{1}{(2^{s-1})^n}\,.$$

Since $0 < \frac{1}{2^{s-1}} < 1$, by Theorem 3.5.47 this geometric series converges. So by Theorem 3.5.56 the series $\zeta(s)$ converges. □

In *MA 1*$^+$ we show that $\zeta(2) = \frac{\pi^2}{6}$. This sum is due to the Swiss mathematician *Leonhard Euler (1707–1783)*. In *MA 1*$^+$ we also show that the sum $\zeta(3)$ is an irrational number; this was proved by the French mathematician *Roger Apéry (1916–1994)* in 1979.

Exercise 3.5.58 *Prove that for $s \le 1$ the series $\zeta(s)$ has sum $+\infty$.*

Exercise 3.5.59 *For which real s does the series $\sum_{n=2}^{\infty} n^{-1}(\log n)^s$ converge?*

Exercise 3.5.60 *Using bound $n^{-2} \le \frac{1}{n(n-1)}$ for $n \ge 2$, give a simple proof of convergence of the series $\sum n^{-2}$.*

• *Using* $\zeta(s)$. We conclude this section on infinite series with three applications of the zeta function in mathematics and physics. The most important connection is to prime numbers. Let (p_n) ($\subset \mathbb{N}$) be the ordering (Proposition 2.2.4) of the set of primes, so that

$$(p_n) = (2,\, 3,\, 5,\, 7,\, 11,\, 13,\, 17,\, 19,\, 23,\, \dots)\,.$$

Theorem 3.5.61 (Euler's infinite product) *For every real $s > 1$ we have*

$$\prod_p \left(1 - \frac{1}{p^s}\right)^{-1} = \prod_{n=1}^{\infty} \left(1 - \frac{1}{(p_n)^s}\right)^{-1} = \lim_{n\to\infty} \prod_{j=1}^{n} \left(1 - \frac{1}{(p_j)^s}\right)^{-1} = \zeta(s)\,.$$

Exercise 3.5.62 *Prove it.*

Exercise 3.5.63 *Could not the value of Euler's infinite product be changed by reordering the terms in it?*

In combinatorics, more precisely in graph theory, we have the following interesting result of A. M. Frieze [35]. Let $n \in \mathbb{N}$ and $w\colon \binom{[n]}{2} \to [0,1]$. A spanning tree T on $[n]$ is a tree (connected graph without cycles) $T = (V, E)$ such that $V \subset [n]$ and $|E| = n - 1$ ($E \subset \binom{[n]}{2}$).

Exercise 3.5.64 *Prove that then $V = [n]$ and T is really "spanning".*

We define the w-weight, or simply the weight, of the spanning tree T by $w(T) = \sum_{e\in E} w(e)$. We denote by $M(w)$ the minimum weight $w(T)$ for T running through all spanning trees on $[n]$. Recall that in probability theory we denote by $\mathbb{E}\,X$ the expectation of the random variable X.

Theorem 3.5.65 (Frieze's) *For n in $\mathbb{N}$ let $w = w_n$ be the random weight $w\colon \binom{[n]}{2} \to [0,1]$, that is, w_n is the product of $\binom{n}{2}$ independent copies of the uniform distribution on the interval $[0,1]$. Then*

$$\lim_{n\to\infty} \mathbb{E}\,M(w_n) = \zeta(3)\,.$$

Finally, when browsing through the lecture notes *Low Temperature Physics* [71] we spotted on page 16 the formula

$$E = \tfrac{\zeta(5/2)\cdot\Gamma(5/2)}{\zeta(3/2)\cdot\Gamma(3/2)} \cdot N \cdot k_{\mathrm{B}} \cdot T \cdot \left(\tfrac{T}{T_{\mathrm{B}}}\right)^{3/2}$$

for the energy of the Bose gas at temperature T.

Chapter 4

Limits of real functions

We begin with Section 4.2 on one-sided limits of functions. Propositions 4.2.7 and 4.2.13 establish various relations between two-sided and one-sided limits. Section 4.3 introduces continuity function at a point. In Proposition 4.3.5 we characterize it by limits, and in Exercise 4.3.6 by Heine's definition. In Proposition 4.3.9 we point out that every function is continuous at every isolated point of its domain. Section 4.4 contains Theorem 4.4.1 on limits of monotone functions, Theorem 4.4.5 on arithmetic of limits of functions, Theorem 4.4.9 on relations of limits of functions and the linear order $(\mathbb{R}^*, <)$, and a generalization of the squeeze theorem for sequences in Theorem 4.4.13. The most interesting result of Section 4.4 is the corrected Cauchy's theorem from [13] in Theorem 4.4.3. In Section 4.5 we present Theorem 4.5.1 on limits of composite functions. Our version is an equivalence.

Later I returned to this chapter and added the extending Section 4.6 on limits of inverse functions. The relation of functional limits and functional inverses was so far not considered in the literature. Thus the section contains original results. We obtain conditions under which $\lim_{x\to A} f(x) = B$ implies $\lim_{y\to B} f^{\langle -1\rangle}(y) = A$. In Theorem 4.6.4 we constrain the definition of limit. In Theorems 4.6.9 and 4.6.12 we constrain the considered function. In Theorem 4.6.17 we give yet another sufficient condition for this inversion of limits.

The last two sections are devoted to asymptotic notation. In Section 4.7 we explain the meaning of symbols O, $\ll$, $\gg$, Ω, Θ, $\asymp$, o, ω and $\sim$. Definition 4.7.4 introduces the notion of asymptoticity of relations on the set of real functions $\mathcal{R}$. Relations $f = O(g)$ (on N), $f = o(g)$ ($x \to A$) and $f \sim g$ ($x \to A$), as we define them, are asymptotic. In the extending Section 4.8 we explain asymptotic expansions of functions and give, without proofs, three examples of them: for $\log(n!)$, for the harmonic numbers h_n and for the probability of connectedness of labeled graphs on n vertices.

4.1 Limits of functions

We extend limits of real sequences (a_n), which are functions of the type

$$a\colon \mathbb{N} \to \mathbb{R}\,,$$

to functions $f\colon M \to \mathbb{R}$ defined on arbitrary sets $M \subset \mathbb{R}$.

• *Deleted neighborhoods and limit points.* Recall that for $A, \varepsilon \in \mathbb{R}$ with $\varepsilon > 0$ we have ε-neighborhoods of points

$$U(A,\,\varepsilon) = (a - \varepsilon,\, a + \varepsilon)\,,$$

and that for $A = \pm\infty$ we have ε-neighborhoods of infinities

$$U(-\infty,\,\varepsilon) = (-\infty,\, -\tfrac{1}{\varepsilon}) \text{ and } U(+\infty,\,\varepsilon) = (\tfrac{1}{\varepsilon},\, +\infty)\,.$$

We define the deleted ε-neighborhood of $A \in \mathbb{R}^*$ by

$$P(A,\,\varepsilon) \equiv U(A,\,\varepsilon) \setminus \{A\}\,.$$

Let $M \subset \mathbb{R}$. An element $L \in \mathbb{R}^*$ is a limit point of M if for every ε we have $P(L, \varepsilon) \cap M \neq \emptyset$. The set of limit points of M is denoted by $L(M)$ $(\subset \mathbb{R}^*)$.

Exercise 4.1.1 *Prove the following proposition.*

Proposition 4.1.2 (on limit points) *Let $M \subset \mathbb{R}$ and $A \in \mathbb{R}^*$. The next four claims are mutually equivalent.*

1. *$A \in L(M)$.*
2. *There is a sequence $(a_n) \subset M \setminus \{A\}$ such that $\lim a_n = A$.*
3. *There is an injective sequence $(a_n) \subset M$ such that $\lim a_n = A$.*
4. *For every $n \in \mathbb{N}$ we have $P(A, \frac{1}{n}) \cap M \neq \emptyset$.*

Exercise 4.1.3 *A set $M \subset \mathbb{R}$ is finite $\iff$ $L(M) = \emptyset$.*

Exercise 4.1.4 *If $M \subset \mathbb{R}$ and $b \in L(M)$ then also $b \in L(M \setminus \{b\})$.*

• *Limits of real functions.* We introduce notation we will often use.

Definition 4.1.5 (notation for real functions) *For $M \subset \mathbb{R}$ we define*

$$\mathcal{F}(M) \equiv \{f = \langle M,\, \mathbb{R},\, G_f\rangle\colon\ f\colon M \to \mathbb{R} \ \text{ and } \ M \subset \mathbb{R}\}$$

(recall Definition 1.1.2) and set

$$\mathcal{R} \equiv \textstyle\bigcup_{M \subset \mathbb{R}} \mathcal{F}(M)\,.$$

For $f \in \mathcal{F}(M)$ we define

$$Z(f) \equiv \{b \in M\colon\ f(b) = 0\} \quad (\subset M)\,.$$

Recall that the domain of any function $f\colon X \to Y$ is $M(f) = X$.

Thus $\mathcal{F}(M)$, for $M \subset \mathbb{R}$, is the set of functions with the definition domain M and range $\mathbb{R}$, and $\mathcal{R}$ is the set of all such functions for all sets of real numbers M. By $Z(f)$ we denote the set of zeros of f. The real empty function is

$$\emptyset_f \equiv \langle \emptyset, \mathbb{R}, \emptyset \rangle \quad (\in \mathcal{R}) .$$

It is an empty function.

Definition 4.1.6 (limits of functions) *Let f be in $\mathcal{R}$, A in $L(M(f))$ and L in $\mathbb{R}^*$. If for every ε there is a δ such that*

$$f[P(A, \delta)] \subset U(L, \varepsilon) , \tag{$*$}$$

we write $\lim_{x \to A} f(x) = L$ and say that f has at A the limit L.

(In Definition 4.6.3 we consider a strengthening of this definition.) Recall that by our definition of image,

$$f[P(A, \delta)] = f[P(A, \delta) \cap M(f)] = \{f(x) \colon\; x \in P(A, \delta) \cap M(f)\} .$$

The limit does not depend on the value $f(A)$ and the function f need not be defined at A. If $A = \pm\infty$ then it in fact cannot be defined at A. For a sequence $(a_n) \subset \mathbb{R}$, which is a function $a \colon \mathbb{N} \to \mathbb{R}$, we have $\lim_{x \to +\infty} a(x) = \lim a_n$.

Exercise 4.1.7 *Find all limit points of the set $\mathbb{N}$ ($\subset \mathbb{R}$).*

If $A = a \in \mathbb{R}$ and $L = b \in \mathbb{R}$, then $\lim_{x \to a} f(x) = b$ means that

$$\forall \varepsilon \, \exists \delta \, \big(x \in M(f) \wedge 0 < |x - a| \le \delta \Rightarrow |f(x) - b| \le \varepsilon\big)$$

(recall that we like $\le$ more than $<$). We stress that

$$\text{if } f \in \mathcal{F}(M) \text{ and } A \notin L(M) \text{ then } \lim_{x \to A} f(x) \text{ is not defined} .$$

For then for some δ we have $P(A, \delta) \cap M = \emptyset$ and $f[P(A, \delta)] = \emptyset$, which means that the above inclusion ($*$) holds for every L and every ε. If $\lim_{x \to A} f(x)$ exists then always $A \in L(M(f))$.

Proposition 4.1.8 (locality of limits) *If $f, g \in \mathcal{R}$, $A \in \mathbb{R}^*$ and for some θ we have $f \,|\, P(A, \theta) = g \,|\, P(A, \theta)$, then*

$$\lim_{x \to A} f(x) = \lim_{x \to A} g(x) ,$$

if one side is defined.

Proof. We can take δ in Definition 4.1.6 such that $\delta \le \theta$. Then $P(A, \delta) \subset P(A, \theta)$ and $f[P(A, \delta)] = g[P(A, \delta)]$. $\square$

Later we show that continuity and derivatives are local too.

Proposition 4.1.9 (unique limits) *If* $\lim_{x\to K} f(x) = L$ *and* $\lim_{x\to K} f(x) = L'$ *then* $L = L'$. *Thus limits of functions are unique.*

Proof. For every ε there is a δ such that the nonempty (!) set $f[P(K,\delta)]$ is contained both in $U(L,\varepsilon)$ and $U(L',\varepsilon)$. Thus $\forall\varepsilon\big(U(L,\varepsilon)\cap U(L',\varepsilon)\neq\emptyset\big)$. By Exercise 2.1.11, $L = L'$. $\square$

We show that limits of restrictions are equal to limits of original functions.

Proposition 4.1.10 (limits of restrictions) *Let* $f \in \mathcal{F}(M)$, X *be any set,* $A \in L(X\cap M)$ *and let* $\lim_{x\to A} f(x) = L$. *Then* $\lim_{x\to A}(f\,|\,X)(x) = L$.

Proof. Let an ε be given. Then there is a δ such that $f[P(A,\delta)] \subset U(L,\varepsilon)$. From $P(A,\delta)\cap(X\cap M) \subset P(A,\delta)\cap M$ we get

$$(f\,|\,X)[P(A,\,\delta)] \subset f[P(A,\,\delta)] \subset U(L,\,\varepsilon)\,.$$

Hence $\lim_{x\to A}(f\,|\,X)(x) = L$. $\square$

This is the first of several results on interactions between limits of functions and operations on $\mathcal{R}$. Operations we consider later are composition and the arithmetic operations of addition, multiplication, and division. Functional inverses are relegated to *MA 1*$^{+}$.

Exercise 4.1.11 *Find a function* $f \in \mathcal{F}(M)$ *and a set* X *such that* $\lim_{x\to A} f(x)$ *does not exist but* $\lim_{x\to A}(f\,|\,X)(x)$ *exists.*

• *Heine's definition of the limit of a function.* It is clear that limits of sequences are particular cases of limits of functions. The German mathematician E. Heine realized that one can go in the opposite direction and reduce the limits of functions to the limits of sequences.

Theorem 4.1.12 (Heine's limit) *Let* $f \in \mathcal{F}(M)$ *and* $K \in L(M)$. *Then*

$$\lim_{x\to K} f(x) = L \iff \forall\,(a_n) \subset M\setminus\{K\}\,\big(\lim a_n = K \Rightarrow \lim f(a_n) = L\big)\,.$$

In words, a function f *defined on* M *has limit* L *at* K *iff for every sequence* (a_n) *in* M *with* $a_n \neq K$ *but with* $\lim a_n = K$ *the sequence* $(f(a_n))$ *has limit* L.

Proof. Implication $\Rightarrow$. Suppose that

$$\lim_{x\to K} f(x) = L\,,$$

that $(a_n) \subset M\setminus\{K\}$ is a sequence with $\lim a_n = K$ and that an ε is given. Then there is a δ such that for every $x \in P(K,\delta)\cap M$ we have $f(x) \in U(L,\varepsilon)$. For this δ there is an n_0 such that for every $n \geq n_0$ we have $a_n \in P(K,\delta)\cap M$. Hence if $n \geq n_0$ then $f(a_n) \in U(L,\varepsilon)$ and $f(a_n) \to L$.

Reverse implication $\neg \Rightarrow \neg$. Suppose that

$$\text{it is not true that } \lim_{x \to K} f(x) = L\,.$$

Then there exists a number ε such that for every δ there exists a point $b = b(\delta) \in P(K, \delta) \cap M$ with

$$f(b) \notin U(L, \varepsilon)\,.$$

We set $\delta = \frac{1}{n}$ and select points

$$b_n \equiv b(\tfrac{1}{n}) \in P(K, \tfrac{1}{n}) \cap M$$

such that $f(b_n) \notin U(L, \varepsilon)$ for every n. Then (b_n) is a sequence in $M \setminus \{K\}$ with $\lim b_n = K$, but the sequence $(f(b_n))$ does not have limit L. The right side of the equivalence does not hold. $\Box$

In the proof of the reverse implication we used the axiom of choice.

Exercise 4.1.13 *How exactly did we use it?*

• *A functional limit.* We get with the help of the identities $x - y = \frac{x^2-y^2}{x+y}$ a $\frac{x}{y} = \frac{1}{y/x}$ that

$$\begin{aligned}
\lim\nolimits_{x \to +\infty} \left(\sqrt{x + \sqrt{x}} - \sqrt{x}\right) &= \lim\nolimits_{x \to +\infty} \tfrac{\sqrt{x}}{\sqrt{x+\sqrt{x}} + \sqrt{x}} \\
&= \lim\nolimits_{x \to +\infty} \tfrac{1}{\sqrt{1+1/\sqrt{x}} + 1} \\
&= \tfrac{1}{\sqrt{1+1/(+\infty)} + 1} = \tfrac{1}{1+1} = \tfrac{1}{2}\,.
\end{aligned}$$

Exercise 4.1.14 *Compute the following limits of functions.*

1. $\lim_{x \to -\infty} \frac{x}{\sqrt{1+x^2-1}}$.
2. $\lim_{x \to +\infty} \frac{1}{\sqrt{1+x}-\sqrt{x}}$.
3. $\lim_{x \to 0} \frac{1}{x}$.
4. $\lim_{x \to -\infty} \frac{1}{x}$.

• *A theorem of A.-L. Cauchy.* The book *Cours d'analyse*, see [13] for English translation, "is a seminal textbook in infinitesimal calculus published by Augustin-Louis Cauchy in 1821" [23]. In the next proposition we present counterexamples to Theorem I in Section 2.3 of it. We illustrate by this limits of functions. Cauchy's theorem, quoted from [13], says the following.

> If the difference $f(x+1) - f(x)$ converges towards a certain limit k, for increasing values of x, then the fraction $\frac{f(x)}{x}$ converges at the same time towards the same limit.

However, we found the following counterexamples.

Proposition 4.1.15 (counterexamples) *Let $k \in \mathbb{R}$ and*

$$(a_n) \subset (0, +\infty)$$

be any sequence such that $a_1 < a_2 < \dots$, $\lim a_n = +\infty$ and $a_n - a_m \notin \mathbb{N}$ for every pair of indices $m < n$ (see Exercise 4.1.16). Then there exists a function $f\colon (0, +\infty) \to \mathbb{R}$ such that

$$f(x+1) - f(x) = k$$

for every $x > 0$, but $f(a_{2n}) = 0$ and $f(a_{2n-1}) = a_{2n-1}$ for every n. We have in particular

$$\lim_{x\to+\infty} (f(x+1) - f(x)) = k$$

but

$$\tfrac{f(a_{2n})}{a_{2n}} = 0 \;\text{ and }\; \tfrac{f(a_{2n-1})}{a_{2n-1}} = 1$$

for every n, hence $\lim_{x\to+\infty} \frac{f(x)}{x}$ does not exist.

Proof. Let k and (a_n) be as stated, and $y \in (0, 1]$ be arbitrary. If $y + m \neq a_n$ for every $m \in \mathbb{N}_0$ and every $n \in \mathbb{N}$, we define the value $f(y) \in \mathbb{R}$ arbitrarily and then for every $l \in \mathbb{N}$ we set

$$f(y + l) \equiv f(y) + lk\,.$$

If $y + m = a_{2n}$ for a unique $m \in \mathbb{N}_0$ and $n \in \mathbb{N}$, we define $f(y) \equiv -mk$ and then for every $l \in \mathbb{N}$ we set

$$f(y + l) \equiv f(y) + lk\,.$$

Finally, if $y + m = a_{2n-1}$ for a unique $m \in \mathbb{N}_0$ and $n \in \mathbb{N}$, we define $f(y) \equiv a_{2n-1} - mk$ and then for every $l \in \mathbb{N}$ we set

$$f(y + l) \equiv f(y) + lk\,.$$

By the assumption on (a_n) this definition of f is correct. It is clear that $M(f) = (0, +\infty)$ and that the function f has the stated properties. □

Exercise 4.1.16 *Give examples of real sequences (a_n) with the properties stated in the proposition.*

The next exercise requires access to [13].

Exercise 4.1.17 *Find an error in Cauchy's proof of Theorem I in Section 2.3 of the* Cours d'analyse ([13]) *by which if $f\colon (0, +\infty) \to \mathbb{R}$ has*

$$\lim_{x\to+\infty} (f(x+1) - f(x)) = k \quad (\in \mathbb{R})\,,$$

then $\lim_{x\to+\infty} \frac{f(x)}{x} = k$.

We fix Cauchy's theorem in Theorem 4.4.3 by adding monotonicity assumption.

4.2 One-sided limits

The complement of any point a to the real axis,

$$\mathbb{R} \setminus \{a\} = (-\infty,\, a) \cup (a,\, +\infty)\,,$$

consists of two separated intervals. In the plane $\mathbb{R}^2$ we can go around a point but in $\mathbb{R}$ this is impossible. So we consider left-sided and right-sided limits of functions. First we define one-sided neighborhoods.

• *One-sided neighborhoods and one-sided limit points.* A left, resp. a right, ε-neighborhood of a point $b \in \mathbb{R}$ is

$$U^-(b,\, \varepsilon) \equiv (b - \varepsilon,\, b],\ \text{resp.}\ U^+(b,\, \varepsilon) \equiv [b,\, b + \varepsilon)\,.$$

A left, resp. a right, deleted ε-neighborhood of a point $b \in \mathbb{R}$ is

$$P^-(b,\, \varepsilon) \equiv (b - \varepsilon,\, b),\ \text{resp.}\ P^+(b,\, \varepsilon) \equiv (b,\, b + \varepsilon)\,.$$

A point $b \in \mathbb{R}$ is a left, resp. a right, limit point of $M \subset \mathbb{R}$ if for every ε we have

$$P^-(b,\, \varepsilon) \cap M \neq \emptyset,\ \text{resp.}\ P^+(b,\, \varepsilon) \cap M \neq \emptyset\,.$$

The set of these points is denoted by $L^-(M)$, resp. $L^+(M)$ ($\subset \mathbb{R}$). A point $b \in \mathbb{R}$ is a two-sided limit point of $M \subset \mathbb{R}$ if for every ε,

$$P^-(b,\, \varepsilon) \cap M \neq \emptyset \,\wedge\, P^+(b,\, \varepsilon) \cap M \neq \emptyset\,.$$

The set of these points is denoted by $L^{\mathrm{TS}}(M)$ ($\subset \mathbb{R}$). Two-sided limit points play a key role in the criterion of local extremes. We do not define one-sided neighborhoods for infinities, nor $\pm\infty$ can be a one-sided limit point of a set.

Exercise 4.2.1 *$b \in L^-(M)$, resp. $b \in L^+(M)$ $\iff$ $\exists\ (a_n) \subset (-\infty, b) \cap M$, resp. $\exists\ (a_n) \subset (b, +\infty) \cap M$, such that* $\lim a_n = b$.

Exercise 4.2.2 *Let $M \subset \mathbb{R}$ and $b \in \mathbb{R}$. Prove the following.*

1. *If $b \in L^-(M)$, then $b \in L(M)$.*
2. *If $b \in L^+(M)$, then $b \in L(M)$.*
3. *If $b \in L(M)$, then $b \in L^-(M)$ or $b \in L^+(M)$.*
4. *In general, it is not true that if $b \in L(M)$ then $b \in L^-(M)$ and $b \in L^+(M)$.*

Let $M \subset \mathbb{R}$. By Exercise 4.1.3 no finite set M has a limit point, the less one-sided limit point, and every infinite M has a limit point. This is not true for one-sided limit points.

Exercise 4.2.3 *Find infinite subsets of $\mathbb{R}$ without one-sided limit points.*

Exercise 4.2.4 *Every infinite and bounded real set has a one-sided limit point.*

• *One-sided limits of functions.* We consider finer variants of limits of functions, the one-sided limits.

Definition 4.2.5 (one-sided limits) *Let f be in $\mathcal{F}(M)$, b be in $L^-(M)$ and L be in $\mathbb{R}^*$. If for every ε there is a δ such that*

$$f[P^-(b,\,\delta)] \subset U(L,\,\varepsilon)\,,$$

we write $\lim_{x\to b^-} f(x) = L$ and say that f has at b the left-sided limit L. Replacing the three signs $^-$ with three signs $^+$ we get the right-sided limit at b that is denoted by $\lim_{x\to b^+} f(x) = L$.

Like the ordinary limit, the one-sided limit of f at b is not defined if b is not the respective one-sided limit point of $M(f)$. The existence of $\lim_{x\to b^\pm} f(x)$ again means that b is the respective one-sided limit point of $M(f)$.

Exercise 4.2.6 *Prove the following proposition.*

Proposition 4.2.7 (one-sided limits) *The following hold.*

1. *If $\lim_{x\to a} f(x) = L$, then $\lim_{x\to a^-} f(x) = L$ or $\lim_{x\to a^+} f(x) = L$.*
2. *If $\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L$, then $\lim_{x\to a} f(x) = L$.*
3. *If $\lim_{x\to a^-} f(x) = K$, $\lim_{x\to a^+} f(x) = L$ and $K \ne L$, then the limit $\lim_{x\to a} f(x)$ does not exist.*

For example, $\lim_{x\to 0} \operatorname{sgn} x$ does not exist because $\lim_{x\to 0^-} \operatorname{sgn} x = -1$ and $\lim_{x\to 0^+} \operatorname{sgn} x = 1$.

Exercise 4.2.8 *Prove the following proposition.*

Proposition 4.2.9 (uniqueness of one-sided limits) *If $\lim_{x\to b^\pm} f(x) = K$ and $\lim_{x\to b^\pm} f(x) = L$ then $K = L$, with equal signs.*

Exercise 4.2.10 *Prove the following proposition.*

For $b \in \mathbb{R}$ let $I^-(b) \equiv (-\infty, b)$ and $I^+(b) \equiv (b, +\infty)$.

Proposition 4.2.11 (Heine's limits) *Suppose that f is in $\mathcal{F}(M)$ and b is in $L^\pm(M)$. Then $\lim_{x\to b^\pm} f(x) = L \iff$ for every sequence $(a_n) \subset M \cap I^\pm(b)$ with $\lim a_n = b$ one has $\lim f(a_n) = L$, with equal signs.*

Sometimes ordinary limits are replaced unnecessarily with one-sided ones. For example, we can see notation $\lim_{x\to 0^+} \log x$. In our notation it suffices and is correct to write just $\lim_{x\to 0} \log x$. Indeed,

$$\lim_{x\to 0} \log x = \lim_{x\to 0^+} \log x = -\infty\,,$$

but $\lim_{x\to 0^-} \log x$ is not defined because

$$0 \notin L^-(M(\log x)) = L^-((0,\, +\infty)) = (0,\, +\infty)\,.$$

We conclude with one more relation between ordinary and one-sided limits. We use it in the proof of Corollary 4.5.5.

Exercise 4.2.12 *Prove the following proposition.*

Proposition 4.2.13 (using restriction) *Let $f \in \mathcal{F}(M)$ and $b \in L^{\pm}(M)$. Then*

$$\lim_{x\to b^{\pm}} f(x) = L \iff \lim_{x\to b} (f \mid I^{\pm}(b))(x) = L\,,$$

with equal signs.

4.3 Continuity at a point

We arrive at an important definition.

Definition 4.3.1 (pointwise continuity) *Let $f \in \mathcal{F}(M)$ and $b \in M$. Then f is continuous at b if for every ε there is a δ such that*

$$f[U(b,\, \delta)] \subset U(f(b),\, \varepsilon)\,.$$

Else we say that f is discontinuous at b.

Exercise 4.3.2 *Let $b \in M(f)$. Explain in detail when f is discontinuous at b.*

For example, $\operatorname{sgn} x$ is discontinuous at 0, but it is continuous at any other point. If $b \notin M(f)$ then f is *neither continuous nor discontinuous* at b. Comparing the above definition with

$$\lim_{x\to b} f(x) = L \iff \forall\varepsilon\exists\delta\,\big(f[P(b,\, \delta)] \subset U(L,\, \varepsilon)\big)\,,$$

we see that the last L is replaced with $f(b)$, and the deleted neighborhood $P(b,\delta)$ with the full neighborhood $U(b,\delta)$.

Proposition 4.3.3 (locality of continuity) *If $f, g \in \mathcal{R}$, $b \in M(f) \cap M(g)$ and there is a θ such that $f = g$ on $U(b,\theta)$ then f is continuous at $b \iff g$ is continuous at b.*

Proof. This is immediate from Definition 4.3.1 because we can take the δ in it such that $\delta \le \theta$. Then $U(b, \delta) \subset U(b, \theta)$ and $f[U(b, \delta)] = g[U(b, \delta)]$. $\Box$

Exercise 4.3.4 *A function f is continuous at b ($\in M(f)$) iff for every ε there exists a δ such that $x \in M(f) \wedge |x - b| \le \delta \Rightarrow |f(x) - f(b)| \le \varepsilon$.*

The continuity of f at b is *not* equivalent to $\lim_{x \to b} f(x) = f(b)$. This only holds for limit points of M.

Proposition 4.3.5 (pointwise continuity) *For any function $f \in \mathcal{F}(M)$ and any point $b \in M \cap L(M)$ the following are mutually equivalent.*

1. *The function f is continuous at b.*
2. *The limit $\lim_{x \to b} f(x) = f(b)$.*
3. *For every sequence $(a_n) \subset M$ with $\lim a_n = b$ we have $\lim f(a_n) = f(b)$.*

Proof. Let f and b be as stated. The implication $1 \Rightarrow 2$. Let f be continuous at b by Definition 4.3.1 and let an ε be given. Thus there is a δ such that $f[U(b, \delta)] \subset U(f(b), \varepsilon)$. We have $b \in L(M)$ and $f[P(b, \delta)] \subset U(f(b), \varepsilon)$. Hence $\lim_{x \to b} f(x) = f(b)$.

The implication $2 \Rightarrow 3$. Suppose that $\lim_{x \to b} f(x) = f(b)$, that $(a_n) \subset M$ has $\lim a_n = b$ and that an ε is given. Thus there is a δ such that

$$f[P(b,\, \delta)] \subset U(f(b),\, \varepsilon)\ . \tag{$*$}$$

We take an n_0 such that $n \ge n_0 \Rightarrow a_n \in U(b, \delta)$. Then also $n \ge n_0 \Rightarrow f(a_n) \in U(f(b), \varepsilon)$: for $a_n \ne b$ we use the inclusion $(*)$, and for $a_n = b$ it is automatic that $f(a_n) = f(b) \in U(f(b), \varepsilon)$. Hence $\lim f(a_n) = f(b)$.

The implication $3 \Rightarrow 1$. We prove the reversal $\neg 1 \Rightarrow \neg 3$. Suppose that f is not continuous at b. Then there is an ε such that $\forall\, \delta\, \exists\, a = a(\delta) \in U(b, \delta) \cap M$ with $f(a) \notin U(f(b), \varepsilon)$. For every $n \in \mathbb{N}$ we *chose* such $a_n \equiv a(\frac{1}{n})$ and get the sequence $(a_n) \subset M$ such that $\lim a_n = b$, but for every n it holds that $f(a_n) \notin U(f(b), \varepsilon)$. Hence $(f(a_n))$ does not converge to $f(b)$ and part 3 does not hold. $\Box$

The last implication again used the axiom of choice. Part 3 describes *Heine's definition of pointwise continuity.*

Exercise 4.3.6 *In Proposition 4.3.5 one can omit in the equivalence $1 \iff 3$ the assumption that $b \in L(M)$. Thus, f is continuous at a point $b \in M(f)$ $\iff$ for every sequence $(a_n) \subset M$ with $\lim a_n = b$ one has $\lim f(a_n) = f(b)$.*

The right side of this equivalence is sometimes taken as the definition of pointwise continuity.

• *Isolated points.* Let $M \subset \mathbb{R}$. The set $M \setminus L(M)$ consists of the so-called <u>isolated points</u> of M.

Exercise 4.3.7 *A point $b \in M$ $(\subset \mathbb{R})$ is an isolated point of M iff for some ε we have $U(b, \varepsilon) \cap M = \{b\}$.*

Exercise 4.3.8 *Let $b \in M \subset \mathbb{R}$. Then b is either a limit point of M or an isolated point of M.*

Proposition 4.3.9 (continuity at isolated points) *Every function $f \in \mathcal{R}$ is continuous at every isolated point of $M(f)$.*

Proof. Let $f \in \mathcal{F}(M)$ and let $b \in M$ be an isolated point of M. By Exercise 4.3.7 there is a δ such that $U(b, \delta) \cap M = \{b\}$. For this δ the inclusion $f[U(b, \delta)] = f[\{b\}] = \{f(b)\} \subset U(f(b), \varepsilon)$ holds for every ε. Hence f is continuous at b by Definition 4.3.1. $\square$

We see that every sequence $(a_n) \subset \mathbb{R}$, understood as a function $a \in \mathcal{F}(\mathbb{N})$, is continuous at every point n of the definition domain $\mathbb{N}$.

Exercise 4.3.10 *Let $f \in \mathcal{F}(M)$ and $b \in M$. Then f is not continuous at b if and only if there exists a sequence $(a_n) \subset M$ such that* $\lim a_n = b$, $\lim f(a_n) = A$ *and $A \neq f(b)$.*

• *One-sided continuity at a point.* A function f is left-continuous at $b \in M(f)$ if for every ε there is a δ such that

$$f[U^-(b,\, \delta)] \subset U(f(b),\, \varepsilon)\,.$$

By replacing the sign $^-$ with the sign $^+$ we get the right-continuity.

Exercise 4.3.11 *A function is continuous at a point iff it is both left- and right-continuous at the point.*

• *Riemann's function.* It is the function $r \in \mathcal{F}(\mathbb{R})$ with values $r(x) = 0$ for $x \in \mathbb{R} \setminus \mathbb{Q}$ and $r(\frac{m}{n}) = \frac{1}{n}$ for fractions $\frac{m}{n}$ in lowest terms.

Proposition 4.3.12 (Riemann's function) *Riemann's function is continuous exactly at irrational numbers.*

Proof. Let $x = \frac{m}{n}$ be a fraction in lowest terms and $\varepsilon \leq \frac{1}{n}$. For every δ there is an irrational $\alpha \in U(x, \delta)$. But $r(\alpha) = 0 \notin U(r(x), \varepsilon) = U(\frac{1}{n}, \varepsilon)$, so that r is discontinuous at x. Let $x \in \mathbb{R} \setminus \mathbb{Q}$ and $\varepsilon \in (0, 1)$ be given. We set

$$M \equiv \{|x - \tfrac{m}{n}| \colon\ \tfrac{m}{n} \in \mathbb{Q} \cap U(x,\, 1) \wedge \tfrac{1}{n} \geq \varepsilon\} \text{ and } \delta \equiv \min(M)\,.$$

By Exercise 4.3.13, δ exists and is positive. For this δ we have $y \in U(x, \delta) \Rightarrow r(y) \in U(r(x), \varepsilon) = U(0, \varepsilon)$ — for every $y \in U(x, \delta)$ we have $r(y) = 0$ or $r(y) = \frac{1}{n} < \varepsilon$. Hence r is continuous at x. $\square$

Exercise 4.3.13 *Show that M is a nonempty finite set of positive real numbers.*

4.4 Arithmetic of limits. Limits and order

We extend some results on limits of sequences to limits of functions.

• *Limits of monotone functions.* Let $f \in \mathcal{F}(M)$ and X be any set. We say that f weakly increases, resp. weakly decreases, on X if

$$\text{for any } x \leq y \text{ in } X \cap M \text{ we have } f(x) \leq f(y), \text{ resp. } f(x) \geq f(y)\,.$$

If f weakly increases or weakly decreases on X, then it is monotone on X. Note that X need not be a subset of M.

Theorem 4.4.1 (limits of monotone functions) *Let f be in $\mathcal{F}(M)$. The following holds.*

1. *If $b \in L^-(M)$ and there is a θ such that f weakly increases on $P^-(b,\theta)$ then*
$$\lim_{x\to b^-} f(x) = \sup(f[P^-(b,\,\theta)])\,.$$
2. *If $+\infty \in L(M)$ and there is a θ such that f weakly increases on $U(+\infty,\theta)$ then*
$$\lim_{x\to +\infty} f(x) = \sup(f[U(+\infty,\,\theta)])\,.$$

Both suprema are taken in the linear order $(\mathbb{R}^*,<)$.

Proof. 1. Let f, M, b and θ be as stated, and let ε be given. We set

$$A \equiv \sup(f[P^-(b,\,\theta)])$$

and take any $a \in U(A,\varepsilon)$ with $a < A$. By the definition of supremum there is a $c \in P^-(b,\theta) \cap M$ such that $a < f(c) \leq A$. We set $\delta \equiv b - c$. For every $d \in M$ with $c < d < b$ it holds that $a < f(c) \leq f(d) \leq A$. Hence, by Exercise 2.1.10, $f(d) \in U(A,\varepsilon)$. Thus

$$f[P^-(b,\,\delta)] \subset U(A,\,\varepsilon)$$

and $\lim_{x\to b^-} f(x) = A$.

2. Let f, M and θ be as stated, and let ε be given. We set

$$A \equiv \sup(f[U(+\infty,\,\theta)])$$

and take any $a \in U(A,\varepsilon)$ with $a < A$. By the definition of supremum there is a $c \in U(+\infty,\theta) \cap M$ such that $a < f(c) \leq A$. We set $\delta \equiv \frac{1}{c}$. For every $d \in M$ with $c < d$ it holds that $a < f(c) \leq f(d) \leq A$. Using Exercise 2.1.10 we get that $f(d) \in U(A,\varepsilon)$. Hence

$$f[U(+\infty,\,\delta)] \subset U(A,\,\varepsilon)$$

and $\lim_{x\to +\infty} f(x) = A$. □

The theorem is not valid for two-sided limits: the function $\mathrm{sgn}\colon \mathbb{R} \to \{-1,0,1\}$ weakly increases on $\mathbb{R}$ but $\lim_{x\to 0} \mathrm{sgn}\, x$ does not exist. We find two-sided limits of monotone functions by reducing them via Proposition 4.2.7 to one-sided limits. These we compute by means of the previous theorem and the next exercise.

Exercise 4.4.2 *Describe further variants of the theorem: for locally weakly decreasing functions and/or the right-sided limit at b, resp. at $-\infty$.*

• *Theorem I in Section 2.3 of Cauchy's* Cours d'analyse [13]. We present this theorem in a corrected form. In Proposition 4.1.15 we gave counterexamples to the original version. Our correction consists in adding the assumption of monotonicity.

Theorem 4.4.3 (Cauchy's Theorem I corrected) *Let $k \in \mathbb{R}$ and let f in $\mathcal{F}((0, +\infty))$ be a monotone function such that*

$$\lim_{x \to +\infty} (f(x+1) - f(x)) = k\,.$$

Then

$$\lim_{x \to +\infty} \tfrac{f(x)}{x} = k\,.$$

Proof. We begin with the case $k = 0$ and weakly increasing function f. The case with weakly decreasing f is similar. Let an ε be given. We take an $h > 0$ such that for every $x \geq h$,

$$-\varepsilon \leq f(x+1) - f(x) \leq \varepsilon\,.$$

Let $n \in \mathbb{N}$ and $x = h, h+1, \ldots, h+n-1$. By summing these inequalities and rearranging the result we get

$$\tfrac{f(h)}{n} - \varepsilon \leq \tfrac{f(h+n)}{n} \leq \tfrac{f(h)}{n} + \varepsilon\,.$$

Let $c \equiv \max(\frac{|f(h)|}{\varepsilon}, \frac{1}{\varepsilon}) + 1$ and $x \in \mathbb{R}$ be such that $x \geq h + c$. We take the unique $n \in \mathbb{N}$ such that $h + n \leq x < h + n + 1$. Then $n \geq c - 1$ and $\frac{|f(h)|}{n} \leq \varepsilon$. By the above inequalities and since f weakly increases,

$$-2\varepsilon \leq \tfrac{f(h)}{n} - \varepsilon \leq \tfrac{f(h+n)}{n} \leq \tfrac{f(x)}{n} \ \text{ and } \ \tfrac{f(x)}{n+1} \leq \tfrac{f(h+n+1)}{n+1} \leq \tfrac{f(h)}{n+1} + \varepsilon \leq 2\varepsilon\,.$$

From $0 < \frac{n}{x} \leq 1$ and $0 < \frac{n+1}{x} \leq 1 + \frac{1}{c} \leq 1 + \varepsilon$ we get

$$-2\varepsilon \leq -2\varepsilon \cdot \tfrac{n}{x} \leq \tfrac{f(x)}{x} \leq 2\varepsilon \cdot \tfrac{n+1}{x} \leq 2\varepsilon(1+\varepsilon)\,.$$

Thus $\lim_{x \to +\infty} \frac{f(x)}{x} = 0 = k$.

We suppose that $k < 0$ and (hence) that the function f weakly decreases. The case of $k > 0$ and weakly increasing f is similar. Let an $\varepsilon < \min(-\frac{k}{2}, 1)$ be given. We take an $h > 0$ such that for every $x \geq h$,

$$k - \varepsilon \leq f(x+1) - f(x) \leq k + \varepsilon\,.$$

Let $n \in \mathbb{N}$ and $x = h, h+1, \ldots, h+n-1$. By summing these inequalities and rearranging the result we get

$$k - \varepsilon + \tfrac{f(h)}{n} \leq \tfrac{f(h+n)}{n} \leq k + \varepsilon + \tfrac{f(h)}{n}\,.$$

Let

$$c \equiv \max\left(\left\{\tfrac{|f(h)|}{\varepsilon}, \tfrac{1}{\varepsilon}, \tfrac{h+1}{\varepsilon}\right\}\right) + 1$$

and $x \in \mathbb{R}$ be such that $x \geq h + c$. We take the unique $n \in \mathbb{N}$ such that $h+n \leq x < h+n+1$. Then $n \geq c-1$ and $\frac{|f(h)|}{n} \leq \varepsilon$. By the above inequalities and since f weakly decreases,

$$\begin{aligned} & k - 2\varepsilon \leq k - \varepsilon + \tfrac{f(h)}{n+1} \leq \tfrac{f(h+n+1)}{n+1} \leq \tfrac{f(x)}{n+1} \\ & \text{and } \tfrac{f(x)}{n} \leq \tfrac{f(h+n)}{n} \leq k + \varepsilon + \tfrac{f(h)}{n} \leq k + 2\varepsilon\,. \end{aligned}$$

From $0 < \frac{n+1}{x} \leq 1 + \frac{1}{c} \leq 1 + \varepsilon$, $k + 2\varepsilon < 0$ and $\frac{n}{x} \geq 1 - \frac{h+1}{c} \geq 1 - \varepsilon > 0$ we get

$$(1+\varepsilon)(k-2\varepsilon) \leq \tfrac{n+1}{x}(k-2\varepsilon) \leq \tfrac{f(x)}{x} \leq \tfrac{n}{x}(k+2\varepsilon) \leq (1-\varepsilon)(k+2\varepsilon)\,.$$

Thus $\lim_{x\to+\infty} \frac{f(x)}{x} = k$. □

Exercise 4.4.4 *Extend this theorem to infinite limits $k = \pm\infty$.*

• *Arithmetic of limits of functions.* We extend some results on arithmetic of limits of sequences to functions. In proofs we use Heine's definition of limits of functions.

Theorem 4.4.5 (arithmetic of functional limits) *Suppose that $f, g \in \mathcal{R}$, $A \in L(M(f) \cap M(g))$, $\lim_{x\to A} f(x) = K$ and $\lim_{x\to A} g(x) = L$. Then*

$$\lim_{x\to A}(f+g)(x) = K + L,\ \lim_{x\to A}(fg)(x) = KL \text{ and } \lim_{x\to A}(f/g)(x) = \tfrac{K}{L}\,,$$

if the right-hand side is not indefinite.

Proof. Let f, g, A, K and L be as stated. We only consider the limit of the ratio of two functions, proofs for sum and product are similar and easier. We assume that $\frac{K}{L}$ is not indefinite. Then $L \neq 0$ and $A \in L(M(f/g))$ (Exercise 4.4.6). Let $(a_n) \subset M(f/g) \setminus \{A\}$ be any sequence with $\lim a_n = A$. The implication $\Rightarrow$ in Heine's definition of limits of functions gives that $\lim f(a_n) = K$ and $\lim g(a_n) = L$. Using Theorem 2.6.2 we get that

$$\lim \tfrac{f(a_n)}{g(a_n)} = \tfrac{\lim f(a_n)}{\lim g(a_n)} = \tfrac{K}{L}\,.$$

Since for every sequence (a_n) as above the sequence

$$\left(\tfrac{f(a_n)}{g(a_n)}\right) = ((f/g)(a_n))$$

has this limit, the implication $\Leftarrow$ in Heine's definition of limits of functions gives that also $\lim_{x\to A}(f/g)(x) = \frac{K}{L}$. □

Exercise 4.4.6 *In the previous proof, why for $L \neq 0$ is $A \in L(M(f/g))$?*

Using Proposition 4.2.13 we easily adapt the previous theorem for one-sided limits.

Exercise 4.4.7 *Deduce from the theorem the next corollary.*

Corollary 4.4.8 (limits of reciprocals 1) *If $g \in \mathcal{R}$, $\lim_{x\to A} g(x) = B$ and $B \ne 0$ then*

$$\lim_{x\to A}(k_1/g)(x) = \lim_{x\to A} \frac{1}{g(x)} = \frac{1}{B}\,.$$

• *Limits of functions and the linear order* $(\mathbb{R}^*, <)$. Recall that for sets $M, N \subset \mathbb{R}$ the notation $M < N$ means that for every $a \in M$ and $b \in N$ we have $a < b$. Recall that for any function f and any set X,

$$f[X] = f[X \cap M(f)] = \{f(x)\colon\ x \in X \cap M(f)\}\,.$$

In the next theorem and proposition we have $f, g \in \mathcal{R}$.

Theorem 4.4.9 (limits versus order 2) *We assume that $\lim_{x\to A} f(x) = K$ and $\lim_{x\to B} g(x) = L$, where possibly $A \ne B$. The following holds.*

1. *If $K < L$ then there is a δ such that $f[P(A,\delta)] < g[P(B,\delta)]$.*
2. *If for every $\delta > 0$ there exist an $x \in P(A,\delta) \cap M(f)$ and a $y \in P(B,\delta) \cap M(g)$ such that $f(x) \ge g(y)$, then $K \ge L$.*

Proof. 1. Since $K < L$, by Exercise 2.1.11 there is an ε such that $U(K,\varepsilon) < U(L,\varepsilon)$. Then by the assumption there is a δ such that $f[P(A,\delta)] \subset U(K,\varepsilon)$ and $g[P(B,\delta)] \subset U(L,\varepsilon)$. Hence $f[P(A,\delta)] < g[P(B,\delta)]$.

2. Part 2 is the contrapositive of the implication in part 1. □

We strengthen the theorem in the same way as Proposition 2.8.5 strengthens Theorem 2.8.1.

Exercise 4.4.10 *Prove the following proposition.*

Proposition 4.4.11 (strengthening Theorem 4.4.9) *Let $\lim_{x\to A} f(x) = K$ and $\lim_{x\to B} g(x) = L$, where possibly $A \ne B$. Then the following hold.*

1. *If $K < L$ then there exist δ and $a, b \in \mathbb{R}$, such that*

$$f[P(A,\,\delta)] < \{a\} < \{b\} < g[P(B,\,\delta)]\,.$$

2. *If for every δ and every $a < b$ in $\mathbb{R}$ there is an $x \in P(A,\delta) \cap M(f)$ and a $y \in P(B,\delta) \cap M(g)$ such that $f(x) \ge a$ or $g(y) \le b$, then $K \ge L$.*

Exercise 4.4.12 *State variants of Theorem 4.4.9 and Proposition 4.4.11 for one-sided limits and prove them.*

• *The squeeze theorem.* We generalize the squeeze theorem and the corollary of it from sequences to functions.

Theorem 4.4.13 (unexpected squeeze 2) *Suppose that $f, g \in \mathcal{F}(M)$, that $\lim_{x\to A} f(x) = b$ is in $\mathbb{R}$ and that $\lim_{x\to A} |f(x) - g(x)| = 0$. Then*

$$\lim_{x\to A} g(x) = b\,.$$

Proof. Let f, g, M, A ($\in L(M)$) and b be as stated, and let an ε be given. We take a δ such that $f[P(A,\delta)] \subset U(b, \frac{\varepsilon}{2})$ and $(f-g)[P(A,\delta)] \subset U(0, \frac{\varepsilon}{2})$. Then for every $x \in P(A,\delta) \cap M$ we have

$$|g(x) - b| \le |g(x) - f(x)| + |f(x) - b| \le \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} = \varepsilon\,.$$

Hence $g[P(A,\delta)] \subset U(b,\varepsilon)$ and $\lim_{x\to A} g(x) = b$. □

Recall that $I(a,b) = \{x \in \mathbb{R}\colon\ \min(a,b) \le x \le \max(a,b)\}$.

Corollary 4.4.14 (standard squeeze 2) *Suppose that $f, g, h \in \mathcal{F}(M)$, that $\lim_{x\to K} f(x) = \lim_{x\to K} g(x) = b$ is in $\mathbb{R}$ and that for every $x \in M$ we have $h(x) \in I(f(x), g(x))$. Then*

$$\lim_{x\to K} h(x) = b\,.$$

Proof. Let f, g, h, M, K and b be as stated. Then $\lim_{x\to K} |f(x) - g(x)| = 0$. Since for every $x \in M$ we have

$$|f(x) - h(x)|,\ |g(x) - h(x)| \le |f(x) - g(x)|\,,$$

we get $\lim_{x\to K} |f(x) - h(x)| = 0$ and $\lim_{x\to K} |g(x) - h(x)| = 0$, and are done by Theorem 4.4.13. □

The advantage of Theorem 4.4.13 over Corollary 4.4.14 is that the theorem easily generalizes to maps between metric spaces.

Exercise 4.4.15 *State and prove this generalization.*

4.5 Limits of composite functions

Composition of functions has no analogue for sequences, and so its interplay with limits of functions yields a genuinely new theorem. It is usually stated as an implication, but we give it a form of an equivalence.

• *Limits of composite functions.* Let $f, g \in \mathcal{R}$. Recall that the composite function $f(g)\colon M(f(g)) \to \mathbb{R}$ has the definition domain

$$M(f(g)) = \{x \in M(g)\colon\ g(x) \in M(f)\}\,.$$

It may be a proper subset of $M(g)$.

Theorem 4.5.1 (limits of composites) *We assume that*

$$\lim_{x\to A} g(x)=K,\ \lim_{x\to K} f(x)=L \ \text{ and that } A\in L(M(f(g)))\,.$$

Then

$$\lim_{x\to A} f(g)(x)=L$$

if and only if one of two conditions holds.

1. *Either $K\notin M(f)$, or $K\in M(f)$ and $f(K)=L$.*
2. *For some θ we have $K\notin g[P(A,\theta)]$.*

If neither condition holds, then $K\in M(f)$, $f(K)\ne L$ and $\lim_{x\to A} f(g)(x)$ does not exist or equals $f(K)$.

Proof. Let A, g, K, f and L be as stated and an ε be given. By the assumption there is a δ' such that (a) $f[P(K,\delta')]\subset U(L,\varepsilon)]$, and a δ such that (b) $g[P(A,\delta)]\subset U(K,\delta')$. Suppose that condition 1 holds. Then inclusion (a) strengthens to $f[U(K,\delta')]\subset U(L,\varepsilon)$ and

$$f(g)[P(A,\,\delta)]=f[\,g[P(A,\,\delta)]\,]\subset f[U(K,\,\delta')]\subset U(L,\,\varepsilon)\,.$$

Hence $\lim_{x\to A} f(g)(x)=L$. Suppose that condition 2 holds. We can take the previous δ such that $\delta\le\theta$, where θ is as in condition 2. Then inclusion (b) strengthens to $g[P(A,\delta)]\subset P(K,\delta')$ and

$$f(g)[P(A,\,\delta)]=f[\,g[P(A,\,\delta)]\,]\subset f[P(K,\,\delta')]\subset U(L,\,\varepsilon)\,.$$

Hence again $\lim_{x\to A} f(g(x))=L$.

Suppose that neither condition holds. Hence K is in $M(f)$ but $f(K)\ne L$, and for every n there is an $a_n\in P(A,\frac{1}{n})\cap M(g)$ such that $g(a_n)=K$. Then $(a_n)\subset M(f(g))\setminus\{A\}$, $\lim a_n=A$ and

$$\lim\, f(g)(a_n)=\lim\, f(g(a_n))=\lim\, f(K)=f(K)\ \ (\ne L)\,.$$

By Theorem 4.1.12 the limit $\lim_{x\to A} f(g(x))$ either does not exist or equals to $f(K)$ (which is not L). □

Condition 1 is satisfied if $K=\pm\infty$. Condition 2 is satisfied if the function g is injective. We get the following corollary of Theorem 4.5.1.

Corollary 4.5.2 (when composite limit works) *Let*

$$\lim_{x\to A} g(x)=K,\ \lim_{x\to K} f(x)=L \ \text{ and } A\in L(M(f(g)))\,.$$

If $K=\pm\infty$ or g is injective, then

$$\lim_{x\to A} f(g)(x)=L\,.$$

Exercise 4.5.3 *Prove Theorem 4.5.1 by means of Heine's definition of limits of functions.*

• *Applications of Theorem 4.5.1.* The next two equivalences are useful tools to determine limits of functions.

Corollary 4.5.4 (shift to 0) *Let $f \in \mathcal{R}$ and $b \in L(M(f))$. Then*

$$\lim_{x \to b} f(x) = L \iff \lim_{x \to 0} f(x+b)(x) = L\,.$$

Proof. Implication $\Rightarrow$. We assume that $\lim_{x \to b} f(x) = L$ and apply Corollary 4.5.2 to the composite limit $\lim_{x \to 0} f(g)(x)$ where $g(x) \equiv x + b$ is injective.

Implication $\Leftarrow$. We assume that $\lim_{x \to 0} f(x+b)(x) = L$ and apply Corollary 4.5.2 to the composite limit $\lim_{x \to b} g(h(x))$ where $g(x) \equiv f(x+b)$ and $h(x) \equiv x - b$ is injective. $\Box$

Corollary 4.5.5 (shift to $0^{\pm}$) *Let $f \in \mathcal{R}$ and $\pm\infty \in L(M(f))$. Then*

$$\lim_{x \to \pm\infty} f(x) = L \iff \lim_{x \to 0^{\pm}} f(\tfrac{1}{x})(x) = L\,,$$

with equal signs. Note that the last limit is one-sided.

Proof. We assume that all signs are $+$, the case with all signs $-$ is similar. Implication $\Rightarrow$. We assume that $\lim_{x \to +\infty} f(x) = L$ and apply Corollary 4.5.2 to the composite limit $\lim_{x \to 0} f(g(x)$ where $g(x) \equiv \frac{1}{x} \,|\, (0, +\infty)$ is injective.

Implication $\Leftarrow$. We assume that $\lim_{x \to 0^+} f(\frac{1}{x})(x) = L$ and apply Corollary 4.5.2 to the composite limit $\lim_{x \to +\infty} g(h)(x)$ where $g(x) \equiv f(\frac{1}{x}) \,|\, (0, +\infty)$ and $h(x) \equiv \frac{1}{x} \,|\, (0, +\infty)$ is injective. $\Box$

Exercise 4.5.6 *If $p(x)$ is a non-constant polynomial, $p(a) = b$ and $f \in \mathcal{R}$ has the limit $\lim_{x \to b} f(x) = L$, then*

$$\lim_{x \to a} f(p)(x) = L\,.$$

4.6 Limits of inverse functions

In this original extending section we investigate the interplay of functional limits and inverses of functions.

• *Why is there nothing in the literature about limits of inverse functions?* Maybe because it is clear that if $f \in \mathcal{R}$ is an injective function and

$$\lim_{x \to A} f(x) = B\,,$$

then in general it does <u>not</u> follow that

$$\lim_{y \to B} f^{\langle -1 \rangle}(y) = A\,.$$

Indeed, the former limit means that all points close to A are sent by f to points close to B. However, also some points not close to A may be sent by f to points close to B. If this happens, the latter inverse limit does not exist.

Exercise 4.6.1 *Describe a simple example of such situation.*

But even in the general situation we can say something nontrivial.

Proposition 4.6.2 (limits of inverses) *Suppose that $f \in \mathcal{F}(M)$ is an injective function, $A \in L(M)$ and that*

$$\lim_{x \to A} f(x) = B\,.$$

Then the following is true.

1. *We have $B \in L(f[M])$ and*

$$\lim_{y \to B} f^{\langle -1 \rangle}(y) = A\,,$$

   if this limit exists.

2. *There exists a set $N \subset M$ such that $B \in L(f[N])$ and*

$$\lim_{y \to B} (f \mid N)^{\langle -1 \rangle}(y) = A\,,$$

Proof. Let f, M and A be as stated. 1. We take any sequence $(a_n) \subset M \setminus \{A\}$ with $a_n \to A$ (which exists because $A \in L(M)$) and consider the sequence $(b_n) \equiv (f(a_n))$ ($\subset f[M]$). By Theorem 4.1.12 we have $b_n \to B$. The injectivity of f implies that $b_n = B$ for at most one n and we get that $B \in L(f[M])$. We may assume that $(b_n) \subset f[M] \setminus \{B\}$. Since $b_n \to B$ and

$$f^{\langle -1 \rangle}(b_n) = a_n \to A\,,$$

Theorem 4.1.12 implies that the limit of $f^{\langle -1 \rangle}(y)$ at B, if it exists, has to be A.

2. Just set

$$N \equiv \{a_n :\ n \in \mathbb{N}\}$$

where $(a_n) \subset M \setminus \{A\}$ is any sequence with $a_n \to A$. $\square$

We present three families of theorems on limits of inverse functions. In the first two we present conditions under which the limit $\lim_{x \to A} f(x) = B$ does imply the limit $\lim_{y \to B} f^{\langle -1 \rangle}(y) = A$.

• *First theorem on limits of inverses.* One approach is to strengthen the definition of functional limit.

Definition 4.6.3 (strong functional limit) *Let $f \in \mathcal{F}(M)$, $A \in L(M)$ and $B \in \mathbb{R}^*$. If for every ε there exist δ and θ such that*

$$P(B,\, \theta) \cap f[M] \subset f[P(A,\, \delta)] \subset U(B,\, \varepsilon)\,,$$

we say that the <u>strong limit</u> of f at A is B and we write $(\lim_{x \to A}) f(x) = B$.

The standard limit in Definition 4.1.6 is strengthened by adding the first inclusion.

Theorem 4.6.4 (limits of inverses 1) *Suppose that $f \in \mathcal{F}(M)$ is an injective function, $A \in L(M)$, $B \in \mathbb{R}^*$ and that*

$$(\lim_{x \to A}) f(x) = B\,.$$

Then

$$\lim_{y \to B} f^{\langle -1 \rangle}(y) = A\,.$$

Proof. Let f, M, A and B be as stated. We have $B \in L(f[M])$ by part 1 of Proposition 4.6.2. Suppose for the contrary that the limit of $f^{\langle -1 \rangle}(y)$ at B is not A. Then there is a sequence $(b_n) \subset f[M] \setminus \{B\}$ such that

$$b_n \to B \wedge a_n \equiv f^{\langle -1 \rangle}(b_n) \to A' \neq A\,.$$

We take a δ_0 such that

$$U(A,\, \delta_0) \cap U(A',\, \delta_0) = \emptyset\,.$$

Since $A \in L(M)$ and the limit of $f(x)$ at A is B, we take a sequence $a_n' \subset M \setminus \{A\}$ such that $a_n' \to A$ and (by Theorem 4.1.12) $f(a_n') \to B$. We take an index $m \in \mathbb{N}$ such that $a_m' \in P(A, \delta_0)$ and $f(a_m') \neq B$. Then we take an ε such that

$$f(a_m') \notin U(B,\, \varepsilon)\,.$$

Now for this ε if δ is such that

$$f[P(A,\, \delta)] \subset U(B,\, \varepsilon)\,,$$

then $\delta < \delta_0$. But for every $\delta < \delta_0$ and every θ the inclusion

$$P(B,\, \theta) \cap f[M] \subset f[P(A,\, \delta)]$$

does not hold because f is injective and

$$a_n \in U(A',\, \delta_0), \text{ hence } a_n \notin P(A,\, \delta), \text{ and } f(a_n) = b_n \in P(B,\, \theta) \cap f[M]$$

for every large n. Thus it is not true that the strong limit of $f(x)$ at A is B, contrary to the assumption. $\square$

Exercise 4.6.5 *Why is $a_n \notin P(A, \delta)$ for $\delta < \delta_0$?*

• *Second theorems on limits of inverses.* Another approach is to restrict the function $f(x)$. By a strongly positive function we mean any function

$$g\colon [0,\, +\infty) \to [0,\, +\infty)$$

such that $g(0) = 0$ and that for every $c > 0$ we have

$$\inf\big(g[(c,\, +\infty)]\big) > 0\,.$$

Definition 4.6.6 (lower regulated functions) *We say that a function f in $\mathcal{F}(M)$ is lower regulated if there is a strongly positive function $g(x)$ such that for every two points $x, y \in M$,*

$$|f(x) - f(y)| \ge g(|x - y|)\,.$$

For example, a function $f \in \mathcal{R}$ is lower regulated if there is a constant $c > 0$ such that

$$|f(x) - f(y)| \ge c|x - y|$$

for every $x, y \in M(f)$.

Exercise 4.6.7 *Every lower regulated function is injective.*

Exercise 4.6.8 *If $f\colon (a, b) \to \mathbb{R}$ is differentiable on (a, b), $|f'| \ge c > 0$ on (a, b) and $M \subset (a, b)$, then the restriction $f \mid M$ is lower regulated.*

Theorem 4.6.9 (limits of inverses 2a) *Let $f \in \mathcal{F}(M)$, $A \in L(M)$, $b \in \mathbb{R}$ and let*

$$\lim_{x \to A} f(x) = b\,.$$

If the function f is lower regulated, then

$$\lim_{y \to b} f^{\langle -1 \rangle}(y) = A\,.$$

Proof. Let f, M, A and b be as stated. In view of part 1 of Proposition 4.6.2 and of Theorem 4.1.12 it suffices to show that

$$\text{if } (b_n) \subset f[M] \setminus \{b\} \text{ satisfies } b_n \to b, \text{ then } f^{\langle -1 \rangle}(b_n) \to A\,.$$

We assume for the contrary that a sequence $(b_n) \subset f[M] \setminus \{b\}$ has $\lim b_n = b$ but $f^{\langle -1 \rangle}(b_n) \not\to A$. Passing to a subsequence and using part 2 of Theorem 2.2.11 we may assume that $\lim f^{\langle -1 \rangle}(b_n) = A'$ with $A' \ne A$. We denote $a_n \equiv f^{\langle -1 \rangle}(b_n)$. Using the assumption, we take any sequence $(a'_n) \subset M \setminus \{A\}$ with $a'_n \to A$. By the assumption and Theorem 4.1.12, $b'_n \equiv f(a'_n) \to b$. There is a constant $c > 0$ such that

$$|a_n - a'_n| \ge c \quad (> 0)$$

for every large n. But, since both $f(a_n) \to b$ and $f(a'_n) \to b$, we have

$$\lim_{n \to \infty} |f(a_n) - f(a'_n)| = 0\,.$$

Setting $x \equiv a_n$ and $y \equiv a'_n$ for sufficiently large n, we get a contradiction with the assumption that f is lower regulated. □

We obtain a similar theorem in the case when $\lim_{x \to A} f(x) = \pm\infty$. By an almost bounded function we mean any function

$$g\colon [0, +\infty) \to (0, +\infty)$$

such that for every $c > 0$ we have

$$\sup\big(g[(c, +\infty)]\big) < +\infty\,.$$

Definition 4.6.10 (upper regulated functions) *We say that a function f in $\mathcal{F}(M)$ is* <u>upper regulated</u> *if there is an almost bounded function $g(x)$ such that for every two points $x, y \in M$,*

$$|f(x) - f(y)| \le g(|x - y|)\,.$$

Exercise 4.6.11 *If $f\colon (a,b) \to \mathbb{R}$ is differentiable on (a,b), $|f'| \le c$ on (a,b) and $M \subset (a,b)$, then the restriction $f \mid M$ is upper regulated.*

Theorem 4.6.12 (limits of inverses 2b) *We assume that $f \in \mathcal{F}(M)$, A is in $L(M)$, $B \in \{-\infty, +\infty\}$ and that*

$$\lim_{x \to A} f(x) = B\,.$$

If the function f is injective and upper regulated, <u>*then*</u>

$$\lim_{y \to B} f^{\langle -1 \rangle}(y) = A\,.$$

Proof. Let f, M, A and B be as stated. We proceed as in the previous proof. We bring to contradiction the assumption that there is a sequence (b_n) such that $b_n \to B$ but

$$\lim f^{\langle -1 \rangle}(b_n) = A' \ne A\,.$$

We denote $a_n \equiv f^{\langle -1 \rangle}(b_n)$. Using the assumption, we take any sequence $(a_n') \subset M \backslash \{A\}$ with $a_n' \to A$. By the assumption and Theorem 4.1.12, $b_n' \equiv f(a_n') \to B$. There is a constant $c > 0$ such that

$$|a_n - a_n'| \ge c \ \ (> 0)$$

for every large n. We use Exercise 4.6.13, pass to a subsequence and may assume that

$$\lim |f(a_n) - f(a_n')| = +\infty\,.$$

Setting $x \equiv a_n$ and $y \equiv a_n'$ for sufficiently large n, we get a contradiction with the assumption that f is upper regulated. $\Box$

Exercise 4.6.13 *If* $\lim a_n = \lim b_n = \pm\infty$ *then (b_n) has a subsequence (c_n) such that* $\lim |a_n - c_n| = +\infty$.

• *Third theorems on limits of inverses.* In these results we associate with a given function $f \in \mathcal{R}$ a relation between the sets $L(M(f))$ and $L(f[M(f)])$.

Definition 4.6.14 (relation $\mathcal{L}(f)$) *Let $f \in \mathcal{F}(M)$. We define a binary relation*

$$\mathcal{L}(f) \subset L(M) \times L(f[M])$$

by putting (A, B) in $\mathcal{L}(f)$ $\iff$ there exists a sequence $(a_n) \subset M \setminus \{A\}$ such that $a_n \to A$ and $f(a_n) \to B$.

For a relation $R \subset X \times Y$ we call an element $x \in X$, resp. $y \in Y$, R-isolated if there is no pair $(x, y') \in R$, resp. $(x', y) \in R$. We call the element R-unique if there is exactly one such pair.

Proposition 4.6.15 (relation $\mathcal{L}(f)$) *Let $f \in \mathcal{F}(M)$ be an injective function. The relation $\mathcal{L}(f)$ has two properties.*

1. *No element $A \in L(M)$, resp. $B \in L(f[M])$, is $\mathcal{L}(f)$-isolated.*
2. *An element $A \in L(M)$, resp. $B \in L(f[M])$, is $\mathcal{L}(f)$-unique $\iff$ the limit*
$$\lim_{x \to A} f(x), \text{ resp. } \lim_{y \to B} f^{\langle -1 \rangle}(y), \text{ exists}.$$

Proof. Let f and M be as stated. 1. Let $A \in L(M)$. We take any sequence $(a_n) \subset M \setminus \{A\}$ with $a_n \to A$. By Theorem 2.4.12 there is a subsequence (a_{m_n}) such that $\lim f(a_{m_n}) = B'$. It follows that $B' \in L(f[M])$ and $(A, B') \in \mathcal{L}(f)$. The argument for B is the same, only f is replaced with $f^{\langle -1 \rangle}$.

2. Let $A \in L(M)$. The implication $\Rightarrow$. We assume that A is $\mathcal{L}(f)$-unique. Thus there is a unique $B \in L(f[M])$ such that for a sequence $(a'_n) \subset M \setminus \{A\}$ we have $a'_n \to A$ and $f(a'_n) \to B$. Now let $(a_n) \subset M \setminus \{A\}$ be any sequence with $a_n \to A$. If $f(a_n) \not\to B$ then by using part 2 of Theorem 2.2.11 we would contradict $\mathcal{L}(f)$-uniqueness of A. Hence $f(a_n) \to B$ and $\lim_{x \to A} f(x) = B$ by Theorem 4.1.12.

The implication $\neg \Rightarrow \neg$. Now we assume that there exist two different elements $B, B' \in L(f[M])$ and two sequences $(a_n), (a'_n) \subset M \setminus \{A\}$ such that $a_n, a'_n \to A$, $f(a_n) \to B$ and $f(a'_n) \to B'$. By Theorem 4.1.12 the limit $\lim_{x \to A} f(x)$ does not exist. $\square$

Exercise 4.6.16 *Prove the next theorem.*

Theorem 4.6.17 (limits of inverses 3a) *Let $f \in \mathcal{F}(M)$ be an injective function. We assume that the limit $\lim_{x \to A} f(x)$ exists for every $A \in L(M)$. Then the relation $\mathcal{L}(f)$ has two properties.*

1. *It is a surjective function from $L(M)$ on $L(f[M])$.*
2. *If this function $\mathcal{L}(f)$ is injective, then it is a bijection and*
$$\lim_{y \to B} f^{\langle -1 \rangle}(y) = \mathcal{L}(f)^{\langle -1 \rangle}(B)$$
for every $B \in L(f[M])$.

We obtain version 3b of this theorem dealing with pointwise continuity in Section 6.7.

4.7 Asymptotic notation

Books on computational complexity, or on analysis of algorithms, intersect with books on mathematical analysis in passages devoted to asymptotic notation. In this section we present our version of it. In fact, what does “asymptotic” mean? See Definition 4.7.4.

• *Asymptotic relations.* The symmetric difference of sets X and Y is

$$X\,\Delta\,Y \equiv (X \setminus Y) \cup (Y \setminus X)\,.$$

Definition 4.7.1 (almost equality) *Functions $f = \langle M(f), \mathbb{R}, G_f\rangle$ and $g = \langle M(g), \mathbb{R}, G_g\rangle$ in $\mathcal{R}$ are almost equal, in symbols $f \doteq g$, if the set*

$$G_f\,\Delta\,G_g$$

is finite.

Exercise 4.7.2 *We have the equivalence that $f \doteq g$ iff*

$$M(f)\,\Delta\,M(g) \text{ and } \{x \in M(f) \cap M(g)\colon\ f(x) \neq g(x)\} \text{ are finite sets}\,.$$

Exercise 4.7.3 *The relation $\doteq$ on the set $\mathcal{R}$ is an equivalence relation.*

Definition 4.7.4 (asymptotic relations) *We say that a relation $\mathcal{A} \subset \mathcal{R} \times \mathcal{R}$ is asymptotic if for any functions f, g, f_0 and g_0 in $\mathcal{R}$ such that $f \doteq f_0$ and $g \doteq g_0$ we have the equivalence*

$$f\,\mathcal{A}\,g \iff f_0\,\mathcal{A}\,g_0\,.$$

This resembles the robustness of properties of sequences in Definition 2.1.20.

• *Asymptotic notation O, $\ll$ and other.* A function $f \in \mathcal{R}$ is bounded on a set N if there is a constant $c \geq 0$ such that $|f(x)| \leq c$ for every $x \in M(f) \cap N$.

Definition 4.7.5 (O and $\ll$) *Let $f, g \in \mathcal{R}$ and $N \subset \mathbb{R}$. If the function f/g is bounded on the set N, we say that*

$$f \text{ is big } O \text{ of } g \text{ on } N \text{ and write } f = O(g) \ (\text{on } N)\,.$$

Notation $f \ll g$ (on N) means the same.

In few exceptional cases we consider a more general form of this notation for functions $f\colon M \to \mathbb{C}$ and $g\colon N \to \mathbb{C}$ with $M, N \subset \mathbb{C}$.

Exercise 4.7.6 *Show that $f = O(g)$ (on N) iff there is a constant $c \geq 0$ such that $|f(x)| \leq c|g(x)|$ for every $x \in N \cap M(f) \cap M(g) \setminus Z(g)$.*

Proposition 4.7.7 (O is asymptotic) *Let $N \subset \mathbb{R}$. <u>Then</u> the relation $\mathcal{A} \subset \mathcal{R}^2$, defined by*

$$f\,\mathcal{A}\,g \iff f = O(g) \text{ (on } N\text{)},$$

is asymptotic.

Proof. Let $N \subset \mathbb{R}$, and let f, g, f_0 and g_0 be in $\mathcal{R}$ and such that $f \doteq f_0$, $g \doteq g_0$ and $f = O(g)$ (on N). We show that also $f_0 = O(g_0)$ (on N). By the assumption there is a constant $c \geq 0$ such that $\left|\frac{f(x)}{g(x)}\right| \leq c$ for every $x \in M(f/g) \cap N$. It follows from the definition of the relation $\doteq$ that the set

$$X \equiv \tfrac{f_0}{g_0}[N] \setminus \tfrac{f}{g}[N]$$

is finite (Exercise 4.7.8). If $X \neq \emptyset$ then we set $d \equiv \max(\{|x|:\ x \in X\})$. For $X = \emptyset$ we set $d \equiv 1$. Then for every $x \in M(f_0/g_0) \cap N$ we have $\left|\frac{f_0(x)}{g_0(x)}\right| \leq \max(c, d)$, as needed. $\square$

Exercise 4.7.8 *Why is the set X finite?*

The next exercise shows that a simplified definition of O, whose versions appear in the literature, is not asymptotic.

Exercise 4.7.9 *Suppose that we define, for $f, g \in \mathcal{R}$ and $N \subset \mathbb{R}$, that $f = O'(g)$ (on N) iff for some $c \geq 0$ we have $|f(x)| \leq c|g(x)|$ for every $x \in N \cap M(f) \cap M(g)$. Show that the relation O' is in general not asymptotic.*

The notation

$$f = g + O(h) \text{ (on } N\text{)}$$

has the <u>error form</u> and means that $f - g = O(h)$ (on N). Notation like

$$\log x = O_\varepsilon(x^\varepsilon) \text{ (on } [1, +\infty))$$

means that the constant c in Exercise 4.7.6 is a function of ε. Notation $f \gg g$ (on N) and $f = \Omega(g)$ (on N) mean that $g \ll f$ (on N). Notation $f = \Theta(g)$ (on N) and $f \asymp g$ (on N) both mean that simultaneously $f \ll g$ (on N) and $g \ll f$ (on N).

Exercise 4.7.10 *Answer the following questions.*

1. *Is $x^2 = O(x^3)$ (on $\mathbb{R} \setminus (-1, 1)$)?*
2. *Is $x^2 = O(x^3)$ (on $\mathbb{R}$)?*
3. *Is $x^3 = O(x^2)$ (on $\mathbb{R}$)?*
4. *Is $x^3 = O(x^2)$ (on $(-20, 20)$)?*
5. *Is $\log x = O(x^{1/3})$ (on $(0, +\infty)$)?*

6. *Is* $\log x = O(x^{1/3})$ *(on* $(1, +\infty)$*)?*

• *Asymptotic notation* o*,* ω*, and* $\sim$*.* Now the definitions employ limits of functions.

Definition 4.7.11 (o and ω) *Let* $f, g \in \mathcal{R}$ *and* $A \in L(M(f/g))$*. If*

$$\lim_{x \to A} \tfrac{f(x)}{g(x)} = 0\,,$$

we say that f *is little o of* g *for* $x \to A$ *and write* $f(x) = o(g(x))$ $(x \to A)$*. Notation* $f(x) = \omega(g(x))$ $(x \to A)$ *means the same.*

Like before, $f = g + o(h)$ $(x \to A)$ means that $f - g = o(h)$ $(x \to A)$.

Definition 4.7.12 ($\sim$) *Let* $f, g \in \mathcal{R}$ *and let* $A \in L(M(f/g))$*. If*

$$\lim_{x \to A} \tfrac{f(x)}{g(x)} = 1\,,$$

we say that f *is asymptotically equal to* g *for* $x \to A$ *and write* $f(x) \sim g(x)$ $(x \to A)$*.*

For example, $x^2 \sim (x-3)^2$ $(x \to +\infty)$.

Proposition 4.7.13 (o and $\sim$ are asymptotic) *Let* $A \in \mathbb{R}^*$*. Then the two relations* $\mathcal{A}, \mathcal{B} \subset \mathcal{R}^2$*, defined by*

$$f\,\mathcal{A}\,g \iff f = o(g) \quad (x \to A) \ \text{ and } \ f\,\mathcal{B}\,g \iff f \sim g \quad (x \to A)\,,$$

are asymptotic.

Proof. Let $A \in \mathbb{R}^*$ and f, g, f_0 and g_0 be functions in $\mathcal{R}$ such that $f \doteq f_0$, $g \doteq g_0$. Let $f = o(g)$ $(x \to A)$. We show that then $f_0 = o(g_0)$ $(x \to A)$ as well. For the relation $\mathcal{B}$ we argue similarly. By the assumption, A is in $L(f/g)$ and $\lim_{x \to A} \frac{f(x)}{g(x)} = 0$. By Exercise 4.7.8 the set $\frac{f_0}{g_0}[\mathbb{R}] \Delta \frac{f}{g}[\mathbb{R}]$ is finite. Hence $A \in L(f_0/g_0)$ and $\lim_{x \to A} \frac{f_0(x)}{g_0(x)} = \lim_{x \to A} \frac{f(x)}{g(x)} = 0$. □

Exercise 4.7.14 *Answer the following questions.*

1. *Is* $x^2 = o(x^3)$ $(x \to +\infty)$*?*
2. *Is* $x^3 = o(x^2)$ $(x \to 0)$*?*
3. *Is* $x^2 = o(x^3)$ $(x \to 0)$*?*
4. *Is* $(x+1)^3 \sim x^3$ $(x \to 1)$*?*
5. *Is* $(x+1)^3 \sim x^3$ $(x \to +\infty)$*?*
6. *Is* $\mathrm{e}^{-1/x^2} = o(x^{20})$ $(x \to 0)$*?*

• *Properties and mutual relations of asymptotic symbols.* We cannot devote to this important topic much space. We restrict to one proposition and ten exercises.

Proposition 4.7.15 (o implies O) *Let $f, g \in \mathcal{R}$. If $f(x) = o(g(x))$ $(x \to A)$, then there is a θ such that*

$$f = O(g) \text{ (on } P(A, \theta)) .$$

Proof. Let f, g and A be as stated. Since $\lim_{x\to A} \frac{f(x)}{g(x)} = 0$, for $\varepsilon = 1$ there is a θ such that for every $x \in M(f/g) \cap P(A, \theta)$,

$$\big|\tfrac{f(x)}{g(x)}\big| = \big|\tfrac{f(x)}{g(x)} - 0\big| < 1 .$$

Hence $f = O(g)$ (on $P(A, \theta)$). $\square$

In the ten exercises, f, g and h are in $\mathcal{R}$, $N \subset \mathbb{R}$ and $A \in \mathbb{R}^*$.

Exercise 4.7.16 *If $g = o(h)$ $(x \to A)$, $f(x) = O(x)$ (on N) and*

$$A \in L\big(M((f \mid N)(g))\big) ,$$

then $f(g) = o(h)$ $(x \to A)$.

Exercise 4.7.17 *If $f = O(h)$ (on N) and $g = O(h)$ (on N), then $f+g = O(h)$ (on N).*

Exercise 4.7.18 *If $f = O(h)$ (on N) and g is bounded on N, then $fg = O(h)$ (on N).*

Exercise 4.7.19 *If $f = O(h)$ (on N) and $\frac{1}{g}$ is bounded on N, then $f/g = O(h)$ (on N).*

Exercise 4.7.20 *If $f = o(h)$ $(x \to A)$, $g = o(h)$ $(x \to A)$ and $A \in L(\frac{f+g}{h})$, then $f + g = o(h)$ $(x \to A)$.*

Exercise 4.7.21 *If $f = o(h)$ $(x \to A)$, g is bounded on $P(A, \theta)$ for some θ and $A \in L(\frac{fg}{h})$, then $fg = o(h)$ $(x \to A)$.*

Exercise 4.7.22 *If $f = o(h)$ $(x \to A)$, $\frac{1}{g}$ is bounded on $P(A, \theta)$ for some θ and $A \in L(\frac{f}{gh})$, then $f/g = o(h)$ $(x \to A)$.*

Exercise 4.7.23 *If $f(x) \sim h(x)$ $(x \to A)$, $g(x) = o(h(x))$ $(x \to A)$ and A is in $L(\frac{f+g}{h})$, then $f(x) + g(x) \sim h(x)$ $(x \to A)$.*

Exercise 4.7.24 *If $f(x) \sim h(x)$ $(x \to A)$, $\lim_{x\to A} g(x) = 1$ and $A \in L(\frac{fg}{h})$ then $f(x)g(x) \sim h(x)$ $(x \to A)$.*

Exercise 4.7.25 *If $f(x) \sim h(x)$ $(x \to A)$, $\lim_{x\to A} g(x) = 1$ and $A \in L(\frac{f}{gh})$ then $f(x)/g(x) \sim h(x)$ $(x \to A)$.*

Notation o, O and $\sim$ was introduced by the German mathematicians *Paul Bachmann (1837–1920)* and *Edmund Landau (1877–1938).* Notation $\ll$, $\gg$ and $\asymp$ is due to the Russian mathematician *Ivan M. Vinogradov (1891–1983).*

• *Famous asymptotics.* The prime number counting function $\pi\colon \mathbb{R} \to \mathbb{N}_0$ is defined by

$$\pi(x) \equiv |(-\infty,\, x] \cap \mathbb{P}|$$

where $\mathbb{P} = \{2, 3, 5, 7, 11, \dots\}$ is the set of prime numbers. In 1896 the French mathematician *Jacques Hadamard (1865–1963)* and, in parallel with him, the Belgian mathematician *Charles Jean de la Vallée Poussin (1866–1962)* proved the famous Prime Number Theorem (PNT):

$$\pi(x) \sim \frac{x}{\log x} \quad (x \to +\infty)\,.$$

In Section 2.4 we introduced for $k, n \in \mathbb{N}$ the number $r_k(n) \in \mathbb{N}_0$ as the size of the largest set $X \subset [n]$ containing no arithmetic progression with length k. In 1975 E. Szemerédi proved the now famous theorem, which we mentioned in Section 2.4, that for every k,

$$r_k(n) = o(n) \quad (n \to +\infty)\,.$$

For $x \in \mathbb{R}$ we define

$$D(x) \equiv |\{(m,\, n) \in \mathbb{N}^2\colon\ mn \le x\}|\,.$$

For $n \in \mathbb{N}$ we denote by $\tau(n)$ the number of divisors of n. For example,

$$\tau(28) = |\{1,\, 2,\, 4,\, 7,\, 14,\, 28\}| = 6\,.$$

Exercise 4.7.26 *Show that $D(x) = \sum_{n=1}^{\lfloor x \rfloor} \tau(n)$.*

The (Dirichlet) divisor problem is the problem to estimate the error in asymptotics of $D(x)$. In 1849 the German mathematician *Peter L. Dirichlet (1805–1859)* proved that

$$D(x) = x\log x + (2\gamma - 1)x + O(\sqrt{x}) \quad (\text{on } [1, +\infty))\,,$$

where γ is Euler's constant. In 1903 the Russian-Ukrainian mathematician *Georgij F. Voronoj (1868–1908)* improved it to

$$D(x) = x\log x + (2\gamma - 1)x + O\big(x^{1/3}\log x\big) \quad (\text{on } [2,\, +\infty))\,.$$

Exercise 4.7.27 *Why not on $[1, +\infty)$ as before?*

The 20th century saw a series of further advances in the divisor problem. The current record holder is the British mathematician *Martin N. Huxley (1944)* who proved in 2003 that for every ε,

$$D(x) = x \log x + (2\gamma - 1)x + O_\varepsilon\big(x^{131/416+\varepsilon}\big) \ \ (\text{on } [1, +\infty))\,.$$

For $n \in \mathbb{N}$ and an algorithm (Turing machine) T for multiplying integers we define $T(n)$ as the smallest $k \in \mathbb{N}$ such that T multiplies any two n-digit numbers in at most k steps. The elementary school algorithm T_{es} works in

$$T_{\mathrm{es}}(n) = O(n^2) \ (\text{on } \mathbb{N})$$

steps. In 1960 the Russian mathematician *Anatolij A. Karacuba (1937–2008)* invented an algorithm T_{K} working in

$$T_{\mathrm{K}}(n) = O\big(n^{\log_2 3}\big) = O\big(n^{1.585\ldots}\big) \ (\text{on } \mathbb{N})$$

steps. In 2021 computer scientists *David Harvey* from Australia and *Joris van der Hoeven (1971)* from the Netherlands discovered an algorithm T_{HH} for multiplying integers with complexity

$$T_{\mathrm{HH}}(n) = O(n \log n) \ \ (\text{on } \mathbb{N})\,.$$

Exercise 4.7.28 *But for $n = 1$ the ratio $\frac{T_{\mathrm{HH}}(n)}{n \log n}$ is not defined?*

4.8 Asymptotic expansions

Asymptotic expansions provide infinite sequences of better and better approximations of the given function.

- *Asymptotic expansions.* We define asymptotic scales.

Definition 4.8.1 (asymptotic scales) *A sequence of functions $(f_n) \subset \mathcal{R}$ is an asymptotic scale for $x \to A$ if A is in $L(\bigcap_{n=1}^\infty M(f_n))$, there is a θ such that $f_n \ne 0$ on $P(A, \theta)$ for every n, and for every n we have*

$$f_{n+1}(x) = o(f_n(x)) \ (x \to A)\,.$$

For example, (x^{-n}) is an asymptotic scale for $x \to +\infty$, and (x^n) for $x \to 0$.

Definition 4.8.2 (asymptotic expansions) *Suppose that $(a_n) \subset \mathbb{R}$, $f \in \mathcal{R}$ and that $(f_n) \subset \mathcal{R}$ is an asymptotic scale for $x \to A$. If for every n we have*

$$f(x) = \textstyle\sum_{i=1}^n a_i f_i(x) + o(f_{n+1}(x)) \ \ (x \to A)\,,$$

we say that $(a_n f_n(x))$ is an asymptotic expansion of $f(x)$ for $x \to A$ and write it symbolically as

$$f(x) \approx \textstyle\sum_{n=1}^\infty a_n f_n(x) \ \ (x \to A)\,.$$

From this definition it follows that for every n there is a $\theta_n > 0$ such that $f = \sum_{i=1}^n a_i f_i + O(f_{n+1})$ (on $P(A, \theta_n)$). Thus we have a sequence of asymptotic approximations $\sum_{i=1}^n a_i f_i$ to f with errors of order f_{n+1}.

Exercise 4.8.3 *Prove it.*

The assumption that (f_n) is an asymptotic scale ensures that as n grows, magnitudes of these errors get smaller and smaller. For fixed $x \in \mathbb{R}$, nothing is assumed about the convergence of the series $\sum a_n f_n(x)$; it typically diverges. Usually it is *not* true that $f(x) = \sum_{n=1}^\infty a_n f_n(x)$.

• *Three asymptotic expansion.* We conclude this chapter with three examples of asymptotic expansions. Their proofs will be given in *MA 1*$^+$. In 1730, the Scottish mathematician *James Stirling (1692–1770)* obtained an asymptotic expansion of $\log(n!)$ for $n \to +\infty$. We state his expansion in a moment. Modern theory of asymptotic expansions starts with the French mathematician *Henri Poincaré (1854–1912)* and his memoir of 1886. For expositions of asymptotic expansions, see [22, 30].

Theorem 4.8.4 (asymptotic expansion of $\log(n!)$**)** *For* $n \in \mathbb{N}$,

$$\log(n!) \approx (n + \tfrac{1}{2}) \log n - n + \tfrac{1}{2} \log(2\pi) + \textstyle\sum_{k=1}^\infty \frac{B_{2k}}{2k(2k-1)} \cdot n^{1-2k} \quad (n \to +\infty)\,.$$

Theorem 4.8.5 (harmonic numbers) *For* $n \in \mathbb{N}$,

$$h_n = \textstyle\sum_{i=1}^n \frac{1}{i} \approx \log n + \gamma + \frac{1}{2n} - \sum_{k=1}^\infty \frac{B_{2k}}{2k} \cdot n^{-2k} \quad (n \to +\infty)\,.$$

In these two expansions, B_k $(\in \mathbb{Q})$ for $k \in \mathbb{N}_0$ denote the Bernoulli numbers. They are defined by the power series expansion

$$\tfrac{x}{\exp x - 1} = \textstyle\sum_{k=0}^\infty \frac{B_k}{k!} \cdot x^k$$

and are named after their discoverer, the Swiss mathematician *Jacob Bernoulli (1655/54–1705).* They satisfy $B_{2k+1} = 0$ for every $k \in \mathbb{N}$ and have initial values $B_0 = 1$, $B_1 = -\frac{1}{2}$, $B_2 = \frac{1}{6}$, $B_4 = -\frac{1}{30}$, $B_6 = \frac{1}{42}$, $B_8 = -\frac{1}{30}$, $B_{10} = \frac{5}{66}$, $B_{12} = -\frac{691}{2730}$, $B_{14} = \frac{7}{6}$ and $B_{16} = -\frac{3617}{510}$.

Exercise 4.8.6 *Deduce from the displayed definitoric expansion a recurrence for* B_k *and show that* $B_{2k+1} = 0$ *for every* $k \in \mathbb{N}$.

In [22] we read that the formula in Theorem 4.8.4 is not exactly the original expansion of Stirling, but a very similar formula due to the French mathematician *Abraham de Moivre (1667–1754).* The asymptotic expansion of harmonic numbers is due to L. Euler.

The third asymptotic expansion is much more recent, due to [60]. We say that a graph $G = (V, E)$ is connected if for every partition $\{A, B\}$ of the vertex set V there is an edge $e \in E$ that intersects both blocks A and B.

Exercise 4.8.7 *Equivalently, a graph G is connected iff for every two vertices $u, v \in V$ there exists a $(k+1)$-tuple of vertices*

$$w = \langle u_0,\, u_1,\, \ldots,\, u_k \rangle, \quad k \in \mathbb{N}_0\,,$$

such that $u_0 = u$, $u_k = v$ and $\{u_{i-1}, u_i\} \in E$ for every $i \in [k]$. We say that w is a walk in G joining u and v.

Theorem 4.8.8 (probability of connectedness) *For $n \in \mathbb{N}$,*

$$\begin{aligned}&\tfrac{1}{2^{n(n-1)/2}} \cdot |\{G = ([n],\, E)\colon\ G \text{ is a connected graph}\}| \\ &\approx 1 - \textstyle\sum_{k=1}^{\infty} t_k \cdot 2^{k(k+1)/2} \cdot \binom{n}{k} 2^{-kn} \quad (n \to +\infty)\,,\end{aligned}$$

where t_k is the number of irreducible tournaments of size k (see below).

As mentioned, this result is due to [60]. Since there are $2^{n(n-1)/2}$ graphs $G = ([n], E)$, the product on the left-hand side equals to the probability that a random graph with the vertex set $[n]$ is connected. By the first term of the expansion, for $n \to +\infty$ this probability goes to 1. What is t_k? A tournament $T = (V, E)$ is a pair of a nonempty finite set V of vertices and an irreflexive relation $E \subset V \times V$ such that for every two distinct vertices $u, v \in V$ exactly one pair of (u, v) and (v, u) is in E. We say that T is irreducible if for every partition $\{A, B\}$ of the vertices V there exist pairs $(a, b) \in A \times B$ and $(c, d) \in B \times A$ such that $(a, b), (c, d) \in E$. Then t_k is the number of irreducible tournaments $T = ([k], E)$. By [61, A054946] the initial values of t_k are

$$(t_1,\, t_2,\, \ldots) = (1,\, 0,\, 2,\, 24,\, 544,\, 22320,\, 1677488,\, \ldots)\,.$$

See [10, 11] for a theory of a calculus by which one can derive asymptotic expansions in certain problems of enumerative combinatorics and theoretical physics.

Chapter 5

Elementary functions

5.1 Basic elementary functions

In this and next two sections we introduce five families of functions $f\colon M \to \mathbb{R}$ with $M \subset \mathbb{R}$. Namely, basic elementary functions (BEF), elementary functions (EF), really basic elementary functions (RBEF), polynomials (POL) and rational functions (RAC).

Definition 5.1.1 (BEF) *Basic elementary functions, or* BEF*, are the following functions.*

1. *The constant functions $k_c(x)$ for $c \in \mathbb{R}$.*
2. *The functions* $\exp x$ *and* $\log x$*.*
3. *The functions a^x for $a > 0$, x^b for $b \in \mathbb{R}$, 0^x and x^m for $m \in \mathbb{Z}$.*
4. *The functions* $\sin x$, $\cos x$, $\tan x$ *and* $\cot x$*.*
5. *The functions* $\arcsin x$, $\arccos x$, $\arctan x$ *and* $\operatorname{arccot} x$*.*

We review them shortly. The functions x^b and x^m differ in definition domains.

• *Constant functions*, or constants, are functions ($c \in \mathbb{R}$)

$$k_c\colon \mathbb{R} \to \mathbb{R},\quad k_c(x) = c\,.$$

Instead of $k_c(x)$ we usually write k_c or just c.

Exercise 5.1.2 *How many constant functions $k_c(x)$ are there?*

• *The exponential function*

$$\exp x = \exp(x) = \mathrm{e}^x\colon \mathbb{R} \to \mathbb{R}$$

is for $x \in \mathbb{R}$ defined by the sum

$$\exp x \equiv \sum_{n=0}^{\infty} \frac{1}{n!} x^n = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \dots\,,$$

where $x^0 = 0^0 \equiv 1$.

Exercise 5.1.3 *For every $x \in \mathbb{R}$ the series $\sum_{n=0}^{\infty} \frac{1}{n!} x^n$ is abscon.*

We prove the exponential identity.

Theorem 5.1.4 (exponential identity) *For every $x, y \in \mathbb{R}$ we have*

$$\exp(x+y) = \exp(x) \cdot \exp(y)\,.$$

Proof. Let $x, y \in \mathbb{R}$. By Theorem 3.5.45 and Exercises 5.1.3 and 2.1.28, the product $\mathrm{e}^x \cdot \mathrm{e}^y$ equals

$$\sum_{n=0}^{\infty} \sum_{k=0}^{n} \frac{x^k}{k!} \cdot \frac{y^{n-k}}{(n-k)!} = \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k} = \sum_{n=0}^{\infty} \frac{(x+y)^n}{n!}$$

which is e^{x+y}. □

Nothing changes in the complex domain. Here are some more properties of $\exp x$.

Exercise 5.1.5 *Prove parts 1–3 of the following proposition.*

Proposition 5.1.6 (properties of e^x) *The following holds.*

1. *For every $x \in \mathbb{R}$, we have $\exp x > 0$ and $\exp(-x) = \frac{1}{\exp x}$. Also, $\exp 0 = 1$.*
2. *For all real $x < y$ we have $\exp x < \exp y$.*
3. *$\lim_{x \to -\infty} \exp x = 0$ and $\lim_{x \to +\infty} \exp x = +\infty$.*
4. *The function $\exp$ is a bijection from $\mathbb{R}$ to $(0, +\infty)$.*

We prove part 4 later in Corollary 6.4.3.

• Euler's number is the sum

$$\mathrm{e} \equiv \exp 1 = \sum_{n=0}^{\infty} \frac{1}{n!} = 2 + \frac{1}{2!} + \frac{1}{3!} + \dots = 2.71828\cdots\,.$$

Exercise 5.1.7 *Show that e is irrational. Hint: multiply the equality $\sum_{j=0}^{\infty} \frac{1}{j!} = \frac{n}{m}$ by $m!$.*

• *(Natural)* logarithm is the inverse of the exponential function,

$$\log\colon (0,\, +\infty) \to \mathbb{R}, \quad \log x \equiv \exp^{-1}(x)\,.$$

Properties of $\log x$ follow from the properties of e^x.

Exercise 5.1.8 *Prove the following proposition.*

Proposition 5.1.9 (properties of $\log x$) *The following holds.*

1. *For all real $x, y > 0$ we have*
$$\log(xy) = \log x + \log y\,.$$
If $x < y$ then $\log x < \log y$. Also, $\log 1 = 0$.

2. $\lim_{x\to 0} \log x = -\infty$ *and* $\lim_{x\to+\infty} \log x = +\infty$.

3. *Logarithm is a bijection from $(0, +\infty)$ to $\mathbb{R}$.*

• *Real exponentiation.* In the expression a^b, we call a the base and b the exponent. We consider two families of real exponentials. One expresses a^b by $\exp(b \log a)$ and extends it by the limit $\lim_{x\to-\infty} \exp x = 0$. The other makes use of iterated multiplication.

Definition 5.1.10 (a^b analytically) *We define the following functions.*

1. *Let $a > 0$. We set*
$$a^x \equiv \exp(x \log a) \quad (\in \mathcal{F}(\mathbb{R}))\,.$$

2. *Let $b > 0$. We set $0^b \equiv 0$ and*
$$x^b \equiv \exp(b \log x) \text{ for } x > 0\,.$$
Thus $x^b \in \mathcal{F}([0, +\infty))$.

3. *Let $b \le 0$. We set*
$$x^b \equiv \exp(b \log x) \quad (\in \mathcal{F}((0, +\infty)))\,.$$

4. *We define*
$$0^x \equiv k_0(x) \,|\, (0, +\infty) \quad (\in \mathcal{F}((0, +\infty))$$

We do not define 0^0 and always $a^b \ge 0$. Odd roots like $\sqrt[3]{x} \equiv x^{1/3}$, $\sqrt[5]{x} \equiv x^{1/5}$, etc. are sometimes defined for every $x \in \mathbb{R}$, for example, $\sqrt[3]{-8} = -2$. We define them only for $x > 0$. Note that in parts 2 and 3 definition domains differ.

Definition 5.1.11 (square roots) *We define the (square) root by*
$$\sqrt{x} \equiv x^{1/2} \quad (\in \mathcal{F}([0, +\infty)))\,.$$

Exercise 5.1.12 *So $\sqrt{x} = \exp(\frac{1}{2} \log x)$, right?*

Exercise 5.1.13 *Let $a \ge 0$. Then $\sqrt{a}$ is the unique number $b \ge 0$ such that $b^2 = a$.*

We proceed to the other family of real exponentials.

Definition 5.1.14 (a^b algebraically) *Let $m \in \mathbb{Z}$, where now $\mathbb{Z}$ is understood as disjoint from $\mathbb{R}$. We define the following functions.*

1. *If $m > 0$ then*
$$x^m \equiv \underbrace{x \cdot x \cdot \ldots \cdot x}_{m \text{ factors}} \ \ (\in \mathcal{F}(\mathbb{R}))\,.$$
2. *We set $x^0 \equiv k_1(x)$ ($\in \mathcal{F}(\mathbb{R})$).*
3. *If $m < 0$ then*
$$x^m \equiv \tfrac{k_1(x)}{x^{-m}} \ \ (\in \mathcal{F}(\mathbb{R} \setminus \{0\}))\,.$$

So now $0^0 \equiv 1$ and a^b may be negative. For $m = 1$ we get the identity function $x^1 = x$ ($\in \mathcal{F}(\mathbb{R})$). This function also arises as the composition $\log(\mathrm{e}^x)$.

Exercise 5.1.15 *Definitions 5.1.10 and 5.1.14 coincide on the intersection.*

Exercise 5.1.16 *Show that*
$$\mathrm{e}^x = \exp x$$
for every $x \in \mathbb{R}$. On the left, we have the real exponentiation with the base $\mathrm{e} = 2.71\ldots$ and exponent x. On the right, we have a value of the exponential function.

Exercise 5.1.17 ($(+\infty)^0$) *Show that for every sequence $(a_n) \subset (0, +\infty)$ with $\lim a_n = +\infty$ and for every nonnegative $A \in \mathbb{R}^*$ there exists a sequence (b_n) such that $\lim b_n = 0$ and*
$$\lim_{n\to\infty} \big(a_n\big)^{b_n} = A\,.$$

• *Exponential identities.* We discuss some well-known and some not so well-known identities for real exponentiation.

Theorem 5.1.18 (three exponential identities) *Let $a, b > 0$. For every real x and y the following holds.*

1. $(a \cdot b)^x = a^x \cdot b^x$.
2. $a^x \cdot a^y = a^{x+y}$.
3. $\big(a^x\big)^y = a^{x \cdot y}$.

Proof. 1. Indeed, $(ab)^x$ equals
$$\exp(x \log(ab)) = \exp(x \log a + x \log b) = \exp(x \log a) \cdot \exp(x \log b) = a^x b^x\,.$$

2. Indeed, $a^x a^y$ equals
$$\exp(x \log a) \exp(y \log a) = \exp(x \log a + y \log a) = \exp((x + y) \log a) = a^{x+y}\,.$$

3. Indeed,

$$(a^x)^y = \exp(y \log(\exp(x \log a))) = \exp(yx \log a) = a^{xy} .$$

$\square$

If we drop the assumption that $a > 0$, the third identity ceases to hold:

$$\left((-1)^2\right)^{\frac{1}{2}} = 1^{\frac{1}{2}} = 1 \neq -1 = (-1)^1 = (-1)^{2 \cdot \frac{1}{2}} ,$$

where we used both Definitions 5.1.10 and 5.1.14 of real exponentiation.

Exercise 5.1.19 *How exactly did we use them?*

A. Tarski conjectured that every identity for real exponentiation, for example

$$x^y \cdot (x^y)^y = x^{y+y^2} ,$$

can be derived from the three previous identities and from other basic properties of addition, multiplication, and exponentiation. In 1981 the British mathematician *Alex Wilkie (1948)* refuted Tarski's conjecture and showed that exponential identities like

$$\begin{aligned} &\left((1+x)^y + (1+x+x^2)^y\right)^x \cdot \left((1+x^3)^x + (1+x^2+x^4)^x\right)^y \\ &= \left((1+x)^x + (1+x+x^2)^x\right)^y \cdot \left((1+x^3)^y + (1+x^2+x^4)^y\right)^x \end{aligned}$$

cannot be derived from the three basic identities. Identities of this kind are now called <u>Wilkie's identities</u>. We return to them in *MA 1*$^+$.

Exercise 5.1.20 *Prove that for every real $x, y > 0$ the stated Wilkie's identity holds. Hint:* $(1 + x) \cdot (1 + x^2 + x^4) = (1 + x^3) \cdot (1 + x + x^2)$.

The next exercise shows that 0^0 is an indefinite expression.

Exercise 5.1.21 *Let $A \in \mathbb{R}^*$ with $A \geq 0$. Then there exist sequences $(a_n) \subset (0, +\infty)$ and $(b_n) \subset \mathbb{R}$ such that* $\lim a_n = \lim b_n = 0$ *and*

$$\lim (a_n)^{b_n} = A .$$

Could A be negative?

However, the definition $0^0 \equiv 1$ is often useful.

• *Cosine and sine.* We define cosine and sine for $t \in \mathbb{R}$ by the sums

$$\cos t \equiv \textstyle\sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n)!} t^{2n} \ \ (0^0 = 1) \ \text{ and } \ \sin t \equiv \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)!} t^{2n+1} .$$

In other words,

$$\cos t = 1 - \tfrac{t^2}{2} + \tfrac{t^4}{24} - \cdots \ \text{ and } \ \sin t = t - \tfrac{t^3}{6} + \tfrac{t^5}{120} - \cdots \ \ (\in \mathcal{F}(\mathbb{R})) .$$

Exercise 5.1.22 *For every $t \in \mathbb{R}$, the series defining $\cos t$ and $\sin t$ are abscon.*

The planar set

$$S \equiv \{(x,\, y) \in \mathbb{R}^2 :\ x^2 + y^2 = 1\}$$

is the unit circle. It has radius 1 and center $(0,0)$. The next theorem, whose proof we postpone to *MA 1*$^+$, decribes the main geometric property of cosine and sine.

Theorem 5.1.23 (runner's) *Let $t \in \mathbb{R}$. A runner starts at the point $(1,0)$ of the track S and runs on S with unit speed. For $t > 0$ she runs counter-clockwise, and for $t \le 0$ clockwise. Then*

$$\textit{in time } |t| \textit{ the runner is at the point } (\cos t,\, \sin t) \in S\,.$$

• *The number π.* One can define it in several ways.

Definition 5.1.24 (the number π) *The number*

$$\pi = 3.14159\dots$$

is twice the minimum number $\alpha > 0$ such that $\cos \alpha = 0$.

The existence of α follows from the continuity of cosine (see Corollary 6.7.6), from Theorem 6.4.1 and from the values $\cos 0 = 1$ and $\cos 2 < 0$ (Exercise 5.1.25).

Or one can define π as half of the circumference of S, which equals the time when the runner passes for the second time through the point $(1,0)$. This is informal because the length of a circular arc will be introduced only in *MA 1*$^+$.

Exercise 5.1.25 *Using Theorem 3.5.31 show that $\cos 1 > 0$ and $\cos 2 < 0$.*

• *More on cosine and sine.* Here are some properties of these functions.

Exercise 5.1.26 *Deduce from Theorem 5.1.23 the next proposition.*

Proposition 5.1.27 (on $\sin x$ and $\cos x$) *The following holds.*

1. *Both functions are 2π-periodic, for every $t \in \mathbb{R}$ we have*

$$\cos(t + 2\pi) = \cos t \ \ \textit{and} \ \ \sin(t + 2\pi) = \sin t\,.$$

2. *Sine increases on the interval $[0, \frac{\pi}{2}]$ from 0 to 1.*

3. *For every $t \in [0, \pi]$ we have*

$$\sin t = \sin(\pi - t)$$

and for every $t \in [0, 2\pi]$ we have

$$\sin t = -\sin(2\pi - t)\,.$$

4. For every $t \in \mathbb{R}$ we have

$$\cos t = \sin(t + \tfrac{\pi}{2}) \ \textit{and} \ \cos^2 t + \sin^2 t = 1\,.$$

5. For every $s, t \in \mathbb{R}$ we have

$$\begin{aligned} \sin(s \pm t) &= \sin s \cdot \cos t \pm \cos s \cdot \sin t \ \textit{and} \\ \cos(s \pm t) &= \cos s \cdot \cos t \mp \sin s \cdot \sin t\,. \end{aligned}$$

• *Euler's formula.* We present an important complex identity relating the three functions $\exp x$, $\sin x$ and $\cos x$. We extend for it series to the complex domain. For a complex sequence

$$(z_n) \subset \mathbb{C}$$

we define the <u>limit</u> $\lim z_n$ to be the unique number $z \in \mathbb{C}$, if it exists, such that for every ε there is an n_0 such that for every $n \geq n_0$ we have $|z_n - z| \leq \varepsilon$.

Exercise 5.1.28 *Let $(z_n) = (a_n + b_n i) \subset \mathbb{C}$ with $\lim z_n = a + bi$. Then*

$$a = \lim a_n \ \textit{and} \ b = \lim b_n\,.$$

We extend the exponential function to $\mathbb{C}$ by using the same formula

$$\mathrm{e}^z = \exp z = \textstyle\sum_{n=0}^{\infty} \frac{z^n}{n!} \equiv \lim\limits_{n \to \infty} \sum_{j=0}^{n} \frac{z^j}{j!} \quad (z \in \mathbb{C})\,.$$

Exercise 5.1.29 *This limit exists for every complex number z.*

Since the form of the series is the same as for the real exponential, the function

$$\mathrm{e}^z \colon \mathbb{C} \to \mathbb{C}$$

extends the real exponential. Recall that i denotes the imaginary unit, $i^2 = -1$.

Theorem 5.1.30 (Euler's formula) *For every $t \in \mathbb{R}$ we have*

$$\exp(it) = \cos t + i \sin t\,.$$

Proof. Let $t \in \mathbb{R}$. Since the natural powers of i form a 4-periodic sequence

$$(i^n) = (i,\, -1,\, -i,\, 1,\, i,\, -1,\, -i,\, 1,\, i,\, \dots)\,,$$

we have for every $n \in \mathbb{N}_0$ that

$$\textstyle\sum_{j=0}^{n} \frac{(it)^j}{j!} = \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k \frac{1}{(2k)!} t^{2k} + i \sum_{l=1}^{\lceil n/2 \rceil} (-1)^{l-1} \frac{1}{(2l-1)!} t^{2l-1}\,.$$

We send $n \to \infty$ and get by Exercise 5.1.28 and by the definitions of $\exp(it)$, $\cos t$ and $\sin t$ Euler's formula. □

• *Another beautiful exponential identity.* In [53, Lemma 17.1] we found a remarkable result on the function e^z. In [53] it serves as a lemma in proofs of limit theorems in probability theory. In the proof we follow roughly [53] and begin with a lemma.

Lemma 5.1.31 (telescoping) *Let $u_1, \ldots, u_n, v_1, \ldots, v_n$ be $2n$ formal variables.* <u>*Then*</u>

$$u_1 u_2 \ldots u_n - v_1 v_2 \ldots v_n = \sum_{i=1}^{n} V_{i-1}(u_i - v_i)U_{i+1}\,,$$

where $V_0 = 1$, $V_j = v_1 v_2 \ldots v_j$ for $j \in [n]$, $U_{n+1} = 1$ and $U_j = u_j u_{j+1} \ldots u_n$ for $j \in [n]$.

Exercise 5.1.32 *Prove the lemma.*

Theorem 5.1.33 (sums and products) *Let $a \in \mathbb{C}$, $(k_n) \subset \mathbb{N}$ and for $n \in \mathbb{N}$ let*

$$\langle a_{n,\,j}:\ j \in [k_n] \rangle \quad (\subset \mathbb{C})$$

be k_n-tuples of complex numbers satisfying three conditions.

1. $\lim_{n\to\infty} \sum_{j=1}^{k_n} a_{n,j} = a$.
2. *There is a constant $c > 0$ such that $\sum_{j=1}^{k_n} |a_{n,j}| \le c$ for every $n \in \mathbb{N}$.*
3. $\lim_{n\to\infty} \sum_{j=1}^{k_n} |a_{n,j}|^2 = 0$.

<u>*Then*</u>

$$\lim_{n\to\infty} \prod_{j=1}^{k_n} \big(1 + a_{n,\,j}\big) = \exp(a)\,.$$

Proof. Using condition 3 we take an $n_0 \in \mathbb{N}$ such that for every $n \ge n_0$ and $i \in [k_n]$ we have $|a_{n,i}| \le \frac{1}{2}$. Using the exponential identity and condition 2 we have for every $n \ge n_0$ and every set $A \subset [k_n]$ that

$$\big|\prod_{j\in A}(1 + a_{n,j})\big| \le \prod_{j\in A}(1 + |a_{n,j}|) \le \exp\big(\sum_{j\in A} |a_{n,j}|\big) \le \mathrm{e}^c$$

and, using Exercise 5.1.34 and condition 2, that

$$\big|\exp\big(\sum_{j\in A} a_{n,j}\big)\big| = \exp\big(\mathrm{re}\big(\sum_{j\in A} a_{n,j}\big)\big) \le \exp\big(\sum_{j\in A} |a_{n,j}|\big) \le \mathrm{e}^c\,.$$

Let $n \ge n_0$ and $i \in [k_n]$. In Lemma 5.1.31 we set $u_i \equiv 1 + a_{n,i}$ and $v_i \equiv \exp(a_{n,i})$. We get with the help of Exercise 5.1.35, the exponential identity, the above bounds and condition 3 that

$$\begin{aligned}&\big|\prod_{j=1}^{k_n}(1 + a_{n,\,j}) - \exp\big(\sum_{j=1}^{k_n} a_{n,\,j}\big)\big| = \big|\sum_{i=1}^{k_n} V_{i-1}(u_i - v_i)U_{i+1}\big| \le \\ &\le \sum_{i=1}^{k_n} |V_{i-1}| \cdot |u_i - v_i| \cdot |U_{i+1}| \le \sum_{i=1}^{k_n} \mathrm{e}^c |a_{n,\,i}|^2 \mathrm{e}^c \to 0, \quad n \to \infty\,.\end{aligned}$$

By condition 1 we get the stated formula. □

Exercise 5.1.34 *For every $z \in \mathbb{C}$ we have $|\exp z| = \exp(\mathrm{re}(z))$.*

Exercise 5.1.35 *For every $z \in \mathbb{C}$ with $|z| \le \frac{1}{2}$ we have* $|\exp z - 1 - z| \le |z|^2$.

Exercise 5.1.36 *Let $a \in \mathbb{C}$. Deduce the infinite product*

$$\prod_{n=1}^{\infty} \left(1 + \tfrac{a}{n}\right)^n = \exp(a)\,.$$

• *Tangent and cotangent.* We define these functions by

$$\tan t \equiv \tfrac{\sin t}{\cos t} \text{ and } \cot t \equiv \tfrac{\cos t}{\sin t} \quad (t \in \mathbb{R})\,.$$

Exercise 5.1.37 *Their definition domains are*

$$M(\tan) = \mathbb{R} \setminus \{\tfrac{1}{2}(2m-1)\pi\colon\ m \in \mathbb{Z}\} \text{ and } M(\cot) = \mathbb{R} \setminus \{m\pi\colon\ m \in \mathbb{Z}\}\,.$$

• *Arcsine (inverse sine) and arccosine (inverse cosine).* These functions

$$\arcsin x\colon [-1,1] \to \mathbb{R} \text{ and } \arccos x\colon [-1,1] \to \mathbb{R}$$

are congruent to the inverses of the restriction $\sin x \,|\, [-\frac{\pi}{2}, \frac{\pi}{2}]$ and $\cos x \,|\, [0, \pi]$.

Exercise 5.1.38 *Recall the meaning of "congruence".*

• *Arctangent (inverse tangent) and arccotangent (inverse cotangent).* These functions

$$\arctan\colon \mathbb{R} \to \mathbb{R} \text{ and } \operatorname{arccot} x\colon \mathbb{R} \to \mathbb{R}$$

are congruent to inverses of the restriction $\tan x \,|\, (-\frac{\pi}{2}, \frac{\pi}{2})$ and $\cot x \,|\, (0, \pi)$.

Exercise 5.1.39 *Thus* $\arctan x$ *is congruent to a bijection between which sets?*

5.2 Elementary functions

We introduce a family of elementary functions. We begin from operations on the set of functions $\mathcal{R}$.

• *Five operations on $\mathcal{R}$.* Recall that $\mathcal{R}$ consists of functions $f\colon M \to \mathbb{R}$ with $M \subset \mathbb{R}$.

Definition 5.2.1 (operations on $\mathcal{R}$) *Let $f, g \in \mathcal{R}$. We define the following operations.*

1. *The sum*

$$f + g\colon M(f) \cap M(g) \to \mathbb{R} \text{ by } (f+g)(x) \equiv f(x) + g(x)\,.$$

2. *The product*

$$fg = f \cdot g\colon M(f) \cap M(g) \to \mathbb{R} \text{ by } (fg)(x) \equiv f(x)g(x)\,.$$

3. *The* <u>*ratio*</u>

$$f/g\colon M(f)\cap M(g)\setminus Z(g)\to\mathbb{R}\ \text{ by }\ (f/g)(x)\equiv\tfrac{f(x)}{g(x)}\,.$$

Recall that $Z(g)=\{x\in M(g)\colon\ g(x)=0\}$.

4. *Recall that the* <u>*composition*</u>

$$f(g)=f\circ g\colon M(f(g))\to\mathbb{R},\quad M(f(g))=\{x\in M(g)\colon\ g(x)\in M(f)\}\,,$$

has values $f(g)(x)\equiv f(g(x))$.

5. *For injective* f *its* <u>*inverse*</u>

$$f^{-1}\colon f[M(f)]\to\mathbb{R},\quad f^{-1}(y)\equiv x\iff f(x)=y\,,$$

is congruent to the inverse in Section 1.1.

Operations in 1–4 are binary. The inverse in 5 is a partial unary operation, it is not defined on non-injective functions. The ratio of two functions is always defined, there is no forbidden division by zero in the functional arithmetic. Recall that for f in $\mathcal{R}$ and any set X we have restriction

$$f\,|\,X\colon M(f)\cap X\to\mathbb{R},\quad (f\,|\,X)(x)\equiv f(x)\,.$$

In Chapter 7 we add a sixth operation on $\mathcal{R}$, the derivative $f\mapsto f'$.

A semiring is a structure

$$\langle X,\,0_X,\,1_X,\,+,\,\cdot\rangle$$

where X is a set, $+$ and $\cdot$ are commutative and associative operations on X, the elements 0_X and 1_X in X are neutral to $+$ and $\cdot$, respectively, and $\cdot$ is distributive to $+$.

Exercise 5.2.2 *Prove the following proposition.*

Proposition 5.2.3 (a semiring on $\mathcal{R}$) *Let $\mathcal{R}$ be as above, $0_{\mathcal{R}}\equiv k_0(x)$ and $1_{\mathcal{R}}\equiv k_1(x)$.* <u>*Then*</u>

$$\mathcal{R}_{\mathrm{smr}}\equiv\langle\mathcal{R},\,0_{\mathcal{R}},\,1_{\mathcal{R}},\,+,\,\cdot\rangle$$

is a semiring.

A function $g\in\mathcal{R}$ is <u>invariant</u> in $\mathcal{R}_{\mathrm{smr}}$ with respect to $+$, resp. $\cdot$, if for every function $h\in\mathcal{R}$ we have

$$h+g=g,\ \text{resp.}\ \ h\cdot g=g\,.$$

The proof of the following proposition characterizing invariant elements in $\mathcal{R}_{\mathrm{smr}}$ is immediate.

Proposition 5.2.4 (the invariant function) *The real empty function $\emptyset_f$ is invariant to $+$ and $\cdot$. If $g \in \mathcal{R}$ with $g \neq \emptyset_f$, then*

$$\emptyset_f + g = \emptyset_f \neq g \text{ and } \emptyset_f \cdot g = \emptyset_f \neq g\,,$$

so that g is invariant neither to $+$ nor to $\cdot$.

• *Differences of functions.* The difference of functions f and g in $\mathcal{R}$ is the function

$$f - g\colon M(f) \cap M(g) \to \mathbb{R},\ \ (f-g)(x) \equiv f(x) - g(x)\,.$$

Exercise 5.2.5 *Always $f - g = f + (k_{-1} \cdot g)$.*

Exercise 5.2.6 *Let $f, g, h \in \mathcal{R}$. Which of the implications in the (in general invalid) equivalence $f + g = h \iff f = h - g$ does gold?*

• *"How elementary, dear Watson!"* [1] We introduce elementary functions, or EF. They are sometimes confused with basic elementary functions, or BEF, of the previous section. Recall that

$$\begin{aligned} \mathrm{BEF} \;\; = \;\; & \{\exp x,\, \log x,\, \sin x,\, \cos x,\, \tan x,\, \cot x,\, \arcsin x,\, \arccos x, \\ & \arctan x,\, \mathrm{arccot}\, x,\, 0^x\} \cup \{k_c(x)\colon\ c \in \mathbb{R}\} \cup \{a^x\colon\ a > 0\} \cup \\ & \cup \{x^b\colon\ b \in \mathbb{R}\} \cup \{x^m\colon\ m \in \mathbb{Z}\}\,. \end{aligned}$$

Definition 5.2.7 (elementary functions 1) *A function $f \in \mathcal{R}$ is elementary if there is a tuple of $n \in \mathbb{N}$ functions*

$$\langle f_1,\, f_2,\, \dots,\, f_n\rangle \in \mathcal{R}^n\,,$$

called a generating word of f, such that $f_n = f$ and for every $i \in [n]$ either $f_i \in \mathrm{BEF}$ or there exist indices $j, k < i$ for which

$$f_i = f_j + f_k \vee f_i = f_j \cdot f_k \vee f_i = f_j / f_k \vee f_i = f_j(f_k)\,.$$

The set of elementary functions is denoted by EF.

Exercise 5.2.8 *Every function f_i in the generating word of f is elementary.*

Said less formally, we get EF from BEF by repeated addition, multiplication, division, and composition. For example, the identity function

$$\mathrm{id}(x) = x \equiv \mathrm{id}_{\mathbb{R}}(x)$$

is elementary because $\mathrm{id}(x) = \log(\exp x)$ or $\mathrm{id}(x) = x^1$ with $1 \in \mathbb{Z}$. Recall that x^1 with $1 \in \mathbb{R}$ is $\mathrm{id}(x) \mid [0, +\infty)$.

[1] By [27] the only "elementary" statement in the work of A. C. Doyle on Sherlock Holmes is found in the story *The Crooked Man.* There we read: " 'Excellent!' I [Watson] cried. 'Elementary,' said he."

Exercise 5.2.9 *The absolute value* $|x|$ *is in* $\mathcal{F}(\mathbb{R})$ *and is elementary.*

Exercise 5.2.10 *What is* $k_1(x)/k_0(x)$*?*

Exercise 5.2.11 *Is the empty function* $\emptyset$ *elementary?*

Exercise 5.2.12 *For every* $f \in \mathrm{EF}$ *there is a unique* $g \in \mathrm{EF}$ *such that* $M(g) = M(f)$ *and* $f + g = k_0 \,|\, M(f)$.

Exercise 5.2.13 *Find functions* $f, g \in \mathrm{EF}$ *such that*

$$M(f) = \mathbb{Z} \ \text{ and } \ M(g) = \mathbb{R} \setminus (\{0\} \cup \{\tfrac{1}{n} \colon \ n \in \mathbb{Z} \setminus \{0\}\})\,.$$

• *Restrictions to intervals.* We show that elementary functions are preserved by restrictions to intervals.

Proposition 5.2.14 (getting intervals) *Let* $a \in \mathbb{R}$. *The functions*

$$f_a \equiv (a - x)^{1/2},\ g_a \equiv (x - a)^{1/2},\ F_a \equiv \log(x - a) \ \text{ and } \ G_a \equiv \log(x - a)$$

are elementary and have respective definition domains

$$(-\infty,\, a],\ [a,\, +\infty),\ (-\infty,\, a) \ \text{ and } \ (a,\, +\infty)\,.$$

Proof. This follows immediately from Exercise 5.2.5, and Definitions 5.1.1, 5.2.1 and 5.2.7. □

Recall the real intervals as given in Proposition 2.8.7:

$$\begin{aligned}\mathcal{I} \ \equiv \ & \{\emptyset,\ \{a\},\ \mathbb{R},\ (a,b),\ (-\infty,a),\ (a,+\infty),\ (a,b],\ [a,b),\ [a,b], \\ & (-\infty,a],\ [a,+\infty) \colon \ a,\, b \in \mathbb{R},\, a < b\}\,.\end{aligned}$$

Proposition 5.2.15 (interval restrictions) *Let* $f \in \mathrm{EF}$ *be an elementary function and* $I \in \mathcal{I}$ *be an interval.* *Then*

$$f \,|\, I \in \mathrm{EF}\,.$$

Proof. It suffices to show that for every interval $I \in \mathcal{I}$ the function

$$h_I(x) \equiv k_0(x) \,|\, I$$

is elementary, because then $f \,|\, I = f + h_I$. If I is $\emptyset$ or $\mathbb{R}$ then the result holds trivially. In the remaining cases it is easy to get h_I as a difference of two functions in Proposition 5.2.14, or as a sum of two such differences. For example,

$$h_{\{a\}} = f_a - f_a + g_a - g_a,\ h_{(-\infty,a]} = f_a - f_a \ \text{ and } \ h_{[a,b)} = F_b - F_b + g_a - g_a\,.$$

□

• *Really basic elementary functions* form a subset of BEF.

Definition 5.2.16 (RBEF) *Really basic elementary functions are*

$$\mathrm{RBEF} \equiv \{\exp x,\, \log x,\, \sin x,\, \arcsin x\} \cup \{k_c(x)\colon\ c \in \mathbb{R}\} \cup \{x^b\colon\ b \in \mathbb{R}\}\,.$$

We show that the other functions in BEF are redundant in generation of EF.

Proposition 5.2.17 (RBEF suffice) *Every function*

$$f \in \mathrm{BEF} \setminus \mathrm{RBEF}$$

has a generating word according to the restricted form of Definition 5.2.7 in which BEF *is replaced with* RBEF.

Proof. First we observe that the identity function is elementary in the restricted sense because

$$x = \mathrm{id}(x) = \log(\exp x)\,.$$

(We cannot use the expression $\mathrm{id}(x) = x^1$ because functions x^m, $m \in \mathbb{Z}$, are not available.) Then we express functions in $\mathrm{BEF} \setminus \mathrm{RBEF}$ by functions in RBEF as follows.

- $\cos x = \sin(x + \frac{\pi}{2}) = \sin(x + k_{\pi/2}(x))$.
- $\tan x = \sin x / \cos x$.
- $\cot x = \cos x / \sin x$.
- $\arccos x = \frac{\pi}{2} - \arcsin x = k_{\pi/2}(x) + k_{-1}(x) \cdot \arcsin x$.
- $\arctan x = \arcsin(x/(1 + x \cdot x)^{1/2})$, see Exercise 5.2.18.
- $\mathrm{arccot}\, x = \frac{\pi}{2} - \arctan x = k_{\pi/2}(x) + k_{-1}(x) \cdot \arctan x$.
- $a^x = \exp(x \cdot \log a) = \exp(x \cdot k_{\log a}(x))$, for real $a > 0$.
- $0^x = k_0(x) + \log x - \log x$.
- $x^m = x \cdot x \cdot \ldots \cdot x$, for $m \in \mathbb{N}$.
- $x^0 = k_1(x)$, for $0 \in \mathbb{Z}$, and
- $x^m = k_1(x)/(x \cdot x \cdot \ldots \cdot x)$ with $-m$ factors, for $m \in \mathbb{Z}$ with $m < 0$.

□

We write $x \cdot x$ instead of the ambiguous x^2 which can be interpreted to be in $\mathcal{F}(\mathbb{R})$ or in $\mathcal{F}([0, +\infty))$, depending on whether $2 \in \mathbb{Z}$ or $2 \in \mathbb{R}$.

Exercise 5.2.18 *Prove the equality of functions*

$$\arctan x = \arcsin\left(\tfrac{x}{(1+x\cdot x)^{1/2}}\right).$$

Hence we have the next simpler definition of elementary functions.

Definition 5.2.19 (elementary functions 2) *In Definition 5.2.7 of* EF*, the set* BEF *may be replaced with the smaller set* RBEF*.*

Exercise 5.2.20 *In Definition 5.2.16, which functions can be further deleted from* $\{x^b\colon\ b \in \mathbb{R}\}$ *so that Proposition 5.2.17 still holds?*

The elementary function $\frac{|x|}{x}$ ($\in \mathcal{F}(\mathbb{R} \setminus \{0\})$) is -1 for $x < 0$, and 1 for $x > 0$. It resembles the function signum $\mathrm{sgn}(x)$ in $\mathcal{F}(\mathbb{R})$, given by $\mathrm{sgn}(x) = -1$ for $x < 0$, $\mathrm{sgn}(0) = 0$ and $\mathrm{sgn}(x) = 1$ for $x > 0$. However, signum is not elementary.

Proposition 5.2.21 (signum is not elementary) *The function* $\mathrm{sgn}\, x \notin \mathrm{EF}$*.*

Proof. By Definition 6.1.1 and Theorem 6.7.17 every elementary function is continuous, but signum is not continuous. □

Maybe one should reconsider the definition of elementary functions so that signum is admitted to them. We return to this matter in *MA 1*$^+$.

Exercise 5.2.22 *Give examples of functions in* EF *which are not differentiable at some points of their definition domains.*

5.3 Polynomials and rational functions

Restricted forms of Definition 5.2.7 produce polynomials and rational functions.

• *Polynomials.* Our definition is not conventional.

Definition 5.3.1 (POL) *A function* $f \in \mathcal{R}$ *is a polynomial if there exists an* n*-tuple of functions*

$$\langle f_1,\, f_2,\, \ldots,\, f_n\rangle \in \mathcal{R}^n\,,$$

so called generating word of f*, such that* $f_n = f$ *and for every* $i = 1, 2, \ldots, n$ *we have* $f_i(x) = k_c(x)$ *for some* $c \in \mathbb{R}$ *or* $f_i(x) = \mathrm{id}(x)$ *or there exist indices* $j, k < i$ *for which* $f_i = f_j + f_k$ *or* $f_i = f_j \cdot f_k$*. We denote the set of polynomials by* POL*.*

Again, not only the last function $f_n = f$ but every function $f_1, f_2, \ldots, f_{n-1}$ in the generating word is a polynomial. In our approach, polynomials arise from the identity function and constants by addition and multiplication. Then it is trivial that the sum and product of two polynomials is again a polynomial because it is built in in the definition. In the standard definition of polynomials this fact becomes a nontrivial, technical result.

However, we have to prove that our polynomials coincide with the standard ones. For $f \in \mathcal{R}$ and $n \in \mathbb{N}$ we define $f^n \equiv f \cdot f \cdot \ldots \cdot f$ with n factors f, and set $f^0 \equiv k_1 \,|\, M(f)$.

Proposition 5.3.2 (on zero-th power) *If $f \in \mathrm{EF}$ then $f^0 \in \mathrm{EF}$.*

Proof. Indeed, $f^0 = f - f + k_1 = f + k_{-1} \cdot f + k_1$. $\Box$

We call the constant function $k_0(x)$ the zero polynomial. We call a function $f \in \mathcal{R}$ a canonical polynomial if for some $n+1$ real numbers $a_0,\ a_1,\ \ldots,\ a_n$, where $n \in \mathbb{N}_0$ and $a_n \neq 0$, we have

$$f = k_{a_0} \cdot \mathrm{id}^0 + k_{a_1} \cdot \mathrm{id}^1 + \cdots + k_{a_n} \cdot \mathrm{id}^n\ .$$

We abbreviate it by writing

$$f(x) = \textstyle\sum_{j=0}^n a_j x^j \ \text{ or } \ f(x) = a_0 + a_1 x + \cdots + a_n x^n\ ,$$

and call the $n+1$-tuple

$$\langle a_0,\ a_1,\ \ldots,\ a_n\rangle$$

the canonical form of f (we show that it is unique). It is clear that the zero polynomial and every canonical polynomial is a polynomial. Now we prove the opposite.

Exercise 5.3.3 *If $f(x) = \sum_{j=0}^n a_j x^j$ is a canonical polynomial, then $|Z(f)| \le n$, that is, $f(x)$ has at most n zeros.*

It follows that $k_0(x)$ differs from every canonical polynomial.

Exercise 5.3.4 *If $f(x) \in \mathrm{POL}$ and $f(b) = 0$ then $f(x) = g(x) \cdot (x - b)$ where $g(x) \in \mathrm{POL}$.*

Theorem 5.3.5 (polynomials) *Every nonzero polynomial is a canonical polynomial. Canonical polynomials have unique canonical forms.*

Proof. Let $f \in \mathrm{POL}$ with $f \neq k_0$ and let

$$\langle f_1,\ f_2,\ \ldots,\ f_n\rangle,\ f = f_n\ ,$$

be the generating word of f by Definition 5.3.1. We prove by induction on n that f is a canonical polynomial. If $n = 1$ then $f = k_c$ or $f = \mathrm{id}$ and it is true. Let $n > 1$. If $f = k_c$ or $f = \mathrm{id}$ we are again done. We assume that there are $j, k \in [n-1]$ such that (i) $f = f_j + f_k$ or (ii) $f = f_j f_k$. In case (i) both f_j and f_k cannot be k_0. If exactly one of them is k_0 then f is a canonical polynomial by induction. If none of f_j and f_k is k_0 then we are done by Exercise 5.3.7. In case (ii) none of f_j and f_k is k_0 and we are done by Exercise 5.3.8.

We prove uniqueness of canonical forms. If $f \in \mathcal{R}$ were a canonical polynomial with two different canonical forms, by Exercise 5.3.9 $k_0 = f - f$ would be a canonical polynomial, which by Exercise 5.3.3 is impossible. $\Box$

Definition 5.3.6 (degree) *The* <u>*degree*</u> *of a nonzero polynomial* $p(x)$ *is the number* $d \in \mathbb{N}_0$ *in the canonical form*

$$p(x) = \textstyle\sum_{i=0}^{d} a_i x^i \,.$$

We denote it by $\deg p(x)$ *or by* $\deg(p(x))$.

Exercise 5.3.7 *Show that the sum of two canonical polynomials is a canonical polynomial or the zero polynomial.*

Exercise 5.3.8 *Show that the product of two canonical polynomials is a canonical polynomial.*

Exercise 5.3.9 *Show that the difference of two canonical polynomials with different canonical forms is a canonical polynomial.*

Exercise 5.3.10 *Prove the following proposition.*

Proposition 5.3.11 (POL is an integral domain) *The structure*

$$\mathrm{POL}_{\mathrm{id}} \equiv \langle \mathrm{POL},\, k_0,\, k_1,\, +,\, \cdot \rangle$$

is an integral domain.

Exercise 5.3.12 *Show that the ring* $\mathrm{POL}_{\mathrm{id}}$ *is isomorphic to the ring* $\mathbb{R}[x]$ *of real polynomials in abstract algebra.*

• *Rational functions.* Rational functions are usually defined as ratios of polynomials. We introduce them by augmenting Definition 5.3.1 with division. Now the resulting functions need not be everywhere defined.

Definition 5.3.13 (RAC) *A function* $f \in \mathcal{R}$ *is* <u>*rational*</u> *if there is a tuple of* $n \in \mathbb{N}$ *functions*

$$\langle f_1,\, f_2,\, \ldots,\, f_n \rangle \in \mathcal{R}^n \,,$$

called a <u>*generating word*</u> *of the rational function* f, *such that* $f_n = f$ *such that for every* $i \in [n]$ *we have* $f_i(x) = k_c(x)$ *for some* $c \in \mathbb{R}$ *or* $f_i(x) = \mathrm{id}(x)$ *or there exist indices* $j, k < i$ *for which*

$$f_i = f_j + f_k \vee f_i = f_j \cdot f_k \vee f_i = f_j / f_k \,.$$

The set of rational functions is denoted as RAC.

As before, every function f_i, $i \in [n]$, in the generating word of a rational function $f = f_n$ is rational. Our rational functions arise from constants and the identity by repeated addition, multiplication and division.

Exercise 5.3.14 *Show that* $\mathrm{POL} \subset \mathrm{RAC}$ *and that the sum, product and ratio of two rational functions is a rational function.*

For instance, $\frac{1}{x} \equiv k_1/\mathrm{id}$ is a rational function and $M(\frac{1}{x}) = \mathbb{R} \setminus \{0\}$. Another rational function is the empty function: $\emptyset = k_1/k_0$ and $M(\emptyset) = \emptyset$. We prove for rational functions an analog of Theorem 5.3.5, but we need for it three lemmas on computing with ratios in $\mathcal{R}$.

• *The arithmetic of ratios.* We begin with addition. Recall the operations on $\mathcal{R}$ introduced in Definition 5.2.1.

Lemma 5.3.15 (the sum) *If f_1, g_1, f_2 and g_2 are in $\mathcal{R}$, then*

$$\tfrac{f_1}{g_1} + \tfrac{f_2}{g_2} = \tfrac{f_1 g_2 + f_2 g_1}{g_1 g_2}\,.$$

Proof. We denote the left-hand side of the equality by F and the right-hand side by G. Then $M(F) = M(f_1/g_1) \cap M(f_2/g_2)$ equals to

$$\big((M(f_1) \cap M(g_1)) \setminus Z(g_1)\big) \cap \big((M(f_2) \cap M(g_2)) \setminus Z(g_2)\big)$$

and $M(G) = (M(f_1 g_2 + f_2 g_1) \cap M(g_1 g_2)) \setminus Z(g_1 g_2)$ equals to

$$\big((M(f_1) \cap M(g_2)) \cap (M(f_2) \cap M(g_1)) \cap (M(g_1) \cap M(g_2))\big) \setminus (Z(g_1) \cup Z(g_2))\,.$$

The equality $Z(g_1 g_2) = Z(g_1) \cup Z(g_2)$ follows from the fact that $\mathbb{R}$, being a field, is an integral domain. Hence $M(F) = M(G)$ because both displayed sets are equal to the set

$$\big(M(f_1) \cap M(f_2) \cap M(g_1) \cap M(g_2)\big) \setminus \big(Z(g_1) \cup Z(g_2)\big)\,.$$

The arithmetic in $\mathbb{R}$ shows that $F(x) = G(x)$ for every x in $M(F) = M(G)$. Hence $F = G$. $\square$

The proof of the second lemma is similar and is left to an exercise.

Exercise 5.3.16 *Prove the next lemma.*

Lemma 5.3.17 (the product) *If f_1, g_1, f_2 and g_2 are in $\mathcal{R}$, then*

$$\tfrac{f_1}{g_1} \cdot \tfrac{f_2}{g_2} = \tfrac{f_1 f_2}{g_1 g_2}\,.$$

The next exercise shows that the third lemma has to be more complicated.

Exercise 5.3.18 *Find four functions f_1, g_1, f_2 and g_2 in $\mathcal{R}$ such that*

$$\tfrac{f_1/g_1}{f_2/g_2} \neq \tfrac{f_1 g_2}{f_2 g_1}\,.$$

Lemma 5.3.19 (the ratio) *If f_1, g_1, f_2 and g_2 are in $\mathcal{R}$, then*

$$\tfrac{f_1/g_1}{f_2/g_2} = \tfrac{f_1 (g_2)^2}{f_2 g_1 g_2}\,.$$

Proof. We again denote the left-hand side of the equality by F and the right-hand side by G. Then $M(F) = (M(f_1/g_1) \cap M(f_2/g_2)) \setminus Z(f_2/g_2)$ equals to

$$\big((M(f_1) \cap M(g_1) \setminus Z(g_1)) \cap (M(f_2) \cap M(g_2) \setminus Z(g_2))\big) \setminus \big(Z(f_2) \setminus Z(g_2)\big)$$

and $M(G) = (M(f_1 g_2^2) \cap M(f_2 g_1 g_2)) \setminus Z(f_2 g_1 g_2))$ equals to

$$\big(M(f_1) \cap M(f_2) \cap M(g_1) \cap M(g_2)\big) \setminus \big(Z(f_2) \cup Z(g_1) \cup Z(g_2)\big)\,.$$

Thus $M(F) = M(G)$ because both displayed sets are equal to

$$\big(M(f_1) \cap M(f_2) \cap M(g_1) \cap M(g_2)\big) \setminus \big(Z(g_1) \cup Z(f_2) \cup Z(g_2)\big)\,.$$

The arithmetic in $\mathbb{R}$ shows that $F(x) = G(x)$ for every x in $M(F) = M(G)$. Hence $F = G$. $\Box$

• *Canonical forms of rational functions.* It follows from Definitions 5.3.1 and 5.3.13 that if $f, g \in \mathrm{POL}$ then $f/g \in \mathrm{RAC}$. Indeed, suppose that

$$\langle f_1,\, f_2,\, \dots,\, f_m = f\rangle \ \text{ and } \ \langle g_1,\, g_2,\, \dots,\, g_n = g\rangle$$

are respective generating words of the polynomials f and g by Definition 5.3.1. Then

$$\langle f_1,\, \dots,\, f_m,\, g_1,\, \dots,\, g_n,\, f/g\rangle$$

is a generating word of the rational function f/g by Definition 5.3.13. We prove the reverse, that every rational function is a ratio of two polynomials.

Theorem 5.3.20 (rational functions) *Let $r \in \mathrm{RAC}$ with $r \neq \emptyset$. Then there exist polynomials p and q such that*

$$p/q = r\,.$$

Hence $q \neq k_0$ and $M(r) = \mathbb{R} \setminus Z(q)$, where $Z(q)$ is a finite set by Exercise 5.3.3. We say that the pair (p, q), written also as p/q, is a canonical form of r.

Proof. Let r be a nonempty rational function. By Definition 5.3.13, it has a generating word

$$\langle f_1,\, f_2,\, \dots,\, f_n = r\rangle\,.$$

We proceed by induction on n. If $n = 1$ then r is a constant or the identity. Thus we have canonical forms $f = k_c/k_1$, $c \in \mathbb{R}$, or id/k_1. Let $n > 1$. If $r = f_n$ is a constant or the identity, we are in the previous case. Else there exist $j, k \in \mathbb{N}$ with $j, k < n$ such that (i) $r = f_j + f_k$ or (ii) $r = f_j f_k$ or (iii) $r = f_j/f_k$. Both f_j and f_k are nonempty, otherwise we would have $r = \emptyset$. By induction, f_j and f_k have canonical forms $f_j = p_1/q_1$ and $f_k = p_2/q_2$. In case (i) we get from Lemma 5.3.15 that r has the canonical form

$$r = f_j + f_k = \tfrac{p_1 q_2 + p_2 q_1}{q_1 q_2}\,.$$

In the case (ii) we get from Lemma 5.3.17 that r has the canonical form

$$r = f_j f_k = \tfrac{p_1 p_2}{q_1 q_2} .$$

In the case (iii) we get from Lemma 5.3.19 that r has the canonical form

$$r = f_j / f_k = \tfrac{p_1 (q_2)^2}{q_1 q_2 p_2} .$$

$\square$

Unlike for polynomials, canonical forms of rational functions are not unique. For example, for $r \equiv \mathrm{id}/\mathrm{id}$ ($\in$ RAC) every pair x^k/x^k, $k \in \mathbb{N}$, is a canonical form. Note that $r \neq k_1/k_1$ because $M(r) = \mathbb{R} \setminus \{0\}$ but $M(k_1/k_1) = \mathbb{R}$.

We introduce a congruence $\sim$ on the set $\mathrm{RAC} \setminus \{\emptyset\}$ by

$$r \sim s \overset{\mathrm{def}}{\iff} r \,|\, M(s) = s \,|\, M(r) .$$

Exercise 5.3.21 *Show that $\sim$ is an equivalence relation.*

For example, $k_1 \sim x/x \sim (x \cdot (x-1))/(x \cdot (x-1))$.

Exercise 5.3.22 *Prove the next proposition.*

Proposition 5.3.23 (RAC as a field) *The structure*

$$\mathrm{RAC}_{\mathrm{FI}} \equiv \langle (\mathrm{RAC} \setminus \{\emptyset\})/\!\sim,\ [k_0]_\sim,\ [k_1]_\sim,\ +,\ \cdot \rangle$$

is a field.

We call this field the field of rational maps.

Exercise 5.3.24 *Show that the field $\mathrm{RAC}_{\mathrm{fi}}$ is isomorphic to the field of rational functions $\mathbb{R}(x)$ used in abstract algebra.*

Chapter 6

Continuous functions

In Section 6.1 we introduce dense and sparse sets and state Blumberg's Theorem 6.1.15: every function

$$f\colon \mathbb{R} \to \mathbb{R}$$

has a continuous restriction $f \mid M$ to a set M dense in $\mathbb{R}$. We prove it in *MA 1*$^+$. In the extending Section 6.2 we prove Sierpiński's Theorem 6.2.4: for a function $f\colon \mathbb{R} \to \mathbb{R}$ the equivalence

$$f \text{ is continuous } \iff f \text{ is sequentially continuous}$$

holds in ZF. In Section 6.3 we show in Theorem 6.3.5 that the set of continuous functions $f\colon \mathbb{R} \to \mathbb{R}$ is in bijection with $\mathbb{R}$. The main result of Section 6.4 is Theorem 6.4.1 which says that continuous functions attain every intermediate value.

Section 6.5 introduces real compact, open and closed sets. We downgrade the minimax theorem to a mere Corollary 6.5.6 of Theorem 6.5.5 by which continuous images of compacts are compact. Basic properties of open and closed sets are discussed. Theorem 6.5.14 characterizes compact sets. We include Baire's Theorem 6.5.27 because it is needed in results on real analytic functions. Section 6.6 is devoted to uniform continuity. By Theorem 6.6.3 any continuous function with compact definition domain is uniformly continuous. Theorem 6.6.10 says that every uniformly continuous function has a unique continuous extension to the closure of the definition domain. In Theorem 6.6.11 we generalize with the help of the extension theorem the minimax theorem to bounded definition domains and UC functions.

Section 6.7 treats interactions between continuity and operations on $\mathcal{R}$. Theorem 6.7.1 concerns arithmetic operations, and Theorem 6.7.3 is devoted to continuity of sums of power series. Theorem 6.7.9 deals with composition. In the culminating Theorem 6.7.12 we treat continuity of inverse functions. The chapter concludes with Theorem 6.7.17: every elementary function is continuous.

6.1 Globally continuous functions

• *Global continuity.* Recall that by Definition 4.3.1 a function $f \in \mathcal{R}$ is continuous at a point $a \in M(f)$ if $\forall \varepsilon \exists \delta\, f[U(a,\delta)] \subset U(f(a),\varepsilon)$.

Definition 6.1.1 (global continuity) *Let $f \in \mathcal{R}$ and $X \subset \mathbb{R}$. We say that f is continuous on X if the function f is continuous at every point $b \in M(f) \cap X$. If $f \in \mathcal{R}$ is continuous on $M(f)$, it is a continuous function.*

We denote the subset of continuous functions in $\mathcal{F}(M)$ by $\mathcal{C}(M)$. We denote by

$$\mathcal{C} \equiv \textstyle\bigcup_{M \subset \mathbb{R}} \mathcal{C}(M)$$

the subset of continuous functions in $\mathcal{R}$. $\mathcal{R} \setminus \mathcal{C}$ are discontinuous functions.

Exercise 6.1.2 *Every function $f \in \mathcal{R}$ with finite domain $M(f)$ is continuous.*

Exercise 6.1.3 *For every number $c \in \mathbb{R}$ the constant function $k_c(x)$ is continuous.*

Exercise 6.1.4 *The identity function $\mathrm{id}(x) = x$ is continuous.*

Proposition 6.1.5 (restriction 1) *Let $f \in \mathcal{C}$ and $X \subset \mathbb{R}$. Then*

$$f \,|\, X \in \mathcal{C}\,.$$

Proof. Let $b \in M(f \,|\, X)$ and let an ε be given. Thus $b \in M(f)$. Since $f \in \mathcal{C}$, there is a δ such that $f[U(b,\delta)] \subset U(f(b),\varepsilon)$. Since $U(b,\delta) \cap M(f) \cap X \subset U(b,\delta) \cap M(f)$, we have

$$(f \,|\, X)[U(b,\, \delta)] \subset f[U(b,\, \delta)] \subset U(f(b),\, \varepsilon)\,.$$

□

• *Sequential continuity.* The next definition is sometimes used as the primary definition of continuity, both pointwise and global.

Definition 6.1.6 (sequential continuity) *Let $f \in \mathcal{F}(M)$.*

1. *If $b \in M$ and for every $(a_n) \subset M$ with $a_n \to b$ we have $f(a_n) \to f(b)$, we say that f is sequentially continuous at b.*
2. *If f is sequentially continuous at every $b \in M$, we say that the function f is sequentially continuous.*

Exercise 4.3.6 established that both definitions of continuity are equivalent:

Proposition 6.1.7 (Heine's formulation) *Let $f \in \mathcal{F}(M)$ and $b \in M$.*

1. *The function f is continuous at b $\iff$ f is sequentially continuous at b.*

2. *The function f is continuous $\iff$ f is sequentially continuous.*

Both implications $\Rightarrow$ are easily established in ZF. Both opposite implications $\Leftarrow$ in general require the axiom of choice.

Exercise 6.1.8 *See article [45] on equivalents of the axiom of choice in analysis and topology.*

In the next section we prove a surprising result due to W. Sierpinski: if $M \subset \mathbb{R}$ is an open set, then the second implication $\Leftarrow$ in the proposition can be proven without the axiom of choice.

• *Dense and sparse sets.* Let $M, N \subset \mathbb{R}$. The set N is dense in the set M if for every point $a \in M$ and every δ we have

$$U(a,\,\delta) \cap M \cap N \neq \emptyset\,.$$

Exercise 6.1.9 *N is dense in M $\iff$ for every point $a \in M$ there is a sequence $(b_n) \subset M \cap N$ such that $\lim b_n = a$.*

Exercise 6.1.10 *Show that both sets $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are dense in $\mathbb{R}$.*

Let $M, N \subset \mathbb{R}$. The set N is sparse in the set M if for every point $a \in M$ and every δ there exist a $b \in M$ and θ such that $U(b, \theta) \subset U(a, \delta)$ and

$$U(b, \delta) \cap M \cap N = \emptyset\,.$$

Exercise 6.1.11 *Show that $N \equiv \{\frac{1}{n}\colon\; n \in \mathbb{N}\}$ is sparse in $M \equiv [0, 1]$.*

Proposition 6.1.12 (density and continuity) *Let $f, g \in \mathcal{C}(M)$, let $N \subset \mathbb{R}$ be dense in M and let $f \,|\, N = g \,|\, N$. Then*

$$f = g\,.$$

Proof. Let $b \in M$ and $(a_n) \subset M \cap N$ have $\lim a_n = b$ (Exercise 6.1.9). Using (H) we have

$$f(b) = f(\lim a_n) = \lim f(a_n) = \lim g(a_n) = g(\lim a_n) = g(b)\,.$$

□

A function $g \in \mathcal{C}$ is a kernel of another function $f \in \mathcal{C}$ if g is a restriction of f and $M(g)$ is dense in $M(f)$. Using the previous proposition we easily reconstruct f from g.

Proposition 6.1.13 (kernels) *Every continuous function f in $\mathcal{C}$ has an at most countable kernel.*

Proof. It suffices to show that every set $M \subset \mathbb{R}$ has an at most countable dense subset N (Exercise 6.1.14). Let $M \subset \mathbb{R}$. We obtain such set N by using the axiom of choice and taking one element from every nonempty intersection $(\alpha, \beta) \cap M$, where $\alpha < \beta$ are fractions. □

Exercise 6.1.14 *Why does such set N suffice?*

• *Blumberg's theorem.* In 1922 H. Blumberg discovered the following theorem.

Theorem 6.1.15 (Blumberg's) *For every function*

$$f\colon \mathbb{R} \to \mathbb{R}$$

there exists a set $M \subset \mathbb{R}$ such that

$$M \text{ is dense in } \mathbb{R} \text{ and } f \,|\, M \text{ is continuous}.$$

Henry Blumberg (1886–1950) was born in Lithuania in the town Žagarė, but the family emigrated to America already in 1891. We prove Blumberg's theorem in *MA 1*$^{+}$.

Exercise 6.1.16 *Find such set M for Riemann's function $r(x)$ (it is defined before Proposition 4.3.12).*

6.2 Sierpiński's theorem

In [46], eight equivalent formulations of the axiom of countable real choice are established. One of them is the following.

Proposition 6.2.1 (AC and sequential continuity) *In* ZF *claims 1 and 2 are equivalent.*

1. *A function $f\colon \mathbb{R} \to \mathbb{R}$ is continuous at a point x iff f is sequentially continuous at x.*
2. *The axiom of countable choice holds for subsets of $\mathbb{R}$.*

Proof. See [46]. □

But already in 1918 the Polish mathematician *Wacław Sierpiński (1882–1969)* circumvented in [70] the axiom of choice for global continuity on open real sets. We present his solution in Theorem 6.2.4.

• *Choice from sets of fractions.* The existence of a selector on the set

$$\mathcal{P}_0(\mathbb{Q}) \equiv \mathcal{P}(\mathbb{Q}) \setminus \{\emptyset\}$$

is a theorem in ZF. In the next proposition we state a more convenient choice principle.

Exercise 6.2.2 *Show that in* ZF *the set* $\mathbb{Q}$ *of fractions has a well ordering.*

Proposition 6.2.3 (rational choice) *We have the theorem*

$$\mathrm{ZF} \vdash \forall F \colon \mathbb{N} \to \mathcal{P}_0(\mathbb{Q})\, \exists f \colon \mathbb{N} \to \mathbb{Q}\, \big(n \in \mathbb{N} \Rightarrow f(n) \in F(n)\big)\,.$$

Proof. Let a map $F \colon \mathbb{N} \to \mathcal{P}_0(\mathbb{Q})$ be given. Using Exercise 6.2.2 we take, in ZF, a well ordering

$$(\mathbb{Q},\, \prec)\,.$$

Then we define $f \colon \mathbb{N} \to \mathbb{Q}$ by $(n \in \mathbb{N})$

$$f(n) \equiv \min_{\prec}(F(n))\,.$$

This is a definition within ZF. □

• *Sierpiński's theorem.* Sierpiński's method for eliminating AC in the proof of equivalence of continuity with sequential continuity uses the density of $\mathbb{Q}$ in any open real set.

Theorem 6.2.4 (W. Sierpiński, 1918) *Let* $M \subset \mathbb{R}$ *be an open set. In* ZF*, i.e. without the axiom of choice, we have the theorem*

$$\mathrm{ZF} \vdash \forall f \in \mathcal{F}(M)\, \big(f \text{ is continuous} \iff f \text{ is sequentially continuous}\big)\,.$$

Proof. Let $f \colon M \to \mathbb{R}$ where $M \subset \mathbb{R}$ is an open set. The implication $\Rightarrow$ is easy to establish in ZF. We assume that f is continuous, $b \in M$ and that $(a_n) \subset M$ is a sequence with $a_n \to b$. Let an ε be given. We take a δ such that

$$f[U(b,\, \delta)] \subset U(f(b),\, \varepsilon)\,.$$

Then there is an n_0 such that $a_n \in U(b, \delta)$ for every $n \geq n_0$. It follows that $f(a_n) \in U(f(b), \varepsilon)$ for the same n. Hence $f(a_n) \to f(b)$ and f is sequentially continuous at b.

We prove, in ZF, the implication $\neg \Rightarrow \neg$. This is the main result of the theorem. We assume that f is not continuous. Thus there exist a point $b \in M$ and an ε such that for every $n \in \mathbb{N}$ there exists a point

$$b_n \in U(b,\, \tfrac{1}{n}) \cap M \text{ such that } |f(b_n) - f(b)| > \varepsilon\,.$$

We consider for $n \in \mathbb{N}$ the sets of fractions

$$A_n \equiv \{\alpha \in U(b,\, \tfrac{1}{n}) \cap M \cap \mathbb{Q} \colon\; |f(\alpha) - f(b)| > \tfrac{\varepsilon}{2}\}\,.$$

If $A_n \neq \emptyset$ for every n, we use Proposition 6.2.3 and obtain in ZF a function

$$\varphi \colon \mathbb{N} \to \mathbb{Q}$$

such that $\varphi(n) \in A_n$ for every $n \in \mathbb{N}$. Then $(a_n) \subset \mathbb{Q} \cap M$ with $a_n \equiv \varphi(n)$ is a sequence such that $a_n \to b$ but $f(a_n) \not\to f(b)$, and f is not sequentially continuous at the point b.

The complementary case is that $A_m = \emptyset$ for some $m \in \mathbb{N}$. This by the triangle inequality means that

$$|f(\alpha) - f(b_m)| > \tfrac{\varepsilon}{2} \text{ for every } \alpha \in U(b,\, \tfrac{1}{m}) \cap M \cap \mathbb{Q}$$

(Exercise 6.2.5). We take large $N \in \mathbb{N}$ such that

$$U(b_m,\, \tfrac{1}{N}) \subset U(b,\, \tfrac{1}{m}) \cap M$$

and define for $n \in \mathbb{N}$ the sets of fractions

$$B_n \equiv U(b,\, \tfrac{1}{N+n-1}) \cap \mathbb{Q}\,.$$

Then $|f(\alpha) - b_m| \ge \frac{\varepsilon}{2}$ for every $n \in \mathbb{N}$ and every fraction $\alpha \in B_n$. Using again Proposition 6.2.3 we obtain in ZF a function

$$\psi\colon \mathbb{N} \to \mathbb{Q}$$

such that $\psi(n) \in B_n$ for every $n \in \mathbb{N}$. Then $(a_n) \subset \mathbb{Q} \cap M$ with $a_n \equiv \psi(n)$ is a sequence such that $a_n \to b_m$ but $f(a_n) \not\to f(b_m)$, and f is not sequentially continuous at the point b_m. $\Box$

Exercise 6.2.5 *Justify the inequality in the proof.*

One might think that in the equivalence of both continuities the axiom of choice could be circumvented altogether by well ordering the real numbers, i.e., by replacing in Exercise 6.2.2 the set $\mathbb{Q}$ with the set $\mathbb{R}$. This is impossible because $\mathbb{R}$ much differs from $\mathbb{Q}$. In 1970 the American mathematician *Robert M. Solovay (1938)* proved in [72] that

if the theory ZF+AC+I is consistent, then so is the theory ZF+LM.

A theory is just a set of formulas. ZF are the axioms of ZF set theory (Section ...), AC is the axiom of choice (Axioms 1.1.30 and ...), I is the axiom that there exists an inaccessible cardinal (Definition ...) and LM is the axiom that every set of real numbers is Lebesgue measurable. Consistency of a theory means that one cannot derive from the formulas of the theory a contradiction; else the theory is contradictory. Thus unless the theory ZF + AC + I is contradictory we cannot prove in ZF that $\mathbb{R}$ has a well ordering; a well ordering of $\mathbb{R}$ yields via a well known construction a set of real numbers that is not Lebesgue measurable. In 1984 the Israeli mathematician *Saharon Shelah (1946)* proved in [69] that the axiom I cannot be removed from Solovay's result.

On the other hand, already in 1938/1940 K. Gödel published in [36, 37], see also [24, Chapter VI. "Jetzt, Mengenlehre"], the result that

if the theory ZF is consistent, then so is the theory ZF + AC.

Thus unless ZF is contradictory, in view of Zermelo's Theorem ... we cannot prove in ZF that a well ordering of $\mathbb{R}$ does not exist.

Is it mind-boggling? Yes, it is. It is the up-to-date answer to the question when for a function $f \in \mathcal{F}(M)$ continuity is equivalent with sequential continuity.

6.3 The cardinality of continuous functions

We show that there exists a bijection between the sets $\mathcal{C}(\mathbb{R})$ and $\mathbb{R}$.

• *The Cantor–Bernstein theorem.* We use this theorem, which we prove in *MA 1*$^+$, to construct the mentioned bijection.

Theorem 6.3.1 (Cantor–Bernstein) *Let X and Y be sets. If there exist two injections, from X to Y and from Y to X, then there exists a bijection from X to Y.*

Exercise 6.3.2 *One can extend the conclusion: there exist two bijections, from X to Y and from Y to X.*

We mentioned G. Cantor earlier. *Felix Bernstein (1878–1956)* was a German mathematician. For example,

$$(m,\, n) \mapsto 2^m 3^n$$

is an injection from $\mathbb{N} \times \mathbb{N}$ to $\mathbb{N}$ and

$$n \mapsto (1,\, n)$$

is an injection from $\mathbb{N}$ to $\mathbb{N} \times \mathbb{N}$. Hence by the C.–B. theorem there exists a bijection from $\mathbb{N} \times \mathbb{N}$ to $\mathbb{N}$. The next exercise describes such bijection.

Exercise 6.3.3 *The function $b\colon \mathbb{N} \times \mathbb{N} \to \mathbb{N}$, where $b(m,n) = (2m-1) \cdot 2^{n-1}$, is a bijection.*

• *How many continuous functions from $\mathbb{R}$ to $\mathbb{R}$ are there?* As many as real numbers. To prove that there exists a bijection between $\mathcal{C}(\mathbb{R})$ and $\mathbb{R}$, by Theorem 6.3.1 it suffices to produce two injections, from $\mathbb{R}$ to $\mathcal{C}(\mathbb{R})$ and from $\mathcal{C}(\mathbb{R})$ to $\mathbb{R}$. The former injection is clear, see Exercise 6.3.6. We describe an injection from $\mathcal{C}(\mathbb{R})$ to $\mathbb{R}$. We begin with an injection from $\mathbb{R}^{\mathbb{N}}$, the set of real sequences, to $\mathbb{R}$.

Proposition 6.3.4 (coding (a_n) as a) *There exists an injection $f\colon \mathbb{R}^{\mathbb{N}} \to \mathbb{R}$.*

Proof. Let X be the twelve-element set of ten digits 0, 1, ..., 9, the decimal point . and the minus sign $-$. We view any real number b as a sequence $(b_m) \subset X$, $m \in \mathbb{N}$. For example,

$$-\pi = (-,\, 3,\, .,\, 1,\, 4,\, 1,\, 5,\, \dots)$$

or this year is

$$(2,\, 0,\, 2,\, 5,\, .,\, 0,\, 0,\, 0,\, \dots)\,.$$

We fix a coding c of the elements in X by pairs of digits: $c(0) = 00$, $c(1) = 01$, $\dots$, $c(9) = 09$, $c(.) = 10$ and $c(-) = 11$. We use the inverse bijection

$$b^{-1} = (u,\, v)\colon \mathbb{N} \to \mathbb{N} \times \mathbb{N}$$

of Exercise 6.3.3 and define the desired injection f by

$$f((a_n)) = f(((a_{n,\,m}))) \equiv 0\,.\,d_1\,d_2\,d_3\,d_4\ \dots\ d_{2l-1}\,d_{2l}\ \dots\quad (\in \mathbb{R})\,,$$

where $d_i \in \{0, 1, \dots, 9\}$ and $d_{2l-1}d_{2l} \equiv c(a_{u(l),v(l)})$. It is clear that f is 1-1. $\square$

Now it is easy to get an injection from $\mathcal{C}(\mathbb{R})$ to $\mathbb{R}$.

Theorem 6.3.5 (the cardinality of $\mathcal{C}(\mathbb{R})$) *The set $\mathcal{C}(\mathbb{R})$ of continuous real functions with domain $\mathbb{R}$ is in bijection with $\mathbb{R}$.*

Proof. It suffices to find an injection $g\colon \mathcal{C}(\mathbb{R}) \to \mathbb{R}^{\mathbb{N}}$ because then the composition $f(g)$ with the injection f in the previous proposition is an injection from $\mathcal{C}(\mathbb{R})$ to $\mathbb{R}$. This is easy, we take any bijection $h\colon \mathbb{N} \to \mathbb{Q}$ (Theorem) and define the value of g on $j = j(x) \in \mathcal{C}(\mathbb{R})$ as the real sequence

$$g(j) \equiv (a_n) \ \text{ where } \ a_n \equiv j(h(n))\,.$$

By Exercise 6.1.10 and Proposition 6.1.12 one can uniquely recover j from (a_n). Hence g is injective. $\square$

Exercise 6.3.6 *Show that the map $a \mapsto k_a$, where k_a is the constant function, is an injection from $\mathbb{R}$ to $\mathcal{C}(\mathbb{R})$.*

Exercise 6.3.7 *For every nonempty set $M \subset \mathbb{R}$ the sets $\mathcal{C}(M)$ and $\mathbb{R}$ are in bijection.*

6.4 Attaining intermediate values

We show that continuous functions map intervals to intervals.

• *Any continuous function attains any intermediate value.* We prove it in the next theorem.

Theorem 6.4.1 (intermediate values) *Let $a < b$, $f \in \mathcal{C}([a,b])$ and $f(a) < c < f(b)$ or $f(a) > c > f(b)$. Then*

$$c = f(d) \text{ for some } d \in (a,b)\,.$$

Proof. Let $f(a) < c < f(b)$, the other case is similar. We set

$$X \equiv \{x \in [a,b]\colon\ f(x) < c\} \text{ and } d \equiv \sup(X) \ \ (\in [a,\, b])\,.$$

The continuity of f implies that $d \in (a,b)$. We show that both $f(d) < c$ and $f(d) > c$ lead to contradiction, and hence $f(d) = c$. Let $f(d) < c$. By the continuity of f at d there is a δ such that for every $x \in U(d,\delta) \cap [a,b]$ it holds that $f(x) < c$. But then X contains numbers larger than d, which is a contradiction. Let $f(d) > c$. By the continuity of f at d there is a δ such that for every $x \in U(d,\delta) \cap [a,b]$ it holds that $f(x) > c$. But then every $x < d$ and close to d lies outside X, which is also a contradiction. $\square$

Exercise 6.4.2 *For every interval $I \subset \mathbb{R}$ and every function $f \in \mathcal{C}(I)$ the image $f[I]$ is an interval.*

Corollary 6.4.3 (the image of $\exp x$) *We have*

$$\exp[\mathbb{R}] = (0,\, +\infty)\,.$$

Thus the function exp *is a bijection from $\mathbb{R}$ to $(0, +\infty)$.*

Proof. Since $\exp > 0$ on $\mathbb{R}$, we have $\exp[\mathbb{R}] \subset (0, +\infty)$. From the limits $\lim_{x\to-\infty} \exp x = 0$ and $\lim_{x\to+\infty} \exp x = +\infty$ (part 3 of Proposition 5.1.6), from the continuity of $\exp x$ (Corollary 6.7.6) and from Theorem 6.4.1 it follows that $(0, +\infty) \subset \exp[\mathbb{R}]$. Thus $\exp[\mathbb{R}] = (0, +\infty)$. The exponential increases and hence it is injective and a bijection. $\square$

Exercise 6.4.4 *Prove the following corollary.*

Corollary 6.4.5 (mountaineering) *A mountain climber starts her ascend at midnight, after 24 hours she reaches the summit and then she descends for 24 hours to the base camp. Show that there exists a moment*

$$t_0 \in [0,\, 24]$$

when she is in both days in exactly the same altitude.

A function $f \in \mathcal{F}(M)$ <u>increases</u>, respectively <u>decreases</u>, on an (arbitrary) set X if for every $x < y$ in $M \cap X$ we have

$$f(x) < f(y), \text{ respectively } f(x) > f(y)\,.$$

Corollary 6.4.6 (continuity and injectivity) *If $I \subset \mathbb{R}$ is an interval and $f \in \mathcal{C}(I)$ is injective, <u>then</u> f either increases or decreases on I.*

Proof. If f neither increases nor decreases then I contains three numbers $a < b < c$ such that $f(a) < f(b) > f(c)$ or $f(a) > f(b) < f(c)$. In the first case, we see by Theorem 6.4.1 that for every d such that

$$f(a),\ f(c) < d < f(b)$$

there exist numbers $x \in (a, b)$ and a $y \in (b, c)$ such that

$$d = f(x) = f(y)\,,$$

which contradicts the injectivity of f. The second case leads to a similar contradiction. □

Now we can easily prove Theorem **??** which we state here as a corollary.

Corollary 6.4.7 (Bolzano–Cauchy) *Suppose that $a < b$, $f \in \mathcal{C}([a, b])$ and that $f(a)f(b) \le 0$. Then*

$$Z(f) \ne \emptyset\,.$$

Proof. If $f(a)f(b) = 0$ then $f(a) = 0$ or $f(b) = 0$. If $f(a)f(b) < 0$ then $f(a) < 0 < f(b)$ or $f(b) < 0 < f(a)$, and $0 = f(c)$ for some $c \in (a, b)$ by Theorem 6.4.1. □

6.5 Compact, open and closed sets

In mathematical analysis, compactness is a fundamental concept. We consider only compact sets in $\mathbb{R}$.

• *Real compact sets.* One possible definition is as follows.

Definition 6.5.1 (compact sets) *A set $M \subset \mathbb{R}$ is compact if every sequence $(a_n) \subset M$ has a convergent subsequence (a_{m_n}) with*

$$\lim a_{m_n} \in M\,.$$

An equivalent definition, which generalizes more easily, uses covers by open sets; see Theorem 6.5.18 below (no proof). We introduce open sets later.

Exercise 6.5.2 *Prove that every interval $[a, b]$ is a compact set.*

Exercise 6.5.3 *Is the empty set compact? Is $\mathbb{R}$ compact?*

Let $(X, \prec)$ be a linear order and $A \subset X$. Recall that $\min(A)$, the minimum of A (relative to $\prec$), is the unique element $a \in A$, if it exists, such that

$$b \succeq a \text{ for every } b \in A\,.$$

For $\preceq$ in place of $\succeq$ we get the maximum $\max(A)$. Statement of the next proposition employs the usual linear order $(\mathbb{R}, <)$. In the proof we use the linear order $(\mathbb{R}^*, <)$.

Proposition 6.5.4 (extremities of compacts) *Any nonempty and compact set $M \subset \mathbb{R}$*

$$\textit{has both } \min(M) \textit{ and } \max(M)\,.$$

Proof. We only prove that the maximum of M exists, the proof for minimum is similar. Let $M \subset \mathbb{R}$, $M \neq \emptyset$, be compact and let

$$A \equiv \sup(M)\,,$$

in the linear order $(\mathbb{R}^*, <)$ (see Proposition 2.1.6). If $A = +\infty$ then there is a sequence $(a_n) \subset M$ with $\lim a_n = +\infty$. But such (a_n) has no convergent subsequence, which contradicts the compactness of M. Thus $A \in \mathbb{R}$. We show that $A \in M$, and then $A = \max(M)$. If $A \notin M$, then we again take a sequence $(a_n) \subset M$ with $\lim a_n = A$. It again has no convergent subsequence with limit in M (the limit of every subsequence of (a_n) is A), which again contradicts the compactness of M. $\square$

• *Continuous images of compacts are compact.* This is an important interplay between compactness and continuity.

Theorem 6.5.5 (images of compacts) *If $f \in \mathcal{C}(M)$ and M is compact <u>then</u> the image*

$$f[M] \ \ (\subset \mathbb{R})$$

is a compact set.

Proof. Let f and M be as stated and $(b_n) \subset f[M]$. Using the axiom of choice we take a sequence $(a_n) \subset M$ such that $f(a_n) = b_n$. It has a subsequence (a_{m_n}) with the limit

$$a \equiv \lim a_{m_n} \ \ (\in M)\,.$$

By (H) we have $\lim f(a_{m_n}) = f(a) \equiv b$, so that $(b_{m_n}) = (f(a_{m_n}))$ has the limit $b \in f[M]$. Hence $f[M]$ is compact. $\square$

We show that continuous functions defined on compact sets attain extremal values. If $f \in \mathcal{R}$ and $b \in M(f)$ is such that

$$f(b) \geq f(x) \text{ for every } x \in M(f)\,,$$

we say that f <u>attains (at b) maximum</u>. If the opposite inequality $\leq$ holds for every $x \in M(f)$, then f <u>attains (at b) minimum</u>.

Corollary 6.5.6 (minima and maxima) *Let $M \subset \mathbb{R}$ be a compact set and let $f \in \mathcal{C}(M)$. <u>Then</u> the function f attains both minimum and maximum.*

Proof. Let M and f be as stated. By Proposition 6.5.4 and Theorem 6.5.5 both $\min(f[M])$ and $\max(f[M])$ exist. Hence f attains both minimum and maximum. $\square$

Exercise 6.5.7 *The continuous functions* $\mathrm{id}\,|\,[0,1)$ *and* $1/(1-x)\,|\,[0,1)$ *do not attain maximum.*

• *Local, global and strict extremes.* Let $f \in \mathcal{R}$ and $b \in M(f)$. We already defined what it means that f attains at b maximum, resp. minimum. We also say that f has at b a global maximum, resp. a global minimum. If for every $x \in M \setminus \{b\}$ the inequality holds as strict, we speak of a strict global maximum, resp. a strict global minimum. These (strict) minima or maxima at b are local if for some δ the restriction $f\,|\,U(b,\delta)$ attains them as global.

Exercise 6.5.8 *Let* $f \in \mathcal{R}$. *Prove the equivalence:* f *attains maximum (or minimum) at every point* $b \in M(f)$ *iff* $f = k_c\,|\,M(f)$ *for some* $c \in \mathbb{R}$.

• *Open and closed sets.* We introduce these important families of real sets.

Definition 6.5.9 *A set* $M \subset \mathbb{R}$ *is open if for every point* $b \in M$ *there exists a* δ *such that* $U(b,\delta) \subset M$. *The set* M *is closed if its complement* $\mathbb{R} \setminus M$ *is open.*

Proposition 6.5.10 (properties of open sets) *Let* $\{U_i:\ i \in I\}$, $I \neq \emptyset$, *be a set system of open sets. Then the following holds.*

1. *Both the empty set* $\emptyset$ *and the whole set* $\mathbb{R}$ *are open sets.*
2. *The union* $\bigcup_{i\in I} U_i$ *is an open set.*
3. *For any finite index set* I *the intersection* $\bigcap_{i\in I} U_i$ *is an open set.*

Proof. 1. This is trivial: there is no $a \in \emptyset$ and $U(a,\delta) \subset \mathbb{R}$ for every a and δ.

2. If $a \in \bigcup_{i\in I} U_i$ then $a \in U_i$ for some $i \in I$ and there is a δ such that $U(a,\delta) \subset U_i$, so that $U(a,\delta) \subset \bigcup_{i\in I} U_i$.

3. If $a \in \bigcap_{i\in I} U_i$ then $a \in U_i$ for every $i \in I$ and for every $i \in I$ there is a δ_i such that $U(a,\delta_i) \subset U_i$. We set

$$\delta \equiv \min(\{\delta_i:\ i \in I\})\,.$$

This minimum exists and is positive because I is finite. Thus $U(a,\delta) \subset U_i$ for every $i \in I$. $\Box$

Exercise 6.5.11 *State and prove analogous properties of closed sets.*

Exercise 6.5.12 *If* $U \subset \mathbb{R}$ *is an open set then every element of* U *is a limit point of* U.

Proposition 6.5.13 (on closed sets) *A set* $M \subset \mathbb{R}$ *is closed* $\iff$ *the limit of every convergent sequence* $(a_n) \subset M$ *lies in* M.

Proof. Implication $\Rightarrow$. Let $M \subset \mathbb{R}$ be a closed set and let $(a_n) \subset M$ be a sequence with

$$\lim a_n = a \in \mathbb{R} \setminus M\,.$$

But then for some δ we have $U(a, \delta) \cap M = \emptyset$. This is not possible because $a_n \to a$ and $a_n \in U(a, \delta)$ for every $n \geq n_0$. Hence $a \in M$.

Reverse implication $\neg \Rightarrow \neg$. Suppose that $M \subset \mathbb{R}$ is not a closed set. Then there is a point $a \in \mathbb{R} \setminus M$ such that for every n there exists

$$a_n \in U(a, \tfrac{1}{n}) \cap M\,.$$

Thus $(a_n) \subset M$ and $\lim a_n = a \notin M$. We used the axiom of choice. $\square$

• *Characterizations of compact sets.* A set $M \subset \mathbb{R}$ is <u>bounded</u> if $M \subset [a, b]$ for some real $a < b$. We characterize real compact sets.

Theorem 6.5.14 (real compact sets) *Let $M \subset \mathbb{R}$. <u>Then</u>*

$$M \textit{ is compact} \iff M \textit{ is bounded and closed}\,.$$

Proof. Implication $\Leftarrow$. Let M be bounded and closed and $(a_n) \subset M$. By Theorem 2.4.12 we have a convergent subsequence (a_{m_n}). M is closed and Proposition 6.5.13 shows that $\lim a_{m_n} \in M$. Hence M is compact.

Contrapositive implication $\neg \Leftarrow \neg$. Suppose that M is not bounded. We then define a sequence $(a_n) \subset M$ such that

$$m \neq n \Rightarrow |a_m - a_n| \geq 1$$

— such sequence has no convergent subsequence. First term a_1 is arbitrary. Suppose that $a_1, a_2, \ldots, a_n$ are defined and $|a_i - a_j| \geq 1$ if $i \neq j$. Since M is not bounded, there exists

$$a_{n+1} \in M \text{ such that } |a_{n+1}| \geq 1 + \max(\{|a_1|,\, |a_2|,\, \ldots,\, |a_n|\})\,.$$

By the Δ-inequality we have $|a_{n+1} - a_i| \geq 1$ for every $i \in [n]$. Repeating it indefinitely we get the required sequence (a_n).

Suppose that M is not closed. By Proposition 6.5.13 there is a sequence $(a_n) \subset M$ with limit in $\mathbb{R} \setminus M$. Every subsequence of (a_n) has the same limit and therefore does not converge in M. $\square$

Exercise 6.5.15 $[a, b] \setminus P(c, \delta)$ *is a compact set.*

Exercise 6.5.16 *Show that the implication $\Rightarrow$ holds more generally in metric spaces, every compact set in a metric space is bounded and closed.*

Exercise 6.5.17 *However, the implication $\Leftarrow$, in general, does not hold in metric spaces. The hint is to consider discrete spaces.*

We do not prove the following theorem.

Theorem 6.5.18 (Heine–Borel) *Let $M \subset \mathbb{R}$. Then M is compact $\iff$ for every system $\{U_i\colon\ i \in I\}$ of open sets the implication holds that*

$$\textstyle\bigcup_{i\in I} U_i \supset M \Rightarrow \text{there is a finite set } J \subset I \text{ such that } \bigcup_{i\in J} U_i \supset M\,.$$

In *MA 1*$^+$ we prove a more general version of this theorem for metric spaces.

• *More on open and closed sets.* Let $A \subset M \subset \mathbb{R}$. Then A is relatively closed in M if there is a closed set $U \subset \mathbb{R}$ such that $A = M \cap U$.

Proposition 6.5.19 (zero sets are relatively closed) *Let $f \in \mathcal{C}$. Then the set*

$$Z(f) = \{x \in M(f)\colon\ f(x) = 0\}$$

is relatively closed in $M(f)$.

Proof. Let $N \equiv M(f) \setminus Z(f)$. Using the continuity of f we see that

$$\text{for every } b \in N \text{ there is a } \delta_b \text{ such that } U(b,\, \delta_b) \cap Z(f) = \emptyset\,.$$

Let $A \equiv \bigcup_{b\in N} U(b, \delta_b)$. By part 2 of Proposition 6.5.10 the set A is open. Thus $\mathbb{R} \setminus A$ is closed and $Z(f) = M(f) \cap (\mathbb{R} \setminus A)$. □

We show that continuous injective images of open sets are open.

Proposition 6.5.20 (images of open sets) *Suppose that $f \in \mathcal{C}$ is injective and both sets $M(f)$ and $M \subset \mathbb{R}$ are open. Then the image $f[M]$ is open.*

Proof. Let $b \in f[M]$ and $a \equiv f^{-1}(b)$ $(\in M(f) \cap M)$. Since $M(f)$ and M are open, so is their intersection and we can take an interval $I \equiv [a - \delta, a + \delta]$ such that $I \subset M(f) \cap M$. Then

$$f(a - \delta) < b < f(a + \delta) \ \text{ or } \ f(a - \delta) > b > f(a + \delta)$$

because $f(a-\delta), f(a+\delta) < b = f(a)$ and $f(a-\delta), f(a+\delta) > b = f(a)$ contradict by Theorem 6.4.1 the injectivity of f. We take any

$$\varepsilon < \min(|f(a + \delta) - b|,\, |f(a - \delta) - b|)\,.$$

Using Theorem 6.4.1 we have $U(b, \varepsilon) \subset f[I] \subset f[M]$. Hence $f[M]$ is open. □

Let $A \subset M \subset \mathbb{R}$. Then A is relatively open in M if there is an open set $B \subset \mathbb{R}$ such that $A = M \cap B$.

Proposition 6.5.21 (preimages of open sets) *Let f be in $\mathcal{C}$ and $M \subset \mathbb{R}$ be open. Then the set*

$$f^{-1}[M] = \{x \in M(f)\colon\ f(x) \in M\} \ \textit{is relatively open in } M(f)\,.$$

Proof. For every $b \in M$ there is an ε_b such that $U(b, \varepsilon_b) \subset M$. For every a in $f^{-1}[M]$ $(\subset M(f))$ there is a δ_a such that for $b \equiv f(a)$ we have

$$f[U(a,\, \delta_a)] \subset U(b,\, \varepsilon_b) \subset M\,.$$

Thus, $U(a, \delta_a) \cap M(f) \subset f^{-1}[M]$. Let

$$B \equiv \textstyle\bigcup_{a \in f^{-1}[M]} U(a,\, \delta_a)\,.$$

By part 2 of Proposition 6.5.10 this is an open set. Since $f^{-1}[M] = M(f) \cap B$, the set $f^{-1}[M]$ is relatively open in $M(f)$. $\square$

We need Propositions 6.5.19 and 6.5.21 in the proof of Theorem 7.6.5.

We describe the structure of any open set. An <u>open interval</u> is any nonempty interval that is an open set. These are exactly the intervals $(a < b)$

$$(-\infty,\, a),\ (a,\, +\infty)\ \text{ and }\ (a,\, b)\,.$$

Theorem 6.5.22 (structure of open sets) *A real set is open $\iff$ it is a union of at most countably many mutually disjoint open intervals.*

Proof. Implication $\Leftarrow$ follows from part 2 of Proposition 6.5.10.

Implication $\Rightarrow$. Let $M \subset \mathbb{R}$ be an open set. If $M = \emptyset$, the claim holds trivially. Let $a \in M$. We define I_a as the inclusion-wise maximal open interval I such that $a \in I \subset M$ (Exercise 6.5.23). By Exercise 6.5.24 we have for any $a, b \in M$ that

$$I_a = I_b\ \text{ or }\ I_a \cap I_b = \emptyset\,.$$

Thus $S \equiv \{I_a\colon\ a \in \mathbb{Q} \cap M\}$ is an at most countable set system of mutually disjoint open intervals such that $\bigcup S = M$ (Exercise 6.5.25). $\square$

Exercise 6.5.23 *Show that the interval I_a exists.*

Exercise 6.5.24 *Explain why $I_a = I_b$ or $I_a \cap I_b = \emptyset$.*

Exercise 6.5.25 *Why $\bigcup S = M$?*

• *The Cantor set.* By the previous theorem any closed set $\mathbb{R} \setminus M$ is a union of the “gaps” separating the intervals I_a. For $|S| = n$ we have at most $n + 1$ gaps. It is not easy to imagine that for countable S the set of gaps may be uncountable. Such closed sets are hard to see by our inner eye. An example is the <u>Cantor set</u> C:

$$C \equiv \textstyle\bigcap_{n=1}^{\infty} C_n\ (\subset [0,\, 1] \equiv C_0)\ \text{ where }\ C_n \equiv \frac{1}{3}C_{n-1} \cup \left(\frac{1}{3}C_{n-1} + \frac{2}{3}\right).$$

Here we denote by $aM + b$ the set $\{ax + b\colon\ x \in M\}$. Thus C is the leftover of the interval $[0, 1]$ obtained deleting first the open middle third $(\frac{1}{3}, \frac{2}{3})$, then by deleting from the rest

$$[0,\, 1] \setminus (\tfrac{1}{3},\, \tfrac{2}{3}) = [0,\, \tfrac{1}{3}] \cup [\tfrac{2}{3},\, 1]$$

the open middle thirds $(\frac{1}{9}, \frac{2}{9})$ and $(\frac{7}{9}, \frac{8}{9})$, and so on.

Exercise 6.5.26 *Prove the following properties of C.*

1. *C is an uncountable set.*
2. *C is closed.*
3. *For every ε there exist k intervals $[a_i, b_i]$, $a_i < b_i$ and $i \in [k]$, such that*

$$\textstyle\sum_{i=1}^{k}(b_i - a_i) \le \varepsilon \ \textit{ and } \ \bigcup_{i=1}^{k}[a_i, b_i] \supset C\,.$$

In this sense C has zero length.

• *Baire's theorem.* The following theorem is due to the French mathematician *René-Louis Baire (1874–1932)*. In the proof we employ closed neighborhoods

$$\overline{U}(b,\,\varepsilon) \equiv [b-\varepsilon,\, b+\varepsilon]\,.$$

These are closed sets.

Theorem 6.5.27 (Baire's) *Suppose that*

$$M = \bigcup_{n=1}^{\infty} M_n$$

where $M \subset \mathbb{R}$ is a nonempty closed set. Then there exists an index n such that the set M_n is not sparse in M.

Proof. Suppose that M is as stated and, for contradiction, that every set M_n is sparse. We take an arbitrary point $b_0 \in M$. Since M_1 is sparse, there is a point $b_1 \in M$ and δ_1 such that $\overline{U}(b_1, \delta_1) \subset U(b_0, 1)$, hence $\delta_1 \le 1$, and

$$M_1 \cap \overline{U}(b_1,\, \delta_1) = \emptyset\,.$$

Suppose that we already defined nested closed neighborhoods

$$\overline{U}(b_1,\, \delta_1) \supset \overline{U}(b_2,\, \delta_2) \supset \cdots \supset \overline{U}(b_n,\, \delta_n)$$

such that $b_i \in M$, $\delta_i \le \frac{1}{i}$ and $M_i \cap \overline{U}(b_i, \delta_i) = \emptyset$ for every $i \in [n]$. Since M_{n+1} is sparse, we can take a point $b_{n+1} \in M$ and $\delta_{n+1} \le \frac{1}{n+1}$ such that

$$\overline{U}(b_{n+1},\, \delta_{n+1}) \subset U(b_n,\, \delta_n) \text{ and } M_{n+1} \cap \overline{U}(b_{n+1},\, \delta_{n+1}) = \emptyset\,.$$

Thus we prolonged the above sequence by $n+1$-st term. Since for every $n \in \mathbb{N}$ the centers $b_n,\, b_{n+1},\, \dots$ lie in $\overline{U}(b_n, \delta_n)$, the sequence (b_n) $(\subset M)$ is Cauchy. We invoke Theorem 2.4.18 and take the limit

$$b \equiv \lim b_n\,.$$

Since M and $\overline{U}(b_n, \delta_n)$ are closed sets, we see that $b \in M$ and $b \in \overline{U}(b_n, \delta_n)$ for every $n \in \mathbb{N}$. Hence $b \notin M_n$ for every $n \in \mathbb{N}$, which is a conrtadiction. □

Exercise 6.5.28 *Suppose that $M \subset \mathbb{R}$ is a nonempty closed set such that for every $b \in M$ and every δ we have $M \cap P(b, \delta) \ne \emptyset$. Then M is uncountable.*

Exercise 6.5.29 *State and prove general version of Baire's theorem for metric spaces.*

6.6 Uniform continuity

• *Uniform continuity.* We define an important strengthening of continuity.

Definition 6.6.1 (uniform continuity) *Let $f \in \mathcal{R}$ and $M \subset \mathbb{R}$. We say that f is uniformly continuous on M if for every ε there is a δ such that for any points $a, b \in M \cap M(f)$,*

$$|a - b| \le \delta \Rightarrow |f(a) - f(b)| \le \varepsilon\,.$$

If $M = M(f)$, we say that f is uniformly continuous, or UC*.*

We denote the subset of uniformly continuous functions in $\mathcal{F}(M)$ by $\mathcal{UC}(M)$ and set

$$\mathcal{UC} \equiv \bigcup_{M\subset\mathbb{R}} \mathcal{UC}(M)\,.$$

Exercise 6.6.2 *Every uniformly continuous function is continuous.*

• *Properties of* UC *functions.* For compact definition domains continuous functions are UC.

Theorem 6.6.3 (continuous ⇒ UC) *If $M \subset \mathbb{R}$ is a compact set then*

$$\mathcal{C}(M) \subset \mathcal{UC}(M), \ \ \textit{hence}\ \mathcal{C}(M) = \mathcal{UC}(M)\,.$$

In words, any continuous function with compact definition domain is uniformly continuous.

Proof. Suppose that $M \subset \mathbb{R}$ is compact and that $f \in \mathcal{F}(M)$ is not uniformly continuous. We prove that there is a point $c \in M$ such that f is not continuous at c. By the assumption there is an ε such that for every δ there exist points $a, b \in M$ such that

$$|a - b| \le \delta \ \text{ and } \ |f(a) - f(b)| > \varepsilon\,.$$

Using the axiom of choice we get two sequences $(a_n), (b_n) \subset M$ such that for every n,

$$|a_n - b_n| \le \tfrac{1}{n} \ \text{ and } \ |f(a_n) - f(b_n)| > \varepsilon\,.$$

Using compactness of M we pass to subsequences (Exercise 6.6.4) and get that $\lim a_n = \lim b_n = c$ for some $c \in M$. Since $|f(a_n) - f(b_n)| > \varepsilon$ for every n, it is not true that $\lim f(a_n) = \lim f(b_n) = f(c)$. By (H) f is not continuous at c. □

Exercise 6.6.4 *Explain the step where we pass to subsequences.*

Exercise 6.6.5 *The functions $f(x) \equiv \frac{1}{x} \,|\, (0,1]$ and $g(x) \equiv \sin(\frac{1}{x}) \,|\, (0,1]$ are continuous but not* UC*.*

Exercise 6.6.6 *Let $M \equiv [0,1] \cap \mathbb{Q}$. Find a function $f \in \mathcal{C}(M) \setminus \mathcal{UC}(M)$.*

The closure of a set $M \subset \mathbb{R}$ is the set

$$\overline{M} \equiv \{b \in \mathbb{R}\colon\ \exists\,(a_n) \subset M \text{ with } \lim a_n = b\} \ \ (\subset \mathbb{R})\,.$$

It is the set $(L(M) \cup M) \setminus \{-\infty, +\infty\}$.

Exercise 6.6.7 *$M \subset \mathbb{R}$ is closed iff $\overline{M} = M$.*

Exercise 6.6.8 *Prove the next proposition.*

Proposition 6.6.9 (UC boundedness) *Let $M \subset \mathbb{R}$ be bounded and let f be in $\mathcal{UC}(M)$. Then*

$$\textit{the image } f[M] \textit{ is bounded},$$

so that f is a bounded function.

The next theorem is important.

Theorem 6.6.10 (UC extensions) *Let $f \in \mathcal{UC}(M)$. The following holds.*

1. *Let $b \in \overline{M}$. For every sequence $(a_n) \subset M$ with $\lim a_n = b$ the limit*

   $$g(b) \equiv \lim f(a_n) \ \ (\in \mathbb{R})$$

   exists and does not depend on the sequence (a_n).

2. *The function*

   $$g\colon \overline{M} \to \mathbb{R}$$

   so defined is the unique UC *extension of f to the closure of M.*

Proof. Let f and M be as stated.

1. Let $b \in \overline{M}$ and $(a_n) \subset M$ be any sequence with $\lim a_n = b$. Let an ε be given. We take a δ such that

$$x,\, y \in M,\ |x - y| \le \delta \Rightarrow |f(x) - f(y)| \le \varepsilon\,.$$

Since (a_n) is Cauchy, there is n_0 such that for every $m, n \ge n_0$ we have $|a_m - a_n| \le \delta$. Thus for the same m and n we have

$$|f(a_m) - f(a_n)| \le \varepsilon$$

and the sequence $(f(a_n))$ is Cauchy. By Theorem 2.4.18 it has a finite limit

$$c \equiv \lim f(a_n)\,.$$

Let $(a_n') \subset M$ be another sequence with $\lim a_n' = b$ and $\lim f(a_n') = c'$. If $c \neq c'$ then

$$(b_n) \equiv (a_1,\, a_1',\, a_2,\, a_2',\, a_3,\, \dots)$$

is a sequence converging to b such that $\lim f(b_n)$ does not exist, which is a contradiction. Hence c does not depend on (a_n) and we can define a function $g\colon \overline{M} \to \mathbb{R}$ by setting

$$g(b) \equiv \lim f(a_n)$$

for any sequence $(a_n) \subset M$ with $a_n \to b$.

2. We prove three results: (i) g extends f, (ii) every continuous extension of f to $\overline{M}$ equals g and (iii) g is UC.

(i) Let $b \in M$. We take the constant sequence $(a_n) = (b, b, \dots) \subset M$. It converges to b and we get that

$$g(b) = \lim f(a_n) = \lim f(b) = f(b)\,.$$

(ii) Let $h \in \mathcal{C}(\overline{M})$ extend f and $b \in \overline{M}$. We take any sequence $(a_n) \subset M$ with $\lim a_n = b$. By (H) and the definition of g we have

$$h(b) = \lim h(a_n) = \lim f(a_n) = g(b)\,.$$

(iii) Let an ε be given. We take a δ such that

$$a, b \in M,\ |a - b| \le \delta \Rightarrow |f(a) - f(b)| \le \tfrac{\varepsilon}{3}\,.$$

Let $a, b \in \overline{M}$ with $|a - b| \le \frac{\delta}{3}$. We take two sequences $(a_n), (b_n) \subset M$ such that $\lim a_n = a$ and $\lim b_n = b$, and take their entries a_m and b_n such that $|a_m - a|, |b_n - b| \le \frac{\delta}{3}$ and $|f(a_m) - g(a)|, |f(b_n) - g(b)| \le \frac{\varepsilon}{3}$. Then

$$\begin{aligned}
&|a_m - b_n| \le |a_m - a| + |a - b| + |b - b_n| \le \tfrac{\delta}{3} + \tfrac{\delta}{3} + \tfrac{\delta}{3} = \delta \text{ and hence}\\
&|g(a) - g(b)| \le |g(a) - f(a_m)| + |f(a_m) - f(b_n)| + |f(b_n) - g(b)| \le\\
&\le \tfrac{\varepsilon}{3} + \tfrac{\varepsilon}{3} + \tfrac{\varepsilon}{3} = \varepsilon\,.
\end{aligned}$$

□

In [50] we rebuild, with the help of this extension theorem, a large part of univariate real analysis so that only functions in the families

$$\mathcal{UC}(M),\ M \subset \mathbb{Q}\,,$$

are used. We conclude this section with a generalization of Corollary 6.5.6.

Theorem 6.6.11 (extremes of UC functions) *Let $M \subset \mathbb{R}$ be bounded and f be in $\mathcal{UC}(M)$. Then there exist points $b, c \in \overline{M}$ such that for every $a \in M$,*

$$f(b) \le f(a) \le f(c)\,.$$

Here $f(b)$ and $f(c)$ are understood to be the values of the extension of f to the closure of M provided by Theorem 6.6.10.

Proof. Let M and f be as stated, and let $g \in \mathcal{C}(\overline{M})$ be the (uniformly) continuous extension of f obtained in Theorem 6.6.10. We prove the existence of the minimum b, for the maximum c the argument is similar. We set

$$B \equiv \inf(f[M]) \ \ (\in \mathbb{R}^*)$$

(Proposition 6.6.9 shows that $B \in \mathbb{R}$). Let $(a_n) \subset M$ be a sequence such that $\lim f(a_n) = B$. Using Theorem 2.4.12, we select a subsequence (b_n) of (a_n) such that $\lim b_n = b$ $(\in \mathbb{R})$. It follows that

$$b \in \overline{M} \ \text{ and } \ g(b) = \lim f(b_n) = \lim f(a_n) = B\,.$$

In particular, $B \in \mathbb{R}$. Since B is a lower bound of $f[M]$, we get that

$$g(b) = B \le f(a) \text{ for every } a \in M\,.$$

$\square$

Exercise 6.6.12 *The inequalities in the theorem hold for every $a \in \overline{M}$.*

6.7 Operations on functions and continuity

We investigate the interplay of the five operations in Definition 5.2.1 with continuity. In this respect most interesting is the inverse $f \mapsto f^{-1}$. We prove that sums of power series are continuous and use it to show that the functions $\exp x$, $\cos x$, and $\sin x$ are continuous. The chapter concludes with the proof of continuity of elementary functions.

• *Arithmetic of continuity.* Recall the operations of sum, product and ratio (division) in Definition 5.2.1.

Theorem 6.7.1 (arithmetic of continuity) *For function $f, g \in \mathcal{R}$ the following holds.*

1. *If f and g are continuous at a point $b \in M(f) \cap M(g)$, <u>then</u>*

   $$f + g \text{ and } fg \text{ are continuous at } b\,.$$

   If f and g are continuous at a point $b \in M(f/g)$, <u>then</u>

   $$f/g \text{ is continuous at } b\,.$$

2. *If $f, g \in \mathcal{C}$ <u>then</u> $f + g, fg, f/g \in \mathcal{C}$.*

Proof. 1. We consider only f/g, arguments for sum and product are similar and easier. Let f, g and b be as stated and let $(a_n) \subset M(f/g)$ have $\lim a_n = b$. By (H) one has $\lim f(a_n) = f(b)$ and $\lim g(a_n) = g(b)$. By Theorem 2.6.2 we have

$$\lim(f/g)(a_n) = \lim \tfrac{f(a_n)}{g(a_n)} = \tfrac{\lim f(a_n)}{\lim g(a_n)} = \tfrac{f(b)}{g(b)} = (f/g)(b)\,.$$

Thus by (H) the function f/g is continuous at b.

2. This follows from the first part. $\square$

Exercise 6.7.2 *Show that* POL $\subset \mathcal{C}$ *and* RAC $\subset \mathcal{C}$, *that is, every polynomial and every rational function is continuous.*

• *Continuity of power series.* We want to prove continuity of all functions introduced in Sections 5.1 and 5.2. Those that were defined by composition and inverses will be discussed later. The continuity of exponential, cosine, and sine follows from the next theorem.

Theorem 6.7.3 (continuity of $\sum_{n=0}^{\infty} a_n x^n$) *Suppose that the real numbers* $a_0, a_1, \ldots,$ *are such that* $\lim |a_n|^{1/n} = 0$. *Then for every* $x \in \mathbb{R}$,

$$S(x) \equiv \textstyle\sum_{n=0}^{\infty} a_n x^n$$

is an abscon series and the sum defines a function $S(x)$ *that is in* $\mathcal{C}(\mathbb{R})$.

Proof. Let the coefficients (a_n) be as stated and $x \in \mathbb{R}$. Then $0 \le |a_n|^{1/n}|x| \le \frac{1}{2}$ for every $n \ge n_0$, so that $|a_n x^n| \le (\frac{1}{2})^n$ for $n \ge n_0$. The series $\sum_{n=0}^{\infty} a_n x^n$ is therefore abscon and converges. With $d \equiv \max(|x|, 1)$ and $|c| \le 1$,

$$\begin{aligned} |S(x+c) - S(x)| &= |c| \cdot \big| \textstyle\sum_{n=1}^{\infty} a_n \sum_{i=1}^{n} \binom{n}{i} c^{i-1} x^{n-i} \big| \\ &\le |c| \cdot \textstyle\sum_{n=1}^{\infty} |a_n| \cdot (2d)^n = O(|c|) \ \ (\text{on } [-1,1])\,. \end{aligned}$$

Thus the function $S(x)$ is continuous at x. □

Exercise 6.7.4 *Explain the displayed computation in the previous proof.*

Exercise 6.7.5 *Prove that* $\lim (n!)^{1/n} = +\infty$.

Corollary 6.7.6 (e^x, cosine and sine) *The functions* e^x, $\cos x$ *and* $\sin x$ *are continuous on* $\mathbb{R}$.

Proof. This follows from the definitions

$$\mathrm{e}^x = \textstyle\sum_{n=0}^{\infty} \frac{x^n}{n!},\ \cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \ \text{ and } \ \sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}\,,$$

from Theorem 6.7.3 and from Exercise 6.7.5. □

Corollary 6.7.7 (tangent and cotangent) *Both functions* $\tan x$ *and* $\cot x$ *are continuous on their definition domains.*

Proof. As we know, $\tan x = \frac{\sin x}{\cos x}$ and $\cot x = \frac{\cos x}{\sin x}$. Thus continuity of these functions follows from Corollary 6.7.6 and part 2 of Theorem 6.7.1. □

• *Restriction and composition.* We treated restrictions in Proposition 6.1.5. Now we supplement it with the point-wise form.

Proposition 6.7.8 (restriction 2) *Let* $f \in \mathcal{R}$ *and* $X \subset \mathbb{R}$. *Then the following holds.*

1. *If f is continuous at $b \in M(f \,|\, X)$ <u>then</u> $f \,|\, X$ is continuous at b.*

2. *If $f \in \mathcal{C}$ <u>then</u> $f \,|\, X \in \mathcal{C}$.*

Proof. 1. Let f, b and X be as stated. We take any sequence

$$(b_n) \subset M(f \,|\, X) = M(f) \cap X$$

with $\lim b_n = b$. By (H) we have that $\lim f(b_n) = f(b)$. Thus $\lim(f \,|\, X)(b_n) = \lim f(b_n) = f(b)$ and (H) gives that $f \,|\, X$ is continuous at b.

2. This follows from part 1. Another proof, using neighborhoods, is given in Proposition 6.1.5. $\Box$

We consider the operation of composition.

Theorem 6.7.9 (continuity of composition) *Let $f, g \in \mathcal{R}$. Then the following holds.*

1. *If g is continuous at $b \in M(f(g))$ and f is continuous at $g(b)$ <u>then</u> $f(g)$ is continuous at b.*

2. *If $f, g \in \mathcal{C}$ <u>then</u> $f(g) \in \mathcal{C}$.*

Proof. 1. Let f, g and b be as stated, and

$$(b_n) \subset M(f(g))$$

be a sequence with $\lim b_n = b$. Then $\lim g(b_n) = g(b)$ by (H). Hence, again by (H), $\lim f(g)(b_n) = \lim f(g(b_n)) = f(g(b)) = f(g)(b)$. (H) shows that $f(g)$ is continuous at b.

2. This follows from the first part. $\Box$

Exercise 6.7.10 *Prove part 1 of Theorem 6.7.9 by means of neighborhoods.*

• *Inverses.* The interplay of inverses and continuity is much more interesting than what we have seen so far. The next exercise shows that, in general, inverses do not preserve continuity.

Exercise 6.7.11 *Let $f \in \mathcal{F}(\mathbb{N}_0)$ be defined by*

$$f(0) \equiv 0,\ f(n) \equiv \tfrac{1}{n} \ \ldots\ n \in \mathbb{N}\,.$$

Show that $f \in \mathcal{C}$ but that the inverse f^{-1} is discontinuous.

On the other hand, in many situations inverses do preserve continuity.

Theorem 6.7.12 (continuity of inverses) *Let $M \subset \mathbb{R}$, $f \in \mathcal{C}(M)$ and f be injective. <u>Then</u> in each of five situations the inverse $f^{-1} \in \mathcal{F}(f[M])$ is continuous.*

1. *When M is a compact set.*

2. *When M is an interval.*

3. *When M is an open set.*

4. *When M is a closed set and f is a monotone function.*

5. *When $M \subset (a, b)$, M is dense in (a, b) and f is monotone and* UC.

Proof. Let M and f be as stated.

1. Let M be compact, $b \in f[M]$ and let

$$(b_n) \subset f[M] \text{ be a sequence with } \lim b_n = b\,.$$

Let $a \equiv f^{-1}(b)$ and $a_n \equiv f^{-1}(b_n)$ $(\in M)$. We show that $\lim a_n = a$, which by (H) proves continuity of f^{-1} at b. We show that every subsequence of (a_n) has a subsequence with the limit a. Part 3 of Theorem 2.2.11 then implies that $\lim a_n = a$. Let (a'_n) be a subsequence of (a_n). We use compactness of M and take a subsequence (a_{m_n}) of (a'_n) with $\lim\, a_{m_n} = c \in M$. By (H) it holds that $\lim\, f(a_{m_n}) = f(c) = b$ because $(f(a_{m_n}))$ is a subsequence of (b_n). Since f is injective, $c = a$.

2. Let M be an interval. Corollary 6.4.6 shows that f increases or decreases. Suppose that f decreases, the case of increasing f is similar. Theorem 6.4.1 says that $f[M]$ is an interval. Let $b \in f[M]$ and an ε be given. We show that f^{-1} is right-continuous at b. This is true when b is the right endpoint of the interval $f[M]$ because then

$$U^+(b,\, \delta) \cap f[M] = \{b\}\,.$$

We assume that b is not the right endpoint of $f[M]$. Since f^{-1} decreases, $a \equiv f^{-1}(b)$ $(\in M)$ is not the left endpoint of the interval M. We take a small ε such that $[a - \varepsilon, a] \subset M$. We set

$$\delta \equiv f(a - \varepsilon) - f(a) = f(a - \varepsilon) - b \ \ (> 0)\,.$$

Theorem 6.4.1 implies that f is a decreasing bijection from $[a-\varepsilon, a]$ to $[b, b+\delta]$, and from $(a - \varepsilon, a]$ to $[b, b + \delta)$. So $[b, b + \delta) \subset f[M]$ and $U^+(b,\, \delta) \cap f[M] = U^+(b,\, \delta) = [b, b + \delta)$. Hence

$$f^{-1}[U^+(b,\, \delta)] = U^-(a,\, \varepsilon) \subset U(a,\, \varepsilon) = U(f^{-1}(b),\, \varepsilon)$$

and f^{-1} is right-continuous at b. The left continuity of f^{-1} at b is proven similarly. By Exercise 4.3.11, f^{-1} is continuous at b.

3. Let M be open, $b \in f(M)$, $a \equiv f^{-1}(b)$ $(\in M)$ and let ε be given. We take ε so small that $U(a, \varepsilon) \subset M$. Proposition 6.5.20 says that the set $f[U(a, \varepsilon)]$ $(\ni b)$ is open. So for some δ we have that $U(b, \delta) \subset f[U(a, \varepsilon)]$. Thus

$$f^{-1}[U(b,\, \delta)] \subset U(a,\, \varepsilon) = U(f^{-1}(b),\, \varepsilon)\,.$$

So f^{-1} is continuous at b by Definition 4.3.1.

4. Let M be closed and f be increasing. For decreasing f we argue similarly. We assume for contradiction that for some $b \in f[M]$ there is a sequence $(b_n) \subset f[M]$ such that

$$\lim b_n = b \text{ but } \lim f^{-1}(b_n) \text{ does not exist or } \neq a \equiv f^{-1}(b)\ (\in M)\,.$$

By part 2 of Theorem 2.2.11 and by Proposition the sequence (b_n) has a decreasing or an increasing subsequence (c_n) such that $\lim f^{-1}(c_n) = B\ (\in \mathbb{R}^*)$ and $B \neq a$. We assume that (c_n) decreases, the case of increasing (c_n) is similar. Then

$$b < \cdots < c_2 < c_1 \text{ and } a < \cdots < f^{-1}(c_2) < f^{-1}(c_1)$$

because both f and f^{-1} increase. By part 2 of Theorem 2.8.1 we have that $B \in [a, f^{-1}(c_1))$. Thus, crucially, $B \in \mathbb{R}$ (here the argument fails for non-monotone f). Even $B \in M$ because M is closed. Due to the continuity of f in B we have that $f(B) = \lim f(f^{-1}(c_n)) = \lim c_n = b = f(a)$. But this contradicts the injectivity of f because $B \neq a$.

5. Let M, a, b and f be as stated. We use Theorem 6.6.10 and continuously extend f to a function $\overline{f}\colon [a,b] \to \mathbb{R}$. By Proposition 2.8.5 and because M is dense in (a,b), the function $\overline{f}$ is strictly monotone and therefore injective. By part 1 or part 2 of this theorem the function $(\overline{f})^{-1}$ is continuous. By Proposition 6.1.5, the restriction $(\overline{f})^{-1} \mid f[M] = f^{-1}$ is continuous. $\Box$

In *MA 1*$^+$ we generalize part 5.

Exercise 6.7.13 *Give examples showing that in part 4 of the theorem neither the closedness of M nor the monotonicity of f can be omitted.*

- *Continuity of elementary functions.* We prove that $\mathrm{EF} \subset \mathcal{C}$.

Exercise 6.7.14 *Prove the following corollary.*

Corollary 6.7.15 (continuity of some BEF) *The functions* $\log x$, $\arccos x$, $\arcsin x$, $\arctan x$ *and* $\operatorname{arccot} x$ *are continuous.*

Proposition 6.7.16 (continuity of x^b) *For every $b \in (0, +\infty)$ the function $x^b\colon [0, +\infty) \to [0, +\infty)$ is continuous.*

Proof. Let $b > 0$ and $x > 0$. We find that x^b is continuous at x using the expression $x^b = \exp(b \log x)$, the continuity of e^x (Corollary 6.7.6), continuity of logarithm (Corollary 6.7.15), continuity of the constant function k_b (Exercise 6.1.3), continuity of the product (Theorem 6.7.1) and continuity of the composition (Theorem 6.7.9). Continuity at $x = 0$ follows with the help of Proposition 4.3.5 from the limit

$$\lim_{x \to 0} x^b = \lim_{x \to 0} \exp(b \log x) = \lim_{y \to -\infty} \exp y = 0 = 0^b\,.$$

Here the second equality follows from Theorem 4.5.1 and from part 2 of Proposition 5.1.9. The third equality follows from part 3 of Proposition 5.1.6. $\Box$

Theorem 6.7.17 (EF $\subset \mathcal{C}$) *Every elementary function is continuous.*

Proof. We proceed by induction on the length of the generating word of the given elementary function f (Definition 5.2.19). If f is a constant function, exponential, logarithm, x^b with non-integral exponent $b > 0$, sine or arcsine, it is continuous by, respectively, Exercise 6.1.3, Corollaries 6.7.6 and 6.7.15, Proposition 6.7.16 and Corollaries 6.7.6 and 6.7.15. If f is a sum, a product, a ratio or a composition of two simpler elementary functions, it is continuous by induction and Theorems 6.7.1 and 6.7.9. $\square$

Chapter 7

Derivatives

We define the derivative at any limit point in the definition domain and consider derivatives both locally at points, and globally as a unary operation on $\mathcal{R}$. This is more general than the standard approach, which defines derivatives only at inner points of domains. Our approach yields natural and general formulas for the derivatives of inverse functions.

Section 7.1 contains basic definitions. Theorem 7.1.9 is our version of the popular criterion of local extremes for functions with arbitrary definition domains. By Proposition 7.1.14, finite derivative implies continuity. If it is additionally nonzero, the point is mapped to a limit point of the image (Proposition 7.1.18 and Exercise 7.1.20). Theorem 7.1.29 describes a function with discontinuous derivative. Another such function is described in Exercise 7.5.11.

Section 7.2 starts with Definition 7.2.1 of standard tangent lines. Theorem 7.2.4 shows that lines touching the graph in a single point are tangents. Definition 7.2.10 formalizes the intuition of a tangent as a limit of secants. By Theorem 7.2.12, standard and limit tangents are equivalent. Theorem 7.2.15 shows that the tangent is a limit of secants whose two intersection points enclose the point of contact.

Section 7.3 is devoted to the arithmetic of derivatives. In Theorem 7.3.1 we investigate local and global derivatives of sums. We consider the relation between f', g' and $(f+g)'$ for any pair of functions $f, g \in \mathcal{R}$; in general, $(f+g)' \neq f' + g'$. Theorem 7.3.5 presents local and global Leibniz formulas for derivatives of products. Theorem 7.3.9 does the same for ratios. Corollaries 7.3.4, 7.3.8, 7.3.11, 7.4.3 and 7.4.7 provide assumptions, under which the respective formulas

$$(f+g)' = f' + g',\ (fg)' = f'g + fg',\ (\tfrac{f}{g})' = \tfrac{f'g - fg'}{g^2},\ (f(g))' = f'(g) \cdot g'$$
$$\text{and } (f^{-1})' = \tfrac{1}{f'(f^{-1})}$$

do hold.

In Section 7.4 we turn to derivatives of composite functions (Theorem 7.4.1) and inverses (Theorem 7.4.4 and Corollary 7.4.6). As before, we give local and global formulas and allow arbitrary definition domains. Proofs use Heine's definition of derivatives.

In Section 7.5 in Theorem 7.5.1 we differentiate power series. By this we get derivatives of the functions $\exp x$, $\sin x$, and $\cos x$. We also find the derivative of $\log x$, but the derivatives of other basic elementary functions (Definition 5.1.1) are left to exercises. On page 231 we summarize these derivatives in a table.

In Section 7.6 we pose Problem 7.6.1: decide if derivatives of elementary functions are always elementary. In Theorem 7.6.5 we prove that a subfamily of so called simple elementary functions is closed to derivatives.

7.1 Local and global derivatives

Why derivatives? Using them we obtain local linear approximations of functions and tangent lines to their graphs. In Chapter 9 we investigate more precise local polynomial approximations. The derivative f' of a function f records properties of f in simpler form. For example, the increase of f is recorded as the positivity of f'. If f is elementary, it is straightforward to compute f' locally. However, see Problem 7.6.1.

• *Local derivatives.* Recall that $\mathcal{F}(M)$ is the set of functions $f\colon M \to \mathbb{R}$ where $M \subset \mathbb{R}$, that $\mathcal{R} = \bigcup_{M\subset\mathbb{R}} \mathcal{F}(M)$, and that $L(M)$ $(\subset \mathbb{R}^*)$ is the set of limit points of M.

Definition 7.1.1 (local derivatives) *Let $f \in \mathcal{F}(M)$ and $b \in M \cap L(M)$. The derivative of the function f at the point b is the limit*

$$f'(b) := \lim_{x\to b} \frac{f(x)-f(b)}{x-b} \quad (\in \mathbb{R}^*)\,.$$

This limit need not exist. If $f'(b)$ exists, then $b \in M(f) \cap L(M(f))$ and we often do not mention it explicitly. The uniqueness of limits of functions implies the uniqueness of derivatives. Our definition is more general than the standard definition, which allows for b only inner points of definition domains.

Exercise 7.1.2 *Prove that also $f'(b) = \lim_{h\to 0} \frac{f(b+h)-f(b)}{h}$.*

Proposition 7.1.3 (derivatives are local) *Let $f, g \in \mathcal{R}$ and b be in $M(f) \cap M(g)$. If $f \,|\, U(b,\theta) = g \,|\, U(b,\theta)$ for some $\theta > 0$, then*

$$f'(b) = g'(b)\,,$$

if either side is defined.

Proof. This is immediate from Proposition 4.1.8 and Definition 7.1.1. □

Unlike in Proposition 4.1.8, the neighborhood $P(b,\theta)$ does not suffice because the definitions of $f'(b)$ and $g'(b)$ involve $f(b)$ and $g(b)$.

Let $f \in \mathcal{R}$ and $b \in M(f)$. If $f'(b) \in \mathbb{R}$, we say that f is <u>differentiable</u> at b. Then f has near b a local approximation by the linear function

$$t(x) = f(b) + f'(x) \cdot (x - b) \colon \mathbb{R} \to \mathbb{R}\,.$$

We say that $t(x)$ is the <u>tangent (line)</u> to $f(x)$ at b. The approximation takes the form

$$f(x) = \underbrace{f(b) + f'(b) \cdot (x - b)}_{\text{the tangent at } b} \;+\; \underbrace{o(x - b)}_{\text{the error}} \;\; (x \to b)\,.$$

Heine's definition of derivatives, abbreviated HDD, is a useful tool in proofs.

Proposition 7.1.4 (HDD) *Let $f \in \mathcal{F}(M)$, $b \in M \cap L(M)$ and $B \in \mathbb{R}^*$. <u>Then</u> $f'(b) = B \iff$ for every sequence $(a_n) \subset M \setminus \{b\}$ with $\lim a_n = b$,*

$$\lim_{n \to \infty} \frac{f(a_n) - f(b)}{a_n - b} = B\,.$$

Proof. This follows from Definition 7.1.1 and Theorem 4.1.12. □

• *One-sided derivatives.* Let $M \subset \mathbb{R}$. Recall that $L^-(M)$ $(\subset \mathbb{R})$ are the left-sided limit points of the set M. Similarly, $L^+(M)$ $(\subset \mathbb{R})$ are the right-sided limit points of M.

Definition 7.1.5 (one-sided derivatives) *Let $f \in \mathcal{F}(M)$ and b be in $M \cap L^-(M)$. The <u>left-sided derivative</u> of the function f at the point b is the left-sided limit*

$$f'_-(b) := \lim_{x \to b^-} \frac{f(x) - f(b)}{x - b} \quad (\in \mathbb{R}^*)\,.$$

Changing the signs $-$ in the indices to $+$, we get the <u>right-sided derivative</u> $f'_+(b)$ of f at b.

Exercise 7.1.6 *The following holds.*

1. *If $f'(a) = A$ then $f'_-(a) = A$ or $f'_+(a) = A$.*
2. *If $f'_-(a) = f'_+(a) = A$ then $f'(a) = A$.*
3. *If $f'_-(a) = A \neq B = f'_+(a)$ then $f'(a)$ does not exist.*

Exercise 7.1.7 *As in Proposition 4.2.13, we can reduce one-sided derivatives to two-sided by restricting the function. State it in detail and prove it.*

• *Zeros of derivatives and extremes.* Let $M \subset \mathbb{R}$. Recall that $a \in \mathbb{R}$ is a two-sided limit point of M if for every δ, both intersections $(a - \delta, a) \cap M$ and $(a, a + \delta) \cap M$ are nonempty. The set of two-sided limit points of M is denoted by $L^{\mathrm{TS}}(M)$ $(\subset \mathbb{R})$.

Exercise 7.1.8 $L^{\mathrm{TS}}(M) \subset L(M)$ *and this inclusion is in general strict.*

Here is the well known criterion of local extremes in terms of vanishing derivatives for general domains.

Theorem 7.1.9 (zeros and extremes) *Let $f \in \mathcal{F}(M)$ and $b \in M \cap L^{\mathrm{TS}}(M)$. If $f'(b) \in \mathbb{R}^* \setminus \{0\}$ then for every δ there exist $c, d \in U(b, \delta) \cap M$ such that*

$$f(c) < f(b) < f(d)$$

—f does not have at b a local extreme.

Proof. Let $f'(b) < 0$, the case when $f'(b) > 0$ is similar. We take an ε such that $U(f'(b), \varepsilon) < 0$. Let a δ be given. By Definition 7.1.1, there is a $\theta \le \delta$ such that for every $x \in P(b, \theta) \cap M$ we have

$$\tfrac{f(x)-f(b)}{x-b} \in U(f'(b), \varepsilon) \text{ and hence } \tfrac{f(x)-f(b)}{x-b} < 0\,.$$

Since $b \in L^{\mathrm{TS}}(M)$, we can take points $c \in (b, b+\theta) \cap M$ and $d \in (b-\theta, b) \cap M$. Then, by the above inequality, $f(c) < f(b)$ and $f(d) > f(b)$. □

Exercise 7.1.10 *In the theorem, $U(b, \theta)$ can be replaced with $P(b, \theta)$.*

Exercise 7.1.11 *Function $f(x) = x \colon [0, 1] \to \mathbb{R}$ has derivatives $f'(0) = f'(1) = 1 \ne 0$ and global extremes at 0 and 1. Does it contradict the theorem?*

We restate the previous theorem in the contrapositive.

Corollary 7.1.12 (on local extremes) *Let $f \in \mathcal{F}(M)$ and let $b \in M$. If the function f has a local extreme at the point b, then (exactly) one of the following three claims holds.*

1. *$b \notin L^{\mathrm{TS}}(M)$.*
2. *$b \in L^{\mathrm{TS}}(M)$ but the derivative $f'(b)$ does not exist.*
3. *$b \in L^{\mathrm{TS}}(M)$ and $f'(b) = 0$.*

• *An example.* We want to find the extremes of the function

$$f(x) = x^2 \colon \mathbb{Q} \to \mathbb{Q}\,.$$

It is easy to see that $f'(x) = 2x \colon \mathbb{Q} \to \mathbb{Q}$. Since every $b \in M(f)$ is a two-sided limit point of $M(f)$ and $f'(b)$ always exists, Corollary 7.1.12 says that f may have a local extreme only at a zero of f'. We have $f'(b) = 0$ iff $b = 0$. So there is a single "suspicious" point $b = 0$. Indeed, f has at 0 a strict global minimum.

Exercise 7.1.13 *What about the function $f(x) = x^2 \colon \mathbb{Z} \to \mathbb{N}_0$?*

• *Derivatives and continuity.* In this passage we show that differentiability at a point is a stronger property than local continuity at that point.

Proposition 7.1.14 (derivatives and continuity) *Let $f \in \mathcal{R}$. If $f'(b) \in \mathbb{R}$ then f is continuous at b.*

Proof. We compute

$$\begin{aligned}\lim_{x\to b} f(x) &= \lim_{x\to b} \left(f(b) + (x-b)\cdot \tfrac{f(x)-f(b)}{x-b}\right)\\ &= \lim_{x\to b} f(b) + \lim_{x\to b}(x-b)\cdot \lim_{x\to b} \tfrac{f(x)-f(b)}{x-b}\\ &= f(b) + 0\cdot f'(b) = f(b)\,.\end{aligned}$$

Note that the function $f(x)$ differs from the function $f(b)+(x-b)\cdot\frac{f(x)-f(b)}{x-b}$ (Exercise 7.1.15). The first equality is therefore nontrivial and follows from Proposition 4.1.8. The second equality follows from Theorem 4.4.5. In the third equality, we use that $f'(b) \neq \pm\infty$. By Proposition 4.3.5, the function f is continuous at the point b. □

Exercise 7.1.15 *We have $f(x) \neq f(b) + (x-b)\cdot \frac{f(x)-f(b)}{x-b}$.*

Exercise 7.1.16 *Show that* $\mathrm{sgn}'(0) = +\infty$.

So an infinite derivative at a point does not imply continuity at that point.

Exercise 7.1.17 *Show that* $(|x|)'_-(0) = -1$ *and* $(|x|)'_+(0) = +1$.

By item 3 in Exercise 7.1.6, $(|x|)'(0)$ does not exist. Of course, continuity at a point does not imply the existence of a derivative at that point.

Proposition 7.1.18 (limit points of images) *Let f be in $\mathcal{F}(M)$. If $f'(b)$ is in $\mathbb{R}\setminus\{0\}$, then*

$$f(b) \in L(f[M])\,.$$

Proof. Let an ε be given. Since $f'(b) \neq 0$, by Definition 7.1.1 there is a δ such that $f(x) \neq f(b)$ for every $x \in P(b,\delta)\cap M$. By Proposition 7.1.14, we can take this δ so small that for the same x, we have $f(x) \in U(f(b),\varepsilon)$. Since $b \in L(M)$, there is a point $a \in P(b,\delta)\cap M$. Then

$$f(a) \in P(f(b),\varepsilon)\cap f[M]\,.$$

Hence $f(b) \in L(f[M])$. □

We use this proposition to differentiate inverse functions.

Exercise 7.1.19 *Proposition 7.1.18 does not hold if $f'(b) = \pm\infty$.*

Exercise 7.1.20 *Proposition 7.1.18 holds if $f'(b) = 0$ and f is non-constant on any neighborhood $U(b,\delta)$.*

Exercise 7.1.21 *Adapt Proposition 7.1.14 for one-sided derivatives and one-sided continuity.*

• *Derivatives of the root.* Recall that $\sqrt{x} = x^{1/2}$ and that $M(\sqrt{x}) = [0, +\infty)$. Let $a \ge 0$. We compute

$$(\sqrt{x})'(a) = \lim_{x \to a} \tfrac{\sqrt{x} - \sqrt{a}}{x - a} = \lim_{x \to a} \tfrac{x-a}{(x-a)(\sqrt{x}+\sqrt{a})} = \lim_{x \to a} \tfrac{1}{\sqrt{x}+\sqrt{a}}\,.$$

The last equality again follows from Proposition 4.1.8. Thus $(\sqrt{x})'(0) = +\infty$, and for $a > 0$ we have $(\sqrt{x})'(a) = \frac{1}{2\sqrt{a}}$. We see that an infinite derivative at a point does not exclude continuity at that point.

Exercise 7.1.22 *Compute* $(\sqrt{x})'_{-}(a)$ *and* $(\sqrt{x})'_{+}(a)$ *for real* $a \ge 0$.

• *Global derivative.* We extend our repertoire of operations on $\mathcal{R}$ in Definition 5.2.1 by a unary operation of (global) derivative. For $f \in \mathcal{R}$ we set

$$D(f) := \{b \in M(f)\colon\ f'(b) \in \mathbb{R}\}\ \ (\subset M(f))\,.$$

Thus $D(f)$ is the subset of the domain where f is differentiable. We formally define a function $f'\colon D(f) \to \mathbb{R}$ by

$$D(f) \ni b \mapsto f'(b) \in \mathbb{R}\,.$$

Definition 7.1.23 (global derivative) *We call the unary operation on* $\mathcal{R}$,

$$\mathcal{R} \ni f \mapsto f' \in \mathcal{R}\,,$$

the global derivative.

$D(f) = M(f')$ may be a proper subset of $M(f)$, which causes troubles. For example, $M(\sqrt{x}) = [0, +\infty)$ but

$$D(\sqrt{x}) = M((\sqrt{x})') = (0,\, +\infty)\,.$$

Let $b \in \mathbb{R}$. The notation $f'(b)$ now becomes a little ambiguous: as the derivative of f at b it may be $\pm\infty$, but as the value of the function f' $(\in \mathcal{R})$ at b it has to be finite. We will use such formulations that the interpretation of the symbol $f'(b)$ will be clear. We will investigate interactions of derivatives with various operations.

Proposition 7.1.24 (derivatives and restrictions) *Let* $f \in \mathcal{R}$*, let* $X \subset \mathbb{R}$*, and let* $M = M(f) \cap X$*. The following holds.*

1. *If* $f'(b)$ *exists and* $b \in M \cap L(M)$*, then* $(f \mid X)'(b) = f'(b)$.
2. *The restriction* $f' \mid M \cap L(M)$ *is a subfunction of* $(f \mid X)'$.

Proof. 1. By Proposition 4.1.10, $(f\,|\,X)'(b)$ is

$$\lim_{x\to b}\tfrac{(f\,|\,X)(x)-(f\,|\,X)(b)}{x-b}=\lim_{x\to b}\big(\tfrac{f(x)-f(b)}{x-b}\,\big|\,X\big)(x)=\lim_{x\to b}\tfrac{f(x)-f(b)}{x-b}=f'(b)\,.$$

2. Let $N=M\cap L(M)$, $g=f'\,|\,N$ and $c\in M(g)$. Then $c\in D(f)\cap N$ and by part 1 we have $(f\,|\,X)'(c)=f'(c)=g(c)$. Hence $g=f'\,|\,N$ is a subfunction of $(f\,|\,X)'$. $\Box$

Exercise 7.1.25 *For every $c\in\mathbb{R}$ we have $k_c'(x)=k_0(x)$.*

Exercise 7.1.26 *For every $c\in\mathbb{R}$ we have $(\mathrm{id}_{\mathbb{R}}(x)+k_c(x))'=k_1(x)$.*

Exercise 7.1.27 *For every $f\in\mathcal{R}$ we have $f\,|\,D(f)\in\mathcal{C}$.*

• *Global one-sided derivatives.* For any function $f\in\mathcal{R}$ we introduce sets

$$\begin{aligned}D_-(f)\ &:=\ \{b\in M(f)\colon\ f_-'(b)\in\mathbb{R}\}\text{ and}\\ D_+(f)\ &:=\ \{b\in M(f)\colon\ f_+'(b)\in\mathbb{R}\}\,.\end{aligned}$$

We define the global left-sided derivative of f by

$$f_-'(x)\colon D_-(f)\to\mathbb{R},\ D_-(f)\ni b\mapsto f_-'(b)\in\mathbb{R}\,,$$

and the (global) right-sided derivative of f by

$$f_+'(x)\colon D_+(f)\to\mathbb{R},\ D_+(f)\ni b\mapsto f_+'(b)\in\mathbb{R}\,.$$

Exercise 7.1.28 *How do $D(f)$, $D_-(f)$ and $D_+(f)$ relate?*

• *Discontinuous derivatives.* We define a function $f\in\mathcal{R}$ such that $M(f)=D(f)\neq\emptyset$ (thus $f\in\mathcal{C}$) but $f'\not\in\mathcal{C}$.

Theorem 7.1.29 (discontinuous derivative) *There exists a function $f\in\mathcal{R}$ such that*

$$M(f)=D(f)\neq\emptyset\ \textit{and}\ f'\not\in\mathcal{C}\,.$$

Proof. Let (a_n) and (b_n) be real sequences going to 0 such that

$$a_1>b_1>a_2>b_2>\cdots>0\ \text{ and }\ a_n-b_n=o(b_n)\ \ (n\to\infty)\,.$$

Let $N=\{0\}\cup\bigcup_{n=1}^{\infty}(b_n,\,a_n)$. We define $f\in\mathcal{F}(N)$ by

$$f(0)=0\ \text{ and }\ f(x)=x-b_n\text{ for }x\in(b_n,\,a_n)\,.$$

Let $x\in(b_n,a_n)$. Then, by Proposition 7.1.3 and Exercise 7.1.26, $f'(x)=1$. We have $0\in L(N)$ and

$$f'(0)=\lim_{x\to 0}\tfrac{f(x)}{x}=0$$

because for every $x \in (b_n, a_n)$ we have (Exercise 7.1.30)

$$\left|\tfrac{f(x)}{x}\right| \le \tfrac{a_n - b_n}{b_n} \to 0 \quad (n \to \infty)\,.$$

Hence $D(f) = N$. Since $0 \in L(N)$, $f'(0) = 0$ and $f'(x) = 1$ for every $x \in N \backslash \{0\}$, the derivative f' is a discontinuous function. $\Box$

Exercise 7.1.30 *Why* $\lim \frac{a_n - b_n}{b_n} = 0$ *as* $n \to \infty$*?*

7.2 Standard and limit tangents

We define two kinds of tangent lines.

• *Standard tangents.* Tangent at a point means a finite derivative at that point.

Definition 7.2.1 (tangents) *Let* $f \in \mathcal{F}(M)$*,* $b \in M \cap L(M)$ *and* $f'(b) \in \mathbb{R}$*. The tangent line to the graph* G_f *at the point* $\langle b, f(b)\rangle$ *is the line*

$$\ell = \{\langle x,\, y\rangle \in \mathbb{R}^2\colon\; x \in \mathbb{R},\, y = f'(b)(x-b) + f(b)\} \quad (\subset \mathbb{R}^2)\,.$$

So ℓ goes through the plane point $\langle b, f(b)\rangle$. We rewrite the equation as

$$y = f'(b)x + f(b) - f'(b)b$$

and see that ℓ has the slope $f'(b)$ $(\in \mathbb{R})$. We view the tangent also as the function $\ell \in \mathcal{F}(\mathbb{R})$ given by

$$\ell(x) = f'(b)(x-b) + f(b) = f'(b)x + f(b) - f'(b)b\,.$$

Exercise 7.2.2 *The function* $\ell(x)$ *is the only linear polynomial such that*

$$f(x) = \ell(x) + o(x-b) \quad (x \to b)\,.$$

Exercise 7.2.3 *Let* $f(x) = \sqrt{x}$*. Determine tangents at the points* $\langle a, \sqrt{a}\rangle$*.*

• *Touching lines are tangents.* We show that every line touching the graph at a single point where the derivative exists is the tangent at that point.

Theorem 7.2.4 (touching lines) *Let* f *be in* $\mathcal{F}(M)$*,* $b \in M \cap L^{\mathrm{TS}}(M)$ *and* $f'(b) \in \mathbb{R}^*$*. If the linear function*

$$l(x) = sx + t$$

has value $l(b) = f(b)$ *and if* $f(x) \ge l(x)$ *for every* $x \in M$*, then* $l(x)$ *is the tangent to* G_f *at* $\langle b,\, f(b)\rangle$*.*

Proof. Let $g(x) = f(x) - l(x)$ $(\in \mathcal{F}(M))$. By Exercise 7.2.5, $g'(b) = f'(b) - s$. If $g'(b) \neq 0$, by Theorem 7.1.9 there is a point $c \in M$ near b such that

$$g(c) < g(b) = 0 \text{ and hence } f(c) < l(c)\,,$$

in contradiction with the assumption. Thus $g'(b) = 0$ and $f'(b) = s$. It follows that $l(x)$ is the tangent line to the graph G_f at the point $\langle b, f(b)\rangle$. $\square$

Exercise 7.2.5 *Show that $g'(b) = f'(b) - s$.*

Exercise 7.2.6 *Modify the theorem for $l(x)$ touching G_f from above.*

• *Non-vertical lines.* Non-vertical lines in the plane are the lines

$$\ell = \{\langle x,\, y\rangle \in \mathbb{R}^2 \colon\ x,\, y \in \mathbb{R},\, y = sx + t\} \ \ (\subset \mathbb{R}^2)\,,$$

where $s, t \in \mathbb{R}$. Note that s and t are unique: to different pairs s, t correspond different lines ℓ. We call s the slope of ℓ. We denote the set of non-vertical lines by $\mathcal{N}$ $(\subset \mathcal{P}(\mathbb{R}^2))$.

Exercise 7.2.7 *The function $p\colon \mathcal{N} \to \mathbb{R}^2$, given by*

$$p(\ell) = \langle p_1(\ell),\, p_2(\ell)\rangle = \langle s,\, t\rangle\,,$$

with s and t as in $\ell = \ell(x) = sx + t$, is a bijection.

Definition 7.2.8 (limits in $\mathcal{N}$) *Let $(\ell_n) \subset \mathcal{N}$, $\ell \in \mathcal{N}$ and let the function $p(\cdot) = \langle p_1(\cdot), p_2(\cdot)\rangle$ be as in Exercise 7.2.7. If*

$$\lim_{n\to\infty} p_1(\ell_n) = p_1(\ell) \ \text{ and } \ \lim_{n\to\infty} p_2(\ell_n) = p_2(\ell)\,,$$

we say that the line ℓ is the limit of lines ℓ_n, and write $\mathrm{Lim}\,\ell_n = \ell$.

Exercise 7.2.9 *Let $A = \langle a, b\rangle$ and $A' = \langle a', b'\rangle$ be in $\mathbb{R}^2$ and $a \neq a'$. There is a unique line $\ell \in \mathcal{N}$ such that $A, A' \in \ell$. It has slope $\frac{b'-b}{a'-a}$.*

We denote this unique non-vertical line going through A and A' by

$$\kappa(A,\, A') \text{ or by } \kappa(a,\, b,\, a',\, b')\,.$$

If $f \in \mathcal{R}$ and $A, A' \in G_f$ with $A \neq A'$, we call the line $\kappa(A, A')$ a secant line of the graph G_f.

• *Limit tangents.* We often read in textbooks and lecture notes that tangents are limits of secants, but the details of this limiting process are never revealed. We explain it here.

Definition 7.2.10 (limit tangents) *Let $f \in \mathcal{F}(M)$, $b \in M \cap L(M)$ and let $\ell \in \mathcal{N}$. The line ℓ is a* <u>*limit tangent*</u> *to the graph G_f at the point $\langle b, f(b)\rangle$, if for every sequence $(a_n) \subset M \setminus \{b\}$ with* $\lim a_n = b$ *we have, by Definition 7.2.8, the limit*

$$\mathrm{Lim}\, \kappa(b,\, f(b),\, a_n,\, f(a_n)) = \ell\,.$$

This definition of tangents does not need derivatives.

Exercise 7.2.11 *If ℓ is a limit tangent to G_f at $\langle b, f(b)\rangle$ then $\langle b, f(b)\rangle \in \ell$.*

We prove that the two definitions of tangents are logically equivalent.

Theorem 7.2.12 (standard and limit tangents) *Let $f \in \mathcal{F}(M)$, $b \in M \cap L(M)$ and let $\ell \in \mathcal{N}$.* <u>*Then*</u> *ℓ is a tangent to G_f at $\langle b, f(b)\rangle$ by Definition 7.2.1 $\iff$ ℓ is a limit tangent to G_f at $\langle b, f(b)\rangle$ by Definition 7.2.10.*

Proof. We prove the implication $\Rightarrow$. Let $f'(b) \in \mathbb{R}$,

$$\ell(x) = f'(b)x + f(b) - f'(b)f(b)$$

and $(a_n) \subset M \setminus \{b\}$ be a sequence with $\lim a_n = b$. We denote

$$c_n = \tfrac{f(a_n)-f(b)}{a_n-b}\,.$$

The secant $\kappa_n = \kappa_n(x)$ through the points $\langle b, f(b)\rangle$ and $\langle a_n, f(a_n)\rangle$ is

$$\kappa_n(x) = \kappa(b,\, f(b),\, a_n,\, f(a_n))(x) = c_n(x-b) + f(b) = c_n x + f(b) - c_n b\,.$$

By HDD we have $\lim c_n = f'(b)$. Thus also $\lim(f(b) - c_n b) = f(b) - f'(b)b$. By Definition 7.2.8, $\mathrm{Lim}\, \kappa_n = \ell$.

We prove the implication $\Leftarrow$. Let $\ell(x) = sx + t$ and let (a_n), c_n and κ_n be as above. We assume that for every such sequence (a_n) we have $\mathrm{Lim}\, \kappa_n = \ell$ by Definition 7.2.8. So always $\lim c_n = s$, $\langle b, f(b)\rangle \in \ell$ by Exercise 7.2.11, and

$$\lim(f(b) - c_n b) = f(b) - sb = t\,.$$

By HDD we have $s = f'(b)$. Hence $t = f(b) - f'(b)b$. So ℓ is a tangent to G_f at $\langle b, f(b)\rangle$ by Definition 7.2.1. $\square$

• *Tangents without points of contact.* We show that the tangent at the point $B \in G_f$ is the limit of secants going through pairs of points in G_f that are separated by B and converge to B.

Exercise 7.2.13 *Prove the next lemma.*

Lemma 7.2.14 (convex combination) *Let $s, v \in \mathbb{R}$ be positive, $\alpha = \frac{s}{s+v}$ and $\beta = \frac{v}{s+v}$, so that $\alpha, \beta > 0$ and $\alpha + \beta = 1$.* <u>*Then*</u> *for every $r, t \in \mathbb{R}$ we have*

$$\frac{r+t}{s+v} = \alpha \cdot \frac{r}{s} + \beta \cdot \frac{t}{v}\ .$$

Theorem 7.2.15 (tangents and secants) *Let $f \in \mathcal{F}(M)$, $b \in L^{\mathrm{TS}}(M) \setminus M$, and $\ell \in \mathcal{N}$. Then two claims are equivalent.*

1. *One can extend f to $M \cup \{b\}$ so that ℓ is a tangent to G_f at $\langle b, f(b)\rangle$.*
2. *For every two sequences $(x_n), (y_n) \subset M$ such that $\lim x_n = \lim y_n = b$ and $x_n < b < y_n$ for every n we have the limit*

$$\mathrm{Lim}\,\kappa(x_n,\, f(x_n),\, y_n,\, f(y_n)) = \ell\,.$$

Proof. We prove the implication $1 \Rightarrow 2$. We assume that f has been extended to b by a value $f(b)$, that $f'(b) \in \mathbb{R}$, and that

$$\ell = \ell(x) = f'(b)x + f(b) - f'(b)b\,.$$

Let $(x_n) \subset M$ with $x_n < b$ and $(y_n) \subset M$ with $b < y_n$ be sequences converging to b. We denote

$$r_n = f(b) - f(x_n),\ s_n = b - x_n,\ t_n = f(y_n) - f(b) \text{ and } v_n = y_n - b\,.$$

We use Lemma 7.2.14 and write the slope u_n of the secant

$$\ell_n = \kappa(x_n,\, f(x_n),\, y_n,\, f(y_n))$$

of G_f as the convex combination

$$u_n = \tfrac{f(y_n)-f(x_n)}{y_n-x_n} = \tfrac{r_n+t_n}{s_n+v_n} = \alpha_n \cdot \tfrac{r_n}{s_n} + \beta_n \cdot \tfrac{t_n}{v_n}$$

of the slopes $\frac{r_n}{s_n}$ and $\frac{t_n}{v_n}$ of the respective secants

$$\kappa(x_n,\, f(x_n),\, b,\, f(b)) \text{ and } \kappa(b,\, f(b),\, y_n,\, f(y_n))\,.$$

Since $\lim \frac{r_n}{s_n} = \lim \frac{t_n}{v_n} = f'(b)$, by Corollary 2.8.10 also $\lim u_n = f'(b)$. Since

$$\ell_n(x) = u_n x + f(x_n) - u_n x_n\,,$$

$\lim u_n = f'(b)$, $\lim x_n = b$ and $\lim f(x_n) = f(b)$ (f is continuous at b due to $f'(b) \in \mathbb{R}$), we have by Definition 7.2.8 that $\mathrm{Lim}\,\ell_n = \ell$.

We prove the implication $\neg 1 \Rightarrow \neg 2$. We assume that f cannot be extended to b by any value $f(b)$ so that ℓ is tangent to G_f at $\langle b, f(b)\rangle$. This means that if we take the unique number $f(b) \in \mathbb{R}$ such that $\langle b,\, f(b)\rangle \in \ell$ and if s is the slope of ℓ, then it is not true that

$$\lim_{x \to b} \tfrac{f(x)-f(b)}{x-b} = s\,.$$

We then obtain pairs of points in M such that their components converge to b from opposite sides, and it is does not hold that the limit of the corresponding secants is ℓ.

The first case is that the extended function f is not continuous at b. By Exercise 7.2.16, there exist sequences $(x_n) \subset M$ with $x_n < b$ and $(y_n) \subset M$ with $b < y_n$ such that $\lim x_n = \lim y_n = b$, $\lim f(x_n) = K$, $\lim f(y_n) = L$, and it is not true that

$$K = L = f(b)\,.$$

If $K \neq L$ then the slopes of the secants

$$\ell_n = \kappa(x_n,\, f(x_n),\, y_n,\, f(y_n))$$

go to $\pm\infty$ and it is not true that $\mathrm{Lim}\,\ell_n = l$. If $K = L \neq f(b)$ then the intersections of the secants ℓ_n with the vertical line $x = b$ converge to a (possibly infinite) point different from $\langle b, f(b)\rangle$. By Exercise 7.2.17, the limit $\mathrm{Lim}\,\ell_n$, if it exists, cannot be a line going through $\langle b, f(b)\rangle$. Again, it is not true that $\mathrm{Lim}\,\ell_n = l$.

The second case is that the extended function f is continuous at b, but it does not hold that

$$\lim_{x\to b} \tfrac{f(x)-f(b)}{x-b} = s\,.$$

Then there is an $A \in \mathbb{R}^* \setminus \{s\}$ and a sequence $(x_n) \subset M \setminus \{b\}$ lying on one side of b such that

$$\lim x_n = b \text{ and } \lim \tfrac{f(x_n)-f(b)}{x_n-b} = A\,.$$

We may assume that $x_n < b$ for every n, the case when always $x_n > b$ is similar. We take any sequence $(y_n) \subset M$ with $b < y_n$ and $\lim y_n = b$. Then $\lim f(y_n) = f(b)$ and by Exercise 7.2.18 we can choose from (y_n) a subsequence (y_{m_n}) such that

$$\lim_{n\to\infty} \tfrac{f(x_n)-f(y_{m_n})}{x_n-y_{m_n}} = A\,.$$

Since $A \neq s$, it is not true that $\mathrm{Lim}\,\ell_n = l$. □

Exercises 7.2.16–7.2.18 are lemmas for the proof. Exercise 7.2.19 shows that secants through pairs of points that converge to $B \in G_f$ from the same side need not converge to the tangent at B.

Exercise 7.2.16 *Let $f \in \mathcal{F}(M)$ and $b \in M \cap L^{\mathrm{TS}}(M)$. If f is not continuous at b, then for some sequences $(x_n), (y_n) \subset M$ with $x_n < b < y_n$ and $\lim x_n = \lim y_n = b$ we have $\lim f(x_n) = K$, $\lim f(y_n) = L$, but $K \neq f(b)$ or $L \neq f(b)$.*

Exercise 7.2.17 *Let $(\ell_n) \subset \mathcal{N}$, $\ell \in \mathcal{N}$, $\langle b, c\rangle \in \ell$, $\mathrm{Lim}\,\ell_n = \ell$ and*

$$(x = b) \cap \ell_n = \{\langle b,\, c_n\rangle\}\,.$$

Then $\lim c_n = c$.

Exercise 7.2.18 *Let sequences (x_n), (y_n), (z_n) and (u_n) be such that $\lim x_n = \lim z_n = b$, $x_n \neq b$, $\lim y_n = \lim u_n = c$ and $\lim \frac{y_n-c}{x_n-b} = A$. Then for some sequence $(m_n) \subset \mathbb{N}$ we have $\lim \frac{y_n-u_{m_n}}{x_n-z_{m_n}} = A$.*

Exercise 7.2.19 *Find a function $f \in \mathcal{F}(M)$ with tangent ℓ at $\langle b, f(b)\rangle$ and two sequences $(x_n), (y_n) \subset M$ such that $b < x_n < y_n$, $\lim x_n = \lim y_n = b$, but it does not hold that*

$$\operatorname{Lim} \kappa(x_n,\, f(x_n),\, y_n,\, f(y_n)) = \ell\,.$$

7.3 Arithmetic of derivatives

Global derivative is a unary operation on $\mathcal{R}$. We describe its interactions with the operations of addition, multiplication, and division. In the next section, we treat composition and inverse. We consider both point-wise and global formulas. Point-wise formulas involve finite and infinite values of derivatives at points. Global formulas involve only finite values of global derivatives.

- *Sums.* We differentiate sums of functions.

Theorem 7.3.1 **($(f+g)'$)** *Let $f, g \in \mathcal{R}$ and $M = M(f) \cap M(g)$.*

1. *If $b \in M \cap L(M)$, $f'(b) = K$, $g'(b) = L$, and $K + L$ is not an indefinite expression, <u>then</u>*
$$(f+g)'(b) = K + L\,.$$
2. *The function $(f' + g') \,|\, L(M)$ is a subfunction of the function $(f+g)'$.*

Proof. 1. Let $h = f + g$, then $b \in M(h) \cap L(M(h))$. By Theorem 4.4.5,

$$\begin{aligned} h'(b) &= \lim_{x\to b} \tfrac{h(x)-h(b)}{x-b} = \lim_{x\to b}\left(\tfrac{f(x)-f(b)}{x-b} + \tfrac{g(x)-g(b)}{x-b}\right) \\ &= \lim_{x\to b} \tfrac{f(x)-f(b)}{x-b} + \lim_{x\to b} \tfrac{g(x)-g(b)}{x-b} = f'(b) + g'(b) = K + L\,. \end{aligned}$$

2. Let $h = (f' + g') \,|\, L(M)$ and

$$c \in M(h) = D(f) \cap D(g) \cap L(M)\,.$$

By part 1, $(f+g)'(c) = f'(c) + g'(c) = h(c)$. It follows that h is a subfunction of $(f+g)'$. □

Exercise 7.3.2 *Adapt the theorem for the difference $f - g$.*

Item 1 of the theorem is standard. Item 2 is new. In our approach to derivatives, the formula

$$(f+g)' = f' + g'$$

in general does not hold. Let $f = k_0 \,|\, (-\infty, 0]$ and $g = k_0 \,|\, [0, +\infty)$. Then $M = M(f) \cap M(g) = \{0\}$ and

$$f' + g' = k_0 \,|\, (-\infty, 0] + k_0 \,|\, [0, +\infty) = k_0 \,|\, \{0\} \neq \emptyset_f = (k_0 \,|\, \{0\})' = (f+g)'\,.$$

The restriction to $L(M)$ therefore cannot be omitted.

Let $f(x) = |x|$ and $g(x) = -|x|$ ($\in \mathcal{F}(\mathbb{R})$). Then $M = M(f) \cap M(g) = \mathbb{R}$, $L(M) = \mathbb{R}^*$,

$$(f' + g') \,|\, L(M) = k_0 \,|\, (\mathbb{R} \setminus \{0\}) \text{ and } (f+g)' = (k_0)' = k_0 \,.$$

Thus $(f' + g') \,|\, L(M)$ may be a proper subfunction of $(f+g)'$.

In an application of item 1 of Theorem 7.3.1 we compute a derivative. Let $f(x) = \mathrm{sgn}(x)$, $g(x) = \sqrt{x}$ and $b = 0$. Then $M = M(f) \cap M(g) = [0, +\infty)$ and

$$\big(\mathrm{sgn}(x) + \sqrt{x}\big)'(0) = \mathrm{sgn}'(0) + (\sqrt{x})'(0) = +\infty + (+\infty) = +\infty \,.$$

Exercise 7.3.3 *What is* $\big(\mathrm{sgn}(x) - \sqrt{x}\big)'(0)$*?*

We find assumptions for the validity of $(f+g)' = f' + g'$. We need it for the proof of Theorem 7.6.5.

Corollary 7.3.4 ($(f+g)' = f' + g'$) *Let $f, g \in \mathcal{R}$ and $M = M(f) \cap M(g)$. If $D(f) = M(f)$, $D(g) = M(g)$ and $M \subset L(M)$, then*

$$(f+g)' = f' + g' \,.$$

Proof. Let $h = (f' + g') \,|\, L(M)$. From the assumptions it follows that $M(h) = M$ and that $h = f' + g'$. Since $M((f+g)') \subset M$, item 2 of Theorem 7.3.1 gives that $f' + g' = h = (f+g)'$. □

• *Products.* We differentiate products of functions.

Theorem 7.3.5 (Two Leibniz formulas) *Let f and g be in $\mathcal{R}$ and $M = M(f) \cap M(g)$.*

1. *Local Leibniz formula. If $b \in M \cap L(M)$, $f'(b) = K$, $g'(b) = L$, one of f and g is continuous at b and if $K \cdot g(b) + f(b) \cdot L$ is defined, then*

$$(fg)'(b) = Kg(b) + f(b)L \,.$$

2. *Global Leibniz formula. The function $(f'g + fg') \,|\, L(M)$ is a subfunction of the function $(fg)'$.*

Proof. 1. Let $h = fg$ and g be continuous at b. The case when f is continuous at b is resolved in Exercise 7.3.6. Then $b \in M(h) \cap L(M(h))$ and, by the assumptions, by Theorem 4.4.5 and by Proposition 4.3.5, we have

$$\begin{aligned} h'(b) &= \lim_{x \to b} \tfrac{f(x)g(x) - f(b)g(b)}{x - b} = \lim_{x \to b} \tfrac{(f(x) - f(b))g(x) + f(b)(g(x) - g(b))}{x - b} \\ &= \lim_{x \to b} \tfrac{f(x) - f(b)}{x - b} \cdot \lim_{x \to b} g(x) + f(b) \lim_{x \to b} \tfrac{g(x) - g(b)}{x - b} = Kg(b) + f(b)L \,. \end{aligned}$$

2. Let $h = (f'g + fg') \,|\, L(M)$ and $c \in M(h)$. Then

$$c \in D(f) \cap D(g) \cap L(M) \,,$$

g is continuous at c because $g'(c) \in \mathbb{R}$, and by the first part we have that $(fg)'(c) = f'(c)g(c) + f(c)g'(c) = h(c)$. Hence h is a subfunction of $(fg)'$. $\Box$

Like in the case of sum, it is not hard to produce examples showing that the restriction to $L(M)$ cannot be omitted and that $(f'g + fg') \,|\, L(M)$ may be a proper subfunction of $(fg)'$.

Exercise 7.3.6 *Solve quickly the case when f is continuous at b.*

We show that the assumption of continuity of f or g at b is substantial.

Exercise 7.3.7 *We define functions $f, g \in \mathcal{F}(\mathbb{R})$ by $f(0) = -\frac{1}{2}$, $g(0) = \frac{1}{2}$ and for $a \neq 0$ by*

$$f(a) = \operatorname{sgn} a \ \textit{ and } \ g(a) = -\operatorname{sgn} a\,.$$

Show that $(fg)'(0)$ does not exist but that at 0 the right-hand side of the local Leibniz formula is $+\infty$.

For future use we obtain a simple form of the global Leibniz formula.

Corollary 7.3.8 ($(fg)' = f'g + fg'$) *Let $f, g \in \mathcal{R}$ and $M = M(f) \cap M(g)$. If $D(f) = M(f)$, $D(g) = M(g)$ and $M \subset L(M)$, then*

$$(fg)' = f'g + fg'\,.$$

Proof. Let $h = (f'g + fg') \,|\, L(M)$. From the assumptions, it follows that $M(h) = M$ and $h = f'g + fg'$. Since $M((fg)') \subset M$, part 2 of Theorem 7.3.5 gives that $f'g + fg' = h = (fg)'$. $\Box$

- *Division.* We differentiate ratios of functions.

Theorem 7.3.9 ($(\frac{f}{g})'$) *Suppose that $f, g \in \mathcal{R}$ and $M = M(f) \cap M(g) \setminus Z(g)$.*

1. *If $b \in M \cap L(M)$, $f'(b) = K$, $g'(b) = L$, g is continuous at b and if*

$$\frac{K \cdot g(b) - f(b) \cdot L}{g(b)^2}$$

is defined, then

$$\left(\frac{f}{g}\right)'(b) = \frac{Kg(b) - f(b)L}{g(b)^2}\,.$$

2. *The function $\frac{f'g - fg'}{g^2} \,|\, L(M)$ is a subfunction of the function $(\frac{f}{g})'$.*

Proof. 1. Let $h = \frac{f}{g}$ and g be continuous at b. Then $b \in M(h) \cap L(M(h))$ and

$$h'(b) = \lim_{x \to b} \frac{f(x)/g(x) - f(b)/g(b)}{x - b} = \lim_{x \to b} \frac{f(x)g(b) - f(b)g(b) + f(b)g(b) - f(b)g(x)}{g(x)g(b)(x - b)}\,.$$

Due to the assumptions, Proposition 4.3.5 and Theorem 4.4.5, this equals

$$\lim_{x\to b}\tfrac{f(x)-f(b)}{x-b}\lim_{x\to b}\tfrac{g(b)}{g(x)g(b)}-\lim_{x\to b}\tfrac{f(b)}{g(x)g(b)}\lim_{x\to b}\tfrac{g(x)-g(b)}{x-b}$$
$$=\tfrac{f'(b)g(b)-f(b)g'(b)}{g(b)^2}\,.$$

2. Let $h=\frac{f'g-fg'}{g^2}\mid L(M)$ and $c\in M(h)$. Then

$$c\in D(f)\cap D(g)\cap L(M)\setminus Z(g)\,,$$

g is continuous at c, because $g'(c)\in\mathbb{R}$, and by the first part we have that $(\frac{f}{g})'(c)=\frac{f'(c)g(c)-f(c)g'(c)}{g(c)^2}=h(c)$. Hence h is a subfunction of $(\frac{f}{g})'$. □

Again, the restriction to $L(M)$ cannot in general be omitted and $\frac{f'g-fg'}{g^2}\mid L(M)$ can be a proper restriction of $(\frac{f}{g})'$.

Exercise 7.3.10 *Show, as in Exercise 7.3.7, that the assumption of continuity of g at b cannot be omitted.*

For later use we again obtain a corollary with a simple form of the formula for derivatives of ratios.

Corollary 7.3.11 ($(\frac{f}{g})'=\frac{f'g-f'g}{g^2}$) *Let $f,g\in\mathcal{R}$ and $M=M(f)\cap M(g)$. If $D(f)=M(f)$, $D(g)=M(g)$ and $M\subset L(M)$ then*

$$\left(\frac{f}{g}\right)'=\frac{f'g-f'g}{g^2}\,.$$

Proof. Let $h=\frac{f'g-fg'}{g^2}\mid L(M)$. From the assumptions it follows that $M(h)=M\setminus Z(g)$ and $h=\frac{f'g-fg'}{g^2}$. Since $M((\frac{f}{g})')\subset M\setminus Z(g)$, part 2 of Theorem 7.3.9 gives that $\frac{f'g-fg'}{g^2}=h=(\frac{f}{g})'$. □

7.4 Composite and inverse functions

Let $f,g\in\mathcal{R}$. Recall that the composite function

$$f(g)\colon M(f(g))\to\mathbb{R},\ \text{ where }\ f(g)(x)=f(g(x))\,,$$

has the domain

$$M(f(g))=\{x\in M(g)\colon\ g(x)\in M(f)\}=g^{-1}[M(f)]\,.$$

Thus $M(f(g))\subset M(g)$ and this inclusion may be proper. Any injection $f\in\mathcal{R}$ has the inverse

$$f^{-1}\colon f[M(f)]\to\mathbb{R}\ \text{ where }\ f^{-1}(y)=x\iff f(x)=y\,.$$

For non-injective f the inverse is not defined.

- *Composites.* We differentiate composite functions.

Theorem 7.4.1 **($(f(g))'$)** *Let $f, g \in \mathcal{R}$ and $M = M(f(g))$.*

1. *If $b \in M \cap L(M)$, $f'(g(b)) = K$, $g'(b) = L$, g is continuous at b and if $K \cdot L \neq 0 \cdot (+\infty), (+\infty) \cdot 0$, then*

$$f(g)'(b) = KL\,.$$

2. *The function $(f'(g) \cdot g') \,|\, L(M)$ is a subfunction of the function $(f(g))'$.*

Proof. 1. Let $(a_n) \subset M \setminus \{b\}$ be any sequence with $\lim a_n = b$. By the assumption, $\lim g(a_n) = g(b)$. We partition (a_n) in two (possibly finite or empty) subsequences (b_n) and (c_n): for every n we have $g(b_n) = g(b)$ and $g(c_n) \neq g(b)$. We show that if $(x_n) = (b_n)$ or $(x_n) = (c_n)$ is an infinite sequence, then

$$\lim_{n\to\infty} \tfrac{f(g)(x_n)-f(g)(b)}{x_n-b} = f'(g(b)) \cdot g'(b)\,.$$

Then by Theorem 2.2.14 it follows that $\lim \frac{f(g)(a_n)-f(g)(b)}{a_n-b} = f'(g(b)) \cdot g'(b)$. By HDD,

$$f(g)'(b) = f'(g(b)) \cdot g'(b)\,.$$

Let the sequence (b_n) be infinite. Then $\lim b_n = b$. By HDD we have

$$g'(b) = \lim \tfrac{g(b_n)-g(b)}{b_n-b} = \lim \tfrac{g(b)-g(b)}{b_n-b} = \lim 0 = 0\,.$$

Thus

$$\lim \tfrac{f(g)(b_n)-f(g)(b)}{b_n-b} = \lim \tfrac{f(g(b))-f(g(b))}{b_n-b} = 0 = f'(g(b)) \cdot 0 = f'(g(b)) \cdot g'(b)\,.$$

Let the sequence (c_n) be infinite. Then $\lim c_n = b$, $\lim g(c_n) = g(b)$ and by Theorem 4.4.5 and HDD we have

$$\begin{aligned}&\lim \tfrac{f(g)(c_n)-f(g)(b)}{c_n-b} = \lim \tfrac{f(g(c_n))-f(g(b))}{c_n-b} = \\ &= \lim \big(\tfrac{f(g(c_n))-f(g(b))}{g(c_n)-g(b)} \cdot \tfrac{g(c_n)-g(b)}{c_n-b}\big) = \lim \tfrac{f(g(c_n))-f(g(b))}{g(c_n)-g(b)} \cdot \lim \tfrac{g(c_n)-g(b)}{c_n-b} = \\ &= f'(g(b)) \cdot g'(b)\,.\end{aligned}$$

2. Let $h = (f'(g) \cdot g') \,|\, L(M)$ and $c \in M(h)$. Then

$$c \in M(f'(g)) \cap D(g) \cap L(M)\,.$$

The functuon g is continuous at c, because $g'(c) \in \mathbb{R}$, and by the first part $(f(g))'(c) = f'(g(c)) \cdot g'(c) = h(c)$. Hence h is a subfunction of $(f(g))'$. $\Box$

Again, the restriction to $L(M)$ cannot be omitted and $(f'(g) \cdot g') \,|\, L(M)$ can be a proper restriction of $f(g)'$.

Exercise 7.4.2 *Part 1 does not hold without the continuity of g at b.*

In a corollary, we simplify the formula for the derivative of composites.

Corollary 7.4.3 ($(f(g))' = f'(g) \cdot g'$) *Let $f, g \in \mathcal{R}$ and $M = M(f(g))$. If $D(f) = M(f)$, $D(g) = M(g)$ and $M \subset L(M)$ then*

$$(f(g))' = f'(g) \cdot g' .$$

Proof. Let $h = (f'(g) \cdot g') \,|\, L(M)$. From the assumptions on f and g it follows that $M(h) = M$ and $h = f'(g) \cdot g'$. Since $M((f(g))') \subset M$, part 2 of Theorem 7.3.5 gives that

$$f'(g) \cdot g' = h = (f(g))' .$$

$\square$

• *Inverses.* We take derivatives of inverse functions. A function $f \in \mathcal{F}(M)$ increases, respectively decreases, at a point $b \in M$ if there exists a δ such that for every x and x' with $b - \delta < x < b < x' < b + \delta$ we have

$$f(x) < f(b) < f(x'), \text{ respectively } f(x) > f(b) > f(x') .$$

Theorem 7.4.4 ($(f^{-1})'$) *Let $f \in \mathcal{F}(M)$ be injective, b be in $M \cap L(M)$, $f'(b)$ in $\mathbb{R}^*$, and f^{-1} be continuous at $c = f(b)$.*

1. *If $f'(b) \in \mathbb{R} \setminus \{0\}$, then*

$$\left(f^{-1}\right)'(c) = \frac{1}{f'(b)} = \frac{1}{f'(f^{-1}(c))} .$$

2. *If $f'(b) = 0$ and f increases, respectively decreases, at b, then*

$$(f^{-1})'(c) = +\infty, \textit{ respectively } -\infty .$$

3. *If $f'(b) = \pm\infty$ and $c \in L(f[M])$, then*

$$(f^{-1})'(c) = 0 .$$

Proof. We take a sequence $(b_n) \subset f[M] \setminus \{c\}$ with $\lim b_n = c$ and set $a_n = f^{-1}(b_n)$. Then $(a_n) \subset M \setminus \{b\}$ and by the continuity of f^{-1} at c we have $\lim a_n = b$.

1. Let $f'(b)$ be finite and nonzero. Then $c \in L(f[M])$ by Proposition 7.1.18. By Theorem 4.4.5 and HDD we have

$$\lim \tfrac{f^{-1}(b_n) - f^{-1}(c)}{b_n - c} = \lim \tfrac{1}{\frac{f(a_n) - f(b)}{a_n - b}} = \tfrac{1}{\lim \frac{f(a_n) - f(b)}{a_n - b}} = \tfrac{1}{f'(b)} .$$

By HDD we have $(f^{-1})'(c) = \frac{1}{f'(b)}$.

2. Let $f'(b) = 0$. Then $c \in L(f[M])$ by Exercise 7.1.20. Suppose that f decreases (respectively increases) at b. Then there is an n_0 such that if $n \geq n_0$ then

$$\tfrac{f(a_n) - f(b)}{a_n - b} < 0 \text{ (respectively } \cdots > 0) .$$

The previous computation and part 5 of Proposition 2.6.4 show that

$$(f^{-1})'(b) = \tfrac{1}{0^-} = -\infty \text{ (respectively } \cdots = +\infty) .$$

3. Let $f'(b) = \pm\infty$ and $c \in L(f[M])$. Then

$$(f^{-1})'(c) = \tfrac{1}{\pm\infty} = 0$$

by part 1. □

Here is a counterexample to part 1 when the continuity of f^{-1} at c is dropped.

Exercise 7.4.5 *Let*

$$M = \big([-1, 1] \setminus \{\tfrac{1}{n}\colon\ n \in \mathbb{N}\}\big) \cup \{1 + \tfrac{1}{n}\colon\ n \in \mathbb{N}\} .$$

We define $f \in \mathcal{F}(M)$ by $f(x) = x$ if $x \neq 1 + \frac{1}{n}$, and by $f(1+\frac{1}{n}) = \frac{1}{n}$. Then f is injective and $f'(0) = 1$, but for $c = f(0) = 0$ the derivative $(f^{-1})'(c) = (f^{-1})'(0)$ does not exist.

We state the formula for the global derivative of inverses separately as a corollary.

Corollary 7.4.6 $\mathbf{((f^{-1})')}$ *Let $f \in \mathcal{R}$ be injective and*

$$M = \{x \in f[M(f)]\colon\ f^{-1} \textit{ is continuous at } x\} .$$

Then the function $\frac{1}{f'(f^{-1})} \,|\, M$ is a subfunction of the function $(f^{-1})'$.

Proof. Let $h = \frac{1}{f'(f^{-1})} \,|\, M$ and $c \in M(h)$. Then

$$c \in M(f'(f^{-1})) \cap M \setminus Z(f'(f^{-1})) ,$$

f^{-1} is continuous at c because $c \in M$, and by part 1 of Theorem 7.4.4 it holds that $(f^{-1})'(c) = \frac{1}{f'(f^{-1}(c))} = h(c)$. Hence h is a subfunction of $(f^{-1})'$. □

We do not need derivatives of inverses for the proof of Theorem 7.6.5, but for completeness, we still provide the corollary with the simple formula for $(f^{-1})'$.

Corollary 7.4.7 $\mathbf{((f^{-1})' = \frac{1}{f'(f^{-1})})}$ *Let $f \in \mathcal{R}$ be injective and $M = f[M(f)]$. If $D(f) = M(f)$, $f^{-1} \in \mathcal{C}$ and $M \subset L(M)$, then*

$$(f^{-1})' = \frac{1}{f'(f^{-1})} .$$

Proof. Let $h = \frac{1}{f'(f^{-1})}$. It follows from the assumptions and from Corollary 7.4.6 that $M(h) = M \setminus Z(f'(f^{-1}))$ and that h is a subfunction of $(f^{-1})'$.

Suppose that $c \in M$ is such that $f'(f^{-1}(c)) = 0$. If $c \in M((f^{-1})')$ then part 1 of Theorem 7.4.1 gives that

$$1 = (\mathrm{id} \,|\, M)'(c) = (f(f^{-1}))'(c) = f'(f^{-1}(c)) \cdot (f^{-1})'(c) = 0 \cdot (f^{-1})'(c)\,,$$

which is impossible. Thus $c \notin M((f^{-1})')$ and we deduce that $M((f^{-1})') \subset M \setminus Z(f'(f^{-1}))$. Hence

$$\tfrac{1}{f'(f^{-1})} = h = (f^{-1})\,.$$

□

7.5 Basic elementary functions

We determine derivatives of functions in BEF (Definition 5.1.1). Constants are easy, by Exercise 7.1.25 we have for every $c \in \mathbb{R}$ the derivative

$$k_c'(x) = k_0(x)\,.$$

• *The exponential function, sine and cosine.* We strengthen Theorem 6.7.3 and differentiate power series. By Theorem 6.7.3, any real numbers a_0, a_1, ... satisfying

$$\lim |a_n|^{1/n} = 0$$

determine for $x \in \mathbb{R}$ an abscon series $\sum_{n\ge 0} a_n x^n$ with sum

$$S(x) = \textstyle\sum_{n\ge 0} a_n x^n \in \mathcal{C}(\mathbb{R})\,.$$

Theorem 7.5.1 (derivatives of power series) *We have*

$$D(S(x)) = M(S(x)) = \mathbb{R} \;\text{ and }\; S'(x) = \textstyle\sum_{n\ge 0}(n+1)a_{n+1}x^n\,.$$

Proof. Let $a_n \in \mathbb{R}$ for $n \in \mathbb{N}_0$ be such that $\lim |a_n|^{1/n} = 0$. We define the function

$$T(x) = \textstyle\sum_{n\ge 0}(n+1)a_{n+1}x^n\,.$$

By Proposition 2.1.29 and the assumption on a_n we have

$$\lim_{n\to\infty} \big((n+1)\cdot|a_{n+1}|\big)^{1/n} = 0\,.$$

Thus $M(T(x)) = \mathbb{R}$. Let $x, c \in \mathbb{R}$ with $c \ne 0$ and $y = \max(\{1, |c|, |x|\})$. Using the identity

$$\tfrac{a^{n+1}-b^{n+1}}{a-b} = \textstyle\sum_{j=0}^n a^j b^{n-j}\,,$$

we get the bound

$$\begin{aligned}
&\big|\tfrac{1}{c}(S(x+c) - S(x)) - T(x)\big| \\
&\le \textstyle\sum_{n\ge 1} |a_{n+1}| \cdot |\sum_{j=0}^n (x+c)^j x^{n-j} - (n+1)x^n| \\
&\le |c| \textstyle\sum_{n\ge 1} |a_{n+1}| \cdot \sum_{j=1}^n \sum_{i=1}^j \binom{j}{i} y^{i-1} y^{n-i} \\
&\le |c| \textstyle\sum_{n\ge 1} |a_{n+1}| \cdot y^{n-1} \cdot \sum_{j=1}^n 2^j \le |c| \cdot \sum_{n\ge 1} |a_{n+1}| \cdot (2y)^{n+1}\,.
\end{aligned}$$

For $c \to 0$ this goes in limit to 0. Hence

$$S'(x) = \lim_{c\to 0} \tfrac{S(x+c)-S(x)}{c} = T(x) = \textstyle\sum_{n\geq 0}(n+1)a_{n+1}x^n\,.$$

$\square$

Exercise 7.5.2 *Explain estimates in the previous proof.*

Corollary 7.5.3 ($\exp x$, $\cos x$ and $\sin x$) *We have the global formulas*

$$(\exp x)' = \exp x,\ (\cos x)' = -\sin x \text{ and } (\sin x)' = \cos x \quad (\in \mathcal{F}(\mathbb{R}))\,.$$

Proof. By the previous theorem, we have ($x \in \mathbb{R}$)

$$\begin{aligned}
(\exp x)' &= \big(\textstyle\sum_{n\geq 0} \frac{1}{n!}x^n\big)' = \sum_{n\geq 0} \frac{(n+1)}{(n+1)!}x^n = \exp x\,,\\
(\cos x)' &= \big(\textstyle\sum_{n\geq 0}(-1)^n \frac{1}{(2n)!}x^{2n}\big)' = \sum_{n\geq 0}(-1)^{n+1} \frac{(2n+2)}{(2n+2)!}x^{2n+1}\\
&= -\sin x \ \text{ and}\\
(\sin x)' &= \big(\textstyle\sum_{n\geq 0}(-1)^n \frac{1}{(2n+1)!}x^{2n+1}\big)' = \sum_{n\geq 0}(-1)^n \frac{(2n+1)}{(2n+1)!}x^{2n} = \cos x\,.
\end{aligned}$$

$\square$

• *Logarithm.* Since $\log x$ is the inverse of $\exp x$ and $(\exp x)' = \exp x$, Corollary 7.4.7 yields the global formula

$$(\log x)' = \frac{1}{(\exp x)'(\log x)} = \frac{1}{\exp(\log x)} = \frac{k_1(x)}{x \,|\, (0,\,+\infty)} = \tfrac{1}{x} \,|\, (0,\,+\infty)\,.$$

We stress that $(\log x)'$ is not the function $\frac{1}{x}$ ($\in \mathcal{F}(\mathbb{R}\setminus\{0\})$).

Exercise 7.5.4 *What is* $(\log|x|)'$*?*

• *Real exponentiation.* In the next two exercises we differentiate specializations of the function a^b.

Exercise 7.5.5 *Let $a, b \in \mathbb{R}$ and $m \in \mathbb{Z}$. Prove the global formulas below.*

1. *For $a > 0$ the derivative $(a^x)' = a^x \cdot \log a$ is in $\mathcal{F}(\mathbb{R})$.*
2. *For $b > 1$ the derivative $(x^b)' = bx^{b-1}$ is in $\mathcal{F}([0,+\infty))$.*
3. *For $b = 1$ the derivative $(x^b)' = k_1(x) \,|\, [0,+\infty)$.*
4. *For $b < 1$ the derivative $(x^b)' = bx^{b-1}$ is in $\mathcal{F}((0,+\infty))$.*
5. *We have $(0^x)' = k_0(x) \,|\, (0,+\infty)$.*
6. *For $m > 0$ the derivative $(x^m)' = mx^{m-1}$ is in $\mathcal{F}(\mathbb{R})$.*
7. *For $m = 0$ the derivative $(x^m)' = k_0(x)$ is in $\mathcal{F}(\mathbb{R})$.*

8. For $m < 0$ the derivative $(x^m)' = mx^{m-1}$ is in $\mathcal{F}(\mathbb{R} \setminus \{0\})$.

Exercise 7.5.6 *Let $b \in (0, 1)$. Prove that $(x^b)'(0) = +\infty$.*

• *Tangent and cotangent.*

Exercise 7.5.7 *Prove the global formulas*

$$(\tan x)' = \frac{1}{\cos^2 x} \ \text{ and } \ (\cot x)' = -\frac{1}{\sin^2 x}\,.$$

• *Inverse trigonometric functions.*

Exercise 7.5.8 *Prove the global formulas below.*

1. *The derivative $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$ is in $\mathcal{F}((-1, 1))$.*
2. *The derivative $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$ is in $\mathcal{F}((-1, 1))$.*
3. *The derivative $(\arctan x)' = \frac{1}{1+x^2}$ is in $\mathcal{F}(\mathbb{R})$.*
4. *The derivative $(\mathrm{arccot}\, x)' = -\frac{1}{1+x^2}$ is in $\mathcal{F}(\mathbb{R})$.*

Exercise 7.5.9 *Show that*

$$(\arcsin x)'(-1) = (\arcsin x)'(1) = +\infty$$

and that

$$(\arccos x)'(-1) = (\arccos x)'(1) = -\infty\,.$$

• *Overview of derivatives of $f \in \mathrm{BEF}$.* We first list f with $D(f) = M(f)$, and then give the three exceptions for which $D(f) \neq M(f)$.

f	$M(f)$	f'	$M(f')$
$\exp x$	$\mathbb{R}$	$\exp x$	$= M(f)$
$\sin x$	$\mathbb{R}$	$\cos x$	$= M(f)$
$\cos x$	$\mathbb{R}$	$-\sin x$	$= M(f)$
$\arctan x$	$\mathbb{R}$	$\frac{1}{x^2+1}$	$= M(f)$
$\operatorname{arccot} x$	$\mathbb{R}$	$-\frac{1}{x^2+1}$	$= M(f)$
a^x for $a \in (0,+\infty)$	$\mathbb{R}$	$a^x \cdot \log a$	$= M(f)$
x^m for $m \in \mathbb{N}$	$\mathbb{R}$	mx^{m-1}	$= M(f)$
x^0 for $0 \in \mathbb{Z}$	$\mathbb{R}$	$k_0(x)$	$= M(f)$
$k_c(x)$ for $c \in \mathbb{R}$	$\mathbb{R}$	$k_0(x)$	$= M(f)$
x^m for negative $m \in \mathbb{Z}$	$\mathbb{R} \setminus \{0\}$	mx^{m-1}	$= M(f)$
$\log(\lvert x\rvert)$, not in BEF	$\mathbb{R} \setminus \{0\}$	$1/x$	$= M(f)$
x^b for $b \in (1,+\infty)$	$[0,+\infty)$	bx^{b-1}	$= M(f)$
x^1 for $1 \in \mathbb{R}$	$[0,+\infty)$	$k_1(x) \mid [0,+\infty)$	$= M(f)$
0^x	$(0,+\infty)$	$k_0(x) \mid (0,+\infty)$	$= M(f)$
x^b for $b \in (-\infty,0]$	$(0,+\infty)$	bx^{b-1}	$= M(f)$
$\log x$	$(0,+\infty)$	$\frac{1}{x} \mid (0,+\infty)$	$= M(f)$
$\tan x$	$\mathbb{R} \setminus \{k\pi + \frac{\pi}{2}\colon\ k \in \mathbb{Z}\}$	$\frac{1}{\cos^2 x}$	$= M(f)$
$\cot x$	$\mathbb{R} \setminus \{k\pi\colon\ k \in \mathbb{Z}\}$	$-\frac{1}{\sin^2 x}$	$= M(f)$
x^b for $b \in (0,1)$	$[0,+\infty)$	bx^{b-1}	$(0,+\infty)$
$\arcsin x$	$[-1,1]$	$\frac{1}{\sqrt{1-x^2}}$	$(-1,1)$
$\arccos x$	$[-1,1]$	$-\frac{1}{\sqrt{1-x^2}}$	$(-1,1)$

The infinite values of pointwise derivatives of the function in BEF are not included in the table. They are as follows.

- $(x^b)'(0) = +\infty$ for $b \in (0,1)$,
- $(\arcsin)'(-1) = (\arcsin)'(1) = +\infty$ and
- $(\arccos)'(-1) = (\arccos)'(1) = -\infty$.

Exercise 7.5.10 *Explain why we cannot merge the twelfth and thirteenth row of the table in one row with the functions x^b for $b \in [1,+\infty)$.*

Theorem 7.1.29 describes a function $f \in \mathcal{R}$ such that $M(f) = D(f) \neq \emptyset$ and $f' \notin \mathcal{C}$. Here is a different, well-known example with $M(f) = D(f) = \mathbb{R}$.

Exercise 7.5.11 *Let $f \in \mathcal{F}(\mathbb{R})$ be given by $f(0) = 0$ and by*

$$f(x) = x^2 \sin\left(\tfrac{1}{x}\right)$$

for $x \neq 0$. Show that $M(f) = D(f) = \mathbb{R}$ and $f' \notin \mathcal{C}$.

• *Derivatives of polynomials and rational functions.* The zero polynomial and polynomials with degree 0 are constant functions. As we know, $k_c'(x) = k_0(x)$. It is easy to see that for degree $d \ge 1$ we have

$$\big(\textstyle\sum_{i=0}^{d} a_i x^i\big)' = \sum_{i=0}^{d-1} (i+1) a_{i+1} x^i \,.$$

Let p and q be polynomials. By Corollary 7.3.11, the derivative of the rational function $p(x)/q(x)$ is again a rational function,

$$\Big(\frac{p(x)}{q(x)}\Big)' = \frac{p'(x)q(x) - p(x)q'(x)}{q(x)^2}\,.$$

See Proposition 8.2.6 for an important property of derivatives of rational functions.

Exercise 7.5.12 *What happens if $q(x) = k_0(x)$?*

Exercise 7.5.13 *Let $k \in \mathbb{Z}$. Find a rational function $r_k(x)$ such that*

$$r_k'(x) = x^k\,,$$

or show that no such rational function exists.

7.6 Simple elementary functions

• *A problem on elementary functions.* We introduced elementary functions in Definitions 5.2.7 and 5.2.19. In Theorem 6.7.17 we established by induction their continuity. It was easy because continuity of functions is preserved under sum, product, ratio, and composition. One might hope to prove similarly with the help of the formulas

$$(f+g)' = f' + g',\ (fg)' = f'g + fg',\ (\tfrac{f}{g})' = \tfrac{f'g - fg'}{g^2} \text{ and } (f(g))' = f'(g)\cdot g'$$

that the derivative of an elementary function is elementary. Except that we know from Sections 7.3 and 7.4 that these formulas in general do not hold, not even for elementary functions. For example, let

$$f(x) = \arcsin x \ \text{ and } \ g(x) = -\arcsin x\,.$$

Then

$$(f(x) + g(x))' = k_0(x) \,|\, [-1,1]$$

which differs from

$$f'(x) + g'(x) = \tfrac{1}{\sqrt{1-x^2}} - \tfrac{1}{\sqrt{1-x^2}} = k_0(x) \,|\, (-1,1)\,.$$

Although $f'(x), g'(x) \in \mathrm{EF}$, we cannot deduce from it that $(f(x)+g(x))' \in \mathrm{EF}$ because $(f(x)+g(x))' \ne f'(x)+g'(x)$. In this case, $(f(x)+g(x))'$ is elementary because

$$(f(x)+g(x))' = k_0(x) \,|\, [-1,1] = \sqrt{1+x} - \sqrt{1+x} + \sqrt{1-x} - \sqrt{1-x}\,,$$

but this justification has nothing to do with the derivatives $f'(x) = \frac{1}{\sqrt{1-x^2}}$ and $g'(x) = -\frac{1}{\sqrt{1-x^2}}$. We pose the following problem.

Problem 7.6.1 (elementary derivatives) *Prove or disprove that*

$$f \in \mathrm{EF} \Rightarrow f' \in \mathrm{EF}\,.$$

• *Simple elementary functions.* We partially solve this problem in affirmative by introducing a relatively large subset SEF of EF and showing that it is closed to derivatives.

Definition 7.6.2 (VBEF & SEF) *Very basic elementary functions are*

$$\mathrm{VBEF} = \{k_c(x)\colon\ c \in \mathbb{R}\} \cup \{\exp x,\, \log x,\, \sin x,\, \mathrm{arcsin}_0\, x\}$$

where

$$\mathrm{arcsin}_0\, x = \arcsin x \,|\, (-1,\, 1)\,.$$

Simple elementary functions, abbreviated SEF*, is the subset of* EF *obtained by replacing in Definition 5.2.19 the starting set of functions* RBEF *with* VBEF*.*

We obtain simple elementary functions by starting from constants, the exponential function, logarithm, sine, and the restricted arkus sine, and repeatedly applying addition, multiplication, division, and composition. The next exercise explains why we do not add to VBEF the restrictions $x^b \,|\, (0, +\infty)$ for real $b > 0$.

Exercise 7.6.3 *Express them in terms of $k_c(x)$, $\exp x$ and $\log x$.*

Exercise 7.6.4 *Show that $\mathrm{arcsin}_0\, x \in \mathrm{EF}$.*

We show that the family of functions SEF is closed to derivatives.

Theorem 7.6.5 (derivatives in SEF) *For every function $f \in \mathrm{SEF}$, the domain $M(f)$ is an open set, $D(f) = M(f)$ and $f' \in \mathrm{SEF}$ as well.*

Proof. First, recall that the identity function $\mathrm{id}(x)$ is in SEF; it equals $\log(\exp x)$. For the constant function $k_c(x)$, we usually write just c. To begin with, we check that every function in VBEF has the three stated properties. Clearly, $M(k_c(x)) = M(\exp x) = M(\sin x) = \mathbb{R}$, $M(\log x) = (0, +\infty)$ and $M(\mathrm{arcsin}_0\, x) = (-1, 1)$ are open sets. The derivatives $k_c'(x) = k_0(x)$, $(\exp x)' = \exp x$,

$$(\log x)' = \tfrac{1}{x} \,|\, (0,\, +\infty) = \tfrac{1}{x} + \log x + (-1) \cdot \log x\,,$$

$(\sin x)' = \cos x = \sin(x + \frac{\pi}{2})$ and

$$(\mathrm{arcsin}_0(x))' = \tfrac{1}{\sqrt{1-x^2}} = 1/\exp(\tfrac{1}{2}\log(1 - x \cdot x))$$

are in SEF. Each has the same domain as the original function.

Let f be in SEF. We proceed by induction on the length of a generating word of f. In the induction step, we show for every case (i) $f = g + h$ or (ii) $f = g \cdot h$ or (iii) $f = g/h$ or (iv) $f = g(h)$, where $g, h \in$ SEF are functions with the three stated properties, that f has these properties as well. We use that $M \subset L(M)$ for every open set $M \subset \mathbb{R}$ (Exercise 6.5.12) and that open sets in $\mathbb{R}$ are closed to finite intersections (part 3 of Proposition 6.5.10). It follows that in case (i) the set

$$M(f) = M(g+h) = M(g) \cap M(h)$$

is open. By Corollary 7.3.4, $f' = g' + h' \in$ SEF and

$$M(f') = M(g') \cap M(h') = M(g) \cap M(h) = M(f)\,.$$

Case (ii) is similar except that we use Corollary 7.3.8.

In cases (iii) and (iv), we use EF $\subset \mathcal{C}$ (Theorem 6.7.17) and two results on open sets and continuous functions: the zero set of a continuous function is relatively closed (Proposition 6.5.19) and the preimage of an open set by a continuous function is relatively open (Proposition 6.5.21). Hence in case (iii) there is a closed set $U \subset \mathbb{R}$ such that

$$\begin{aligned} M(f) &= (M(g) \cap M(h)) \setminus Z(h) = M(g) \cap M(h) \cap (\mathbb{R} \setminus Z(h)) \\ &= M(g) \cap M(h) \cap \big(\mathbb{R} \setminus (M(h) \cap U)\big) \\ &= M(g) \cap M(h) \cap \big((\mathbb{R} \setminus M(h)) \cup (\mathbb{R} \setminus U)\big) \\ &= M(g) \cap M(h) \cap (\mathbb{R} \setminus U) \end{aligned}$$

and this is an open set. By Corollary 7.3.11, $f' = \frac{g'h - gh'}{h^2} \in$ SEF and

$$\begin{aligned} M(f') &= (D(g) \cap M(h) \cap M(g) \cap D(h)) \setminus Z(h^2) \\ &= M(g) \cap M(h) \setminus Z(h) \\ &= M(f)\,. \end{aligned}$$

Finally, in case (iv) the set $M(f) = h^{-1}[M(g)]$ is open. By Corollary 7.4.3, $f' = g'(h) \cdot h' \in$ SEF and

$$\begin{aligned} M(f') &= h^{-1}[M(g')] \cap M(h') = h^{-1}[M(g)] \cap M(h) \\ &= h^{-1}[M(g)] = M(g(h)) \\ &= M(f)\,. \end{aligned}$$

$\square$

Chapter 8

Mean value theorems

In Section 8.1 we meet three mean value theorems: Rolle's Theorem 8.1.1, Lagrange's Theorem 8.1.6 and Cauchy's Theorem 8.1.11. The first two theorems are generalized in Theorems 8.1.5 and 8.1.8. The extending Sections 8.2–8.4 are devoted to three applications of mean value theorems. In Section 8.2 we show by means of Rolle's theorem that the sequence

$$(\log n) = (0,\, \log 2,\, \log 3,\, \dots)$$

is not P-recurrent. In Theorem 8.2.10 we actually prove a more general result. In Sections 8.3 and 8.4 we show with the help of Lagrange's theorem in two effective ways that real transcendental numbers exist.

In Section 8.5 in Theorem 8.5.5 and Proposition 8.5.7 we obtain by means of the first derivative results on monotonicity of functions. Theorems 8.5.12 and 8.5.13 are l'Hospital rules for computing functional limits of the type $\frac{0}{0}$ and $\frac{\infty}{\infty}$. In Section 8.6 in Proposition 8.6.4 we determine by the sign of the derivative

$$(f')'(b)$$

the type of the local extreme of f at b. In Theorem 8.6.11 we prove that any convex or concave function f defined on a set $M \subset \mathbb{R}$ with no minimum and no maximum has on $L^{\pm}(M)$ finite one-sided derivatives. Thus such f is continuous. In Theorem 8.6.16 we determine by the sign of $(f')'(b)$ convexity/concavity of a function f defined on an interval. Definition 8.6.19 introduces inflection points. Theorems 8.6.22 and 8.6.23 provide necessary and sufficient conditions for their existence.

In Section 8.7 we give thirteen steps for determining the main geometric features of the graph of a function. Step 0 places the function in the hierarchy

$$\mathrm{SEF} \subset \mathrm{EF} \subset \mathcal{R}\,.$$

We demonstrate the procedure of thirteen steps on three examples for functions $\operatorname{sgn} x$, $\tan x$ and $\arcsin\left(\frac{2x}{x^2+1}\right)$.

8.1 Rolle, Lagrange and Cauchy

The three theorems named after these mathematicians establish various relations between values of functions and their derivatives. In these theorems, $a < b$ are real numbers.

• *Rolle's theorem.* We begin with this theorem and deduce the other two theorems from it.

Theorem 8.1.1 (Rolle 1) *Let $f \in \mathcal{C}([a,b])$ and $f(a) = f(b)$. If $f'(c) \in \mathbb{R}^*$ for every $c \in (a,b)$, <u>then</u> $f'(c) = 0$ for some $c \in (a,b)$.*

Proof. If $f(x)$ is constant, we are done as $f'(c) = 0$ for every $c \in (a,b)$. Else, by Corollary 6.5.6, there exists $c \in (a,b)$ such that f has at c a global extreme. Since $c \in L^{\mathrm{TS}}([a,b])$ and $f'(c)$ exists, by Corollary 7.1.12 we have $f'(c) = 0$. □

Michel Rolle (1652–1719) was a French mathematician.

Exercise 8.1.2 *Let $f(x)$ be the absolute value $|x|$ restricted to $[-1,1]$. Then $f(1) = f(-1) = 1$ and f is continuous, but $f'(c) \neq 0$ for every $c \in (-1,1)$. Which assumption of Rolle's theorem is not satisfied?*

The well known solution of Exercise 5.3.3 employs the algebraic division of polynomials with remainder. Now we show an analytic solution.

Corollary 8.1.3 (the number of roots) *Let f be a nonzero polynomial. The set of zeros $Z(f)$ is finite and*

$$|Z(f)| \leq \deg f\,.$$

Proof. We argue by contradiction. Let g be a nonzero polynomial with the minimum degree $d \in \mathbb{N}_0$ such that $|Z(g)| > \deg g = d$. So there exist $d+1$ real numbers $a_1 < a_2 < \cdots < a_{d+1}$ such that

$$g(a_1) = g(a_2) = \cdots = g(a_{d+1}) = 0\,.$$

We have $d \geq 1$ because nonzero constant polynomials have no roots. By Theorem 8.1.1 there exist d real numbers b_i such that

$$a_1 < b_1 < a_2 < b_2 < a_3 < \cdots < a_{d+1} \;\text{ and }\; g'(b_1) = g'(b_2) = \cdots = g'(b_d) = 0\,.$$

As we know from the end of Section 7.5, g' is a nonzero polynomial with degree $d-1$. We have obtained a contradiction with the minimality of d. □

The next version of Rolle's theorem does not need derivatives; it applies to the function in Exercise 8.1.2.

Proposition 8.1.4 (Rolle 2) *Let $f \in \mathcal{C}([a,b])$ and $f(a) = f(b)$. Then there exists $c \in (a,b)$ such that for every ε there exist $d_1, d_2 \in (a,b)$ satisfying*

$$d_1 < c < d_2,\ d_2 - d_1 \le \varepsilon \ \text{ and } \ f(d_1) = f(d_2)\,.$$

Geometrically, some point $\langle c, f(c)\rangle \in G_f$ can be arbitrarily tightly enclosed by pairs of intersection points of horizontal secants.

Proof. If $f(x)$ is constant, the result trivially holds. Else, let f attain at $c \in (a,b)$ the maximum value (for the minimum value, we would argue similarly). We distinguish three cases. (i) $f(x)$ is constantly $f(c)$ on $[c-\delta, c]$ ($\subset [a,b]$) or (ii) $f(x)$ is constantly $f(c)$ on $[c, c+\delta]$ ($\subset [a,b]$) or (iii) $f(x)$ attains values smaller than $f(c)$ arbitrarily close to c on both sides of c. In case (i), respectively (ii), we replace c with $c - \frac{\delta}{2}$, respectively $c + \frac{\delta}{2}$, and then c clearly has the stated property. In case (iii), we keep c and find the required points d_1 and d_2 by means of Theorem 6.4.1. $\Box$

The third version of Theorem 8.1.1 generalizes domains $[a,b]$ to compact sets. Recall that $L^{\mathrm{TS}}(M)$ denotes the set of two-sided limit points of M.

Theorem 8.1.5 (Rolle 3) *Let $M \subset \mathbb{R}$ be a compact set with $M \cap L^{\mathrm{TS}}(M) \ne \emptyset$ and let $f \in \mathcal{C}(M)$ satisfy two conditions.*

1. *The restriction $f \,|\, (M \setminus L^{\mathrm{TS}}(M))$ is constant.*
2. *We have $f'(c) \in \mathbb{R}^*$ for every $c \in M \cap L^{\mathrm{TS}}(M)$.*

Then $f'(c) = 0$ for some $c \in M \cap L^{\mathrm{TS}}(M)$.

Proof. If $f(x)$ is constant, then $f'(c) = 0$ for every $c \in M \cap L^{\mathrm{TS}}(M)$ by the assumptions and Theorem 7.1.9. Else, if $f(x)$ is not constant, by the assumptions and Corollary 6.5.6, f attains at some $c \in M \cap L^{\mathrm{TS}}(M)$ global extreme. Using condition 2 and Theorem 7.1.9, we get $f'(c) = 0$. $\Box$

In Theorem 8.1.1 we have $M = [a,b]$, $M \cap L^{\mathrm{TS}}(M) = (a,b)$ and $M \setminus L^{\mathrm{TS}}(M) = \{a,b\}$. To show a different application of Theorem 8.1.5, we set

$$N \equiv \{(-1)^n \tfrac{1}{n}:\ n \in \mathbb{N}\}\,.$$

The theorem then applies with $M = \{0\} \cup N$ and any constant function f in $\mathcal{F}(M)$. Now $M \cap L^{\mathrm{TS}}(M) = \{0\}$ and $M \setminus L^{\mathrm{TS}}(M) = N$.

• *Lagrange's theorem.* This mean value theorem has an interesting geometric interpretation and generalization.

Theorem 8.1.6 (Lagrange 1) *Let $f \in \mathcal{C}([a,b])$. If $f'(c) \in \mathbb{R}^*$ for every c in (a,b), then*

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

for some $c \in (a,b)$.

Proof. Let $z \equiv \frac{f(b)-f(a)}{b-a}$. Then the function

$$g(x) \equiv f(x) - z(x-a) \quad (\in \mathcal{C}([a,b]))$$

satisfies the assumptions of Theorem 8.1.1, especially $g(a) = g(b)$ $(= f(a))$. Thus $g'(c) = f'(c) - z = 0$ for some $c \in (a,b)$, and $f'(c) = z$. □

Geometrically, there exists a point $C \equiv \langle c, f(c)\rangle \in G_f$ with $a < c < b$ such that the tangent to G_f at C is parallel to the secant $\kappa(a, f(a), b, f(b))$. *Joseph-Louis Lagrange (1736–1813)* was a French mathematician, physicist, and astronomer of Italian origin.

Exercise 8.1.7 *Let $a < b$ be real numbers. For $f \in \mathcal{R}$ let $R(f)$ be Rolle's theorem for f — if $f \in \mathcal{C}([a,b])$, $f(a) = f(b)$ and $f'(c) \in \mathbb{R}^*$ for every $c \in (a,b)$, then $f'(c) = 0$ for some $c \in (a,b)$. Similarly, for $f \in \mathcal{R}$ let $L(f)$ be Lagrange's theorem for f — if $f \in \mathcal{C}([a,b])$ and $f'(c) \in \mathbb{R}^*$ for every $c \in (a,b)$, then $f'(c) = \frac{f(b)-f(a)}{b-a}$ for some $c \in (a,b)$. Explain why we know that the logical equivalence "Rolle's theorem $\iff$ Lagrange's theorem", formally*

$$\big(\forall f \in \mathcal{R}\colon\ R(f)\big) \iff \big(\forall f \in \mathcal{R}\colon\ L(f)\big)\,,$$

holds before we even start proving both theorems.

Tangent lines to graphs are of two kinds. Let $f \in \mathcal{F}(M)$, $b \in M$ and let $f'(b) \in \mathbb{R}$. We consider the tangent

$$l(x) \equiv f'(b)(x-b) + f(b) \quad (\in \mathcal{F}(\mathbb{R}))$$

to G_f at $B \equiv \langle b, f(b)\rangle$. If for some δ we have

$$\forall x \in U(b,\,\delta) \cap M\colon\ l(x) \ge f(x) \text{ or } \forall x \in U(b,\,\delta) \cap M\colon\ l(x) \le f(x)\,,$$

we say that ℓ is a touching tangent (see Theorem 7.2.4). If no such δ exists, we call ℓ a cutting tangent. In the latter case, G_f contains points arbitrarily close to B that lie below and above the tangent. In the former case, all points in G_f sufficiently close to B lie on the same side of the tangent. The second version of Lagrange's theorem yields for any slope r close to the slope of the secant a touching tangent with slope r.

Theorem 8.1.8 (Lagrange 2) *Let $f \in \mathcal{C}([a,b])$, $f'(c) \in \mathbb{R}^*$ for every c in (a,b) and let $s \equiv \frac{f(b)-f(a)}{b-a}$. We assume that G_f is not a straight plane segment. Then there exists δ such that for every $r \in (s-\delta,\, s+\delta)$ the graph G_f*

has a touching tangent with slope r at some point $\langle c, f(c)\rangle$ with $c \in (a,b)$.

Proof. Let T $(\subset \mathbb{R}^2)$ be the straight segment joining the points $\langle a, f(a)\rangle$ and $\langle b, f(b)\rangle$. We know that $G_f \setminus T \ne \emptyset$. So there is a point $\langle d, f(d)\rangle \in G_f$ with $a < d < b$ lying above or below T. We assume the latter case. In the former case, we argue similarly. Let s_1, respectively s_2, be the slope of the straight plane segment joining the points $\langle a, f(a)\rangle$ and $\langle d, f(d)\rangle$, respectively $\langle d, f(d)\rangle$ and $\langle b, f(b)\rangle$. Then $s_1 < s < s_2$ and we take any δ such that

$$s_1 < s - \delta < s + \delta < s_2 .$$

We show that for every $r \in (s-\delta, s+\delta)$ the graph G_f has a touching tangent with slope r at some interior point. Let $r \in (s - \delta, s + \delta)$. For $t \in \mathbb{R}$, we define the linear function $l_t(x) \equiv rx + t$ $(\in \mathcal{F}(\mathbb{R}))$ and set (in the linear order $\langle \mathbb{R}^*, <\rangle$)

$$u \equiv \sup(\{t \in \mathbb{R}\colon\ f(x) \geq l_t(x) \text{ for every } x \in [a,\, b]\}) .$$

By Corollary 6.5.6, the function f has a global minimum. So $u \in \mathbb{R}$. Clearly, $f(x) \geq l_u(x)$ for every $x \in [a, b]$ and for some $x = c \in [a, b]$, the equality occurs (Exercise 8.1.9). If $c = a$ or $c = b$, then by the choice of δ and r, we have $f(d) < l_u(d)$. Hence $c \in (a, b)$. By Theorem 7.2.4, the line $l_u(x)$ is a touching tangent to G_f at $\langle c, f(c)\rangle$. By the definition of $l_u(x)$, the slope is r. $\square$

Exercise 8.1.9 *Prove the claim about $f(x)$ and $l_u(x)$.*

Exercise 8.1.10 *What happens if the graph G_f is a straight plane segment?*

• *Cauchy's theorem.* A new feature is that this mean value theorem involves two functions.

Theorem 8.1.11 (Cauchy) *Let $f, g \in \mathcal{C}([a, b])$, $g(b) \neq g(a)$ and let $f'(c) \in \mathbb{R}^*$ and $g'(c) \in \mathbb{R}$ for every $c \in (a, b)$; so now $g'(c) \neq \pm\infty$. Then*

$$f'(c) = \frac{f(b) - f(a)}{g(b) - g(a)} \cdot g'(c)$$

for some $c \in (a, b)$.

Proof. Let $z \equiv \frac{f(b)-f(a)}{g(b)-g(a)}$. Then the function

$$h(x) \equiv f(x) - z(g(x) - g(a)) \quad (\in \mathcal{C}([a,\, b]))$$

satisfies the assumptions of Theorem 8.1.1, especially $h(a) = h(b) = f(a)$. Hence there exists $c \in (a, b)$ with $h'(c) = f'(c) - zg'(c) = 0$. So $f'(c) = zg'(c)$. $\square$

Exercise 8.1.12 *Where in the proof would be $g'(c) = \pm\infty$ problematic? Modify the statement of the theorem to accommodate the values $g'(c) = \pm\infty$.*

8.2 The sequence $(\log n)$ is not P-recurrent

This section is based on the preprint [49] of the author. We show with the help of Theorem 8.1.1 that the sequence

$$(\log n) = (0,\, \log 2,\, \log 3,\, \dots)$$

does not satisfy any recurrence relation from the class of so called P-recurrences. We begin with the definition of these recurrences.

• P-*recurrent sequences.* These sequences generalize C-recurrent (constantly recurrent) sequences. A C-recurrent sequence $(a_n) \subset \mathbb{R}$ satisfies for some $k \in \mathbb{N}$ real coefficients $c_1,\, c_2,\, \dots,\, c_k$, not all of them zero, and every integer $n \ge k$ the relation

$$\textstyle\sum_{i=1}^k c_i a_{n-i+1} = c_1 a_n + c_2 a_{n-1} + \cdots + c_k a_{n-k+1} = 0\,.$$

A well known C-recurrent sequence is the Fibonacci sequence

$$(F_n) = (1,\, 1,\, 2,\, 3,\, 5,\, 8,\, 13,\, 21,\, 34,\, \dots)\,.$$

P-recurrent sequences allow polynomials $c_i(n)$ instead of the constants c_i. So every C-recurrent sequence is P-recurrent.

Definition 8.2.1 (P-recurrence) *Let $(a_n) \subset \mathbb{R}$. We say that the sequence (a_n) is P-recurrent if there exist $k \ge 1$ polynomials $p_1,\, p_2,\, \dots,\, p_k$ in* POL*, not all of them zero, such that for every integer $n \ge k$ we have*

$$\textstyle\sum_{i=1}^k p_i(n) \cdot a_{n-i+1} = p_1(n)a_n + p_2(n)a_{n-1} + \cdots + p_k(n)a_{n-k+1} = 0\,.$$

For instance,

$$(a_n) = (n!) = (1,\, 2,\, 6,\, 24,\, 120,\, 720,\, 5040,\, \dots)$$

is P-recurrent: for every $n \ge 2$ we have

$$1 \cdot a_n + (-n) \cdot a_{n-1} = 0\,.$$

The next two exercises describe two ways to show that Definition 8.2.1 with $n \ge k$ relaxed to $n \ge n_0 \ge k$ again yields P-recurrent sequences.

Exercise 8.2.2 *Any relation*

$$\textstyle\sum_{i=1}^k p_i(n) \cdot a_{n-i+1} = 0$$

holding for every $n \ge n_0 \ge k$ can be viewed as a P*-recurrence of an order $l \ge k$ holding for every $n \ge l$.*

Exercise 8.2.3 *In any relation*

$$\textstyle\sum_{i=1}^k p_i(n) \cdot a_{n-i+1} = 0$$

holding for every $n \ge n_0 \ge k$, we can replace the coefficients $p_i(x)$ with other polynomials $q_i(x)$ such that the new relation holds for every $n \ge k$.

See the book [73] for uses of P-recurrent sequences in enumerative combinatorics. In this section, we use Theorem 8.1.1 to prove the following.

Theorem 8.2.4 (logs are not P-recurrent, [49]) *The sequence*

$$(\log n) = (0,\ \log 2,\ \log 3,\ \dots)$$

is not P*-recurrent.*

• *Poles.* We prove Theorem 8.2.4 with the help of poles. Let $f \in \mathcal{F}(M)$ and $b \in L(M)$. If $\lim_{x\to b} f(x) \in \mathbb{R}$, then b is a <u>regular point</u> of f. If

$$f(x) \sim c(x-b)^{-k} \quad (x \to b)$$

for some $c \in \mathbb{R}\setminus\{0\}$ and $k \in \mathbb{N}$, we say that b is a <u>pole</u> of f of <u>order</u> k.

Exercise 8.2.5 *Let $f, g \in \mathcal{R}$, $b \in L(M(f+g))$, b be a pole of f of order k and let b be a regular point of g or a pole of g of order $l < k$. Then b is a pole of $f+g$ of order k.*

Proposition 8.2.6 (poles of derivatives) *Let $r(x) \in$ RAC and let $b \in \mathbb{R}$. <u>Then</u> b is a regular point of $r'(x)$ or a pole of $r'(x)$ with order at least 2.*

Proof. If $r(x) = k_0(x)$, then $r'(x) = k_0(x)$ and b is a regular point of $r(x)'$. We assume that $r(x) \ne k_0(x)$, use Theorem 5.3.20 and write $r(x)$ as a ratio of two polynomials, $r(x) = p(x)/q(x)$. Using Exercise 5.3.4 we get factorizations

$$p(x) = p_0(x)\cdot(x-b)^k \ \text{ and } \ q(x) = q_0(x)\cdot(x-b)^l\,,$$

where $p_0(x), q_0(x) \in$ POL are such that $p_0(b), q_0(b) \ne 0$ and $k, l \in \mathbb{N}_0$. We have

$$r(x) = r_0(x)\cdot(x-b)^m\,,$$

where $r_0(x) = p_0(x)/q_0(x) \in$ RAC is such that $b \in M(r_0)$ and $r_0(b) \ne 0$, and $m = k - l \in \mathbb{Z}$. If $m \ge 0$ then b is a regular point of $r'(x)$. Let $m < 0$. Using results on derivatives in the previous chapter we get

$$r'(x) = r_0'(x)\cdot(x-b)^m + r_0(x)\cdot m(x-b)^{m-1}\,.$$

By Exercise 8.2.5, b is a pole of $r'(x)$ of order $1 - m \ge 2$. □

• *Nonzero functions.* A function $f \in \mathcal{R}$ is <u>nonzero</u> if $f(b) \ne 0$ for some $b \in M(f)$. Recall that $Z(f) = \{b \in M(f):\ f(b) = 0\}$.

Proposition 8.2.7 (zeros of nonzero restrictions) *Let $f \in \mathcal{R}$ be a nonzero restriction of a rational function. <u>Then</u> $Z(f)$ is finite.*

Proof. We can assume that $f(x) \in \mathrm{RAT}$ and is nonzero. By Theorem 5.3.20, $f(x) = p(x)/q(x)$ where $p(x)$ and $q(x)$ are polynomials. Clearly, $Z(f) = Z(p)$. Since $f(x)$ is nonzero, $p(x) \neq k_0(x)$. By Exercise 5.3.3 or Corollary 8.1.3, the set $Z(p) = Z(f)$ is finite. $\Box$

Exercise 8.2.8 *The square of a nonzero function is a nonzero function.*

Corollary 8.2.9 (nonzero derivative) *Let $r(x) \in \mathrm{RAC}$, $c_1, c_2, \dots, c_k$ be $k \in \mathbb{N}$ real numbers, not all of them zero, and let*

$$f(x) = r(x) + \textstyle\sum_{i=1}^{k} c_i \log(x - i + 1)\,.$$

Then $f'(x)$ is a nonzero restriction of a rational function.

Proof. We may assume that $c_k \neq 0$. Clearly,

$$f'(x) = \big(r'(x) + \textstyle\sum_{i=1}^{k} \frac{c_i}{x-i+1}\big) \,|\, (k-1, +\infty)$$

is a restriction of a rational function. Let $b := k - 1$. If b is a regular point of $r'(x)$, then by Exercise 8.2.5 b is a pole of $f'(x)$ of order 1. Else the same exercise and Proposition 8.2.6 give that b is a pole of $f'(x)$ of order at least 2. In either case we see that $f'(x)$ is nonzero. $\Box$

- *Proof of Theorem 8.2.4.* We prove the following stronger result.

Theorem 8.2.10 (finitely many zeros) *Every function of the form*

$$f(x) = r(x) \,|\, (c, +\infty) + \textstyle\sum_{j=1}^{k} p_j(x) \log(x - j + 1)\,,$$

where $r(x) \in \mathrm{RAC}$, $c \in \mathbb{R}$, $k \in \mathbb{N}$, $p_j(x) \in \mathrm{POL}$ and some $p_j(x)$ is nonzero, has finitely many zeros.

Proof. For every such function $f = f(x)$ we define the degree as

$$\deg(f) = \min\big(\textstyle\sum_{\substack{j\in[k] \\ p_j(x)\neq k_0(x)}} \deg(p_j)\big)$$

where the minimum is taken over all representations of $f(x)$ in the stated form. We argue by contradiction and consider a function $f_0(x)$ with the minimum degree and infinitely many zeros. We take an infinite and strictly monotone sequence $(a_n) \subset Z(f_0)$ (Exercise 8.2.11). We may assume that for every $n \in \mathbb{N}$, the interval (a_n, a_{n+1}), respectively (a_{n+1}, a_n), is contained in $M(f_0)$ (Exercise 8.2.12). Using Theorem 8.1.1 we get a sequence $(b_n) \subset Z(f_0')$ such that

$$a_1 < b_1 < a_2 < b_2 < a_3 < \dots, \text{ respectively } a_1 > b_1 > a_2 > b_2 > a_3 > \dots$$

(Exercise 8.2.13). In particular, the set $Z(f_0')$ is infinite. This is a contradiction. If $\deg(f_0) = 0$, we contradict Proposition 8.2.7 and Corollary 8.2.9. If $\deg(f_0) > 0$, we contradict the minimality of $\deg(f_0)$ because $f_0'(x)$ is a function of the considered form that has infinitely many zeros and has degree smaller than $f_0(x)$ (Exercise 8.2.14). $\Box$

Exercise 8.2.11 *How do we select in the infinite set $Z(f)$ an increasing, or a decreasing, sequence (a_n)?*

Exercise 8.2.12 *Why can we assume that the gaps between consecutive terms in (a_n) are contained in $M(f)$?*

Exercise 8.2.13 *How do we exactly apply Rolle's theorem to f and (a_n) so that we get the interleaving zeros (b_n) of f'?*

Exercise 8.2.14 *Show that for* $\deg(f) > 0$ *the derivative f' has the considered form and* $\deg(f') < \deg(f)$.

Proof of Theorem 8.2.4. Suppose, for the contrary, that the sequence $(\log n)$ is P-recurrent. Then there exist $k \ge 1$ polynomials $p_1(x)$, $p_2(x)$, $\dots$, $p_k(x)$, not all equal to $k_0(x)$, such that for every $n \ge k$,

$$\textstyle\sum_{j=1}^{k} p_j(n)\log(n-j+1) = 0\,.$$

Hence the function

$$f(x) = \textstyle\sum_{j=1}^{k} p_j(x)\log(x-j+1)$$

has infinitely many zeros: $Z(f) \supset \{k, k+1, \dots\}$. This contradicts Theorem 8.2.10 (Exercise 8.2.15). $\square$

Exercise 8.2.15 *Check that* $\sum_{j=1}^{k} p_j(x)\log(x-j+1)$ *belongs to the family of functions considered in Theorem 8.2.10.*

Exercise 8.2.16 *Generalize Theorem 8.2.4 to sequences* $(\log(n+c))$ *for any real* $c > -1$.

Exercise 8.2.17 *Prove by means of Rolle's theorem the following proposition.*

Proposition 8.2.18 *Let $I \subset \mathbb{R}$ be a nontrivial interval, $f \in \mathcal{C}(I)$, $f'(x) \in \mathbb{R}^*$ for every $x \in I$ and let $Z(f)$ be infinite. Then $Z(f')$ is infinite.*

8.3 Cantor's transcendental numbers

We prove by means of Theorem 8.1.6 the existence of transcendental numbers. In the next section we give a simpler argument.

• *Recursive real numbers.* For simplicity of notation, we only work with real numbers in $[0,1]$. We denote by $(n)_{10}$ the natural number $n \in \mathbb{N}$ written in the base 10 as a word over the alphabet $\{0,1,\dots,9\}$. For example, $(2^{10})_{10} = 1024$. Let

$$\mathbb{N}_{10} := \{(n)_{10} \colon\ n \in \mathbb{N}\}\,.$$

Exercise 8.3.1 *The map* $\mathbb{N} \ni n \mapsto (n)_{10} \in \mathbb{N}_{10}$ *is injective.*

Algorithms cannot work directly with natural numbers; they work with their codes, which are words over a finite alphabet. An example of these codes are the words $(n)_{10}$.

Definition 8.3.2 (recursive reals) *If a map*

$$\mathcal{A}\colon \mathbb{N}_{10} \to \{0,\, 1, \ldots, 9\}$$

is given by an algorithm, that is by a Turing machine, we call the sum

$$\kappa(\mathcal{A}) := \textstyle\sum_{n \ge 1} \mathcal{A}((n)_{10}) \cdot 10^{-n} \quad (\in [0,\, 1])$$

a recursive real number in the interval $[0, 1]$.

Exercise 8.3.3 *Show that some real numbers in* $[0, 1]$ *are not recursive.*

• *Algebraic and transcendental numbers.* We remind these two sets of numbers. We denote by $\mathbb{Q}[x]$ the set of rational polynomials; $\mathbb{Q}[x]$ is a subset of POL obtained in Definition 5.3.1 by allowing only rational constant functions, that is, functions $k_c(x)$ for $c \in \mathbb{Q}$. The set $\mathbb{Z}[x]$ of integral polynomials is defined similarly.

Definition 8.3.4 (algebraic numbers) *A complex number number (see Appendix A.5)* $\alpha \in \mathbb{C}$ *is algebraic if* $p(\alpha) = 0$ *for a nonzero polynomial* $p(x) \in \mathbb{Q}[x]$.

Exercise 8.3.5 *All fractions and roots* $\sqrt{n}$, $n \in \mathbb{N}_0$, *are algebraic numbers.*

Exercise 8.3.6 *The polynomial* $p(x)$ *in Definition 8.3.4 can always be modified so that its degree is preserved and* (i) *it remains rational but becomes monic (with leading coefficient* 1*) or* (ii) *becomes integral.*

Algebraic numbers for which both forms (i) and (ii) are simultaneously achievable, that is, roots of monic integral polynomials, are called algebraic integers.

Exercise 8.3.7 *Which fractions are algebraic integers?*

Exercise 8.3.8 *Is the golden ratio* $\phi \equiv \frac{1+\sqrt{5}}{2}$ *an algebraic integer?*

Exercise 8.3.9 *How is the golden ratio related to the Fibonacci numbers?*

Definition 8.3.10 (transcendental numbers) *We say that a complex number is transcendental if it is not algebraic.*

• *Cantor's proof.* In 1870s, G. Cantor found a simple proof of the existence of (real) transcendental numbers. This was a great achievement of the nascent set theory, which we recall here.

Theorem 8.3.11 (Cantor) *Real transcendental numbers form an uncountable set.*

Proof. Let T ($\subset \mathbb{R}$) be the set of real transcendental numbers. We assume for the contrary that T is at most countable. Exercise 8.3.12 shows that the set $A \cap \mathbb{R}$ of real algebraic numbers is countable. It follows that the set

$$\mathbb{R} = T \cup (A \cap \mathbb{R})$$

is countable. This contradicts Corollary 1.6.4 that $\mathbb{R}$ is an uncountable set. $\Box$

Exercise 8.3.12 *Show that the set A of algebraic numbers is countable.*

• *An effective version of Cantor's proof.* We extract from the previous proof an algorithm

$$\mathcal{A}\colon \mathbb{N}_{10} \to \{0,\, 1,\, \ldots,\, 9\}$$

producing a recursive transcendental number $\kappa(\mathcal{A})$. We define $\mathcal{A}$ with the help of the following proposition, which in turn we obtain by means of Theorem 8.1.6. The proposition is an effective version, for integral polynomials, of the fact that any nonzero value of a continuous function has a neighborhood on which the function does not vanish.

Proposition 8.3.13 ($p(x) \neq 0$ on I) *Let*

$$p(x) = a_n x^n + \cdots + a_1 x + a_0,\ n \in \mathbb{N}_0,\ a_i \in \mathbb{Z} \text{ and } a_n \neq 0\,,$$

be a nonzero integral polynomial with degree n. Let $\alpha = a/10^k$, for $a \in \mathbb{Z}$ and $k \in \mathbb{N}_0$, be a decimal fraction such that $p(\alpha) \neq 0$. Let

$$b = (n+1)^2 \cdot \max(\{|a_0|,\, |a_1|,\, \ldots,\, |a_n|\}) \cdot (|a|+1)^n \ \text{ and } \ l = kn + b \ \ (\in \mathbb{N})\,.$$

<u>Then</u> $p(x) \neq 0$ for every real number $x \in [\alpha, \alpha + 10^{-l}]$.

Proof. Since α has denominator 10^k and $p(x)$ has integral coefficients and degree n, the assumption that $p(\alpha) \neq 0$ implies that $|p(\alpha)| \geq 10^{-kn}$. For every $x \in (\alpha, \alpha+1]$, Theorem 8.1.6 gives a $y \in (\alpha, x)$ such that

$$p(x) = p(\alpha) + p'(y) \cdot (x - \alpha)\,.$$

By Exercise 8.3.14, $|p'(y)| \leq b$ where b is defined above. If $x - \alpha \leq 10^{-l} = 10^{-kn-b} < 10^{-kn}/b$ then

$$|p(x)| \geq |p(\alpha)| - |p'(y)| \cdot (x - \alpha) > 10^{-kn} - b \cdot \tfrac{10^{-kn}}{b} = 0\,,$$

as stated. $\Box$

Exercise 8.3.14 *Prove the estimate $|p'(y)| \leq b$.*

Theorem 8.3.15 (effective Cantor's proof) *Cantor's proof yields an algorithm* $\mathcal{A}\colon \mathbb{N}_{10} \to \{0, 1, \ldots, 9\}$ *such that the real number*

$$\kappa := \kappa(\mathcal{A}) = \textstyle\sum_{n \geq 1} \mathcal{A}((n)_{10}) \cdot 10^{-n}$$

is not a root of any nonzero integral polynomial. Hence κ *is a recursive real transcendental number.*

Proof. It is clear that there is an algorithm $\mathcal{B}\colon \mathbb{N}_{10} \to \bigcup_{n=1}^{\infty} \mathbb{Z}^n =: Z$ such that

$$\mathcal{B}[\mathbb{N}_{10}] = \{(a_0,\, a_1,\, \ldots,\, a_n) \in Z\colon\; n \in \mathbb{N}_0 \wedge a_n \neq 0\}$$

— $\mathcal{B}$ lists all nonzero integral polynomials by the tuples of their coefficients. For simplicity of notation we take the elements of Z directly; in practice they are encoded as words over a finite alphabet. For $m \in \mathbb{N}$ we set

$$p_m(x) := \textstyle\sum_{j=0}^{n_m} a_{j,m} x^j \;\text{ where }\; (a_{0,\,m},\, a_{1,\,m},\, \ldots,\, a_{n_m,\,m}) = \mathcal{B}((m)_{10})\,.$$

The algorithm $\mathcal{A}$ calls the algorithm $\mathcal{B}$ and generates a sequence of decimal fractions

$$\alpha_1 = \tfrac{z_1}{10^{k_1}} = \tfrac{0}{10^0} = 0,\; \alpha_2 = \alpha_1 + \tfrac{z_2}{10^{k_2}},\; \alpha_3 = \alpha_2 + \tfrac{z_3}{10^{k_3}},\; \ldots$$

such that $k_j \in \mathbb{N}_0$, $0 = k_1 < k_2 < \ldots$ (so that the denominator of α_j is 10^{k_j}), $z_j \in \{0, 1, \ldots, 10^{k_j - k_{j-1}} - 1\}$ for every $j \in \mathbb{N}$ (with $k_0 := 0$), and that for every $m \in \mathbb{N}$ we have

$$\forall x \in \big[\alpha_m,\, \alpha_m + 10^{-k_m}\big]\colon p_1(x) p_2(x) \ldots p_{m-1}(x) \neq 0\,,$$

where for $m = 1$ the product is defined as 1.

Suppose that $m \in \mathbb{N}$ and that $\mathcal{A}$ already generated $\alpha_1,\, \alpha_2,\, \ldots,\, \alpha_m$. To generate the next decimal fraction α_{m+1}, the algorithm $\mathcal{A}$ gets the polynomial $p_m(x)$ by calling $\mathcal{B}$ and takes a large number $k \in \mathbb{N}$ such that

$$k > k_m \text{ and } 10^{k-k_m} > \deg(p_m)\,.$$

By Exercise 5.3.3 there is a number $j \in \{0, 1, \ldots, 10^{k-k_m} - 1\}$ such that

$$p_m\big(\alpha_m + \tfrac{j}{10^k}\big) \neq 0\,.$$

By Exercise 8.3.16, $\mathcal{A}$ can get this j. Then $\mathcal{A}$ invokes Proposition 8.3.13 with $p(x) = p_m(x)$ and $\alpha = \alpha_m + \frac{j}{10^k}$, gets (from Proposition 8.3.13) the number $l \in \mathbb{N}$ and computes the numbers

$$k_{m+1} := \max(k,\, l),\; z_{m+1} := j \cdot 10^{k_{m+1}-k} \;\text{ and }\; \alpha_{m+1} := \alpha_m + \tfrac{z_{m+1}}{10^{k_{m+1}}}\,.$$

Since the interval $I := \big[\alpha_{m+1}, \alpha_{m+1} + 10^{-k_{m+1}}\big]$ is contained in both intervals

$$\big[\alpha_m,\, \alpha_m + 10^{-k_m}\big] \text{ and } \big[\alpha,\, \alpha + 10^{-l}\big]\,,$$

the value $p_j(x) \neq 0$ for every $j = 1, 2, \ldots, m$ and every $x \in I$, as required.

For $j \in \mathbb{N}$, we denote by $\overline{z_j}$ the word obtained by padding in front of $(z_j)_{10}$ zeros to the length $k_j - k_{j-1}$. For example, if $k_j - k_{j-1} = 5$ and $z_j = (z_j)_{10} = 27$, then $\overline{z_j} = 00027$. The final outputs

$$\mathcal{A}((n)_{10}) \in \{0,\, 1,\, \ldots,\, 9\},\ n \in \mathbb{N}\,,$$

of $\mathcal{A}$ are defined by the equality of two infinite words over $\{0, 1, \ldots, 9\}$:

$$\mathcal{A}(1)\,\mathcal{A}(2)\,\cdots = \overline{z_1}\,\overline{z_2}\,\cdots\,.$$

Since

$$\kappa = \kappa(\mathcal{A}) = \textstyle\sum_{n=1}^{\infty} \mathcal{A}((n)_{10}) \cdot 10^{-n} \in \bigcap_{m=1}^{\infty} \left[\alpha_m,\, \alpha_m + 10^{-k_m}\right],$$

the value $p_m(\kappa) \neq 0$ for every $m \in \mathbb{N}$. The number κ is transcendental. $\square$

Exercise 8.3.16 *How does the algorithm $\mathcal{A}$ select the number j?*

A natural idea is to attempt to extend the construction and obtain a recursive real number that is not a zero of any nonzero function in $\mathrm{EF}_{\mathbb{Q}}$. This subfamily of elementary functions is generated by rational constants $k_c(x)$, $c \in \mathbb{Q}$, and rational powers x^b, $b \in \mathbb{Q} \cap (0, +\infty)$.

8.4 Liouville's transcendental numbers

The French mathematician and physicist *Joseph Liouville (1809–1882)* was the first who proved, in 1844, that transcendental numbers exist. In this section we explain his method. It produces very simple recursive real transcendental numbers.

- *Liouville's inequality.* This is another application of Theorem 8.1.6.

Theorem 8.4.1 (Liouville's inequality) *Let $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ be algebraic. Then there exist $n \in \mathbb{N}$ and $c > 0$ such that for every $p/q \in \mathbb{Q}$ with $q > 0$ we have*

$$\left|\alpha - \frac{p}{q}\right| \geq \frac{c}{q^n}\,.$$

Proof. Using Exercise 8.3.6 we take a nonzero $f(x) \in \mathbb{Z}[x]$ with the minimum degree $n = \deg(f) \geq 2$ such that $f(\alpha) = 0$. Let $I = [\alpha - 1, \alpha + 1]$. If $\frac{p}{q} \in \mathbb{Q} \setminus I$ then

$$\left|\alpha - \tfrac{p}{q}\right| \geq 1 \geq 1/q^n\,.$$

If $\frac{p}{q} \in I \cap \mathbb{Q}$ then $\frac{p}{q} \neq \alpha$ because α is irrational. By Theorem 8.1.6 there is a real number x lying between α and $\frac{p}{q}$ such that (recall that $f(\alpha) = 0$)

$$f(\alpha) - f(\tfrac{p}{q}) = f'(x)(\alpha - \tfrac{p}{q}) \text{ and therefore } \left|\alpha - \tfrac{p}{q}\right| = |f(p/q)|\,/\,|f'(x)|\,.$$

Crucially, $f(\frac{p}{q}) \neq 0$: if $f(\frac{p}{q}) = 0$ then $g(x) = f(x)/(x - p/q)$ would be a rational polynomial with $g(\alpha) = 0$ (Exercise 8.4.2) and $\deg g = \deg f - 1$, in contradiction with the minimality of $\deg f$. Thus $f(\frac{p}{q}) \neq 0$ and, as we already argued in the previous section, we have $|f(\frac{p}{q})| \geq q^{-n}$. We take a real number $d > 0$ such that $|f'(y)| \leq d^{-1}$ for every $y \in I$ (Exercise 8.4.3). Then

$$\left|\alpha - \tfrac{p}{q}\right| \geq d/q^n \,.$$

We get Liouville's inequality with the constant $c = \min(1, d)$. $\square$

Exercise 8.4.2 *Why is α a root of $g(x)$?*

Exercise 8.4.3 *How do we get the constant d?*

• *Liouville's transcendental numbers.* The following recursive real transcendental number is constructed by Liouville's method.

Corollary 8.4.4 (λ is transcendental) *The real number*

$$\lambda \equiv \textstyle\sum_{n \geq 1} 10^{-n!} = 0.110001000000000000000001000\ldots$$

is recursive and transcendental.

Proof. It is clear that λ is recursive: the j-th decimal digit of λ after the decimal point is 0 or 1, and it is 1 iff $j = n!$ for some $n \in \mathbb{N}$. The number λ is irrational because it does not have eventually periodic decimal expansion (Exercise 8.4.5). For $m \in \mathbb{N}$ let $q_m = 10^{m!}$ and let $z_m \in \mathbb{N}$ be defined by

$$\textstyle\sum_{n=1}^{m} \frac{1}{10^{n!}} = \frac{z_m}{10^{m!}} \,.$$

Clearly, $q_m \geq 2$. Then, using a geometric series, we get the bound

$$|\lambda - \tfrac{z_m}{q_m}| = \textstyle\sum_{n=m+1}^{\infty} \frac{1}{10^{n!}} \leq \frac{1}{10^{(m+1)!}} \frac{1}{1 - 10^{-(m+1)!}} < 2q_m^{-m-1} \,.$$

Fractions $\frac{z_m}{q_m}$ violate Liouville's inequality for λ (Exercise 8.4.6) and λ is transcendental. $\square$

Exercise 8.4.5 *Show that every rational number has an eventually periodic decimal expansion.*

Exercise 8.4.6 *Explain why fractions $\frac{z_m}{q_m}$ violate for large mLiouville's inequality.*

Exercise 8.4.7 *What is the complexity of the natural algorithm $\mathcal{L}$ that computes the decimal expansion of λ?*

Exercise 8.4.8 *For every $k \in \mathbb{N}$, $k \geq 2$, the sum $\sum k^{-n!}$ is transcendental.*

8.5 Monotonicity and l'Hospital rules

We employ Theorem 8.1.6 to determine intervals of monotonicity of functions, and we prove l'Hospital's rules by which one can compute limits. First, we look at the interaction of finite and infinite derivatives.

• *Dependence of finite and infinite derivatives.* Equal signs of finite derivatives force the same sign for infinite derivatives. This application of Theorem 8.1.6 is not so well known. We assume that $a < b$ are real numbers.

Theorem 8.5.1 (same signs of derivatives) *Let $f \in \mathcal{C}([a,b])$. If $f'(c) \in \mathbb{R}^*$ for every $c \in (a,b)$ and every finite $f'(c)$ is nonnegative, then $f'(c) \geq 0$ for every $c \in (a,b)$.*

Proof. Suppose for the contrary that $f'(c) = -\infty$ for some $c \in (a,b)$. It follows that there exist numbers c_0 and c_1 such that $a < c_0 < c < c_1 < b$ and $f(c_0) > f(c) > f(c_1)$. By Theorem 8.1.6 there is a point $c_2 \in (c_0, c_1)$ such that

$$\tfrac{f(c_1)-f(c_0)}{c_1-c_0} = f'(c_2) \geq 0\,.$$

Thus $f(c_1) \geq f(c_0)$, which is a contradiction. □

Exercise 8.5.2 *State and prove the symmetric version of the theorem that forbids the derivative $f'(c) = +\infty$.*

Exercise 8.5.3 *Find $f \in \mathcal{F}([-1,1])$ such that $f'(c) = 1$ for every $c \in [-1,1] \setminus \{0\}$ and $f'(0) = -\infty$. Can f be continuous?*

• *Derivatives and intervals of monotonicity.* For any $M \subset \mathbb{R}$, the set

$$M^0 := \{a \in M \colon\ \exists\, \delta \colon U(a,\delta) \subset M\}$$

is the *interior* of M. The interior of an interval arises by deleting the endpoints. We introduce the following abbreviations.

Definition 8.5.4 (holding on) *Let $A \in \mathbb{R}^*$, $f \in \mathcal{F}(M)$ and X be any set. We say that*

$$\underline{f \geq A \text{ (holds) on } X}$$

if $f(b) \geq A$ for every $b \in M \cap X$. Similarly for the notation $f \leq A$, $f > A$, $f < A$, $f = a$ and $f \neq a$ (holding) on X.

Theorem 8.5.5 (global monotonicity) *Suppose that $I \subset \mathbb{R}$ is a nontrivial interval, $f \in \mathcal{C}(I)$ and $f'(c) \in \mathbb{R}^*$ for every $c \in I^0$. The following holds.*

1. *If $f' \geq 0$ on I^0 then f weakly increases on I.*
2. *If $f' \leq 0$ on I^0 then f weakly decreases on I.*

3. *If $f' > 0$ on I^0 <u>then</u> f increases on I.*

4. *If $f' < 0$ on I^0 <u>then</u> f decreases on I.*

Proof. We only prove the last claim and leave the rest to Exercise 8.5.6. Let $f' < 0$ on I^0 and let $x < y$ be in I. By Theorem 8.1.6 there is a point $z \in (x, y)$ (hence $z \in I^0$) such that

$$\tfrac{f(y)-f(x)}{y-x} = f'(z) < 0\,.$$

Thus $f(x) > f(y)$ and f decreases. □

Exercise 8.5.6 *Prove parts 1–3 of the theorem.*

Proposition 8.5.7 (local monotonicity) *Let $f \in \mathcal{F}(M)$ and $b \in M$.*

1. *If $f'_-(b) < 0$, there is a δ such that $f > f(b)$ on $P^-(b, \delta)$.*

2. *If $f'_-(b) > 0$, there is a δ such that $f < f(b)$ on $P^-(b, \delta)$.*

3. *If $f'_+(b) < 0$, there is a δ such that $f < f(b)$ on $P^+(b, \delta)$.*

4. *If $f'_+(b) > 0$, there is a δ such that $f > f(b)$ on $P^+(b, \delta)$.*

In each case, the set $P^\pm(b, \delta) \cap M$ is infinite.

Proof. We only prove the last claim and leave the rest to Exercise 8.5.8. Let $f'_+(b) > 0$. Then there is a δ such that for every $x \in P^+(b, \delta) \cap M$,

$$\tfrac{f(x)-f(b)}{x-b} > 0 \text{ and therefore } f(x) > f(b)\,.$$

The set $P^+(b, \delta) \cap M$ is infinite because b is a right limit point of M. □

Exercise 8.5.8 *Prove parts 1–3 of the proposition.*

Exercise 8.5.9 *Can we say that in each of the four cases in the proposition the set $f[P^\pm(b, \delta)]$ is infinite?*

• *Extensions of derivatives.* Let $a < b$ be real numbers. The next proposition has the same assumptions as Theorem 8.1.6 (Lagrange's mean value theorem).

Proposition 8.5.10 (extending derivatives) *Let $f \in \mathcal{C}([a, b])$ and suppose that $f'(c) \in \mathbb{R}^*$ for every $c \in (a, b)$. <u>Then</u>*

$$a,\, b \in L(D(f)),\ f'(a) = \lim_{x \to a} f'(x) \ \text{ and } \ f'(b) = \lim_{x \to b} f'(x)\,,$$

if the limits exist.

Proof. We show that $b \in L(D(f))$ and that if the limit $B = \lim_{x\to b} f'(x)$ exists, then $f'(b)$ exists and $f'(b) = B$. For the point a and the derivative $f'(a)$ we argue similarly.

By Theorem 8.1.6, $(c, b) \cap D(f) \neq \emptyset$ for every $c \in (a, b)$. Thus $b \in L(D(f))$. Let $\lim_{x\to b} f'(x) = B$ and an ε be given. Then $f'[(b-\delta, b)] \subset U(B, \varepsilon)$ for some δ. By Theorem 8.1.6, for every $x \in (b-\delta, b)$ there is a $y \in (x, b)$ such that

$$\tfrac{f(x)-f(b)}{x-b} = f'(y) \in U(B, \varepsilon)\,.$$

Hence $f'(b) = B$. □

Exercise 8.5.11 *Find a function $f \in \mathcal{C}([a, b])$ such that for every $c \in (a, b]$ the derivative $f'(c)$ exists but the limit $\lim_{x\to b} f'(x)$ does not exist.*

• *Two l'Hospital rules* (HR). We show a method to compute $\lim_{x\to A} \frac{f(x)}{g(x)}$ of the forms $\frac{0}{0}$ and $\frac{\pm\infty}{\pm\infty}$. It is based on the transformation

$$\tfrac{f(x)}{g(x)} = \tfrac{f(x)/x}{g(x)/x}\,.$$

We begin with a local result.

Theorem 8.5.12 (local HR) *Let $f, g \in \mathcal{R}$, $b \in L(M(f/g)) \cap M(f) \cap M(g)$ and let $f(b) = g(b) = 0$. Then*

$$\lim_{x\to b} \frac{f(x)}{g(x)} = \frac{f'(b)}{g'(b)} \quad (\in \mathbb{R}^*)$$

if the last ratio is defined.

Proof. We assume that $f'(b)$ and $g'(b)$ exist and that $\frac{f'(b)}{g'(b)}$ is not an indefinite expression. Then by Theorem 4.4.5 we have

$$\lim_{x\to b} \tfrac{f(x)}{g(x)} = \lim_{x\to b} \tfrac{\frac{f(x)-f(b)}{x-b}}{\frac{g(x)-g(b)}{x-b}} = \tfrac{\lim_{x\to b} \frac{f(x)-f(b)}{x-b}}{\lim_{x\to b} \frac{g(x)-g(b)}{x-b}} = \tfrac{f'(b)}{g'(b)}\,.$$ □

For example,

$$\lim_{x\to 0} \tfrac{\sin x}{x} = \tfrac{\cos 0}{k_1(0)} = \tfrac{1}{1} = 1 \text{ and } \lim_{x\to 0} \tfrac{\exp x - 1}{x} = \tfrac{\exp 0}{k_1(0)} = \tfrac{1}{1} = 1\,.$$

We continue with a global result. We assume that $a < b$ are real numbers.

Theorem 8.5.13 (global HR) *Let $f, f', g, g' \in \mathcal{F}((a, b))$ and let $g' \neq 0$ on (a, b). Suppose that*

$$\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0 \text{ or } \lim_{x\to a} g(x) = \pm\infty\,.$$

Then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)} \quad (\in \mathbb{R}^*)$$

if the last limit exists.

Proof. Note that $f, g \in \mathcal{C}$. 1. Let

$$\lim_{x \to a} f(x) = \lim_{x \to a} g(x) = 0 \text{ and } \lim_{x \to a} \tfrac{f'(x)}{g'(x)} = K\,.$$

Let an ε be given. We define $f(a) = g(a) = 0$, then $f, g \in \mathcal{C}([a, b))$. Theorem 8.1.1 implies that $g \neq 0$ on (a, b) (Exercise 8.5.14). There is a δ such that

$$(f'/g')[(a,\, a+\delta)] \subset U(K,\, \varepsilon)\,.$$

By Theorem 8.1.11, for every $x \in (a, a+\delta)$ there is a $y \in (a, x)$ such that

$$\tfrac{f(x)}{g(x)} = \tfrac{f(x)-f(a)}{g(x)-g(a)} = \tfrac{f'(y)}{g'(y)} \in U(K, \varepsilon)\,.$$

Hence $\lim_{x \to a} \frac{f(x)}{g(x)} = K$.

2. We prove this case later by means of integrals. □

Exercise 8.5.14 *Why can we assume that* $g \neq 0$ *on* (a, b)*?*

Exercise 8.5.15 *Let* $\varepsilon > 0$*. Compute* $\lim_{x \to 0} x^{\varepsilon} \log x$*.*

Marquis *Guillaume de l'Hospital (1661–1704)* published in 1696 the historically first textbook of differential calculus *Analyse des Infiniment Petits pour l'Intelligence des Lignes Courbes.*

Exercise 8.5.16 *Adapt the global* HR *for the limit at* b *and for the definition domains* $P(a, \delta)$ *and* $U(A, \delta) = U(\pm\infty, \delta)$*.*

• *A non-example.* We show that the assumption that $M(f) = M(g)$ is an interval is substantial. For any interval $I \subset \mathbb{R}$ we set $I_{\mathbb{Q}} := I \cap \mathbb{Q}$.

Theorem 8.5.17 (a non-example) *There exist functions*

$$f,\, g \in \mathcal{F}((0,\, 1)_{\mathbb{Q}})$$

such that $\lim_{x \to 0} f(x) = \lim_{x \to 0} g(x) = 0$*,* $D(f) = D(g) = (0, 1)_{\mathbb{Q}}$*,* $f' = g' = 1$ *on* $(0, 1)_{\mathbb{Q}}$*, hence* $\lim_{x \to 0} \frac{f'(x)}{g'(x)} = 1$*, but that*

$$\lim_{x \to 0} \frac{f(x)}{g(x)} = 0\,.$$

Proof. Let $(c_n) \subset (0, 1)$ be a sequence of irrational numbers such that

$$c_0 = 1 > c_1 > c_2 > \cdots > 0,\ \lim c_n = 0 \text{ and } \lim \tfrac{c_{n-1}}{c_n} = 1\,.$$

For $\alpha \in (c_n, c_{n-1})_{\mathbb{Q}}$, $n \in \mathbb{N}$, we set

$$f(\alpha) = c_n^2 + \alpha - c_n \text{ and } g(\alpha) = \alpha\,.$$

The graph G_f consists of short, straight "segments" starting on the parabola $y = x^2$ and each has slope 1. The other function is the restriction $g(x) = \mathrm{id}(x) \,|\, (0,1)_{\mathbb{Q}}$. It is clear that $f' = g' = k_1(x) \,|\, (0,1)_{\mathbb{Q}}$ (Exercise 8.5.18) and therefore

$$\lim_{x\to 0} \tfrac{f'(x)}{g'(x)} = \lim_{x\to 0} \tfrac{1}{1} = 1\,.$$

For $\alpha \in (c_n, c_{n-1})_{\mathbb{Q}}$ we have

$$0 < \tfrac{f(\alpha)}{g(\alpha)} \le \tfrac{c_n^2 + c_{n-1} - c_n}{c_n} = c_n + \tfrac{c_{n-1}}{c_n} - 1 \to 0 \ \ (n \to \infty)\,.$$

Thus $\lim_{x\to 0} \frac{f(x)}{g(x)} = 0$. $\square$

Exercise 8.5.18 *Why are the derivatives of f and g constantly 1?*

8.6 Second and higher order derivatives

We begin with the definition of higher-order derivatives.

• *Derivatives of order $k \in \mathbb{N}_0$.* We simply iterate the operation of derivative in Definition 7.1.23. Recall that $\mathcal{R}$ is the set of functions $f\colon M \to \mathbb{R}$ with $M \subset \mathbb{R}$.

Definition 8.6.1 ($f^{(k)}(x)$) *Let $k \in \mathbb{N}$. We define a unary operation $f^{(k)}$ on $\mathcal{R}$ by the iteration,*

$$\mathcal{R} \ni f \mapsto f^{(k)} \equiv (\ldots((f')')' \ldots)' \in \mathcal{R}\,,$$

with k applications of derivative. It is the derivative of order k. We set $f^{(0)} := f$ and write f' for $f^{(1)}$, f'' for $f^{(2)}$ and f''' for $f^{(3)}$.

For example,

$$(x \sin x)'' = (\sin x + x\cos x)' = 2\cos x - x \sin x\,.$$

Exercise 8.6.2 *What is the difference between $f''(b)$ and $(f')'(b)$?*

Exercise 8.6.3 *Determine the sequences of functions*

$$\big((\sin x)^{(n)}\big) \ \ \textit{and} \ \ \big((\tfrac{1}{x})^{(n)}\big),\ n \in \mathbb{N}_0\,.$$

• *Derivatives of derivatives and extremes.* We can read the types of local extremes from the signs of the elements $(f')'(b)$ $(\in \mathbb{R}^*)$, $b \in \mathbb{R}$.

Proposition 8.6.4 ($(f')'(b)$ and local extremes) *Let $f \in \mathcal{R}$. Suppose that*

$$U(b,\,\delta) \subset D(f),\ f'(b) = 0 \ \ \textit{and that}\ (f')'(b) \in \mathbb{R}^*\,.$$

If $(f')'(b) < 0$ then f has at b a strict local maximum. If $(f')'(b) > 0$ then f has at b a strict local minimum.

Proof. Let f, b and δ be as stated and let $(f')'(b) > 0$ (the case with $(f')'(b) < 0$ is similar). By parts 2 and 4 of Proposition 8.5.7 there is a $\theta < \delta$ such that for every $x \in P^-(b,\theta)$ (respectively $x \in P^+(b,\theta)$) we have $f'(x) < 0 = f'(b)$ (respectively $f'(x) > 0$). By Theorem 8.5.5 the function f decreases on $[b-\theta, b]$ and increases on $[b, b+\theta]$. Hence f has at b a strict local minimum. $\square$

Exercise 8.6.5 *Show that under the assumptions of the previous proposition and with $f''(b) = 0$, it is possible that f does not have at b a local extreme.*

Exercise 8.6.6 *Find examples for Proposition 8.6.4 where $(f')'(b) = \pm\infty$.*

• *Convexity and concavity.* Convex graphs of functions bulge downward, and concave graphs upward.

Definition 8.6.7 (convex and concave) *A function $f \in \mathcal{F}(M)$ is convex, respectively concave, if for any three points $a < b < a'$ in M we have*

$$f(b) \le sb + c, \text{ respectively } f(b) \ge sb + c\,,$$

where

$$y = sx + c \text{ is the secant } \kappa(a,\ f(a),\ a',\ f(a')) \text{ of } G_f\,.$$

If these inequalities hold as strict, we speak of strict convexity, respectively strict concavity, of f.

So, the strict convexity of f means that the middle point $(b, f(b)) \in G_f$ always lies below the secant line of G_f passing through the extreme points $(a, f(a))$ and $(a', f(a'))$. For convex f, the point $(b, f(b))$ may also lie on the secant. Similarly for concavity.

Exercise 8.6.8 *Function $f(x) = x^2$ is strictly convex. Function $f(x) = |x|$ is convex, but not strictly convex. Function $f(x) = \log x$ is strictly concave.*

Exercise 8.6.9 *(Strict) convexity and (strict) concavity are preserved under restrictions of functions.*

Exercise 8.6.10 *A function f is (strictly) convex $\iff$ $-f$ is (strictly) concave.*

• *Convexity, concavity and continuity.* Convexity and concavity force the existence of one-sided derivatives. We say that a set $M \subset \mathbb{R}$ is end-free if it has neither minimum nor maximum.

Theorem 8.6.11 (existence of $f'_\pm$) *Let $f \in \mathcal{F}(M)$ be a convex, respectively concave, function defined on a nonempty and end-free set $M \subset \mathbb{R}$. Then, with equal signs, for every $b \in M \cap L^\pm(M)$ there exists finite one-sided derivative*

$$f'_\pm(b) \in \mathbb{R}\,,$$

and the function

$$f'_{\pm}\colon M \cap L^{\pm}(M) \to \mathbb{R}$$

weakly increases, respectively weakly decreases.

Proof. We prove that any convex function f has at every point

$$b \in M \cap L^{-}(M)$$

finite left-sided derivative $f'_{-}(b) \in \mathbb{R}$, and then that the function f'_{-} weakly increases. The remaining three cases are treated similarly. By part 1 of Theorem 4.4.1, the limit

$$\lim_{x \to b^-} \tfrac{f(x)-f(b)}{x-b} = f'_{-}(b)$$

exists and is finite because for any $a \in M$ with $a < b$ the function

$$g(x) \equiv \tfrac{f(x)-f(b)}{x-b} \,\Big|\, [a,\, b)$$

weakly increases, and

$$g \le \tfrac{f(b')-f(b)}{b'-b}$$

on $[a, b)$ for any fixed $b' \in M$ with $b' > b$—such number b' exists because $\max(M)$ does not exist. These two properties of the function $g(x)$ easily follow from the convexity of f because

$$\tfrac{f(x)-f(b)}{x-b}$$

is the slope of the secant $\kappa(x, f(x), b, f(b))$ of G_f, and similarly for

$$\tfrac{f(b')-f(b)}{b'-b}\,.$$

We prove monotonicity of f'_{-}. For every two points $b < b'$ in $M \cap L^{-}(M)$, the inequality $f'_{-}(b) \le f'_{-}(b')$ follows again from the convexity of f. For every $x, y \in M$ with $x < b < y < b'$ we have two inequalities between slopes

$$\tfrac{f(x)-f(b)}{x-b} \le \tfrac{f(y)-f(b)}{y-b} \le \tfrac{f(y)-f(b')}{y-b'}\,,$$

hence

$$\tfrac{f(x)-f(b)}{x-b} \le \tfrac{f(y)-f(b')}{y-b'}\,.$$

This inequality is preserved in the limit transitions $x \to b^-$ and $y \to (b')^-$, and we get

$$f'_{-}(b) \le f'_{-}(b')\,.$$

□

But the two-sided derivative $f'(b)$ may not always exist for concave or convex function f because it may be that $f'_{-}(b) \ne f'_{+}(b)$. Consider, for example, the function $|x|$ at the point 0.

Corollary 8.6.12 (implied continuity) *Let $M \subset \mathbb{R}$ be a nonempty end-free set and $f \in \mathcal{F}(M)$ be convex or concave. Then $f \in \mathcal{C}(M)$.*

Proof. We show that f is right-continuous at every point $b \in M$. The left-continuity is proven similarly. It follows by Exercise 4.3.11 that f is continuous at b. If $b \in M$ but is not the right limit point of M, the function f is right-continuous at b trivially. If

$$b \in M \cap L^+(M)\,,$$

then by the previous theorem the one-sided derivative $f'_+(b) \in \mathbb{R}$ exists. Thus by Exercise 7.1.21, f is right-continuous at b. □

Exercise 8.6.13 *Prove the next proposition.*

Proposition 8.6.14 (derivatives at endpoints) *Let $M \subset \mathbb{R}$, $b = \max(M)$, $b \in L^-(M)$ and let $f \in \mathcal{F}(M)$ be convex or concave. Then there exist the one-sided and possibly infinite derivative*

$$f'_-(b) \in \mathbb{R}^*\,.$$

The same holds if we replace max with min and the sign $-$ with $+$.

Exercise 8.6.15 *Is it true that if $I \subset \mathbb{R}$ is a nontrivial interval and $f \in \mathcal{F}(I)$ is a convex or concave function, then f is continuous?*

• *Convexity, concavity and f''.* Convex and concave parts of the graph G_f can be determined by means of f''. Recall Definition 8.5.4.

Theorem 8.6.16 (convexity, concavity and $(f')'(b)$) *Let I be a nontrivial real interval, $f \in \mathcal{C}(I)$ and $D(f) \supset I^0$. We suppose that for every c in I^0 the derivative $(f')'(c) \in \mathbb{R}^*$ exists. The following holds.*

1. *If $f'' \ge 0$ on I^0 then f is convex.*
2. *If $f'' > 0$ on I^0 then f is strictly convex.*
3. *If $f'' \le 0$ on I^0 then f is concave.*
4. *If $f'' < 0$ on I^0 then f is strictly concave.*

In the proof we use the next lemma whose proof we leave as an exercise.

Exercise 8.6.17 *Prove the next lemma.*

Lemma 8.6.18 (on slopes) *Let (a,a'), (b,b') and (c,c') be in $\mathbb{R}^2$, $a < b < c$ and*

$$\tfrac{b'-a'}{b-a} \le \tfrac{c'-b'}{c-b}\,.$$

Then the point (b,b') lies below or on the line $\kappa(a,a',c,c')$. We get analogous results if the inequality $\le$ is replaced with any of the three inequalities $\{<, \ge, >\}$.

Proof of Theorem 8.6.16. Let f and I be as stated and let $f'' \ge 0$ on I^0, the remaining three cases are treated similarly. Let the three points $a < b < c$ be in I. By Theorem 8.1.6 there exist numbers $y \in (a, b)$ and $z \in (b, c)$ such that

$$s = \tfrac{f(b)-f(a)}{b-a} = f'(y) \text{ and } t = \tfrac{f(c)-f(b)}{c-b} = f'(z)\,.$$

By Theorem 8.5.5, f' weakly increases on I^0 because $f'' \ge 0$ on I^0. From $y < z$ it follows that $s = f'(y) \le f'(z) = t$. By Lemma 8.6.18 the point $(b, f(b))$ lies below or on the line $\kappa(a, f(a), c, f(c))$. So f is convex by Definition 8.6.7. $\square$

• *Inflection points.* At an inflection point the graph of a function crosses the tangent line.

Definition 8.6.19 (inflection points) *Let $f \in \mathcal{F}(M)$, $b \in M \cap L^{\mathrm{TS}}(M)$, $\ell(x)$ in $\mathcal{F}(\mathbb{R})$ be the tangent to G_f at $(b, f(b))$ and $z \in \{0, 1\}$. If there is a δ such that*

$$(-1)^z(f-\ell) \le 0 \text{ on } (b-\delta,\, b) \text{ and } (-1)^z(f-\ell) \ge 0 \text{ on } (b,\, b+\delta)\,,$$

we call $(b, f(b))$ an inflection point of G_f. If these inequalities hold as strict, we call $(b, f(b))$ a strict inflection point of G_f.

We see that the tangent line at an inflection point is a particular case of the cutting tangent of Section 8.1.

Exercise 8.6.20 *The origin $(0, 0)$ is a strict inflection point of the graph of the function $f(x) = x^3$.*

Exercise 8.6.21 *Which points of the graph of the constant function $k_1(x)$ are inflection points?*

Theorem 8.6.22 (no inflection) *Suppose that $f \in \mathcal{R}$, $D(f) \supset U(b, \delta)$, that the derivative $(f')'(b) \in \mathbb{R}^*$ exists and that $(f')'(b) \neq 0$. Then*

$$(b,\, f(b)) \text{ is not an inflection point of } G_f\,.$$

Proof. Let $(f')'(b) > 0$, the case with $(f')'(b) < 0$ is similar. Let ℓ $(\subset \mathbb{R}^2)$ be the tangent to G_f at $(b, f(b)$. By Proposition 8.5.7 there is a $\theta \le \delta$ such that for every $x \in (b-\theta, b)$ and every $x' \in (b, b+\theta)$ we have inequalities

$$f'(x) < f'(b) < f'(x')\,.$$

Let $x \in (b-\theta, b)$, $x' \in (b, b+\theta)$ and let s and t be the respective slopes of the secants

$$\kappa(x,\, f(x),\, b,\, f(b)) \text{ and } \kappa(b,\, f(b),\, x',\, f(x'))$$

of G_f. The inequalities and Theorem 8.1.6 give that

$$s < f'(b) < t\,.$$

Thus both points $(x, f(x))$ and $(x', f(x'))$ lie above ℓ. The condition in Definition 8.6.19 is not satisfied. $\square$

We obtain a sufficient condition for existence of inflection points.

Theorem 8.6.23 (inflection exists) *Let $f \in \mathcal{R}$, $M(f'') \supset U(b, \delta)$ and z be in $\{0, 1\}$. If*

$$(-1)^z f'' \geq 0 \text{ on } (b - \delta,\, b) \text{ and } (-1)^z f'' \leq 0 \text{ on } (b,\, b + \delta)$$

then

$$(b, f(b)) \text{ is an inflection point of } G_f\,.$$

If these inequalities hold strictly then $(b, f(b))$ is a strict inflection point of G_f.

Proof. Let f, b, δ and z be as stated. We assume that $f'' \leq 0$ on $(b - \delta, b)$ and $f'' \geq 0$ on $(b, b + \delta)$, the other three cases are similar. We have $M(f') \supset U(b, \delta)$ and denote the tangent line to G_f at $(b, f(b))$ by ℓ $(\subset \mathbb{R}^2)$. By Theorem 8.5.5 the derivative f' weakly decreases, respectively increases, on $[b - \delta, b]$, respectively $[b, b + \delta]$. Thus for every $x \in [b - \delta, b)$ and $x' \in (b, b + \delta]$,

$$f'(x) \geq f'(b) \leq f'(x')\,.$$

By Theorem 8.1.6,

$$\tfrac{f(b) - f(x)}{b - x} \geq f'(b) \leq \tfrac{f(x') - f(b)}{x' - b}\,.$$

So the slopes of the lines

$$\kappa(x,\, b,\, f(x),\, f(b)) \text{ and } \kappa(b,\, x',\, f(b),\, f(x'))$$

are at least the slope $f'(b)$ of ℓ. So the point $(x, f(x))$ lies below or on ℓ, and $(x', f(x'))$ above or on ℓ. Hence $(b, f(b))$ is an inflection point. $\square$

8.7 How to draw graphs of functions

We describe thirteen steps for determining main geometric features of the graph of a function. First we define asymptotes.

- *Asymptotes.* The graph gets arbitrarily close to these lines.

Definition 8.7.1 (vertical asymptotes) *If for $f \in \mathcal{F}(M)$ and $b \in L^-(M)$ we have*

$$\lim_{x \to b^-} f(x) = \pm\infty\,,$$

we call the line $x = b$ a left vertical asymptote of f. By replacing the two signs $^-$ by two signs $^+$ we obtain the right vertical asymptotes of f.

Exercise 8.7.2 *The axis y is both a left and right vertical asymptote of $\frac{1}{x}$. It is a right vertical asymptote of* $\log x$.

Definition 8.7.3 (asymptotes at infinity) *Let s, b be in $\mathbb{R}$, f in $\mathcal{F}(M)$ and let $\pm\infty \in L(M)$. If, with equal signs,*

$$\lim_{x\to\pm\infty}(f(x) - sx - b) = 0\,,$$

we call the line $\ell(x) = sx + b$ an asymptote (of f) at $\pm\infty$.

Exercise 8.7.4 *The line*

$$y = sx + b$$

is an asymptote of a function f at $\pm\infty$ $\iff$

$$\lim_{x\to\pm\infty}\frac{f(x)}{x} = s \ \textit{ and } \ \lim_{x\to\pm\infty}(f(x) - sx) = b$$

(equal signs).

Exercise 8.7.5 *Find the asymptote of $\frac{1}{x}$ at $+\infty$ and at $-\infty$.*

Definition 8.7.1 and Exercise 8.7.4 imply that asymptotes are unique.

• *Drawing graphs of functions.* We recall from Definition 5.2.19 that an elementary function is obtained from constant functions $\{k_c(x)\colon\ c \in \mathbb{R}\}$ and functions

$$\exp x,\ \log x,\ \sin x,\ \arcsin x \text{ and } x^b \text{ for } b \in (0, +\infty) \setminus \mathbb{N}\,,$$

by repeated addition, multiplication, division, and composition. In simple elementary functions we omit from the initial generators the functions $\arcsin x$ and x^b. Let $F \in \mathcal{R}$. We determine the main geometric features of G_F.

Step 0. *Is it (simple) elementary?* Is $F \in$ EF? Is $F \in$ SEF? Memberships of F in EF and SEF have bearing on $M(F)$, the continuity of F and $D(F)$.

Step 1. *The definition domain.* We find $M(F)$ ($\subset \mathbb{R}$). If $F \in$ EF, we start from the domains

$$M(\mathrm{e}^x) = M(\sin x) = M(k_c) = \mathbb{R}\,,$$

$M(x^b) = [0, +\infty)$, $M(\log x) = (0, +\infty)$ and $M(\arcsin x) = [-1, 1]$, and use the domain relations

$$M(f + g) = M(fg) = M(f) \cap M(g),\ \ M(f/g) = M(f) \cap M(g) \setminus Z(g)$$

and

$$M(f(g)) = \{x \in M(g)\colon\ g(x) \in M(f)\}\,.$$

Step 2. *Is it special?* Is F even ($F(-x) = F(x)$), odd ($F(-x) = -F(x)$) or c-periodic ($F(c + x) = F(x)$)?

Exercise 8.7.6 *Define these families of functions in more detail.*

Step 3. *Derivatives and continuity.* We determine for each point a in $M(F)$ if F is continuous at a, and if $F'(a) \in \mathbb{R}^*$ exists and is finite. By this we determine F'. If $F \in \mathrm{EF}$ then $F \in \mathcal{C}$ by Theorem 6.7.17. If $F \in \mathrm{SEF}$ then $D(F) = M(F)$ by Theorem 7.6.5.

Step 4. *Limits.* At any point $a \in M(F)$ where F is discontinuous we investigate one-sided limits. We also investigate limits of F at the elements of $L(M(F)) \setminus M(F)$. For example, $\lim_{x\to 0^{\pm}} \operatorname{sgn} x = \pm 1$. Or $\lim_{x\to -\infty} \exp x = 0$ and $\lim_{x\to +\infty} \exp x = +\infty$.

Step 5. *Intersections of the graph with coordinate axes.* These are the points $(x, 0)$ $(\in \mathbb{R}^2)$ where $x \in Z(F)$, plus the point $(0, F(0))$ if $0 \in M(f)$.

Step 6. *One-sided derivatives.* At any point $a \in M(F)$ where $F'(a)$ does not exist, we investigate $F'_-(a)$ and $F'_+(a)$. Proposition 8.5.10 is relevant. For example, it implies that $(|x|)'_-(0) = -1$ and $(|x|)'_+(0) = 1$. However, these one-sided derivatives of the absolute value are easily computed directly.

Step 7. *Maximal intervals of monotonicity and extremes.* We find all inclusion-wise maximal intervals I $(\subset \mathbb{R})$ on which (that is, on $I \cap M(F)$) the function F is monotone. For elementary functions, Theorem 8.5.5 can be often used. More ambitiously, one can try to determine all inclusion-wise maximal subsets of $M(F)$ where F is monotone, we call them

$$\underline{\text{maximal domains of monotonicity}}\,,$$

or MDM. For functions more general than elementary, this can be difficult. We consider MDM only in Exercise 8.7.8. We find local and global extremes of F. For this Corollary 6.5.6, Theorem 7.1.9, and Proposition 8.6.4 are relevant.

Step 8. *The image.* The image of F is the set $F[M(F)]$ $(\subset \mathbb{R})$.

Step 9. *Maximal intervals of convexity and concavity.* We find all inclusion-wise maximal intervals I $(\subset \mathbb{R})$ on which F is convex or concave. For elementary functions, Theorem 8.6.16 can usually be used. We will not consider the harder problem of finding all inclusion-wise maximal subsets of $M(f)$ where F is convex or concave.

Step 10. *Inflection points.* We find these points of G_F. Theorems 8.6.22 and 8.6.23 are relevant.

Step 11. *Asymptotes.* We find asymptotes of F. Definitions 8.7.1 and 8.7.3, and Exercise 8.7.4 are relevant.

Step 12. *Sketching the graph.* Using hand, the computer or the Internet we sketch the usually uncountable set

$$G_F = \{(a,\, F(a))\colon\ a \in M(F)\} \ \ (\subset \mathbb{R}^2)\,.$$

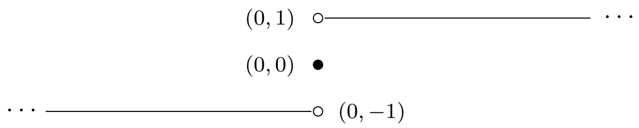
• *First example.* Let

$$F = F(x) \equiv \operatorname{sgn} x\,.$$

Recall that $F(x) = -1$ for $x < 0$, $F(x) = 1$ for $x > 0$ and $F(0) = 0$. **Step 0.** $F \notin$ EF by Proposition 5.2.21. **Step 1.** $M(F) = \mathbb{R}$. **Step 2.** The function F is odd. **Step 3.** F is continuous at every point $x \neq 0$ and $F'(x) = 0$ for every $x \neq 0$. At the point 0 the function F is discontinuous and $F'(0) = +\infty$. **Step 4.** We have the one-sided limits $\lim_{x\to 0^-} F(x) = -1$ and $\lim_{x\to 0^+} F(x) = 1$. Also, $\lim_{x\to -\infty} F(x) = -1$ and $\lim_{x\to +\infty} F(x) = 1$. **Step 5.** G_F intersects both coordinate axes only at the origin $(0,0)$. **Step 6.** Since $F'(x)$ exists for every $x \in \mathbb{R}$, there is nothing to compute: always

$$F'_-(x) = F'_+(x) = F'(x)\,.$$

Step 7. We see from the definition of F that F weakly increases on $\mathbb{R}$, that x is a global minimum, respectively maximum, of F iff $x < 0$, respectively $x > 0$, and that F has no strict local or global extreme. **Step 8.** The image of F is $\{-1, 0, 1\}$. **Step 9.** The maximal interval of convexity, respectively concavity, of F is the interval $(-\infty, 0]$, respectively $[0, +\infty)$. **Step 10.** The function F has no strict inflection point but it has inflection at every point $(x, F(x))$ with $x \neq 0$. At the origin $(0,0)$ the graph G_F does not have tangent. **Step 11.** F has no vertical asymptotes. The axis x is the asymptote of F at $-\infty$ and at $+\infty$. **Step 12.** A sketch of the graph G_F of the function $F(x) = \operatorname{sgn} x$ is



• *Second example.* Let

$$F = F(x) \equiv \tan x = \tfrac{\sin x}{\cos x} = \tfrac{\sin x}{\sin(x+\pi/2)}\,.$$

Step 0. The last ratio shows that $F \in$ SEF. **Step 1.**

$$\begin{aligned} M(F) &= M(\sin x) \cap M(\cos x) \setminus Z(\cos x) = \mathbb{R} \setminus Z(\cos x) \\ &= \mathbb{R} \setminus \left\{n\pi + \tfrac{\pi}{2}:\ n \in \mathbb{Z}\right\} = \textstyle\bigcup_{n\in\mathbb{Z}} \left(\pi n - \tfrac{\pi}{2},\, \pi n + \tfrac{\pi}{2}\right). \end{aligned}$$

Step 2. The function F is π-periodic because

$$\sin(\pi + x) = -\sin x \ \text{ and } \ \cos(\pi + x) = -\cos x\,.$$

It is an odd function because sine is odd and cosine is even. **Step 3.** By Theorems 6.7.17 and 7.6.5, the function F is continuous and $D(F) = M(F)$. By Exercise 7.5.7,

$$F'(x) = \tfrac{1}{\cos^2 x}\,.$$

Step 4. For $n \in \mathbb{Z}$ let $b_n \equiv \pi n + \frac{\pi}{2}$. Then

$$\lim_{x\to b_n^-} F(x) = +\infty \ \text{ and } \ \lim_{x\to b_n^+} F(x) = -\infty\,.$$

The limits of $F(x)$ at $\pm\infty$ do not exist. **Step 5.** The graph G_F intersects the axis y at the point $(0,0)$, and the axis x at the points

$$(b_n - \tfrac{\pi}{2},\, 0) = (\pi n,\, 0),\ n \in \mathbb{Z}\,,$$

Step 6. $D(F) = M(F)$ and there is nothing to compute. **Step 7.** Since

$$F'(x) = \tfrac{1}{\cos^2 x} > 0 \text{ on } M(F)\,,$$

the function F increases on every interval

$$(\pi n - \tfrac{\pi}{2},\, \pi n + \tfrac{\pi}{2}),\ n \in \mathbb{Z}\,.$$

These are the maximal intervals of monotonicity. F has no extremes. **Step 8.** Theorem 6.4.1 and the infinite limits in step 4 show that the image of F is $\mathbb{R}$. **Step 9.** By Corollaries 7.3.11 and 7.5.3, and Theorem 7.6.5 we have

$$F''(x) = \tfrac{2\sin x}{\cos^3 x} \text{ and } M(F'') = M(F') = M(F)\,.$$

Since

$$F'' < 0 \text{ on } (\pi n - \tfrac{\pi}{2}, \pi n) \text{ and } F'' > 0 \text{ on } (\pi n, \pi n + \tfrac{\pi}{2})\,,$$

the function F is strictly concave, respectively convex,

$$\text{on } (\pi n - \tfrac{\pi}{2}, \pi n], \text{ respectively } [\pi n, \pi n + \tfrac{\pi}{2}), \text{ for } n \in \mathbb{Z}\,.$$

These are the maximal intervals of convexity and concavity. **Step 10.** Due to the sign of F'' in the previous step, the inflection points of F are exactly $(b_n - \frac{\pi}{2}, 0) = (\pi n, 0)$, $n \in \mathbb{Z}$, and are strict. **Step 11.** The limits in step 4 show that every line

$$x = b_n = \pi n + \tfrac{\pi}{2},\ n \in \mathbb{Z}\,,$$

is both right and left vertical asymptote of F. There is no asymptote at $\pm\infty$. **Step 12.** Use, for example, the drawing calculator at `https://www.desmos.com/calculator`.

- *Third example.* We roughly follow the lecture notes [17, pp. 193–194]. Let

$$F = F(x) \equiv \arcsin\left(\tfrac{2x}{1+x^2}\right).$$

Step 0. The last expression shows that $F \in \mathrm{EF}$. We will see that $D(F) \neq M(F)$. Therefore, by Theorem 7.6.5, $F \notin \mathrm{SEF}$. Hence $F \in \mathrm{EF} \setminus \mathrm{SEF}$. **Step 1.** $M(F) = \mathbb{R}$ because $M(\arcsin x) = [-1, 1]$ and $2|x| \leq 1 + x^2$ for every $x \in \mathbb{R}$ because

$$x^2 \pm 2x + 1 = (x \pm 1)^2 \geq 0\,.$$

Step 2. The function F is odd because the functions $\sin x$, $\arcsin x$, and $\frac{2x}{1+x^2}$ are odd. It is not periodic. **Step 3.** By Theorem 6.7.17 the function F is continuous. Using part 1 of Exercise 7.5.8, part 2 of Theorem 7.4.1 (not

Corollary 7.4.3), Corollary 7.3.11, Exercise 7.1.25, Corollaries 7.3.4 and 7.3.8, and part 6 of Exercise 7.5.5 we get that

$$D(F) = \left\{x \in \mathbb{R}\colon\ \tfrac{2x}{1+x^2} \ne \pm 1\right\} = \mathbb{R} \setminus \{-1, 1\} = M(F) \setminus \{-1,\, 1\}$$

(in step 6 we show that the derivatives $F'(-1)$ and $F'(1)$ do not exist) and that

$$\begin{aligned} F'(x) &= \tfrac{1}{\sqrt{1-(2x/(1+x^2))^2}} \cdot \tfrac{2\cdot(1+x^2)-2x\cdot 2x}{(1+x^2)^2} = 2 \cdot \tfrac{(1-x^2)/(1+x^2)^2}{|(1-x^2)/(1+x^2)|} \\ &= 2 \cdot \tfrac{1-x^2}{|1-x^2|} \cdot \tfrac{1}{1+x^2} = \tfrac{2\cdot \mathrm{sgn}(1-x^2)}{1+x^2} \mid D(F)\,. \end{aligned}$$

Step 4. Clearly,

$$\lim_{x\to-\infty} F(x) = \lim_{x\to+\infty} F(x) = \arcsin 0 = 0$$

because $\frac{2x}{1+x^2} \to 0$ for $x \to \pm\infty$. **Step 5.** The graph G_F intersects both axes only at the origin $(0,0)$. **Step 6.** It is clear that $\lim_{x\to 1^\pm} F'(x) = \mp 1$. So Proposition 8.5.10 gives that $F'_\pm(1) = \mp 1$. Since $F(x)$ is odd, $F'_\pm(-1) = \pm 1$. By part 3 of Exercise 7.1.6, the derivatives $F'(-1)$ and $F'(1)$ do not exist. **Step 7.** Since

$$F' < 0 \text{ on } (-\infty, -1),\ F' > 0 \text{ on } (-1, 1) \text{ and } F' < 0 \text{ on } (1, +\infty)\,,$$

Theorem 8.5.5 implies that

$$F \text{ decreases on } (-\infty, -1], \text{ increases on } [-1, 1] \text{ and decreases on } [1, +\infty)\,.$$

These are the maximal intervals of monotonicity. Also $F(x) < 0$ for $x < 0$, $F(x) > 0$ for $x > 0$, and $F(0) = 0$. Considering these domains of monotonicity, the zero limits in step 4 and the fact that F is odd, we see that $F(-1) = -\frac{\pi}{2}$ is the strict global minimum, that $F(1) = \frac{\pi}{2}$ is the strict global maximum, and that F has no other (local or global) extreme. **Step 8.** Using Theorem 6.4.1 we get the image $F[M(F)] = [-\frac{\pi}{2}, \frac{\pi}{2}]$. **Step 9.** Using Exercise 7.1.25, part 1 (or part 2) of Theorem 7.3.9, Corollary 7.3.4 and part 6 of Exercise 7.5.5 we get that

$$F''(x) = \tfrac{-4x\cdot \mathrm{sgn}(1-x^2)}{(1+x^2)^2} \mid D(F)\,.$$

Since $F'' < 0$ on $(-\infty, -1)$, $F'' > 0$ on $(-1, 0)$, $F'' < 0$ on $(0, 1)$ and $F'' > 0$ on $(1, +\infty)$, Theorem 8.6.16 gives that F is strictly concave on $(-\infty, -1]$, strictly convex on $[-1, 0]$, strictly concave on $[0, 1]$ and strictly convex on $[1, +\infty)$. These are the maximal intervals of convexity and concavity. **Step 10.** Because of the sign of F'' and since at the points ± 1 the graph G_F does not have tangents, by Theorems 8.6.22 and 8.6.23 the point $(0, 0)$ is the only inflection point of G_F. **Step 11.** By the limits in step 4, the x-axis is an asymptote of F at $-\infty$ and at $+\infty$. The function F has no vertical asymptote. **Step 12.** Use, for example, the drawing calculator at `https://www.desmos.com/calculator`.

Exercise 8.7.7 *Draw in steps 0–12 the graph of the Riemann function*

$$F = F(x) \equiv r(x)\,.$$

Recall that $r(\alpha) = 0$ *for* $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ *and* $r(\frac{p}{q}) = \frac{1}{q}$ *if the fraction* $\frac{p}{q}$ *is in lowest terms.*

Exercise 8.7.8 *Find for* $r(x)$ *maximal domains of monotonicity.*

Exercise 8.7.9 *Draw in steps 0–12 the graph of the function*

$$F = F(x) \equiv x^x \quad (= \mathrm{e}^{x \log x})\,.$$

Chapter 9

Taylor polynomials

This chapter is based on lecture 9

`https://kam.mff.cuni.cz/~klazar/MAI24_pred9.pdf`

given on April 18, 2024. In Section 9.1 we define for a function $f(x)$ its Taylor polynomial $T_n^{f,b}(x)$ with order n and center b. In our approach, this polynomial has degree at most n and is determined by the condition that

$$f(x) = T_n^{f,b}(x) + o((x-b)^n) \ \ (x \to b)\,.$$

By Proposition 9.1.4 it is unique and by Theorem 9.1.9 if the definition domain of $f(x)$ and its derivatives is an interval $[b, c)$, then the coefficients of $T_n^{f,b}(x)$ are given by the standard formula. In Propositions 9.1.12 and 9.1.14 and Theorem 9.1.15 we demonstrate that in general the coefficient of the quadratic term in $T_n^{f,b}(x)$ is independent of the second derivative $f''(x)$.

Section 9.2 is devoted to standard Taylor polynomials of functions defined on intervals. In Propositions 9.2.1, 9.2.3, 9.2.4 and 9.2.5 we determine Taylor polynomials of functions e^x, $\cos x$, $\sin x$, $(1+x)^a$, $\log(1+x)$, $\log(\frac{1}{1-x})$, $\arctan x$, $\arcsin x$ and $\arccos x$. We show how to compute by Taylor polynomials limits of the type

$$\lim_{x\to 0} \frac{f(x)}{g(x)}$$

and in Theorem 9.3.16 and Proposition 9.3.25 we describe arithmetic of Taylor polynomials. Theorem 10.1.27 gives an example of a nonzero and $\mathcal{C}^\infty(\mathbb{R})$ function whose all derivatives at 0 are 0.

In Section 10.1 in Theorems 10.1.3 and 10.1.7, the simple and the advanced Taylor theorem, we give formulas for the Taylor remainder, which is the difference of the function and its Taylor polynomial. We define, in our approach, the Taylor series of a function. In Theorem 10.1.13 we derive Maclaurin series of the functions e^x, $\cos x$ and $\sin x$. Theorem 10.1.15 completely determines the domains where Newton's binomial series sums to the real power $(1+x)^a$.

The extending Section 10.3 is devoted to functions representable as sums of real power series. ...

9.1 Taylor polynomials

We consider local polynomial approximations of functions.

• *Linear and constant approximations.* Let $M \subset \mathbb{R}$. Recall Definition 4.7.11 of the asymptotic notation $o(\cdot)$ and recall that if a function $f(x) \in \mathcal{F}(M)$ is differentiable at a point b ($\in M \cap L(M)$), which means that the derivative $f'(b) \in \mathbb{R}$ exists, then $f(x)$ is approximated near b as

$$f(x) = f(b) + f'(b)(x-b) + o(x-b) \ \ (x \to b)\,.$$

The linear function

$$\ell(x) \equiv f(b) + f'(b)(x-b) = f'(b)x + f(b) - f'(b)b$$

is the tangent (line) to the graph G_f at the point $(b, f(b))$. This approximation is in fact equivalent to the differentiability.

Exercise 9.1.1 $f(x) = f(b) + c(x-b) + o(x-b)$ *(*$x \to b$*)* $\iff$ $f'(b) = c$.

In an even simpler situation a function $f(x) \in \mathcal{F}(M)$ continuous at b is approximated near the point b ($\in L(M) \cap M$) as

$$f(x) = f(b) + o(1) \ \ (x \to b)$$

by the constant function $k_{f(b)}(x)$.

Exercise 9.1.2 $f(x) = f(b) + o(1)$ *(*$x \to b$*)* $\iff$ f *is continuous at* b.

• *Approximation definition of Taylor polynomials.* We strengthen the above linear and constant approximations to polynomial approximations. By Definition 5.3.1, polynomials are the $\mathcal{F}(\mathbb{R})$ functions that arise from constant functions $k_c(x)$, $c \in \mathbb{R}$, and the identity function $\mathrm{id}(x)$ by repeated addition and multiplication. It is clear that for every $n \in \mathbb{N}_0$ and real numbers $a_0,\ a_1,\ \dots,\ a_n$ the function

$$\textstyle\sum_{j=0}^{n} a_j(x-b)^j \ \ \big(= \sum_{j=0}^{n} k_{a_j}(x) \cdot \prod_{i=1}^{j}(\mathrm{id}(x) + k_{-1}(x) \cdot k_b(x))\big)$$

is a polynomial. The next definition is inspired and motivated by so called Peano derivatives, see survey articles [31, 78] on them.

Definition 9.1.3 (Taylor polynomials) *Let* $n \in \mathbb{N}_0$*,* $M \subset \mathbb{R}$*,* $b \in M \cap L(M)$ *and let* $f(x) \in \mathcal{F}(M)$*. If* $a_0,\ a_1,\ \dots,\ a_n$ *are* $n+1$ *real numbers such that*

$$f(x) = \textstyle\sum_{j=0}^{n} a_j(x-b)^j + o((x-b)^n) \ \ (x \to b)\,,$$

we say that

$$\textstyle\sum_{j=0}^{n} a_j(x-b)^j \text{ is a } \underline{\text{Taylor polynomial}}$$

of the function $f(x)$ *with order* n *and center* b*.*

In the definition we assume that $b \in M(f)$ but we do not need the value $f(b)$.

Proposition 9.1.4 (uniqueness) *Taylor polynomials are unique.*

Proof. Let, for the contrary, $n \in \mathbb{N}_0$, $f \in \mathcal{F}(M)$, $b \in M \cap L(M)$, and $p(x) = \sum_{j=0}^{n} a_j(x-b)^j$ and $q(x) = \sum_{j=0}^{n} b_j(x-b)^j$ be two distinct polynomials such that

$$f(x) = p(x) + o((x-b)^n) \text{ and } f(x) = q(x) + o((x-b)^n) \quad (x \to b)\,.$$

Equivalently,

$$p(x) = f(x) + o((x-b)^n) \text{ and } q(x) = f(x) + o((x-b)^n) \quad (x \to b)\,.$$

Subtracting we get

$$\textstyle\sum_{j=m}^{n} c_j(x-b)^j = o((x-b)^n) \quad (x \to b)\,,$$

for some $m \in \mathbb{N}_0$ with $m \le n$ and $c_m \ne 0$. This is an impossible asymptotic equality. $\square$

The unique Taylor polynomial of a function f with order n and center b, if it exists, is denoted by

$$T_n^{f,\,b}(x) \quad (\in \mathrm{POL})\,.$$

Exercise 9.1.5 *Let $T_n^{f,b}(x) = \sum_{j=0}^{n} a_j(x-b)^j$. If $m \in \mathbb{N}_0$ with $m \le n$, then $T_m^{f,b}(x) = \sum_{j=0}^{m} a_j(x-b)^j$.*

Changing the variable we can move the center to 0.

Proposition 9.1.6 (moving to 0) *Let $n \in \mathbb{N}_0$, $M \subset \mathbb{R}$, $b \in M \cap L(M)$ and $f \in \mathcal{F}(M)$. Suppose that the Taylor polynomial*

$$T_n^{f,b}(x) = \textstyle\sum_{j=0}^{n} a_j(x-b)^j$$

exists. Then the composite function

$$g(x) \equiv f(x+b) = f \circ (\mathrm{id} + k_b)$$

has the Taylor polynomial

$$T_n^{g,\,0}(x) = \textstyle\sum_{j=0}^{n} a_j x^j\,.$$

Proof. Then $0 \in M(g) \cap L(M(g))$, because $M(g) = M - b$, and using Theorem 4.5.1 we get

$$\lim_{x \to 0} \tfrac{g(x) - \sum_{j=0}^{n} a_j x^j}{x^n} = \lim_{x \to 0} \tfrac{f(x+b) - \sum_{j=0}^{n} a_j (x+b-b)^j}{(x+b-b)^n} = \lim_{y \to b} \tfrac{f(y) - \sum_{j=0}^{n} a_j (y-b)^j}{(y-b)^n} = 0\,.$$

$\square$

• *Classical Taylor polynomials.* In the next theorem we obtain the classical formula for coefficients of Taylor polynomials of $f(x)$ in terms of values of derivatives $f^{(j)}(x)$. In the proof we use an auxiliary proposition.

Proposition 9.1.7 (Taylor polynomials of f and f') *Let $n \in \mathbb{N}$, $b < c$ be in $\mathbb{R}$ and let $f, f' \in \mathcal{F}([b,c))$.*

$$\textit{If } T_{n-1}^{f',b}(x) = \sum_{j=0}^{n-1} a_j (x-b)^j \ \underline{\textit{then}}\ T_n^{f,b}(x) = f(b) + \sum_{j=1}^{n} \frac{a_{j-1}}{j}(x-b)^j .$$

Proof. Let n, b, c and f be as stated. Suppose that the assumption of the implication holds. We denote the last displayed polynomial by $p(x)$. Then $p'(x) = T_{n-1}^{f',b}(x)$. By the first case of Theorem 8.5.13 and the assumption,

$$\lim_{x\to b} \tfrac{f(x)-p(x)}{(x-b)^n} = \lim_{x\to b} \tfrac{(f(x)-p(x))'}{((x-b)^n)'} = n^{-1} \lim_{x\to b} \tfrac{f'(x)-T_{n-1}^{f',b}(x)}{(x-b)^{n-1}} = 0 .$$

Hence $p(x) = T_n^{f,b}(x)$ by Proposition 9.1.4. $\square$

Exercise 9.1.8 *Why is in $p(x)$ the term $f(b)$? Does not the computation via* HR 2 *work also for the simpler polynomial $q(x) \equiv \sum_{j=1}^{n} \frac{1}{j} a_{j-1}(x-b)^j$? Then $q'(x) = T_{n-1}^{f',b}(x)$ too.*

For $n = 1$ we have

$$T_n^{f,b}(x) = f(b) + f'(b)(x-b)$$

whenever the Taylor polynomial exists. However, in the next passage we show that for $n \geq 2$ the coefficient of $(x-b)^2$ in $T_n^{f,b}(x)$ has in general nothing to do with $f''(b)$. In the following theorem we describe situation when $T_n^{f,b}(x)$ has the classical coefficients $\frac{1}{j!} f^{(j)}(b)$, $j = 0, 1, \dots, n$. We assume that $n \geq 2$ because the cases $n = 0, 1$ are resolved, for a general definition domain $M(f)$, by Exercises 9.1.2 and 9.1.1, respectively.

Theorem 9.1.9 (classical Taylor polynomials) *Let $n \in \mathbb{N}$ with $n \geq 2$, $b < c$ be in $\mathbb{R}$ and let f, f', $\dots$, $f^{(n)}$ be in $\mathcal{F}([b,c))$. $\underline{\textit{Then}}$ $T_n^{f,b}(x)$ exists and*

$$T_n^{f,b}(x) = \sum_{j=0}^{n} \frac{1}{j!} f^{(j)}(b) \cdot (x-b)^j .$$

Proof. Let n, b, c and f be as stated. We proceed by induction on $n \geq 2$. Let $n = 2$. By Exercise 9.1.1,

$$T_1^{f',b}(x) = f'(b) + f''(b)(x-b) .$$

Hence by Proposition 9.1.7,

$$\begin{aligned} T_2^{f,b}(x) &= f(b) + \tfrac{f'(b)}{1}(x-b) + \tfrac{f''(b)}{2}(x-b)^2 \\ &= \tfrac{f^{(0)}(b)}{0!}(x-b)^0 + \tfrac{f'(b)}{1!}(x-b) + \tfrac{f''(b)}{2!}(x-b)^2 . \end{aligned}$$

Let $n > 2$. By the inductive assumption,

$$T_{n-1}^{f',\,b}(x) = \textstyle\sum_{j=0}^{n-1} \frac{1}{j!} f^{(j+1)}(b)(x-b)^j\;.$$

Hence by Proposition 9.1.7,

$$T_n^{f,\,b}(x) = f(b) + \textstyle\sum_{j=1}^{n} \frac{1}{j} \cdot \frac{1}{(j-1)!} f^{(j)}(b)(x-b)^j = \sum_{j=0}^{n} \frac{1}{j!} f^{(j)}(b)(x-b)^j\;.$$

□

Exercise 9.1.10 *Show that the previous theorem and proposition hold for definition domains* $(c, b]$ *with* $c < b$ *and* $U(b, \delta)$*.*

Exercise 9.1.11 *Suppose that* $p(x) = \sum_{j=0}^{n} a_j x^j$ *is a polynomial and* $m \in \mathbb{N}_0$*. What is* $T_m^{p,0}(x)$*?*

• *Non-classical Taylor polynomials.* Definition 9.1.3 is non-standard, Taylor polynomials are usually defined by the formula in Theorem 9.1.9. We illustrate by several examples differences between our and standard definition.

Proposition 9.1.12 (no derivatives) *Let* $n \geq 2$ *and* b*,* c *and* f *be as in Theorem 9.1.9. Let* $M \subset [b, c)$ *be any set such that* $b \in M \cap L(M)$*, and let*

$$g \equiv f \,|\, M\;.$$

Then, no matter if for $j \geq 2$ *the derivatives* $g^{(j)}(b)$ *exist or not, we have*

$$T_n^{g,\,b}(x) = T_n^{f,\,b}(x) = \sum_{j=0}^{n} \frac{1}{j!} f^{(j)}(b)(x-b)^j\;.$$

Proof. By Theorem 9.1.9, Proposition 9.1.4 and Definition 9.1.3. □

Thus the Taylor polynomial $T_n^{g,b}(x)$ may exist even if none of the derivatives $g^{(j)}(b)$, $j \geq 2$, exists. However, always $g^{(0)}(b) = g(b) = f(b)$ and $g'(b) = f'(b)$.

Exercise 9.1.13 *We define*

$$f \in \mathcal{F}(\{\tfrac{1}{n}:\ n \in \mathbb{N}\} \cup \{0\}) \text{ by } f(0) \equiv 0 \text{ and } f(\tfrac{1}{n}) \equiv n^{-4}\;.$$

Find $T_n^{f,0}(x)$ *for every* $n \in \mathbb{N}$*. What are the derivatives* $f^{(j)}(0)$ *for* $j \in \mathbb{N}_0$*?*

Definition domains of functions g without derivatives $g^{(j)}(b)$, $j \geq 2$, in Proposition 9.1.12 are typically sparse. However, it is not hard to find a function f with the following properties.

1. Functions f and f' are defined on $\mathbb{R}$.
2. $T_2^{f,0}(x) = 0 + 0x + 0x^2$.

3. The derivative $(f')'(0)$ does not exist.

Proposition 9.1.14 (an example of such f) *The function f, defined by*

$$f(0) \equiv 0 \text{ and } f(x) \equiv x^3 \sin(x^{-3}) \text{ for } x \neq 0\,,$$

has properties 1–3.

Proof. Since $f'(x) = 3x^2 \sin(x^{-3}) - 3x^{-1}\cos(x^{-3})$ for $x \neq 0$, and $f'(0) = 0$ by the definition of derivative, f has property 1. The limit

$$\lim_{x \to 0} x^{-2}(f(x) - 0) = 0$$

shows that f has property 2. Since f' is unbounded on any neighborhood of zero, f has property 3. $\square$

Recall that for every real interval I we set

$$I_{\mathbb{Q}} \equiv I \cap \mathbb{Q}\,.$$

We modify the construction in Theorem 8.5.17 and get the next theorem. It shows that the value $f''(b)$ is unrelated to the coefficient of $(x-b)^2$ in our Taylor polynomials.

Theorem 9.1.15 (independence of a_2 and $f''(0)$) *For any $c \in \mathbb{R}$ there is a function f such that*

$$f,\, f',\, f'' \in \mathcal{F}([0,\, 1]_{\mathbb{Q}}),\; T_2^{f,\,0}(x) = \textstyle\sum_{j=0}^{2} 0x^j \text{ and } f''(0) = c\,.$$

Proof. Let

$$(c_n) \subset (0,\, 1)$$

be a sequence of irrational numbers such that

$$c_0 \equiv 1 > c_1 > c_2 > \cdots > 0,\ \lim c_n = 0 \text{ and } \lim \tfrac{c_{n-1}}{c_n} = 1\,.$$

We set $f(0) \equiv 0$ and for $\alpha \in (c_n, c_{n-1}]_{\mathbb{Q}}$, $n \in \mathbb{N}$, we define

$$f(\alpha) \equiv c_n^3 + \tfrac{c}{2} \cdot \alpha^2 - \tfrac{c}{2} \cdot c_n^2\,.$$

Thus the graph G_f consists of short pierced parabolic segments starting on the cubic $y = x^3$. Clearly, $f'(\alpha) = c\alpha$ on $(0, 1]_{\mathbb{Q}}$ and it is easy to see that also $f'(0) = 0$. Hence $f''(\alpha) = c$ on $[0, 1]_{\mathbb{Q}}$. For $\alpha \in (c_n, c_{n-1})_{\mathbb{Q}}$ we have

$$0 < \tfrac{f(\alpha)}{\alpha^2} \le \tfrac{c_n^3 + (c/2)c_{n-1}^2 - (c/2)c_n^2}{c_n^2} = c_n + \tfrac{c}{2}\big(\tfrac{c_{n-1}}{c_n}\big)^2 - \tfrac{c}{2} \to 0 \ \ (n \to \infty)\,.$$

Therefore $f(x) = o(x^2)$ $(x \to 0)$ and $T_2^{f,0}(x) = 0 + 0x + 0x^2$. $\square$

Exercise 9.1.16 *Extend the previous construction to the definition domain $[-1, 1]_{\mathbb{Q}}$ and the center $b = 0$.*

9.2 Examples of Taylor polynomials

We derive several classical formulas for Taylor polynomials. For simplicity of notation, we set the center to $b = 0$.

• *Taylor polynomials of the exponential, cosine, and sine.* Taylor polynomials are partial sums of power series defining these functions.

Proposition 9.2.1 (e^x, $\cos x$ and $\sin x$) *We have the following Taylor polynomials with order $n \in \mathbb{N}_0$ and center 0.*

1. *If $f(x) \equiv \mathrm{e}^x$ then*

$$T_n^{f,0}(x) = \sum_{j=0}^{n} \frac{1}{j!} \cdot x^j \,.$$

2. *If $f(x) \equiv \cos x$ then*

$$T_{2n}^{f,0}(x) = \sum_{j=0}^{n} (-1)^j \frac{1}{(2j)!} \cdot x^{2j} \,.$$

3. *If $f(x) \equiv \sin x$ then*

$$T_{2n+1}^{f,0}(x) = \sum_{j=0}^{n} (-1)^j \frac{1}{(2j+1)!} \cdot x^{2j+1} \,.$$

Proof. These formulas follow from Theorem 9.1.9, Corollary 7.5.3 and Exercise 8.6.3. □

Returning to the definitions in Section 5.1 we see that for each of the three functions $f(x) \in \{\mathrm{e}^x, \cos x, \sin x\}$ and every $a \in \mathbb{R}$ we have, remarkably,

$$f(a) = \lim_{n \to \infty} T_n^{f,0}(a) \,.$$

Exercise 9.2.2 *Show that the function $f(x) \equiv \frac{1}{1-x}$ does not have this property.*

• *More examples of Taylor polynomials.* For any $a \in \mathbb{R}$ and $j \in \mathbb{N}_0$ we define the generalized binomial coefficient as $\binom{a}{0} \equiv 1$, and for $j > 0$ as

$$\binom{a}{j} \equiv \frac{a(a-1)\dots(a-j+1)}{j!} \,.$$

Proposition 9.2.3 ($(1+x)^a$) *If $n \in \mathbb{N}_0$, $a \in \mathbb{R}$ and $f(x) \equiv (1+x)^a$, then*

$$T_n^{f,0}(x) = \sum_{j=0}^{n} \binom{a}{j} x^j \,.$$

Proof. This is immediate from Theorem 9.1.9 and derivatives ($j \in \mathbb{N}$)

$$\big((1+x)^a\big)^{(j)} = a(a-1)\dots(a-j+1)\cdot(1+x)^{a-j}\,,$$

which follow from parts 2–4 of Exercise 7.5.5. $\square$

The following logarithmic expansions are often used.

Proposition 9.2.4 ($\log(1+x)$ and $\log(\frac{1}{1-x})$) *Suppose that $n \in \mathbb{N}_0$. If $f(x) \equiv \log(1+x)$ and $g(x) \equiv \log(\frac{1}{1-x})$ <u>then</u>*

$$T_n^{f,0}(x) = \sum_{j=1}^{n}(-1)^{j-1}j^{-1}\cdot x^j \quad \textit{and} \quad T_n^{g,0}(x) = \sum_{j=1}^{n} j^{-1}\cdot x^j\,.$$

Proof. The first formula follows from Theorem 9.1.9 and derivatives ($j \in \mathbb{N}$)

$$\big(\log(1+x)\big)^{(j)} = (-1)^{j-1}\cdot(j-1)!\cdot(1+x)^{-j}\,|\,(-1,\,+\infty)\,.$$

The second formula follows from the first: $\log(\frac{1}{1-x}) = -\log(1+(-x))$. $\square$

Taylor polynomials of inverse trigonometric functions are derived with the help of Proposition 9.1.7.

Proposition 9.2.5 ($\arctan x$, $\arcsin x$ and $\arccos x$) *Let $n \in \mathbb{N}_0$.*

1. *If $f(x) \equiv \arctan x$ <u>then</u>*

$$T_{2n+1}^{f,0}(x) = \sum_{j=0}^{n}(-1)^j\frac{1}{2j+1}\cdot x^{2j+1}\,.$$

2. *If $f(x) \equiv \arcsin x$ <u>then</u>*

$$T_{2n+1}^{f,0}(x) = \sum_{j=0}^{n}\frac{1}{2j+1}(-1)^j\binom{-\frac{1}{2}}{j}\cdot x^{2j+1} = \sum_{j=0}^{n}\frac{1}{2j+1}\binom{j-\frac{1}{2}}{j}\cdot x^{2j+1}\,.$$

3. *If $f(x) \equiv \arccos x$ <u>then</u>*

$$T_{2n+1}^{f,0}(x) = \frac{\pi}{2} - \sum_{j=0}^{n}\frac{1}{2j+1}\binom{j-\frac{1}{2}}{j}\cdot x^{2j+1}\,.$$

Proof. 1. For $|x| < 1$ we have by part 3 of Exercise 7.5.8 and Theorem 3.5.47 that

$$(\arctan x)' = \tfrac{1}{1+x^2} = \textstyle\sum_{j=0}^{\infty}(-1)^j x^{2j}\,.$$

Exercise 9.2.6 gives that for every $n \in \mathbb{N}_0$,

$$\textstyle\sum_{j=0}^{\infty}(-1)^j x^{2j} = \sum_{j=0}^{n}(-1)^j x^{2j} + o(x^{2n}) \quad (x \to 0)\,.$$

Thus by Proposition 9.1.4 we have for every $n \in \mathbb{N}_0$ that

$$T_{2n}^{f',0}(x) = \textstyle\sum_{j=0}^{n}(-1)^j x^{2j}$$

and the result follows by Proposition 9.1.7.

2. By part 1 of Exercise 7.5.8 we have

$$(\arcsin x)' = \tfrac{1}{\sqrt{1-x^2}} = (1-x^2)^{-1/2}\,.$$

Thus Proposition 9.2.3 gives for every $n \in \mathbb{N}_0$ that

$$T_{2n}^{f',0}(x) = \textstyle\sum_{j=0}^{2n}(-1)^j \binom{-1/2}{j} x^{2j}\,.$$

We are done by Proposition 9.1.7 and Exercise 9.2.7.

3. The computation is similar to the previous one. □

Exercise 9.2.6 *Prove that for every $n \in \mathbb{N}_0$,*

$$\textstyle\sum_{j=0}^{\infty} x^j = \sum_{j=0}^{n} x^j + o(x_n) \quad (x \to 0)\,.$$

Exercise 9.2.7 *Prove that for every $j \in \mathbb{N}_0$ and $a \in \mathbb{R}$,*

$$(-1)^j \tbinom{-a}{j} = \tbinom{a+j-1}{j}\,.$$

• *Limits* $\lim_{x\to b} \frac{f(x)}{g(x)}$. We show how to compute them with the help of Taylor polynomials. By Proposition 9.1.6 we can restrict ourselves to the center $b = 0$. An important auxiliary result concerns the limit

$$\lim_{x\to 0} \frac{p(x)\,|\,M}{q(x)\,|\,M}$$

of the ratio of restricted polynomials. Restrictions to sets M results from our approach, we consider Taylor polynomials of functions with arbitrary definition domains.

Proposition 9.2.8 (ratios of restricted polynomials) *Let $n \in \mathbb{N}_0$, $M \subset \mathbb{R}$ with $0 \in L(M)$ and let*

$$L \equiv \lim_{x\to 0} \frac{\sum_{j=0}^{n} a_j x_j \,|\, M + o(x^n)}{\sum_{j=0}^{n} b_j x_j \,|\, M + o(x^n)} \quad (x \to 0)\,,$$

where we assume that not all b_j are zero. Let $m \in \mathbb{N}_0$ with $m \le n$ be the minimum index such that $a_m \neq 0$ or $b_m \neq 0$. If $b_m = 0$, let $l \in \mathbb{N}_0$ with $m < l \le n$ be the minimum index such that $b_l \neq 0$. The following holds.

1. *If $b_m \neq 0$ then L exists and*

$$L = \frac{a_m}{b_m}\,.$$

2. If $b_m = 0$, so that $a_m \neq 0$, and $0 \notin L^{\mp}(M)$, <u>then</u> L exists and, with equal signs,

$$L = (\pm 1)^{l-m} \cdot (\operatorname{sgn} a_m) \cdot (\operatorname{sgn} b_l) \cdot (+\infty)\,.$$

3. If $b_m = 0$ and $0 \in L^{\mathrm{TS}}(M)$, <u>then</u> L exists if $l - m$ is even and

$$L = (\operatorname{sgn} a_m) \cdot (\operatorname{sgn} b_l) \cdot (+\infty)\,.$$

If $l - m$ is odd <u>then</u> L does not exist.

Proof. 1. We define empty sums as 0 and have

$$L = \lim_{x \to 0} \tfrac{a_m + \sum_{j=m+1}^{n} a_j x^{j-m} + o(x^{n-m})}{b_m + \sum_{j=m+1}^{n} b_j x^{j-m} + o(x^{n-m})} = \tfrac{a_m}{b_m}\,.$$

2. Now

$$L = \lim_{x \to 0} \tfrac{a_m + o(1)}{b_l x^{l-m} + o(x^{l-m})} \mid M = \lim_{x \to 0^{\pm}} \tfrac{a_m + o(1)}{b_l x^{l-m} + o(x^{l-m})}$$

and this equals to the stated product of signs and $+\infty$.

3. If $l - m$ is even, for $x \to 0$ the factor x^{l-m} goes to 0 through positive values and L is the stated product of signs and $+\infty$. If $l - m$ is odd then the two limits $\lim_{x \to 0^{\pm}} \frac{\cdots}{\cdots}$ are two different infinities and L does not exist. □

We mention a simple case of the proposition for deleted neighborhoods of 0.

Corollary 9.2.9 ($M = P(0, \delta)$) *For $M = P(0, \delta)$ the following holds in the previous proposition.*

1. If $b_m \neq 0$ <u>then</u>

$$L = \frac{a_m}{b_m}\,.$$

2. If $b_m = 0$, so that $a_m \neq 0$, and $l - m$ is even <u>then</u>

$$L = (\operatorname{sgn} a_m) \cdot (\operatorname{sgn} b_l) \cdot (+\infty)\,.$$

3. If $b_m = 0$ and $l - m$ is odd <u>then</u> L does not exist.

Proof. Cases 1 and 3 of the proposition occur and case 2 does not occur. □

We illustrate each case of the corollary by an example. Using Taylor polynomials with order 3, we get by case 1 that

$$\lim_{x \to 0} \tfrac{\sin(2x) - 2\sin x}{\cos(2x) - \cos x} = \lim_{x \to 0} \tfrac{(2x - 8x^3/6) - 2(x - x^3/6) + o(x^3)}{(1 - 4x^2/2) - (1 - x^2/2) + o(x^3)} = \lim_{x \to 0} \tfrac{-x^3 + o(x^3)}{-3x^2/2 + o(x^3)}$$

equals $\frac{0}{-3/2} = 0$. Taylor polynomials with order 4 show by case 3 that

$$\lim_{x \to 0} \tfrac{\sin(2x) - 2\sin x}{\cos(2x) - \cos x + 3x^2/2} = \lim_{x \to 0} \tfrac{-x^3 + o(x^4)}{(1 - 4x^2/2 + 16x^4/24) - (1 - x^2/2 + x^4/24) + 3x^2/2 + o(x^4)}$$
$$= \lim_{x \to 0} \tfrac{-x^3 + o(x^4)}{5x^4/8 + o(x^4)}$$

does not exist. Finally, Taylor polynomials with order 5 show by case 2 that

$$\lim_{x\to 0}\frac{\sin(2x)-2\sin x}{\arctan x - x + x^3/3} = \lim_{x\to 0}\frac{-x^3+x^5/4+o(x^5)}{x^5/5+o(x^5)} = -\infty\,.$$

Exercise 9.2.10 *Compute*

$$\lim_{x\to 0}\frac{\sqrt{1+x}-1-x/2}{\log(1+x)-x}\,.$$

Exercise 9.2.11 *Compute*

$$\lim_{x\to 0}\frac{\sqrt{1+x}-\sqrt{1+3x}}{\sqrt[3]{1+x}-\sqrt[3]{1+2x}}\,.$$

Exercise 9.2.12 *Compute*

$$\lim_{x\to 0}\frac{\arcsin x - x}{\sin x - x}\,.$$

9.3 *Arithmetic of Taylor polynomials

In view of the usefulness of Taylor polynomials we investigate their interactions with the six operations on the set of functions $\mathcal{R}$, namely with addition, multiplication, division, composition, inverse and (global) derivative. For details of these operations see Definitions 5.2.1 and 7.1.23. Like for continuity, the most complicated and interesting operation turns out to be inverse. We devote to it the last Section 10.6. We begin with the arithmetic of the error terms $o((x-b)^n)$.

• *Arithmetic of errors.* We leave these properties of $o((x-b)^n)$ $(x \to b)$ as exercises. We assume that $n \in \mathbb{N}_0$, $b \in \mathbb{R}$ and $x \to b$.

Exercise 9.3.1 *Always* $o((x-b)^n) + o((x-b)^n) = o((x-b)^n)$.

Exercise 9.3.2 *Let* $m \in \mathbb{N}_0$ *and* $a \in \mathbb{R}$. *If* $m \geq n+1$ *then*

$$a(x-b)^m = o((x-b)^n)\,.$$

Exercise 9.3.3 *Let* $m \in \mathbb{N}_0$ *and* $a \in \mathbb{R}$. *Then*

$$a(x-b)^m \cdot o((x-b)^n) = o((x-b)^n)\,.$$

Exercise 9.3.4 *For* $m, n \in \mathbb{Z}$, $a \in \mathbb{R}$ *and* $x \to b$ *we actually have*

$$a(x-b)^m \cdot o((x-b)^n) = o((x-b)^{m+n})\,.$$

• *Addition and multiplication* of Taylor polynomials is easy.

Proposition 9.3.5 (addition, multiplication) *Let $n \in \mathbb{N}_0$, $f, g \in \mathcal{F}(M)$ and $b \in M \cap L(M)$. Suppose that the Taylor polynomials $T_n^{f,b}(x)$ and $T_n^{g,b}(x)$ exist. We write*

$$p(x) = T_n^{f,\,b}(x) = \textstyle\sum_{j=0}^n a_j(x-b)^j \quad \text{and} \quad q(x) = T_n^{g,\,b}(x) = \textstyle\sum_{j=0}^n b_j(x-b)^j\,.$$

Then the following holds.

1. *The Taylor polynomial $T_n^{f+g,b}(x)$ exists and*

$$T_n^{f+g,\,b}(x) = T_n^{f,\,b}(x) + T_n^{g,\,b}(x) \quad (\in \mathrm{POL})\,.$$

2. *The Taylor polynomial $T_n^{fg,b}(x)$ exists and*

$$T_n^{fg,\,b}(x) = \sum_{\substack{i,j\in\mathbb{N}_0 \\ i+j\le n}} a_i b_j (x-b)^{i+j} \equiv r(x) \quad (\in \mathrm{POL})\,.$$

Proof. Let n, f, g, b, $p(x)$, a_j, $q(x)$ and b_j be as stated, and let $x \to b$.

1. By the assumption and Exercise 9.3.1 we have

$$f(x)+g(x) = p(x)+o((x-b)^n)+q(x)+o((x-b)^n) = p(x)+q(x)+o((x-b)^n)\,.$$

Hence, by Proposition 9.1.4,

$$T_n^{f+g,\,b}(x) = T_n^{f,\,b}(x) + T_n^{g,\,b}(x)\,.$$

2. By Exercises 9.3.1 and 9.3.2, $p(x)q(x) = r(x)+o((x-b)^n)$. Hence by the assumption and Exercises 9.3.1 and 9.3.3,

$$\begin{aligned} f(x)g(x) &= (p(x)+o((x-b)^n))\cdot(q(x)+o((x-b)^n)) \\ &= p(x)q(x)+o((x-b)^n) = r(x)+o((x-b)^n)\,. \end{aligned}$$

Proposition 9.1.4 shows that $T_n^{fg,b}(x) = r(x)$. □

For example, if $f(x) \equiv \cos x$, $g(x) \equiv \sqrt{1+x}$, $b = 0$ and $n = 4$, then

$$\begin{aligned} T_n^{f+g,\,b}(x) &= (1+1)+(0+\tfrac{1}{2})x+(-\tfrac{1}{2}-\tfrac{1}{8})x^2+(0+\tfrac{1}{16})x^3+(\tfrac{1}{24}-\tfrac{5}{128})x^4 \\ &= 2+\tfrac{1}{2}x-\tfrac{5}{8}x^2+\tfrac{1}{16}x^3+\tfrac{1}{384}x^4 \end{aligned}$$

Exercise 9.3.6 *Determine the Taylor polynomial of* $\arctan x + \sin x$ *with order* 3 *and center* 0.

For example, if $f(x) \equiv \exp x$, $g(x) \equiv \log(\frac{1}{1-x})$, $b = 0$ and $n = 2$, then

$$T_n^{fg,\,b}(x) = 1\cdot 0+(1\cdot 1+1\cdot 0)x+(1\cdot\tfrac{1}{2}+1\cdot 1+\tfrac{1}{2}\cdot 0)x^2 = x+\tfrac{3}{2}x^2\,.$$

Exercise 9.3.7 *Determine the Taylor polynomial of* $\arctan x \cdot \sin x$ *with order* 3 *and center* 0.

• *The division* of Taylor polynomials is more complicated, but manageable.

Theorem 9.3.8 (division 1) *Let $n \in \mathbb{N}_0$, $f \in \mathcal{F}(M)$ and $b \in M \cap L(M)$. Suppose that the Taylor polynomial $T_n^{f,b}(x)$ exists,*

$$p(x) \equiv T_n^{f,\,b}(x) = \textstyle\sum_{j=0}^n a_j(x-b)^j$$

and that $a_0 = f(b) \neq 0$. <u>Then</u> for $n = 0$ we have

$$T_n^{1/f,\,b}(x) = T_0^{1/f,\,b}(x) = a_0^{-1}\,,$$

and for $n \geq 1$ we get

$$\begin{aligned} T_n^{1/f,\,b}(x) &= \frac{1}{a_0}\left(1 + \sum_{\substack{k\in[n],\, e_1,\, e_2,\, \dots,\, e_k\in\mathbb{N}\\ e\equiv e_1+e_2+\cdots+e_k\leq n}} (-1)^k a'_{e_1}a'_{e_2}\dots a'_{e_k}(x-b)^e\right)\\ &\equiv r(x), \;\; \textit{where } a'_j \equiv a_j/a_0\,. \end{aligned}$$

Proof. Let n, f, b, $p(x)$ and a_j be as stated, and let $x \to b$. We suppose that $a_0 \neq 0$ and $n = 0$. Using the identity $\frac{1}{1+x} = 1 - \frac{x}{1+x}$ we get that

$$\tfrac{1}{f(x)} = \tfrac{1}{a_0+o(1)} = a_0^{-1}\tfrac{1}{1+o(1)} = a_0^{-1}(1+o(1)) = a_0^{-1} + o(1)\,.$$

Thus $T_0^{1/f,b}(x) = a_0^{-1}$. For $n \geq 1$ we use the identity $\frac{1}{1+x} = \sum_{k=0}^n (-1)^k x^k + \frac{(-x)^{n+1}}{1+x}$. By Exercises 9.3.1–9.3.3 we have

$$\begin{aligned} \tfrac{1}{f(x)} &= \tfrac{1}{p(x)+o((x-b)^n)} = \tfrac{1}{a_0}\cdot\tfrac{1}{1+\sum_{j=1}^n a'_j(x-b)^j+o((x-b)^n)}\\ &\ \tfrac{1}{a_0}\cdot\Big(1+\textstyle\sum_{k=1}^n(-1)^k\big(\sum_{j=1}^n a'_j(x-b)^j + o((x-b)^n)\big)^k +\\ &\ +o((x-b)^n)\Big)\\ &= \tfrac{1}{a_0}\cdot\Big(1+\textstyle\sum_{k=1}^n(-1)^k\big(\sum_{j=1}^n a'_j(x-b)^j\big)^k\Big) + o((x-b)^n)\\ &= r(x) + o((x-b)^n)\,. \end{aligned}$$

Proposition 9.1.4 shows that $T_n^{1/f,b}(x) = r(x)$. □

For example, if $n = 3$, $f(x) \equiv 2 + \log(1+x)$ and $b = 0$, then $p(x) = 2 + x - \frac{1}{2}x^2 + \frac{1}{3}x^3$, $a'_1 = \frac{1}{2}$, $a'_2 = -\frac{1}{4}$, $a'_3 = \frac{1}{6}$ and

$$\begin{aligned} T_3^{1/f,\,0}(x) &= \tfrac{1}{2}\big(1 - \tfrac{1}{2}x + \tfrac{1}{4}x^2 - \tfrac{1}{6}x^3 + \big(\tfrac{1}{2}\big)^2x^2 - 2\big(\tfrac{1}{2}\big)\big(\tfrac{1}{4}\big)x^3 - \big(\tfrac{1}{2}\big)^3x^3\big)\\ &= \tfrac{1}{2} - \tfrac{1}{4}x + \tfrac{1}{4}x^2 - \tfrac{13}{48}x^3\,. \end{aligned}$$

Exercise 9.3.9 *Suppose that $T_3^{f,b}(x) = 5 + x + x^2 + x^3$. Find $T_3^{1/f,b}(x)$.*

• *Laurent Taylor polynomials.* In this passage we extend Theorem 9.3.8 to the situation when $a_0 = f(b) = 0$.

Definition 9.3.10 (Laurent polynomials) *A (real) Laurent polynomial with center b ($\in \mathbb{R}$) is a rational function of the form*

$$l(x) = \sum_{j=m}^{n} a_j (x-b)^j$$

where $m \le n$ are integers and $a_j \in \mathbb{R}$.

The adjective "Laurent" refers to the French mathematician *Pierre A. Laurent (1813–1854)* who introduced power series expansions with negative exponents.

Exercise 9.3.11 *What is $M(l(x))$?*

Definition 9.3.12 (Laurent Taylor polynomials) *Let $m, n \in \mathbb{Z}$ with $m \le n$, $M \subset \mathbb{R}$, $b \in L(M)$ and let $f(x) \in \mathcal{F}(M)$. If a_j for $j = m, m+1, \ldots, n$ are $n - m + 1$ real numbers such that*

$$f(x) = \textstyle\sum_{j=m}^{n} a_j (x-b)^j + o((x-b)^n) \quad (x \to b)\,,$$

we say that

$$T^{f,\,b}_{m,\,n}(x) \equiv \textstyle\sum_{j=m}^{n} a_j (x-b)^j \text{ is a } \underline{\text{Laurent Taylor polynomial}}$$

of the function $f(x)$ with orders m, n and center b.

Clearly, for $n \in \mathbb{N}_0$ always

$$T^{f,\,b}_{0,\,n}(x) = T^{f,\,b}_{n}(x)\,,$$

if either side is defined.

Exercise 9.3.13 *Show that the Laurent polynomial $T^{f,b}_{m,n}(x)$ is uniquely determined by the parameters m, n, f and b.*

We generalize Theorem 9.3.8 as follows.

Theorem 9.3.14 (division 2) *Let $m, n \in \mathbb{Z}$ with $m \le n$, $f \in \mathcal{F}(M)$ and let $b \in L(M)$. Suppose that the Laurent Taylor polynomial $T^{f,b}_{m,n}(x)$ exists,*

$$p(x) \equiv T^{f,\,b}_{m,\,n}(x) = \textstyle\sum_{j=m}^{n} a_j (x-b)^j$$

and that not all coefficients a_j are zero. Let $l \in \{m, m+1, \ldots, n\}$ be the minimum index such that $a_l \ne 0$. Then for $l = n$ we have

$$T^{1/f,\,b}_{-n,\,-n}(x) = a_l^{-1}(x-b)^{-l} = a_n^{-1}(x-b)^{-n}\,,$$

and for $l < n$ we get, with $N \equiv n - l$,

$$T^{1/f,\,b}_{-l,\,n-2l}(x) = \frac{(x-b)^{-l}}{a_l}\left(1 + \sum_{\substack{k\in[N],\, e_1,\, e_2,\, \ldots,\, e_k \in \mathbb{N} \\ e \equiv e_1 + e_2 + \cdots + e_k \le N}} (-1)^k a'_{e_1} a'_{e_2} \ldots a'_{e_k} (x-b)^e\right)$$

where $a'_j \equiv a_{l+j}/a_l$.

Proof. These formulas follow from Exercises 9.3.1–9.3.4, and from Theorem 9.3.8 and its proof by means of the factorization

$$p(x) = a_l(x-b)^l \cdot \textstyle\sum_{j=0}^{N} a'_j(x-b)^j = a_l(x-b)^l \cdot \big(1 + \sum_{j=1}^{N} a'_j(x-b)^j\big)\,.$$

□

For example, suppose that $m = -2$, $n = 3$, $b = 0$ and $f(x) \equiv \frac{1}{x} + \sin x$. We want to find $T^{1/f,0}_{?,?}(x)$. We have

$$T^{f,\,0}_{m,\,n}(x) = x^{-1} + x + \tfrac{1}{6}x^3 = x^{-1} \cdot \big(1 + x^2 + \tfrac{1}{6}x^4\big)\,.$$

Therefore $l = -1$, $N = 4$, $a'_1 = 0$, $a'_2 = 1$, $a'_3 = 0$, $a'_4 = \frac{1}{6}$ and

$$T^{1/f,\,0}_{1,\,5}(x) = x\big(1 - 1x^2 - \tfrac{1}{6}x^4 + 1^2x^4\big) = x - x^3 + \tfrac{5}{6}x^5$$

Exercise 9.3.15 *Find $T^{1/f,0}_{?,?}(x)$ if $m = 0$, $n = 3$ and $f(x) = \arctan x$.*

• *Composition* of Taylor polynomials is similar to division.

Theorem 9.3.16 (composition) *Let $n \in \mathbb{N}_0$, $f, g \in \mathcal{R}$, $b \in M(g) \cap L(M(g))$ and $g(b) \in M(f) \cap L(M(f))$. Suppose that the Taylor polynomials $T^{f,g(b)}_n(x)$ and $T^{g,\,b}_n(x)$ exist. We write*

$$p(x) \equiv T^{f,\,g(b)}_n(x) = \textstyle\sum_{j=0}^{n} a_j(x-g(b))^j,\ \ q(x) \equiv T^{g,\,b}_n(x) = \sum_{j=0}^{n} b_j(x-b)^j\,,$$

so that $b_0 = g(b)$. <u>Then</u> for $n = 0$ we have

$$T^{f\circ g,\,b}_n(x) = T^{f\circ g,\,b}_0(x) = a_0\,,$$

and for $n \geq 1$ we get

$$T^{f\circ g,\,b}_n(x) = a_0 + \sum_{\substack{k\in[n],\, e_1,\, e_2,\, \dots,\, e_k\in\mathbb{N}\\ e\equiv e_1+e_2+\cdots+e_k\leq n}} a_k b_{e_1} b_{e_2} \dots b_{e_k}(x-b)^e \equiv r(x)\,.$$

Proof. Let n, f, g, b, $p(x)$, a_j, $q(x)$ and b_j be as stated, and let $x \to b$. For $n = 0$ the formula is trivial. Let $n \geq 1$. Using Exercises 9.3.1–9.3.3 we get from

$$f(y) = p(y) + o((y-g(b))^n)\ \ (y \to g(b))$$

by substituting for y on the left-hand side the function $g(x)$ and on the right-hand side the expression $q(x) + o((x-b)^n)$ $(x \to b)$ that for $x \to b$ we indeed have

$$
\begin{aligned}
f(g(x)) &= p\big(q(x)+o((x-b)^n)\big)+o\big(\big(q(x)+o((x-b)^n)-g(b)\big)^n\big) \\
&= a_0+\textstyle\sum_{k=1}^n a_k\big(\sum_{j=0}^n b_j(x-b)^j+o((x-b)^n)-g(b)\big)^k+ \\
&\quad +o\big(\big(\textstyle\sum_{j=0}^n b_j(x-b)^j+o((x-b)^n)-g(b)\big)^n\big) \\
&= a_0+\textstyle\sum_{k=1}^n a_k\big(\sum_{j=1}^n b_j(x-b)^j+o((x-b)^n)\big)^k+ \\
&\quad +o\big(\big(\textstyle\sum_{j=1}^n b_j(x-b)^j+o((x-b)^n)\big)^n\big) \\
&= a_0+\textstyle\sum_{k=1}^n a_k\big(\sum_{j=1}^n b_j(x-b)^j\big)^k+o((x-b)^n) \\
&= r(x)+o((x-b)^n)\,.
\end{aligned}
$$

Proposition 9.1.4 shows that $T_n^{f\circ g,b}(x)=r(x)$. □

Exercise 9.3.17 *Why does the substitution transform*

$$o((y-g(b))^n)\quad (y\to g(b))$$

into $o((q(x)+o((x-b)^n)-g(b))^n)$ $(x\to b)$*?*

Exercise 9.3.18 *Why*

$$o\big(\big(\textstyle\sum_{j=1}^n b_j(x-b)^j+o((x-b)^n)\big)^n\big)=o((x-b)^n)\quad (x\to b)\,?$$

For example, if $n=3$, $b=0$ and $f(x)=g(x)\equiv\sin x$, then $g(b)=0$, $a_0=b_0=0$, $a_1=b_1=1$, $a_2=b_2=0$, $a_3=b_3=-\frac{1}{6}$ and

$$T_3^{f\circ f,\,0}(x)=a_0+a_1b_1x+a_1b_3x^3+a_3b_1^3x^3=x-\tfrac{1}{3}x^3\,.$$

In another example we determine in two way the Taylor polynomial

$$T_2^{\sin(\cos x),\,0}(x)\,.$$

Using Theorems 9.3.16 and 9.1.9 we have $n=2$, $f(x)=\sin x$, $g(x)=\cos x$, $b=0$, $g(b)=1$, $a_0=\sin 1$, $a_1=\cos 1$, $a_2=-\frac{1}{2}\sin 1$, $b_0=1$, $b_1=0$, $b_2=-\frac{1}{2}$ and

$$T_2^{\sin(\cos x),\,0}(x)=a_0+a_1b_2x^2+a_2b_1^2x^2=\sin 1-\tfrac{1}{2}(\cos 1)x^2\,.$$

Or, directly without composing, we set

$$F(x)\equiv\sin(\cos x)$$

and using only Theorem 9.1.9 and formulas for differentiation we get, of course, the same: $F'(x)=-\cos(\cos x)\cdot\sin x$ and $F''(x)=-\sin(\cos x))\cdot\sin x-\cos(\cos x)\cdot\cos x$, so that

$$T_2^{\sin(\cos x),\,0}(x)=F(0)+F'(0)x+\tfrac{1}{2}F''(0)x^2=\sin 1-\tfrac{1}{2}(\cos 1)x^2\,.$$

Exercise 9.3.19 *Let* $f(x) \equiv T_5^{\sin x, 0}(x)$ *and* $g(x) \equiv T_5^{\arcsin x, 0}(x)$. *Check by means of the formula in Theorem 9.3.16 that*

$$T_5^{f \circ g,\, 0}(x) = x + o(x^5)\,.$$

• *Global derivative.* Let $n \in \mathbb{N}$. One might think that if the Taylor polynomials $T_{n-1}^{f',\, b}(x)$ and $T_n^{f,\, b}(x)$ exist, then they satisfy relation

$$T_{n-1}^{f',\, b}(x) = \left(T_n^{f,\, b}\right)'\,.$$

However, we already gave examples showing that for general definition domains this does not hold.

Proposition 9.3.20 (general domains are bad) *For every* $c \in \mathbb{R}$ *there exists a function* f *such that* $f, f', f'' \in \mathcal{F}([0,1]_{\mathbb{Q}})$,

$$T_2^{f,\, 0}(x) = 0 + 0x + 0x^2$$

and $f'(x) = cx \mid [0,1]_{\mathbb{Q}}$. *Thus*

$$T_1^{f',\, 0}(x) = 0 + cx \neq 0 + 0x = \left(T_2^{f,\, 0}(x)\right)'\,.$$

Proof. We take the function f defined in the proof of Theorem 9.1.15. The first equality follows from Exercise 9.1.1:

$$T_1^{f',\, 0}(x) = f'(0) + (f')'(0)x = 0 + cx\,.$$

$\square$

In contrast with the four previous operations on $\mathcal{R}$, global derivative does not yield any relation between Taylor polynomials of f and f'. Recall, however, Proposition 9.1.7. It implies that on intervals we do have

$$\left(T_n^{f,b}(x)\right)' = T_{n-1}^{f',b}(x)\,.$$

Corollary 9.3.21 (intervals are good) *Let* $n \in \mathbb{N}$, $b < c$ *be in* $\mathbb{R}$ *and let* $f, f' \in \mathcal{F}([b,c))$. *Suppose that* $T_{n-1}^{f',b}(x)$ *exists.* *<u>Then</u> also* $T_n^{f,b}(x)$ *exists and*

$$\left(T_n^{f,\, b}\right)' = T_{n-1}^{f',\, b}(x)\,.$$

Proof. This is immediate from Proposition 9.1.7. $\square$

Exercise 9.3.22 *Extend this result to definition domains* $(c, b]$ *with* $c < b$ *and* $U(b, \delta)$.

• *Computing Taylor polynomials by algebra.* We give more convenient algebraic versions of formulas in Proposition 9.3.5 and Theorems 9.3.8 and 9.3.16. We work in the polynomial ring

$$\mathrm{POL}_{\mathrm{id}} = \langle \mathrm{POL},\, k_0(x),\, k_1(x),\, +,\, \cdot \rangle$$

which is an integral domain.

Definition 9.3.23 (mod x^k) *For $p(x), q(x) \in \mathrm{POL}$ and $k \in \mathbb{N}_0$ we write*

$$p(x) = q(x) \pmod{x^k}$$

and say that the polynomials $p(x)$ and $q(x)$ are <u>*equal modulo*</u> *x^k, if in the canonical form of the polynomial $p(x) - q(x)$ the coefficients of x^0, x^1, …, x^{k-1} are zero.*

In other words, (canonical forms of) $p(x)$ and $q(x)$ have identical coefficients of x^j for $j < k$. We also write

$$p(x) \bmod x^k \quad (\in \mathrm{POL})$$

to denote the unique polynomial $q(x)$ such that $\deg q(x) < k$ and $p(x) = q(x)$ $\pmod{x^k}$.

Exercise 9.3.24 *Prove the next proposition.*

Proposition 9.3.25 (computing mod x^k) *Let $k \in \mathbb{N}_0$ and let $p(x)$, $p_0(x)$, $q(x)$ and $q_0(x)$ be polynomials such that*

$$p(x) = q(x) \pmod{x^k} \;\; \textit{and} \;\; p_0(x) = q_0(x) \pmod{x^k}.$$

<u>*Then*</u>

$$p(x) + p_0(x) = q(x) + q_0(x) \pmod{x^k} \;\; \textit{and} \;\; p(x)p_0(x) = q(x)q_0(x) \pmod{x^k}.$$

The algebraic versions of Proposition 9.3.5 and Theorems 9.3.8 and 9.3.16 are as follows; the trivial addition is omitted. For simplicity of notation, we consider only the center $b = 0$. For brevity we do not consider extensions to Laurent Taylor polynomials.

Proposition 9.3.26 (multiplication mod x^{n+1}) *Let $p(x), q(x) \in \mathrm{POL}$ and $n \in \mathbb{N}_0$ be such that $n \ge \deg p(x), \deg q(x)$.* <u>*Then*</u>

$$(p(x) + o(x^n)) \cdot (q(x) + o(x^n)) = \big(p(x) \cdot q(x) \bmod x^{n+1}\big) + o(x^n) \quad (x \to 0).$$

Proof. This follows from Exercises 9.3.1–9.3.3. □

Exercise 9.3.27 *Explain in detail what the proposition and the next two theorems exactly say. What is the precise meaning of the $o(\cdot)$ terms?*

Theorem 9.3.28 (division mod x^{n+1}) *Let $p(x) \in \mathrm{POL}$ with $a_0 \equiv p(0) \neq 0$ and let $n \in \mathbb{N}_0$ be such that $n \geq \deg p(x)$. Then*

$$\frac{1}{p(x)+o(x^n)} = \Big(\frac{1}{a_0}\sum_{k=0}^{n}\big(1-a_0^{-1}p(x)\big)^k \bmod x^{n+1}\Big) + o(x^n) \quad (x \to 0)\,.$$

Proof. Using the identity $\frac{1}{1+x} = \sum_{k=0}^{n}(-1)^k x^k + \frac{(-x)^{n+1}}{1+x}$ and Exercises 9.3.1–9.3.3 we get that

$$\begin{aligned} \tfrac{1}{p(x)+o(x^n)} &= \tfrac{1}{a_0}\Big(\sum_{k=0}^{n}\big(1-a_0^{-1}p(x)+o(x^n)\big)^k + \tfrac{(1-a_0^{-1}p(x))^{n+1}}{1+x}\Big) \\ &= \Big(\tfrac{1}{a_0}\sum_{k=0}^{n}\big(1-a_0^{-1}p(x)\big)^k \bmod x^{n+1}\Big) + o(x^n) \quad (x \to 0)\,. \end{aligned}$$

□

Theorem 9.3.29 (composition mod x^{n+1}) *Let $p(x)$ and $q(x)$ be in POL and let $n \in \mathbb{N}_0$ be such that $n \geq \deg p(x), \deg q(x)$. We write $p(x) = \sum_{j=0}^{n} a_j x^j$. Then*

$$\begin{aligned} &(p(y)+o((y-q(0))^n)) \circ (q(x)+o(x^n)) \quad (y \to q(0),\, x \to 0) \\ &= \Big(\sum_{k=0}^{n} a_k \cdot q(x)^k \bmod x^{n+1}\Big) + o(x^n) \quad (x \to 0)\,. \end{aligned}$$

Proof. Using Exercises 9.3.1–9.3.3 we get that $(y \to g(0),\, x \to 0)$

$$\begin{aligned} &(p(y)+o((y-g(0))^n)) \circ (q(x)+o(x^n)) \\ &= \textstyle\sum_{k=0}^{n} a_k\big(q(x)+o(x^n)\big)^k + o\big(\big(q(x)-q(0)+o(x^n)\big)^n\big) \\ &= \big(\textstyle\sum_{k=0}^{n} a_k \cdot q(x)^k \bmod x^{n+1}\big) + o(x^n) \quad (x \to 0)\,. \end{aligned}$$

□

• *Changing the center of a polynomial.* This is another technique for obtaining new Taylor polynomials from old ones.

Proposition 9.3.30 (changing the center) *Let $b \in \mathbb{R}$, $n \in \mathbb{N}$ and let a_j for $j = 0, 1, \ldots, n$ be real numbers. Then we have equality of polynomials*

$$\sum_{j=0}^{n} a_j x^j = \sum_{i=0}^{n} b_i (x-b)^i \quad \text{where} \quad b_i = \sum_{k=0}^{n-i} a_{k+i}\binom{k+i}{i} b^k$$

Proof. We obtain this identity by expanding $x^j = ((x-b)+b)^j$ via binomial theorem (Exercise 2.1.28) and then renaming the difference $j-i$ by k. □

For example,

$$1 + x + x^2 = 1 + ((x-1)+1) + ((x-1)+1)^2 = 3 + 3(x-1) + (x-1)^2\,.$$

Exercise 9.3.31 *Change the center* 0 *of* $1+x+x^2+x^3$ *to* -1.

• *Examples.* We first determine the Taylor polynomial of $\frac{1}{\cos x}$ with order 6 and center 0. We set $p(x) \equiv -\frac{1}{2}x^2 + \frac{1}{24}x^4 - \frac{1}{720}x^6$ and using Theorem 9.3.28 we get

$$\begin{aligned}
&\tfrac{1}{\cos x} = \tfrac{1}{1+p(x)+o(x^6)} = \left(\textstyle\sum_{k=0}^{3}(-p(x))^k \bmod x^7\right) + o(x^6) = \\
&= 1 - (-\tfrac{1}{2}x^2 + \tfrac{1}{24}x^4 - \tfrac{1}{720}x^6) + (-\tfrac{1}{2}x^2)^2 + 2(-\tfrac{1}{2}x^2)\tfrac{1}{24}x^4 - \\
&- (-\tfrac{1}{2}x^2)^3 + o(x^6) = 1 + \tfrac{1}{2}x^2 + \tfrac{5}{24}x^4 + \tfrac{61}{720}x^6 + o(x^6)\,.
\end{aligned}$$

We restricted the exponent to $k \le 3$ because $p(x)^k = 0 \pmod{x^7}$ if $k \ge 4$.

Next we determine the Taylor polynomial of $\sqrt{1+\sin x}$ with order 5 and center 0. We set $q(x) \equiv x - \frac{1}{6}x^3 + \frac{1}{120}x^5$ and using Theorem 9.3.29 we get

$$\begin{aligned}
&\sqrt{1+\sin x} = \left(1 + \textstyle\sum_{j=1}^{5}\binom{1/2}{j}\cdot q(x)^j \bmod x^6\right) + o(x^5) = \\
&= 1 + \tfrac{1}{2}q(x) - \tfrac{1}{8}(x^2 - 2x\tfrac{1}{6}x^3) + \tfrac{1}{16}(x^3 - 3x^2\tfrac{1}{6}x^3) - \\
&- \tfrac{15}{16\cdot 24}x^4 + \tfrac{105}{32\cdot 120}x^5 + o(x^5) = 1 + \tfrac{1}{2}x - \tfrac{1}{8}x^2 - \tfrac{1}{48}x^3 + \tfrac{16-15}{16\cdot 24}x^4 + \\
&+ \tfrac{16-120+105}{32\cdot 120}x^5 + o(x^5) = \\
&= 1 + \tfrac{1}{2}x - \tfrac{1}{8}x^2 - \tfrac{1}{48}x^3 + \tfrac{1}{384}x^4 + \tfrac{1}{3840}x^5 + o(x^5)\,.
\end{aligned}$$

Exercise 9.3.32 *Why are these coefficients so simple? Please, work in the spirit of Exercise 2.5.2. (I admit that I took help from the Internet.)*

Finally, we determine the Taylor polynomial of the function $\tan x$ with order 6 and center 0. Using Proposition 9.3.26 and the above Taylor polynomial of $\frac{1}{\cos x}$ we get that

$$\begin{aligned}
&\tan x = \tfrac{\sin x}{\cos x} = \sin x \cdot \tfrac{1}{\cos x} = \\
&\left((x - \tfrac{1}{6}x^3 + \tfrac{1}{120}x^5)\cdot(1 + \tfrac{1}{2}x^2 + \tfrac{5}{24}x^4 + \tfrac{61}{720}x^6) \bmod x^7\right) + o(x^6) = \\
&= x + \tfrac{1}{3}x^3 + \tfrac{1-10+25}{120}x^5 + o(x^6) = x + \tfrac{1}{3}x^3 + \tfrac{2}{15}x^5 + 0x^6 + o(x^6)\,.
\end{aligned}$$

There is a formula for these coefficients in terms of the Bernoulli numbers.

Exercise 9.3.33 *Show that every coefficient of* x^{2n} *in Taylor polynomials of* $\tan x$ *is zero.*

Exercise 9.3.34 *Find the Taylor polynomial with order* 5 *and center* 0 *of the function*

$$\sqrt[3]{1+\sin x}\,.$$

• *Algebraic complexity of computations of Taylor polynomials.* Since this book is based on lectures for students in the School of Computer Science of MFF UK (Faculty of Mathematics and Physics of Charles University in Prague), we briefly discuss algorithmic aspects of computing Taylor polynomials. We do not consider the problem to compute the Taylor polynomial of a given function, but the problem to compute the Taylor polynomial of the sum, product, … of two functions if their Taylor polynomials are known. We consider just the *arithmetic complexity* of a computation, which is the number of arithmetic operations it does. The formulas in Theorems 9.3.8 and 9.3.16 are explicit but do not provide efficient algorithms. Therefore we turn to formulas in Theorems 9.3.28 and 9.3.29. For simplicity of notation we treat only the case $b = 0$, but everything could be easily extended to general center $b \in \mathbb{R}$.

Theorem 9.3.35 (computing Taylor polynomials) *Let $n \in \mathbb{N}$, $f, g \in \mathcal{R}$ and let the Taylor polynomials*

$$T_n^{f,\,0}(x) = \sum_{j=0}^{n} a_j x^j \ \text{ and } \ T_n^{g,\,0}(x) = \sum_{j=0}^{n} b_j x^j$$

exist and be given. The following holds.

1. *The coefficients of $T_n^{f+g,0}(x)$ can be computed from $a_0, \ldots, a_n, b_0, \ldots, b_n$ in $n+1$ arithmetic operations.*
2. *The coefficients of $T_n^{fg,0}(x)$ can be computed from $a_0, \ldots, a_n, b_0, \ldots, b_n$ in $O(n^3)$ ($n \in \mathbb{N}$) arithmetic operations.*
3. *For $a_0 \neq 0$ the coefficients of $T_n^{1/f,0}(x)$ can be computed from $a_0, \ldots, a_n$ in $O(n^5)$ ($n \in \mathbb{N}$) arithmetic operations.*
4. *Let in addition $g(0) = 0$. The coefficients of $T_n^{f\circ g,b}(x)$ can be computed from $a_0, \ldots, a_n, b_0, \ldots, b_n$ in $O(n^5)$ ($n \in \mathbb{N}$) arithmetic operations.*

Proof. Parts 1 and 2 are left for Exercise 9.3.36. 3. Let $a_0 \neq 0$ and

$$T_n^{1/f,\,0}(x) = \textstyle\sum_{j=0}^{n} c_j x^j\,.$$

We compute $c_0, \ldots, c_n$ from $a_0, \ldots, a_n$ by means of the formula in Theorem 9.3.28. Let $k \in \mathbb{N}$, $k \leq n$, be fixed. Using the algorithm of part 2 we compute the k-fold product of polynomials in the summand

$$\big(1 - a_0^{-1} \textstyle\sum_{j=0}^{n} a_j x^j\big)^k \bmod x^{n+1}$$

in $O(kn^3)$ ($n \in \mathbb{N}$) arithmetic operations. Then using the algorithm of part 1 we compute the product of a_0^{-1} with the sum of $n+1$ polynomials (with degrees at most n) modulo x^{n+1} in Theorem 9.3.28 in

$$n + 2 + n(n+1) + \textstyle\sum_{k=1}^{n} O(kn^3) = O(n^5)$$

arithmetic operations.

4. Let $g(0) = 0$ and

$$T_n^{f\circ g,0}(x) = \textstyle\sum_{j=0}^{n} c_j x^j\,.$$

We compute $c_0, \ldots, c_n$ from $a_0, \ldots, a_n$, $b_0, \ldots, b_n$ by means of the formula in Theorem 9.3.29. Using the algorithm of part 2 we compute the product of a_k with the k-fold product of polynomials in the summand

$$a_k \cdot \left(\textstyle\sum_{j=0}^{n} b_j x^j\right)^k \bmod x^{n+1}$$

in $n+1+O(kn^3) = O(kn^3)$ $(n \in \mathbb{N})$ arithmetic operations. Then using the algorithm of part 1 we compute the sum of $n+1$ polynomials (with degrees at most n) modulo x^{n+1} in Theorem 9.3.29 in

$$n(n+1) + \textstyle\sum_{k=1}^{n} O(kn^3) = O(n^5)$$

arithmetic operations. □

Exercise 9.3.36 *Prove parts 1 and 2 of the theorem.*

See Chapter 2 of the monograph [15] for more efficient polynomial arithmetic.

Chapter 10

Real analytic functions

10.1 Taylor series

In this section we derive explicit formulas for Taylor remainder, which is the difference of a function and its Taylor polynomial. We define Taylor series and determine sums of Taylor series for several elementary functions.

• *Taylor remainders.* We already computed with them a lot, but only in the $o((x-b)^n)$ form.

Definition 10.1.1 (Taylor remainder) *Suppose that a function $f \in \mathcal{R}$ has the Taylor polynomial $T_n^{f,b}(x)$, so that $b \in M(f) \cap L(M(f))$. Then the function*

$$R_n^{f,\,b}(x) \equiv f(x) - T_n^{f,\,b}(x) \quad (\in \mathcal{F}(M(f)))$$

is the Taylor remainder of f with order n and center b.

Exercise 10.1.2 *We have*

$$R_n^{f,\,b}(x) = o((x-b)^n) \quad (x \to b)\,.$$

This asymptotics of the Taylor remainder is ineffective, it provides no explicit bound on it. We improve upon it by obtaining explicit formulas for $R_n^{f,b}(x)$.

• *Simple Taylor theorem.* First we give a simple form of the Taylor theorem. Then we explain why a more complicated but stronger form is needed, and state and prove this form.

Theorem 10.1.3 (simple Taylor theorem) *Suppose that $n \in \mathbb{N}_0$, $I \subset \mathbb{R}$ is a nonempty open interval, $p > 0$ is a real number and that f, f', …, $f^{(n+1)}$ are in $\mathcal{F}(I)$. Then for every two distinct points $b, x \in I$ there is a point c between them such that*

$$\left|R_n^{f,\,b}(x)\right| = \frac{|f^{(n+1)}(c)|}{n! \cdot p} \cdot |x-c|^{n+1-p} \cdot |x-b|^p\,.$$

Proof. Let n, I, p and f be as stated. We take distinct points $b, x \in I$ and first assume that $b < x$. We define auxiliary functions

$$\varphi(t) \equiv (x-t)^p \mid [b,\, x]$$

and (with $0^0 \equiv 1$)

$$F(t) \equiv f(x) - \textstyle\sum_{i=0}^{n} \frac{1}{i!} f^{(i)}(t) \cdot (x-t)^i \mid [b,\, x]\,.$$

We have $\varphi, F \in \mathcal{C}([b,x])$, $\varphi(x) = 0$, $\varphi(b) = (x-b)^p$, $F(x) = 0$ and $F(b) = R_n^{f,b}(x)$. Also,

$$\begin{aligned} F'(t) &= -f'(t) - \textstyle\sum_{i=1}^{n} \left(\frac{1}{i!} f^{(i+1)}(t) \cdot (x-t)^i - \frac{1}{(i-1)!} f^{(i)}(t) \cdot (x-t)^{i-1} \right) \\ &= -\frac{1}{n!} f^{(n+1)}(t) \cdot (x-t)^n\,. \end{aligned}$$

By Cauchy's Theorem 8.1.11 there exists a point $c \in (b, x)$ such that

$$\frac{R_n^{f,\,b}(x)}{(x-b)^p} = \frac{F(x)-F(b)}{\varphi(x)-\varphi(b)} = \frac{F'(c)}{\varphi'(c)} = \frac{f^{(n+1)}(c)\cdot(x-c)^n}{n!\cdot p(x-c)^{p-1}}\,.$$

Solving this equation for $|R_n^{f,b}(x)|$ we get the stated formula for the (absolute value of) Taylor remainder.

Let $x < b$. We argue as before, with the modified auxiliary function

$$\varphi(t) \equiv (t-x)^p \mid [x,\, b]$$

and the same $F(t)$ (restricted to $[x, b]$). Now we get

$$\frac{R_n^{f,\,b}(x)}{(b-x)^p} = \frac{F(x)-F(b)}{\varphi(x)-\varphi(b)} = \frac{F'(c)}{\varphi'(c)} = -\frac{f^{(n+1)}(c)\cdot(x-c)^n}{n!\cdot p(c-x)^{p-1}}\,.$$

Solving this equation for $|R_n^{f,b}(x)|$ we get again the stated formula. □

The formula for $|R_n^{f,b}(x)|$ in the previous and the next theorem is called the Schlömilch remainder. It bears the name of the German mathematician *Oskar X. Schlömilch (1823–1901).* To get a simple formula, we sacrificed the sign of $R_n^{f,b}(x)$ (we do not need it for determining if $R_n^{f,b}(x) \to 0$ as $n \to \infty$). With some effort it can be recovered.

Exercise 10.1.4 *State the formula in the previous theorem in a more precise form as* $R_n^{f,b}(x) = \cdots$.

Below, for complete treatment of the binomial series we need the Schlömilch remainder with non-integral values of the exponent p. For $p \in \mathbb{N}$ we get simple well known formulas including the sign of $R_n^{f,b}(x)$.

Corollary 10.1.5 (Lagrange and Cauchy remainders 1) *Let* $n \in \mathbb{N}_0$, $I \subset \mathbb{R}$ *be a nonempty open interval and let* $f,\, f',\, \dots,\, f^{(n+1)}$ *be in* $\mathcal{F}(I)$. *Then the following holds.*

1. *For every two distinct points $b, x \in I$ there is a point c between them such that we have the* <u>*Lagrange remainder*</u>

$$R_n^{f,\,b}(x) = \frac{f^{(n+1)}(c)}{(n+1)!} \cdot (x-b)^{n+1}\,.$$

2. *For every two distinct points $b, x \in I$ there is a point c between them such that we have the* <u>*Cauchy remainder*</u>

$$R_n^{f,\,b}(x) = \frac{f^{(n+1)}(c)}{n!} \cdot (x-c)^n (x-b)\,.$$

Proof. These remainders follow from the previous proof if we set $p = n+1$ and $p = 1$, respectively. Then we can take (the restriction of) $\varphi(t) \equiv (x-t)^p$ in both cases $x < b$ and $b < x$ and we get the stated formulas. $\Box$

Theorem 10.1.3 has the drawback that in some cases, when we need it, it does not apply. For example, we would like to set in the Maclaurin series

$$(1+x)^a = \textstyle\sum_{n=0}^{\infty} \binom{a}{n} x^n$$

the variable x to -1 and deduce for every real $a > 0$ the summation

$$\textstyle\sum_{n=0}^{\infty} \binom{a}{n} (-1)^n = 0\,,$$

but Theorem 10.1.3 does not give it because finite derivatives

$$\big((1+x)^a\big)^{(n)}(-1)$$

do not exist for large n.

Exercise 10.1.6 *Prove the above summation in the case when $a \in \mathbb{N}$.*

• *Advanced Taylor theorem.* We fix this problem with the following theorem.

Theorem 10.1.7 (advanced Taylor theorem) *Let $n \in \mathbb{N}_0$, let $b \neq x$ and $p > 0$ be real numbers, and let I be the closed interval with endpoints b and x. Suppose that $f \in \mathcal{C}(I)$, that $M(f^{(i)}) \supset I \setminus \{x\}$ for $i = 1, 2, \ldots, n+1$ and that for every $i = 1, 2, \ldots, n$ we have*

$$\lim_{t \to x} f^{(i)}(t) \cdot (t-x)^i = 0\,.$$

<u>*Then*</u> *there is a point $c \in I^0$ such that*

$$\big|R_n^{f,\,b}(x)\big| = \frac{|f^{(n+1)}(c)|}{n! \cdot p} \cdot |x-c|^{n+1-p} \cdot |x-b|^p\,.$$

Proof. Let n, b, x, p and f be as stated. We argue as in the proof of Theorem 10.1.3 and use the same functions $\varphi(t)$ and $F(t)$ in $\mathcal{F}(I)$, with a small modification. The function $\varphi(t)$ is as in that proof. The function $F(t)$ is defined on $I \setminus \{x\}$ by the formula in that proof and we set $F(x) \equiv 0$. It follows from the assumptions that then $F \in \mathcal{C}(I)$. Else, we argue as in that proof. $\square$

The advanced Taylor theorem, or rather the proof of it, again provides us Lagrange and Cauchy remainders.

Exercise 10.1.8 *Prove the next corollary.*

Corollary 10.1.9 (Lagrange and Cauchy remainders 2) *Let $n \in \mathbb{N}_0$, let $b \neq x$ be real numbers, and let I be the closed interval with endpoints b and x. Suppose that $f \in \mathcal{C}(I)$, that $M(f^{(i)}) \supset I \setminus \{x\}$ for $i = 1, 2, \ldots, n+1$ and that for every $i = 1, 2, \ldots, n$ we have*

$$\lim_{t \to x} f^{(i)}(t) \cdot (t-x)^i = 0\,.$$

Then the following holds.

1. *There is a point $c \in I^0$ such that we have the Lagrange remainder*

$$R_n^{f,\,b}(x) = \frac{f^{(n+1)}(c)}{(n+1)!} \cdot (x-b)^{n+1}\,.$$

2. *There is a point $c \in I^0$ such that we have the Cauchy remainder*

$$R_n^{f,\,b}(x) = \frac{f^{(n+1)}(c)}{n!} \cdot (x-c)^n (x-b)\,.$$

• *Taylor series.* We define the Taylor series of a function as the union of its Taylor polynomials of all orders. By Exercise 9.1.5 such definition is correct — the coefficient a_j does not depend on n.

Definition 10.1.10 (Taylor series) *Let $f \in \mathcal{F}(M)$ and $b \in M \cap L(M)$. Suppose that for every $n \in \mathbb{N}_0$ there exists the Taylor polynomial*

$$T_n^{f,\,b}(x) = \sum_{j=0}^{n} a_j (x-b)^j\,.$$

Then we call the series ($x \in \mathbb{R}$)

$$T^{f,\,b}(x) \equiv \sum_{n=0}^{\infty} a_n (x-b)^n$$

the Taylor series of $f(x)$ with center b. In the case when $b = 0$ we often speak of Maclaurin series of $f(x)$.

Taylor polynomials and Taylor series are named after the English mathematician *Brook Taylor (1685–1731)*. Maclaurin series refer to the Scottish mathematician *Colin Maclaurin (1698–1746)*. We shall investigate for which $x \in \mathbb{R}$ we have $T^{f,b}(x) = f(x)$. Thus we introduce the set

$$\begin{aligned} E(T^{f,b}(x)) &\equiv \{a \in \mathbb{R}\colon \text{ the series } T^{f,b}(a) \text{ converges and } T^{f,b}(a) = f(a)\} \\ &= \{a \in \mathbb{R}\colon \lim_{n\to\infty} R_n^{f,\,b}(a) = 0\}\,. \end{aligned}$$

For example, always $b \in E(T^{f,b}(x))$.

Exercise 10.1.11 *Why? Why do we have in the formula for the Taylor series that* $(b-b)^0 = 0^0 = 1$*?*

We remind the standard notation that ($M \subset \mathbb{R}$)

$$\mathcal{C}^\infty(M) \equiv \{f \in \mathcal{R}\colon\ f^{(j)} \in \mathcal{F}(M) \text{ for every } j \in \mathbb{N}_0\}\,.$$

If $M \not\subset L(M)$ then $\mathcal{C}^\infty(M) = \emptyset$. Theorem 9.1.9 has an immediate corollary.

Corollary 10.1.12 (classical Taylor series) *Let $f \in \mathcal{C}^\infty(M)$ where $M \subset \mathbb{R}$ is a nonempty open set. Then for every $b \in M$ the function f has the Taylor series with center b and*

$$T^{f,\,b}(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(b)}{n!} \cdot (x-b)^n\,.$$

• *Taylor series of* $\exp x$*,* $\cos x$ *and* $\sin x$*.* We determine sums of Taylor series of several elementary functions and begin with $\exp x$, $\cos x$ and $\sin x$. These functions are in $\mathcal{C}^\infty(\mathbb{R})$ and their Taylor series are therefore given for every center $b \in \mathbb{R}$ by the formula in Corollary 10.1.12.

Theorem 10.1.13 (e^x, $\cos x$ and $\sin x$) *Let $f(x) \in \{\exp x, \cos x, \sin x\}$. Then*

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(b)}{n!} \cdot (x-b)^n$$

for every $b, x \in \mathbb{R}$.

Proof. Let $n \in \mathbb{N}$, $b, x \in \mathbb{R}$ with $b < x$ and $f(x) = \exp x$. Using Lagrange's remainder in Theorem 10.1.3 we get

$$R_n^{f,\,b}(x) = (n+1)!^{-1} \cdot \mathrm{e}^c \cdot (x-b)^{n+1} \ \text{ for some } \ c = c(n) \in (b,\,x)\,.$$

Thus for fixed b and x we have $\lim_{n\to\infty} R_n^{f,b}(x) = 0$ and the sum of $T^{f,b}(x)$ equals $f(x) = \exp x$. For $x < b$ and/or $f(x)$ equal to $\cos x$ or $\sin x$ we proceed in a similar way. □

Note that the definitoric series of $\exp x$, $\cos x$ and $\sin x$ in Section 5.1 are their Taylor series with center 0:

$$\exp(a) = T^{\exp x,\,0}(a) = \textstyle\sum_{n=0}^{\infty} \frac{a^n}{n!} \ \text{ for every } \ a \in \mathbb{R}\,,$$

and similarly

$$\cos(a) = T^{\cos x,\,0}(a) = \textstyle\sum_{n=0}^{\infty} \frac{(-1)^n a^{2n}}{(2n)!},\ \sin(a) = T^{\sin x,\,0}(a) = \textstyle\sum_{n=0}^{\infty} \frac{(-1)^n a^{2n+1}}{(2n+1)!}$$

for every $a \in \mathbb{R}$.

Exercise 10.1.14 *All series in the theorem are abscon.*

• *Binomial series.* The next Maclaurin series of the real power $(1+x)^a$ with $x, a \in \mathbb{R}$ (see Definitions 5.1.10 and 5.1.14) is called the binomial series. It is due to *Isaac Newton (1642/43–1727)* who was one of the main creators of modern science ([81]).

Theorem 10.1.15 (binomial series) *Let $a \in \mathbb{R}$ and $f(x) \equiv (1+x)^a$. Then*

$$T^{f,\,0}(x) = \sum_{n=0}^{\infty} \binom{a}{n} \cdot x^n\,.$$

For $x \in \mathbb{R}$ we have

$$T^{f,\,0}(x) = (1+x)^a \iff x \in E(a) \equiv E(T^{f,0}(x))$$

and the set $E(a)$ is as follows.

1. *If $a \in \mathbb{N}_0$, then $E(a) = \mathbb{R}$ and the binomial series is a finite sum. If $a \notin \mathbb{N}_0$, then the binomial series is an infinite series.*
2. *If $a > 0$ but $a \notin \mathbb{N}$, then $E(a) = [-1, 1]$.*
3. *If $a \in (-1, 0)$, then $E(a) = (-1, 1]$.*
4. *If $a \le -1$, then $E(a) = (-1, 1)$.*

Proof. The form of $T^{f,0}(x)$ follows from Proposition 9.2.3. Part 1 is easy, parts 2–4 are harder. If $a \in \mathbb{N}_0$ then $\binom{a}{n} = 0$ for $n > a$ and the series is a finite sum. The equality $T^{f,0}(x) = (1+x)^a$ for every $x \in \mathbb{R}$ then follows from the binomial theorem (Exercise 2.1.28). If $a \notin \mathbb{N}_0$ then $\binom{a}{n} \ne 0$ for every $n \in \mathbb{N}_0$.

In the rest of the proof we assume that $a \in \mathbb{R} \setminus \mathbb{N}_0$. It is not hard to show (Exercise 10.1.17) that for some constant $c < 0$,

$$\tbinom{a}{n} \gg n^c \ \ (n \in \mathbb{N})\,.$$

This bound and comparison with the geometric series give

$$E(a) \subset [-1, 1] \text{ for every } a \in \mathbb{R} \setminus \mathbb{N}_0\,.$$

We show that

$$(-1,1) \subset E(a) \text{ for every } a \in \mathbb{R} \setminus \mathbb{N}_0\,.$$

As we know, $0 \in E(a)$. If $x \in (-1,1)$, $x \neq 0$, then by using Corollary 10.1.5 with $b=0$, $f(t)=(1+t)^a$ and the Cauchy remainder we get

$$\left|R_n^{f,\,0}(x)\right| = (n+1)\left|\tbinom{a}{n+1}\right| \cdot (1+c)^{a-n-1} \cdot |x-c|^n \cdot |x| \to 0,\ n \to \infty\,.$$

Indeed, as for the first factor, by Exercise 10.1.18

$$\tbinom{a}{n+1} \ll n^d \ \ (n \in \mathbb{N})$$

for some constant $d>0$. The number $c=c(n) \in (-1,1)$ lies between 0 and x, so that

$$(1+c)^{a-1} \le \max\left((1-|x|)^{a-1},\, 1\right)$$

and

$$0 < \tfrac{|x-c|}{1+c} \le \tfrac{|x|-|c|}{1-|c|} = 1 - \tfrac{1-|x|}{1-|c|} \le 1 - \tfrac{1-|x|}{1} = |x| < 1\,.$$

We see that $R_n^{f,\,0}(x)$ goes exponentially fast to 0 with $n \to \infty$. Thus $T^{f,0}(x) = f(x)$ for every $x \in (-1,1)$.

It remains to determine for $a \in \mathbb{R} \setminus \mathbb{N}_0$ if $\pm 1 \in E(a)$. Below we obtain for every $a \in \mathbb{R} \setminus \mathbb{N}_0$ the asymptotics

$$\left|\tbinom{a}{n}\right| = \Theta(1) \cdot n^{-a-1} \ \ (n \in \mathbb{N})\,.$$

The factor $\Theta(1)$ is a function $g\colon \mathbb{N} \to (0,+\infty)$ such that

$$c \le g(n) \le d \text{ for every } n \in \mathbb{N} \text{ and some constants } 0 < c < d\,.$$

We postpone the proof of the asymptotics and finish the determination of the set $E(a)$. Let $x=1$. Then we have series

$$\textstyle\sum_{n=0}^{\infty} \binom{a}{n}\,.$$

For $a \le -1$ and $n \in \mathbb{N}_0$ we have the bound

$$\left|\tbinom{a}{n}\right| \ge \left|\tbinom{-1}{n}\right| = 1$$

and the series diverges. Thus $1 \notin E(a)$ for $a \le -1$. Let $a > -1$, $a \notin \mathbb{N}_0$ and $x=1$. Using Corollary 10.1.5 with $b=0$, $f(t)=(1+t)^a$ and the Lagrange remainder we get

$$\left|R_n^{f,\,0}(1)\right| = \left|\tbinom{a}{n+1}\right| \cdot (1+c)^{a-n-1} \cdot 1^{n+1} \to 0,\ n \to \infty\,,$$

because $0 < c = c(n) < 1$, $|\binom{a}{n}| = \Theta(1) \cdot n^{-a-1}$ and $-a-1 < 0$. Thus $1 \in E(a)$ for $a > -1$.

Let $x=-1$. Then we have series

$$\textstyle\sum_{n=0}^{\infty} \binom{a}{n}(-1)^n\,.$$

Despite the appearance it is not an alternating series because for $n > a+1$ the sign of the summand does not change. For $a < 0$ we have $-a-1 > -1$ and by the asymptotics of $|\binom{a}{n}|$ the series diverges because $\zeta(s)$ diverges for $s \le 1$ (Exercise 3.5.58). Thus $-1 \notin E(a)$ for $a < 0$. Let $a > 0$, $a \notin \mathbb{N}$ and $x = -1$. We use Theorem 10.1.7 (with the Schlömilch remainder) with $b = 0$, $f(t) = (1+t)^a$ and any $p \in (0, a)$. We can use this theorem because $f \in \mathcal{C}([-1, 0])$ and

$$\lim_{t \to x} f^{(i)}(t) \cdot (t-x)^i = \lim_{t \to -1} f^{(i)}(t) \cdot (t+1)^i = 0$$

for every $i \in \mathbb{N}$, as $a > 0$ and

$$f^{(i)}(t) = a(a-1)\dots(a-i+1) \cdot (1+t)^{a-i}\,.$$

We get

$$\left|R_n^{f,\,0}(-1)\right| = \tfrac{n+1}{p}\left|\tbinom{a}{n+1}\right| \cdot (1+c)^{a-n-1} \cdot (c+1)^{n+1-p} \cdot 1^p \to 0,\ n \to \infty\,,$$

because the first factor is $\Theta(1) \cdot n^{-a}$ with $a > 0$, we have $c = c(n) \in (-1, 0)$ and the product $(1+c)^{a-p}$ of the second and third factor lies in $(0, 1)$ as $a - p > 0$. Thus $-1 \in E(a)$ for $a > 0$. This completes the proof of parts 2–4.

We derive the postponed asymptotics of $|\binom{a}{n}|$ ($n \in \mathbb{N}$). We assume that $a \in \mathbb{R} \setminus \mathbb{N}_0$. For $m \in \mathbb{N}_0$ we have

$$S(a,\,m) \equiv \tbinom{a}{m}^{-1} \cdot \tbinom{a}{m+1} = \tfrac{a-m}{m+1} = -1 + \tfrac{a+1}{m+1}\,.$$

Since $\log(1+x) = x + O(x^2)$ ($|x| \le \frac{1}{2}$), we have

$$\log(|S(a,m)|) = -\tfrac{a+1}{m+1} + O_a(m^{-2}) \quad (m \ge 2|a|+2)\,.$$

We set $N \equiv \lceil 2|a|+2 \rceil$ and $K \equiv |\prod_{m=0}^{N-1} S(a,m)|$. For every $n \ge N+1$ we have

$$\begin{aligned}\log\left(\left|\tbinom{a}{n}\right|\right) \quad &= \quad \log\left(\left|\textstyle\prod_{m=0}^{n-1} S(a,\,m)\right|\right) = \log K - (a+1)\textstyle\sum_{m=N}^{n-1} \tfrac{1}{m+1} + \\ &\quad + O_a\left(\textstyle\sum_{m=N}^{n-1} m^{-2}\right).\end{aligned}$$

We recall the asymptotics

$$\textstyle\sum_{m=0}^{n-1} \tfrac{1}{m+1} = \log n + \gamma + O(n^{-1}) \quad (n \in \mathbb{N})$$

in Theorem 3.5.24. Hence, in view of convergence of $\zeta(2)$,

$$\log\left(\left|\tbinom{a}{n}\right|\right) = -(a+1)\log n + O(1) \quad (n \in \mathbb{N})\,.$$

Applying $\exp x$ we get by Exercise 10.1.16 that

$$\left|\tbinom{a}{n}\right| = \Theta(1) \cdot n^{-a-1} \quad (n \in \mathbb{N})\,.$$

$\square$

Exercise 10.1.16 *Show that* $\mathrm{e}^{O(1)} = \Theta(1)$.

Exercise 10.1.17 *For every* $a \in \mathbb{R} \setminus \mathbb{N}_0$ *there is a* $c < 0$ *such that*

$$\tbinom{a}{n} \gg n^c \quad (n \in \mathbb{N})\,.$$

Exercise 10.1.18 *For every* $a \in \mathbb{R}$ *there is a* $d > 0$ *such that*

$$\tbinom{a}{n} \ll n^d \quad (n \in \mathbb{N})\,.$$

Exercise 10.1.19 *Which of the binomial series are abscon?*

Exercise 10.1.20 *Find* $E(F^{f,b}(x))$ *with* $f(x) = (1+x)^a$ *for general center* b.

• *Nice summations.* Using the binomial series we obtain families of summation identities.

Exercise 10.1.21 *Prove the following corollary.*

Corollary 10.1.22 (sums of binomial series) *Let* $a \in \mathbb{R}$. *The next identities hold.*

1. *If* $a \in \mathbb{N}_0$ *then* $(1+x)^a = \sum_{n=0}^{\infty} \binom{a}{n} x^n = \sum_{n=0}^{a} \binom{a}{n} x^n$ *for every* $x \in \mathbb{R}$.
2. *If* $a \le -1$ *then* $(1+x)^a = \sum_{n=0}^{\infty} \binom{a}{n} x^n$ *for every* $x \in (-1,1)$.
3. *If* $-1 < a < 0$ *then* $(1+x)^a = \sum_{n=0}^{\infty} \binom{a}{n} x^n$ *for every* $x \in (-1,1]$.
4. *If* $a > 0$ *then* $(1+x)^a = \sum_{n=0}^{\infty} \binom{a}{n} x^n$ *for every* $x \in [-1,1]$.

• *Logarithmic series.* We determine Maclaurin series of two functions based on logarithm.

Theorem 10.1.23 (Maclaurin series of logarithms) *Let* $f(x) \equiv \log(1+x)$ *and* $g(x) \equiv \log(\frac{1}{1-x})$. *Then*

$$T^{f,0}(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \cdot x^n \text{ and } T^{f,0}(x) = \log(1+x) \iff x \in (-1,1]\,.$$

Also,

$$T^{g,0}(x) = \sum_{n=1}^{\infty} \frac{1}{n} \cdot x^n \text{ and } T^{g,0}(x) = \log\left(\tfrac{1}{1-x}\right) \iff x \in [-1,1)\,.$$

Proof. It suffices to consider just the function $f(x)$ because the result for $g(x)$ follows from the identity

$$\log\left(\tfrac{1}{1-x}\right) = -\log(1+(-x)) \quad (x \in [-1,\, 1))\,.$$

The form of $T^{f,0}(x)$ follows from Proposition 9.2.4. For $x \in (-1,1)$, $x \neq 0$, we use Corollary 10.1.5 with $b = 0$, $f(t) = \log(1+t)$ and the Cauchy remainder ($n \in \mathbb{N}$): with some $c = c(n)$ between 0 and x we have

$$\left|R_n^{f,\,0}(x)\right| = \tfrac{(n-1)!}{(1+c)^{n+1}} \cdot \tfrac{1}{n!} \cdot |x-c|^n \cdot |x| \to 0,\ n \to \infty\,,$$

because we know from the proof of Theorem 10.1.15 that $0 < \frac{|x-c|}{1+c} \le |x| < 1$.

For $x = 1$ we use the same bound: with some $c = c(n) \in (0,1)$, $n \in \mathbb{N}$, we have again

$$\left|R_n^{f,\,0}(1)\right| = \tfrac{1}{(1+c)^{n+1}} \cdot \tfrac{1}{n} \cdot (1-c)^n \cdot 1 < n^{-1} \to 0,\ n \to \infty\,.$$

Finally, for $x > 1$ the series obviously diverges. □

Exercise 10.1.24 *Which of these series are abscon?*

Now we easily prove the first summation in Theorem 3.5.33.

Corollary 10.1.25 (alternating harmonic series) *We have*

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = \log 2\,.$$

Proof. It is the instance $x = 1$ of the Maclaurin series for $\log(1+x)$. □

• *Series for inverse trigonometric functions.* We determine Maclaurin series of the functions $\arctan x$ and $\arcsin x$. Their derivatives get more and more complicated. Therefore we use other techniques instead of Taylor remainders to determine $E(T^{\arctan x, 0}(x))$ and $E(T^{\arcsin x, 0}(x))$.

Theorem 10.1.26 (Two more Maclaurin series) *Let* $f(x) \equiv \arctan x$ *and* $g(x) \equiv \arcsin x$*. Then*

$$T^{f,\,0}(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} \cdot x^{2n-1} \ \textit{and}\ T^{f,\,0}(x) = \arctan x \iff x \in [-1,1]\,.$$

Also,

$$T^{g,\,0}(x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{\binom{-\frac{1}{2}}{n-1}}{2n-1} \cdot x^{2n-1} \ \textit{and}\ T^{g,\,0}(x) = \arcsin x \iff x \in [-1,1]\,.$$

Proof. Both Maclaurin series follow from Proposition 9.2.5. It is easy to see that both $T^{f,0}(x)$ and $T^{g,0}(x)$ absolutely converge on $(-1,1)$. Since

$$f'(x) = \tfrac{1}{1+x^2} = \textstyle\sum_{n=0}^{\infty}(-1)^n x^{2n}$$

for every $x \in (-1,1)$, using results of the next section we see that the derivative

$$\big(f(x) - T^{f,\,0}(x)\big)' = 0 \text{ on } (-1,1)\,.$$

Thus $(f(x)\,|(-1,1) = T^{f,0}(x) + k_c(x)\,|(-1,1)$ for some c. Since $f(0) = 0 = T^{f,0}(0)$, we get that $c = 0$ and $(-1,1) \subset E(T^{f,0}(x))$. A similar argument using that

$$g'(x) = \big(1-x^2\big)^{-1/2} = \textstyle\sum_{n=0}^{\infty}(-1)^n\binom{-1/2}{n}x^{2n}$$

for every $x \in (-1,1)$ shows that $(-1,1) \subset E(T^{g,0}(x))$. It is easy to see that

$$E(T^{f,\,0}(x)),\ E(T^{g,\,0}(x)) \subset [-1,\,1]$$

because for $|x| > 1$ both Maclaurin series diverge (also, $M(\arcsin x) = [-1,1]$). Both points ± 1 belong to both sets $E(T^{f,0}(x))$ and $E(T^{g,0}(x))$ due to Abel's Theorem 10.3.20, see Corollary 10.3.25. $\square$

• *A function $f \in \mathcal{C}^\infty(\mathbb{R})$ with $E(T^{f,0}(x)) = \{0\}$.* We give an example of such function. Recall that if $f \in \mathcal{C}^\infty(\mathbb{R})$ then by Theorem 9.1.9 for every $b \in \mathbb{R}$ the Taylor series

$$T^{f,\,b}(x)$$

exists. Trivially, always $b \in E(T^{f,b}(x))$. We have seen several examples of functions when the set $E(T^{f,0}(x))$ is a nontrivial interval. Now we define a function for which this set is just $\{0\}$.

Theorem 10.1.27 (minimal $E(T^{f,0}(x))$) *Let $F \in \mathcal{F}(\mathbb{R})$ be given by*

$$F(0) \equiv 0 \ \text{ and } \ F(x) \equiv \exp\big(-\tfrac{1}{x^2}\big)\ \ldots\ x \neq 0\,.$$

Then

$$F \in \mathcal{C}^\infty(\mathbb{R}),\ T^{F,\,0}(x) = \textstyle\sum_{n=0}^{\infty} 0x^n \ \text{ and } E(T^{F,\,0}(x)) = \{0\}\,.$$

Proof. Let $\mathbb{R}_0 \equiv \mathbb{R} \setminus \{0\}$ and

$$G(x) \equiv F(x)\,|\,\mathbb{R}_0 = \exp\big(-\tfrac{1}{x^2}\big)\,.$$

We prove by induction on $n \in \mathbb{N}_0$ that $G^{(n)} \in \mathcal{F}(\mathbb{R}_0)$. We actually prove by induction that for every $n \in \mathbb{N}_0$,

$$G^{(n)}(x) = \exp(-\tfrac{1}{x^2})\textstyle\sum_{j=0}^{k_n} a_{n,\,j}x^{-j} \quad (\in \mathcal{F}(\mathbb{R}_0))\,,$$

where $k_n \in \mathbb{N}_0$ and $a_{n,j} \in \mathbb{Z}$. For $n = 0$ this holds with $k_0 = 0$ and $a_{n,0} = 1$. Then, by the results on derivatives in Chapter 7,

$$\begin{aligned} G^{(n+1)}(x) = (G^{(n)}(x))' &= \exp(-\tfrac{1}{x^2})2x^{-3}\textstyle\sum_{j=0}^{k_n} a_{n,\,j}x^{-j} + \\ &\quad + \exp(-\tfrac{1}{x^2})\textstyle\sum_{j=0}^{k_n}(-j)a_{n,\,j}x^{-j-1} \\ &= \exp(-\tfrac{1}{x^2})\textstyle\sum_{j=0}^{k_{n+1}} a_{n+1,\,j}x^{-j}\,, \end{aligned}$$

where $k_{n+1} \in \mathbb{N}_0$ and $a_{n+1,j} \in \mathbb{Z}$. It is clear that for every $n \in \mathbb{N}_0$ and every $x \in \mathbb{R}_0$,

$$F^{(n)}(x) = G^{(n)}(x)\,.$$

We show by induction on $n \in \mathbb{N}_0$ that the derivative $F^{(n)}(0)$ exists and is 0. For $n = 0$ this follows from the definition of $F(x)$. Suppose that it holds for an $n \in \mathbb{N}_0$. By Exercise 10.1.28, the above form of $G^{(n)}(x)$ and Theorem 4.4.5,

$$F^{(n+1)}(0) = \lim_{x\to 0} \tfrac{F^{(n)}(x)-F^{(n)}(0)}{x} = \lim_{x\to 0} x^{-1}G^{(n)}(x) = 0\,.$$

We see that $F \in \mathcal{C}^\infty(\mathbb{R})$ and $F^{(n)}(0) = 0$ for every $n \in \mathbb{N}_0$. By Theorem 9.1.9,

$$T^{F,\,0}(x) = 0 + 0x + 0x^2 + \cdots\,.$$

The sum of this series is for $x \in \mathbb{R}$ the constant zero function $k_0(x)$. But $k_0(x) \neq F(x)$ for every $x \neq 0$, and therefore $E(T^{F,0}(x))$ is just $\{0\}$. $\Box$

Exercise 10.1.28 *For every $n \in \mathbb{N}_0$ it is true that*

$$\lim_{x\to 0} x^{-n}\exp(-\tfrac{1}{x^2}) = 0\,.$$

10.2 *Formal power series

In this section we develop the theory of formal power series to the extent sufficient for obtaining in Theorem 10.4.1 rough asymptotics of the numbers op_n of nonempty ordered partitions of $[n]$ (Definition 10.2.1).

• *Ordered partitions.* Let $k \in \mathbb{N}$ and $n \in \mathbb{N}_0$. An ordered partition of $[n]$ with k parts is any k-tuple

$$\overline{A} = \langle A_1,\, A_2,\, \dots,\, A_k\rangle$$

of (possibly empty) mutually disjoint sets A_i such that $\bigcup_{i=1}^k A_i = [n]$. Recall that $[n] = \{1, 2, \dots, n\}$ and $[0] = \emptyset$. We define

$$\mathrm{OP}(k,\, n) \equiv \{\text{ordered partitions of } [n] \text{ with } k \text{ parts}\} \text{ and } \mathrm{op}_{k,\,n} \equiv |\mathrm{OP}(k,\, n)|\,.$$

For example, $\mathrm{op}_{2,2} = 4$ because

$$\mathrm{OP}(2,\, 2) = \{\langle\emptyset,\, [2]\rangle,\, \langle[2],\, \emptyset\rangle,\, \langle\{1\},\, \{2\}\rangle,\, \langle\{2\},\, \{1\}\rangle\}\,.$$

A nonempty ordered partition $\overline{A}$ has all parts $A_i \neq \emptyset$.

We are interested in the asymptotics of the sequence (op_n) of the following counting numbers.

Definition 10.2.1 (op_n and $\mathrm{OP}()n)$ *Let $n \in \mathbb{N}$. We denote by op_n the number of nonempty ordered partitions of $[n]$ with any number of parts. We set $\mathrm{op}_0 \equiv 1$. We denote the set of nonempty ordered partitions of $[n]$ by $\mathrm{OP}(n)$ ($n \in \mathbb{N}$). Thus $\mathrm{op}_n = |\mathrm{OP}(n)|$.*

For example, $\mathrm{op}_2 = 3$ because the nonempty ordered partitions of $[2]$ are

$$\langle [2] \rangle,\ \langle \{1\},\ \{2\} \rangle \ \text{ and } \ \langle \{2\},\ \{1\} \rangle\,.$$

Definition 10.2.2 (multinomial coefficients) *We suppose that $k \in \mathbb{N}$ and that $n, n_i \in \mathbb{N}_0$ for $i \in [k]$ are such that $n_1 + n_2 + \cdots + n_k = n$. We define the multinomial coefficient $\binom{n}{n_1, n_2, \ldots, n_k}$ by*

$$\binom{n}{n_1,\ n_2,\ \ldots,\ n_k} \equiv \frac{n!}{n_1! \cdot n_2! \cdot \ldots \cdot n_k!}\,.$$

Thus $\binom{n}{k} = \binom{n}{k, n-k}$ for every $n, k \in \mathbb{N}_0$.

Exercise 10.2.3 *Prove the next proposition.*

Proposition 10.2.4 (counting OP) *With the notation of the previous definition we have*

$$|\{\langle A_1,\ A_2,\ \ldots,\ A_k \rangle \in \mathrm{OP}(k,\ n)\colon\ |A_i| = n_i,\ i \in [k]\}| = \binom{n}{n_1,\ n_2,\ \ldots,\ n_k}\,.$$

For any set X and $n \in \mathbb{N}_0$ we introduce the notation

$$\binom{X}{n} \equiv \{A\colon\ A \subset X \wedge |A| = n\}\,.$$

Thus $\left|\binom{[n]}{k}\right| = \binom{n}{k}$ for every $n, k \in \mathbb{N}_0$.

Exercise 10.2.5 *What is $\binom{X}{0}$?*

The following exercise shows that the sequence

$$\left(\tfrac{1}{n!} \cdot \mathrm{op}_n\colon\ n \in \mathbb{N}\right)$$

is supermultiplicative.

Exercise 10.2.6 *Let $m, n \in \mathbb{N}$. Define an injection*

$$\iota\colon \mathrm{OP}(m) \times \mathrm{OP}(n) \times \binom{[m+n]}{m} \to \mathrm{OP}(m+n)\,.$$

• *Formal power series.* These are just real sequences, but indexed by $\mathbb{N}_0$ instead of $\mathbb{N}$. They are endowed with arithmetic operations.

Definition 10.2.7 (formal power series) *We write*

$$\mathbb{R}[[x]] \equiv \{A\colon\ A \textit{ is a map from } \mathbb{N}_0 \textit{ to } \mathbb{R}\}$$

for the set of formal power series*, or* fps*, with one formal variable x and real coefficients $a_n \equiv A(n)$, $n \in \mathbb{N}_0$. We write the elements $A = A(x)$ in $\mathbb{R}[[x]]$ as formal sums*

$$\textstyle\sum_{n\ge 0} a_n x^n\,.$$

Instead of the constant term *a_0x^0 we usually write just a_0, and instead of a_1x^1 we write just a_1x.*

We define the sum and product of two fps as

$$\textstyle\sum_{n\ge 0} a_n x^n + \sum_{n\ge 0} b_n x^n \equiv \sum_{n\ge 0}(a_n + b_n)x^n$$

and

$$\textstyle\sum_{n\ge 0} a_n x^n \cdot \sum_{n\ge 0} b_n x^n \equiv \sum_{n\ge 0}\big(\sum_{k=0}^n a_k b_{n-k}\big)x^n\,.$$

The neutral elements in $\mathbb{R}[[x]]$ are

$$0 = 0_{\mathbb{R}[[x]]} \equiv 0x^0 + 0x^1 + \dots \ \text{ and } \ 1 = 1_{\mathbb{R}[[x]]} \equiv 1x^0 + 0x^1 + 0x^2 + \dots\,.$$

Exercise 10.2.8 *In every (commutative unital) ring R_{ri} the structure*

$$\langle R^\times,\, 1_R,\, \cdot\rangle$$

is an Abelian group.

Proposition 10.2.9 (ring of fps) *The structure*

$$\mathbb{R}[[x]]_{\mathrm{ri}} \equiv \langle \mathbb{R}[[x]],\, 0_{\mathbb{R}[[x]]},\, 1_{\mathbb{R}[[x]]},\, +,\, \cdot\rangle$$

is a ring. Its units are exactly the fps *with nonzero constant terms.*

Proof. Commutativity and associativity of addition and multiplication of fps are left for Exercise 10.2.10. Neutrality of $0_{\mathbb{R}[[x]]}$ and $1_{\mathbb{R}[[x]]}$ is easy to check. We treat in detail the distributive law, additive inverses and units.

In $\mathbb{R}[[x]]_{\mathrm{ri}}$ the distributive law holds:

$$\begin{aligned}
&\textstyle\sum_{n\ge 0} a_n x^n \cdot \big(\sum_{n\ge 0} b_n x^n + \sum_{n\ge 0} c_n x^n\big) = \\
&= \textstyle\sum_{n\ge 0} a_n x^n \cdot \sum_{n\ge 0}(b_n + c_n)x^n = \\
&= \textstyle\sum_{n\ge 0} \sum_{j=0}^n a_j(b_{n-j} + c_{n-j})x^n = \\
&= \textstyle\sum_{n\ge 0} \sum_{j=0}^n (a_j b_{n-j} + a_j c_{n-j})x^n = \dots = \\
&= \textstyle\sum_{n\ge 0} a_n x^n \cdot \sum_{n\ge 0} b_n x^n + \sum_{n\ge 0} a_n x^n \cdot \sum_{n\ge 0} c_n x^n\,.
\end{aligned}$$

The additive inverse of $\sum_{n\ge 0} a_n x^n$ is $\sum_{n\ge 0}(-a_n)x^n$. If $A(x)$ is a fps that has zero constant term, then so has every multiple of it and $A(x)$ is not a unit.

Let $A(x) = \sum_{n\geq 0} a_n x^n$ be a fps with $a_0 \neq 0$. We show that there is a fps $\sum_{n\geq 0} b_n x^n$ such that

$$\textstyle\sum_{n\geq 0} a_n x^n \cdot \sum_{n\geq 0} b_n x^n = a_0 b_0 + (a_0 b_1 + a_1 b_0)x + \cdots = 1\,.$$

This is equivalent with the system

$$\textstyle\sum_{j=0}^{n} a_j b_{n-j} = c_n,\ n \in \mathbb{N}_0\,,$$

where b_n are unknowns, $c_0 = 1$ and $c_n = 0$ for $n > 0$. The solution starts with $b_0 \equiv a_0^{-1}$, determined by the 0-th equation. If b_0, b_1, …, b_n are already determined, we compute b_{n+1} from the $(n+1)$-st equation:

$$\textstyle b_{n+1} \equiv -a_0^{-1} \sum_{j=0}^{n} a_{n+1-j} b_j\,.$$

Hence $A(x)$ is a unit. □

Exercise 10.2.10 *Show that the operations* $+$ *and* $\cdot$ *on* $\mathbb{R}[[x]]$ *are commutative and associative.*

• *Formal exponential.* Formal exponential series is as important as its functional version e^x.

Definition 10.2.11 (formal exponential) *The* fps

$$\mathrm{e}_{\mathrm{fps}}^{x} \equiv \sum_{n\geq 0} \frac{1}{n!} \cdot x^n$$

is the formal exponential.

Exercise 10.2.12 *Show that in the ring* $\mathbb{R}[[x,y]]_{\mathrm{ri}}$ *of* fps *with two formal variables x and y the identity*

$$\mathrm{e}_{\mathrm{fps}}^{x+y} = \mathrm{e}_{\mathrm{fps}}^{x} \cdot \mathrm{e}_{\mathrm{fps}}^{y}$$

holds.

• *Infinite series in* $\mathbb{R}[[x]]_{\mathrm{ri}}$. We review limit transitions and sums of infinite series in $\mathbb{R}[[x]]_{\mathrm{ri}}$. For a fps $A(x) = \sum_{n\geq 0} a_n x^n$ and $m \in \mathbb{N}_0$ we introduce the notation

$$[x^m]\, A(x) \equiv a_m$$

for the coefficient of x^m in $A(x)$.

Definition 10.2.13 (formal limit) *Let* $(A_n(x)) \subset \mathbb{R}[[x]]$ *and let* $A(x)$ *be a fps. If for every* $m \in \mathbb{N}_0$ *there is an* $n_0 = n_0(m)$ *in* $\mathbb{N}_0$ *such that for every* $n \geq n_0$ *we have*

$$[x^m]\, A_n(x) = [x^m]\, A(x)\,,$$

we say that the sequence of fps $(A_n(x))$ *has the formal limit* $A(x)$ *and we write* $\mathrm{flim}\, A_n(x) = A(x)$ *or* $\mathrm{flim}_{n\to\infty} A_n(x) = A(x)$.

Exercise 10.2.14 *Formal limits are unique.*

For a series of fps $\sum_{n=1}^{\infty} A_n(x)$ we define its sum as the formal limit

$$\mathrm{flim}_{n\to\infty} \sum_{j=1}^{n} A_j(x)\,,$$

if this formal limit of partial sums exists. We denote the sum, which is a fps $A(x)$, again by $\sum_{n=1}^{\infty} A_n(x)$. A very convenient property of formal convergence is that the necessary condition of convergence (Proposition 3.5.13) is also sufficient.

Theorem 10.2.15 (NCC becomes SCC) *A series $\sum_{n=1}^{\infty} A_n(x) \subset \mathbb{R}[[x]]$ has a sum* $\iff$

$$\mathrm{flim}_{n\to\infty} A_n(x) = 0_{\mathbb{R}[[x]]}\,.$$

Proof. Let $(n \in \mathbb{N})$

$$A_n(x) = \sum_{m\ge 0} a_{m,\,n} x^m\,.$$

We prove implication $\Rightarrow$. Let $m \in \mathbb{N}_0$. By the assumption the sequence

$$\big(\sum_{j=1}^{n} a_{m,\,j}:\ n \in \mathbb{N}\big)$$

is eventually constant. Thus the sequence $(a_{m,1}, a_{m,2}, \dots)$ is eventually zero and $\mathrm{flim}\, A_n(x) = 0$.

We prove implication $\Leftarrow$. Let $m \in \mathbb{N}_0$. Since the sequence $(a_{m,1}, a_{m,2}, \dots)$ is eventually zero, the above displayed sequence of partial sums eventually constantly equals a_m. Thus

$$\mathrm{flim}_{n\to\infty} \sum_{j=1}^{n} A_j(x) = \sum_{m\ge 0} a_m x^m\,.$$

$\square$

An important application of sums of series of fps is the formula for sums of formal geometric series.

Proposition 10.2.16 (formal geometric series) *Suppose that the fps $A(x)$ has zero constant term. Then the inverse to the unit $1 - A(x)$ equals to the sum of the formal geometric series with the quotient $A(x)$,*

$$\big(1_{\mathbb{R}[[x]]} - A(x)\big)^{-1} = \sum_{n=0}^{\infty} A(x)^n = 1_{\mathbb{R}[[x]]} + \sum_{n=1}^{\infty} A(x)^n \quad (\in \mathbb{R}[[x]])\,.$$

Proof. It is easy to see that

$$\mathrm{flim}\, A(x)^n = 0\,.$$

Therefore by the previous theorem the formal geometric series has a sum $B(x) \equiv 1 + \sum_{n=1}^{\infty} A(x)^n$ $(\in \mathbb{R}[[x]])$. We have

$$\begin{aligned}
(1 - A(x)) \cdot B(x) &= 1 - A(x) + \textstyle\sum_{n=1}^{\infty} A(x)^n - A(x) \cdot \sum_{n=1}^{\infty} A(x)^n \\
&= \textstyle\sum_{n=0}^{\infty} A(x)^n - \sum_{n=1}^{\infty} A(x)^n \\
&= A(x)^0 = 1
\end{aligned}$$

(Exercise 10.2.17). Thus $B(x) = (1 - A(x))^{-1}$ $\square$

Exercise 10.2.17 *Justify the previous computation in detail.*

• *Exponential generating functions.* Back to combinatorics. A useful family of fps are exponential generating functions, or EGF. These are fps of the form

$$\sum_{n\geq 0} \frac{a_n}{n!} \cdot x^n$$

where $a_n \in \mathbb{R}$. Often even $a_n \in \mathbb{N}_0$ and the coefficients a_n count some combinatorial structures of size n.

Theorem 10.2.18 (product formula for EGF) *Let $k \in \mathbb{N}$. Suppose that the $k+1$ real sequences (a_n) and $(b_{n,i})$ with $n \in \mathbb{N}_0$ and $i \in [k]$ are for every $n \in \mathbb{N}_0$ bound by the convolution*

$$a_n = \sum_{\substack{\overline{A}\in \mathrm{OP}(k,\,n)\\ \overline{A}=\langle A_1,\,\dots,\,A_k\rangle}} \prod_{i=1}^{k} b_{|A_i|,\,i}\,.$$

Then the corresponding EGF *are related by the product*

$$\sum_{n\geq 0} \frac{a_n}{n!} \cdot x^n = \prod_{i=1}^{k} \sum_{n\geq 0} \frac{b_{n,\,i}}{n!} \cdot x^n\,.$$

Proof. This follows from the formula in Proposition 10.2.4:

$$\begin{aligned}
\textstyle\sum_{n\geq 0} \frac{a_n}{n!} \cdot x^n &= \textstyle\sum_{n\geq 0} \frac{1}{n!} \big(\sum_{\substack{\overline{A}\in \mathrm{OP}(k,\,n)\\ \overline{A}=\langle A_1,\,\dots,\,A_k\rangle}} \prod_{i=1}^{k} b_{|A_i|,\,i} \big) \cdot x^n \\
&= \textstyle\sum_{n\geq 0} \frac{1}{n!} \big(\sum_{\substack{n_1,\,\dots,\,n_k\in\mathbb{N}_0\\ n_1+\cdots+n_k=n}} \sum_{\substack{\overline{A}\in \mathrm{OP}(k,\,n)\\ \overline{A}=\langle A_1,\,\dots,\,A_k\rangle\\ |A_i|=n_i}} \prod_{i=1}^{k} b_{|A_i|,\,i} \big) \cdot x^n \\
&= \textstyle\sum_{n\geq 0} \frac{1}{n!} \big(\sum_{\substack{n_1,\,\dots,\,n_k\in\mathbb{N}_0\\ n_1+\cdots+n_k=n}} \binom{n}{n_1,\,\dots,\,n_k} \prod_{i=1}^{k} b_{n_i,\,i} \big) \cdot x^n \\
&= \textstyle\sum_{\substack{n,\,n_1,\,\dots,\,n_k\in\mathbb{N}_0\\ n_1+\cdots+n_k=n}} \prod_{i=1}^{k} \frac{b_{n_i,\,i}}{n_i!} \cdot x^{n_i} = \prod_{i=1}^{k} \sum_{n_i\geq 0} \frac{b_{n_i,\,i}}{n_i!} \cdot x^{n_i}\,.
\end{aligned}$$

□

Corollary 10.2.19 (EGF of ordered partitions) *In the ring $\mathbb{R}[[x]]_{\mathrm{ri}}$ of formal power series the identity*

$$\sum_{n\geq 0} \frac{\mathrm{op}_n}{n!} \cdot x^n = \frac{1}{2-\mathrm{e}^x_{\mathrm{fps}}}$$

holds.

Proof. For $k \in \mathbb{N}$ and $n \in \mathbb{N}_0$ we denote by $a_{k,n}$ the number of nonempty ordered partitions of $[n]$ with k parts; $a_{k,0} = 0$ for every k. For $k \in \mathbb{N}$, $n \in \mathbb{N}_0$ and $i \in [k]$ we set $b_{k,n,i} \equiv 1$ if $n > 0$ and $b_{k,0,i} \equiv 0$. Then

$$a_{k,\,n} = \sum_{\substack{\overline{A} \in \mathrm{OP}(k,\,n) \\ \overline{A} = \langle A_1,\,\dots,\,A_k \rangle}} \prod_{i=1}^{k} b_{k,\,|A_i|,\,i}$$

for every $k \in \mathbb{N}$ and every $n \in \mathbb{N}_0$. We therefore use the previous theorem, on the third line below, and get

$$\begin{aligned}
\sum_{n\ge 0} \tfrac{\mathrm{op}_n}{n!} \cdot x^n &= 1 + \sum_{n\ge 1} \tfrac{\mathrm{op}_n}{n!} \cdot x^n \\
&= 1 + \sum_{n\ge 1} \sum_{k\ge 1} \tfrac{a_{k,\,n}}{n!} \cdot x^n = 1 + \sum_{k\ge 1} \sum_{n\ge 0} \tfrac{a_{k,\,n}}{n!} \cdot x^n \\
&= 1 + \sum_{k\ge 1} \prod_{i=1}^{k} \sum_{n\ge 0} \tfrac{b_{k,\,n,\,i}}{n!} \cdot x^n = 1 + \sum_{k\ge 1} \left(\mathrm{e}^x_{\mathrm{fps}} - 1\right)^k \\
&= \tfrac{1}{1-(\mathrm{e}^x_{\mathrm{fps}}-1)} = \tfrac{1}{2-\mathrm{e}^x_{\mathrm{fps}}}\,.
\end{aligned}$$

On the fourth line we used Proposition 10.2.16. □

Exercise 10.2.20 *Justify the computation in the previous proof in detail.*

• *Formal Dirichlet series.* For the next exercise we regard the zeta series $\zeta(s) = \sum_{n\ge 1} \frac{1}{n^s}$ as a formal Dirichlet series, with the formal variable s. For $n \in \mathbb{N}$ with $n > 1$, let m_n be the number of k-tuples ($k \in \mathbb{N}$)

$$\langle n_1,\, n_2,\, \dots,\, n_k \rangle \in (\mathbb{N} \setminus \{1\})^k$$

such that $\prod_{i=1}^{k} n_i = n$, and let $m_1 \equiv 1$. For example, $m_6 = 3$ because 6 can be factorized as 6, $2 \cdot 3$ and $3 \cdot 2$.

Exercise 10.2.21 *Show that in the ring of formal Dirichlet series the identity*

$$\sum_{n\ge 1} \frac{m_n}{n^s} = \frac{1}{2 - \zeta(s)}$$

holds.

10.3 *Real analytic functions

In this section, we investigate functions that can be expressed as sums of power series (with the center $b \in \mathbb{R}$). These are familiar series

$$\textstyle\sum_{n=0}^{\infty} a_n (x-b)^n,\ a_n \in \mathbb{R}\,,$$

which look like Taylor series. The numbers $a_0,\, a_1,\, \dots$ are the coefficients of the power series and $x \in \mathbb{R}$. Coefficients of Taylor series come from Taylor polynomials, but now we allow a_n to be arbitrary real numbers.

• *Real analytic functions.* They are sums of power series and are precisely defined as follows.

Definition 10.3.1 (RAF) *We say that a function $f \in \mathcal{R}$ is real analytic if the following holds.*

1. *The definition domain $M(f)$ is an open set.*
2. *For every point $b \in M(f)$ there exist real numbers δ and $a_0,\ a_1,\ \ldots$ such that $U(b, \delta) \subset M(f)$ and for every $x \in U(b, \delta)$ we have*

$$f(x) = \textstyle\sum_{n=0}^{\infty} a_n (x-b)^n\,.$$

The set of real analytic functions is denoted by RAF.

For example, later we show that the function $\exp(-1/x^2)$ with the definition domain

$$(-\infty,\, 0) \cup (0,\, +\infty)$$

is real analytic.

Definition 10.3.2 (radius of convergence) *By the radius of convergence of a power series $S(x) = \sum_{n=0}^{\infty} a_n(x-b)^n$ we mean the element*

$$R = R(S(x)) \equiv \frac{1}{\limsup_{n\to\infty} |a_n|^{1/n}} \quad (\in [0,\, +\infty) \cup \{+\infty\})\,,$$

where we set $\frac{1}{0} \equiv +\infty$.

For example,

$$R(T^{\log(1+x),\,0}(x)) = R\big(\textstyle\sum_{n=1}^{\infty} (-1)^{n-1} n^{-1} x^n\big) = 1$$

because $\lim n^{1/n} = 1$. One can describe by the radius of convergence of a power series the set of $x \in \mathbb{R}$ where it converges. In this description we get the intervals $(b, b) = \emptyset$ and $(-\infty, +\infty) \equiv \mathbb{R}$.

Theorem 10.3.3 (interval of convergence) *Let $S(x) = \sum_{n=0}^{\infty} a_n(x-b)^n$ be a power series with center b. We define the real set*

$$I = I(S(x)) \equiv \{x \in \mathbb{R}\colon\ \text{the power series } S(x) \text{ converges}\}\,.$$

Then

$$\text{either } I = \{b\} \text{ or } I \text{ is a nontrivial interval}.$$

In the latter case either $I = \mathbb{R}$ or I is a bounded interval and b is its midpoint. The interior of I is

$$I^0 = (b - R(S(x)),\, b + R(S(x)))$$

and for every $x \in I^0$ the power series $S(x)$ absolutely converges. We call I the interval of convergence of $S(x)$.

Proof. Let R, where $R \geq 0$ is real or $R = +\infty$, be the radius of convergence of the power series $S(x)$ and I be its interval of convergence. If $R = 0$ then for every large $c > 0$ we have $|a_n| \geq c^n$ for infinitely many n. Thus for every $x \neq b$ we have

$$|a_n(x-b)^n| \geq 1$$

for infinitely many n and $S(x)$ diverges. So for $R = 0$ we have

$$I = \{b\} \text{ and } I^0 = \emptyset = (b-R,\, b+R) = (b,\, b)\,.$$

If $R = +\infty$ then for every small $c > 0$ we have $|a_n| \leq c^n$ for every large n. Thus for every $x \in \mathbb{R}$ and every small $c > 0$ we have

$$|a_n(x-b)^n| \leq c^n$$

for every large n and, by comparison with the geometric series, $S(x)$ absolutely converges. In particular, for $R = +\infty$ we have

$$I = \mathbb{R} = (b-R,\, b+R) = (-\infty,\, +\infty)\,.$$

We consider the remaining case when $0 < R < +\infty$. Let $x \in \mathbb{R}$ be such that $r \equiv |x-b| < R$. We show that the power series $S(x)$ absolutely converges. We take a small δ such that $r + \delta < R$. From the definition of R we have that $|a_n| \leq (r+\delta)^{-n}$ for every large n. Hence

$$|a_n(x-b)^n| \leq \left(\tfrac{r}{r+\delta}\right)^n$$

for every large n and $S(x)$ absolutely converges by comparison with the geometric series. We finally show that if $r \equiv |x-b| > R$ then the power series $S(x)$ diverges. We take a small δ such that $r - \delta > R$. From the definition of R we have that $|a_n| \geq (r-\delta)^{-n}$ for infinitely many n. Hence

$$|a_n(x-b)^n| \geq \left(\tfrac{r}{r-\delta}\right)^n > 1$$

for infinitely many n and $S(x)$ diverges. Hence for $0 < R < +\infty$ we have

$$(b-R,\, b+R) = I^0 \subset I \subset [b-R,\, b+R]\,,$$

$S(x)$ absolutely converges on the former interval and the proof of the theorem is complete. $\square$

As we saw in the case of the binomial series, it may not be easy to determine if the points $b \pm R$ belong to I.

Exercise 10.3.4 *Intervals of convergence of power series* $A(x) = \sum_{n=0}^{\infty} a_n x^n$ *and* $B(x) = \sum_{n=0}^{\infty} a_n (x-b)^n$ *are related by*

$$I(B(x)) = I(A(x)) + b = \{a + b\colon\ a \in I(A(x))\}\,.$$

Exercise 10.3.5 *Let $T^{f,b}(x)$ be a Taylor series. What is the relation between the sets*

$$E(T^{f,\,b}(x)) \;\; and \;\; I(T^{f,\,b}(x))\,?$$

Exercise 10.3.6 *Find four power series with respective intervals of convergence*

$$(-1,\,1),\;\; (-1,\,1],\;\; [-1,\,1),\;\; and \;\; [-1,\,1]\,.$$

Exercise 10.3.7 *Find the radii of convergence of the power series with center* 0 *defining the functions* $\exp x$, $\cos x$ *and* $\sin x$.

• *Sum functions, power series and fps.* Suppose that $A(x) = \sum_{n=0}^{\infty} a_n(x-b)^n$ is a power series with center b. The <u>sum function</u> of $A(x)$ is the function

$$F_A\colon I(A(x)) \to \mathbb{R},\;\; F_A(x) \equiv \textstyle\sum_{n=0}^{\infty} a_n(x-b)^n\,.$$

We also consider the function

$$F_A^0\colon I(A(x))^0 \to \mathbb{R},\;\; F_A^0 \equiv F_A \,|\, I(A(x))^0\,.$$

In the next passage we prove that F_A^0 is real analytic. Pringsheim's Theorem 10.3.26 below shows that in general real analyticity cannot be extended to endpoints of the interval of convergence.

Notation like

$$A(x) = \textstyle\sum_{n\ge 0} a_n x^n \;\text{ or }\; A(x) = \sum_{n=0}^{\infty} a_n x^n \quad (a_n \in \mathbb{R})$$

has two meanings. We can understand it as a power series with center 0, i.e. as sequences

$$(a_0,\, a_1x,\, a_2x^2,\, \dots) \;\text{ with }\; x \in \mathbb{R}$$

that define on $I(A(x))$ the sum function $F_A(x)$. Or we can understand it as a fps (formal power series) in $\mathbb{R}[[x]]$, i.e. just as the sequence of real coefficients

$$(a_0,\, a_1,\, a_2,\, \dots)\,.$$

We work simultaneously with both interpretations. For example, we speak of the radius of convergence of a fps and so on.

Exercise 10.3.8 *Explain why a fps $\sum_{n\ge 0} a_n x^n$ is more basic than the corresponding sum function $F_A(x) = \sum_{n=0}^{\infty} a_n(x-b)^n$, $x \in I$.*

• *Power series and real analytic functions.* We expect that the F_A^0 sum function of a power series is real analytic and we prove it.

Theorem 10.3.9 (real analyticity of F_A^0) *Suppose that a power series*

$$A(x) = \textstyle\sum_{n=0}^{\infty} a_n(x-b)^n$$

has a nontrivial interval of convergence I. <u>Then</u> the sum function

$$F_A^0\colon I^0 \to \mathbb{R},\ F_A^0(x) \equiv \sum_{n=0}^{\infty} a_n(x-b)^n\,,$$

defined on the interior of I is real analytic.

Proof. Let $c \in I^0$, $c \neq b$, and let δ be such that $U(c,\delta) \subset I^0$. Then for every $x \in U(c,\delta)$ we have

$$\begin{aligned} F_A^0(x) &= \textstyle\sum_{n=0}^{\infty} a_n(x-b)^n = \sum_{n=0}^{\infty} a_n(x-c+(c-b))^n \\ &= \textstyle\sum_{n=0}^{\infty} a_n \sum_{j=0}^{n} \binom{n}{j}(x-c)^j(c-b)^{n-j} \\ &= \textstyle\sum_{j=0}^{\infty} \big(\sum_{n,\,n\geq j} \binom{n}{j} a_n(c-b)^{n-j}\big)\cdot(x-c)^j \\ &\equiv \textstyle\sum_{j=0}^{\infty} b_j(x-c)^j,\ \text{ with } b_j \equiv \sum_{n,\,n\geq j} \binom{n}{j} a_n(c-b)^{n-j}\,. \end{aligned}$$

On the second line, we used the binomial theorem. It remains to justify the change of order of summation between the second and third line, and the convergence of series defining the new coefficients b_j. We leave the former for Exercise 10.3.10 and turn to the latter. Since $|c-b| < R(S(x))$, there is a constant $d \in (0,1)$ such that for every large n,

$$\big|a_n(c-b)^n\big| \leq d^n\,.$$

Also, $\binom{n}{j} = O(n^j)$ $(n \geq j)$. Thus for any $d_0 \in (d,1)$ we have

$$\big|\tbinom{n}{j}(c-b)^{-j}\cdot a_n(c-b)^n\big| \leq d_0^n$$

for every sufficiently large n $(\geq j)$. Comparison with the geometric series shows that the series defining b_j absolutely converges. □

Exercise 10.3.10 *Justify the change of order of summation in the proof.*

• *Arithmetic operations on real analytic functions.* We investigate the interplay between real analyticity and the operations of addition, multiplication and division on $\mathcal{R}$.

Proposition 10.3.11 (addition of power series) *Suppose that*

$$A(x) = \textstyle\sum_{n\geq 0} a_n x^n \text{ and } B(x) = \sum_{n\geq 0} b_n x^n \quad (\in \mathbb{R}[[x]])$$

are fps *with positive radii of convergence R_A and R_B, respectively. Let $C(x) \equiv A(x) + B(x) = \sum_{n\geq 0}(a_n+b_n)x^n$ as a sum of* fps. <u>*Then*</u>

$$F_A(x) + F_B(x) = C(x)$$

for every $x \in (-R,\,R)$ where $R = \min(R_A, R_B)$.

Proof. Let $A(x)$, $B(x)$, R_A, R_B, $C(x)$ and R be as stated, and let $x \in (-R, R)$. By linear combination of series (Proposition 3.5.15),

$$F_A(x) + F_B(x) = \textstyle\sum_{n=0}^{\infty} a_n x^n + \sum_{n=0}^{\infty} b_n x^n = \sum_{n=0}^{\infty} (a_n + b_n) x^n = C(x)\,.$$

□

Theorem 10.3.12 (addition in RAF) *If $f, g \in \mathrm{RAF}$ then $f + g \in \mathrm{RAF}$.*

Proof. Suppose that f, g are real analytic functions and $h \equiv f + g$. The domain $M(h) = M(f) \cap M(g)$ is an open set. Let $b \in M(h)$. Since $f, g \in \mathrm{RAF}$, there is a δ such that $U(b, \delta) \subset M(h)$ and for every $x \in U(b, \delta)$ we have

$$f(x) = \textstyle\sum_{n=0}^{\infty} a_n (x-b)^n \text{ and } g(x) = \sum_{n=0}^{\infty} b_n (x-b)^n\,.$$

By linear combination of series (Proposition 3.5.15) we have for the same x that

$$h(x) = f(x) + g(x) = \textstyle\sum_{n=0}^{\infty} (a_n + b_n)(x-b)^n\,.$$

Thus h is real analytic. □

Proposition 10.3.13 (multiplying power series) *Suppose that*

$$A(x) = \textstyle\sum_{n\ge 0} a_n x^n \textit{ and } B(x) = \sum_{n \ge 0} b_n x^n \quad (\in \mathbb{R}[[x]])$$

are fps with positive radii of convergence R_A and R_B, respectively. Let $C(x) \equiv A(x) \cdot B(x) = \sum_{n\ge 0} \big(\sum_{j=0}^{n} a_j b_{n-j}\big) x^n$ as a product of fps. *Then*

$$F_A(x) \cdot F_B(x) = C(x)$$

for every $x \in (-R,\, R)$ where $R = \min(R_A, R_B)$.

Proof. Let $A(x)$, $B(x)$, R_A, R_B, $C(x)$ and R be as stated, and let $x \in (-R, R)$. By Theorem 10.3.3 both series

$$F_A(x) = \textstyle\sum_{n=0}^{\infty} a_n x^n \text{ and } F_B(x) = \sum_{n=0}^{\infty} b_n x^n$$

absolutely converge. By Theorem 3.5.45 we have

$$F_A(x) \cdot F_B(x) = \textstyle\sum_{n=0}^{\infty} \big(\sum_{j=0}^{n} a_j b_{n-j}\big) x^n = C(x)\,.$$

□

Theorem 10.3.14 (products in RAF) *If $f, g \in \mathrm{RAF}$ then $f \cdot g \in \mathrm{RAF}$.*

Proof. Suppose that f, g are real analytic functions and $h \equiv fg$. The domain $M(h) = M(f) \cap M(g)$ is an open set. Let $b \in M(h)$. Since $f, g \in \mathrm{RAF}$, there is a δ such that $U(b, \delta) \subset M(h)$ and for every $x \in U(b, \delta)$ we have

$$f(x) = \textstyle\sum_{n=0}^{\infty} a_n (x-b)^n \ \text{ and } \ g(x) = \textstyle\sum_{n=0}^{\infty} b_n (x-b)^n \,.$$

Since these series absolutely converge, by Theorem 3.5.45 we have for the same x that

$$h(x) = f(x) \cdot g(x) = \textstyle\sum_{n=0}^{\infty} \big(\sum_{j=0}^{n} a_j b_{n-j} \big) (x-b)^n \,.$$

Thus h is real analytic. $\square$

Proposition 10.3.15 (reciprocals of power series 1) *Suppose that*

$$A(x) = \textstyle\sum_{n \geq 0} a_n x^n \quad (\in \mathbb{R}[[x]])$$

is a fps with $a_0 \neq 0$ *and positive radius of convergence* R_A*. Let* $B(x) \equiv A(x)^{-1} = \frac{1}{A(x)} = \sum_{n=0}^{\infty} b_n x^n$ *be the* fps *multiplicative inverse of* $A(x)$*. Then*

$$\frac{1}{F_A(x)} = B(x)$$

for every $x \in (-R,\, R)$ *where* $0 < R \leq R_A$*.*

Proof. Let $A(x)$, R_A and $B(x)$ be as stated. More or less by Proposition 10.2.16 we have

$$B(x) = \textstyle\sum_{n=0}^{\infty} b_n x^n = a_0^{-1} \sum_{k \geq 0} (-1)^k (a_1' x + a_2' x^2 + \dots)^k \quad (\in \mathbb{R}[[x]])$$

where $a_j' = a_0^{-1} a_j$. Our task is to deduce from this representation of $B(x)$ that $B(x)$ has a positive radius of convergence. Comparing the coefficients we have for every $n \in \mathbb{N}$ the formula

$$b_n = a_0^{-1} \textstyle\sum_{k \geq 1} (-1)^k \sum_{\substack{m_1,\, \dots,\, m_k \in \mathbb{N} \\ m_1 + \dots + m_k = n}} a_{m_1}' \dots a_{m_k}' \,.$$

Since

$$\limsup_{n \to \infty} |a_n'|^{1/n} = \limsup_{n \to \infty} |a_n|^{1/n} = \tfrac{1}{R_A} < +\infty \,,$$

there is a real constant $c \geq \frac{1}{R_A} > 0$ such that $|a_n'| \leq c^n$ for every $n \in \mathbb{N}$. In fact, we may set

$$c \equiv \sup\big(\{|a_n'|^{1/n} \colon\ n \in \mathbb{N}\}\big) \,.$$

Hence we bound b_n for every $n \in \mathbb{N}$ by

$$\begin{aligned}
|b_n| &\leq |a_0^{-1}| \textstyle\sum_{k \geq 1} \sum_{\substack{m_1,\, \dots,\, m_k \in \mathbb{N} \\ m_1 + \dots + m_k = n}} c^{m_1} \dots c^{m_k} \\
&= |a_0^{-1}| c^n \textstyle\sum_{k=1}^{n} [x^n] \big(\frac{x}{1-x}\big)^k = |a_0^{-1}| c^n \sum_{k=1}^{n} [x^{n-k}] (1-x)^{-k} \\
&= |a_0^{-1}| c^n \textstyle\sum_{k=1}^{n} (-1)^{n-k} \binom{-k}{n-k} = |a_0^{-1}| c^n \sum_{k=1}^{n} \binom{n-1}{n-k} \\
&= |a_0^{-1}| c^n \textstyle\sum_{j=0}^{n-1} \binom{n-1}{j} \leq |a_0^{-1}| (2c)^n \,.
\end{aligned}$$

Thus $\limsup |b_n|^{1/n} \leq 2c$ and $B(x)$ has a positive radius of convergence $R \leq \frac{1}{2c} \leq R_A$. $\Box$

We need this result for general center b.

Corollary 10.3.16 (reciprocals of power series 2) *Let $b \in \mathbb{R}$ and let*

$$A(x) = \textstyle\sum_{n \geq 0} a_n (x-b)^n$$

be a power series with center b, $a_0 \neq 0$ and positive radius of convergence R_A. Let $B(x) \equiv (\sum_{n \geq 0} a_n x^n)^{-1} = \sum_{n=0}^{\infty} b_n x^n$ be the fps *multiplicative inverse.* *<u>Then</u>*

$$\frac{1}{F_A(x)} = \sum_{n=0}^{\infty} b_n (x-b)^n$$

for every $x \in (b-R,\, b+R)$ where $0 < R \leq R_A$.

Proof. This follows from the previous proposition by replacing x appropriately with $x - b$. $\Box$

Exercise 10.3.17 *Prove the corollary in detail.*

Theorem 10.3.18 (division in RAF) *If $f, g \in$ RAF <u>then</u> $f/g \in$ RAF.*

Proof. In view of Theorem 10.3.14 and the identity

$$f/g = f \cdot \tfrac{1}{g}$$

it suffices to show that if $g \in$ RAF then also $\frac{1}{g} \in$ RAF.

Suppose that g is a real analytic function and $h \equiv \frac{1}{g}$. The domain $M(h) = M(g) \setminus Z(g)$ is an open set by the proof of Theorem 7.6.5. Let $b \in M(h)$. Since $g \in$ RAF, there is a δ such that $U(b, \delta) \subset M(h)$ and for every $x \in U(b, \delta)$ we have

$$g(x) = \textstyle\sum_{n=0}^{\infty} b_n (x-b)^n\,.$$

Since $b \notin Z(g)$, we have $b_0 \neq 0$. Hence by Corollary 10.3.16 there is a $\theta \in (0, \delta)$ and a power series $C(x)$ with center b such that

$$\tfrac{1}{g(x)} = C(x)$$

for every $x \in U(b, \theta)$, as we need. $\Box$

• *Composition in* RAF *and real analyticity of simple elementary functions.*

Theorem 10.3.19 (SEF $\subset$ RAF**)** *Every simple elementary function is real analytic.*

Proof.

$\square$

• *Abel's theorem.* This classical theorem is due to the Norwegian mathematician *Niels H. Abel (1802–1829).* We state it in not completely standard way.

Theorem 10.3.20 (Abel's) *For every power series*

$$A(x) = \textstyle\sum_{n=0}^{\infty} a_n(x-b)^n$$

the sum function $F_A\colon I(A(x)) \to \mathbb{R}$ is continuous.

In the proof we use an inequality due also to N. H. Abel.

Proposition 10.3.21 (Abel's inequality) *Let $n \in \mathbb{N}$ and a_1, ..., a_n, b_1, ..., b_n be $2n$ real numbers such that $b_1 \ge b_2 \ge \cdots \ge b_n \ge 0$. Then*

$$\left|\sum_{j=1}^{n} a_j b_j\right| \le \max\left(\left\{\left|\textstyle\sum_{j=1}^{m} a_j\right|:\ m \in [n]\right\}\right) \cdot b_1 \equiv A \cdot b_1 \,.$$

Proof. For $m \in [n]$ we set $A_m \equiv \sum_{j=1}^{m} a_j$ and define $A_0 = b_{n+1} \equiv 0$. Using in the right moment the triangle inequality we get

$$\begin{aligned} \left|\textstyle\sum_{j=1}^{n} a_j b_j\right| &= \left|\textstyle\sum_{j=1}^{n}(A_j - A_{j-1})b_j\right| = \left|\textstyle\sum_{j=1}^{n} A_j(b_j - b_{j+1})\right| \\ &\le \textstyle\sum_{j=1}^{n} |A_j|(b_j - b_{j+1}) \le A \cdot \sum_{j=1}^{n}(b_j - b_{j+1}) \\ &= A \cdot b_1 \,. \end{aligned}$$

$\square$

Proof of Theorem 10.3.20. We may assume that the center $b = 0$ and that $I \equiv I(A(x)) \ne \{0\}$. Let $y \in I$, we want to show that

$$\lim_{x \to y} F_A(x) = F_A(y) \,.$$

We may assume that $y > 0$. First we show that

$$\lim_{x \to y^-} F_A(x) = F_A(y) \,.$$

Let an $\varepsilon > 0$ be given. Since the series $\sum_{n=0}^{\infty} a_n y^n$ converges, we can take an $N \in \mathbb{N}$ such that

$$\left|\textstyle\sum_{j=m}^{n} a_j y^j\right| \le \tfrac{\varepsilon}{2}$$

whenever $n \ge m \ge N$. Then we take a small $\delta \in (0, y)$ such that for every $x \in (y - \delta, y)$ we have

$$\left|\textstyle\sum_{j=0}^{N} a_j y^j - \sum_{j=0}^{N} a_j x^j\right| \le \tfrac{\varepsilon}{2} \,.$$

Now for every $x \in (y-\delta, y)$ and every $n > N$ we have by Abel's inequality, used on the third line below, that

$$\begin{aligned}
&\left|\textstyle\sum_{j=0}^{n} a_j y^j - \sum_{j=0}^{n} a_j x^j\right| \\
&\le \left|\textstyle\sum_{j=0}^{N} a_j y^j - \sum_{j=0}^{N} a_j x^j\right| + \left|\textstyle\sum_{j=N+1}^{n} a_j y^j - \sum_{j=N+1}^{n} a_j x^j\right| \\
&\le \tfrac{\varepsilon}{2} + \left|\textstyle\sum_{j=N+1}^{n} a_j y^j \cdot \left(1 - \left(\frac{x}{y}\right)^j\right)\right| \le \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} \cdot 1 = \varepsilon\,.
\end{aligned}$$

Thus $|F_A(y) - F_A(x)| \le \varepsilon$ for every $x \in (y-\delta, y)$ and we get the left-sided limit. If $y = \max(I)$, we are done. If $y \ne \max(I)$ then $y \in I^0$ and we prove the above limit by the argument used in the proof of Theorem 6.7.3 (Exercise 10.3.23). □

Thus, unlike real analyticity, continuity of the sum function always extends to endpoints of the interval of convergence. Usually one understands by Abel's theorem only the left-sided continuity of F_A. There is a version of Abel's theorem for complex power series.

Exercise 10.3.22 *Justify the reductions at the beginning of the proof.*

Exercise 10.3.23 *Prove in detail that the sum function of a power series is continuous at every interior point of the interval of convergence.*

Exercise 10.3.24 *Deduce by Abel's theorem the implication*

$$\textstyle\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n} x^n = \log(1+x) \text{ on } (1-\delta, 1) \Rightarrow \sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n} = \log 2\,.$$

Similarly we get the second summation in Theorem 3.5.33.

Corollary 10.3.25 (two applications) *We have, with equal signs,*

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} (\pm 1)^{2n-1} = \arctan(\pm 1) = \pm \frac{\pi}{4}$$

and

$$\sum_{n=1}^{\infty} (-1)^{n-1} \frac{\binom{-1/2}{n-1}}{2n-1} (\pm 1)^{2n-1} = \arcsin(\pm 1) = \pm \frac{\pi}{2}\,.$$

Proof. Since both functions are odd, it suffices to consider only the case with sign $+$. By Theorem 10.1.26,

$$\textstyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} x^{2n-1} = \arctan x \text{ on } (-1, 1)\,.$$

The series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1}$ converges by Theorem 3.5.31. Thus

$$\textstyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} 1^n = \lim_{x \to 1} \arctan x = \arctan 1 = \frac{\pi}{4}$$

by Abel's theorem. The proof of the second summation is very similar. □

• *Pringsheim's theorem.* This theorem is due to the German mathematician *Alfred Pringsheim (1850–1941)* who was the father in law of the German novelist *Thomas Mann (1875–1955).* Pringsheim's theorem tells us that for power series with nonnegative coefficients real analyticity does not extend to endpoints of the interval of convergence.

Theorem 10.3.26 (Pringsheim's) *Suppose that a power series*

$$A(x) = \textstyle\sum_{n=0}^{\infty} a_n(x-b)^n$$

has all coefficients $a_n \ge 0$ *and has a nontrivial and bounded interval of convergence* I *with* $I^0 = (c, d)$. *Then there does not exist real analytic function*

$$f(x)\colon (d-\delta,\, d+\delta) \to \mathbb{R} \ \text{ with } \ \delta \in (0,\, d-c)$$

such that

$$f(x) = A(x) \ \text{ on } \ (d-\delta,\, d)\,.$$

For the proof we need the next auxiliary proposition which brings us back to the beginning of the book.

Proposition 10.3.27 (nonnegative double sum) *Let*

$$f\colon \mathbb{N}_0^2 \to [0,\, +\infty)$$

be an infinite table (matrix) with nonnegative real entries. Then we always have the equality of double sums

$$\sum_{m=0}^{\infty}\sum_{n=0}^{\infty} f(m,\, n) = \sum_{n=0}^{\infty}\sum_{m=0}^{\infty} f(m,\, n) \quad (\in [0,\, +\infty) \cup \{+\infty\})\,.$$

Proof. You proved it already in Exercise ... . □

The following proof of Pringsheim's theorem is taken from [34, pp. 240–242].

Proof of Theorem 10.3.26. We may assume that the center $b = 0$. Let R $(\in (0, +\infty))$ be the radius of convergence of $A(x) = \sum_{n=0}^{\infty} a_n x^n$. We assume for the contrary that there is a $\delta \in (0, R)$ and a real analytic function

$$f\colon (R-\delta,\, R+\delta) \to \mathbb{R}$$

such that $A(x) = f(x)$ for every number $x \in (R-\delta, R)$. Let $h \in (0, \frac{1}{3}\delta)$ and $z \equiv R - h$ (> 0). By the proof of Theorem 10.3.9 we have the expansion $f(x) = \sum_{n=0}^{\infty} b_n(x-z)^n$ around z with the coefficients

$$b_n = \textstyle\sum_{m=n}^{\infty} \binom{m}{n} a_m z^{m-n}\,.$$

In particular, $b_n \geq 0$ for every $n \in \mathbb{N}_0$. Below we use Proposition 10.3.27 on the third line and the binomial theorem on the fourth line and get the contradiction

$$\begin{array}{rcl}
f(R+h) & = & \sum_{n=0}^{\infty} b_n (2h)^n \\
& = & \sum_{n=0}^{\infty} \big(\sum_{m=n}^{\infty} \binom{m}{n} a_m z^{m-n} \big) (2h)^n \\
& = & \sum_{n=0}^{\infty} \big(\sum_{m=0}^{\infty} \binom{m}{n} a_m z^{m-n} \big) (2h)^n \\
& = & \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \binom{m}{n} a_m z^{m-n} (2h)^n \\
& = & \sum_{m=0}^{\infty} \sum_{n=0}^{m} \binom{m}{n} a_m z^{m-n} (2h)^n \\
& = & \sum_{m=0}^{\infty} a_m (R+h)^m = A(R+h)
\end{array}$$

because the number $R+h$ lies outside the interval $I(A(x))$. □

For complex power series Pringsheim's theorem plays an even more important role.

Exercise 10.3.28 *Justify the reduction at the beginning of the proof.*

• *The Pringsheim–Boas theorem, without proof.* A. Pringsheim had published another theorem in [63] in 1893. Forty years later, the American mathematician *Ralph P. Boas Jr. (1912–1992)* discovered that Pringsheim's proof was fallacious and gave a correct proof, see [9]. The theorem says that moderate growth of all Taylor coefficients suffices for a function to be real analytic. For the sake of brevity we relegate the proof to *MA 1*$^+$.

Theorem 10.3.29 (Pringsheim–Boas) *Let $a < b$ be in $\mathbb{R}$ and $f \in \mathcal{C}^\infty((a,b))$. For $j \in \mathbb{N}_0$ and $t \in (a,b)$ we set $a_j(t) \equiv \frac{1}{j!} f^{(j)}(t)$ and define*

$$\rho(t) \equiv \frac{1}{\limsup_{j \to \infty} |a_j(t)|^{1/j}} \,.$$

If $\rho(t) \geq \delta > 0$ for every $t \in (a,b)$, then f is real analytic.

Exercise 10.3.30

10.4 *Asymptotics of ordered partitions

After the preparation in Section 10.2 (definition of op_n and the formula in Corollary 10.2.19) and in the previous section (Pringsheim's theorem) we finally obtain the rough asymptotics of the numbers op_n of nonempty ordered partitions of $[n]$ (Definition 10.2.1).

• *Asymptotics of* op_n. It is as follows.

Theorem 10.4.1 (rough asymptotics of op_n) *These numbers satisfy*

$$\lim_{n \to \infty} \left(\frac{\mathrm{op}_n}{n!} \right)^{1/n} = \frac{1}{\log 2} \,.$$

Equivalently (Exercise 10.4.3),

$$\mathrm{op}_n = (\log 2)^{-n+o(n)} \cdot n! \ \ (n \to \infty)\,.$$

We know from Corollary 10.2.19 the formula for the EGF of the numbers op_n, but this is only a formal relation in the ring $\mathbb{R}[[x]]_{\mathrm{ri}}$. We convert it to a functional relation by means of the next proposition whose proof makes use of Pringsheim's theorem, Proposition 10.3.15 and Theorem 10.3.18.

Proposition 10.4.2 (determining R) *Let $a > 0$ and*

$$A(x) = \textstyle\sum_{n\ge 0} a_n x^n \ \ (\in \mathbb{R}[[x]])$$

be a fps *with $R(A(x)) > a$ such that $F_A(x) \ne 0$ on $(-a, a)$, hence $a_0 \ne 0$, but $F_A(a) = 0$. We further assume that the* fps *multiplicative inverse*

$$B(x) = A(x)^{-1} = \tfrac{1}{A(x)} = \textstyle\sum_{n\ge 0} b_n x^n \ \ (\in \mathbb{R}[[x]])$$

has all coefficients $b_n \ge 0$. Then its radius of convergence $R(B(x)) = a$.

Proof. If $a_0 = 0$ then $F_A(0) = 0$, hence $a_0 \ne 0$ and by Proposition 10.2.9 we have the fps inverse $B(x)$. By Proposition 10.3.15 the fps $B(x)$ has positive radius of convergence R_B. Since the finite limit

$$\lim_{x \to a} \tfrac{1}{F_A(x)}$$

does not exist, Proposition 10.3.15 implies that $0 < R_B \le a$. Strict inequality $R_B < a$ would contradict, with the help of the real analytic function

$$f(x) = \tfrac{1}{F_A(x)} \,|\, U(R_B,\, a - R_B)$$

obtained via Theorem 3.1.17, Pringsheim's Theorem 10.3.26. Hence $R_B = a$. □

Proof of Theorem 10.4.1. Using Corollary 10.2.19 and the previous proposition with $a = \log 2$ and the sum function $F_A(x) = 2 - \mathrm{e}^x$ we get that the EGF

$$\textstyle\sum_{n\ge 0} \frac{\mathrm{op}_n}{n!} \cdot x^n$$

has the radius of convergence $R = \log 2$. Thus

$$\limsup_{n\to\infty} \left(\tfrac{\mathrm{op}_n}{n!}\right)^{1/n} = \tfrac{1}{\log 2}\,.$$

We know by Exercise 10.2.6 that the numbers $\frac{\mathrm{op}_n}{n!}$ are supermultiplicative. Using Corollary 2.5.8 we deduce that

$$\lim_{n\to\infty} \left(\tfrac{\mathrm{op}_n}{n!}\right)^{1/n} = \tfrac{1}{\log 2}\,.$$

□

Exercise 10.4.3 *The two statements of the theorem are equivalent.*

10.5 *Arnol'd's limits

In the booklet [5, p. 21], see [6] for an English translation, the Russian mathematician *Vladimir I. Arnol'd (1937–2010)* challenges the reader to compute the following limit.

> Here is an example of a problem that people like Barrow, Newton or Huygens would solve in a couple of minutes, but contemporary mathematicians, in my opinion, are unable to solve it quickly (in any case I have not yet seen a mathematician who could cope with it quickly): evaluate
>
> $$\lim_{x\to 0}\frac{\sin(\tan x)-\tan(\sin x)}{\arcsin(\arctan x)-\arctan(\arcsin x)}\,.$$

Indeed, for a long time I could not do anything with the limit. In one July night in 2025 I only laboriously computed that the numerator and denominator are both

$$-\tfrac{1}{30}x^7+o(x^7)\ \ (x\to 0)\,,$$

so that the limit equals 1. But then a solution occurred to me. Limits of the form

$$\lim_{x\to 0}\frac{f(x)-g(x)}{g^{\langle -1\rangle}(x)-f^{\langle -1\rangle}(x)}=1\,,$$

like Arnol'd's, follow from the next result on inverse fps (formal power series).

Theorem 10.5.1 (fps and its inverse) *For $n\in\mathbb{N}$ there exist real polynomials*

$$p_n(x_1,\,x_2,\,\dots,\,x_n)$$

with n variables such that the following holds. If $A(x)=x+\sum_{n\ge 2}a_nx^n$ is a fps *and*

$$B(x)=A^{\langle -1\rangle}(x)=x+\sum_{n\ge 2}b_nx^n$$

is the fps *inverse of $A(x)$, <u>then</u> $b_2=-a_2$ and for every $n\ge 3$ we have*

$$b_n=-a_n+p_{n-2}(a_2,\,a_3,\,\dots,\,a_{n-1})\,.$$

The mentioned polynomials are functions $p_n\colon\mathbb{R}^n\to\mathbb{R}$ given as finite sums of <u>monomials</u>, products of the form

$$ax_1^{m_1}x_2^{m_2}\dots x_n^{m_n}$$

where $a\in\mathbb{R}$, $m_i\in\mathbb{N}_0$ and x_i are variables ranging in $\mathbb{R}$. The fps $A(x)$ and $B(x)$ satisfy

$$A(B(x))=B(A(x))=x.$$

We treat inverse fps in the next passage where we prove the theorem. Now we easily deduce from it Arnol'd's limits in general form.

Corollary 10.5.2 (Arnol'd's limits) *Suppose that*

$$f,\, g\colon (-\delta,\, \delta) \to \mathbb{R}$$

are two injective and real analytic functions such that $f \neq g$, $f(0) = g(0) = 0$ and $f'(0) = g'(0) = 1$. Then

$$\lim_{x\to 0} \frac{f(x) - g(x)}{g^{\langle -1\rangle}(x) - f^{\langle -1\rangle}(x)} = 1\,.$$

Proof. Let

$$f(x) = x + \textstyle\sum_{n\geq 2} a_n x^n \;\text{ and }\; g(x) = x + \sum_{n\geq 2} b_n x^n$$

be the expansions of the functions around 0 and $m \in \mathbb{N}$ with $m \geq 2$ be minimum such that $a_m \neq b_m$. If $m = 2$ then by Theorem 10.5.1 we have

$$\lim_{x\to 0} \tfrac{f(x)-g(x)}{g^{\langle -1\rangle}(x)-f^{\langle -1\rangle}(x)} = \lim_{x\to 0} \tfrac{a_2x^2-b_2x^2+o(x^2)}{-b_2x^2+a_2x^2+o(x^2)} = 1\,.$$

Let $m \geq 3$. Since $a_2 = b_2,\, \ldots,\, a_{m-1} = b_{m-1}$, by Theorem 10.5.1 these equalities hold also for the coefficients of $x^2,\, \ldots,\, x^{m-1}$ in the expansions of $f^{\langle -1\rangle}(x)$ and $g^{\langle -1\rangle}(x)$. Using Theorem 10.5.1 we again compute

$$\begin{aligned}
&\lim_{x\to 0} \tfrac{f(x)-g(x)}{g^{\langle -1\rangle}(x)-f^{\langle -1\rangle}(x)} = \\
&= \lim_{x\to 0} \tfrac{a_mx^m-b_mx^m+o(x^m)}{-b_mx^m+p_{m-2}(b_2,\,\ldots,\,b_{m-1})x^m+a_mx^m-p_{m-2}(a_2,\,\ldots,\,a_{m-1})x^m+o(x^m)} = \\
&= \lim_{x\to 0} \tfrac{a_mx^m-b_mx^m+o(x^m)}{-b_mx^m+a_mx^m+o(x^m)} = 1\,.
\end{aligned}$$

□

• *Inverses of* fps. We did not consider this operation on $\mathbb{R}[[x]]$ in Section 10.2 and we treat it briefly now. We look at it more thoroughly in the next final section.

Definition 10.5.3 (inverse fps) *Let $A(x)$ and $B(x)$ be two* fps *with zero constant terms. We say that $B(x)$ is the inverse of $A(x)$, and we write $B(x) = A^{\langle -1\rangle}(x)$, if*

$$A(B(x)) = x\,.$$

Exercise 10.5.4 *If $B(x) = A^{\langle -1\rangle}(x)$ then also $B(A(x)) = x$.*

Proposition 10.5.5 (inverse fps) *The inverse of a* fps *$A(x)$ is unique. It exists $\iff$ $[x^0]A(x) = 0$ and $[x^1]A(x) \neq 0$.*

Proof. It is easy to see that if $[x^0]A(x) \neq 0$ or if $[x^1]A(x) = 0$ then the inverse of $A(x)$ does not exist. Suppose that $A(x) = \sum_{n\geq 0} a_n x^n$ with $a_0 = 0$ and $a_1 \neq 0$. We are looking for a fps $B(x) = \sum_{n\geq 1} b_n x^n$ such that

$$A(B(x)) = \textstyle\sum_{n\geq 1} a_n(b_1x + b_2x^2 + \ldots)^n = x\,.$$

This relation is equivalent with the system of equations

$$a_1b_1 = 1,\; a_1b_2 + a_2b_1^2 = 0,\; \ldots,\; a_1b_n + q_n(a_1,\, \ldots,\, a_n,\, b_1,\, \ldots,\, b_{n-1}) = 0,\; \ldots$$

where $n \geq 3$ and $q_n(\ldots)$ is a real polynomial with $2n-1$ variables. It is clear that the system has a unique solution (b_n) in terms of (a_n): the sequence (b_n) begins with $b_1 = a_1^{-1}$ and if $b_1,\, b_2,\, \ldots,\, b_n$ are already determined, then

$$b_{n+1} = -a_1^{-1} \cdot q_{n+1}(a_1,\, \ldots,\, a_{n+1},\, b_1,\, \ldots,\, b_n)\,.$$

□

Proof of Theorem 10.5.1. In view of the previous proof this proof is easy. We assume $a_1 = b_1 = 1$ and get from the system that $b_2 = -a_1^{-1}a_2b_1^2 = -a_2$. For $n \geq 3$ we see that the n-th equation in the system is in more detail

$$a_1b_n + a_nb_1^n + r_n(a_1,\, \ldots,\, a_{n-1},\, b_1,\, \ldots,\, b_{n-1}) = 0$$

where $r_n(\ldots)$ is a real polynomial with $2n-2$ variables. We get by induction

$$\begin{aligned} b_n &= -a_1^{-1}a_nb_1^n - a_1^{-1}r_n(a_1,\, \ldots,\, a_{n-1},\, b_1,\, \ldots,\, b_{n-1}) \\ &= -a_n - r_n\big(1,\, a_2,\, \ldots,\, a_{n-1}, \\ &\qquad 1,\, -a_2,\, -a_3 + p_1(a_2),\, \ldots,\, -a_{n-1} + p_{n-3}(a_2,\, \ldots,\, a_{n-2})\big) \\ &= -a_n + p_{n-2}(a_2,\, a_3,\, \ldots,\, a_{n-1}) \end{aligned}$$

where $p_j(\ldots)$ are real polynomials with j variables. □

10.6 *Inverses of Taylor polynomials and series

Chapter 11

Newton integral

Chapter 12

Riemann integral

Chapter 13

Henstock–Kurzweil integral

Chapter 14

Applications of integrals

Appendix A

Auxiliary notions and notation

In Section A.1 we review notation and notions related to logic and set theory. Definition A.1.3 of (ordered) k-tuples, which builds on Kuratowski's pairs in Definition A.1.1, is of interest. In Section A.2 we develop naive ZFC set theory to the extent sufficient for the introduction of natural numbers in Section 1.2. Section A.3 is devoted to literary ZFC set theory with classes; this is our understanding of set theory. In Section A.4, we tackle the most difficult question of all: How do we know that a (mathematical) theorem is true. Section A.5 introduces, in the spirit of the first chapter, the field $\mathbb{C}$ of complex numbers. Metric spaces are recalled in Section A.6.

A.1 Logical and set-theoretic notation

We review some notation from mathematical logic and set theory. Recall the Fraktur hand form of Latin letters:

$$\mathfrak{a},\ \mathfrak{A},\ \mathfrak{b},\ \mathfrak{B},\ \mathfrak{c},\ \mathfrak{C},\ \mathfrak{d},\ \mathfrak{D},\ \mathfrak{e},\ \mathfrak{E},\ \mathfrak{f},\ \mathfrak{F},\ \mathfrak{g},\ \mathfrak{G},\ \mathfrak{h},\ \mathfrak{H},\ \mathfrak{i},\ \mathfrak{I},\ \mathfrak{j},\ \mathfrak{J},\ \mathfrak{k},\ \mathfrak{K},\ \mathfrak{l},\ \mathfrak{L},\ \mathfrak{m},\ \mathfrak{M},\ \mathfrak{n},\ \mathfrak{N},$$

$$\mathfrak{o},\ \mathfrak{O},\ \mathfrak{p},\ \mathfrak{P},\ \mathfrak{q},\ \mathfrak{Q},\ \mathfrak{r},\ \mathfrak{R},\ \mathfrak{s},\ \mathfrak{S},\ \mathfrak{t},\ \mathfrak{T},\ \mathfrak{u},\ \mathfrak{U},\ \mathfrak{v},\ \mathfrak{V},\ \mathfrak{w},\ \mathfrak{W},\ \mathfrak{x},\ \mathfrak{X},\ \mathfrak{y},\ \mathfrak{Y},\ \mathfrak{z}\ \text{and}\ \mathfrak{Z}\,.$$

The Greek alphabet is used in mathematical notation more frequently:

$$\alpha,\ \beta,\ \Gamma,\ \gamma,\ \Delta,\ \delta,\ \varepsilon,\ \zeta,\ \eta,\ \Theta,\ \theta,\ \vartheta,\ \iota,\ \kappa,\ \Lambda,\ \lambda,\ \mu,\ \nu,\ \Xi,\ \xi,\ o,\ \Pi,\ \pi,\ \rho,\ \Sigma,\ \sigma,$$

$$\tau,\ \Upsilon,\ \upsilon,\ \Phi,\ \phi,\ \varphi,\ \chi,\ \Psi,\ \psi,\ \Omega\ \text{and}\ \omega$$

— alpha, beta, gamma, delta, epsilon, zeta, eta, theta, iota, kappa, lambda, mu, nu, xi, omicron, pi, rho, sigma, tau, upsilon, phi, chi, psi, omega; the omitted capital letters, such as A for α, B for β, Z for ζ or H for η, are identical with their Latin form.

We use $=$ mostly as a set-theoretic equality, which is governed by the axiom of extensionality (Axiom A.2.1), but also as a defining equality. In [44], the logic of equality is nicely explained. The symbols $:=$ and $=:$ mean defining equality; for example, $[n] := \{1, 2, \dots, n\}$ or $\{0, 1, 2, \dots\} =: \mathbb{N}_0$. We define $\mathbb{N} := \{1, 2, \dots\}$. By $\neq$ we denote non-equality.

As for logical connectives, we write $\neg$ for the negation ("it is not true that ..."), $\vee$ for the disjunction ("... or ..."), $\wedge$ or & for the conjunction ("... and ..."), $\to$ for the implication ("if ... then ...") and $\leftrightarrow$ for the equivalence ("... if and only if ..."). Sometimes we abbreviate the last phrase by " ... iff ...". We write $\exists$ ("there is ...") and $\forall$ ("for every ...") for the existential and the general quantifier, respectively. For example,

$$\exists\, x\colon\ x > 0 \ \text{ and } \ \forall\, a\colon\ a \neq b$$

mean, respectively, that there is an element in the set S (understood from the context) that is greater than zero, and that every element in S differs from the element b. We use symbols $\Rightarrow$ and $\Leftarrow$ for metamathematical implications. The symbol $\iff$ expresses metamathematical equivalence.

We use standard set-theoretic notation: $\in$ for the membership ("... is an element of ..."), $\notin$ for the non-membership, $\cup$ for the union ($a \cup b = \{c\colon\ c \in a \vee c \in b\}$), $\cap$ for the intersection ($a \cap b = \{c \in a\colon\ c \in b\}$) and $\setminus$ for the set difference ($a \setminus b = \{c \in a\colon\ c \notin b\}$). By $\emptyset$ we denote the empty set, the unique set with no elements. Recall that for a set A its sum is the set

$$\bigcup A = \{x\colon\ \exists\, y\colon\ x \in y \wedge y \in A\}\,.$$

Similarly, for a nonempty set A its intersection is the set

$$\bigcap A = \{x\colon\ \forall\, y\colon\ y \in A \to x \in y\}\,.$$

In this book, we use the set theory called ZFC *set theory with classes*, which is explained in the next section. There we will see that $\bigcap \emptyset$ is the proper class of all sets.

The following definition of ordered pairs is standard.

Definition A.1.1 *Let x and y be sets. Their ordered pair is the set*

$$[x,\, y] := \{\{x\},\, \{x,\, y\}\}\,.$$

This definition is due to *Kazimierz Kuratowski (1896–1980)* in 1921. With the help of the axiom of extensionality, it is easy to prove the basic property of ordered pairs.

Proposition A.1.2 *For every four sets a, b, c, d, we have*

$$[a,\, b] = [c,\, d] \iff a = c \wedge b = d\,.$$

One standardly generalizes ordered pairs to ordered triples, quadruples, ... by repeated application of pair:

$$[a,\, b,\, c] := [a,\, [b,\, c]],\ \ [a,\, b,\, c,\, d] := [a,\, [b,\, [c,\, d]]],\ \ldots\,.$$

These and similar definitions have the defect, apparently often not noticed, that the arity of the tuple cannot be determined solely from it; context is needed. It is not clear, for example, if the set $[a, b, c, d]$ means an ordered pair of the sets a and $[b, c, d]$, or an ordered triple of the sets a, b, and $[c, d]$, or an ordered quadruple of the sets a, b, c, and d. We therefore prefer an alternative definition of (ordered) tuples.

Definition A.1.3 *Let $x_1,\, x_2,\, \ldots,\, x_k$ be $k \ge 2$ sets. We define their k-tuple as the set*

$$\langle x_1,\, x_2,\, \ldots,\, x_k\rangle := \{[\{\emptyset\},\, x_1],\, [\{\{\emptyset\}\},\, x_2],\, \ldots,\, [\{\{\ldots\{\emptyset\}\ldots\}\},\, x_k]\}\,.$$

Such tuples determine their arity.

Proposition A.1.4 *Let $k, l \in \mathbb{N} \setminus \{1\}$, let $a = \langle x_1,\, x_2, \ldots, x_k\rangle$, and let $b = \langle y_1,\, y_2, \ldots, y_l\rangle$. Then*

$$a = b \iff k = l \wedge x_1 = y_1 \wedge x_2 = y_2 \wedge \cdots \wedge x_k = y_k\,.$$

A.2 Naive ZFC set theory

The main purpose of this section is to develop enough of ZFC set theory so that in Section 1.2 natural numbers can be rigorously defined by means of it. At the end of the section, we explain our position on the ontology of sets.

• *Set formulas.* We consider the infinite alphabet

$$A = \{u,\, v,\, w,\, x,\, y,\, z,\, x_0,\, x_1,\, x_2,\, \ldots\}$$

and strings over it. A string s over A is any finite nonempty sequence

$$s = a_1\, a_2\, \ldots\, a_n \ \text{ with } \ a_i \in A\,.$$

• ZFC *axioms.*

Axiom A.2.1 (of extensionality)

Axiom A.2.2 (foundation) *Every nonempty set a has an element x that is $\in$-minimal among all elements of a. Said by a quasi-formula,*

$$\forall a\, (a \ne \emptyset \to (\exists x(x \in a \wedge x \cap a = \emptyset)))\,.$$

This axiom excludes sets a such that $a \in a$, or more generally, any finite cycle $x_1 \in x_2 \in \cdots \in x_k \in x_1$. It also ensures that every chain of set memberships x_1, $x_2 \in x_1$, $x_3 \in x_2 \in x_1$, $\dots$ is, in fact, finite.

Axiom A.2.3 (of separation)

Definition A.2.4 *A set x is <u>inductive</u> if $\emptyset \in x$ and for every set $y \in x$ also $y \cup \{y\} \in x$.*

Axiom A.2.5 (of infinity) *There exists an inductive set.*

It is not hard to write this axiom as a set formula.

A.3 Literary ZFC set theory with classes

Leon Henkin (1921–2006)

A.4 How do we know that a theorem is true?

A.5 Complex numbers

We extend the hierarchy of numeric domains

$$\mathbb{N}_0 - \mathbb{Z} - \mathbb{Q} - \mathbb{R}$$

built in Chapter 1 by the field of complex numbers $\mathbb{C}$. We keep the schema of Sections 1.2–1.5 and obtain the algebraic characterization of $\mathbb{C}$ in Theorem A.5.2 as the unique, up to isomorphism, i-field. Now the linear order leaves the scene.

• *i-fields.* Recall Definition 1.4.1 of fields and recall that every field homomorphism $f\colon F \to K$ is injective. Indeed, if $f(a) = f(b)$ for $a \neq b$, then $f(a) - f(b) = f(a - b) = 0_K$ with $a - b \neq 0_F$ and

$$\begin{aligned} 1_K &= f(1_F) = f((a-b)\cdot(a-b)^{-1}) = f((a-b))\cdot f((a-b)^{-1}) \\ &= 0_K \cdot f((a-b)^{-1}) = 0_K \end{aligned}$$

by part 1 of Proposition 1.3.8, which contradicts the definition of a field. We speak instead of a (field) <u>embedding</u> f of F in K. Or we say that K <u>extends</u> F.

We suppose that the reader is familiar with the basics of vector spaces over fields, especially with bases and dimensions. If $f\colon F \to K$ is an embedding, we can view K as a vector space over F; for $a \in F$ and $b \in K$, the scalar multiplication works by

$$a \cdot b := f(a) \cdot b\,.$$

We denote the dimension of this vector space by $[K : F]$ ($\in \mathbb{N} \cup \{\infty\}$).

Definition A.5.1 *A field F is an* <u>*i-field*</u> *if it extends the field $\mathbb{R}$ and has degree $[F : \mathbb{R}] = 2$.*

The main goal of this section is to prove the next characterization theorem for complex numbers.

Theorem A.5.2 *There exists an i-field. Every two i-fields are isomorphic.*

We prove the former claim in Proposition A.5.7, and the latter claim in Proposition A.5.10. As in the four previous numeric domains, we have the following class.

Corollary A.5.3 *The class*

$$\text{COMPLEX NUMBERS} := \{x\colon\ x \text{ is an } i\text{-field}\}$$

contains the "standard" complex numbers $\mathbb{C}$ and every two sets in it are isomorphic as fields.

• *The set $\mathbb{C}$ of complex numbers.* We define this set.

Definition A.5.4 *The set of* <u>*complex numbers*</u> *is the Cartesian square*

$$\mathbb{C} = \mathbb{R}^2 := \mathbb{R} \times \mathbb{R}\,.$$

We use the standard notation $a + bi := \langle a, b\rangle \in \mathbb{C} = \mathbb{R} \times \mathbb{R}$. If $z = a + bi$, we write $\mathrm{re}(z) := a$ for the <u>*real part*</u> *and $\mathrm{im}(z) := b$ for the* <u>*imaginary part*</u> *of z. For better readability, we sometimes replace $a + bi$ with $a \oplus bi$.*

It is often useful to identify $\mathbb{C}$ with the Euclidean plane $\mathbb{R}^2$ and employ geometric arguments.

• *The arithmetic on $\mathbb{C}$ and the algebraic structure $\mathbb{C}$.* We define, in the standard way, addition and multiplication on $\mathbb{C}$. Then we introduce the algebraic structure $\mathbb{C}$.

Recall the arithmetic on $\mathbb{R}$ introduced in Section 1.5. In the next definition, the operations on the right side of $:=$ are in $\mathbb{R}$.

Definition A.5.5 *Let $w = a + bi$ and $z = c + di$ be complex numbers. We define their sum $+$ and product $\cdot$ as follows.*

1. *We set $w + z := (a + c) \oplus (b + d)i$.*
2. *We set $w \cdot z := (ac - bd) \oplus (ad + bc)i$.*

Definition A.5.6 *The algebraic structure of* <u>*complex numbers*</u>

$$\mathbb{C} = \langle \mathbb{C},\, 0_{\mathbb{C}},\, 1_{\mathbb{C}},\, +,\, \cdot\rangle$$

consists of the set $\mathbb{C} = \mathbb{R}^2$ in Definition A.5.4, the elements $0_{\mathbb{C}} := 0 + 0i$ and $1_{\mathbb{C}} := 1 + 0i$, and the operations $+$ and $\cdot$ in Definition A.5.5.

We usually write just 0 and 1 instead of $0_{\mathbb{C}}$ and $1_{\mathbb{C}}$.

• $\mathbb{C}$ *is an i-field.* We prove the first claim in Theorem A.5.2.

Proposition A.5.7 *The algebraic structure*

$$\mathbb{C} = \langle \mathbb{R}^2,\, 0,\, 1,\, +,\, \cdot \rangle$$

introduced in Definition A.5.6 is an i-field.

Proof. The neutrality of 0 and 1 to + and $\cdot$, respectively, follows easily from Definition A.5.5. From this definition and the fact that $\mathbb{R}$ is a field, it follows that + is commutative and associative, and that $\cdot$ is commutative. We show that $\cdot$ is associative. Let $z_1 = a + bi$, $z_2 = c + di$, and $z_3 = e + fi$ be complex numbers. Then, indeed,

$$\begin{aligned}
(z_1 \cdot z_2) \cdot z_3 &= ((ac - bd) \oplus (ad + bc)i) \cdot (e \oplus fi) \\
&= ((ac - bd)e - (ad + bc)f) \oplus ((ac - bd)f + (ad + bc)e)i \\
&= (a(ce - df) - b(de + cf)) \oplus (a(cf + de) - b(df - ce))i \\
&= (a \oplus bi) \cdot ((ce - df) \oplus (de + cf)i) = z_1 \cdot (z_2 \cdot z_3)\,.
\end{aligned}$$

Similarly,

$$\begin{aligned}
z_1 \cdot (z_2 + z_3) &= (a \oplus bi) \cdot ((c + e) \oplus (d + f)i) \\
&= (a(c + e) - b(d + f)) \oplus (a(d + f) + b(c + e))i \\
&= ((ac - bd) \oplus (ad + bc)i) + ((ae - bf) \oplus (af + be)i) \\
&= z_1 \cdot z_2 + z_1 \cdot z_3
\end{aligned}$$

and we see that $\cdot$ is distributive to +.

Let $z = a + bi \in \mathbb{C}$. Then

$$(a \oplus bi) + (-a \oplus (-b)i) = 0 \oplus 0i = 0$$

and, if $a \neq 0$ or $b \neq 0$ and $c := (a^2 + b^2)^{-1}$,

$$(a \oplus bi) \cdot (ac \oplus (-bc)i) = (a^2 + b^2)c \oplus (-abc + bac)i = 1 \oplus 0i = 1\,.$$

Hence z has an additive inverse, and if $z \neq 0$, it has a multiplicative inverse. We have proven that $\mathbb{C}$ is a field. It remains to show that $\mathbb{C}$ is an extension of $\mathbb{R}$ with degree 2. It is easy to see that

$$f\colon \mathbb{R} \to \mathbb{C},\ f(a) = a \oplus 0i\,,$$

is a field embedding, and that $\{1_{\mathbb{C}}, \iota\}$, where $\iota := 0 + 1i$, is a linear basis of the vector space $\mathbb{C}$ over $\mathbb{R}$. □

• *The field* $\mathbb{C}$ *is completely normed.* In Sections 1.4 and 1.5, we used linear orders on the fields $\mathbb{Q}$ and $\mathbb{R}$ to norm them by absolute values. In $\mathbb{C}$ we cannot have any linear order satisfying the two order axioms, but we can still define a norm. Namely, using Proposition 1.5.27, for $z \in \mathbb{C}$ we set

$$|z| = |a + bi| := \sqrt{a^2 + b^2} \ \ (\in \mathbb{R}_{\geq 0})\,.$$

Proposition A.5.8 *The map* $|\cdots|\colon \mathbb{C} \to \mathbb{R}_{\geq 0}$ *is a complete norm. It means that it has four properties.*

1. $|z| = 0 \iff z = 0$.
2. $|w \cdot z| = |w| \cdot |z|$.
3. $|w + z| \leq |w| + |z|$.
4. *Every sequence* $(z_n) \subset \mathbb{C}$ *that is Cauchy with respect to* $|\cdots|$ *has a limit.*

Proof. 1 is immediate from the definition. 2 follows from the identity ($a, b, c, d \in \mathbb{R}$)

$$(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2\,.$$

We give two proofs for the triangle inequality (TI) in 3.

The algebraic proof goes as follows. If $w = z = 0$, TI is trivial. We assume that $w \neq 0$. We take w^{-1} out from TI via multiplicativity, cancel it, and we see that it suffices to prove, for every $z \in \mathbb{C}$, that

$$|1 + z| \leq |1| + |z| = 1 + |z|\,.$$

Let $z = a + bi$. After squaring, the last inequality is equivalent to

$$(1 + a)^2 + b^2 \leq 1 + 2\sqrt{a^2 + b^2} + a^2 + b^2\,.$$

After an algebraic rearrangement, this is equivalent to the true inequality

$$2a \leq 2\sqrt{a^2 + b^2}\,.$$

Hence, TI holds.

The proof via plane geometry. It is easy to see that TI in $\mathbb{C}$ is equivalent to TI in the Euclidean plane $\mathbb{R}^2$, and we prove the latter. For two distinct points $A, B \in \mathbb{R}^2$ we denote by AB the straight segment joining them, by $\overline{AB}$ the line going through them, and by $|AB|$ ($\in \mathbb{R}_{>0}$) the length of AB. We show in **three steps** that for any three non-collinear points $A, B, C \in \mathbb{R}^2$ we have

$$|AB| < |AC| + |CB|\,.$$

If A, B, and C are collinear, TI is trivial. In **step 1**, we show that if A, B, C form a triangle with right angle at B, then $|AB|, |BC| < |AC|$. This is immediate from the Pythagorean theorem

$$|AB|^2 + |BC|^2 = |AC|^2\,.$$

In **step 2**, we assume for the triangle ABC that $|AB| \geq |AC|, |CB|$ and show that the heel D of the height from C to $\overline{AB}$ lies in AB. Suppose for the contrary that $D \notin AB$ and lies in the half-line going from A through B; the symmetric case is similar. We consider the triangle ADC with right angle at D. Step 1 gives

$$|AB| < |AD| < |AC|\,,$$

contrary to the assumption. Finally, in **step 3** we take the same triangle ABC with AB at least as long as any of the other two sides and prove that $|AB| < |AC| + |CB|$; this will prove TI for $\mathbb{R}^2$. By steps 1 and 2, the heel D lies inside AB. We consider the triangles ADC and CDB with right angles at D. By step 1,

$$|AB| = |AD| + |DB| < |AC| + |CB|\,.$$

4. Let $(z_n) \subset \mathbb{C}$ be Cauchy (see Definition 1.4.21). For $\lim z_n = w$ see Definition 1.4.22. We write $z_n = a_n + b_n i$ and since $|\mathrm{re}(z)|, |\mathrm{im}(z)| \le |z|$ for every $z \in \mathbb{C}$, we see that $(a_n) \subset \mathbb{R}$ and $(b_n) \subset \mathbb{R}$ are Cauchy sequences in $\mathbb{R}$. By Theorem 1.4.28 we have limits

$$a = \lim a_n \ \text{ and } \ b = \lim b_n \quad (\in \mathbb{R})\,.$$

The inequality $|z| \le |\mathrm{re}(z)| + |\mathrm{im}(z)|$ implies that

$$\lim z_n = \lim(a_n + b_n i) = a + bi\,.$$

$\Box$

• *The i-fields are mutually isomorphic.* To conclude the proof of Theorem A.5.2, we employ a lemma.

Lemma A.5.9 *In every i-field F there exists an element $\iota \in F$ such that*

$$\iota^2 = \iota \cdot \iota = -1_F\,.$$

Proof. Let F be an *i*-field. We may assume that $\mathbb{R}$ is a subfield of F. Since $[F : \mathbb{R}] = 2$, we have $F \setminus \mathbb{R} \ne \emptyset$ and take an element $\alpha \in F \setminus \mathbb{R}$. Then the set $\{1_F, \alpha\}$ is linearly independent over $\mathbb{R}$, but $\{1_F, \alpha, \alpha^2\}$ is linearly dependent. It follows that we have an identity

$$a1_F + b\alpha + c\alpha^2 = 0\,,$$

where $a, b, c \in \mathbb{R}$ and $c \ne 0$. Equivalently,

$$(\alpha + b/2c)^2 = b^2/4c^2 - a/c =: d \quad (\in \mathbb{R})\,.$$

If $d \ge 0$, using Proposition 1.5.27 we deduce the contradiction that $\alpha \in \mathbb{R}$. Thus $d < 0$ and again using Proposition 1.5.27, we see that the element

$$\iota := \tfrac{\alpha + b/2c}{\sqrt{-d}} \quad (\in F)$$

has the property that $\iota^2 = -1_F$. $\Box$

Recall Proposition 1.2.7.

Proposition A.5.10 *Suppose that*

$$F = \langle F,\, 0_F,\, 1_F,\, +,\, \cdot \rangle \;\; \textit{and} \;\; G = \langle G,\, 0_G,\, 1_G,\, \oplus,\, \odot \rangle$$

are two i-fields. <u>*Then*</u> *F and G are isomorphic, which means that there exists a bijection $f\colon F \to G$ with two properties.*

1. *For every $a, b \in F$, we have $f(a+b) = f(a) \oplus f(b)$.*
2. *For every $a, b \in F$, we have $f(a \cdot b) = f(a) \odot f(b)$.*

Proof. Using the previous lemma, we take elements $\iota \in F$ and $\kappa \in G$ such that $\iota^2 = -1_F$ and $\kappa^2 = -1_G$. For the simplicity of notation, we assume that $\mathbb{R}$ is a subfield of F, and that $g\colon \mathbb{R} \to G$ is a field embedding. It follows that

$$B = \{1_F,\, \iota\} \;\text{ and }\; B' = \{1_G,\, \kappa\}$$

is a linear basis of F over $\mathbb{R}$ and G over $\mathbb{R}$, respectively. For any $a \in F$ we write $a = u \cdot 1_F + v \cdot \iota \in F$ with $u, v \in \mathbb{R}$. We define a map $f\colon F \to G$ by

$$f(a) := g(u) \odot 1_G \oplus g(v) \odot \kappa\,.$$

We show that f is the desired isomorphism. Since B and B' are bases, f is a bijection. Let $a, b \in F$ with

$$a = u \cdot 1_F + v \cdot \iota \;\text{ and }\; b = u' \cdot 1_F + v' \cdot \iota, \;\text{ where }\; u,\, v,\, u',\, v' \in \mathbb{R}\,.$$

Then

$$\begin{aligned} f(a+b) &= f((u+u') \cdot 1_F + (v+v') \cdot \iota) = g(u+u') \odot 1_G \oplus g(v+v') \odot \kappa \\ &= g(u) \odot 1_G \oplus g(v) \odot \kappa \oplus g(u') \odot 1_G \oplus g(v') \odot \kappa = f(a) \oplus f(b) \end{aligned}$$

and

$$\begin{aligned} f(a \cdot b) &= f((u \cdot u' - v \cdot v') \cdot 1_F + (u \cdot v' + v \cdot u') \cdot \iota) \\ &= g(u \cdot u' - v \cdot v') \odot 1_G \oplus g(u \cdot v' + v \cdot u') \odot \kappa \\ &= (g(u) \odot g(u') \ominus g(v) \odot g(v')) \odot 1_G \oplus \\ &\quad \oplus (g(u) \odot g(v') \oplus g(v) \odot g(u')) \odot \kappa \\ &= (g(u) \odot 1_G \oplus g(v) \odot \kappa) \odot (g(u') \odot 1_G \oplus g(v') \odot \kappa) \\ &= f(a) \odot f(b)\,. \end{aligned}$$

In these computations, we used the definitions of f, ι and κ, the fact that $g\colon \mathbb{R} \to G$ is a field homomorphism, and properties of the operations $+$, $\cdot$, $\oplus$, and $\odot$, like the distributive law. $\square$

The proof of Theorem A.5.2 is complete.

A.6 Metric spaces

We review the basics of metric spaces. They generalize the real line. Many results on real numbers can be extended to metric spaces.

Definition A.6.1 *A metric space is a pair*

$$\langle X,\, d\rangle,\ d\colon X \times X \to [0,\, +\infty)\,,$$

such that $X \neq \emptyset$ and the map d, called a metric or a distance, has three properties.

1. $d(x, y) = 0$ *iff* $x = y$.
2. $d(x, y) = d(y, x)$.
3. *The triangle inequality* $d(x, y) \leq d(x, z) + d(z, y)$ *holds.*

Let $\langle X, d\rangle$ be a metric space, $b \in X$ and $r > 0$ be a real number. The ball in the space, centered at b and with the radius r, is the set

$$B = B(b,\, r) = \{a \in X\colon\ d(a,\, b) < r\}\,.$$

Always $B \neq \emptyset$ because $b \in B$. A set $Y \subset X$ is open if for every $b \in Y$ there exists an $r > 0$ such that $B(b, r) \subset Y$. The set Y is closed if the complement $X \setminus Y$ is open. The set Y is bounded if for some $c \geq 0$ we have $d(x, y) \leq c$ for every two points $x, y \in Y$. It is easy to prove that Y is bounded iff for some, equivalently every, point $b \in X$ there is a radius $r > 0$ such that $Y \subset B(b, r)$.

A sequence of points $(b_n) \subset X$ converges to the limit $b \in X$ if for every ε there is an n_0 such that for every $n \geq n_0$ we have

$$d(b,\, b_n) < \varepsilon\,.$$

The set $Y \subset X$ is compact if every sequence $(a_n) \subset Y$ has a convergent subsequence with the limit in Y.

Appendix B

Solutions to exercises

1 Four numeric domains

Exercise 1.1.4 Implication $\Rightarrow$ is trivial. We prove $\Leftarrow$. Suppose that $G_f = G_g$. Then A is the set of elements that appear as the first components of the pairs in G_f. Similarly for C. We deduce that $A = C$. Thus $A = C$ and $G_f = G_g$, which means that f and g are congruent.

Exercise 1.1.5 Let C be a partial function from A to B. We consider the set $A' = \{a \in A\colon\ \exists\, b \in B\colon a\, C\, b\}$. Then $\langle A', B, C\rangle$ is a function from A' to B.

Exercise 1.1.6 This is immediate from Exercise 1.1.4.

Exercise 1.1.7 In general, neither equality holds.

Exercise 1.1.8 If f and g are functions and f is empty, then $G_f = \emptyset \subset G_g$.

Exercise 1.1.10 Suppose that elements a and b are neutral to a commutative operation o on a set X. Then $b = a + b = b + a = a$.

Exercise 1.1.11 There are $3^5 = 243$ such words.

Exercise 1.1.12 Let $u = u_1 \dots u_l$, $v = v_1 \dots v_m$, and $w = w_1 \dots w_n$ be words over X, with $l, m, n \in \omega$. We show that the concatenations $z = (uv)w = z_1 \dots z_{l+m+n}$ and $z' = u(vw) = z'_1 \dots z'_{l+m+n}$ are equal. Let $i \in [l+m+n]$. If $i \in [l]$, then $z_i = u_i = z'_i$. If $l+1 \le i \le l+m$, then $z_i = v_{i-l} = z'_i$. Finally, if $l+m+1 \le i \le l+m+n$, then $z_i = w_{i-l-m} = z'_i$.

Exercise 1.1.13 If $X = \{0, 1\}$, $u = 0$, and $v = 1$, then $uv = 01 \ne 10 = vu$.

Exercise 1.1.14 If and only if $X = Y$.

Exercise 1.1.15 For injective, constant, and identity functions, it is true. It is not true for surjective and bijective functions.

Exercise 1.1.17 Let $c \in Y$ be an arbitrary element. We take $b \in X$ such that $f(b) = c$. Then $f(a)\, q\, c = f(a)\, q\, f(b) = f(a\, p\, b) = f(b) = c$. If f is not onto, we cannot argue about the elements in $Y \setminus f[X]$.

Exercise 1.1.18 Let $c, d \in Y$ be any elements and $a := f^{-1}(c)$, $b := f^{-1}(d)$. Then $f^{-1}(c\, q\, d) = f^{-1}(f(a)\, q\, f(b)) = f^{-1}(f(a\, p\, b)) = a\, p\, b = f^{-1}(c)\, p\, f^{-1}(d)$.

Exercise 1.1.19 Let $a, b \in X$ be any elements. Then $g(f)(a\,p\,b) = g(f(a\,p\,b)) = g(f(a)\,q\,f(b)) = g(f(a))\,r\,g(f(b)) = g(f)(a)\,r\,g(f)(b)$.

Exercise 1.1.20 It is not a problem. Both symbols may appear at the same time only when f is injective, and then their meanings coincide.

Exercise 1.1.21 Since $f^{-1}\colon f[X] \to X$ is bijective, we have $(f^{-1})^{-1}\colon X \to f[X]$. Hence for $f[X] \neq Y$ the functions $(f^{-1})^{-1}$ and f differ as sets, but they are congruent. We have $((f^{-1})^{-1})^{-1}\colon f[X] \to X$. Hence $((f^{-1})^{-1})^{-1}$ and f^{-1} are equal as sets.

Exercise 1.1.23 Yes, they are.

Exercise 1.1.24 Let f and g be injective and $f(g)(x) = f(g)(y)$. Thus $f(g(x)) = f(g(y))$ and $g(x) = g(y)$. Thus $x = y$. Let $g\colon X \to Y$ and $f\colon Y \to B$ be onto, and let $b \in B$. Thus, there is $y \in Y$ such that $f(y) = b$. Thus, there is $x \in X$ such that $g(x) = y$. Thus, $f(g)(x) = f(g(x)) = f(y) = b$ and $f(g)$ is onto. If $g\colon X \to Y$ and $f\colon A \to B$ are nonempty and surjective, and $Y \cap A = \emptyset$, then $f(g)\colon \emptyset \to B$ is not onto.

Exercise 1.1.25 The range of $f(g(h))$ and the range of $f(g)(h)$ is equal to the range R of f. We show that $f(g(h))$ and $f(g)(h)$ have the same graphs. Indeed, both graphs are the pairs $\langle x, y\rangle \in M(h) \times R$ such that there exist $a \in M(g)$ and $b \in M(f)$ such that $h(x) = a$, $g(a) = b$, and $f(b) = y$.

Exercise 1.1.26 We set $Y = h[X]$, $\langle X, Y, G_g\rangle = \langle X, Y, G_h\rangle$, and $f\colon Y \to Z$ is given by $f(y) = y$.

Exercise 1.1.27 If f is a bijection, then $g = f^{-1}$ has the required properties. Let $g\colon Y \to X$ be as stated. Since $g(f)(x) = g(f(x)) = x$, the function f is injective. Since $M(g) = Y$ and $f(g)(y) = f(g(y)) = y$, the function f is onto.

Exercise 1.1.28 Those with an empty or one-element definition domain.

Exercise 1.1.29 These mean $\bigcup_{i\in\mathbb{N}} A_i$, $\bigcup_{i\in\mathbb{N}} A_i$, and $\bigcap_{i\in\omega} A_i$, respectively.

Exercise 1.1.31 For $1 \Rightarrow 2$, consider the set system $\{A_Z\colon\ Z \in X\}$ with $A_Z = Z$ and define $Y = S[X]$. To see $2 \Rightarrow 1$, for the given set system $\{A_i\colon\ i \in I\}$, $A_i \neq \emptyset$, consider the set X given by $a \in X$ iff $a = \{i\} \times A_i$ for some $i \in I$, and define $S(i) = y$ where $Y \cap X = \{\langle i, y\rangle\}$. For $1 \Rightarrow 3$, consider the set system $\{A_b\colon\ b \in B\}$ with $A_b = f^{-1}[\{b\}]$ and define $g = S$. Finally, for $3 \Rightarrow 1$ we take for the given set system $\{A_i\colon\ i \in I\}$, $A_i \neq \emptyset$, the sets $A = \bigcup_{i\in I}\{i\} \times A_i$ and $B = I$, the surjection $f\colon A \to B$ given by $f(\langle i, x\rangle) = i$, and define $S(i) = x$ where $g(i) = \langle i, x\rangle$.

Exercise 1.1.32 Consider the set $X = \{\{1, 2\}, \{2, 3\}, \{3, 1\}\}$.

Exercise 1.1.35 Let R be an equivalence relation on a set $A \neq \emptyset$ (for $A = \emptyset$ everything trivially holds) and $[a]_R$ be an equivalence block. Clearly, $a \in [a]_R$. Thus the elements in A/R are nonempty and $\bigcup A/R = A$. Let $a, b \in A$ and $c \in [a]_R \cap [b]_R$. By the transitivity and symmetry of R we have aRb. Thus $[a]_R = [b]_R$ and the elements of A/R are pairwise disjoint.

If $b, c \in [a]_R$ then bRa and cRa. Thus bRc. If bRc, $b \in [a]_R$ and $c \in [a']_R$ then bRa, cRa', so that aRa'. Hence $[a]_R = [a']_R$ and b, c are in a common block.

Exercise 1.1.36 Let X be a partition of $Y \neq \emptyset$ and $R = Y/X$. For $y \in Y$ we take a block $Z \in X$ with $y \in Z$. So yRy and R is reflexive. For $y, y' \in Y$ with yRy' there is a block $Z \in X$ such that $y, y' \in Z$. So also $y'Ry$ and R is symmetric. Let $y, y', y'' \in Y$ with yRy' a $y'Ry''$. Thus there exist blocks $Z, Z' \in X$ such that $y, y' \in Z$

and $y', y'' \in Z'$. But then $Z \cap Z' \neq \emptyset$ and therefore $Z = Z'$. Hence yRy'' and R is transitive. It follows from the definition that $x, y \in Z \in X$ iff $x(Y/X)y$.

Exercise 1.1.37 Let R, $A \neq \emptyset$ and B be as stated. We prove the first equality. We know that $C = A/R$ is a partition of A, thus $S = A/C$ is an equivalence relation on A. We show that $S = R$. Let $a, b \in A$. Then aSb iff there is a block $D \in C$ such that $a, b \in D$. As we know from Exercise 1.1.35, a, b lie in a common block of C iff aRb. Hence $S = R$.

We prove the second equality. We know that $S = A/B$ is an equivalence relation on A. Thus $C = A/S$ is a partition of A. We show that $C = B$. Again, $a, b \in A$ lie in a common block of C iff aSc. This is the case iff a, b lie in a common block of B. Hence $C = B$.

Exercise 1.1.39 Suppose that R is a linear order and a, b are two elements such aRb and bRa. Transitivity gives aRa, which is a contradiction.

Exercise 1.1.40 Always $a \leq a$ because $a = a$. The transitivity of $\leq$ follows from the transitivity of $<$. If a, b are two distinct elements, then $a < b$ or $b < a$ by trichotomy. Thus $\leq$ is dichotomic. If a, b are two elements such that $a \leq b$ and $b \leq a$, then $a \neq b$ would give $a < b$ and $b < a$, in contradiction with the previous exercise. Thus $a = b$ and $\leq$ is weakly asymmetric.

Exercise 1.1.41 Neither $a < b \wedge a = b$ nor $b < a \wedge a = b$ holds because $<$ is irreflexive. Nor $a < b \wedge b < a$ holds by Exercise 1.1.39.

Exercise 1.1.42 This follows from the trichotomy of $<$: if it is not the case that $a < a'$, then $a = a'$ or $a' < a$.

Exercise 1.1.43. When m and n are maxima of B, then both $n \leq m$ and $m \leq n$. Hence $m = n$. Same for minima.

Exercise 1.1.44 Proceed by induction on the size of the set.

Exercise 1.1.46 This follows from uniqueness of maxima and minima.

Exercise 1.1.47 Let $c = \sup(B)$. Then c is an upper bound of B. Let $a < c$. Since c is the minimum upper bound of B, the element a is not an upper bound of B and there exists the stated b. In the other way, let c have the stated properties. They say that c is the smallest upper bound of B, so that $c = \min(H(B)) = \sup(B)$.

Similarly one proves the equivalence that $c \in A$ is an infimum of B iff $c \leq b$ for every $b \in B$ & for every $a \in A$ with $c < a$ there is a $b \in B$ such that $b < a$.

Exercise 1.1.50 See Theorem 1.2.24.

Exercise 1.1.51 $\min(\mathbb{Z})$ does not exist because $n - 1 < n$ for every $n \in \mathbb{Z}$.

Exercise 1.1.52 If a linear order $\langle A, < \rangle$ is a well ordering then such elements a_n do not exist because $\{a_n\colon\ n \in \mathbb{N}\}$ would not have minimum. Suppose that $\langle A, < \rangle$ is not a well ordering. We take any $a_1 \in A$. Then we take any $a_2 \in A$ such that $a_2 < a_1$. Then any $a_3 \in A$ such that $a_3 < a_2$, and continue in this manner. In this way, we define the elements a_n. We use AC.

Exercise 1.1.54 See Exercise 1.1.40.

Exercise 1.1.55 The irreflexivity and transitivity of strict inclusion are easy to check.

Exercise 1.2.2 See Exercise 1.1.10.

Exercise 1.2.3 Additions of 1_S to $0_S < 1_S$ yield infinitely many elements $0_S < 1_S < 1_S + 1_S < \cdots$. For any $m \in \mathbb{N}$, addition and multiplication modulo m on $\mathbb{N}$ yield a finite semiring, actually a ring.

Exercise 1.2.4 If $a = b$, then clearly $a + c = b + c$.

Exercise 1.2.7 This follows easily from Exercises 1.1.17 and 1.2.2.

Exercise 1.2.9 It is clear that with the new element ∞, both operations remain commutative, and that 0_S and 1_S remain neutral. The three identities for the associativity of $+$ and $\cdot$, and for the distributive law, remain valid: if ∞ appears, then both sides are ∞.

Exercise 1.2.11 Neutrality of 1, neutrality of 0, the distributive law, neutrality of 1, and neutrality of 0.

Exercise 1.2.15 Each of the three stated sets is an element of every inductive set.

Exercise 1.2.20 This follows from parts 2 and 3 of Proposition 1.2.18.

Exercise 1.2.22 Axiom A.2.2 of foundation forbids any set x such that $x \in x$.

Exercise 1.2.26 $f(n) = \lfloor n/2 \rfloor$, i.e. $f(n) = m$ if $n \in \{2m, 2m + 1\}$.

Exercise 1.2.28 This follows from the definition of $+$.

Exercise 1.2.30 $3 \cdot 4 = 3 \cdot 3 + 3 = (3 \cdot 2 + 3) + 3 = ((3 \cdot 1 + 3) + 3) + 3 = (((3 \cdot 0 + 3) + 3) + 3) + 3 = ((3 + 3) + 3) + 3$. By the definition of $+$, this equals 12.

Exercise 1.2.37 The equality follows, respectively, from the definition of $+$, from the definition of $\cdot$, from the definition of -1, from $n0 = 0n = 0$, from $n + 0 = 0 + n = n$, and from the definition of $+$.

Exercise 1.2.40 This follows from the equality $3 + 2 = 5$.

Exercise 1.2.41 These follow easily from the property of subtraction in Proposition 1.2.38. As an example, we prove the last identity: $((l - m) + n) + (m - n) = (l - m) + (n + (m - n)) = (l - m) + m = l$.

Exercise 1.2.43 Divide k by the highest possible power of 2. Set $f(k - 1) = \langle l, m \rangle$.

Exercise 1.2.49 The set $\{f(n)\colon\ n \in \omega\}$ would violate the axiom of foundation.

Exercise 1.2.50 Use induction on $n \in \omega$ and the fact that $n + 1 = n \cup \{n\}$.

Exercise 1.2.55 This is the second order axiom.

Exercise 1.2.58 It follows from Theorem 1.2.24 and Propositions 1.2.36 and 1.2.57.

Exercise 1.3.2 If $\alpha + \beta = 0_R$ and $\alpha + \gamma = 0_R$ then the neutrality of 0_R, and associativity and commutativity of $+$ give that $\gamma = (\alpha + \beta) + \gamma = (\alpha + \gamma) + \beta = \beta$.

Exercise 1.3.3 These properties of additive inverses follow from the previous exercise and from the properties of $+$. For example, $(a + b) + ((-a) + (-b)) = (a + (-a)) + (b + (-b)) = 0_R + 0_R = 0_R$, and therefore $-(a + b) = (-a) + (-b)$.

Exercise 1.3.4 If $1_R < 0_R$, then $0_R < -1_R$ by additive inverses and the first order axiom. But then $0_R = 0_R \cdot (-1_R) < (-1_R) \cdot (-1_R) = 1_R$ by the ring second order axiom, which is a contradiction.

Exercise 1.3.5 Let $f\colon R \to R'$ be a ring homomorphism. Then $0_{R'} + 1_{R'} = 1_{R'} = f(1_R) = f(0_R + 1_R) = f(0_R) + f(1_R) = f(0_R) + 1_{R'}$ and we have equality $0_{R'} + 1_{R'} = f(0_R) + 1_{R'}$. Adding $-1_{R'}$ to it, we obtain $0_{R'} = f(0_R)$. Thus if $a + b = 0_R$, we apply f and obtain $f(a) + f(b) = f(0_R) = 0_{R'}$. By Exercise 1.3.2, $f(-a) = f(b) = -f(a)$.

Exercise 1.3.9 For every $m \in \mathbb{N}$, addition and multiplication modulo m on ω yield a finite simple ring R_m.

Exercise 1.3.10 Additive inverse, the neutrality of 0_R, the distributive law, the associativity of $+$, additive inverse, and the neutrality of 0_R.

Exercise 1.3.12 These are immediate from the definition and from the fact that $-0_R = 0_R$.

Exercise 1.3.13 As for the first equality, $a - (b + c) = a + (-(b + c)) = a + ((-b) + (-c)) = (a + (-b)) + (-c) = (a - b) - c$, where we used Exercise 1.3.3. The second equality is similar.

Exercise 1.3.14 -1 and 1.

Exercise 1.3.16 $\langle 0, m\rangle = n$ with $m, n \in \omega$ never holds because all elements of the nonempty left-hand side set are nonempty, but always $n = \emptyset$ or $\emptyset \in n$.

Exercise 1.3.17 It follows from the definition of an ordered pair and from the fact that every natural number is HF.

Exercise 1.3.19 Irreflexivity is trivial from the definition. Let $k < l < m$ be in $\mathbb{Z}$. If all three are in $-\mathbb{N}$ or in ω, then $k < m$ follows from the transitivity of $<$ in $\mathbb{N}_0$. The remaining case is $k \in -\mathbb{N}$ and $m \in \omega$, and then $k < m$ follows from the definition. If $m, l \in \mathbb{Z}$ and are distinct, then both are in $-\mathbb{N}$ or in ω and are compared by the trichotomy of $<$ in $\mathbb{N}_0$, or one is in $-\mathbb{N}$ and the other in ω, and m and l are compared from the definition. Thus $<$ is trichotomic. This linear order is not a well ordering.

Exercise 1.3.23 -2, -5, -6, $-3 < -2$, and $-10 < 1$.

Exercise 1.3.28 This is immediate from the definitions of $0_{\mathbb{Z}}$, $1_{\mathbb{Z}}$, and the operations $+$ and $\cdot$ on $\mathbb{Z}$.

Exercise 1.3.31 For example, $-((1_R + 1_R) + 1_R)$.

Exercise 1.3.37 It is, consider the minimum positive element, where two isomorphisms would differ.

Exercise 1.3.38 Reflexivity and symmetry are obvious. Transitivity: if $m \ominus n \sim m' \ominus n'$ and $m' \ominus n' \sim m'' \ominus n''$, then $m + n' = n + m'$ and $m' + n'' = n' + m''$; hence $m + n' + m' + n'' = n + m' + n' + m''$, which implies by canceling $n' + m'$ that $m + n'' = n + m''$ and $m \ominus n \sim m'' \ominus n''$.

Exercise 1.3.39 We show it here in detail only for $\cdot$. Let $m \ominus n \sim m' \ominus n'$ and $k \ominus l \sim k' \ominus l'$. Then $m + n' = n + m'$, $k + l' = l + k'$, and $m \ominus n \cdot k \ominus l = mk + nl \ominus ml + nk$, where the variables may be primed. Then $mk + nl + m'l' + n'k' = ml + nk + m'k' + n'l' \iff m(k-l) + n(l-k) + m'(l'-k') + n'(k'-l') = 0 \iff (m+n')(k-l) + (n+m')(l-k) = 0 \iff (m + n' - n - m')(k - l) = 0 \iff 0 \cdot (k - l) = 0 \iff \mathrm{T}$.

Exercise 1.3.41 Here we only prove the associativity of $\cdot$. $(m \ominus n \cdot m' \ominus n') \cdot m'' \ominus n''$ is $(mm' + nn')m'' + (mn' + nm')n'' \ominus (mn' + nm')m'' + (mm' + nn')n''$. On the other hand, $m \ominus n \cdot (m' \ominus n' \cdot m'' \ominus n'')$ is $(m'm'' + n'n'')m + (m'n'' + n'm'')n \ominus (m'n'' +$

$n'm'')m + (m'm'' + n'n'')n$. By inspection, on the left-hand sides of the two $\ominus$, we have the same quadruple of cubic monomials, and the same holds for the right-hand sides of the two $\ominus$.

Exercise 1.3.42 Irreflexivity and transitivity are easy to see. As for trichotomy, let $k\ominus l \not\sim m\ominus n$ be two distinct difference integers. Thus $k+n \neq l+m$. Now $k+n < l+m$ gives $k \ominus l < m \ominus n$, and $k + n > l + m$ gives $k \ominus l > m \ominus n$.

Exercise 1.3.43 Let $m \ominus n$, $m' \ominus n'$, and $m'' \ominus n''$ be three difference integers with $m\ominus n < m'\ominus n'$, so that $m+n' < n+m'$. Thus $m+m''+n'+n'' < n+n''+m'+m''$ and $m \ominus n + m'' \ominus n'' < m' \ominus n' + m'' \ominus n''$, which proves the first order axiom.

Suppose that $m'' \ominus n'' > 0 \ominus 0$, so that $m'' - n'' > 0$. Then $m \ominus n \cdot m'' \ominus n''$ is $A = mm''+nn''\ominus mn''+nm''$, and $m'\ominus n'\cdot m''\ominus n''$ is $B = m'm''+n'n''\ominus m'n''+n'm''$. Now $A < B \iff mm'' + nn'' + m'n'' + n'm'' < mn'' + nm'' + m'm'' + n'n'' \iff m(m'' - n'') + n(n'' - m'') + m'(n'' - m'') + n'(m'' - n'') < 0 \iff (m + n' - n - m')(m'' - n'') < 0 \iff$ T, because $m + n' - n - m' < 0$ and $m'' - n'' > 0$. This proves the second order axiom.

Exercise 1.4.2 See part 1 of Proposition 1.3.8.

Exercise 1.4.3 If $a \cdot b = 0_F$ and $a, b \in F$ are nonzero, then the previous exercise yields $1_F = b^{-1} \cdot a^{-1} \cdot a \cdot b = b^{-1} \cdot a^{-1} \cdot 0_F = 0_F$, which is a contradiction.

Exercise 1.4.4 Let $a, b, c \in F \setminus \{0_F\}$ be such that $a \cdot b = 1_F = a \cdot c$. Multiplying the last equality by b we get that $b = (b \cdot a) \cdot c = 1_F \cdot c = c$.

Exercise 1.4.5 This follows from the previous exercise. For example, $(a \cdot b) \cdot (a^{-1} \cdot b^{-1}) = \cdots = 1_F$, so that $(a \cdot b)^{-1} = a^{-1} \cdot b^{-1}$.

Exercise 1.4.10 The first two properties are straightforward. We prove the last two. The former follows from the identity $x \cdot x^{-1} = 1_F$ by the multiplicativity of $|\cdot|$. The latter follows by applying the TI on the sum $x = (x + y) + (-y)$.

Exercise 1.4.11 $d(x, y) = |x - y| \geq 0_F$ is trivial, as is $d(x, y) = d(y, x)$ because $|z| = |-z|$. TI $d(x, y) \leq d(x, z) + d(z, y)$ follows from applying TI for $|\cdot|$ to the sum $x - y = (x - z) + (z - y)$.

Exercise 1.4.13 $1_F^{-1} = 1_F$, the neutrality of 1_F, and $(a \cdot b^{-1}) \cdot (b \cdot a^{-1}) = 1_F$.

Exercise 1.4.14 $(a \cdot b^{-1}) \cdot (c \cdot d^{-1})^{-1} = a \cdot b^{-1} \cdot c^{-1} \cdot d = (a \cdot d) \cdot (b \cdot c)^{-1}$.

Exercise 1.4.16 The implication $1 \Rightarrow 2$ follows from the inequality $x \leq |x|$. The opposite implication follows from the inequality $|x| \leq \max(-x, x)$. Suppose that 1 holds and that $x \in F$ is nonzero. We take $m \in \omega$ such that $1_F/|x| \leq f_F(m)$, and set $n = m + 1$; then 3 holds. Let 3 hold. Then $|x| < f_F(n)$ and 1 holds.

Exercise 1.4.20 Suppose that $\emptyset \neq X \subset F$ and that X is lower-bounded. Then $\inf(X) = -\sup(-X)$.

Exercise 1.4.23 Let $(a_n) \subset F$ be Cauchy. We take $m \in \mathbb{N}$ such that if $n, n' \geq m$, then $|a_n - a_{n'}| \leq 1_F$. Let $b = \max(\{|a_1|, \ldots, |a_m|\})$. Then for every $n \in \mathbb{N}$ we have, by TI, that $|a_n| \leq 1_F + b$.

Exercise 1.4.24 Let $\lim a_n = a$ and $\lim a_n = b$, where $a, b \in F$ are two elements. If $a \neq b$, we set $e = |a - b|/3_F$ $(> 0_F)$ and take $n_0 \in \mathbb{N}$ such that if $n \geq n_0$, then $|a_n - a|, |a_n - b| \leq e$. Then TI yields the contradiction that $e \leq e(2_F/3_F)$.

Exercise 1.4.25 See Theorem 2.4.18.

Exercise 1.4.29 Reflexivity and symmetry are clear. We prove transitivity. Let $a/b \sim c/d$ and $c/d \sim e/f$. Thus $ad = bc$ and $cf = de$. But then $adf = bcf = bde$ and $adf = bde$. The $d \neq 0$ can be canceled ($\mathbb{Z}$ is a domain) and $af = be$. Hence $a/b \sim e/f$.

Exercise 1.4.31 $\frac{m}{n} \sim \frac{-m}{-n}$.

Exercise 1.4.32 Let $\alpha \in \mathbb{Q}$ and $p_\alpha = \frac{m}{n} \in \alpha$ have $n > 0$ and minimum $|m| + |n|$. It follows that p_α is in lowest terms. The function $\alpha \mapsto p_\alpha$ is the desired bijection. It is bijective and unique because two different elements in Z_0 are $\not\sim$. We prove it. If $\frac{k}{l}, \frac{m}{n} \in Z_0$ are in lowest terms and $\frac{k}{l} \sim \frac{m}{n}$ then $kn = ml$. Thus any prime power dividing k divides m and vice versa. The fundamental theorem of arithmetic gives $k = m$. Thus $l = n$.

Exercise 1.4.37 Let $\frac{a}{b} \sim \frac{a'}{b'}$ and $\frac{c}{d} \sim \frac{c'}{d'}$ be equivalent protofractions, so that $ab' = ba'$ and $cd' = dc'$. Then

$$\tfrac{a}{b} + \tfrac{c}{d} = \tfrac{ad+bc}{bd} \sim \tfrac{a'd'+b'c'}{b'd'} = \tfrac{a'}{b'} + \tfrac{c'}{d'}$$

because

$$(ad + bc)b'd' = ab'dd' + cd'bb' = a'd'bd + b'c'bd = (a'd' + b'c')bd\,.$$

Similarly,

$$\tfrac{a}{b} \cdot \tfrac{c}{d} = \tfrac{ac}{bd} \sim \tfrac{a'c'}{b'd'} = \tfrac{a'}{b'} \cdot \tfrac{c'}{d'}$$

because

$$acb'd' = ab'cd' = ba'c'd = a'c'bd\,.$$

Exercise 1.4.39 $\frac{0}{1} \not\sim \frac{1}{1}$ because $0 \cdot 1 \neq 1 \cdot 1$.

Exercise 1.4.41 It is easy to check that F is injective and preserves $+$, $\cdot$, and additive inverses. Also, $F(1) = 1_{\mathbb{Q}}$.

Exercise 1.4.43 If $m \cdot n = 0$, then $F(m) \cdot F(n) = 0_{\mathbb{Q}}$ in the field $\mathbb{Q}$ and we use Exercise 1.4.3.

Exercise 1.4.45 Because for every $m \in \mathbb{Z}$ we have $(-m)^2 = m^2$.

Exercise 1.4.46 If m^2 is even, then m is even because the squares of odd numbers are odd.

Exercise 1.4.48 For fractions $s, r > 0$ with $s^2 > 2$, the inequality $(s - r)^2 > 2$ holds if $s^2 - 2 > 2sr - r^2$. Thus, for example, if $0 < r < \frac{s^2-2}{2s}$. For fractions r, s with $s^2 < 2$, $s > 0$ and $r \in (0, 1)$ (then $r^2 < r$), the inequality $(s+r)^2 < 2$ holds if $2sr + r^2 < 2 - s^2$. Thus, for example, if $0 < r < \min(\{1, \frac{2-s^2}{2s+1}\})$.

Exercise 1.4.55 If $x > 0_F$ but $x^{-1} < 0_F$, then using the second order axiom we get $1_F = x^{-1} \odot x < 0_F$, which contradicts Exercise 1.3.4. The second claim follows from the second order axiom at once.

Exercise 1.4.58 It is. Two different isomorphisms would yield a non-identical automorphism of the field $\mathbb{Q}$. From it we would obtain a non-identical automorphism of the domain $\mathbb{Z}$, which cannot exist by induction.

Exercise 1.5.3 Reflexivity and symmetry of $\sim$ are trivial. The transitivity easily follows from the triangle inequality.

Exercise 1.5.5 Every integer is HMC, even HF. Every fraction is a countable set of 2-tuples of integers and is therefore HMC. Since the set of fractions is countable, it is HMC.

Exercise 1.5.7 This follows from the previous exercise.

Exercise 1.5.9 The relation is irreflexive by definition. It is transitive because set inclusion is transitive. Let $X, Y \subset \mathbb{Q}$ be two distinct cuts. If neither $X \subset Y$ nor $Y \subset X$ holds, we take distinct fractions $\alpha \in X \setminus Y$ and $\beta \in Y \setminus X$. Then neither $\alpha < \beta$ nor $\alpha > \beta$ holds, by property (ii) of cuts. This is a contradiction.

Exercise 1.5.11 Replace the sum of cuts with their intersection, and argue as for suprema.

Exercise 1.5.13 Since every Cauchy sequence is bounded, $\Phi(a_n)$ has property (i) of cuts. Properties (ii) and (iii) follow easily from the definition of Φ.

Exercise 1.5.17 Argue as in part 1.

Exercise 1.5.21 For example, $(1, 0, 0, 0, \dots) \odot (0, 1, 0, 0, \dots) = \overline{0}$.

Exercise 1.5.24 If m and n are so large that $|a_m - a_n| \le e'$ and $|a_m|, |a_n| \ge e$, then $|b_m - b_n| = |a_m^{-1}| \cdot |a_n^{-1}| \cdot |a_m - a_n| \le e'e^{-2}$.

Exercise 1.5.25 $\Phi(\alpha, \alpha, \dots) = F(\alpha)$.

Exercise 1.5.28 $a^2 = b^2$ is equivalent with $(a - b)(a + b) = 0_F$.

Exercise 1.5.29 If $a = 0_F$, then $a^2 = 0_F \ge 0_F$. If $a > 0_F$, then $a^2 > 0_F \cdot a = 0_F$ by the second order axiom. If $a < 0_F$, then $0_F < -a$ by the first order axiom and $0_F = 0_F \cdot (-a) < (-a) \cdot (-a) = a^2$.

Exercise 1.5.30 With the help of Proposition 1.5.27, in the complete ordered field F we can define $a <_F b \iff \exists c\colon (c \ne 0_F \wedge c^2 = b - a)$.

Exercise 1.5.32 Suppose that X is dense in F and that $\alpha \in F$. For every $n \in \mathbb{N}$, we select an $x_n \in X$ such that $|\alpha - x_n| \le 1_F / f_F(n)$. Since F is Archimedean, it follows that $\lim x_n = \alpha$. Suppose that X is not dense in F. Then there exist $\alpha, e \in F$ with $e > 0_F$ such that $|\alpha - x| \ge e$ for every $x \in X$. Then, clearly, there is no sequence $(x_n) \subset X$ such that $\lim x_n = \alpha$.

Exercise 1.5.36 See Theorem 2.6.2.

Exercise 1.5.38 See Theorem 2.6.2.

Exercise 1.5.40 See Theorem 2.8.1.

Exercise 1.6.2 A number $\gamma \in \mathbb{R}$ is in the intersection iff $a_n \le \gamma \le b_n$ for every n. This is true iff γ is both an upper bound of all a_n, and a lower bound of all b_n. And this is true iff $\gamma \in [\alpha, \beta]$.

Exercise 1.6.3 Under this assumption, $\alpha = \beta$ $(= c)$.

Exercise 1.6.5 If $\mathbb{R}$ is countable, then there is a bijection $f\colon \omega \to \mathbb{R}$. We have the bijection g given by $\mathbb{N} \ni n \mapsto g(n) = n - 1 \in \omega$. Then the composite map $f(g)$ is a surjection from $\mathbb{N}$ to $\mathbb{R}$.

Exercise 1.6.6 We divide I_n into thirds and take a third not containing a_{n+1}.

Exercise 1.7.2 The union of all open intervals is X. If I and J are open intervals, then so is $I \cap J$.

Exercise 1.7.7 This follows from the trichotomy of $<_X$.

Exercise 1.7.9 By adding the mentioned comparisons for infinities, the irreflexivity, transitivity, and trichotomy are preserved.

Exercise 1.7.12 In $\mathbb{R}'$, the element ∞ is the limit of any sequence $(a_n) \subset \mathbb{R}$ that converges from the left side to 0 in $\mathbb{R}$.

Exercise 1.7.15 10. The mentioned inequality turns into the equality $C = C$ after the addition of C. It is easy to check that in other cases, the inequality is preserved (if the involved expressions are defined). 11. The mentioned inequality turns into the equality $C = C$ after multiplying by C. It is easy to check that in other cases, the inequality is preserved (if the involved expressions are defined).

Exercise 1.7.16 $-\infty$, undefined, $-\infty$, and $-\infty$,

Exercise 1.7.19 $+\infty$, $+\infty$, 0, and undefined.

2 Limits of real sequences

Exercise 2.1.1 See Proposition 1.4.9.

Exercise 2.1.2 The irreflexivity of $<$ is clear. If $A < B$ and $B < K$, and one of A, B, and K is infinity, then (i) $A = -\infty$ and $B \in \mathbb{R}$, or (ii) $B \in \mathbb{R}$ and $K = +\infty$. In both cases $A < K$, and transitivity holds. Let $A, B \in \mathbb{R}^*$ be distinct and one of them be infinity. Then one of $-\infty < +\infty$, $a < +\infty$ or $-\infty < a$ occurs, and $<$ is trichotomic.

Exercise 2.1.4 $+\infty$, $-\infty$, $-\infty$, and undefined.

Exercise 2.1.7 Just $\emptyset$ and $\{-\infty\}$.

Exercise 2.1.10 Neighborhoods of points and infinities are intervals.

Exercise 2.1.11 For $A, B \in \mathbb{R}$ we may take any $\varepsilon < \frac{B-A}{2}$. If $A = -\infty$ and $B = +\infty$, we may take any ε. If $A = -\infty$ and $B \in \mathbb{R}$, we may take any $\varepsilon < \frac{1}{|B|+1}$; similarly if $A \in \mathbb{R}$ and $B = +\infty$.

Exercise 2.1.12 This is immediate from the definition of neighborhoods.

Exercise 2.1.13 This is again immediate from the definition of neighborhoods.

Exercise 2.1.15 It follows from the equivalence $x \in U(a, \varepsilon) \iff |x - a| < \varepsilon$ and the inequality $\varepsilon/2 < \varepsilon$.

Exercise 2.1.16 Suppose that $L = \lim a_n$. Then for any given ε there is a k such that for every $n \ge k$ we have $a_n \in U(L, \varepsilon)$. Then for every $n \ge \sum_{i=1}^{k} m_i$ we have $b_n \in U(L, \varepsilon)$. Hence $\lim b_n = L$. In the proof of the opposite implication, we use that (a_n) is a subsequence of (b_n).

Exercise 2.1.18 $\lim a_n = +\infty \iff$ for every $c > 0$ there exists n_0 such that if $n \ge n_0$, then $a_n \ge c$. The proof is similar to the $-\infty$ case.

Exercise 2.1.21 If (a_n) and (b_n) satisfy $a_n = b_n$ for every $n \ge m$ and $\lim a_n = L$, then also $\lim b_n = L$. This follows from the fact that the index n_0 in the definition of limit can be taken $\ge m$.

Exercise 2.1.22 We only prove 3, the proofs for 1 and 2 are similar. Suppose that X is as stated, that $a_n \neq b_n$ for only finitely many n and that $(a_n) \in \bigcap X$. Thus for every $Y \in X$ we have $(a_n) \in Y$. Since every Y is robust, also $(b_n) \in Y$. Hence $(b_n) \in \bigcap X$. So $\bigcap X$ is robust.

Exercise 2.1.23 The second and fifth properties are not robust; the others are robust.

Exercise 2.1.25 Let $k \in \mathbb{N}$, $\varepsilon = \frac{1}{k}$, and $n_0 < \lceil\frac{1}{\varepsilon}\rceil + 1 = k + 1$. Thus $n_0 \leq k$ and $\frac{1}{n_0} \notin U(0, \varepsilon) = (-\frac{1}{k}, \frac{1}{k})$ because $\frac{1}{n_0} \geq \frac{1}{k}$.

Exercise 2.1.27 Clearly, $\frac{\sqrt[3]{n}-\sqrt{n}}{\sqrt[4]{n}} = \frac{n^{-1/6}-1}{n^{-1/4}} \to \frac{0-1}{0^+} = \frac{-1}{0^+} = -\infty$, due to positivity of $n^{-1/4}$.

Exercise 2.1.28 The coefficient of the monomial $a^j b^{n-j}$ is the number of ways to obtain it: we choose j factors $a + b$ in the product $(a + b)^n$ from which we pick the number a, and pick b from the remaining $n - j$ factors. There are $\binom{n}{j}$ ways to do it because there are $\binom{n}{j}$ j-element subsets of $[n]$.

Exercise 2.1.30 It follows by negating the limit $n^{1/n} \to 1$.

Exercise 2.1.31 In any ordered field F, the second order axiom and the transitivity of $\geq$ imply that if $a \geq b \geq 0_F$ and $c \geq d \geq 0_F$, then $ac \geq bd$.

Exercise 2.1.33 It is not.

Exercise 2.1.34 One replaces the given sequence $a_1 \geq a_2 \geq \dots$ with the sequence $-a_1 \leq -a_2 \leq \cdots$.

Exercise 2.1.35 This follows from the identity $|a_n - 0| = ||a_n| - 0|$ $(= |a_n|)$.

Exercise 2.2.1 Reflexivity follows from setting $m_n = n$. Transitivity is clear when one views subsequences as obtained by omitting terms in original sequences.

Exercise 2.2.2 For example, $(0, 1, 0, 1, \dots)$ and $(1, 0, 1, 0, \dots)$.

Exercise ?? Hint: if $\iota_n = 1\,2\dots n$ is the identical permutation, then the desired m-tuple is $\iota_{k-1} \ominus \iota_{k-1} \ominus \cdots \ominus \iota_{k-1}$ with $l - 1$ copies of ι_{k-1}.

Exercise ?? The generalization says that in any linear order $(X, <)$ every sequence (a_n) has a monotone subsequence. The same proof works.

Exercise 2.2.7 Let (b_n) and (a_n) be the stated sequences and let an ε be given. Thus there is an n_0 such that $n \geq n_0 \Rightarrow a_n \in U(L, \varepsilon)$. Then there is an n_1 such that $n \geq n_1$ $\Rightarrow m_n \geq n_0$. Then for every $n \geq n_1$ we have that $b_n = a_{m_n} \in U(L, \varepsilon)$ and $\lim b_n = L$.

Exercise 2.2.8 It is easy to see that the sequence $(m_n) \subset \mathbb{N}$ witnessing $(b_n) \preceq^* (a_n)$ has an increasing subsequence.

Exercise 2.2.12 Let B_0 and B_1 be the supports of the two subsequences in part 1 of Theorem 2.2.11. Since their limits differ, both sets $B_0 \setminus B_1$ and $B_1 \setminus B_0$ are infinite.

Exercise 2.2.13 If a sequence converges then every subsequence of it has the same finite limit and the right-hand side does not hold. If a sequence diverges then it has no limit or an infinite limit. In the former case it has, by part 1 of Theorem 2.2.11, two subsequences with different limits. In the latter case the sequences itself has limit $\pm\infty$.

Exercise 2.2.16 We can take only every other element of B_0.

Exercise 2.2.17 Now in the proof of Theorem 2.2.15 the subsequence corresponding to B_0 converges.

Exercise 2.2.18 Using Exercise 2.2.12 we take in (a_n) two disjoint subsequences (b_n) and (c_n) with different limits, and then proceed as in the proof of Theorem 2.2.15.

Exercise 2.3.2 By Corollary 2.2.10.

Exercise 2.3.8 For every $n \geq 2$, it is true that $\tau(n) \geq 2$ because 1 and n always divide n. For infinitely many n, namely for the prime numbers, the equality holds.

Exercise 2.3.9 One can reduce parts 3 and 4 to parts 1 and 2 by means of the identity $\liminf a_n = -\limsup(-a_n)$.

Exercise 2.3.10 We let m run in $\mathbb{N}$ and compose (a_n) of segments S_m: $(a_n) = S_1 S_2 \dots$, where S_m runs through the numbers $-m$, $-m + \frac{1}{m}$, $-m + \frac{2}{m}$, $\dots$, m.

Exercise 2.3.11 No, it is not because $L(a_n) \cap \mathbb{R}$ is always a closed set.

Exercise 2.3.12 $L(a_n) = \{0, +\infty\}$.

Exercise 2.4.1 The implication $\Leftarrow$ is trivial. The implication $\Rightarrow$ follows from the transitivity of $\leq$. Similarly for the rest of the exercise.

Exercise 2.4.2 If such c exists then for every n it holds that $-c \leq a_n \leq c$ and (a_n) is bounded both from below and from above. Suppose that (a_n) is bounded by the definition, so that $d \leq a_n \leq c$ for every n and some numbers d and c. Then $|a_n| \leq \max(|d|, |c|)$ for every n.

Exercise 2.4.3 The last three concerning boundedness.

Exercise 2.4.5. Suppose that (a_n) weakly increases for every $n \geq m$ and that $b_n = a_n$ for every $n \geq n_0$. Then (b_n) weakly increases for every $n \geq \max(m, n_0)$. Same for weakly decreasing tails.

Exercise 2.4.6 For example, if (a_n) weakly decreases, then for every n, the implication $a_m > a_n \Rightarrow m < n$ holds. Hence, (a_n) goes down. Similarly, for weakly increasing sequences.

Exercise 2.4.7 For example, $(1, 0, 2, 1, 3, 2, 4, 3, 5, \dots)$ goes up, but no tail is monotone.

Exercise 2.4.8 A sequence $(a_n) \subset \mathbb{R}$ is quasi-monotone iff

$$\big(\forall\, l\, \exists\, m\colon n \geq m \Rightarrow a_n \geq a_l\big) \vee \big(\forall\, l\, \exists\, m\colon n \geq m \Rightarrow a_n \leq a_l\big)\ .$$

Exercise 2.4.11 Suppose for example that (a_n) goes up starting from $n = m$ and that $b_n = a_n$ for every $n \geq n_0$. Then (b_n) goes up from $n = \max(m, n_0)$.

Exercise 2.4.13 We reduce it to part 2 by switching from the sequence (a_n) to the sequence $(-a_n)$.

Exercise 2.4.14 It is easy to see that the limit c of this subsequence satisfies the inequalities $a \leq c \leq b$.

Exercise 2.4.16 Suppose that (a_n) is Cauchy and (b_n) is such that $b_n = a_n$ for $n \geq n_0$. If for a given ε for every $m, n \geq n_1$ it holds that $|a_m - a_n| \leq \varepsilon$, then for every $m, n \geq \max(n_0, n_1)$ it holds that $|b_m - b_n| \leq \varepsilon$. Hence (b_n) is Cauchy.

Exercise 2.4.17 Let (a_n) be Cauchy. Then there is an n_0 such that for every $m, n \geq n_0$ one has that $|a_m - a_n| \leq 1$. By TI it holds for every n that $|a_n| \leq 1 + \max(\{|a_1|, \dots, |a_{n_0}|\})$. Hence (a_n) is bounded.

Exercise 2.4.19 Take for example the sequence $(1, 1.4, 1.41, 1.414, \dots)$ of truncations of the decimal expansion of $\sqrt{2}$.

Exercise 2.4.20 We used it via the application of the B.–W. theorem whose proof uses the theorem on limits of monotone sequences. This theorem requires the existence of suprema and infima of sets of real numbers.

Exercise 2.5.1 black

Exercise 2.5.2 Back then, there was no AI, no Internet, no mobile phones, no computers that one could use in everyday life ... In the fall of that Orwell year, I started my study of mathematics (and "cybernetics") at Charles University in Prague ... Now my parents are both dead, but back then my father was a young, energetic man of 50 and my mother was a nice young lady of 47 ... Do I wish I could go back to 1984? On one hand, yes, on the other hand, I see how stupid I was back then ...

Exercise 2.5.4 We have $n = m + m + \cdots + m + l$, with $k + 1$ summands.

Exercise 2.5.5 To prove the first lemma, see the properties of the exponential function, especially continuity. We prove the second lemma. If $A = +\infty$, then there is a sequence $(x_n) \subset X$ with $\lim x_n = +\infty$. Then $\lim \exp(x_n) = +\infty = \sup(Y)$, so that $\sup(Y) = \exp(A)$. Let $A \in \mathbb{R}$. Since $x \leq A$ for every $x \in X$, also $y = \exp x \leq \exp A$ for every $y \in Y$. If $a < \exp A$, then $\log a < A$ and there is $x \in X$ such that $\log a < x \leq A$. Then $a < \exp x \leq \exp A$ and $\exp x \in Y$. Thus $\sup(Y) = \exp A$.

Exercise 2.5.10 Let $u \neq \emptyset$. If $v \in [n]^*$ is r-sparse and $|v| \geq (\binom{n}{r}(|u|-1)+1)r$ then by the pigeonhole principle you can find $|u|$ intervals $\dots I_1 \dots I_2 \dots \dots I_{|u|} \dots$ in v such that $|I_i| = r$ and that all I_i use the same r-element set of letters. Taking appropriate terms from these intervals, one per each I_i, you easily build a copy of u in v.

Exercise 2.5.12 We prove the upper bound $\mathrm{ex}(abab, n) \leq 2n - 1$ by induction on n. For $n = 1$ it holds. Suppose that $v \in [n]^*$ with $n \geq 2$ is a 2-sparse (there is no immediate repetition in v) word not containing $abab$. Two closest occurrences of some $j \in [n]$ (if there is any repetition) show that some $i \in [n]$ occurs in u only once. Deleting this occurrence and possibly one more term of v, we get a 2-sparse word $v' \in ([n] \setminus \{i\})^*$. Clearly, $abab \not\preceq v'$. Hence $|v| \leq |v'| + 2 \leq 2(n-1) - 1 + 2 = 2n - 1$. On the other hand, words $12 \dots (n-1)n(n-1) \dots 21$ show that the upper bound is tight.

Exercise 2.5.14 These are trivialities.

Exercise 2.5.17 An automorphism sending a vertex u to a vertex v induces a bijection from the set of edges incident with u to the the set of edges incident with v.

Exercise 2.5.18 In a path starting at v, we always have a finite number of possibilities for the next edge. An automorphism sending a vertex u to a vertex v induces a bijection from the set of paths starting at u to the set of paths starting at v.

Exercise 2.5.19 For the first edge of a path we have r possibilities. For every next edge we have only at most $r - 1$ possibilities because we cannot backtrack.

Exercise 2.5.21 Let $G = \langle \mathbb{Z}^2, E \rangle$ be the 4-regular graph of the square lattice. Let $E' = \{\{\langle a, b \rangle, \langle a+1, b \rangle\}\colon\ a, b \in \mathbb{Z},\ a \text{ and } b \text{ have the same parity}\}$ $(\subset E)$. Then $H =$

$G \setminus E' = \langle \mathbb{Z}^2, E \setminus E' \rangle$ is the hexagonal tiling graph. It is clearly 3-regular and if $\overline{u}, \overline{v} \in \mathbb{Z}^2$ are two vertices, then the shift by the vector $\overline{v} - \overline{u}$, followed if needed by a vertical reflection, is an automorphism sending $\overline{u}$ to $\overline{v}$.

Exercise 2.5.23 Classically, C_n is the number of noncrossing matchings $\langle [2n], E \rangle$.

Exercise 2.5.25 We identify e_3 and e_4 as the unique two edges crossing the bisector of the segment $\langle 2m, 0 \rangle \langle 2m+1, 0 \rangle$.

Exercise 2.5.26 A word $u \in S_m$ has m possibilities for the first letter, (for every choice of the first letter it has) $m-1$ possibilities for the second letter, ...

Exercise 2.5.29 If p_1, p_2, ..., p_k are forbidden permutations, then it may happen that for some i and j, p_i is not $\ominus$-irreducible and p_j is not $\oplus$-irreducible. Then we do not know if the $\ominus$- and $\oplus$-sum of two permutations avoiding every permutation p_i still has this property.

Exercise 2.6.1 Since $|-b| = |b|$, it suffices to prove the first inequality. We apply to $a = (a+b) + (-b)$ the standard TI and rearrange the result.

Exercise 2.6.3 1. Let $|a_n| \le d$ for every n, $L = -\infty$ and a $c < 0$ be given. It is clear that for every large n one has $b_n \le c - d$. Thus for every large n we have that $a_n + b_n \le d + c - d = c$. Hence $a_n + b_n \to -\infty$. The case $L = +\infty$ is similar.

2. Let $|a_n| \le d$ for every n, $b_n \to 0$ and an ε be given. Clearly, for every large n we have $|b_n| \le \frac{\varepsilon}{d}$. So for every large n, we have $|a_n b_n| \le d \cdot \frac{\varepsilon}{d} = \varepsilon$. Hence $a_n b_n \to 0$.

3. Let a_n, c, $L = +\infty$ and b_n be as stated and let a $d > 0$ be given. One has for every large n that $b_n \ge \frac{d}{c}$. So for every large n it holds that $a_n b_n \ge c \cdot \frac{d}{c} = d$. Hence $a_n b_n \to +\infty = L$. The other case is similar.

4. Let $|a_n| \le d$ for every n, $b_n \to \pm\infty$ and an ε be given. For every large n, we have $|b_n| \ge \frac{d}{\varepsilon}$. So for every large n we have $\left|\frac{a_n}{b_n}\right| = |a_n| \cdot \frac{1}{|b_n|} \le d \frac{1}{d/\varepsilon} = \varepsilon$. Hence $\frac{a_n}{b_n} \to 0$.

5. Let a_n, c and b_n be as stated and a $d > 0$ be given. For every large n, one has $0 < b_n \le \frac{c}{d}$. So for every large n we have $\frac{a_n}{b_n} \ge \frac{c}{c/d} = d$. Hence $\frac{a_n}{b_n} \to +\infty$.

6. Let a_n, c, $L = -\infty$ and b_n be as stated and let a $d < 0$ be given. For every large n, one has $b_n \le dc$. So for every large n we have $\frac{b_n}{a_n} \le \frac{dc}{c} = d$. Hence $\frac{b_n}{a_n} \to -\infty = L$. The other case is treated similarly.

Exercise 2.6.6 Then if $\lim \frac{a_n}{b_n}$ exists, it equals 0.

Exercise 2.7.1 This inequality is equivalent to the inequality $(\sqrt{a} - \sqrt{b})^2 \ge 0$.

Exercise 2.7.4 Instead of k-variable function f, we take $k+1$-variable function, where the new variable is reserved for the index n.

Exercise 2.7.5 $F_n \ge n - 1$ for every $n \in \mathbb{N}$.

Exercise 2.7.9 For every ε there is δ such that $f[U(b_1, \delta) \times \cdots \times U(b_1, \delta)] \subset U(f(\overline{b}), \varepsilon)$. The proof is similar to that of Proposition 4.3.5.

Exercise 2.7.12 The proof is similar to that of Proposition 2.7.2.

Exercise 2.7.14 We need to show that $F_{n+1}F_{n-1} - F_n^2 = F_{n+1}^2 - F_{n+2}F_n$. This is equivalent to $F_{n+1}(F_{n-1} - F_{n+1}) = F_n(F_n - F_{n+2})$. Since the first bracket is $-F_n$ and the second one is $-F_{n+1}$, the identity holds.

Exercise 2.7.19 This infinite continued fraction leads to the f-recurrent sequence $(a_n) = (0, 1, \frac{1}{2}, \frac{2}{3}, \dots)$ for the function $f = f(x) = \frac{1}{1+x}$. Similarly to Proposition 2.7.15, we show that $\lim a_n = \frac{1}{2}(\sqrt{5} - 1) = 1/\phi$.

Exercise 2.8.2 The set of pairs of sequences $\{\langle (a_n), (b_n)\rangle :\ \exists\, n_0\, \forall\, m, n \ge n_0\colon a_m < b_n\}$ is a proper subset of the set $\{\langle (a_n), (b_n)\rangle :\ \exists\, n_0\, \forall\, n \ge n_0\colon a_n < b_n\}$.

Exercise 2.8.3. For example, $(a_n) = (\frac{1}{n})$ and $(b_n) = (0, 0, \dots)$.

Exercise 2.8.4. Let (a_n), (b_n), K and L be as stated. We take a number c such that $K < c < L$. By Exercise 2.1.10 there is an ε such that $U(K, \varepsilon) < U(c, \varepsilon) < U(L, \varepsilon)$. Then we take any two numbers $a, b \in U(c, \varepsilon)$ such that $a < b$. For every large m and n we have that $a_m \in U(K, \varepsilon)$ and $b_n \in U(L, \varepsilon)$. Hence $a_m \le a$ and $b \le b_n$.

Reversal of this implication is: if for every n_0 and every real numbers $a < b$ there exist m and n with $m, n \ge n_0$ such that $a_m > a$ or $b_n < b$, then $K \ge L$.

Exercise 2.8.8 Any singleton $\{a\}$ is such an interval.

Exercise 2.8.11 Let $\lim a_n = -\infty$, $b_n \le a_n$ for every large n and let a $c < 0$ be given. Then for every large n one has that $b_n \le a_n \le c$. Thus $b_n \le c$ and $\lim b_n = -\infty$. The case of the limit $+\infty$ is similar.

3 Series

Exercise 3.1.14 The first absolute value is at most $\frac{\varepsilon}{3}$ because $X' \subset X_0$. The i-th absolute value in the sum is at most $2^{-i}\frac{\varepsilon}{3}$ because $Z_i' \subset Z_i''$. The third absolute value is at most $\frac{\varepsilon}{3}$ because $Y' \subset \{Z_1, \dots, Z_n\}$.

Exercise 3.1.15 Reflexivity is witnessed by the identity bijection, symmetry by the inverse bijection and transitivity by the composition of two bijections.

Exercise 3.1.16 Let $R = \sum_{x\in X} r(x)$ and $R' = \sum_{x\in Y} s(x)$ be congruent AK series. If X and Y are finite, the equality of their sums is trivial. Suppose that they are infinite and that $f\colon X \to Y$ is a bijection proving that $R \sim R'$. Let $g\colon \mathbb{N} \to X$ be any bijection. Then $S(R') = \lim \sum_{i=1}^{n} s(f(g)(i)) = \lim \sum_{i=1}^{n} r(g(i)) = S(R)$ because $f(g)$ is a bijection from $\mathbb{N}$ to Y.

Exercise 3.1.18 The first absolute value is at most $\frac{\varepsilon}{3}$ because $Z' \subset W$. The second absolute value is at most $\frac{\varepsilon}{3}$ because $X' \subset X''$. The third absolute value is at most $\frac{\varepsilon}{3}$ because $Y' \subset Y''$.

Exercise 3.1.19 Let $Q = \sum_{x\in X} r(x)$, $Q' = \sum_{x\in X'} r'(x)$, $R = \sum_{x\in Y} s(x)$ and $R' = \sum_{x\in Y'} s'(x)$ be as stated and let $f\colon X \to X'$, $g\colon Y \to Y'$ be bijections witnessing that $Q \sim Q'$ and $R \sim R'$. Thus for every $x \in X$ and $y \in Y$ we have that $r(x) = r'(f(x))$ and $s(y) = s'(g(y))$. Let $Z \equiv X \times \{0\} \cup Y \times \{1\}$ and $W \equiv X' \times \{0\} \cup Y' \times \{1\}$. We consider $Q + R = \sum_{z\in Z} t(z)$ and $Q' + R' = \sum_{z\in W} t'(z)$. We define the bijection $h\colon Z \to W$ by $h(z) \equiv (f(x), 0)$ if $z = (x, 0)$, and by $h(z) \equiv (g(y), 1)$ if $z = (y, 1)$. Then it follows that for every $z \in Z$ we have $t(z) = t'(h(z))$. Hence $Q + R \sim Q' + R'$.

Exercise 3.1.21 The first bound by $\frac{\varepsilon}{3}$ comes from the inclusion $Z \subset X'' \times Y''$. The second bound is $|(r + \delta)\theta| \le \frac{\varepsilon}{3}$. The third bound is $|\delta s| \le \frac{\varepsilon}{3}$.

Exercise 3.1.22 We proceed as in the previous exercise, with the modifications that $Z \equiv X \times Y$, $W \equiv X' \times Y'$ and that the bijection $h\colon X \times Y \to X' \times Y'$ is given by $h((x, y)) \equiv (f(x), g(y))$. Then for every $(x, y) \in X \times Y$ we have that $r(x)s(y) = r'(f(x))s'(g(y))$ because $r(x) = r'(f(x))$ and $s(y) = s'(g(y))$. Hence $Q \cdot R \sim Q' \cdot R'$.

Exercise 3.1.24 For $R = \sum_{x \in X} r(x)$, $R' = \sum_{y \in Y} s(y)$ and $R'' = \sum_{z \in Z} t(z)$ the bijection sends $(x, 0)$ to $((x, 0), 0)$, $((y, 0), 1)$ to $((y, 1), 0)$ and $((z, 1), 1)$ to $(z, 1)$.

Exercise 3.1.25 For R and R' as in the previous exercise the bijection sends (x, y) to (y, x).

Exercise 3.1.26 For R, R' and R'' as in Exercise 3.1.24 the bijection sends $(x, (y, z))$ to $((x, y), z)$.

4 Infinite series. Elementary functions

Exercise 3.5.3 If $A = a > 0$ and $B = +\infty$, we take the described series $\sum a_n$ and its subseries $\sum a_{2n-1}$. Similarly if $A \le 0$ and $B = \pm\infty$.

Exercise 3.5.4 As the next solution shows, if we change in a series $\sum a_n$ finitely many summands and get $\sum a_n'$ then there is an m and a c such that for every $n \ge m$ it holds that $s_n' = s_n + c$. Then finite $\lim s_n$ exists iff finite $\lim s_n'$ exists.

Exercise 3.5.5 Let $\sum a_n$ and $\sum b_n$ be convergent series and let there be an m such that $b_n = a_n$ for $n \ne m$ and $b_m = a_m + c$ with $c \ne 0$. Let (s_n) and (t_n) be respective partial sums. Then $t_n = s_n$ for $n < m$ and $t_n = s_n + c$ for $n \ge m$, so that $\sum b_n = \lim t_n = c + \lim s_n = c + \sum a_n$.

Exercise 3.5.6 It is, for example, the series $1 - 1 + \frac{1}{2} - \frac{1}{2} + \cdots$ with sum 0 we encountered in the first lecture. Its subseries $1 + \frac{1}{2} + \ldots$ has sum $+\infty$.

Exercise 3.5.7 Let (s_n) be partial sums of $a_1 + a_2 + \ldots$, and (t_n) be partial sums of $a_m + a_{m+1} + \ldots$. Then $t_n = s_{n+m-1} - a_1 - \cdots - a_{m-1}$.

Exercise 3.5.8 Partial summands weakly increase (resp. decrease) for $n \ge n_0$.

Exercise 3.5.9 We have $\lim s_n = \lim n = +\infty$.

Exercise 3.5.14 The first equality follows from the definition of partial sums, the second from Theorem 2.6.2, the third from the assumption and Proposition 2.2.3, and the fourth one is trivial.

Exercise 3.5.16 $\sum (-1)^{n+1} n^{-2} = \sum n^{-2} - 2 \sum (2n)^{-2} = (1 - \frac{1}{2}) \sum n^{-2} = \frac{1}{2} \sum n^{-2} = \frac{\pi^2}{12}$.

Exercise 3.5.18 $\prod_{j=1}^{n} (1 + \frac{1}{j}) = \prod_{j=1}^{n} \frac{j+1}{j} = n + 1$ and the infinite product is $+\infty$.

Exercise 3.5.20 The standard NCC is that $\lim a_n = 1$.

Exercise 3.5.21 Due to monotonicity, we have $\lim a_n = L$. All subsequences have this limit and hence $L = +\infty$.

Exercise 3.5.22 Let (s_n) and (t_n) be partial sums of both series. There is a c such that for every $n \ge n_0$ we have $s_n \ge c + t_n$. But $\lim(c + t_n) = c + \lim t_n = c + (+\infty) = +\infty$, and the one-policeman theorem shows that also $\lim s_n = +\infty$.

Exercise 3.5.25 Let $n \ge 2$. We assume that $1 + \frac{1}{2} + \cdots + \frac{1}{n} = m$ ($\in \mathbb{N}$) and deduce a contradiction. Following the hint we write every denominator $j = 1, 2, \ldots, n$ in the form $j = a(j) \cdot 2^{b(j)}$ where $a(j) \in \mathbb{N}$ is odd and $b(j) \in \mathbb{N}_0$. For $j_0 = 2^k$, where $k \in \mathbb{N}$ is the largest number with $2^k \le n$, this expression takes the form $j_0 = 1 \cdot 2^k$. For every $j \in [n] \setminus \{j_0\}$ it holds that $b(j) < k$. Hence $1 + \frac{1}{2} + \cdots + \frac{1}{n} = \frac{a+b}{a \cdot 2^k}$, where $a \equiv a(1)a(2)\ldots a(n) \in \mathbb{N}$ is an odd number and $b \in \mathbb{N}$ is even, because it is the sum of $n - 1$ even numbers. The numerator $a + b$ is therefore odd and the power $2^k \ge 2$ in the denominator cannot be canceled. Therefore we cannot have $\frac{a+b}{a\cdot 2^k} = m$. The same argument shows that for no $n \ge 2$ we have that $h_n = \frac{k}{l}$ with odd l.

Exercise 3.5.26 ??? — presently no solution is known.

Exercise 3.5.28 One changes the series $\sum a_n$ to $\sum(-a_n)$.

Exercise 3.5.32 If a series $a_1+a_2+\dots$ and $m \in \mathbb{N}$ are such that $|a_m| \ge |a_{m+1}| \ge \dots$, $\lim a_n = 0$ and $(-1)^{n-m}a_n \ge 0$ for every $n \ge m$, then $\sum a_n$ has the sum $s \in \mathbb{R}$ and we have $\sum_{i=1}^{n} a_i \ge s \ge \sum_{i=1}^{n+1} a_i$ for every $n \ge m$ such that $n-m$ is even. The proof is similar to that of Theorem 3.5.31.

Exercise 3.5.36 Consider the series $\sum a_n \equiv 1-1+1-1+1-1+\dots$ and the sequence $S \equiv (2,2,\dots)$. The S-grouping $0+0+\dots$ has the sum 0 but the sum $\sum a_n$ does not exist.

Exercise 3.5.37 Consider the series $\sum a_n \equiv 1-1+\frac{1}{2}+\frac{1}{2}-\frac{1}{2}-\frac{1}{2}+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}-\frac{1}{3}-\frac{1}{3}-\frac{1}{3}+\dots$ and the sequence $S \equiv (2,4,6,\dots)$. The S-grouping $0+0+\dots$ has the sum 0 but the sum $\sum a_n$ does not exist.

Exercise 3.5.40 We have a constant $c > 0$ such that $\sum_{j=1}^{n} |a_j| \le c$ for every n. Let $f\colon \mathbb{N} \to \mathbb{N}$ be any bijection and $n \in \mathbb{N}$. Then $\sum_{j=1}^{n} |a_{f(j)}| \le \sum_{j=1}^{N} |a_j| \le c$, where $N \in \mathbb{N}$ is such that $[N] \supset f[\,[n]\,]$. Thus the series $\sum a_{f(n)}$ is abscon.

Exercise 3.5.41 Suppose that $\sum a_n$ is abscon. Then we have a constant $c > 0$ such that $\sum_{j=1}^{n} |a_j| \le c$ for every n. Let $B \subset \mathbb{N}$ be infinite, $f\colon \mathbb{N} \to B$ be the ordering of B and $n \in \mathbb{N}$. Then $\sum_{j=1}^{n} |a_{f(j)}| \le \sum_{j=1}^{N} |a_j| \le c$, where $N \in \mathbb{N}$ is such that $[N] \supset f[\,[n]\,]$. Thus the subseries $\sum a_{f(n)}$ is abscon.

Exercise 3.5.44 We use the comparison criterion with $\sum b_n$ being the stated series and $\sum a_n \equiv \sum \frac{c}{n(n+1)} = \sum c(\frac{1}{n} - \frac{1}{n+1})$, where c is such that $|c_n| \le c$ for every n.

Exercise 3.5.46 The coefficient of x^n in $F(x)G(x)$ is $\sum_{j=0}^{n} a_j b_{n-j}$.

Exercise 3.5.48 This follows from the equality $q^m + q^{m+1} + \dots = q^m \cdot (1+q+\dots)$.

Exercise 3.5.49 Every converging one, so iff $q \in (-1,1)$.

Exercise 3.5.51 $\sum a_n \equiv \sum \frac{1}{n}$ has $\lim a_n^{1/n} = 1$ and $\sum a_n = +\infty$. $\sum a_n \equiv \sum \frac{1}{n^2}$ has also $\lim a_n^{1/n} = 1$ but $\sum a_n$ converges.

Exercise 3.5.53 Use the same two series from the previous exercise.

Exercise 3.5.54 For example, $\frac{1}{2}+\frac{1}{3}+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{2^3}+\frac{1}{3^3}+\dots < +\infty$, but $\frac{1}{2}(\frac{3}{2})^n \to +\infty$.

Exercise 3.5.58 This is immediate from the divergence of the harmonic series.

Exercise 3.5.59 Iff $s > 1$, again by CCC.

Exercise 3.5.60 The series $\sum_{n=2}^{\infty} \frac{1}{n(n-1)}$ converges because $\frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}$.

Exercise 3.5.62 By the Fundamental Theorem of Arithmetic, every natural number is expressed in a unique way as a product of prime powers. Let $s > 1$ and $n \in \mathbb{N}$. Using the formula for sum of a geometric series, we have

$$0 \le \prod_{j=1}^{n} \left(1 - \frac{1}{(p_j)^s}\right)^{-1} - \sum_{m=1}^{p_n} \frac{1}{m^s} \le \sum_{m > p_n} \frac{1}{m^s} \to 0 \quad (n \to \infty)\,.$$

Exercise 3.5.63 No it could not, it is an infinite product $\prod_{n=1}^{\infty} a_n$ such that $a_n > 1$ for every n.

Exercise 3.5.64 It is easy to prove by induction that any tree with $k \in \mathbb{N}$ vertices has $k-1$ edges.

Exercise 3.4.5 This follows from definitions.

Exercise 3.4.7 If $A = 0$ then $B = -\infty = \lim_{n\to\infty} \sum_{j=1}^{n} b_{\rho(j)}$ and

$$\lim_{n\to\infty} \exp(\textstyle\sum_{j=1}^{n} b_{\rho(j)}) = 0 = |A|\,.$$

Similarly, if $A = \pm\infty$ then $B = +\infty = \lim_{n\to\infty} \sum_{j=1}^{n} b_{\rho(j)}$ and the displayed limit equals $+\infty = |A|$.

Exercise 3.4.8 One can show that $\prod_{n=1}^{\infty} a_n$ has an reordering with no infinite product iff $a_n \neq 0$ for every n and the following two conditions hold. (i) If $a_n < 0$ for infinitely many n, then it is not the case that $\sum_{|a_n|<1} \log(|a_n|) = -\infty$ and $\sum_{|a_n|\ge 1} \log(|a_n|) < +\infty$. (ii) If $a_n < 0$ for finitely many n, then $\sum \log(a_{z_n}) = -\infty$ and $\sum \log(a_{k_n}) = +\infty$.

Exercise 4.1.1 Let M and A be as stated and let part 1 hold, so that $A \in L(M)$ by the given definition. We chose for every n an $a_n \in P(A, \frac{1}{n}) \cap M$ and get a sequence $(a_n) \subset M \setminus \{A\}$ such that $\lim a_n = A$. Hence part 2 holds. For every m there is an ε such that $a_1, \dots, a_m \notin U(A, \varepsilon)$. So we can choose from (a_n) an injective subsequence and part 3 holds. Suppose that part 3 holds and let $(b_n) \subset M$ be an injective sequence with $\lim b_n = A$. For given n we have $b_m \in U(A, \frac{1}{n})$ for every large m. From these for only one m it holds that $b_m = A$, hence $P(A, \frac{1}{n}) \cap M \neq \emptyset$ and part 4 holds. It is clear that part 4 implies part 1.

Exercise 4.1.3 If $M \subset \mathbb{R}$ is finite, then it is bounded and neither $-\infty$ nor $+\infty$ is a limit point of M. Also, for every b there is a δ such that $P(b, \delta) \cap M = \emptyset$. Hence $L(M) = \emptyset$. If $M \subset \mathbb{R}$ is infinite, then we can choose an injective sequence $(a_n) \subset M$. By part 1 of Theorem 2.2.11, (a_n) has a subsequence (b_n) with $\lim b_n = L$. Then, by part 3 of Proposition 4.1.2, $L \in L(M)$.

Exercise 4.1.4 This is immediate from part 2 of Proposition 4.1.2.

Exercise 4.1.7. $L(\mathbb{N}) = \{+\infty\}$.

Exercise 4.1.11 For instance $A = 0$, $M = \{\pm\frac{1}{n} \mid n \in \mathbb{N}\}$, $X = \{\frac{1}{n} \mid n \in \mathbb{N}\}$, $f = 0$ on $M \setminus X$ and $f = 1$ on X.

Exercise 4.1.13 We select an element from each set $\{x \in P(K, \frac{1}{n}) \cap M\colon\ f(x) \notin U(L, \varepsilon)\}$, $n \in \mathbb{N}$.

Exercise 4.1.14. 1. Due to the transformation ($x < 0$, hence $x = -|x|$) $\frac{x}{\sqrt{1+x^2}-1} = \frac{1}{-\sqrt{1/x^2+1}-1/|x|}$ we get for $x \to -\infty$ the limit $\frac{1}{-\sqrt{1/(+\infty)+1}-0} = -1$.

2. The transformation $\frac{1}{\sqrt{1+x}-\sqrt{x}} = \sqrt{1+x} + \sqrt{x}$ gives for $x \to +\infty$ the limit $\sqrt{1+(+\infty)} + \sqrt{+\infty} = +\infty$.

3 a 4. These limits are trivial, the first does not exist and the second equals 0.

Exercise 4.1.16 For example, $(a_n) = (n\sqrt{2})$ or, more generally, $(a_n) = (n\alpha)$ where $\alpha > 0$ is any irrational number.

Exercise 4.1.17 The equality $\frac{f(x)}{x} = \frac{f(h)}{x} + (1-\frac{h}{x})(k+\alpha)$ ((2) on p. 36 of [13]), where $x = h + n$ for large fixed $h > 0$, $n \in \mathbb{N}$ and $-\varepsilon < \alpha < \varepsilon$, holds only on the discrete set $x \in \{h + n \colon n \in \mathbb{N}\}$, but we need it to hold on $x \in (x_0, +\infty)$. Cauchy regards "x as a variable quantity which converges towards the limit ∞." It is not too surprising that [13] has no clear and precise definition of the limit of a function at a (finite or infinite) point.

Exercise 5.1.2 As many as real numbers, $k_c \mapsto c$ is a bijection from the set of constants to $\mathbb{R}$.

Exercise 5.1.3 We take an m such that $m \ge 2|x|$. Then for every $n \ge m$ we have that $|\frac{x^n}{n!}| \le (\frac{|x|^m}{m!}) \cdot (\frac{1}{2})^{n-m} = \frac{1}{m!}(2|x|)^m \cdot (\frac{1}{2})^n$. Then we use geometric series.

Exercise 5.1.5 1. $\exp 0 = 1$ is trivial and the rest follows from the exponential identity. 2. For $x < y$ we have that $\mathrm{e}^y - \mathrm{e}^x = \mathrm{e}^x(\mathrm{e}^{y-x} - 1) > 0$, due to the exponential identity. 3. For $x > n$ it holds that $\mathrm{e}^x > n$, so that $\lim_{x \to +\infty} \mathrm{e}^x = +\infty$. Also, $\lim_{x \to -\infty} \mathrm{e}^x = \frac{1}{\lim_{x \to +\infty} \mathrm{e}^x} = \frac{1}{+\infty} = 0$.

Exercise 5.1.7 For contradiction, let $\sum_{j=0}^{\infty} \frac{1}{j!} = \frac{n}{m}$ with $n, m \in \mathbb{N}$. Following the hint we get that $r \equiv \sum_{j>m} \frac{m!}{j!} = n \cdot (m-1)! - \cdots \in \mathbb{N}$. This is impossible because $0 < r \le \frac{1}{m+1} \sum_{j=0}^{\infty} \frac{1}{(m+2)^j} = \frac{m+2}{(m+1)^2} < 1$.

Exercise 5.1.8 1. $\log 1 = 0$ follows from $\exp 0 = 1$. By flipping the graph over the line $y = x$ we get from the increasing function $\exp x$ the increasing function $\log x$. For $x, y > 0$ we have by the exponential identity the equality $\exp(\log x + \log y) = x \cdot y$, so that $\log x + \log y = \log(xy)$. 2. These limits are again obtained by flipping the graph of the exponential over $y = x$. 3. This follows from part 4 of the previous proposition.

Exercise 5.1.12 No, $\sqrt{x} = x^{1/2} \ne \exp(\frac{1}{2} \log x)$. The former function has at 0 value 0, but the latter function is not defined at 0.

Exercise 5.1.13 For $a = 0$ it is trivial as $\sqrt{0} = 0$ and the only solution of $x^2 = 0$ is $x = 0$. If $a > 0$ then $\sqrt{a} = a^{1/2} = \exp(\frac{1}{2} \log a)$ and $(\sqrt{a})^2 = \exp(2 \cdot \frac{1}{2} \log a) = a$ by the exponential identity. The factorization $x^2 - a = (x + \sqrt{a})(x - \sqrt{a})$ then shows that $\pm\sqrt{a}$ are the only solutions of the equation $x^2 = a$.

Exercise 5.1.15 We begin with a^x, $a > 0$. For $x = 0$ we have that $\exp(x \log a) = \exp 0 = 1$. For $x \in \mathbb{N}$ it holds that $\exp(x \log a) = \exp(\log a + \cdots + \log a)$, with x factors $\log a$. By the exponential identity this equals $\exp(\log a) \cdot \ldots \cdot \exp(\log a) = a \cdot \ldots \cdot a$, with x factors a. For every $x \in \mathbb{N}$ it holds due to the exponential identity that $\exp((-x) \log a) = \frac{1}{\exp(x \log a)}$. Thus a^x agrees with x^m. We continue with x^b. Let $b \in \mathbb{N}$, $x > 0$. Then again $\exp(b \log x) = \exp(\log x + \cdots + \log x) = \exp(\log x) \cdot \ldots \cdot \exp(\log x) = x \cdot \ldots \cdot x$, with b factors x. Also $0^b = 0 = 0 \cdot \ldots \cdot 0$. Let $b = 0$ and $x > 0$. Then $x^b = 1$. Let $b \in \mathbb{Z}$ with $b < 0$ and $x > 0$. Then we again get by the exponential identity that $x^b = \frac{1}{x^{-b}}$. Hence x^b agrees with x^m. Finally, $0^x = 0$ for $x \in \mathbb{N}$ also agrees with x^m.

Exercise 5.1.16 By Definition 5.1.10 one has that $\mathrm{e}^x = \exp(x \log \mathrm{e})$. Since $\mathrm{e} = \exp 1$ and $\log x$ is inverse to $\exp x$, it equals to $\exp(x \log(\exp 1)) = \exp(x \cdot 1) = \exp x$.

Exercise 5.1.17 Note that $a_n = \mathrm{e}^{\log a_n}$. For $A = 0$ we set $b_n = -\frac{1}{\sqrt{\log(2+|a_n|)}}$. For $0 < A < +\infty$ we set $b_n = \frac{\log A}{\log(2+|a_n|)}$. For $A = +\infty$ we set $b_n = \frac{1}{\sqrt{\log(2+|a_n|)}}$.

Exercise 5.1.19 $(-1)^2 = 1$ and $(-1)^1 = 1$ are computed by the latter definition, $1^{\frac{1}{2}} = 1$ by the former.

Exercise 5.1.20 With $A \equiv 1+x$, $B \equiv 1+x+x^2$, $C \equiv 1+x^3$ and $D \equiv 1+x^2+x^4$, for which by the hint $AD = BC \equiv E$, we should show that $(A^y+B^y)^x \cdot (C^x+D^x)^y = (A^x+B^x)^y \cdot (C^y+D^y)^x$. Equivalently, that $E^{xy}(1+(\frac{B}{A})^y)^x(1+(\frac{C}{D})^x)^y = E^{xy}(1+(\frac{A}{B})^x)^y(1+(\frac{D}{C})^y)^x$. But this holds because $\frac{B}{A} = \frac{D}{C}$ and $\frac{C}{D} = \frac{A}{B}$, and multiplication is commutative.

Exercise 5.1.21 For $A = 0$ we set $a_n \equiv \frac{1}{n^n}$ and $b_n \equiv \frac{1}{n}$. For $0 < A < 1$ we set $a_n \equiv A^n$ and $b_n \equiv \frac{1}{n}$. For $A = 1$ we set $a_n = b_n \equiv \frac{1}{n}$. For $1 < A < +\infty$ we set $a_n \equiv \frac{1}{A^n}$ and $b_n \equiv -\frac{1}{n}$. For $A = +\infty$ we set $a_n \equiv \frac{1}{n^n}$ and $b_n \equiv -\frac{1}{n}$. No, it could not, $a^b < 0$ only for $b \in \mathbb{Z} \setminus \{0\}$.

Exercise 5.1.22 Proceed as in Exercise 5.1.3.

Exercise 5.1.25 By the theorem, $\cos 1 \ge 1 - \frac{1^2}{2!} = \frac{1}{2}$ and $\cos 2 \le 1 - \frac{2^2}{2!} + \frac{2^4}{4!} = -\frac{1}{3}$.

Exercise 5.1.26 1. The runner runs one lap in time 2π and gets in the same position. 2. This is the behavior of the y-coordinate of the runner in the first quarter of the lap. 3. The track is symmetric according to the y-axis, and according to the origin $(0,0)$. 4. The counter-clockwise rotation of S around the origin by $\frac{\pi}{2}$ is equivalent to the exchange of the coordinate axes. The second relation says that the points on S have distance 1 from the origin. 5. Search the Internet for pictures for "geometric proof of summation formulae for sine and cosine".

Exercise 5.1.28 This follows from the bounds $|a-a_n|, |b-b_n| \le |a+bi-z_n|$.

Exercise 5.1.29 It follows from the fact that the metric space $(\mathbb{C}, |\cdot|)$ is complete (every Cauchy sequence in it converges) and the fact that the sequence $(\sum_{j=0}^{n} \frac{z^j}{j!})$, $n \in \mathbb{N}$, is Cauchy, which is easy to prove by means of (complex) geometric series.

Exercise 5.1.32 Iterate the identity $u_1u_2\dots u_n - v_1v_2\dots v_n = (u_1-v_1)u_2\dots u_n + v_1(u_2\dots u_n - v_2\dots v_n)$.

Exercise 5.1.34 $\mathrm{e}^{a+bi} = \mathrm{e}^a\mathrm{e}^{bi}$.

Exercise 5.1.35 Using geometric series we have $|\exp z - 1 - z| \le \frac{1}{2}\sum_{n\ge 2}|z|^n = \frac{|z|^2}{2(1-|z|)} \le |z|^2$.

Exercise 5.1.36 For every $n \in \mathbb{N}$ we set $k_n = n$ and $a_{n,j} = \frac{a}{n}$, and use the theorem.

Exercise 5.1.37 We know from the properties of cosine and sine what zeros they have.

Exercise 5.1.38 By the definition of inverses in Section 1.1, the functional inverse of $\sin x \,|\, [-\frac{\pi}{2}, \frac{\pi}{2}]$ goes from $[-1,1]$ to $[-\frac{\pi}{2}, \frac{\pi}{2}]$, but $\arcsin\colon [-1,1] \to \mathbb{R}$. So arcsine is not equal to this inverse and is only congruent to it, in the sense of congruence of functions in Definition 1.1.3. Similarly for arccosine.

Exercise 5.1.39 Between $\mathbb{R}$ and $(-\frac{\pi}{2}, \frac{\pi}{2})$.

Exercise 5.2.2 This follows from the commutativity, associativity and distributivity of the operations $+$ and $\cdot$ on $\mathbb{R}$ and from the fact that the operation of intersection of two sets enjoys these properties too: $M \cap N = N \cap M$, $(M \cap N) \cap P = M \cap (N \cap P)$ and $M \cap (N \cap P) = (M \cap N) \cap (M \cap P)$. In $\mathbb{R}$ the number 0, respectively 1, is neutral to addition, respectively multiplication, and always $\mathbb{R} \cap M = M$. No function $f \in \mathcal{R}$ with $M(f) \ne \mathbb{R}$ has additive or multiplicative inverse.

Exercise 5.2.5 The functions $f-g$ and $f+f_{-1}\cdot g$ have equal values and also equal definition domains: $M(f)\cap M(g)=M(f)\cap(\mathbb{R}\cap M(g))$.

Exercise 5.2.6 We set $g=h\equiv\emptyset_f$ and see that $\Rightarrow$ does not hold. Similarly, $f=g\equiv\emptyset_f$ show that $\Leftarrow$ does not hold.

Exercise 5.2.8 For every $i\in[n]$ the initial segment $\langle f_1,f_2,\dots,f_i\rangle$ is a generating word of f_i.

Exercise 5.2.9 $|x|=(\mathrm{id}\cdot\mathrm{id})^{1/2}$.

Exercise 5.2.10 It is the empty function $\emptyset$.

Exercise 5.2.11 Yes, it is, for example by the previous exercise.

Exercise 5.2.12 $g\equiv k_{-1}\cdot f$.

Exercise 5.2.13 For instance $f(x)\equiv\sqrt{\sin(\pi x)}+\sqrt{-\sin(\pi x)}$ and $g(x)\equiv\frac{1}{\sin(\pi/x)}$; here π is k_π and x is the identity id.

Exercise 5.2.18 On $(-\frac{\pi}{2},\frac{\pi}{2})$ we have $\tan x=\frac{\sin x}{\sqrt{1-\sin^2 x}}$ and $\sin x=\frac{\tan x}{\sqrt{1+\tan^2 x}}$. Hence $\arctan x=\arcsin(\sin(\arctan x))=\arcsin\big(\frac{x}{\sqrt{1+x^2}}\big)$.

Exercise 5.2.20 x^b for $b\le 0$ and $b\in\mathbb{N}$ $(\subset\mathbb{R})$.

Exercise 5.2.22 For instance $\arcsin x$, $|x|$ or $\arcsin(\sin x)$.

Exercise 5.3.3 Suppose that p has the canonical from $\sum_{j=0}^{n}a_jx^j$ and that n is minimum. We show by induction on $\deg p\equiv n$ that $|Z(p)|\le n$. For $n=0$ it holds, then $p=k_{a_0}$ with $a_0\ne 0$ and p has no zero. Let $n>0$. If p has no zero, the inequality holds. Let $a\in Z(p)$. Then we divide with remainder the polynomial p in the above canonical form by the polynomial $x-a=\mathrm{id}-k_a$ and get the expression $p=(x-a)q$, for some canonical polynomial q with degree at most $n-1$. For every $b\ne a$ with $p(b)=0$ we have $q(b)=0$. By induction, $|Z(p)|=1+|Z(p)\setminus\{a\}|\le 1+|Z(q)|\le 1+\deg q\le 1+(n-1)=n$.

Exercise 5.3.4 We proved it in the previous proof.

Exercise 5.3.7 Let $f=\sum_{j=0}^{m}a_jx^j$ and $g=\sum_{j=0}^{n}m_jx^j$. Then $f+g=\sum_{j=0}^{p}(a_j+b_j)x^j$, where $p\equiv\max(m,n)$, if $j>m$ then $a_j\equiv 0$ and if $j>n$ then $b_j\equiv 0$, is a canonical polynomial, provided that not all sums a_j+b_j are zero. If they are all zero, then $f+g$ is the zero polynomial.

Exercise 5.3.8 Let $f=\sum_{j=0}^{m}a_jx^j$ and $g=\sum_{j=0}^{n}m_jx^j$. Then the product $fg=\sum_{j=0}^{m+n}(\sum_{i=0}^{j}a_ib_{j-i})x^j$ is a canonical polynomial.

Exercise 5.3.9 Let $f=\sum_{j=0}^{m}a_jx^j$ and $g=\sum_{j=0}^{n}m_jx^j$ be such that $m\ne n$ or $a_j\ne b_j$ for some $j\le\min(m,n)$. It follows that $f-g=\sum_{j=0}^{p}(a_j-b_j)x^j$, where $p\equiv\max(m,n)$, if $j>m$ then $a_j\equiv 0$ and if $j>n$ then $b_j\equiv 0$, is a canonical polynomial because not all differences a_j-b_j is zero.

Exercise 5.3.10 By Proposition 5.2.3 it only remains to prove that additive inverses exist and that the product of two nonzero polynomials is nonzero. The former is provided by Exercise 5.2.12 and the latter by Exercise 5.3.8.

Exercise 5.3.12 The map that sends k_0 to the abstract zero polynomial, and a canonical polynomial $f = \sum_{j=0}^{n} a_j x^j$ to the abstract polynomial $\sum_{j=0}^{n} a_j x^j$ in $\mathbb{R}[x]$, is the required isomorphism.

Exercise 5.3.14 This is a trivial consequence of the definitions.

Exercise 5.3.16 Now the domain of the left-hand side is $\big(M(f_1) \cap M(g_1) \setminus Z(g_1)\big) \cap \big(M(f_2) \cap M(g_2) \setminus Z(g_2)\big)$, and of the right-hand side it is $(M(f_1) \cap M(f_2)) \cap (M(g_1) \cap M(g_2)) \setminus (Z(g_1) \cup Z(g_2))$. They both are equal to $M(f_1) \cap M(f_2) \cap M(g_1) \cap M(g_2) \setminus (Z(g_1) \cup Z(g_2))$.

Exercise 5.3.18 Let $f_1 = f_2 = g_1 \equiv k_1$ and $g_2 \equiv \mathrm{id}$. Then $\frac{f_1/g_1}{f_2/g_2}$ is $\mathrm{id}\,|\,\mathbb{R} \setminus \{0\}$ but $\frac{f_1 g_2}{f_2 g_1}$ is id.

Exercise 5.3.21 The reflexivity and symmetry of $\sim$ are clear. We prove transitivity. Let $r \sim s$ and $s \sim t$. We take these rational functions in canonical forms: $r = \frac{a}{b}$, $s = \frac{c}{d}$ and $t = \frac{e}{f}$. By the assumption $r = s$ on $M(r) \cap M(s) = \mathbb{R} \setminus Z(bd)$ and $s = t$ on $\mathbb{R} \setminus Z(df)$. Thus $r = t$ on $\mathbb{R} \setminus (Z(bd) \cup Z(df)) = (\mathbb{R} \setminus Z(bf)) \setminus Z(d) = (M(r) \cap M(t)) \setminus Z(d)$. By the continuity of r and t on $M(r) \cap M(t) \cap Z(d)$ the functions r and t are equal on the whole $M(r) \cap M(t)$. Hence $r \sim t$.

Exercise 5.3.22 Let $r, s, r', s' \in \mathrm{RAC} \setminus \{\emptyset\}$. It is not hard to see, using continuity of rational functions, that $r \sim r'$ and $s \sim s'$ imply that also $r + s \sim r' + s'$ and $r \cdot s \sim r' \cdot s'$. Thus addition and multiplication of equivalence blocks of rational functions is correctly defined. Neutrality of $[k_0]_\sim$ and $[k_1]_\sim$ to $+$ and $\cdot$, respectively, is immediate. The commutativity and associativity of addition and multiplication and the distributive law in $\mathrm{RAC_{fi}}$ follow from the arithmetic in $\mathbb{R}$. Additive and multiplicative inverses require equivalence blocks. Let $r = p/q \in \mathrm{RAC} \setminus \{\emptyset\}$ have the canonical form (p, q). Then $[(-p)/q]_\sim$ is the additive inverse: the sum is $[k_0/q]_\sim = [k_0]_\sim$. If $p \neq k_0$ then $[q/p]_\sim$ is the multiplicative inverse: the product is $[pq/qp]_\sim = [k_1]_\sim$.

Exercise 5.3.24 Recall that the abstract rational functions $\mathbb{R}(x)$ are the equivalence classes $[\frac{p}{q}]_\sim$ where $p, q \in \mathbb{R}[x]$, $q \neq 0$, are abstract (real) polynomials and the equivalence relation (congruence) $\sim$ on $\mathbb{R}[x] \times (\mathbb{R}[x] \setminus \{0\})$ is given by $\frac{p_1}{q_1} \sim \frac{p_2}{q_2}$ iff $p_1 q_2 = p_2 q_1$ (with the product and equality in $\mathbb{R}[x]$). It is not too hard to show that the map sending the zero rational function $[k_0]_\sim \in \mathrm{RAC}$ to the zero abstract rational function $[\frac{0}{1}]_\sim$ in $\mathbb{R}(x)$, and the nonzero rational function $[r]_\sim = [p/q]_\sim$, where $(p, q) \in (\mathrm{POL} \setminus \{k_0\})^2$ is a canonical form of $r \in (\mathrm{RAC} \setminus \{\emptyset\}) \setminus \{[k_0]_\sim\}$, to the nonzero abstract rational function $[\frac{p}{q}]_\sim \in \mathbb{R}(x) \setminus \{0\}$, is an isomorphism of fields. The notation $p = \sum_{j=0}^{n} a_j x^j$, where $n \in \mathbb{N}_0$, $a_j \in \mathbb{R}$ and $a_n \neq 0$, is interpreted in two ways: as the canonical form of a nonzero polynomial in POL, and as a nonzero abstract polynomial in $\mathbb{R}[x]$.

5 Limits of functions. Asymptotic notation

Exercise 4.2.1 Let $b \in L^-(M)$. For every $n \in \mathbb{N}$ we chose from $P^-(b, \frac{1}{n}) \cap M$ a point a_n and get the required sequence (a_n). If $b \notin L^-(M)$ then for some ε we have that $P^-(b, \varepsilon) \cap M = \emptyset$ and the required sequence does not exist. For $L^+(M)$, the proof is similar.

Exercise 4.2.2 The first two implications follow from the fact that any one-sided deleted neighborhood is contained in the two-sided one. We prove the third implication. Let b be a limit point of M. Then there is a sequence $(a_n) \subset M \setminus \{b\}$ such that $\lim a_n = b$. It has a subsequence (a_{m_n}) such that for every n it holds that $a_{m_n} > b$

or for every n it holds that $a_{m_n} < b$. Thus b is a one-sided limit point of M. 4. For instance $0 \in L([0,1))$ but $0 \notin L^-([0,1))$.

Exercise 4.2.3 It is, for example, the set $\mathbb{N} \subset \mathbb{R}$.

Exercise 4.2.4 We find in the set a strictly monotone sequence.

Exercise 4.2.6 1. Here a is a limit point of M. Thus, as we know, it is the left or right limit point of M. 2. This follows from the inclusion $P(a,\delta) \subset P^-(a,\delta') \cup P^+(a,\delta'')$ for $\delta = \min(\delta',\delta'')$. 3. If $\lim_{x\to a} f(x) = A$, it suffices to take ε so small that $U(A,\varepsilon)$ and $U(K,\varepsilon)$, or $U(A,\varepsilon)$ and $U(L,\varepsilon)$, are disjoint and we get a contradiction.

Exercise 4.2.8 The proof is similar to the proof of Proposition 4.1.9.

Exercise 4.2.10 The proof is similar to the proof of Theorem 4.1.12.

Exercise 4.2.12 This follows from the equality $f[P^\pm(b,\delta)] = (f \mid I^\pm(b))[P(b,\delta)]$.

Exercise 4.3.2 There is a sequence $(b_n) \subset M(f)$ such that $\lim b_n = b$, but $\lim f(b_n)$ does not exist or is not equal to $f(b)$. Or, by Exercise 4.3.10, iff there is a sequence $(b_n) \subset M(f)$ with $\lim b_n = b$ such that $\lim f(b_n) = A \neq f(b)$.

Exercise 4.3.4 The solution of the inequalities $|x-b| < \delta$, respectively $|f(x)-f(b)| < \varepsilon$, is exactly $U(b,\delta)$, respectively $U(f(b),\varepsilon)$. Non-strict inequalities are equivalent: decrease δ or ε a little.

Exercise 4.3.6 If $b \in M \setminus L(M)$, there is a δ such that $U(b,\delta) \cap M(f) = \{b\}$. Then the only sequences in $M(f)$ with the limit b are the eventually constant sequences (a_n) with $a_n = b$ for $n \geq n_0$. Then $\lim f(a_n) = f(b)$ which agrees with continuity of f in such point b (trivially or by Proposition 4.3.9).

Exercise 4.3.7 If $b \in M$ is isolated, it is not limit and there is a ε such that $M \cap P(b,\varepsilon) = \emptyset$. Then $M \cap U(b,\varepsilon) = \{b\}$. If $b \in M$ is not isolated, it is limit and for every ε we have that $M \cap P(b,\varepsilon) \neq \emptyset$. Hence for every ε we have that $M \cap U(b,\varepsilon) \neq \{b\}$.

Exercise 4.3.8 This follows from the definition of isolated points.

Exercise 4.3.10 This is immediate from Heine's definition of continuity at a point and from part 3 of Theorem 2.2.11.

Exercise 4.3.11 Let f be continuous at $b \in M(f)$ and let an ε be given. Then for some δ it holds that $f[U(b,\delta)] \subset U(f(b),\varepsilon)$. Thus $f[U^-(b,\delta)] \subset U(f(b),\varepsilon)$ and $f[U^+(b,\delta)] \subset U(f(b),\varepsilon)$ because $U^-(b,\delta)$ and $U^+(b,\delta)$ are contained in $U(b,\delta)$. So f is both left- and right-continuous at b.

Let f be both left- and right-continuous at $b \in M(f)$ and let an ε be given. Then there exist δ' and δ'' such that $f[U^-(b,\delta')] \subset U(f(b),\varepsilon)$ and $f[U^+(b,\delta')] \subset U(f(b),\varepsilon)$. We set $\delta \equiv \min(\delta',\delta'')$. We get that $f[U(b,\delta)] \subset U(f(b),\varepsilon)$, because $U(b,\delta) \subset U^-(b,\delta') \cup U^+(b,\delta'')$. Hence f is continuous at b.

Exercise 4.3.13 Every number in M is positive because x is irrational. $U(x,1)$, which is an interval of length 2, contains only finitely many fractions with bounded denominators. Hence M is a finite set. But $U(x,1)$ contains at least one integer and so $M \neq \emptyset$.

Exercise 4.4.2 Let $f \in \mathcal{F}(M)$. For $b \in L^-(M)$ and f that weakly decreases on $P^-(b,\delta)$ we replace supremum with infimum. For $b \in L^+(M)$ and f that weakly increases, resp. weakly decreases, on $P^+(b,\delta)$ we take infimum, resp. supremum. For

$+\infty \in L(M)$ and f that weakly decreases on $U(+\infty, \delta)$ we replace supremum with infimum. For $-\infty \in L(M)$ and f that weakly increases, resp. weakly decreases, on $U(-\infty, \delta)$ we take infimum, resp. supremum.

Exercise 4.4.4 Let $k = -\infty$, hence f weakly decreases, and let a $c < -1$ be given. We take an $h > 0$ such that for every $x \ge h$ we have $f(x+1) - f(x) \le c$. Let $n \in \mathbb{N}$ and $x = h, h+1, \dots, h+n-1$. By summing these inequalities and rearranging the result we get $\frac{f(h+n)}{n} \le c + \frac{f(h)}{n}$. Let $d \equiv \max(|f(h)|, 2(h+1)) + 1$ and $x \in \mathbb{R}$ be such that $x \ge h+d$. We take the unique $n \in \mathbb{N}$ such that $h+n \le x < h+n+1$. Then $n \ge |f(h)|$. By these inequalities and since f weakly decreases, we have $\frac{f(x)}{n} \le \frac{f(h+n)}{n} \le c + \frac{f(h)}{n} \le c+1$. Since $\frac{n}{x} \ge 1 - \frac{h+1}{d} \ge \frac{1}{2}$ and $c+1 < 0$, we have $\frac{f(x)}{x} \le (c+1)\frac{n}{x} \le \frac{c+1}{2}$. Hence $\lim_{x \to +\infty} \frac{f(x)}{x} = -\infty$. For $k = +\infty$ the argument is similar.

Exercise 4.4.6 $M(f/g) = M(f) \cap M(g) \setminus Z(g)$ and there is a δ such that $Z(g) \cap U(A, \delta) = \emptyset$.

Exercise 4.4.7 This is a particular case of part 3 of the theorem with $f = k_1$.

Exercise 4.4.10 1. The previous proof of the theorem is easily modified, for $K < L$ there is an ε and real numbers a, b such that $U(K, \varepsilon) < \{a\} < \{b\} < U(L, \varepsilon)$. 2. Again the reversal of an implication.

Exercise 4.4.12 The ordinary limits are only changed to one-sided. The proofs are basically reductions to ordinary limits by means of Proposition 4.2.13.

Exercise 4.4.15 First we generalize the notion of limits from $\mathcal{R}$ to maps between MS (metric space(s)). If $f\colon M \to N$, where (M, d) and (N, e) are MS, and $a \in M$, $b \in N$, then $\lim_{x \to a} f(x) = b$ means that for every ε there is a δ such that if $x \in M$ satisfies $0 < d(x, a) \le \delta$ then $e(f(x), b) \le \delta$. Theorem 4.4.13 generalizes as follows. If $a \in M$, $b \in N$ and $f, g\colon M \to N$ are maps between MS such that $\lim_{x \to a} f(x) = b$ and $\lim_{x \to a} e(f(x), g(x)) = 0$ (this is a map from the MS M to the MS $(\mathbb{R}, |x-y|)$), then also $\lim_{x \to a} g(x) = b$.

Exercise 4.5.3 Let a sequence $(a_n) \subset M(f(g)) \setminus \{A\}$ have $\lim a_n = A$. By Heine's definition of limits of functions (HDLF), $\lim g(a_n) = K$. Suppose that condition 1 holds. Then for the n with $g(a_n) = K$ we have that $f(g)(a_n) = f(g(a_n)) = f(K) = L$. If there are infinitely many n such that $g(a_n) \ne K$ then for the corresponding subsequence (a_{m_n}) we have by HDLF that $\lim f(g(a_{m_n})) = L$. Thus $\lim f(g)(a_n) = L$ and HDLF says that $\lim_{x \to A} f(g)(x) = L$. Suppose that condition 2 holds. Then, deleting from $(g(a_n))$ finitely many terms, we may assume that $(g(a_n)) \subset M(f) \setminus \{K\}$. By HDLF, $\lim f(g(a_n)) = L$. Again by HDLF, $\lim_{x \to A} f(g)(x) = L$. The case that none of the conditions holds is resolved via HDLF already in the original proof.

Exercise 4.5.6 The inner function $p(x)$ now in general is not injective, but we have the finiteness bound $|p^{-1}(y)| \le \deg p$ for every $y \in \mathbb{R}$, and Theorem 4.5.1 can be used because condition 2 holds.

Exercise 4.6.1 For example, take $M \equiv \{\frac{1}{n}\colon\ n \in \mathbb{N}\} \cup \{1 + \frac{1}{n}\colon\ n \in \mathbb{N}\}$ and define $f \in \mathcal{F}(M)$ by $f(\frac{1}{n}) = f(1 + \frac{1}{n}) \equiv \frac{1}{n}$.

Exercise 4.6.5 Because $a_n \in U(A', \delta_0)$, for $\delta < \delta_0$ we have $P(A, \delta) \subset U(A, \delta_0)$ and $U(A, \delta_0) \cap U(A', \delta_0) = \emptyset$.

Exercise 4.6.7 If $x \ne y$ then $|f(x) - f(y)| \ge g(|x-y|) > 0$ and $f(x) \ne f(y)$.

Exercise 4.6.8 Let $x < y$ be in M. By Lagrange's theorem we have for some $d \in (x, y)$ that

$$|f(y) - f(x)| = |f'(d)| \cdot |y - x| \geq c|y - x| .$$

Exercise 4.6.11 This is similar to the previous proof, only the inequality is reversed.

Exercise 4.6.13 It is easy to see that there is an increasing sequence $(m_n) \subset \mathbb{N}$ such that $|a_n - b_{m_n}| \geq n$ for every $n \in \mathbb{N}$ (the subsequence of (b_n) is running fast ahead of (a_n)).

Exercise 4.6.16 Both properties follow easily from Proposition 4.6.15.

Exercise 4.7.2 A pair (a, b) is in $G_f \Delta G_g$ iff $a \in M(f)\Delta M(g)$ or $f(a) \neq g(a)$.

Exercise 4.7.3 Reflexivity and symmetry of $\doteq$ are clear. The transitivity follows from the fact that for any three sets A, B and C we have $A\Delta C \subset A\Delta B \cup B\Delta C$.

Exercise 4.7.6 This is just an explicit unfolding of the boundedness of $\frac{f}{g}$ on N.

Exercise 4.7.8 Because if $f \doteq f_0$ and $g \doteq g_0$, then $\frac{f}{g} \doteq \frac{f_0}{g_0}$.

Exercise 4.7.9 If $f = O'(g)$ (on N) and $a \in M(f) \cap M(g) \cap N \cap Z(g)$, then necessarily $a \in Z(f)$. Thus if we change f to f_0 by replacing $f(a) = 0$ with any nonzero value $f_0(a)$, then $f_0 \neq O'(g)$ (on N). Hence O' in general does not stand change in a single value of the function and is not an asymptotic relation.

Exercise 4.7.10 1. Yes. 2. No (problem near 0). 3. No (problem near $\pm\infty$). 4. Yes. 5. No (problem near 0). 6. Yes.

Exercise 4.7.14 1. Yes. 2. Yes. 3. No. 4. No. 5. Yes. 6. Yes.

Exercise 4.7.16 This follows from the fact that the limit $\lim_{x\to A} \frac{u(x)}{v(x)} = 0$ is preserved when we multiply the numerator $u(x)$ by a bounded factor.

Exercise 4.7.17 We assume that, for a constant $c \geq 0$, for every $x \in M(f) \cap M(h) \cap N \setminus Z(h)$ it holds that $|\frac{f}{h}(x)| \leq c$, and the same holds with g in place of f. Hence it holds for every $x \in M(f) \cap M(g) \cap M(h) \cap N \setminus Z(h)$ that $|\frac{f+g}{h}(x)| \leq |\frac{f}{h}(x)| + |\frac{g}{h}(x)| \leq 2c$.

Exercise 4.7.18 For f we have a bound as previously and for every $x \in M(g) \cap N$ it holds that $|g(x)| \leq c$. Hence it holds for every $x \in M(f) \cap M(g) \cap M(h) \cap N \setminus Z(h)$ that $|\frac{fg}{h}(x)| = |\frac{f}{h}(x)| \cdot |g(x)| \leq c^2$.

Exercise 4.7.19 This is similar to the previous exercise.

Exercise 4.7.20 We have that $\lim_{x\to A} \frac{f}{h}(x) = 0$ and $\lim_{x\to A} \frac{g}{h}(x) = 0$. Hence the limit $\lim_{x\to A} \frac{f+g}{h}(x)$, which is defined because A is a limit point of $M((f + g)/h)$, equals $\lim_{x\to A} \frac{f}{h} + \lim_{x\to A} \frac{g}{h} = 0 + 0 = 0$.

Exercise 4.7.21 Similarly, from $\lim_{x\to A} \frac{f}{h}(x) = 0$ and a bound that $|g(x)| \leq c$ for every $x \in M(g) \cap P(A, \theta)$ we easily deduce that also $\lim_{x\to A} \frac{fg}{h}(x) = 0$.

Exercise 4.7.22 This is similar to the previous exercise.

Exercise 4.7.23 Like before we get from the assumptions on f, g, h and A that $\lim_{x\to A} \frac{f+g}{h}(x) = \lim_{x\to A} \frac{f}{h}(x) + \lim_{x\to A} \frac{g}{h}(x) = 1 + 0 = 1$.

Exercise 4.7.24 We get from the assumptions on f, g, h and A that $\lim_{x\to A} \frac{fg}{h}(x) = \lim_{x\to A} \frac{f}{h}(x) \cdot \lim_{x\to A} g(x) = 1 \cdot 1 = 1$.

Exercise 4.7.25 This is similar to the previous exercise.

Exercise 4.7.26 For $k \in \mathbb{N}$ with $k \le x$ the number of pairs $(m, n) \in \mathbb{N}^2$ with $mn = k$ equals $\tau(k)$.

Exercise 4.7.27 Because $\lim_{x \to 1^+} x^{1/3} \log x = 0$.

Exercise 4.7.28 No problem in our definition of big O, only we get no upper bound on $T_{\mathrm{HH}}(1)$.

Exercise 4.8.3 This is an application of Proposition 4.7.15.

Exercise 4.8.6 Moving the denominator $\exp x - 1$ to the right, we get for $n \in \mathbb{N}_0$ that the coefficient $\sum_{k=0}^{n-1} \frac{B_k}{k!} \cdot \frac{1}{(n-k)!}$ of x^n equals 0 for $n \neq 1$, and 1 for $n = 1$. So for $n = 1$ we get that $B_0 = 1$, and for $n \ge 2$ that $B_{n-1} = -\frac{1}{n} \sum_{k=0}^{n-2} \binom{n}{k} B_k$. The second claim follows from the identity $f(-x) = f(x)$ for the formal power series $f(x) \equiv \frac{x}{\mathrm{e}^x - 1} + \frac{x}{2}$.

Exercise 4.8.7 Suppose that G is connected by the new equivalent definition and that $\{A, B\}$ is a partition of V. We take any two vertices $u \in A$ and $v \in B$. The walk joining u and v provides the edge that intersects both A and B. If G is not connected by the new equivalent definition, we take any two (distinct) vertices $u, v \in V$ that cannot be joined by any walk, define A as the set of vertices that can be reached from u by a walk, and set $B \equiv V \setminus A$ $(\ni v)$. Then $\{A, B\}$ is a partition of V such that no edge in E intersects both A and B. Hence G is not connected by the original definition.

6 Continuous functions

Exercise 6.1.2 Use Proposition 4.3.9, every point of the definition domain is isolated.

Exercise 6.1.3 For every $x \in \mathbb{R}$, δ and ε we have $k_a[U(x, \delta)] = \{a\} \subset U(k_a(x), \varepsilon) = U(a, \varepsilon)$.

Exercise 6.1.4 For every $a \in \mathbb{R}$ and given ε it suffices to set $\delta = \varepsilon$ because $x[U(a, \delta)] = U(a, \delta)$.

Exercise 6.1.9 Let N be dense in M and $a \in M$. Using the axiom of choice we take for every n a point b_n in $N \cap U(a, \frac{1}{n})$ and get a sequence $(b_n) \subset N$ with $\lim b_n = a$. If $(b_n) \subset N$ has $\lim b_n = a \in M$ then for every δ for every large n it holds that $b_n \in U(a, \delta)$.

Exercise 6.1.10 Let $a < b$ be in $\mathbb{R}$. We take $n \in \mathbb{N}$ so large that $\frac{2\sqrt{2}}{n} < b - a$. It follows that for some $m \in \mathbb{Z}$ we have $\frac{m}{n} \in (a, b)$, and that for some $m \in \mathbb{Z} \setminus \{0\}$ we have $\frac{m\sqrt{2}}{n} \in (a, b)$. Thus every nontrivial interval contains both a fraction and an irrational number. Hence $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are dense in $\mathbb{R}$.

Exercise 6.1.11 Let $0 \le a < b \le 1$ and $n \in \mathbb{N}$ be maximum such that $\frac{1}{n} \ge \frac{a+b}{2}$. Then the nontrivial interval $(\max(\{a, \frac{1}{n+1}\}), \frac{a+b}{2})$ is contained in (a, b) and is disjoint to N.

Exercise 6.1.14 Because then $f \,|\, N$ is an at most countable kernel of f.

Exercise 6.1.16 $M = \mathbb{R} \setminus \mathbb{Q}$.

Exercise 6.2.2 In ZF we easily define a bijection $f \colon \mathbb{N}_0 \to \mathbb{Q}$. Now $\mathbb{N}_0 = \omega$ is well ordered by the relation $\in$.

Exercise 6.2.5 If $A_m = \emptyset$ then $|\alpha - b| \le \frac{\varepsilon}{2}$ for every $\alpha \in U(b_m, \frac{1}{m}) \cap M \cap \mathbb{Q}$. If we had for any of these fractions α that $|\alpha - b_m| \le \frac{\varepsilon}{2}$, we would have $|b_m - b| \le |b_m - \alpha| + |\alpha - b| \le \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$, which is not the case.

Exercise 6.3.2 The inverse of any bijection is a bijection.

Exercise 6.3.3 The function s is onto $\mathbb{N}$ because every natural number is a product of an odd number and a power of two. These expressions are unique: if $(2k-1)2^{l-1} = (2m-1)2^{n-1}$ then $l = n$, else 2 would divide an odd number, so also $k = m$. Hence s is injective.

Exercise 6.3.6 By Exercise 6.1.2 constants are continuous. Clearly, $k_a = k_b$ implies that $a = b$.

Exercise 6.3.7 An injection from $\mathbb{R}$ to $\mathcal{C}(M)$ is given by restrictions of constant functions. To get an injection from $\mathcal{C}(M)$ to $\mathbb{R}^{\mathbb{N}}$ we use Proposition 6.1.13.

Exercise 6.4.2 Let $a < c < b$ with $a, c \in f[I]$. Then by Theorem also $c \in f[I]$. Hence $f[I]$ is an interval.

Exercise 6.4.4 We prove that for every two continuous functions $f, g\colon [0,1] \to \mathbb{R}$ satisfying $f(0) = g(1) = 0$ and $f(1) = g(0) = 1$ there is a $t \in (0,1)$ such that $f(t) = g(t)$. We set $h \equiv f - g\colon [0,1] \to \mathbb{R}$ and use the theorem on intermediate values.

Exercise 6.5.2 This follows from Theorems 2.4.12 and 2.8.1.

Exercise 6.5.3 The empty set is trivially compact. The set $\mathbb{R}$ is not compact.

Exercise 6.5.7 Continuity of both functions is easy to show. Neither function has maximum because for every $x \in [0,1)$ and every $y \in (x,1)$ the value at y is greater than that at x.

Exercise 6.5.8 If f attains maximum or minimum everywhere, then for every a, b in $M(f)$ we have both $f(a) \le f(b)$ and $f(a) \ge f(b)$, and f is constant. Clearly, constant functions attain maximum and minimum everywhere.

Exercise 6.5.11 $\emptyset$ and $\mathbb{R}$ are closed sets, and closed sets are closed to finite unions and arbitrary intersections. The proof follows from passing to complements and using de Morgan formulas.

Exercise 6.5.12 This is clear from the fact that if $b \in M$ then $U(b, \delta) \subset M$ for some δ, and therefore $P(b, \theta) \subset M$ for every $\theta \le \delta$.

Exercise 6.5.15 $[a,b] \setminus P(c,\delta) = [a,b] \cap (\mathbb{R} \setminus ((c-\delta, c) \cup (c, c+\delta)))$ which is a closed and bounded set.

Exercise 6.5.16 The arguments in the proof of Theorem 6.5.14 easily extend to metric spaces.

Exercise 6.5.17 Any discrete metric space (X, d) is closed and bounded, but if X is infinite then X is not compact because sequences $(x_n) \subset X$ with $x_m \ne x_n$ for $m \ne n$ do not have convergent subsequences.

Exercise 6.5.23 I_a is the union of all open intervals that contain a and are contained in M.

Exercise 6.5.24 The union of two intersecting open intervals is an open interval.

Exercise 6.5.25 Because for every $a \in M$ there is a $b \in M \cap \mathbb{Q}$ near a such that $a \in I_b$.

Exercise 6.5.26 C is closed because it is an intersection of closed sets. It is uncountable because it is exactly the set of those points in $[0,1]$ whose 3-adic expansions use only digits 0 and 2. C has "length" 0 because $1 - \frac{1}{3} - \frac{2}{9} - \frac{4}{27} - \frac{8}{81} - \cdots = 0$.

Exercise 6.5.28 The stated property of points of M means that every singleton set $\{b\}$, $b \in M$, is sparse in M. If M were at most countable, the union $M = \bigcup_{b \in M}\{b\}$ would contradict Baire's theorem.

Exercise 6.5.29 Baire's theorem for metric spaces concerns complete spaces (X, d). In such space every Cauchy sequence of points converges. A set $Y \subset X$ is sparse (in X) if every ball $B \subset X$ has a subball $B' \subset B$ such that $Y \cap B' = \emptyset$. The theorem then says that if (X, d), $X \neq \emptyset$, is complete and $X = \bigcup_{n=1}^{\infty} X_n$, then there is an index n such that X_n is not sparse. The proof is similar to the one we gave in the case $(X, d) = (M, |x - y|)$ for nonempty closed $M \subset \mathbb{R}$.

Exercise 6.6.2 Let $f\colon M \to \mathbb{R}$ be uniformly continuous, $c \in M$ and an ε be given. We take for this ε the δ guaranteed by the uniform continuity. Then certainly $f[U(c,\delta)] \subset U(f(c), \varepsilon)$, so that f is continuous at c.

Exercise 6.6.4 If $(a_n), (b_n) \subset M$ for a compact set M, we have convergent subsequences (a_{k_n}) and (b_{m_n}). For simpler notation we denote them again by (a_n) and (b_n).

Exercise 6.6.5 Let $a_n \equiv \frac{1}{n}$ and $b_n \equiv \frac{2}{\pi(2n-1)}$. Then $\lim(a_n - a_{n+1}) = \lim(b_n - b_{n+1}) = 0$ but for every n one has that $f(a_{n+1}) - f(a_n) = 1$ and $f(b_{n+1}) - f(b_n) = 2$.

Exercise 6.6.6 For example $f \equiv 0$ on $[0, \frac{1}{\sqrt{2}}] \cap \mathbb{Q}$ and $f \equiv 1$ on $[\frac{1}{\sqrt{2}}, 1] \cap \mathbb{Q}$.

Exercise 6.6.7 If M is closed then the set of finite limits of sequences in M coincides with M. If M is not closed then a point outside of M is in $\overline{M}$.

Exercise 6.6.8 We take a δ for $\varepsilon = 1$ in the UC property of f, and a finite set $N \subset M$ with the property that for every $a \in M$ there is a $b_a \in N$ with $|a - b_a| \le 1$. Then for every $a \in M$ we have $|f(a)| \le |f(a) - f(b_a)| + |f(b_a)| \le 1 + \max(\{|f(x)|\colon\ x \in N\})$.

Exercise 6.6.12 This follows easily from the theorem on the relation of limits of functions and ordering.

Exercise 6.7.2 It follows from the arithmetic of continuity, from the definition of polynomials and rational functions, and from continuity of constants and identity.

Exercise 6.7.4 In the difference $a_n(x+c)^n - a_n x^n$ we use the binomial theorem, cancel $a_n x^n$ and take out c. The triangle inequality gives that $|a_n \sum_{i=1}^{n} \binom{n}{i} c^{i-1} x^{n-i}|$ is at most $|a_n| \sum_{i=1}^{n} \binom{n}{i} |c|^{i-1} |x|^{n-i}$. We replace the numbers $|c|$ and $|x|$ by d which is not smaller and is at least 1, so that we can increase $i-1$ to i. Then we bound the inner sum by $(d+d)^n$.

Exercise 6.7.5 It suffices to show that for every $c > 0$ we have that $\lim \frac{c^n}{n!} = 0$. The sequence $(\frac{c^n}{n!})$ is nonnegative and, for large n, decreasing. Therefore it has the limit $d \ge 0$ and there is an m such that $d = \inf(\{\frac{c^n}{n!}\colon\ n \ge m\})$. Suppose for the contrary that $d > 0$. Then for large enough $n \ge m$ one has that $\frac{d(n+1)}{c} > \frac{c^n}{n!}$. Hence $d > \frac{c^{n+1}}{(n+1)!}$, which is a contradiction, and $d = 0$.

Exercise 6.7.10 Let $b \in M(f(g))$ and let an ε be given. There is a δ such that $f[U(g(b), \delta)] \subset U(f(g)(b), \varepsilon)$. There is a θ such that $g[U(b, \theta)] \subset U(g(b), \delta)$. Hence $f(g)[U(b, \theta)] \subset \ldots \subset U(f(g)(b), \varepsilon)$ and $f(g)$ is continuous at b according to the definition.

Exercise 6.7.13 The function f given as $f(x) = x$ on $(0, 1)$ and with the value $f(2) = 1$ is continuous and increasing but the inverse $f^{-1}\colon (0, 1] \to (0, 1) \cup \{2\}$ is not continuous at 1. The function f with the values $f(0) = 1$ and $f(n) = 1 - \frac{1}{n}$ for $n \in \mathbb{N}$ has the closed definition domain $\mathbb{N}_0 \subset \mathbb{R}$ and is injective and continuous, but the inverse $f^{-1}\colon \{1 - \frac{1}{n}\colon\ n \in \mathbb{N}\} \cup \{1\} \to \mathbb{N}_0$ is not continuous at 1.

Exercise 6.7.14 $\log x$ is continuous by any of parts 2–4 of the theorem, $\arccos x$ and $\arcsin x$ by any of parts 1, 2 and 4, and $\arctan x$ and $\operatorname{arccot} x$ by any of parts 2–4.

7 Derivatives

Exercise 7.1.2 Use Corollary 4.5.4.

Exercise 7.1.6 It is an instance of Proposition 4.2.7.

Exercise 7.1.7 Let $f \in \mathcal{F}(M)$ and $b \in M \cap L^{\pm}(M)$. Then we have the equivalence that $f'_{\pm}(b) = A \iff (f \mid I^{\pm}(b))'(b) = A$, with equal signs. In fact, it is an instance of Proposition 4.2.13.

Exercise 7.1.8 The first claim follows easily from definitions. 0 is a limit point of the interval $(0, 1)$ but it is not its two-sided limit point.

Exercise 7.1.10 The displayed inequalities show that $c, d \neq b$.

Exercise 7.1.11 No, it does not. Neither 0 nor 1 is a two-sided limit point of the domain $[0, 1]$.

Exercise 7.1.13 Now $M(f) = \mathbb{Z}$ and $L^{\mathrm{TS}}(M(f)) = \emptyset$. By Corollary 7.1.12, every point b in $M(f)$ is "suspicious". In fact, f has both local minimum and local maximum at every point of its domain.

Exercise 7.1.15 The function $f(x)$ is defined at b, the other function is not. This is the only point where they differ.

Exercise 7.1.16 $\operatorname{sgn}'_{-}(0) = \lim_{x \to 0^-} \frac{-1-0}{x-0} = +\infty$. Similarly $\operatorname{sgn}'_{+}(0) = +\infty$. By item 2 in Exercise 7.1.6, $\operatorname{sgn}'(0) = +\infty$.

Exercise 7.1.17 $(|x|)'_{-}(0) = \lim_{x \to 0^-} \frac{-x-0}{x-0} = -1$ and similarly for $(|x|)'_{+}(0)$.

Exercise 7.1.19 For example, $\operatorname{sgn}'(0) = +\infty$ and $\operatorname{sgn} 0 = 0 \notin L(\operatorname{sgn}[\mathbb{R}]) = \emptyset$.

Exercise 7.1.20 Use continuity of the function f at b.

Exercise 7.1.21 If $f \in \mathcal{R}$ and $f'_{-}(b) \in \mathbb{R}$ then f is left-continuous at b, and similarly for $f'_{+}(b) \in \mathbb{R}$. Proofs are the same as for Proposition 7.1.14.

Exercise 7.1.22 The derivative $(\sqrt{x})'_{-}(0)$ is not defined because 0 is not a left limit point of the definition domain $[0, +\infty)$. Remaining one-sided derivatives are equal to the ordinary ones.

Exercise 7.1.25 For every $x \neq b$ we have $\frac{k_c(x)-k_c(b)}{x-b} = \frac{0}{x-b} = 0$. Hence $k'_c = k_0$.

Exercise 7.1.26 For every a we have $\lim_{x \to a} \frac{x+c-(a+c)}{x-a} = 1$.

Exercise 7.1.27 This is what Proposition 7.1.14 says.

Exercise 7.1.28 $D(f) \subset D_-(f) \cup D_+(f)$ and this inclusion is in general strict.

Exercise 7.1.30 This is exactly the assumption that $a_n - b_n = o(b_n)$ $(n \to \infty)$.

Exercise 7.2.2 This follows from the limit $\lim_{x\to b} \frac{f(x)-f(b)}{x-b} = f'(b)$.

Exercise 7.2.3 For $a > 0$ the equation is $y = \frac{x}{2\sqrt{a}} + \frac{\sqrt{a}}{2}$. At $\langle 0, 0\rangle$ the tangent is not defined.

Exercise 7.2.5 Use the definition of derivatives or results on derivatives of sums of functions.

Exercise 7.2.6 The required modifications are straightforward.

Exercise 7.2.7 We only need to show that for every non-vertical line $\ell \in \mathcal{N}$ there exists a unique pair $\langle s, t\rangle \in \mathbb{R}^2$ such that $\ell = \{\langle x, sx+t\rangle\colon\ x \in \mathbb{R}\}$. Suppose that $\langle s', t'\rangle \in \mathbb{R}^2$ is another pair determining ℓ. Then $t = s0+t = s'0+t' = t'$, hence $t = t'$. From $s + t = s1 + t = s'1 + t' = s' + t'$ we get $s = s'$.

Exercise 7.2.9 The system $sa + t = b$ & $sa' + t = b'$, with given $a \ne a'$, b, b', and unknowns s and t, has a unique solution whose component s is given by the stated formula.

Exercise 7.2.11 We assume that $\ell(x) = sx+t$ is a limit tangent to G_f at $\langle b, f(b)\rangle$. We take any sequence $(b_n) \subset M(f)\setminus\{b\}$ such that $b_n \to b$. Let the line $\kappa(b_n, f(b_n), b, f(b))$ be given as $y = s_n x + t_n$. Then $s_n \to s$, $t_n \to t$ and always $s_n b + t_n = f(b)$. Hence $f(b) = s_n b + t_n \to sb + t$. We see that $sb + t = f(b)$ and $\langle b, f(b)\rangle \in \ell$.

Exercise 7.2.13 This is a straightforward check of a rational identity.

Exercise 7.2.16 Let b, M and f be as stated. Using Heine's definition of pointwise continuity and Theorem 2.2.11, we take a sequence (x_n) converging to b from one side and such that $\lim f(x_n) \ne f(b)$. Then we take any sequence (y_n) converging to b from the other side and such that $\lim f(y_n)$ exists.

Exercise 7.2.17 Let $\ell_n(x) = s_n x+t_n$ and $\ell(x) = sx+t$. By the assumption, $\lim s_n = s$ and $\lim t_n = t$. But $c = sb + t$ and $c_n = s_n b + t_n$, so that $\lim c_n = \lim(s_n b + t_n) = (\lim s_n)\cdot b + \lim t_n = sb + t = c$.

Exercise 7.2.18 Let d_n be the infimum of the ε such that $\frac{y_n - c}{x_n - b} \in U(A, \varepsilon)$. We may clearly assume that always $y_n \ne c$ or that $A = 0$. Then we easily define by induction an increasing sequence $(m_n) \subset \mathbb{N}$ such that always $\frac{y_n - u_{m_n}}{x_n - z_{m_n}} \in U(A, d_n + \frac{1}{n})$. We are done because $\lim d_n = 0$.

Exercise 7.2.19 We set $b = 0$, $M = \mathbb{R}$, $f(x) = x^2 \sin(\frac{1}{x})$ for $x \ne 0$, $f(0) = 0$, $x_n = \frac{2}{(4n-1)\pi}$ and $y_n = \frac{2}{(4n-3)\pi}$, for n running in $\mathbb{N}$. Then $f'(0) = 0$ and $x_n, y_n \to 0$, but the secants $\kappa(x_n, f(x_n), y_n, f(y_n))$ have slopes $\frac{y_n^2 + x_n^2}{y_n - x_n} \gg \frac{n^2}{n^2} = 1$ $(n \in \mathbb{N})$.

Exercise 7.3.2 The adaptation is straightforward.

Exercise 7.3.3 $(\mathrm{sgn}(x) - \sqrt{x})'(0) = \lim_{x\to 0} \frac{1-\sqrt{x}}{x} = \lim_{x\to 0} \frac{1/\sqrt{x}-1}{\sqrt{x}} = \frac{+\infty}{0^+} = +\infty$.

Exercise 7.3.6 Use that $fg = gf$.

Exercise 7.3.7 Thus $f'(0) = +\infty$, $g'(0) = -\infty$ and $f'(0)g(0) + f(0)g'(0) = (+\infty)\frac{1}{2} + (-\frac{1}{2})(-\infty) = +\infty$. On the other hand, $(fg)(x) = -1$ for $x \neq 0$ and $(fg)(0) = -\frac{1}{4}$ give $(fg)'_-(0) = +\infty$, $(fg)'_+(0) = -\infty$ and $(fg)'(0)$ does not exist.

Exercise 7.3.10 We modify the previous exercise and change the value $f(0)$ to $f(0) = \frac{1}{2}$. Then $\frac{f'(0)g(0)-f(0)g'(0)}{g(0)^2} = ((+\infty)\cdot\frac{1}{2} - \frac{1}{2}\cdot(-\infty))/\frac{1}{4} = +\infty$, but $(f/g)(x) = -1$ for $x \neq 0$ and $(f/g)(0) = 1$. The derivative $(f/g)'(0)$ again does not exist.

Exercise 7.4.2 With g not continuous at b the formula $(f(g))'(b) = f'(g(b))\cdot g'(b)$ may not hold in the sense that the right-hand side is defined but the left-hand side is undefined. We set $f(x) \equiv x^2$, take $g(x)$ as the modified signum with the value $g(0) = \frac{1}{2}$ and $b \equiv 0$ ($M = \mathbb{R} \subset L(M) = \mathbb{R}^*$). Then $f'(g(b)) = (2x)(\frac{1}{2}) = 1$, $g'(b) = +\infty$ and the right-hand side is $1\cdot(+\infty) = +\infty$. However, $f(g)(x) = 1$ for every $x \neq 0$ and $f(g)(0) = \frac{1}{4}$, so that $(f(g))'(b)$ does not exist.

Exercise 7.4.5 All stated properties of f, except the last one, are very easy to see. Clearly, $f^{-1}(c) = f^{-1}(0) = 0$. We show that $(f^{-1})'(0)$ does not exist by providing two sequences $(y_n) \subset f[M]\setminus\{c\}$ with $y_n \to c$ such that the two limits $\lim \frac{f^{-1}(y_n)-f^{-1}(c)}{y_n-c} = \lim \frac{f^{-1}(y_n)}{y_n}$ are different. For the first sequence $(y_n) \equiv (\frac{1}{n})$ the limit is $\lim \frac{1+1/n}{1/n} = +\infty$. For the second sequence $(y_n) \equiv (\frac{2}{2n+1})$ the limit is $\lim \frac{2/(2n+1)}{2/(2n+1)} = 1$.

Exercise 7.5.2 For $n = 0$ we have $\frac{1}{c}(a_n(x+c)^n - a_n x^n) = 0$. For $n \geq 1$ this difference equals $a_n \sum_{j=0}^{n-1}(x+c)^j x^{n-1-j}$. We use the transformation $\sum_{j=0}^{n}(x+c)^j x^{n-j} - (n+1)x^n = c\sum_{j=1}^{n}\sum_{i=1}^{j}\binom{j}{i}c^{i-1}x^{n-i}$ (by the binomial theorem), where $n \geq 1$. Then we use the triangle inequality and the estimates $|c|, |x| \leq y$. Finally, the sum of binomial coefficients in the j-th row of the Pascal triangle is 2^j and $y^{n-1} \leq y^{n+1}$.

Exercise 7.5.4 $(\log|x|)' = \frac{1}{x}$ $(\in \mathcal{F}(\mathbb{R}\setminus\{0\}))$.

Exercise 7.5.5 1. The derivative $(a^x)' = (\exp((x\log a))' = \cdots = a^x\cdot\log a$ follows from Corollaries 7.4.3 and 7.5.3, and from the derivative $(k_c(x)\cdot\mathrm{id}(x))' = k_c(x)$. 2. For $b > 1$ and $x > 0$ we get the derivative from the expression $x^b = \exp(b\log x)$; for $x = 0$ we use the definition of the derivative and the limit

$$\lim_{x\to 0} x^{-1}\exp(b\log x) = \lim_{x\to 0}\exp((b-1)\log x) = \lim_{y\to-\infty}\exp y = 0\,.$$

3. This is trivial. 4. For $b < 1$ we use the expression $x^b = \exp(b\log x)$. 5. This is trivial. 6. We use the Leibniz formula. 7. This is trivial. 8. We use the expression $x^m = \frac{1}{x^{-m}}$ and the derivative of ratios.

Exercise 7.5.6 Now we have the limit

$$\lim_{x\to 0} x^{-1}\exp(b\log x) = \lim_{x\to 0}\exp((b-1)\log x) = \lim_{y\to+\infty}\exp y = +\infty\,.$$

Exercise 7.5.7 The derivative of $\tan x = \frac{\sin x}{\cos x}$ follows from the derivatives $(\sin x)' = \cos x$ and $(\cos x)' = -\sin x$, from the identity $\sin^2 x + \cos^2 x = k_1(x)$ and from Corollary 7.3.11. Similarly for $\cot x$.

Exercise 7.5.8 Let $f \equiv \sin x\,|\,[-\frac{\pi}{2}, \frac{\pi}{2}]$. Then $\arcsin x$ is the inverse of f. By part 1 of Theorem 7.4.4 we have for $a \in (-1, 1)$ that

$$(\arcsin x)'(a) = \tfrac{1}{f'(\arcsin a)} = \tfrac{1}{(\cos x\,|\,[-\frac{\pi}{2},\frac{\pi}{2}])(\arcsin a)} = \tfrac{1}{(\sqrt{1-f(\arcsin a)^2})} = \tfrac{1}{\sqrt{1-a^2}}\,.$$

We can differentiate $\arctan x$ similarly or we can use the expression in terms of acsine in Exercise 5.2.18. Similarly for the derivatives of $\arccos x$ and $\operatorname{arccot} x$.

Exercise 7.5.9 This is an application, at the points -1 and 1, of part 2 of Theorem 7.4.4 to the inverse of $\sin x \,|\, [-\frac{\pi}{2}, \frac{\pi}{2}]$.

Exercise 7.5.10 Because for $0 \in \mathbb{R}$ we have $x^0 = k_1(x) \,|\, (0, +\infty)$.

Exercise 7.5.11 For $x \neq 0$ we have $f'(x) = 2x \sin(\frac{1}{x}) - \cos(\frac{1}{x})$; $f'(0) = \lim_{x \to 0} \frac{1}{x} \cdot x^2 \sin(\frac{1}{x}) = \lim_{x \to 0} x \sin(\frac{1}{x}) = 0$. This is a discontinuous function.

Exercise 7.5.12 Both sides of the equality are $\emptyset$, the empty rational function.

Exercise 7.5.13 If $k \neq -1$ then $r_k(x) = \frac{x^{k+1}}{k+1} + k_c(x)$. For $k = -1$ no rational function $r_{-1}(x)$ with $r_{-1}'(x) = 1/x$ exists by Proposition 8.2.6.

Exercise 7.6.3 $\exp(b \log x)$.

Exercise 7.6.4 This follows from Proposition 5.2.15.

8 Applications of mean value theorems

Exercise 8.1.2 $f'(0)$ does not exist.

Exercise 8.1.7 If we assume the left-hand side of the equivalence, then we can deduce from it the right-hand side with the help of the reduction of Lagrange's theorem to Rolle's that is described in the proof of Theorem 8.1.6. If we assume the right-hand side of the equivalence, then the left-hand side follows because Rolle's theorem is a particular case of Lagrange's.

Exercise 8.1.9 If $f(x) < l_u(x)$ for some $x \in [a, b]$, then by the continuity of $l_t(x)$ in t (for fixed x) we would have $f(x) < l_{u'}(x)$ for every $u' \in (u - \varepsilon, u]$. This is a contradiction with the definition of u as a supremum. We get, by increasing u a bit, a similar contradiction if $f(x) > l_u(x)$ for every $x \in [a, b]$.

Exercise 8.1.10 Then every tangent line to G_f has slope $s = \frac{f(b)-f(a)}{b-a}$.

Exercise 8.1.12 If $f(a) = f(b)$ and $g'(c) = \pm\infty$ then the product $z \cdot g'(c) = 0 \cdot (\pm\infty)$ is indefinite. However, one can modify the statement of the theorem by distinguishing the cases $f(b) \neq f(a)$ and $f(b) = f(a)$. In the former case we can allow $g'(c) \in \mathbb{R}^*$, and in the latter case we have a $c \in (a, b)$ such that $f'(c) = 0$.

Exercise 8.2.2 Add the dummy zero coefficients $p_{k+1}(x) = 0, \ldots, p_{n_0}(x) = 0$.

Exercise 8.2.3 Multiply the coefficients $p_i(x)$ by the polynomial $(x - k)(x - k - 1) \ldots (x - n_0 + 1)$.

Exercise 8.2.5 Use Exercise 4.7.23.

Exercise 8.2.8 If $f \in \mathcal{R}$ is nonzero, then $f(b) \neq 0$ for some $b \in M(f)$. Then $f^2 = f \cdot f$, $M(f^2) = M(f) \cap M(f) = M(f)$, $b \in M(f^2)$ and $f^2(b) = f(b)^2 \neq 0$.

Exercise 8.2.11 First we select an injective sequence. Then we use Proposition.

Exercise 8.2.12 The denominator of $r(x)$ has finitely many zeros $\{z_1 < z_2 < \cdots < z_l\}$ in $(k-1, +\infty)$, and a gap (z_{i-1}, z_i), $i \in [l+1]$, between them, where $z_0 = k - 1$ and $z_{l+1} = +\infty$, contains a tail $(a_r, a_{r+1}, \ldots)$ of (a_n).

Exercise 8.2.13 Suppose that (a_n) decreases. Then we can apply to any restiction $f \mid [a_{n+1}, a_n]$ Rolle's theorem, because f is differentiable on $[a_{n+1}, a_n]$ and $f(a_{n+1}) = f(a_n) = 0$. We get a point $b_n \in (a_{n+1}, a_n)$ with $f'(b_n) = 0$. Similarly for increasing sequence (a_n).

Exercise 8.2.14 Clearly, $r'(x) \in \mathrm{RAC}$. Suppose that $p_j(x) \neq 0$. Then we have $(p_j(x)\log(x-j+1))' = p_j'(x)\log(x-j+1) + p_j(x) \cdot \frac{1}{x-j+1} \mid (j-1, +\infty)$. The latter summand is absorbed in $(r(x) \mid (c, +\infty))'$. If $\deg p_j = 0$, the former summand disappears. If $\deg p_j > 0$, then in the former summand we have $\deg p_j' = \deg p_j - 1$.

Exercise 8.2.16 We just replace the function $\log(x-j+1)$ with $\log(x-j+1+c)$.

Exercise 8.2.17 Proceed as in the proof of Theorem 8.2.10.

Exercise 8.3.1 This follows from uniqueness of expansions of natural numbers in base 10. See Theorem ... .

Exercise 8.3.3 The set of recursive real numbers is countable.

Exercise 8.3.5 Any fraction $\alpha = \frac{a}{b}$ is a root of $x - \alpha$ and of $bx - a$. The number $\sqrt{n}$ is a root of $x^2 - n$.

Exercise 8.3.6 We get form (i) by dividing the polynomial by the leading coefficient. We get form (ii) by multiplying it by the product of denominators of its coefficients.

Exercise 8.3.7 Exactly the integers $\mathbb{Z}$.

Exercise 8.3.8 It is, $\phi^2 - \phi + 1$.

Exercise 8.3.9 We have the (Binet) formula $F_n = \frac{1}{\sqrt{5}}\big(\phi^n - \psi^n\big)$ where ψ is the other root of $x^2 - x + 1$.

Exercise 8.3.12 Every nonzero complex polynomial with degree $d \in \mathbb{N}_0$ has at most d complex roots (basically Exercise 5.3.3). Since the set of integer polynomials is countable, the set of algebraic numbers is also countable because it is a countable union of finite sets. But the set $\mathbb{R}$ of real numbers is uncountable (Theorem) and therefore also the set of real transcendental numbers is uncountable, in particular nonempty.

Exercise 8.3.14 $|(\sum_{j=0}^n a_j x^j)'(x)|$, where $x \in [\alpha, \alpha+1$ with $\alpha = \frac{a}{10^k}$ for $a \in \mathbb{Z}$ and $k \in \mathbb{N}_0$, is by the triangle inequality at most $\sum_{j=0}^n j|a_j|(|a|+1)^{j-1}$. This is at most $(n+1)^2 \max(\dots)(|a|+1)^n$. We replace n by $n+1$ in order to have only positive factors in the bound, and $n-1$ by n in order to have nonnegative exponents.

Exercise 8.3.16 The algorithm computes the values $p_m(\alpha_m + \frac{j}{10^k})$ for $j = 0, 1, \dots, 10^{k-k_m} - 1$, which are always fractions, and selects the first j for which the value is nonzero. Such j exists because $p_m(x)$ is a nonzero integral polynomial with sufficiently small degree.

Exercise 8.4.2 In the division $g(x) = \frac{f(x)}{x - \frac{p}{q}}$ the polynomial $f(x)$ loses one multiplicity of the root $\frac{p}{q}$, but $\alpha \neq \frac{p}{q}$ as it is irrational.

Exercise 8.4.3 The constant d exists because f' is continuous and I is compact.

Exercise 8.4.5 Let $\frac{a}{b}$ with $a \in \mathbb{Z}$ and $b \in \mathbb{N}$ be any fraction. We may assume that $a \in \mathbb{N}$. Recall the school algorithm for computing the decimal expansion of $\frac{a}{b}$, for example $\frac{1}{7} = 1 : 7 = 0.142\ldots$, with the residues $1, 3, 2, 6, \ldots$. Once a residue repeats, the residues and the expansion start to repeat.

Exercise 8.4.6 Let $n \in \mathbb{N}$ and $c > 0$ be arbitrary. We take $m \in \mathbb{N}$ large enough so that $\frac{2}{q_m^{(m+1)/2}} < c$ and $\frac{m+1}{2} \ge n$. Then $|\lambda - \frac{z_m}{q_m}| < \frac{2}{q_m^{m+1}} = \frac{2}{q_m^{(m+1)/2}} \cdot \frac{1}{q_m^{(m+1)/2}} < \frac{c}{q_m^n}$ and Liouville's inequality is violated.

Exercise 8.4.7 Algorithm $\mathcal{L}$ determines for every input $n \in \mathbb{N}$ if $n = m!$ for some $m \in \mathbb{N}$. If it is the case, then $\mathcal{L}$ outputs the digit $\mathcal{L}(n) = 1$, and else it outputs 0. In more detail, $\mathcal{L}$ multiplies the numbers $1, 2, \ldots, m$ as long as $m! \le n$. Since $m! \ge 2^{m-1}$, $\mathcal{L}$ knows the digit $\mathcal{L}(n)$ at the latest for $m \le \log_2(n+2) + 1 \le \log_2(6n)$. Multiplying two numbers $\le n$ takes time $O(\log_2(n+1)^2) = O(\log_2(6n)^2)$. Thus $\mathcal{L}$ computes the decimal digit $\mathcal{L}(n)$ ($\in \{0, 1\}$) in time $O(\log_2(6n)^3)$, which is time polynomial in the size of n (number of its digits). This (probably) cannot be achieved by the algorithm $\mathcal{A}$ in Theorem 8.3.15. In any case, the description of $\mathcal{A}$ is much more complex than that of $\mathcal{L}$.

Exercise 8.4.8 The proof is very similar to the proof of Corollary 8.4.4.

Exercise 8.5.2 In the assumptions we just change $\ge$ to $\le$. In the conclusion we change, of course, $-\infty$ to $+\infty$ The proof of this version is similar to the original one.

Exercise 8.5.3 We define $f(x) \equiv x + 1$ for $-1 \le x < 0$, $f(0) \equiv 0$ and $f(x) \equiv x - 1$ for $0 < x \le 1$.

Exercise 8.5.6 These proofs are very similar to the one shown.

Exercise 8.5.8 These proofs are very similar to the one shown.

Exercise 8.5.9 We cannot. The signum function $\operatorname{sgn} x$ shows, for $b \equiv 0$, that these sets may have just one element.

Exercise 8.5.11 Take, for example, $k_0(x) \,|\, [a, b]$ and, to get $f(x)$, for any sequence $a < a_1 < a_2 < \cdots < b$ with $\lim a_n = b$ deform the flat G_{k_0} around every point $(a_n, 0)$ by a small upward bump. Thus a function $f \in \mathcal{F}([a, b])$ arises that satisfies the assumptions of the proposition. Each bump is so low that the value $f'(b) = k_0'(b) = 0$ remains, but at the same time it is so steep that, say, $f'(a_n) = 1$ for every n. The bumps are also so narrow that $f'\big(\frac{a_n + a_{n+1}}{2}\big) = 0$. Then $f'(b) = 0$ but the limit $\lim_{x \to b} f'(x)$ does not exist.

Exercise 8.5.15 We compute by part 2 of HR 2 on the interval $(0, 1)$ that

$$\lim_{x \to 0} x^{\varepsilon} \log x = \lim_{\ldots} \tfrac{\log x}{x^{-\varepsilon}} = \tfrac{1}{-\varepsilon} \lim_{x \to 0^+} \tfrac{1/x}{x^{-\varepsilon - 1}} = \tfrac{1}{-\varepsilon} \lim_{x \to 0^+} x^{\varepsilon} = 0\,.$$

Exercise 8.5.16 We replace $f(x)$ with $f(a + b - x)$. The definition domain $P(b, \delta)$ is the union $(b - \delta, b) \cup (b, b + \delta)$. We move from the interval $U(+\infty, \delta) = (\frac{1}{\delta}, +\infty)$ and $x \to +\infty$ to the interval $(0, \delta)$ and $y \to 0$ by means of the substitution $y = \frac{1}{x}$.

Exercise 8.5.18 Because of the definitions of the functions $f(x)$, $g(x)$ and the derivative.

Exercise 8.6.2 No, the former vale is always in $\mathbb{R}$, but the latter element may be $\pm\infty$.

Exercise 8.6.3 The former sequence is 4-periodic: $(\overline{\sin x, \cos x, -\sin x, -\cos x})$. The n-th term of the latter sequence is $(\frac{1}{x})^{(n)} = (-1)^n n! x^{-n-1}$.

Exercise 8.6.5 Consider $f(x) \equiv x^3$ and $b \equiv 0$.

Exercise 8.6.6 We set $b \equiv 0$ and define $g(x) \equiv \sqrt{x}$ for $x \geq 0$ and $g(x) \equiv -\sqrt{-x}$ for $x \leq 0$. Then the function

$$f(x) \equiv \textstyle\int_{-\delta}^{x} g(t)\,\mathrm{d}t \colon U(0,\,\delta) \to \mathbb{R}$$

has at 0 a strict local minimum, $f' = g$ on $U(0, \delta)$ and $(f')'(0) = g'(0) = +\infty$.

Exercise 8.6.8 The strict convexity of x^2 follows from Theorem 8.6.16 because $(x^2)'' = 2 > 0$. The claim about $|x|$ is clear from the graph which is a union of two half-lines. The strict concavity of $\log x$ follows from Theorem 8.6.16 because $(\log x)'' = -x^{-2} < 0$.

Exercise 8.6.9 This is logically clear from the definition.

Exercise 8.6.10 This is clear from the definition by applying the symmetry $(x, y) \mapsto (x, -y)$ of the plane.

Exercise 8.6.13 The argument, based on Theorem 4.4.1, is the same as in the proof of Theorem 8.6.11. Only the upper bound is now not available.

Exercise 8.6.15 It is not true because the one-sided derivative at the endpoint may be infinite, and then continuity at the point is not guaranteed. For example, the function $f \in \mathcal{F}([0, 1])$, given as $f(x) = 0$ for $x \neq 1$ and $f(1) = 1$, is convex.

Exercise 8.6.17 It follows that (c, c') lies above or on the line going through the first two points. Thus the second point (b, b') lies below or on the line going through the first point (a, a') and the third point (c, c').

Exercise 8.6.20 Indeed, the tangent at $(0, 0)$ is the x-axis.

Exercise 8.6.21 Every point of the graph is an inflection point.

Exercise 8.7.2 These functions have one-sided limits $\pm\infty$ at 0.

Exercise 8.7.4 Let $\lim_{x\to+\infty}(f(x) - sx - b) = 0$. By adding the limit $\lim_{x\to+\infty} b = b$ we get that $\lim_{x\to+\infty}(f(x) - sx) = b$. Dividing by the limit $\lim_{x\to+\infty} x = +\infty$ we get that $\lim_{x\to+\infty}(\frac{f(x)}{x} - s) = 0$, thus $\lim_{x\to+\infty} \frac{f(x)}{x} = s$.

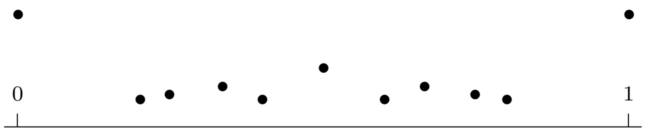
Suppose that $\lim_{x\to+\infty} \frac{f(x)}{x} = s$ and $\lim_{x\to+\infty}(f(x) - sx) = b$. Subtracting from the latter limit the limit $\lim_{x\to+\infty} b = b$, we get that $\lim_{x\to+\infty}(f(x) - sx - b) = 0$.

Exercise 8.7.5 It is the axis x.

Exercise 8.7.6 $F \in \mathcal{F}(M)$ is even, resp. odd, if $M = -M$ $(= \{-x :\ x \in M\})$ and for every $x \in M$ we have $F(-x) = F(x)$, resp. $F(-x) = -F(x)$. The function F is c-periodic ($c \in \mathbb{R}$) if $M = M \pm c$ $(= \{x \pm c :\ x \in M\})$ and for every $x \in M$ we have $F(x + c) = F(x)$.

Exercise 8.7.7 **0.** $F \notin \mathrm{EF}$. **1.** $M(F) = \mathbb{R}$. **2.** F is an even and 1-periodic function. The only periods are integers. **3.** From Proposition 4.3.12 we know that F is continuous exactly at irrational numbers. We show that $F'(\alpha)$ does not exist for any $\alpha \in \mathbb{R}$. For rational $\alpha = \frac{m}{n}$ in lowest terms, $F(\alpha) = \frac{1}{n}$ and $F(x) = 0$ for x arbitrarily close to α both from the left and the right. This gives differential ratios at α going to $+\infty$ on the left, and to $-\infty$ on the right, of α. If α is irrational then these zero values of $F(x)$ give diff. ratios at α equal to 0 and arbitrarily close to α. But by the theorem of Dirichlet there exist infinitely many different fractions $\frac{m}{n}$ in lowest terms such that $|\alpha - \frac{m}{n}| < \frac{1}{n^2}$. These fractions give at α diff. ratios in absolute value $> \frac{1/n}{1/n^2} = n \to +\infty$. **4.** It is not hard to see that $\lim_{x\to\alpha} F(x) = 0$ for every $\alpha \in \mathbb{R}$ and that the limits $\lim_{x\to\pm\infty} F(x)$ do not exist. **5.** The function F intersects the axis x exactly at the points $(\alpha, 0)$ for irrational α, and the axis y at the point $(0, 1)$. **6.** We show that for irrational $\alpha \in \mathbb{R}$ the derivative $F'_+(\alpha)$ does not exist, and that $F'_+(\alpha) = -\infty$ for $\alpha \in \mathbb{Q}$. Suppose that α is irrational and that $\beta \to \alpha^+$ via irrational β. Then $\frac{F(\beta)-F(\alpha)}{\beta-\alpha} = 0$. For rational Dirichlet's approximations β of α we get $\frac{F(\beta)-F(\alpha)}{\beta-\alpha} \to -\infty$. Thus $F'_+(\alpha)$ does not exist. Let $\alpha = \frac{k}{l}$ be rational and in lowest terms. For irrational β we get $\frac{F(\beta)-F(\alpha)}{\beta-\alpha} = -\frac{1}{l} \cdot \frac{1}{\beta-\alpha} \to -\frac{1}{l} \cdot \frac{1}{0^+} = -\infty$. For rational $\beta = \frac{m}{n}$ in lowest terms we similarly get $\frac{F(\beta)-F(\alpha)}{\beta-\alpha} = \frac{n^{-1}-l^{-1}}{\beta-\alpha} \to \frac{0-l^{-1}}{0^+} = -\infty$. Similarly, $F'_-(\alpha)$ does not exist for irrational $\alpha \in \mathbb{R}$ and $F'_-(\alpha) = +\infty$ for $\alpha \in \mathbb{Q}$.

7. Since F is not monotone on any non-trivial interval, the only maximal intervals of monotonicity are the singletons $\{a\}$, $a \in \mathbb{R}$. As for the extremes, the function F has at every irrational α non-strict global minimum, and at every $\alpha \in \mathbb{Q}$ strict local maximum. Also, at every $\alpha \in \mathbb{Z}$ the function F has non-strict global maximum. F has no other (local or global) extreme. **8.** The image of F is the set $\{\frac{1}{n}\colon\ n \in \mathbb{N}\} \cup \{0\}$. **9.** Like in step 7, maximal intervals of convexity and concavity are the singletons $\{a\}$, $a \in \mathbb{R}$. **10.** Since there are no tangents, there are no inflection points. **11.** F has no asymptotes. **12.** Here are eleven topmost points of G_F in $[0, 1] \times \mathbb{R}$:



Exercise 8.7.8 The description of MDMs of $F(x) = r(x)$ is not simple. It suffices to describe those on which F weakly increases. Since F is even, from them we easily get those on which F weakly decreases. It is easy to see that the MDMs on which F is constant is exactly the set $\{\alpha \in \mathbb{R}\colon\ \alpha \text{ is irrational}\}$, together with the sets $\{\frac{m}{n} \in \mathbb{Q}\colon\ m \in \mathbb{Z} \wedge \mathrm{GCD}(m, n) = 1\}$, $n \in \mathbb{N}$.

We describe the MDMs where F weakly increases and is not constant. We introduce sequences of fractions in lowest terms on which F is constant. Let $n \in \mathbb{N}$. A $C(n, -\infty)$-sequence is a left-infinite sequence of fractions $S = (\cdots < \frac{m_3}{n} < \frac{m_2}{n} < \frac{m_1}{n})$ such that S contains all lowest terms fractions $\frac{m}{n} \le \frac{m_1}{n}$. We set $l(S) \equiv \frac{m_1}{n}$ to be the last term of S. Similarly, $S = (\frac{m_1}{n} < \frac{m_2}{n} < \frac{m_3}{n} < \dots)$ is a $C(n, +\infty)$-sequence if S contains all lowest terms fractions $\frac{m}{n} \ge \frac{m_1}{n}$. We set $f(S) \equiv \frac{m_1}{n}$ to be the first term of S. For $k \in \mathbb{N}$ with $k \ge 2$ we say that $S = (\frac{m_1}{n} < \cdots < \frac{m_k}{n})$ is a $C(n, k)$ sequence if S contains all lowest terms fractions $\frac{m}{n}$ with $\frac{m_1}{n} \le \frac{m}{n} \le \frac{m_k}{n}$. Again, $f(S)$ and $l(S)$ are the first and last term of S, respectively. Note that for every S the restriction $F \mid S$ is constantly $\frac{1}{n}$ and that every S is internally MDM, no $\alpha \in \mathbb{R}$ lying inside S but not in S can be added to S because then $F \mid S \cup \{\alpha\}$ is not monotone.

We introduce similar sequences of fractions in lowest terms on which F increases. An $I(-\infty)$-sequence is a left-infinite sequence of fractions $S = (\cdots < \frac{m_3}{n_3} < \frac{m_2}{n_2} < \frac{m_1}{n_1})$ such that $\cdots > n_3 > n_2 > n_1$ and there is no fraction $\frac{m}{n}$ in lowest terms with $\frac{m_{i+1}}{n_{i+1}} < \frac{m}{n} < \frac{m_i}{n_i}$ and $n_{i+1} > n > n_i$. We set $l(S) \equiv \frac{m_1}{n_1}$ to be the last term of S. We similarly define finite k-term, $k \in \mathbb{N}$ with $k \ge 2$, $I(k)$-sequences $S = (\frac{m_1}{n_1} < \cdots < \frac{m_k}{n_k})$ and their first and last terms $f(S)$ and $l(S)$. Again every such S is internally MDM. Note that that there is no $I(+\infty)$-sequence. An $I(-\infty)$-sequence can run to the left to $-\infty$, an example is given by the sequence $(\cdots < -\frac{n^2+1}{n} < \cdots < -\frac{5}{2} < -\frac{2}{1})$.

Now we describe the structure of any MDM X ($\subset \mathbb{R}$) of F on which F weakly increases and is not constant. It has the form

$$X = AB_1B_2 \dots B_kC$$

where A is the part going from $-\infty$, $k \in \mathbb{N}_0$, $B_1B_2\dots B_k$ is the alternating part consisting of finite segments, which is empty for $k = 0$, and C is the part going to $+\infty$. For the set A we have three possibilities: (i) $A = C(n, -\infty)$ for some $n \in \mathbb{N}$, (ii) $A = I(-\infty)$ going to the left to $-\infty$ or (iii) $A = \{\alpha \in \mathbb{R}\colon\ \alpha \text{ is irrational and } \alpha < c\}$ for some $c \in \mathbb{R}$. In $B_1B_2\dots B_k$, constant and increasing rational sequences alternate. The set C is always a $C(n, +\infty)$-sequence.

In case (i), we have $k = 2j - 1$ with $j \in \mathbb{N}$, B_1, B_3, $\dots$ are $I(k_1)$-, $I(k_3)$-, $\dots$ sequences and B_2, B_4, $\dots$ are $C(n_2, k_2)$-, $C(n_4, k_4)$-, $\dots$ sequences such that $l(A) = f(B_1)$, $l(B_1) = f(B_2)$, $\dots$, $l(B_{k-1}) = f(B_k)$, and C is a $C(n_{k+1}, +\infty)$-sequence such that $l(B_k) = f(C)$. The concatenation $AB_1B_2\dots B_kC$ is done in such a way that the listed equal first and last elements are identified. In case (ii), we have $k = 2j$ with $j \in \mathbb{N}_0$, the alternating part may be empty, and the constant and increasing rational sequences alternate in it in the other order. In case (iii), we have the subcase (iv) with c irrational and the subcase (v) with $c \in \mathbb{Q}$. In subcase (iv) we have $k = 2j - 1$ with $j \in \mathbb{N}$, B_1 is an $I(-\infty)$-sequence with infimum c, B_3, B_5, $\dots$ are finite increasing rational sequences and B_2, B_4, $\dots$ are finite constant rational sequences. In subcase (v) we may have both $k = 2j - 1$ with $j \in \mathbb{N}$ and $k = 2j$ with $j \in \mathbb{N}_0$, but $f(B_1)$, respectively $f(C)$, equals c, and in $B_1B_2\dots B_k$ finite constant and increasing rational sequences alternate in the appropriate order.

Exercise 8.7.9 **0.** $F \in \text{SEF}$. **1.** $M(F(x)) = M(\log x) = (0, +\infty)$. **2.** This function is not of a special form. **3.** We have $F \in \mathcal{C}$ and $D(F) = M(F)$ because $F \in \text{SEF}$. The derivative is $F'(x) = (1 + \log x)F(x) = (1 + \log x)x^x$. **4.** $\lim_{x\to 0} F(x) = \cdots = 1$ because $\lim_{x\to 0} x \log x = 0$. Clearly, $\lim_{x\to+\infty} F(x) = +\infty$. **5.** G_F is disjoint to both coordinate axes. **6.** Since $F \in \text{SEF}$, we have $D(F) = M(F)$ and there is nothing to compute. **7.** We equate the F' found in step 3 to 0 and get that $F' < 0$ on $(0, \frac{1}{\mathrm{e}})$ and $F' > 0$ on $(\frac{1}{\mathrm{e}}, +\infty)$. Thus the maximal intervals of monotonicity of F are the intervals $(0, \frac{1}{\mathrm{e}}]$ and $[\frac{1}{\mathrm{e}}, +\infty)$. On the former F decreases and on the latter it increases. At $x = \frac{1}{\mathrm{e}}$ the function F has a strict global minimum with the value $1/\mathrm{e}^{1/\mathrm{e}}$. It follows that F has no other (local or global) extreme. In particular, there is no local maximum: if $x \in (0, \frac{1}{\mathrm{e}}]$ then $F(y) > F(x)$ for every $y \in (0, x)$, and if $x \in [\frac{1}{\mathrm{e}}, +\infty)$ then $F(y) > F(x)$ for every $y \in (x, +\infty)$. **8.** By steps 4 and 7 and continuity of F the image of F is the interval $[1/\mathrm{e}^{1/\mathrm{e}}, +\infty)$. **9.** We have $F''(x) = (\frac{1}{x} + (1 + \log x)^2)x^x$. Since $F'' > 0$ on $(0, +\infty)$, F is convex on its definition domain. **10.** It follows from the previous step that F has no inflection. **11.** It is clear that F has no vertical asymptotes. Since $\lim_{x\to+\infty} \frac{F(x)}{x} = +\infty$, by Exercise 8.7.4 the function F does not have asymptote at $+\infty$. **12.** `https://www.desmos.com/calculator`.

9 Taylor polynomials and series. Real analytic functions

Exercise 9.1.1 If $f(x) = f(b) + c(x - b) + o(x - b)$ $(x \to b)$, then, by the definition of the symbol o, we have $\lim_{x\to b} \frac{f(x)-(f(b)+c(x-b))}{x-b} = 0$. This by arithmetic of functional limits is equivalent with $\lim_{x\to b} \frac{f(x)-f(b)}{x-b} = c$. In the other directions, if $\lim_{x\to b} \frac{f(x)-f(b)}{x-b} = c$, then arithmetic of functional limits gives that

$$\lim_{x\to b} \frac{f(x)-(f(b)+c(x-b))}{x-b} = 0\,.$$

Hence $f(x) = f(b) + c(x - b) + o(x - b)$ $(x \to b)$.

Exercise 9.1.2 If f is continuous at b, then $\lim_{x\to b} f(x) = f(b)$ (we are assuming that b is not an isolated point of M). Arithmetic of functional limits gives that $\lim_{x\to b} \frac{f(x)-f(b)}{1} = 0$. Thus $f(x) = f(b) + o(1)$ $(x \to b)$. The opposite implication is proven similarly.

Exercise 9.1.5 This follows from the fact that if $m < n$ then $\sum_{j=m+1}^{n} a_j(x-b)^j + o((x-b)^n) = o((x-b)^m)$.

Exercise 9.1.8 The assumption of HR 2 requires that the limit is of the type $\frac{0}{0}$.

Exercise 9.1.10 l'Hospital rule HR 2 works for these one-sided neighborhoods,

Exercise 9.1.11 $T_m^{p,0} = \sum_{j=0}^{m} a_j x^j$, where for $j > n$ we set $a_j \equiv 0$.

Exercise 9.1.13 The coefficients a_j in $T_n^{f,0}(x) = \sum_{j=0}^{n} a_j x^j$ are $a_0 = a_1 = a_2 = a_3 = 0$, $a_4 = 1$ and $a_j = 0$ for $j \ge 5$. As for the derivatives, $f^{(0)}(0) = f(0) = 0$, $f'(0) = 0$ and $f^{(j)}(0)$ does not exist for any $j \ge 2$ because $L(D(f)) = L(\{0\}) = \emptyset$.

Exercise 9.2.2 Now $T_n^{f,0}(x) = \sum_{j=0}^{n} x^j$ and $\lim T_n^{f,0}(2) = +\infty$, but $f(2) = -1$.

Exercise 9.2.6 For $|x| < 1$ we have $\sum_{j=0}^{\infty} x^j = \sum_{j=0}^{n} x^j + \frac{x^{n+1}}{1-x}$.

Exercise 9.2.7 Just multiply $(-1)^j$ and the numerator of the generalized binomial coefficient on the left-hand side.

Exercise 9.2.10

Exercise 9.3.1

Exercise 9.3.2

Exercise 9.3.3

Exercise 9.3.6 This Taylor polynomial equals $(0+0) + (1+1)x + (0+0)x^2 + (-\frac{1}{3} - \frac{1}{6})x^3 = 2x - \frac{1}{2}x^3$.

Exercise 9.3.7

Exercise 9.3.9

Exercise 9.3.32 Near 0 we have $\sqrt{1 + \sin x} = \sin(\frac{x}{2}) + \cos(\frac{x}{2})$.

Exercise 9.3.33 $\tan x$ is an odd function.

Exercise 9.3.34

Exercise 10.1.8

Exercise 10.2.5 $\{\emptyset\}$.

Exercise 10.2.6 Let $X \in \binom{[m+n]}{m}$. We define the maps $\kappa\colon [m] \to [m+n]$ and $\lambda\colon [n] \to [m+n]$ by $\kappa(j) \equiv$ the j-th element of X (in the standard ordering of $\mathbb{N}$), and similarly by $\lambda(j) \equiv$ the j-th element of $[m+n] \setminus X$. Now we can define the injection ι by

$$\langle\langle A_1, \ldots, A_k\rangle, \langle B_1, \ldots, B_l\rangle, X\rangle \mapsto \langle \kappa[A_1], \lambda[B_1], \kappa[A_2], \lambda[B_2], \ldots\rangle\,.$$

Exercise 10.5.4

10 Primitives of UC functions

Exercise For example, $F(x) \equiv 2\sqrt{x}$ and $f(x) \equiv 1/\sqrt{x}$. Then $F' = f$ but $M(F) \neq M(f)$.

14 More applications of Riemann integrals

Exercise For example, $g(t) = (\cos t, \sin t)$ for t in $[0, \pi]$ and $h(x) = (x, f(x))$ for x in $[-1, 1]$.

Exercise For example, $f(t) = t(\overline{b} - \overline{a}) + \overline{a}$ for $t \in [0, 1]$.

Exercise For example, $\varphi(t) = (t, 2\cos(10\pi t), 2\sin(10\pi t))$ for $t \in [0, 1]$.

Exercise These are exactly constant maps from intervals $[a, b]$ to $\mathbb{R}^d$.

Exercise This length is $b - a$.

Exercise If $c \in (a, b)$ is such that $\varphi(c) \notin \overline{a}\overline{b}$, then $\|\varphi(c) - \varphi(a)\| + \|\varphi(b) - \varphi(c)\| > \|\overline{b} - \overline{a}\|$.

Exercise

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